

Reimagining Gravity: Generalized Symmetries, Double Copy, and Spinning Black Holes

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The Caltech logo, consisting of the word "Caltech" in a bold, orange, sans-serif font.

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To my mother, who gave me a canvas of imagination.

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ABSTRACT

General relativity is a century-old subject. Yet modern explorations through generalized symmetries, scattering amplitudes, and effective field theory have raised open problems, motivating a reassessment of conventional views on gravitation, spacetime, and spin. First, do generalized symmetries exist in dynamical gravity as a low-energy effective field theory? Second, is there a field-theoretic explanation for the tree-level double copy relationship between general relativity and Yang-Mills theory? Third, what are the exact equations of motion or Lagrangians of four-dimensional spinning black holes in external fields, in their point-particle effective theory?

In this dissertation, three different perspectives on gravity are developed to shed light on each of these puzzles.

First, we view gravity as a gauge theory of the Lorentz group. This refers to the vielbein formulation of gravity. It implies that the effective field theory of gravity, at the fully nonlinear level, exhibits a one-form symmetry valued in the center of the local Lorentz group. The charged objects are the Wilson loops of the spin connection, i.e., spin holonomies in various representations. The symmetry operator is a topological operator associated with a certain area measured in Planck units. Their linking admits an elegant physical interpretation in classical gravitation: the symmetry operator materializes a tetradic cousin of cosmic string that induces quantized spin precession angles as a gravitational Aharonov-Bohm effect. Notably, our construction identifies a new symmetry of the standard model at scales below the lightest neutrino mass. The absence of global symmetries in quantum gravity suggests that this gravitational one-form symmetry is either gauged or explicitly broken, the latter of which mandates the existence of fermions.

Second, we investigate how far gravity can be viewed as a gauge theory of diffeomorphisms. Based on an established result on Born-Infeld theory, we point out that color-kinematics duality at the equations of motion level mandates treating the diffeomorphism algebra formally like a gauge algebra in the internal, fiberwise sense of Yang-Mills theory. We show that this seemingly radical prescription admits a concrete construction within a mathematically consistent framework: field theory of a dynamical frame field. We demonstrate that color-kinematics duality systematically defines diffeomorphism gauge connec-

tion, diffeomorphism covariant derivative, and even diffeomorphism Wilson line. In this context, we revisit the teleparallel formulation of gravity and evaluate its relevance to double copy in three or four dimensions. The Misner string is reinterpreted as a topological classical solution in any diffeomorphism gauge theory, endable on Newman-Unti-Tamburino charges.

Third, we view four-dimensional gravity as a nonlinear interaction between self-dual and anti-self-dual parts. We begin by reviewing the simplicity of self-dual gravity in the context of double copy, Plebański's second heavenly equation, and Mason-Newman or Lax pair formulations. This motivates the program of understanding full gravity by perturbing away from the simpler self-dual sector, an insight often advocated by twistor theorists. Chiral formulations of gravity are discussed in this context, via both Lorentz and diffeomorphism gauge theory perspectives. In the former, we review Plebański gravity. In the latter, we derive a Chalmers-Siegel version of teleparallel gravity that perturbs away from self-dual gravity in the Mason-Newman characterization. We then show how this view on four-dimensional gravity can be applied to the derivation of four-dimensional spinning black hole solutions as well as their dynamics in point-particle effective theory.

We begin by a pedagogical demonstration of the simplicity of self-dual backgrounds. The motion of a relativistic charged particle in the background of a self-dual dyon is shown to be maximally superintegrable by being isomorphic to the hydrogen atom. Similarly, the motion of a relativistic particle in a self-dual black hole background is maximally superintegrable by being isomorphic to the Kepler problem.

A precise relation between these two backgrounds is established by the Kerr-Schild double copy. We derive a Kerr-Schild metric for the self-dual Taub-Newman-Unti-Tamburino solution by double copying a gauge potential of the self-dual dyon. A simple complexified coordinate transformation is constructed to explicitly show that this new metric is diffeomorphic to the well-known Gibbons-Hawking instanton metric.

By taking this result as a crucial lemma, it is then shown that the Kerr metric represents the nonlinear superposition of self-dual and anti-self-dual Taub-Newman-Unti-Tamburino solutions. This elevates the Newman-Janis algorithm to a rigorous derivation of the Kerr metric and explicates its origin. More generally, the five-parameter family of solutions including Kerr-Newman and

Kerr-Taub-Newman-Unti-Tamburino black holes represents systems of Taub-Newman-Unti-Tamburino instantons and chiral dyons.

Based on this realization, we propose a probe counterpart of the Newman-Janis algorithm, which uniquely constrains the effective equations of motion of Kerr and Kerr-Newman black holes in external gravitational and electromagnetic fields within the self-dual sector. It is shown that these effective equations of motion are maximally superintegrable in the backgrounds of self-dual black holes, via dynamical Newman-Janis shifts of conserved charges. This identifies an exactly solvable subsector of the general-relativistic spinning black hole binary problem. It also identifies a hidden symmetry as an infrared principle that can pinpoint black holes among all massive spinning objects in the point-particle effective theory, within the self-dual sector.

In the Lagrangian formulation, we construct a unique class of effective world-line actions for Kerr and Kerr-Newman black holes in generic, non-self-dual external fields. We provide their explicit all-orders formulae in terms of expansions in curvature and spin length. This is facilitated by developing an “in-in” formalism for all-orders geodesic deviation as an alternative to the Synge framework.

A universal framework arises for spinning-particle mechanics in which spin is added to spacetime as the imaginary part while chirality and holomorphy are inherently linked. We demonstrate how exact spinning black hole equations of motion are derived beyond the self-dual sector.

Finally, we derive the classical Compton scattering amplitudes of spinning black holes, directly from the explicit effective point-particle actions by implementing a chiral worldline perturbation theory in which the plus-helicity spin exponentiation is Lagrangian-level manifest to all orders. This provides an invariant characterization of the effective dynamics of spinning black holes. Our amplitudes are free of spurious poles and exhibit correct residues on physical factorization channels, for both same-helicity and mixed-helicity configurations. Physically, our computation views the Kerr black hole as a dynamical Taub-Newman-Unti-Tamburino pair and systematically understands it by expanding around the self-dual Taub-Newman-Unti-Tamburino worldline.

These explorations spark a new approach to spinning black hole dynamics that makes maximal use of the simplicity and the manifest Newman-Janis property of the self-dual sector, finding applications to post-Minkowskian gravity.

PUBLISHED CONTENT AND CONTRIBUTIONS

Chapters III, V, VI, VII, VIII, IX, X, XI, XII, and XIII of this thesis are based on the published articles and publicly available preprints [1–9], for all of which I served as the primary author. Chapter II is a review written specifically for this thesis as an original synthesis of known formulations of gravity; Sections 2.5.3 and 2.5.6 go beyond review and contain original constructions and calculations developed by the author. Chapters IV and XIV are based on original research conducted by the author, unpublished at the time of submission for this thesis. Appendices A and B are expository reviews that draw on known results as well as research conducted by the author, independently or in collaboration.

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- [8] J.-H. Kim, “Asymptotic spinspace-time,” *Phys. Rev. D* **111** no. 10, (2025) 105011, [arXiv:2309.11886 \[hep-th\]](#).
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INTRODUCTION

1.1 Gravitation from the Infrared to the Ultraviolet

General relativity (GR) is a century-old subject. It stands as one of the great triumphs of modern physics, not only for its theoretical breakthroughs that reformed our conception of time, space, and gravitation but also for its remarkable experimental success. Its earliest victories came from explaining the anomalous perihelion precession of Mercury and the bending of starlight by the Sun, dramatically confirmed during the 1919 solar eclipse [1–3]. Later experiments further consolidated this success, confirming the gravitational redshift and time delay [4, 5]. The ability to correctly predict such “relativistic clock effects” is necessary for the precision of the global positioning system, on which many of our everyday activities heavily depend [6]. In stronger gravitational fields, striking indirect evidence for gravitational radiation was provided by detecting a decrease in the orbital period of a binary pulsar [7, 8], leading to the 1993 Nobel Prize in physics. Eventually, the direct detection of gravitational waves by LIGO and Virgo opened an entirely new observational window onto dynamical spacetime [9, 10], as is recognized by the 2017 Nobel Prize.

Black holes have provided another arena of triumph. Penrose’s singularity theorem established black hole formation as a robust prediction of GR [11]. A supermassive compact object, named Sagittarius A*, was discovered at the center of the Milky Way by monitoring stellar orbits over nearly three decades [12, 13]. These achievements were honored with the 2020 Nobel Prize. More recently, horizon-scale black-hole imaging has provided further consistency checks on the idea that the astrophysical black holes we observe are, to excellent approximation, described by the Kerr metric [14].

Together, these observations throughout a century have made GR the most successful classical theory of gravity, a theory whose geometric vision continues to withstand increasingly stringent tests [15].

Now turning to the history of the theoretical side, it should be noted that GR has been derived and illuminated from two complementary perspectives. To one tradition, GR is Einstein’s profound *geometrization* of gravity [16]. To

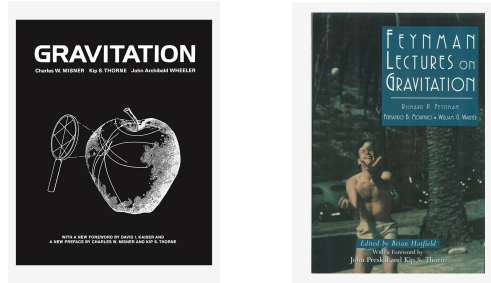


Figure 1.1: Textbooks on GR representing the two schools: the relativists and the field theorists.

another, GR is the unique long-range interacting theory of a massless spin-2 field [17–19]. The former refers to Einstein’s original derivation, which describes a highly nonlinear pathway shaped by paradoxes, thought experiments, and geometric inspirations. The latter refers to the field theorists’ rederivation of Einstein’s artwork *from scratch*, which claims to provide a much more streamlined and principled approach.

The former view, which we may refer to as the *relativist* tradition, could be well-represented by the textbook [20] by Misner, Wheeler, and Thorne. It is written in its characteristic visual and geometric style, providing a guided tour from the concepts of metric and connection to Riemann curvature and Einstein tensor. Gravitation is the dynamical geometry of spacetime itself. Matter tells spacetime how to curve, and spacetime tells matter how to move.

The latter view, which we may refer to as the *field theorist* tradition, is demonstrated in the textbooks written by Feynman [21] and Weinberg [22]. Starting from basic assumptions, one constructs a relativistic quantum field theory (QFT) of a spin-2 field and discovers that its unique classical nonlinear completion happens to be Einstein’s theory [17–19]. The geometrical visions of Einstein are deconstructed into logical necessities demanded for a consistent QFT. For example, instead of imagining space elevators and falling apples, the weak equivalence principle is derived by stipulating the existence of a Lorentz-invariant and gauge-invariant S-matrix containing a massless spin-2 particle and the standard soft-pole factorization of scattering amplitudes [23].

The boundary between the relativist school and the field theorist school is perhaps not strict these days; the best modern work often moves fluently between both, and this bilingual ability is beneficial. Yet the contrast could be sometimes useful to draw, as it highlights two very different instincts about what GR is.

The relativist approach to GR could be illuminated in the broader context of formulating and comprehending physical laws and phenomena in the geometrical language.¹ For gauge theories, such a geometric paradigm had been put forward by Wu, Yang, and Mills [24–28] by the late twentieth century, which had sparked fruitful developments in both physics [29–32] and mathematics [33–38]. Notably, it provided a unified geometrization of the three fundamental forces *except gravity*: electromagnetism, the weak, and the strong.

Regarding this separation, it is worth clarifying that the geometry of Yang-Mills (YM) theory is quite different than that of GR. YM theory governs a connection on a principal fiber bundle over a fixed spacetime, whereas GR governs a dynamical pseudo-Riemannian manifold as the spacetime itself. The redundancies involved in YM theory describe *internal* local transformations and refer to a particular, narrower geometric semantics that demands structures such as fiber bundles [24–32, 39]. The redundancies involved in GR are diffeomorphisms, which are *external* transformations that shift the very points of the base manifold.

It is sometimes said that GR is a gauge theory of local translations or diffeomorphisms [21, 40]. Strictly speaking, this is correct only in the colloquial or broader sense that it involves redundancies: the freedom to choose coordinates.

Despite this crucial discrepancy between gravity and the other three forces, historically, there had been aspirations toward reformulating GR as a gauge theory in the narrower sense [41–61], envisioning a unified geometrization of all fundamental forces.

The field theorist assessment of GR connects to the effective field theory (EFT) of gravity. An EFT is a QFT whose validity is restricted to a certain range of energy scales [62]. A paradigmatic example of an EFT is Fermi’s theory of weak interactions, valid at energies below the W-boson mass. Although nonrenormalizable, it gives a systematically improvable low-energy expansion whose breakdown merely signals the appearance of new short-distance degrees of freedom, namely the W and Z bosons. In the EFT, the effect of these mediator particles is encapsulated and hidden in a four-fermion contact interaction vertex. Another paradigmatic example is chiral perturbation theory, an EFT of pions below the quark confinement scale.

¹For instance, Misner, Thorne, and Wheeler [20], pp. 370–371, draw an analogy between gauge theory and gravity from the perspective of exterior calculus, through diagrams titled “Structure of Electrodynamics/Geometrodynamics in Outline Form.”

In the same spirit, perturbative quantum GR is understood as a low-energy EFT of the massless spin-2 field, i.e., the *graviton* [63–66]. Its perturbative nonrenormalizability, detected via the need of higher-derivative counterterms in loop calculations [67, 68], signals its own breakdown at the Planck scale, $M_{\text{Pl}} \sim 10^{19} \text{GeV} \sim \hbar/(10^{-35} \text{m})$. Crucially, however, this does not obstruct its use as a predictive low-energy QFT. Quantum corrections are systematically organized by a derivative expansion, equivalently by the powers of the low-energy scale over the Planck scale, E/M_{Pl} . The EFT framework separates universal infrared predictions from the unknown ultraviolet physics, ensuring finite predictability at each fixed order in the low-energy expansion.

When thinking along these lines, we are inevitably led to the ultimate question of quantum gravity. If GR breaks down at the Planck scale, what can be the ultraviolet (UV) theory of quantum gravity? More generally, what is the full space of low-energy EFTs that admits consistent UV completions into quantum gravity? Motivated by explorations through string theory, the swampland program [69] has identified and proposed a collection of properties that any consistent theory of quantum gravity should universally satisfy. They are known as swampland conjectures, stating, e.g., that no global symmetries are allowed [70–74], gravity is the weakest among all forces [75], and any charge that is not forbidden is mandatory [71, 73]. The swampland conjectures aim to tell whether a low-energy EFT admits a consistent UV completion into quantum gravity (lies in the “landscape”) or not (lies in the “swampland”).

The no global symmetry conjecture should concern not only ordinary symmetries acting on local operators but also *generalized symmetries* [76–84], including higher-form, noninvertible, and higher-group symmetries. In particular, p -form symmetries are symmetries that act on p -dimensional extended operators. Maxwell theory, for example, exhibits a global one-form symmetry acting on Wilson loops. The no global symmetry conjecture states that this one-form symmetry in the infrared (IR) will be lost in the regime where gravity becomes fully dynamical and quantum. As a result, it must be either gauged or explicitly broken. The latter implies the existence of charged particles, as Wilson lines are endable on electric charges. Ref. [85] gives a related discussion.

The existence of a p -form symmetry amounts to having a topological operator of codimension $(p+1)$. This is the modern definition of symmetry due to Ref. [76]. This topological operator establishes a Ward identity by linking with

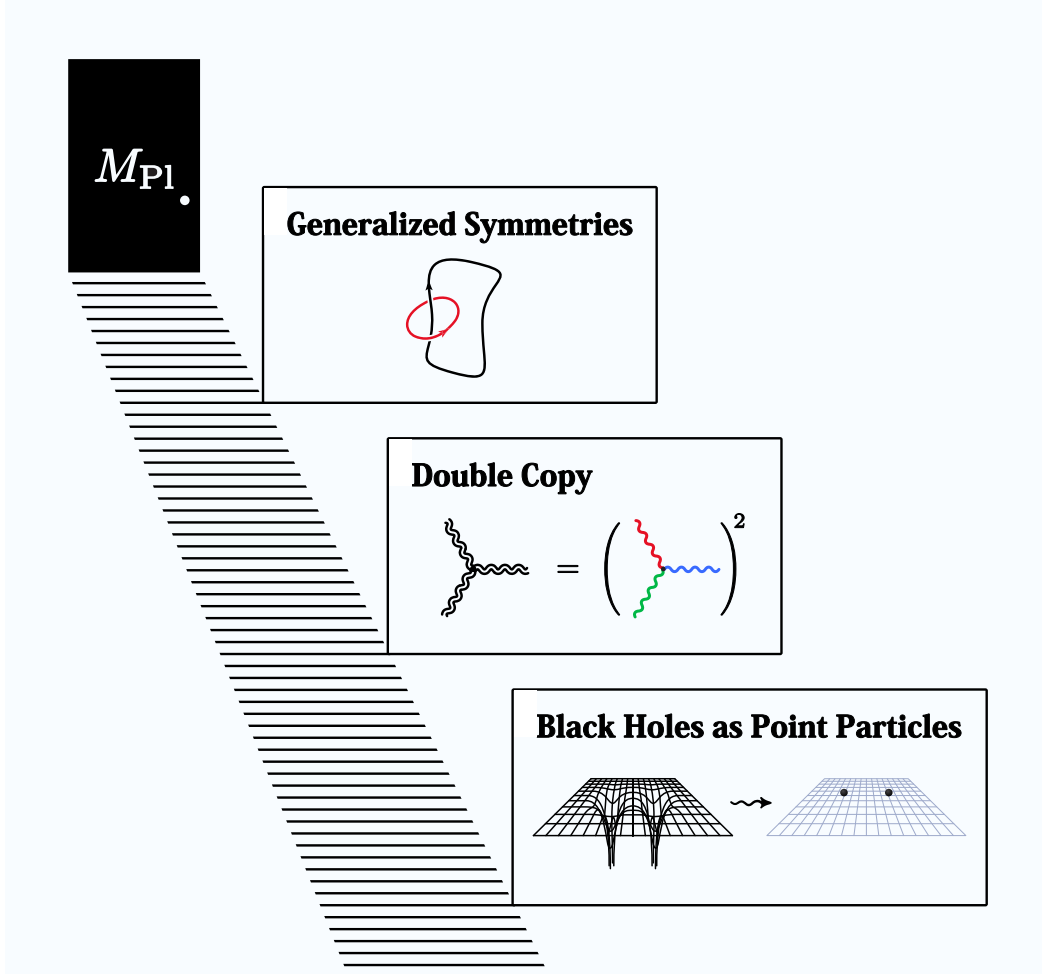


Figure 1.2: Topics in GR from the IR to the UV.

the p -dimensional charged operator. For the above example of Wilson loops in Maxwell theory, the topological operator arises by integrating the Hodge dual of the field strength over closed codimension two surfaces.

Returning to the low-energy regime below the Planck scale, one learns that two interesting topics have emerged from modern investigations of gravity through the lenses of scattering amplitudes and EFT: see Fig. 1.2. The first is *double copy*, relevant to pure GR as a perturbative QFT [86–90]. The second is *point-particle effective theory of black holes* [91–98], which zooms out to a further IR scale.

We may want to describe the latter first: point-particle effective theory of black holes. This topic has found fruitful practical applications in describing the classical gravitational dynamics of astrophysical bodies, relevant to the gravitational-wave physics taking place at LIGO and Virgo.

The general-relativistic two-body problem stands notoriously unsolved. Decades of research have devised various approximation schemes such as post-Newtonian (PN) [99–113], post-Minkowskian (PM) [114–125], and self-force (SF) [126, 127] expansions, while utilizing numerical relativity [128–130], effective one-body formalism [131, 132], and EFT methods [91–97, 114, 133–148].

How could an effective field theorist approach the general-relativistic two-body problem? Take a black hole binary, for instance. In the inspiral phase, the distance between the black holes, say b , is much larger than their typical Schwarzschild radius, say Gm . This separation of scales, $b \gg Gm$, facilitates an effective theory that replaces the curved, dynamical geometry describing the black hole binary with two point particles put on a flat Minkowski space with fields, as illustrated in Fig. 1.2. This is the point-particle effective theory [91–98], organized in the PM expansion in Gm/b . This effective theory integrates out scales smaller than the Schwarzschild radii while taking the exact general relativistic description of the black hole binary geometry as the UV theory.

As is well-reviewed in Ref. [143], the EFT approach to PM gravity establishes an efficient computational framework by importing state-of-the-art techniques in scattering amplitudes such as recursion [149, 150], generalized unitarity [151–153], and double copy [86–89]. Especially, the double copy provides radical shortcuts to gravitational amplitudes.

Our attention now turns to the double copy. The double copy is a remarkable property of perturbative quantum GR, observed and established at the level of scattering amplitudes so far [86–89]. Roughly speaking, it states that gravity is somehow the “square” of the strong force: “ $\text{GR} = (\text{YM})^2$.” It suggests that the entire nonlinearity of gravitational waves—usually represented in terms of an infinite collection of vertices of all valences in the graviton Feynman rules via the Einstein-Hilbert action—is in fact encoded in a sole *cubic* vertex.

The double copy has an important string-theoretic precursor, known as Kawai-Lewellen-Tye (KLT) relations [86]. Later, Bern, Carrasco, and Johansson (BCJ) [87, 88] showed that a purely field-theoretic formulation is viable. Their construction is based on the idea of color-kinematics (CK) duality: YM amplitudes can be represented as sums over cubic graphs in which the kinematic numerators obey the same algebraic relations (Jacobi identity) as the color factors. Replacing the color factors by a second copy of kinematic numerators then yields the graviton amplitudes. See Ref. [89] for a detailed review.

Cheung and Mangan [90] demonstrated that tree-level CK duality can be made manifest to all multiplicities if one identifies the relevant kinematic algebra and current at the level of classical equations of motion (EoM), in the framework of Berends-Giele (BG) recursion [154] (perturbative expansion [155–157]). When applied to GR, one encounters a surprisingly bold conjecture: one may be able to recast Einstein’s equations in the form of YM equations.

The physical significance is twofold. For the geometers, it proposes that there indeed can exist a unified geometric formulation that puts all fundamental forces on the same footing, not as a mere analogy or a formal mathematical similarity but a precise, quantitative correspondence leading to concrete computational outputs. In this sense, the double copy fixes a unique goalpost for the geometrization program and also provides explicit criteria for testing it. For field theory practitioners, the double copy is a powerful computational tool for gravitational amplitudes that circumvents the pain involved in using the Einstein-Hilbert action.

The nonperturbative aspects of double copy are far less explored [158, 159]. Yet correspondences between exact classical solutions in Maxwell theory and GR have been discussed under the name of “classical double copy.” In particular, a version known as Kerr-Schild (KS) double copy [160] ignited a modern discourse on KS metrics [161–175] and established a systematic use [164, 176] of its effective linearization [177, 178] property. For example, the KS double copy identifies the Schwarzschild solution as the gravitational counterpart of the electric monopole (Coulomb solution) [160]. Similarly, the electromagnetic field of the Kerr-Newman solution [179, 180], dubbed “ $\sqrt{\text{Kerr}}$ ” solution [98], double copies to the Kerr solution [160]. See also Refs. [181–185] for alternative constructions using double KS metrics, Weyl tensor, or twistor space.

To sum up, we have provided an overview of GR, outlining its experimental successes as well as its systematic theoretical organization from the IR to the UV. While introducing both the geometric and field-theoretic perspectives, we have encountered various topics as illustrated in Fig. 1.2. To question the UV, one utilizes the swampland conjectures with generalized symmetries. To predict gravitational phenomena in the IR for LIGO and Virgo, one utilizes the point-particle effective theory of black holes as well as the efficient shortcuts provided by amplitudes methods, such as the double copy. Having traveled a century, we now turn to remaining puzzles, which are the subject of this thesis.

1.2 Open Puzzles in Gravitation

Modern explorations of gravity through generalized symmetries, scattering amplitudes, and EFT have raised open problems. They motivate a reassessment of conventional views on gravitation, spacetime, and spin. Below, we state three.

1. Do generalized symmetries exist in dynamical gravity as a low-energy EFT?
2. Is there a field-theoretic explanation for the tree-level double copy relation between GR and YM theory?
3. What is the exact point-particle effective theory of four-dimensional spinning black holes in external fields?

The objective of this thesis is to answer these questions partially, or at least provide new approaches that can be useful or insightful. Our conclusions are summarized below. The progress on Problem 2 may be limited, but nevertheless, it served as an important element for tackling Problems 1 and 3 by providing unique perspectives and inspirations.

1. We found at least one generalized symmetry in dynamical gravity as a low-energy EFT [186]. This identifies a new symmetry of nature at scales below the lightest neutrino mass.
2. We did not succeed in finding such an explanation. However, we were able to make a useful observation that gives a geometrical picture to the double copy for Born-Infeld (BI) theory, in connection with teleparallel gravity.
3. We identified a hidden symmetry in the IR (point-particle effective theory) that uniquely characterizes spinning black holes among all massive spinning objects, yet within the self-dual (SD) sector.

Beyond the SD sector, we constructed minimal proposals for the exact—meaning all orders in spin and coupling—effective actions of Kerr and (nonabelian) Kerr-Newman black holes in backgrounds with gauge and gravitational fields, in the conservative sector. We obtained their explicit formulae by developing a new formalism for all-orders geodesic deviation. We proposed a chiral approach to spinning black hole effective dynamics that maximizes the utility of the simplicity in the SD sector.

Below, we define the above three problems in detail and clarify the necessary or sufficient conditions for their resolution.

1.2.1 Generalized Symmetries in Dynamical Gravity

**Do generalized symmetries exist in dynamical gravity
as a low-energy EFT?**

Goal Find generalized symmetries in pure GR (with higher-curvature corrections) by treating it as a low-energy EFT of gravity. There may or may not be more than one. This problem presumes working in the regime where the dimension and topology of spacetime are strictly fixed. The gravitational path integral will not implement the sum over spacetime topologies.

Problem solved if

- (a) The charged operators and their transformation behavior are identified,
- (b) The topological symmetry operator is identified,
- (c) The Ward identity between the two is concretely established in either the covariant (path integral) or the canonical (Hamiltonian) formalisms.

We may focus on higher-form symmetries for simplicity, although the possibilities of finding more general symmetries should not be precluded. A degree of concreteness may be desired or demanded, in which case one has to construct the charged operators and the symmetry operator explicitly in terms of the local degrees of freedom of the gravitational EFT.

Significance

- The no global symmetry conjecture implies that any global symmetry observed in a low-energy EFT must either be gauged or explicitly broken. In the latter scenario, one could argue that specific types of matter fields are required to break the gravitational higher-form symmetry.
- If found in $d = 4$ spacetime dimensions, one can be reporting a new symmetry in our universe that is unbroken below some mass scale.
- It may shed a new light on the definition of gravitational charge in general relativity, which has been a classic subject to relativists [187–196] (see, e.g., Refs. [197, 198] for recent attempts).

1.2.2 Double Copy

Is there a field-theoretic explanation for the tree-level double copy relationship between YM theory and GR?

Problem solved if

- (a) Vacuum Einstein's equations with no cosmological constant can be perturbatively reformulated as YM equations on flat background with the same spacetime dimension $d > 2$ whose gauge algebra is replaced with a Lie algebra $\mathfrak{g}_{\mathcal{P}}$ that may or may not be infinite-dimensional.

This appears as a bold claim, but it is indeed a valid conjecture for explaining the BCJ duality [87–89] according to Ref. [90]'s formulation. The mystery Lie algebra $\mathfrak{g}_{\mathcal{P}}$ is referred to as the *kinematic algebra* of YM theory. It is plausible that $\mathfrak{g}_{\mathcal{P}}$ can be infinite-dimensional, as it eventually has to take momenta and polarizations as the adjoint indices to construct the kinematic numerators. The reformulation should be invertible on external (asymptotically free) states.

Problem solved also if

- (b) YM equations can be perturbatively reformulated as bi-adjoint scalar (BAS) equations by replacing one of its Lie algebras with $\mathfrak{g}_{\mathcal{P}}$.

Here, flat spacetime is assumed with dimension $d > 2$. A BAS field is a scalar field $\Phi^{a\tilde{a}}$ carrying two adjoint indices $a = 1, \dots, \dim \mathfrak{g}$ and $\tilde{a} = 1, \dots, \dim \tilde{\mathfrak{g}}$ for two Lie algebras $\mathfrak{g}$ and $\tilde{\mathfrak{g}}$. Their structure constants f^a_{bc} and $\tilde{f}^{\tilde{a}}_{\tilde{b}\tilde{c}}$ must satisfy the Jacobi identity by definition. The BAS equations are

$$\text{BAS:} \quad \square \Phi^{a\tilde{a}} = \frac{1}{2} f^a_{bc} \tilde{f}^{\tilde{a}}_{\tilde{b}\tilde{c}} \Phi^{b\tilde{b}} \Phi^{c\tilde{c}}. \quad (1.1)$$

Problem solved iff one is able to generate tree-level BCJ numerators [87, 88] for YM theory and GR to any number of external legs.

Significance

- Will bring immense practical benefit by radically simplifying gluon/graviton perturbation theory, summarizing all interactions into a single universal cubic vertex that merely states/squares the structure constant of $\mathfrak{g}_{\mathcal{P}}$.
- Might bring an unforeseen geometric interpretation of GR that understands gravity in the same fashion as the strong force, not merely as an analogy.

1.2.3 Effective Dynamics of Spinning Black Holes

What is the exact point-particle effective theory of four-dimensional spinning black holes in external fields?

Background The Kerr solution provides an excellent approximation to real-world astrophysical black holes [14]. While the point-particle effective theory of non-spinning black holes stands rather well-established [91–95, 199], the point-particle effective theory of spinning black holes [96–98] is far from complete.

Problem solved iff

- (a) The point-particle effective action of the Kerr black hole in gravitational backgrounds is determined to all orders in spin length and coupling.
- (b) The point-particle effective EoM of the Kerr black hole in gravitational backgrounds are determined to all orders in spin length and coupling.

This problem represents a simple enough yet fundamental curiosity: what is the law of motion governing black holes? Concretely, the EoM of Kerr can be given in the following form:

$$\mathbf{Kerr} : \quad \dot{x}^\mu = ?, \quad \frac{Dp_\mu}{d\tau} = ?, \quad \frac{Dy^\mu}{d\tau} = ? \quad (1.2)$$

In the effective theory, the Kerr black hole is represented by a massive spinning particle. In Eq. (1.2), x^μ and p_μ are the position and momentum of this particle while $y^\mu = *S^{\mu\nu}p_\nu/(-p^2)$ is the particle's Pauli-Lubanski spin pseudovector normalized in the units of length, known as *spin length pseudovector*. Namely, y^μ recasts the ring radius $a = S/M$ as a four-pseudovector. Specifying the time evolution of x^μ , p_μ , and y^μ as in Eq. (1.2) uniquely determines the classical dynamics of the Kerr black hole in the effective theory.

Previous Results At the linear order in the gravitational coupling (linearized gravity), the most general point-particle effective action for a relativistic massive spinning body was established by Levi and Steinhoff [97]. In our presentation, their interaction action is

$$\int \sum_{\ell=2}^{\infty} \frac{C_\ell}{\ell!} p_\mu *^\ell R^\mu_{\rho_1\rho_2\sigma;\rho_3;\dots;\rho_\ell}(x) y^{\rho_1} y^{\rho_2} y^{\rho_3} \dots y^{\rho_\ell} \dot{x}^\sigma d\tau, \quad (1.3)$$

where $*^\ell R^\mu_{\nu\rho\sigma}$ denotes the Riemann tensor acted by the Hodge star ℓ times.

The dynamical variables in Eq. (1.3) are x^μ , p_μ , and y^μ . The constants C_ℓ are Wilson coefficients capturing spin-induced mass 2^ℓ -pole moments $C_\ell |p||y|^\ell$ (electric for even ℓ and magnetic for odd ℓ).

For the Kerr black hole, classic works [200–205] had established that the 2^ℓ -pole moment of the Kerr black hole is Ma^ℓ , which implies $C_\ell = 1$. This “unity” of multipole coefficients has also been derived via modern amplitudes methods [98, 206–209].

At the quadratic order in the coupling, however, Arkani-Hamed, Huang, and Huang (AHH) [206] reported a difficulty in fixing the effective theory from amplitudes methods, formulated in terms of a mixed-helicity Compton amplitude. Since then, a substantial body of works [98, 206–225] has been devoted to the problem. In works [98, 226], the Newman-Janis (NJ) property [202] of the Kerr solution has been identified as an important clue.

Criteria

- (a) The Kerr effective action should reduce to the Levi-Steinhoff [97] action at the leading order in the gravitational coupling.
- (b) The Kerr effective EoM should reduce to an all-orders-in-spin extension of Mathisson-Papapetrou-Dixon (MPD) [227–229] equations (cf. Refs. [230–238]) for $C_\ell = 1$ at the leading order in the gravitational coupling.
- (c) The Kerr effective action or EoM may reflect the NJ property, which will assign certain meanings to the complex combinations $x^\mu \pm iy^\mu$.

Problem solved if

- (a) An IR principle (e.g., symmetry) is shown to constrain the effective theory entirely.
- (b) All Wilson coefficients are determined by matching with the UV theory, which is GR. This entails solving black hole perturbation theory [220–223, 239, 240].
- (c) One is able to predict the gravitational Compton amplitudes of the Kerr black hole to any graviton multiplicities and helicities.

Significance

- Quenches a fundamental curiosity in classical gravitation and relativistic spinning-particle mechanics.

- Is directly relevant to the relativistic two-body problem for spinning black holes.
- Is directly relevant to precision PM calculations if one is interested in the large spin or spin-resummed regime. The gravitational Compton amplitude, for instance, can be directly fed into the 2PM two-body eikonal.

Related Problems

- Investigate spinning black holes with electromagnetic or YM hairs. The abelian $\sqrt{\text{Kerr}}$ solution is the Maxwell analog of the Kerr black hole by classical double copy, whose trivial embedding into YM theory is the nonabelian $\sqrt{\text{Kerr}}$ solution [160]. These are the $G_N \rightarrow 0$ limit of the Kerr-Newman solution [98, 179] and its nonabelian generalization [241]. Determine the point-particle effective action or EoM of abelian/nonabelian $\sqrt{\text{Kerr}}$ or Kerr-Newman black holes in external fields.
- The Newman-Janis algorithm (NJA) [202] is a solution-generating technique that wishes to add angular momentum to four-dimensional black holes. In a sense, it “derives” the Kerr metric by applying an imaginary displacement “ $+ia$ ” to the Schwarzschild metric. This procedure was originally introduced as an ad hoc trick without a clear physical basis [202]. However, it holds a historical significance as the very crucial method that enabled the derivation of the very Kerr-Newman solution [179]. Figure out whether the NJA is a mere mathematical trick or an indication of a deeper physical structure (cf. Refs. [242–251], Refs. [252–259], Refs. [98, 206–209], and Refs. [260, 261]). If there is a physical content, investigate if it can help determine the point-particle effective theory of spinning black holes.
- The relativistic two-body problem is notoriously hard to solve. In particular, one does not expect to solve the relativistic two-body problem for a spinning binary formed by random stellar objects. Black holes, however, are believed to be special as “the most perfect macroscopic objects” in the universe (in the words of Chandrasekhar [262]). Do spinning black holes exhibit a certain ideal property that allows room for integrability in their binary problem, under particular assumptions or in specific regimes? If so, how is that property stated in the point-particle effective theory?

1.3 Plan of This Thesis

This thesis is organized as follows.

- Part 0. Reimagining Gravity
- Part 1. Generalized Symmetries (PROBLEM 1 in Sec. 1.2.1)
- Part 2. Double Copy (PROBLEM 2 in Sec. 1.2.2)
- Part 3. Spinning Black Holes (PROBLEM 3 in Sec. 1.2.3)

The contents of each part are summarized below.

- Part 0. Chap. II reviews, derives, and envisions various formulations of GR in general and four dimensions: metric-based formulations (Sec. 2.1), vielbein formulation (Sec. 2.2), teleparallel formulation (Sec. 2.3), SD gravity and its simplicity (Sec. 2.4), chiral and related formulations (Sec. 2.5). This reimagines gravity as a Lorentz gauge theory, a diffeomorphism gauge theory, or a tapestry of SD and anti-self-dual (ASD) parts.
- Part 1. Chap. III constructs a one-form symmetry in dynamical gravity. This identifies a new symmetry of our nature at scales below the lightest neutrino mass. It views gravity as a Lorentz gauge theory. It introduces a tetradic cousin of cosmic string as a local Lorentz vortex.
- Part 2. Chap. IV investigates the idea of diffeomorphism gauge theory as an implication of EoM level CK duality. It wishes to view gravity as a diffeomorphism gauge theory, but results are only exact in three dimensions or in the SD sector in four dimensions. It illuminates the Misner string as a diffeomorphism vortex.
- Part 3. Chap. V offers a pedagogical demonstration of the simplicity of SD black holes. The motion of a relativistic test particle in the backgrounds of a SD black hole is maximally superintegrable by hidden symmetries, reincarnating the Kepler problem (hydrogen atom).

Chap. VI derives a previously unknown KS metric for the SD Taub-Newman-Unti-Tamburino (NUT) black hole by classical double copy. An explicit diffeomorphism proves its equivalence with a previously known metric due to Gibbons and Hawking.

Chap. VII shows that the Kerr metric represents a pair of SD and ASD Taub-NUT black holes. This elevates the NJA to a rigorous derivation of the Kerr metric with a definite physical origin. Real black holes are derived by nonlinear superpositions of SD and ASD building blocks (Taub-NUT instantons and chiral dyons).

The systematic program of investigating four-dimensional black holes—as *background geometries*—in terms of SD and ASD building blocks is established.

The systematic program of investigating four-dimensional black holes—as *dynamical objects* in point-particle effective theory—in terms of SD and ASD building blocks is proposed.

To this end, Chap. VIII carries out an essential development on all-orders geodesic deviation as a preliminary. A new formalism based on tangent bundle is established as an alternative to Synge calculus. Explicit expressions for the geodesic deviation equation (GDE) and its Lagrangian formulation are obtained up to tenth order.

Chap. IX introduces the probe counterpart of NJA, which Wick rotates the all-orders GDE into a part of exact spinning-particle EoM. The point-particle effective action of the Kerr black hole is completely constrained in the SD sector for hidden symmetries, which identifies a physical IR principle.

Chap. X applies the probe counterpart of NJA to derive a candidate exact point-particle effective action of the Kerr black hole in gravitational backgrounds.

Chap. XI applies the probe counterpart of NJA to derive a candidate point-particle effective action of the Kerr-Newman black hole in Einstein-Maxwell backgrounds.

Chap. XII establishes a framework for spinning-particle mechanics that unifies spacetime and spin into a complex four-manifold, resurrecting an idea that had been envisioned by Newman [252–259].

Chap. XIII demonstrates and outlines the derivation of exact spinning black hole EoM beyond the SD sector in the spinspace-time framework.

An integrable subsector is identified for the general-relativistic binary problem of Kerr or Kerr-Newman black holes, again exhibiting the hidden symmetries of the Kepler problem.

Chap. XIV gives a Lagrangian derivation of the Compton scattering amplitudes of spinning black holes for all helicity configurations. A chiral perturbation theory is developed, achieving Kerr by expanding around SD Taub-NUT.

REIMAGINING GRAVITY

According to its original roots, the term relativity refers to the investigation into how physics appear in different frames of reference. Newtonian mechanics was built upon the Galilean notion of relativity [274], postulating absolute time but relative space. Special relativity was born from the conceptual clash between Galilean relativity and Maxwell’s electrodynamics [275], leading to a unification of space and time. GR then fully embraced the principle of general covariance, investigating how physical laws appear in arbitrary coordinate systems on curved spacetimes.

On the one hand, the freedom to change one’s description of physics is a blessing. By using different coordinate systems or observers, one can manifest or highlight different aspects of the same physical system. On the other hand, the same freedom can turn into a trap leading into a maze of fictitious or redundant structures.

Consider the Lagrangian $\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - g\phi(\partial\phi)^2 - \frac{1}{2}g^2\phi^2(\partial\phi)^2$, where g is a small coupling constant. Apparently, it defines a perturbative QFT where particles interact via cubic and quartic vertices. This, however, turns out to be a mirage. Computation shows that scattering amplitudes all vanish, despite the fact that the number of Feynman diagrams explodes with the number of external legs. This demonstrates the falseness inherent in the Feynman diagram representation of scattering amplitudes.

The resolution is simple. Take $\mathcal{L} = -\frac{1}{2}(\partial\Phi)^2$, which evidently defines a free theory. Our earlier quartic Lagrangian is reproduced by putting $\Phi = \phi + g\phi^2/2$, which is a *field redefinition* that preserves the external scattering states. The *S-matrix equivalence theorem* [276–280] states that scattering amplitudes are physical observables that are invariant under such field redefinitions.

The above demonstration is a typical example well-known in the field theory community [281]. To the eyes of a geometer, it is interesting to note that field redefinitions could be thought of as coordinate transformations in the “field space” [282–287]. In particular, $\Phi = \phi + g\phi^2/2$ is a local change of coordinate on a one-dimensional line. The free Lagrangian $\mathcal{L} = -\frac{1}{2}(\partial\Phi)^2$ is how

physics appears to an “inertial observer” (nice field basis), while $\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - g\phi(\partial\phi)^2 - \frac{1}{2}g^2\phi^2(\partial\phi)^2$ is how a “noninertial observer” (bad field basis) sees the physics. The latter is plagued by virtual interacting particles as fictitious, “noninertial” artifacts creating the false impression of an interacting QFT.

In short, a field basis declares one’s choice of coordinates on the field space. Scattering amplitudes are physical observables independent of such choice. In this view, the subjectivity involved in Feynman rules/diagrams can be seen as a higher-level reincarnation of GR’s general covariance.

At this moment, we may want to describe how CK duality, introduced in Sec. 1.1, could be revisited in this perspective. Again, the angle is the treachery of Feynman diagrams. Scattering amplitudes are gauge-invariant physical observables. The physical meaning of Lehmann–Symanzik–Zimmermann (LSZ) reduction [288] is bulk-to-boundary propagation (cf. Ref. [289]): particles escaping to the boundary are on-shell and physical, while particles remaining in the bulk are off-shell and virtual. The bulk physics is field basis dependent.

Take the usual YM Lagrangian, which describes a quartic perturbation theory. The computation of a tree-level gluon scattering amplitude $\mathcal{A}$ via Feynman diagrams implements a sum over tree graphs, where each vertex carries degree three or four. Crucially, this diagrammatic representation of $\mathcal{A}$ is field basis dependent. By a field redefinition that preserves the external states, the Feynman rules can change arbitrarily while leaving the scattering amplitudes invariant. As a result, one can obtain the same $\mathcal{A}$ by imposing a totally different rule for how virtual gluons interact in the bulk, while keeping the external on-shell gluons the same. This demonstrates the falseness of virtual gluons in the Feynman diagram representation of scattering amplitudes.

CK duality claims that there may exist a particularly nice coordinate system (“observer”) on the field space of YM theory, such that the virtual gluons only interact via a universal cubic vertex composed of two Lie algebra factors. The one is the structure constant of the very color Lie algebra, such as $\mathfrak{su}(N)$. The other is the structure constant of the mystery Lie algebra $\mathfrak{g}_{\mathcal{P}}$, which takes the kinematic data (momenta and polarizations) as “indices.”¹ In this sense it is said that YM theory exhibits a duality between color and kinematics.

¹To be precise, CK duality states that for each triplet of cubic graphs $(\Gamma_i, \Gamma_j, \Gamma_k)$ whose color factors are related by the Jacobi identity $c_i + c_j + c_k = 0$, the same relation holds for the numerator factors as $n_i + n_j + n_k = 0$ [87–89]. The simplest explanation for this fact is the existence of a Lie algebra $\mathfrak{g}_{\mathcal{P}}$ that constructs the BCJ numerators (n_i, n_j, n_k) .

Once such a cubic representation of a gluon amplitude is obtained, one performs the diagram-wise operation of discarding the color factor and doubling the kinematic factor. Remarkably, this produces a graviton amplitude. This is the very double copy correspondence between YM theory and GR.

The implication is that there may exist a particularly nice choice of a “coordinate system on the field space of gravity,” such that the virtual gravitons exhibit only a cubic interaction through a universal vertex that simply squares the structure constant of $\mathfrak{g}_?$.

Note how scattering amplitudes serve as concrete, invariant anchors when navigating the vast sea of field spaces.

In this chapter, we would like to explore the field spaces for gravity. This is an investigation into how GR appears in various frames of descriptions, each manifesting or highlighting different aspects of the same physics: gravitation.

2.1 Gravity as Metric Geometry

According to the standard textbook view, GR is a geometrical theory where the metric is the “foundation of all” [20]. At the risk of boring the reader, let us quickly review how the metric determines all geometrical structures in GR: Levi-Civita connection, Riemann tensor, Ricci tensor, and Ricci scalar.

A metric g on a manifold $\mathcal{M}$ is a symmetric rank-two tensor field $g_{\mu\nu} = g_{\nu\mu}$. It defines the Levi-Civita connection ∇ as the unique torsion-free affine connection on the tangent bundle $T\mathcal{M}$ compatible with it: for arbitrary vector fields U , V , and W ,

$$\nabla_V W - \nabla_W V = [V, W], \quad (2.1a)$$

$$\nabla_U(g(V, W)) = g(\nabla_U V, W) + g(V, \nabla_U W). \quad (2.1b)$$

Here, $[V, W]$ denotes the Lie bracket between vector fields,

$$[V, W] = \left(V^\rho \partial_\rho W^\mu - W^\rho \partial_\rho V^\mu \right) \partial_\mu. \quad (2.2)$$

The Christoffel symbols are the connection coefficients $\Gamma^\mu_{\nu\rho} = \langle dx^\mu, \nabla_{\partial_\rho} \partial_\nu \rangle$ of the Levi-Civita connection in the coordinate frame. By plugging in the coordinate basis vector fields, Eq. (2.1) translates to $\Gamma^\mu_{\rho\sigma} = \Gamma^\mu_{\sigma\rho}$ and $\partial_\rho g_{\mu\nu} = g_{\mu\kappa} \Gamma^k_{\nu\rho} + g_{\kappa\nu} \Gamma^\kappa_{\mu\rho}$. The solution is unique:

$$\Gamma^\mu_{\rho\sigma} = \frac{1}{2} (g^{-1})^{\mu\nu} \left(-\partial_\nu g_{\rho\sigma} + \partial_\sigma g_{\nu\rho} + \partial_\rho g_{\nu\sigma} \right). \quad (2.3)$$

Although nonstandard, we denote the inverse metric as $(g^{-1})^{\mu\nu}$ for clarity.

The curvature tensor of an affine connection ∇ is defined as

$$R(V, W)U = \nabla_V \nabla_W U - \nabla_W \nabla_V U - \nabla_{[V, W]} U. \quad (2.4)$$

By plugging in coordinate basis vector fields, this translates to

$$R^\mu{}_{\nu\rho\sigma} = \partial_\rho \Gamma^\mu{}_{\nu\sigma} - \partial_\sigma \Gamma^\mu{}_{\nu\rho} + \Gamma^\mu{}_{\kappa\rho} \Gamma^\kappa{}_{\nu\sigma} - \Gamma^\mu{}_{\kappa\sigma} \Gamma^\kappa{}_{\nu\rho}. \quad (2.5)$$

The Ricci tensor, Ricci scalar, and Einstein tensor are then defined as

$$\text{Ric}_{\nu\sigma} = R^\mu{}_{\nu\mu\sigma}, \quad \text{Ric} = \text{Ric}_{\mu\nu} (g^{-1})^{\mu\nu}, \quad G_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2} \text{Ric} g_{\mu\nu}. \quad (2.6)$$

If ∇ is Levi-Civita, Eq. (2.5) defines the Riemann tensor $R^\mu{}_{\nu\rho\sigma}$. In this case, it follows that the Riemann and Ricci tensors exhibit the index symmetries

$$R_{\mu\nu\rho\sigma} = R_{[\mu\nu][\rho\sigma]} = R_{[\rho\sigma][\mu\nu]}, \quad \text{Ric}_{\mu\nu} = \text{Ric}_{\nu\mu}, \quad (2.7)$$

where indices are raised and lowered by $(g^{-1})^{\mu\nu}$ and $g_{\mu\nu}$. The Riemann tensor satisfies the algebraic and differential Bianchi identities:

$$R^\mu{}_{[\nu\rho\sigma]} = 0, \quad \nabla_{[\kappa} R^\mu{}_{\nu|\rho\sigma]} = 0. \quad (2.8)$$

2.1.1 Einstein-Hilbert Action

The Einstein-Hilbert action reads

$$S[g] = \frac{1}{2\kappa^2} \int d^d x \sqrt{-\det g} \text{Ric}[g], \quad (2.9)$$

which governs a nondegenerate Lorentzian-signature metric $g_{\mu\nu}$ as a sole dynamical symmetric tensor field given to a d -dimensional spacetime manifold. We have assumed vanishing cosmological constant for simplicity and identified

$$\kappa = \sqrt{8\pi G_N} \quad (2.10)$$

as the gravitational coupling. Variation of Eq. (2.9) derives the vacuum Einstein's equations,

$$\text{Ric}_{\mu\nu}[g] - \frac{1}{2} \text{Ric}[g] g_{\mu\nu} = 0. \quad (2.11)$$

Here, $\text{Ric}_{\mu\nu}[g]$ and $\text{Ric}[g]$ denote the Ricci tensor and the Ricci scalar of the metric g , respectively.

2.1.2 Landau-Lifshitz Equations

In the perturbation theory point of view, the metric formulation of GR is practically a nightmare. Expanding the metric $g_{\mu\nu}$ around a flat background produces an infinite number of vertices with all valences, each of which describes a complicated expression.

Regarding this matter, it is a well-known wisdom that utilizing the densitized inverse metric (“gothic metric”) tends to be helpful:

$$\mathfrak{g}^{\mu\nu} = \sqrt{-\det g} (g^{-1})^{\mu\nu}. \quad (2.12)$$

The gothic metric is utilized in the Landau-Lifshitz (LL) formulation of Einstein’s equations [290], nicely reviewed in Section 6.1 of Poisson and Will [291]. Note that Eq. (2.12) is a local, algebraic field redefinition: a simple change of coordinates on the field space.

Consider the tensor density

$$H^{\mu\nu\alpha\beta} = \mathfrak{g}^{\mu\alpha} \mathfrak{g}^{\nu\beta} - \mathfrak{g}^{\mu\beta} \mathfrak{g}^{\nu\alpha}, \quad (2.13)$$

which exhibits the same $\boxplus$ -tableaux index symmetry as the Riemann tensor. Computation shows that the Einstein tensor, when densitized and index-raised, can be identically rewritten as

$$(-2 \det g) G^{\mu\alpha} = \partial_\nu \partial_\beta H^{\mu\nu\alpha\beta} - \tau_{\text{LL}}^{\mu\alpha}, \quad (2.14)$$

where $\tau_{\text{LL}}^{\mu\alpha}$ describes a complicated expression involving $\mathfrak{g}^{\mu\nu}$, $\mathfrak{g}^{\mu\nu}_{,\rho}$, and $(\mathfrak{g}^{-1})_{\mu\nu}$. Explicitly, $\frac{1}{2} \tau_{\text{LL}}^{\mu\alpha}$ is given by

$$\begin{aligned} & \left(\frac{1}{4} \mathfrak{g}^{\mu\alpha} (\mathfrak{g}^{-1})_{\gamma\delta} \mathfrak{g}^{\rho\gamma}_{,\sigma} \mathfrak{g}^{\sigma\delta}_{,\rho} + \frac{1}{2} \mathfrak{g}^{\mu\kappa}_{,\rho} (\mathfrak{g}^{-1})_{\kappa\lambda} \mathfrak{g}^{\lambda\alpha}_{,\sigma} \mathfrak{g}^{\rho\sigma} - \mathfrak{g}^{\rho(\mu} \mathfrak{g}^{\alpha)\kappa}_{,\nu} (\mathfrak{g}^{-1})_{\kappa\lambda} \mathfrak{g}^{\lambda\nu}_{,\rho} \right) \\ & + \frac{1}{8} \left((\mathfrak{g}^{-1})_{\kappa(\rho} (\mathfrak{g}^{-1})_{\sigma)\lambda} - \frac{1}{d-2} (\mathfrak{g}^{-1})_{\kappa\lambda} (\mathfrak{g}^{-1})_{\rho\sigma} \right) \left(2 \mathfrak{g}^{\mu(\gamma} \mathfrak{g}^{\delta)\alpha} - \mathfrak{g}^{\mu\alpha} \mathfrak{g}^{\gamma\delta} \right) \mathfrak{g}^{\kappa\lambda}_{,\gamma} \mathfrak{g}^{\rho\sigma}_{,\delta} \\ & + \frac{1}{2} \left(\mathfrak{g}^{\mu\alpha}_{,\nu} \mathfrak{g}^{\nu\beta}_{,\beta} - \mathfrak{g}^{\mu\nu}_{,\nu} \mathfrak{g}^{\alpha\beta}_{,\beta} \right). \end{aligned} \quad (2.15)$$

The identity in Eq. (2.14) implies that Eq. (2.11) is equivalent to

$$\partial_\nu \partial_\beta H^{\mu\nu\alpha\beta} = \tau_{\text{LL}}^{\mu\alpha}. \quad (2.16)$$

Eq. (2.16) is the LL formulation of vacuum Einstein’s equations, which we might simply refer to as LL equations.

It should also be clear that a stress-energy pseudotensor of the gravitational field, conserved in terms of the partial derivative, can be constructed with the aid of $\tau_{\text{LL}}^{\mu\alpha}$ [290, 291].

The LL formulation gets along well with the harmonic (de Donder) gauge,

$$\partial_\mu \mathfrak{g}^{\mu\alpha} = 0 \iff \Gamma^\mu_{\rho\sigma} (g^{-1})^{\rho\sigma} = 0. \quad (2.17)$$

This kills the last line in Eq. (2.15), for instance. Still, the perturbation theory due to the LL equations in Eq. (2.16) describes an infinite number of vertices of all valences, as can be seen by plugging in the expansion $\mathfrak{g}^{\mu\alpha} = \eta^{\mu\alpha} + h^{\mu\alpha}$.

2.1.3 Cheung-Remmen Action

Notably, Cheung and Remmen [292] provided a cubic, first-order Lagrangian formulation of GR that employs the gothic metric $\mathfrak{g}^{\mu\nu} = \mathfrak{g}^{\nu\mu}$ together with an auxiliary field $\mathfrak{A}^\mu_{\rho\sigma} = \mathfrak{A}^\mu_{\sigma\rho}$:

$$S[\mathfrak{g}, \mathfrak{A}] = -\frac{1}{2\kappa^2} \int d^d x \left(\mathfrak{A}^\mu_{\rho\sigma} \partial_\mu \mathfrak{g}^{\rho\sigma} + \left(\mathfrak{A}^\mu_{\nu\rho} \mathfrak{A}^\nu_{\mu\sigma} - \frac{1}{d-1} \mathfrak{A}^\mu_{\mu\rho} \mathfrak{A}^\nu_{\nu\sigma} \right) \mathfrak{g}^{\rho\sigma} \right). \quad (2.18)$$

Integrating out $\mathfrak{A}$ shows that Eq. (2.18) is exactly equivalent to

$$S[\mathfrak{g}] = \frac{1}{2\kappa^2} \int d^d x \left(\frac{1}{2} (\mathfrak{g}^{-1})_{\mu\nu} \partial_\rho \mathfrak{g}^{\mu\sigma} \partial_\sigma \mathfrak{g}^{\nu\rho} - \frac{1}{4} (\mathfrak{g}^{-1})_{\mu\nu} (\mathfrak{g}^{-1})_{\rho\sigma} \mathfrak{g}^{\kappa\lambda} \partial_\kappa \mathfrak{g}^{\mu\rho} \partial_\lambda \mathfrak{g}^{\nu\sigma} + \frac{1}{4} \frac{1}{d-2} (\mathfrak{g}^{-1})^{\kappa\lambda} (\partial_\kappa \log \det \mathfrak{g}) (\partial_\lambda \log \det \mathfrak{g}) \right). \quad (2.19)$$

Brute-force calculation verifies that Eq. (2.19) is equivalent to the Einstein-Hilbert action in Eq. (2.9) upon plugging in Eq. (2.12) as an algebraic field redefinition. It is also insightful to note that a further algebraic field redefinition, $\mathfrak{A}^\mu_{\rho\sigma} = \mathfrak{B}^\mu_{\rho\sigma} - \delta^\mu_{(\rho} \mathfrak{B}^{\kappa}_{\sigma)\kappa}$, brings Eq. (2.18) to a form reminiscent of the metric-affine formulation [293] of GR, as is pointed out in Ref. [292].

2.2 Gravity as Lorentz Gauge Theory

The metric formulation of GR describes the dynamics of a pseudo-Riemannian manifold $(\mathcal{M}, g)$ in terms of the Einstein-Hilbert action. The vielbein formulation of GR offers a different view in which the metric is a derived object.

Coframe A coframe e is a local trivialization of the cotangent bundle $T^*\mathcal{M}$. Explicitly, one thinks of a collection of differential one-forms $e^A = e^A_\mu dx^\mu$ on $\mathcal{M}$ for $A = 0, 1, \dots, (d-1)$. Suppose $|e| := \det e$ is nonzero at any point in $\mathcal{M}$.

In a pseudo-Riemannian manifold $(\mathcal{M}, g)$, an orthonormal coframe refers to a coframe that represents the metric g as

$$g_{\mu\nu} = \eta_{AB} e^A_\mu e^B_\nu, \quad (2.20)$$

where $\eta_{AB} = \text{diag}(-1, +1, \dots, +1)$ describes a constant matrix.

This representation, however, is redundant. Two orthonormal coframes represent the same metric by the equivalence

$$e^A{}_\mu \sim \Omega^A{}_B e^B{}_\mu, \quad (2.21)$$

where $\Omega^A{}_B$ is an orthogonal matrix assigned to each point in $\mathcal{M}$. This describes a redundancy involved in the description of pseudo-Riemannian manifolds by the orthonormal coframe. Eq. (2.21) is referred to as a local Lorentz transformation.

The vielbein formulation aims to view GR as governing a dynamical coframe e subject to the equivalence relation in Eq. (2.21). This takes e as fundamental, while the pseudo-Riemannian metric g is a derived structure via Eq. (2.20). The vielbein formulation may also be called the coframe formulation of GR.

Since Eq. (2.21) describes a fiberwise transformation, the coframe formulation of GR conforms to the grammar of gauge theory in the precise YM sense: GR is a gauge theory of the Lorentz group, although the diffeomorphism redundancy has to be taken into account eventually. The frame indices $A, B, \dots$ are the *internal* indices, raised and lowered by η_{AB} and its inverse η^{AB} .

Note that there exist various choices for the local Lorentz group. For instance, it could be the orthogonal group $O(1, d-1)$, the identity component of $SO(1, d-1)$, or the spin group $\text{Spin}(1, d-1)$. For concreteness and the perturbative context, we may presume $SO(1, d-1)$ below, in which case the internal epsilon tensor $\epsilon_{A_1 \dots A_d}$ becomes additionally available as an invariant structure.

Frame A frame E is a local trivialization of the tangent bundle $T\mathcal{M}$. Explicitly, it is a collection of vector fields $E_A = E^\mu{}_A \partial_\mu$ on $\mathcal{M}$ for $A = 0, 1, \dots, (d-1)$. Suppose $|E| := \det E$ does not vanish at any point in $\mathcal{M}$.

Suppose a frame subject to the equivalence relation

$$E^\mu{}_A \sim E^\mu{}_B (\Omega^{-1})^B{}_A, \quad (2.22)$$

for $\Omega^A{}_B$ valued in a Lorentz group. In a pseudo-Riemannian manifold $(\mathcal{M}, g)$, an orthonormal frame refers to such a frame such that

$$g(E_A, E_B) = g_{\mu\nu} E^\mu{}_A E^\nu{}_B = \eta_{AB}. \quad (2.23)$$

We can take it as the dual of the orthonormal coframe introduced in Eq. (2.20):

$$e^A{}_\mu E^\mu{}_B = \delta^A{}_B, \quad E^\mu{}_A e^A{}_\nu = \delta^\mu{}_\nu, \quad (2.24)$$

so the two local Lorentz transformations in Eqs. (2.21) and (2.22) are linked.

Spin Connection Next, let ∇ be the Levi-Civita connection of the pseudo-Riemannian metric g . The spin connection coefficients are the connection coefficients of ∇ in the orthonormal frame:

$$\gamma^A_{BC} = \langle e^A, \nabla_{E_C} E_B \rangle = \left(-\partial_\rho e^A{}_\nu + e^A{}_\mu \Gamma^\mu{}_{\nu\rho} \right) E^\nu{}_B E^\rho{}_C. \quad (2.25)$$

As a one-form, the spin connection in the current context refers to

$$\gamma^A{}_B = \gamma^A{}_{BC} e^C. \quad (2.26)$$

Under Eq. (2.21), the spin connection transforms as

$$\gamma^A{}_B \sim \Omega^A{}_C \gamma^C{}_D (\Omega^{-1})^D{}_B + \Omega^A{}_C d(\Omega^{-1})^C{}_B. \quad (2.27)$$

Certainly, Eq. (2.27) is the very behavior of a connection one-form, Lie algebra valued. The gauge-covariant exterior derivative of a vector-valued gauge-covariant differential form α^A is defined as

$$D\alpha^A = d\alpha^A + \gamma^A{}_B \wedge \alpha^B, \quad (2.28)$$

while it is easy to extend this definition to differential forms valued in any representation of the local Lorentz group.

The torsion-free property of ∇ translates to the equation

$$0 = De^A = de^A + \gamma^A{}_B \wedge e^B, \quad (2.29a)$$

which can also be directly shown from Eq. (2.25). The metric-preserving property of ∇ translates to the equation

$$0 = D\eta_{AB} = \gamma_{AB} + \gamma_{BA}, \quad (2.29b)$$

meaning that $\gamma^A{}_B$ is valued in the adjoint of Lorentz.

Curvature The curvature two-form associated with the spin connection is

$$R^A{}_B = d\gamma^A{}_B + \gamma^A{}_C \wedge \gamma^C{}_B = \frac{1}{2} R^A{}_{B\rho\sigma} dx^\rho \wedge dx^\sigma. \quad (2.30)$$

It follows that $R^A{}_{B\rho\sigma}$ is the Riemann tensor in Eq. (2.5) with two of its indices converted to the local Lorentz frame. Under the local Lorentz transformation in Eq. (2.21), this transforms as

$$R^A{}_B \sim \Omega^A{}_C R^C{}_D (\Omega^{-1})^D{}_B, \quad (2.31)$$

assuming that $\Omega^A{}_B$ is single-valued.

For any vector-valued, gauge-covariant differential form α^A , it holds that

$$D^2\alpha^A = R^A{}_B \wedge \alpha^B. \quad (2.32)$$

With this understanding, Eq. (2.29a) implies

$$0 = D^2e^A = R^A{}_B \wedge e^B = \frac{1}{3!} \left(3R^A{}_{[\nu\rho\sigma]} \right) dx^\nu \wedge dx^\rho \wedge dx^\sigma, \quad (2.33a)$$

while Eq. (2.29b) implies

$$0 = D^2\eta_{AB} = R_{AB} + R_{BA}. \quad (2.33b)$$

Eq. (2.33a) derives the algebraic Bianchi identity in Eq. (2.8). Eq. (2.33b) states that the curvature two-form is Lorentz-valued.

To derive the differential Bianchi identity, consider computing $D^3\alpha^A$ as $D^2(D\alpha^A)$ and $D(D^2\alpha^A)$. By equating the two, one obtains

$$0 = DR^A{}_B = \frac{1}{3!} \left(3D_{[\nu} R^A{}_{B|\rho\sigma]} \right) dx^\nu \wedge dx^\rho \wedge dx^\sigma. \quad (2.34)$$

Einstein Tensor The internal Hodge dual, $\star$, is defined via contracting the internal indices with the internal epsilon tensor. A useful identity follows that

$$\begin{aligned} \text{Eins}_A &:= G^\mu{}_A \left(\frac{1}{(d-1)!} \varepsilon_{\mu\rho_1\cdots\rho_{d-1}} dx^{\rho_1} \wedge \cdots \wedge dx^{\rho_{d-1}} \right) \\ &= -\frac{1}{(d-3)!} \star R_{AB_1\cdots B_{d-3}} \wedge e^{B_1} \wedge \cdots \wedge e^{B_{d-3}}. \end{aligned} \quad (2.35)$$

Here, we have denoted the dynamical volume form as

$$\varepsilon_{\mu_1\cdots\mu_d} = |e| \epsilon_{\mu_1\cdots\mu_d}, \quad (2.36)$$

where $\epsilon_{\mu_1\cdots\mu_d}$ is the permutation symbol taking values in $\{+1, -1, 0\}$ such that $\epsilon_{01\cdots(d-1)} = +1$. To reiterate, $\epsilon_{\mu_1\cdots\mu_d}$ is the “bare” Levi-Civita symbol whose values are merely constants, whereas $\varepsilon_{\mu_1\cdots\mu_d}$ is dynamical and gives the diffeomorphism-invariant integration measure. Namely,

$$\varepsilon = \frac{1}{d!} \epsilon_{A_1\cdots A_d} e^{A_1} \wedge \cdots \wedge e^{A_d} = \sqrt{-\det g} d^d x. \quad (2.37)$$

Note that we have named the differential form in Eq. (2.35) as Eins_A , which may be called the Einstein $(d-1)$ -form. Another notable identity in this framework is the covariant d -form equation,

$$0 = D\text{Eins}_A = \left((\nabla_\mu G^\mu{}_\nu) E^\nu{}_A \right) |e| d^d x, \quad (2.38)$$

which encodes the covariant conservation of the Einstein tensor. This can be seen by combining Eq. (2.34) with the metric- and $\star$ -preserving nature of D .

2.2.1 Electric and Magnetic Equations, I

Eq. (2.35) shows that the vacuum Einstein's equations can be stated in the language of differential forms as $\star R_{AB_1 \dots B_{d-3}} \wedge e^{B_1} \wedge \dots \wedge e^{B_{d-3}} = 0$. From physical grounds, it is helpful to juxtapose this equation with the algebraic Bianchi identity in Eq. (2.33a):

$$\star R_{AB_1 \dots B_{d-3}} \wedge e^{B_1} \wedge \dots \wedge e^{B_{d-3}} = 0, \quad R^A{}_B \wedge e^B = 0. \quad (2.39)$$

These are $(d-1)$ -form and 3-form equations, respectively. Note how these two equations involve the same number of derivatives (on the coframe) but differ by involving a Hodge star. The first is a dynamical equation stating the absence of stress-energy. The second is an identity if R is defined as a derived object from e . By analogy, we may identify these as the electric and magnetic vacuum equations of GR, respectively. The reasoning is shown below.

In Maxwell theory, the abelian gauge connection A serves as the electromagnetic potential. The curvature $F = dA$ is the electromagnetic field strength. This is the Wu-Yang [25] dictionary. The source-free Maxwell's equations are

$$d \star F = 0, \quad dF = 0, \quad (2.40)$$

which are $(d-1)$ -form and 3-form equations, respectively. These two equations involve the same number of derivatives but differ via the Hodge star $\star$ acting on spacetime indices. The first is a dynamical equation stating the absence of electric charges. The second equation encodes the absence of magnetic charges, which is an identity if F is defined as a derived object from A .

It could be helpful to elaborate more on the physical semantics. Let us do so via the magnetic equations. Consider the following sequences of differential forms. For electromagnetism, we have

$$A \xrightarrow{d} F \xrightarrow{d} dF = 0. \quad (2.41)$$

For gravity, we have

$$e^A \xrightarrow{d} -\gamma^A{}_B \wedge e^B \xrightarrow{d} -R^A{}_B \wedge e^B = 0. \quad (2.42)$$

The Bianchi identity $dF = 0$ is automatic when the field strength F is derived from the potential A . Earlier, we have sketched a similar intuition around Eq. (2.8). The metric g can be thought of as describing the gravitational potential. The Levi-Civita connection Γ describes gravitational acceleration

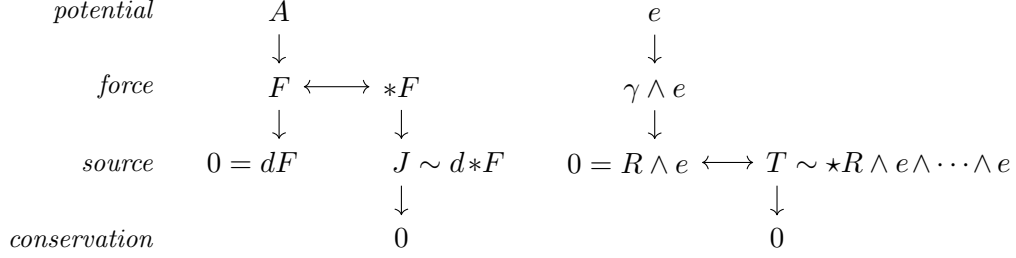


Figure 2.1: Structure of electromagnetism and geometrodynamics in outline form. The vertical direction corresponds to differentiation. The horizontal direction describes Hodge duality.

($\sim 9.8\text{m/s}^2$) and frame-dragging forces measured in a laboratory. The Riemann tensor R realizes the idea of gravitational field in a more invariant fashion, encoding tidal effects. When the potential g defines everything, the algebraic Bianchi identity is automatic. Interestingly, the sequence in Eq. (2.42) makes this intuition mathematically precise in terms of the identity $d^2 = 0$ of exterior calculus while identifying the coframe e as the gravitational potential.

In fact, this seems to be exactly what Misner, Thorne, and Wheeler [20] aim to envision in their diagrams mentioned in Sec. 1.1 (Chapter 15, “Bianchi Identities and the Boundary of a Boundary”). In Fig. 2.1, we have provided our version of their diagrams while being totally schematic about indices, signs, proportionality factors, etc. For simplicity, we have assumed the case where electric sources are present but not magnetic sources.

In the presence of both electric and magnetic sources, the Maxwell’s equations (on a d -dimensional background spacetime) are

$$d*F = J^\mu \left(\frac{1}{(d-1)!} \varepsilon_{\mu\rho_1\dots\rho_{d-1}} dx^{\rho_1} \wedge \dots \wedge dx^{\rho_{d-1}} \right), \quad (2.43a)$$

$$dF = J^{\star\mu_1\dots\mu_{d-3}} \left(\frac{1}{(d-3)!3!} \varepsilon_{\mu_1\dots\mu_{d-3}\rho_1\rho_2\rho_3} dx^{\rho_1} \wedge dx^{\rho_2} \wedge dx^{\rho_3} \right). \quad (2.43b)$$

The electric current is a $(d-1)$ -form or equivalently a vector. The magnetic current is a 3-form or equivalently a $(d-3)$ -vector.

In the presence of ordinary stress-energy $T_{\mu\nu}$, the Einstein’s equations are

$$\begin{aligned} & -\frac{1}{\kappa^2} \frac{1}{(d-3)!} \star R_{AB_1\dots B_{d-3}} \wedge e^{B_1} \wedge \dots \wedge e^{B_{d-3}} \\ & = T^\mu{}_A \left(\frac{1}{(d-1)!} \varepsilon_{\mu\rho_1\dots\rho_{d-1}} dx^{\rho_1} \wedge \dots \wedge dx^{\rho_{d-1}} \right), \end{aligned} \quad (2.44a)$$

as can be seen from Eq. (2.35). Here, we have converted one spacetime index of $T_{\mu\nu}$ into an internal index via the coframe/frame. It becomes clear that the natural definition of *magnetic stress-energy* will be (see, e.g., [294])

$$\begin{aligned} & -\frac{1}{\kappa^2} R_{AB} \wedge e^B \\ & = T^{\star\mu_1 \dots \mu_{d-3}}{}_A \left(\frac{1}{(d-3)!3!} \varepsilon_{\mu_1 \dots \mu_{d-3} \rho_1 \rho_2 \rho_3} dx^{\rho_1} \wedge dx^{\rho_2} \wedge dx^{\rho_3} \right). \end{aligned} \quad (2.44b)$$

Note how Eq. (2.44) provides the inhomogeneous version of Eq. (2.39). The electric stress-energy is recast as a $(d-1)$ -form carrying one internal index, while the magnetic stress-energy is formulated as a 3-form also carrying one internal index.

2.2.2 Coframe Formulation of General Relativity (Second-Order)

We are now ready to describe Lagrangian formulations. Consider the action

$$S[e] = \frac{1}{\kappa^2} \int_{\mathcal{M}} \frac{1}{2!(d-2)!} \epsilon_{A_1 \dots A_d} R^{A_1 A_2}[e] \wedge e^{A_3} \wedge \dots \wedge e^{A_d}, \quad (2.45)$$

which takes an orthonormal coframe e^A as the sole dynamical field. Here, $R^{A_1 A_2}[e]$ is the curvature two-form associated with the coframe e^A via Eqs. (2.25) and (2.30). Eq. (2.45) is manifestly invariant under diffeomorphisms on $\mathcal{M}$ as well as the local Lorentz transformations in Eq. (2.21).

Eq. (2.45) is the action for the coframe formulation of GR. By plugging in Eq. (2.30), one equates Eq. (2.45) with the Einstein-Hilbert action in Eq. (2.9). Variation of Eq. (2.45) with respect to e derives the vacuum Einstein's equations in the form stated in Eq. (2.39).

Note that the cosmological constant Λ can be readily implemented by adding the following term in the integrand: a multiple of the dynamical volume form,

$$-\frac{1}{\kappa^2} \Lambda \left(\frac{1}{d!} \epsilon_{A_1 \dots A_d} e^{A_1} \wedge \dots \wedge e^{A_d} \right) = -\frac{1}{\kappa^2} \Lambda |e| d^d x. \quad (2.46)$$

2.2.3 Coframe Formulation of General Relativity (First-Order)

Now consider the action

$$S[e, \gamma] = \frac{1}{\kappa^2} \int_{\mathcal{M}} \frac{1}{2!(d-2)!} \epsilon_{A_1 \dots A_d} R^{A_1 A_2}[\gamma] \wedge e^{A_3} \wedge \dots \wedge e^{A_d}, \quad (2.47)$$

which takes e^A and $\gamma^A{}_B$ as *independent* dynamical fields. Here, e^A is an orthonormal coframe while $\gamma^A{}_B$ is a Lorentz-valued connection. Here, $R^{A_1 A_2}[\gamma]$ is viewed as a sole functional of γ defined by the formula in Eq. (2.30).

Eq. (2.47) is quadratic in γ . Hence γ can be exactly integrated out. By using the Palatini variation identity [295], one finds that the variation of Eq. (2.47) with respect to γ imposes

$$D(e^{A_3} \wedge \cdots \wedge e^{A_d}) = 0. \quad (2.48)$$

Given e is nondegenerate, it can be seen that Eq. (2.48) is algebraically equivalent to the vanishing of torsion, $De^A = 0$, in Eq. (2.29a). As a result, γ is dynamically equated to the spin connection of e . Therefore, Eq. (2.47) is equivalent to Eq. (2.45).

Eq. (2.47) is a first-order action of GR, known as the vielbein-Palatini action. Note that the variation of Eq. (2.47) with respect to e immediately derives $\star R_{AB_1 \cdots B_{d-3}}[\gamma] \wedge e^{B_1} \wedge \cdots \wedge e^{B_{d-3}} = 0$.

Since the top form in Eq. (2.46) describes a sole functional of the coframe, e , the cosmological constant is straightforwardly incorporated in the Palatini action in Eq. (2.47).

2.3 Gravity as Frame Geometry

The coframe formulation of GR involves two sorts of redundancies: diffeomorphisms and local Lorentz transformations. Can there be a formulation that focuses on the diffeomorphism side, rather than local Lorentz? A formulation known as teleparallel gravity [296–307] will achieve this idea, which views GR as a theory of a dynamical frame field E .

Magnetic Equation Revisited It could be instructive to begin our exposition from the magnetic equation of gravity in Eq. (2.39), i.e., the algebraic Bianchi identity. The sequence of differential forms in Eq. (2.42) describes

$$-de^A = \gamma^A_B \wedge e^B, \quad -dde^A = R^A_B \wedge e^B. \quad (2.49)$$

Let us examine the content of these equations in the internal frame.

First of all, a well-known gymnastics [29] unfolds as

$$de^C(E_A, E_B) = \langle \iota_{E_A} de^C, E_B \rangle = \langle \mathcal{L}_{E_A} e^C, E_B \rangle = -\langle e^C, [E_A, E_B] \rangle, \quad (2.50)$$

where we have used the Cartan magic formula : on any differential form α , the Lie derivative $\mathcal{L}_V$ with respect to a vector field V is given by

$$\mathcal{L}_V \alpha = d\iota_V \alpha + \iota_V d\alpha, \quad (2.51)$$

where ι_V denotes the interior product with respect to V .

Given a frame E , the *anholonomy coefficients* are defined as

$$\Omega^C_{AB} = \langle e^C, [E_A, E_B] \rangle, \quad (2.52)$$

measuring the noncoordinate nature of E . Eq. (2.50) implies that the anholonomy coefficients are related to the spin connection coefficients in Eq. (2.25) as

$$\frac{1}{2} \Omega^C_{AB} e^A \wedge e^B = -de^C = \gamma^C_B \wedge e^B = \gamma^C_{BA} e^A \wedge e^B. \quad (2.53)$$

That is,

$$\Omega^C_{AB} = \gamma^C_{BA} - \gamma^C_{AB} = -2\gamma^C_{[AB]}. \quad (2.54)$$

Second of all, the three-form equation in Eq. (2.49) can be now written as

$$d\left(\frac{1}{2} \Omega^D_{AB} e^A \wedge e^B\right) = \frac{1}{3!} \left(3R^D_{[CAB]}\right) e^A \wedge e^B \wedge e^C. \quad (2.55)$$

For any two-form α and vector fields U, V, W , it holds that [308]

$$d\alpha(U, V, W) = \left(U[\alpha(V, W)] + \alpha(U, [V, W]) \right) + \text{cyclic}, \quad (2.56)$$

which sums over the three cyclic permutations of (U, V, W) . Here, $U[f] = \langle df, U \rangle = U^\mu \partial_\mu f$ denotes the action of the vector field U as a differential operator on the scalar field f . Applying Eq. (2.56) to Eq. (2.55), we find

$$R^D_{[CAB]} = E_{[C}[\Omega^D_{AB}] + \Omega^D_{[C|E} \Omega^E_{|AB]} = \langle e^D, [E_{[C}, [E_A, E_B]] \rangle. \quad (2.57)$$

This shows that the algebraic Bianchi identity has secretly encoded the Jacobi identity of the Lie bracket between vector fields:

$$[E_A, [E_B, E_C]]^\mu + [E_B, [E_C, E_A]]^\mu + [E_C, [E_A, E_B]]^\mu = 0. \quad (2.58)$$

Note that this result can also be directly derived from Eqs. (2.1a) and (2.4):

$$\begin{aligned} & R(E_A, E_B) E_C + \text{cyclic} \\ &= \nabla_{E_A} \nabla_{E_B} E_C - \nabla_{E_B} \nabla_{E_A} E_C - \nabla_{[E_A, E_B]} E_C + \text{cyclic}, \\ &= \nabla_{E_A} (\nabla_{E_B} E_C - \nabla_{E_C} E_B) - \nabla_{[E_A, E_B]} E_C + \text{cyclic}, \\ &= \nabla_{E_A} ([E_B, E_C]) - \nabla_{[E_B, E_C]} E_A + \text{cyclic}, \\ &= [E_A, [E_B, E_C]] + \text{cyclic}. \end{aligned} \quad (2.59)$$

To sum up, we have shown that the magnetic equation of gravity can be equivalently formulated as Eq. (2.58), based on the Lie bracket between the frame fields. Note that another name for the Lie algebra of vector fields on $\mathcal{M}$ is the diffeomorphism algebra $\text{diff}(\mathcal{M})$. In Sec. 4.5, we will use this mathematical fact to interpret Eq. (2.58) as the Bianchi identity for a “gauge theory of the diffeomorphism group.”

Diffeomorphism Yang-Mills Equations How about the electric equation of GR in Eq. (2.39), $\star R_{AB_1 \dots B_{d-3}} \wedge e^{B_1} \wedge \dots \wedge e^{B_{d-3}} = 0$? One could wonder if it could be equivalent to

$$[E^B, [E_B, E_A]]^\mu = 0, \quad (2.60)$$

which traces back to Mason and Newman [309]. In the modern context of double copy, this is a fantastic proposal suggesting that the double copy problem of GR is solved by $\mathbf{g}_{\mathcal{P}} = \text{diff}(\mathcal{M})$; see Chap. IV.

Unfortunately, the electric equation of GR fails to be Eq. (2.60). To address this explicitly, we may now want to switch to the Lagrangian formalism.

2.3.1 Topological Limit

Our point of departure is the Palatini action in Eq. (2.47), which splits as

$$S[e, \gamma] = \frac{1}{\kappa^2} \int_{\mathcal{M}} L_1[e, \gamma] + \frac{1}{\kappa^2} \int_{\mathcal{M}} L_2[e, \gamma]. \quad (2.61)$$

Here, $L_1[e, \gamma]$ and $L_2[e, \gamma]$ are top forms given as

$$L_1[e, \gamma] = \frac{1}{2!(d-2)!} \epsilon_{A_1 \dots A_d} d\gamma^{A_1 A_2} \wedge e^{A_3} \wedge \dots \wedge e^{A_d}, \quad (2.62a)$$

$$L_2[e, \gamma] = \frac{1}{2!(d-2)!} \epsilon_{A_1 \dots A_d} \gamma^{A_1 C} \wedge \gamma^{C A_2} \wedge e^{A_3} \wedge \dots \wedge e^{A_d}. \quad (2.62b)$$

Our goal is to reformulate the Palatini action such that the frame field E is taken as one of the fundamental degrees of freedom. We find it instructive to first focus on the first part, Eq. (2.62a). With hindsight, we refer to this case as the topological limit.²

In the topological limit, the Lorentz connection is merely a Lagrange multiplier. By discarding a total derivative, $L_1[e, \gamma]$ in Eq. (2.62a) is brought to

$$\left(\frac{1}{(d-3)!} \star \gamma_{A_3 \dots A_d} \wedge e^{A_4} \wedge \dots \wedge e^{A_d} \right) \wedge de^{A_3}. \quad (2.63)$$

The identity explored in Eq. (2.50), $E^\mu_C de^C = -\frac{1}{2} [E_A, E_B]^\mu e^A \wedge e^B$, then brings Eq. (2.63) to

$$\frac{1}{2} B^{AB}{}_\mu [E_A, E_B]^\mu d^d x, \quad (2.64)$$

²The decoupling between Eqs. (2.62a) and (2.62b) naturally arises by implementing a rescaling of the Lorentz connection, $\gamma^A_B \mapsto \kappa^2 \gamma^A_B$. Amusingly, this mimics a typical move [310, 311] that one encounters in a different context: see Sec. 2.5.2.

where we have employed an algebraic field redefinition,

$$B^{AB}{}_{\mu} d^d x = \left(-\frac{1}{(d-3)!} \star \gamma_{CC_4 \dots C_d} \wedge e^{C_4} \wedge \dots \wedge e^{C_d} \wedge e^A \wedge e^B \right) e^C{}_{\mu}. \quad (2.65)$$

According to Sec. 4.5.3, Eq. (2.64) describes a diffeomorphism BF theory. Its saddles describe *coordinate* frame fields, free of anholonomy: $[E_A, E_B] = 0$.

2.3.2 Frame Formulation of General Relativity (First-Order)

Now let us incorporate the second part of the Palatini action in Eq. (2.62b) to derive the full action. First of all, Eq. (2.65) boils down to

$$B^{AB}{}_{\mu} = \left(-3\gamma^{[AB}{}_{K} \delta^{K]}{}_C \right) |e| e^C{}_{\mu}. \quad (2.66a)$$

The inverse transformation is

$$\gamma^{AB}{}_C = -|E| \left(B^{AB}{}_{\mu} E^{\mu}{}_C + \frac{2}{d-2} \delta^{[A}{}_C B^{B]D}{}_{\rho} E^{\rho}{}_D \right). \quad (2.66b)$$

Direct computation shows that

$$\begin{aligned} L_2[e, \gamma] &= \frac{1}{2} \left(\gamma_{ACB} \gamma^{ABC} - \gamma^{CA}{}_C \gamma^D{}_{AD} \right) |e| d^d x, \\ &= -\frac{1}{2} \mathcal{G}^{\mu\nu}_{ABCD}[E] B^{AB}{}_{\mu} B^{CD}{}_{\nu} d^d x, \end{aligned} \quad (2.67)$$

where we have defined

$$\mathcal{G}^{\mu\nu}_{A_1 A_2 B_1 B_2}[E] = 4|E| \left(E^{\mu}{}_{[B_1} \eta_{B_2][A_1} E^{\nu}{}_{A_2]} - \frac{1}{d-2} E^{\mu}{}_{[A_1} \eta_{A_2][B_1} E^{\nu}{}_{B_2]} \right). \quad (2.68)$$

Collecting all the results so far, we conclude that the Palatini action in Eq. (2.47), up to discarding a total derivative, is equivalent to

$$S[B, E] = \frac{1}{\kappa^2} \int d^d x \left(\frac{1}{2} B^{AB}{}_{\mu} [E_A, E_B]^{\mu} - \frac{1}{8} \mathcal{G}^{\mu\nu}_{ABCD}[E] B^{AB}{}_{\mu} B^{CD}{}_{\nu} \right), \quad (2.69)$$

where $\mathcal{G}^{\mu\nu}_{ABCD}[E]$ is defined in Eq. (2.68). To clarify, Eq. (2.69) is a first-order action of GR that governs two dynamical fields. The first is a frame field, $E^{\mu}{}_A$. The second is an auxiliary field, $B^{AB}{}_{\mu}$.

We should clarify on the gauge redundancies as well. A careful analysis will tell us that there are two. The first kind describes the transformation

$$\begin{aligned} \delta_{\xi} E^{\mu}{}_A &= \xi^{\rho} \partial_{\rho} E^{\mu}{}_A - E^{\rho}{}_A \partial_{\rho} \xi^{\mu}, \\ \delta_{\xi} B^{AB}{}_{\mu} &= \xi^{\rho} \partial_{\rho} B^{AB}{}_{\mu} + B^{AB}{}_{\rho} \partial_{\mu} \xi^{\rho} + B^{AB}{}_{\mu} \partial_{\rho} \xi^{\rho}. \end{aligned} \quad (2.70)$$

Clearly, Eq. (2.70) will be interpreted as the action of an infinitesimal diffeomorphism. It states that $E^{\mu}{}_A$ is a vector while $B^{AB}{}_{\mu}$ is a covector density of weight +1. It is important that $B^{AB}{}_{\mu}$ is a density, since otherwise terms such as $B^{AB}{}_{\mu} [E_A, E_B]^{\mu} d^d x$ cannot be diffeomorphism invariant.

Note that Eq. (2.70) induces

$$\delta_\xi |E| = -|E|^2 \partial_\mu (|E|^{-1} \xi^\mu). \quad (2.71)$$

The action in Eq. (2.69) changes by the boundary term,

$$\frac{1}{\kappa^2} \int d^d x \partial_\rho \left[\xi^\rho \left(\frac{1}{2} B^{AB}{}_\mu [E_A, E_B]^\mu - \frac{1}{8} \mathcal{G}_{ABCD}^{\mu\nu} [E] B^{AB}{}_\mu B^{CD}{}_\nu \right) \right], \quad (2.72)$$

which vanishes as the local parameter ξ^ρ dies off at the infinities.

The second kind is subtler and is more nontrivial to see. It describes the transformation

$$\begin{aligned} \delta_\vartheta E^\mu{}_A &= -E^\mu{}_B \vartheta^B{}_A, \\ \delta_\vartheta B^{AB}{}_\mu &= 2\vartheta^{[A}{}_C B^{C|B]}{}_\mu + \frac{1}{|E|} \left(\partial_\mu \vartheta^{AB} + 2E^\rho{}_C \partial_\rho \vartheta^{C[A} e^{B]}{}_\mu \right), \end{aligned} \quad (2.73)$$

where $\vartheta^{AB} = -\vartheta^{BA}$. Under Eq. (2.73), $\frac{1}{2} B^{AB}{}_\mu [E_A, E_B]^\mu$ changes by

$$\partial_\mu \left(E^\mu{}_A E^\rho{}_B \partial_\rho \vartheta^{AB} \right) - B^{AB}{}_\rho E^\rho{}_C E^\mu{}_A \partial_\mu \vartheta^C{}_B, \quad (2.74)$$

whereas $-\frac{1}{8} \mathcal{G}_{ABCD}^{\mu\nu} [E] B^{AB}{}_\mu B^{CD}{}_\nu$ changes by $+B^{AB}{}_\rho E^\rho{}_C E^\mu{}_A \partial_\mu \vartheta^C{}_B$. Hence the full action in Eq. (2.69) changes by the boundary term,

$$\frac{1}{\kappa^2} \int d^d x \partial_\mu \left(E^\mu{}_A E^\rho{}_B \partial_\rho \vartheta^{AB} \right), \quad (2.75)$$

which vanishes provided the derivative $\partial_\rho \vartheta^{AB}$ dies off at the infinities.

Crucially, this gauge redundancy is not enjoyed by the topological limit but emerges in the full theory defined by Eq. (2.69). It can be seen that Eq. (2.73) precisely arises from the local Lorentz transformation given in Eqs. (2.21) and (2.27) (cf. Ref. [306]). Note how it is significantly non-manifest in Eq. (2.69).

2.3.3 Frame Formulation of General Relativity (Second-Order)

The action in Eq. (2.69) is quadratic in the auxiliary field B . Hence it can be exactly integrated out as

$$B^{AB}{}_\rho = \frac{1}{2|E|} \left(\Omega^{ABC} + \Omega^{BCA} - \Omega^{CAB} \right) e_{C\rho} - \frac{1}{|E|} 2\Omega^C{}_C{}^{[A} e^{B]}{}_\rho. \quad (2.76)$$

Plugging in Eq. (2.76) to Eq. (2.69) gives

$$S[E] = -\frac{1}{2\kappa^2} \int \frac{d^d x}{|E|} \left(\frac{1}{4} \Omega^{ABC} \Omega_{ABC} + \frac{1}{2} \Omega^{BAC} \Omega_{ABC} - \Omega^A{}_{AC} \Omega^B{}_B{}^C \right), \quad (2.77)$$

where $\Omega^C{}_{AB}$ is viewed as the functional of E defined by Eq. (2.52).

Eq. (2.77) establishes a formulation of GR that takes the frame field $E^\mu{}_A$ as the sole dynamical field. We will refer to this as the *frame formulation of GR*.

Equations of Motion As remarked earlier, we see that Einstein's equations cannot be formulated as Eq. (2.60) as explored by Mason and Newman [309]. The frame formulation of GR does enjoy the local Lorentz symmetry as shown in Eq. (2.75). In contrast, Eq. (2.60) is not invariant under local Lorentz transformations. Explicitly, one can simply vary the action in Eq. (2.77) to directly find that the EoM disagree with Eq. (2.60).

In fact, it is not clear whether one can integrate equations like Eq. (2.60) into an action. Naively, one could envision a YM-like action that squares $[E_A, E_B]^\mu$, in which case the spacetime index μ needs to be contracted. This crucially requires an invariant inner product (a Killing form) on the diffeomorphism algebra. Note also how $\mathcal{G}_{ABCD}^{\mu\nu}[E]$ in Eq. (2.69) is dynamical.

Derivation by Bootstrap To reiterate, our analysis shows that the frame formulation of GR enjoys local Lorentz symmetry, although non-manifest. While the frame field E^μ_A provides d^2 components in d dimensions, local Lorentz symmetry subtracts $d(d-1)/2$ degrees of freedom, retrieving the usual number $d(d+1)/2$ in the metric formulation; see also Ref. [306].

In fact, it seems that the action in Eq. (2.77) can be derived from the bottom-up as the *unique* action of a dynamical frame field E^μ_A that is invariant under both diffeomorphisms in Eq. (2.70) and local Lorentz transformations in Eq. (2.73) while being quadratic in the anholonomy coefficients Ω^A_{BC} . First, there are three quadratic contractions of the anholonomy coefficients: $\Omega^{ABC}\Omega_{ABC}$, $\Omega^{BAC}\Omega_{ABC}$, and $\Omega^A_{AC}\Omega^B_B$. Second, diffeomorphism invariance demands placing an overall $|E|^{-1}$ in the Lagrangian. Note that

$$\delta_\xi \Omega_{ABC} = \xi^\mu \partial_\mu \Omega_{ABC}, \quad \delta_\xi (|E|^{-1}) = \partial_\mu (|E|^{-1} \xi^\mu). \quad (2.78)$$

Third, one examines local Lorentz invariance with fuzzy factors a_1, a_2, a_3 :

$$\begin{aligned} & \delta_\vartheta \left[\frac{1}{2|E|} \left(a_1 \Omega^{ABC} \Omega_{ABC} + a_2 \Omega^{BAC} \Omega_{ABC} + a_3 \Omega^A_{AC} \Omega^B_B \right) \right] \\ & \sim \vartheta_{AB} \left[(a_2 - 2a_1) \partial_\mu \left(\frac{E^\mu_C \Omega^{ABC}}{|E|} \right) + (a_3 + 2a_2) \left(\frac{E_A [\Omega^C_{CB}]}{|E|} \right) \right], \end{aligned} \quad (2.79)$$

where $\sim$ means we discarded a total derivative in a way that removes derivatives on the local parameter ϑ^{AB} . Here, the following identity can be helpful:

$$\Omega^C_{CA} = \partial_\mu (|E|^{-1} E^\mu_A) = \text{div}_\varepsilon E_A, \quad \mathcal{L}_{E_A} \varepsilon = \Omega^C_{CA} \varepsilon. \quad (2.80)$$

Apparently, Eq. (2.79) pinpoints $-4a_1 = -2a_2 = a_3$ as the unique solution to this bootstrap. Taking $(a_1, a_2, a_3) = (-1/4, -1/2, 1)$ reproduces Eq. (2.77).

Teleparallel Gravity In the literature, the action in Eq. (2.77) is known to provide the “teleparallel equivalent of GR” [304–307]. A *teleparallel theory* refers to a field theory of a dynamical frame E^μ_A . A frame amounts to a notion of absolute parallelism, meaning a flat connection known as Weitzenböck connection [296]. The Weitzenböck connection declares that a vector $E^\mu_A(x_1)v^A$ at point x_1 is parallel to $E^\mu_A(x_2)v^A$ at point x_2 : the internal index A serves as a “global” index, so the parallel propagator from point x_1 to x_2 is defined as $E^\mu_A(x_2)e^A_\nu(x_1)$. “Teleparallelism” is another name for “absolute parallelism.”

Teleparallel gravity was one of the approaches put forward by Einstein himself during his late-stage research on unified field theories. See Refs. [297–307] and also a remark made in Mason and Newman [309], Sec. 5.

Given our clarifications above, we may prefer the name “frame formulation of GR” over “teleparallel formulation of GR,” in fact: local Lorentz redundancy is secretly present.

Flat Connection Formulation Provided a frame E^μ_A and its inverse e^A_μ , the Weitzenböck connection is defined by the following connection coefficients:

$$\tilde{\Gamma}^\mu_{\nu\rho} = E^\mu_A \partial_\rho e^A_\nu. \quad (2.81)$$

It is easy to show that the Weitzenböck connection is flat, metric-preserving, and torsionful:

$$0 = \partial_\rho \tilde{\Gamma}^\mu_{\nu\sigma} - \partial_\sigma \tilde{\Gamma}^\mu_{\nu\rho} + \tilde{\Gamma}^\mu_{\kappa\rho} \tilde{\Gamma}^\kappa_{\nu\sigma} - \tilde{\Gamma}^\mu_{\kappa\sigma} \tilde{\Gamma}^\kappa_{\nu\rho}, \quad (2.82a)$$

$$0 = \partial_\rho g_{\mu\nu} - g_{\mu\kappa} \tilde{\Gamma}^\kappa_{\nu\rho} - g_{\kappa\nu} \tilde{\Gamma}^\kappa_{\mu\rho}, \quad (2.82b)$$

$$\tilde{T}^\mu_{\rho\sigma} = \tilde{\Gamma}^\mu_{\sigma\rho} - \tilde{\Gamma}^\mu_{\rho\sigma} = -E^\mu_A \Omega^A_{BC} e^B_\rho e^C_\sigma. \quad (2.82c)$$

By using these facts, Eq. (2.77) can be rewritten as

$$-\frac{1}{2\kappa^2} \int \sqrt{-\det g} d^d x \left(\frac{1}{4} \tilde{T}^{\mu\nu\rho} \tilde{T}_{\mu\nu\rho} + \frac{1}{2} \tilde{T}^{\nu\mu\rho} \tilde{T}_{\mu\nu\rho} - \tilde{T}^\mu_{\mu\rho} \tilde{T}^\nu_{\nu\rho} \right), \quad (2.83)$$

where internal indices are completely “hidden.” Eq. (2.83) is the commonly known formula for the teleparallel equivalent of GR. Yet, to truly regard it as a functional of $g_{\mu\nu}$ and $\tilde{\Gamma}^\mu_{\rho\sigma}$ as fundamental field variables, we might be imposing the flatness and metric-preserving conditions in Eqs. (2.82a) and (2.82b) via Lagrange multipliers.

2.4 Simplicity of Self-Dual Gravity

In the remaining parts of this chapter, we specialize in the phenomenologically relevant case of $d = 4$ spacetime dimensions. It is known that four-dimensional gravitation admits a remarkably simpler subsector: SD gravity [309, 310, 312–326]. In particular, Penrose’s nonlinear graviton theorem [312, 313] establishes that SD gravity is classically integrable. This realization has served as one of the motivations behind chiral approaches to gravity that aim to understand the full dynamics of four-dimensional GR by perturbing away from SD gravity.

This section aims to provide a brief sketch of SD gravity, from the angle of double copy [327]. We review two senses in which SD gravity can be “derived” from SDYM theory: Plebański formulation [314] and Mason-Newman formulation [309].

2.4.1 Self-Dual Yang-Mills and Double Copy

First of all, we define SDYM theory and SD gravity.

YM theory governs the dynamics of a connection on a principal fiber bundle over spacetime. In YM theory, the curvature two-form F^a satisfies

$$D * F^a = 0, \quad D F^a = 0. \quad (2.84)$$

These are the *electric and magnetic equations* of YM theory. The former is a dynamical equation. The latter is the Bianchi identity and is identically satisfied when the gauge connection is the foundation of all.

Via complexification, SDYM theory studies configurations such that

$$*F^a = +iF^a, \quad (2.85)$$

where $*$ denotes the Hodge dual acting on spacetime indices. Notably, such SD configurations are automatically on-shell: $D * F^a = i D F^a = 0$. Also, Eq. (2.85) describes a first-order differential equation on the gauge potential A^a while Eq. (2.84) describes second-order differential equations. In these senses, SDYM theory identifies a simpler subsector inside complexified YM theory.

Next, recall the electric and magnetic equations of GR, identified in Eq. (2.39). In four dimensions, we have

$$\star R^A_B \wedge e^B = 0, \quad R^A_B \wedge e^B = 0. \quad (2.86)$$

Again, the former is a dynamical equation while the latter is identically satisfied when the metric/coframe is the foundation of all.

Via complexification, SD gravity studies configurations such that

$$\star R^A{}_B = +iR^A{}_B, \quad (2.87)$$

where $\star$ is the internal Hodge dual.³ Notably, such SD configurations are automatically on-shell: $\star R^A{}_B \wedge e^B = iR^A{}_B \wedge e^B = 0$. Moreover, Eq. (2.87) describes a first-order differential equation on the coframe e^A while Eq. (2.86) describes second-order differential equations. In these senses, SD gravity already identifies a simpler subsector inside GR.

Remarkably, it turns out that at the heart of SDYM theory on flat spacetime there lies an infinite-dimensional Lie algebra, and by “doubling” it one derives SD gravity. In this sense there exists a concrete connection between the SD sectors of YM theory and GR:

$$(\text{SDYM})^2 = \text{SDGR}. \quad (2.88)$$

The astute reader will point out a slight mismatch in the parallel between SDYM theory and SD gravity implied above: while Eq. (2.84) is differential on the curvature F that is subject to Eq. (2.85), Eq. (2.86) is algebraic on R that is subject to Eq. (2.87). This discomfort indeed detects an important point in correctly grasping the double copy relation in Eq. (2.88), as will be revisited in Eq. (2.113).

We shall also clarify on the meaning of complexification in the current context. The discourse on complex spacetimes traces back to Penrose [312, 313] Newman [258, 328], and Plebański [314, 315];⁴ see Newman [329] and Flaherty [330] for nice reviews. The celebrated works [331, 332] of Plebański and Demiański also demonstrate how one naturally arrives at the idea of complexified coordinate transformations and spacetimes.

Penrose [312], for instance, argued that the notion of a quantum spacetime is inherently complex-geometrical. The spacetime of a single graviton in a helicity eigenstate is half-flat, meaning either the SD or ASD curvature is vanishing. This necessitates complexification and leads to a holomorphic metric (not Kähler [312]) geometry.

³Note that SD gravity is readily formulated without a need for the coframe/frame. In the pure metric formulation of GR, the SD gravity equations are $\star R_{\mu\nu\rho\sigma} = +iR_{\mu\nu\rho\sigma}$, where the spacetime Hodge dual $\star$ acts either on the first set $[\mu\nu]$ of antisymmetric indices or the second set $[\rho\sigma]$.

⁴Newman [258, 328] and Plebański [314, 315] coined the term “heaven” for referring to SD spacetimes.

To clarify, the “barred” coordinates or indices never enter in this discussion [312, 329, 330]. Mathematically speaking, this means to work in the holomorphic, complexified category of manifolds so that the coordinate transformations (transition functions) are holomorphic maps [261, 331, 332]. One then studies holomorphic metrics or gauge fields solving complexified field equations.

2.4.2 Plebański Formulation: Second Heavenly Equation

SDYM theory has been well-studied in the lightcone gauge [333–335].

Flat spacetime $\mathbb{M} = (\mathbb{C}^4, \eta)$ is the complexified Minkowski space, i.e., the space $\mathbb{C}^4$ equipped with a holomorphic metric η that exhibits Lorentzian signature on the real section. Let $S^+\mathbb{M}$ and $S^-\mathbb{M}$ be the SD and ASD spinor bundles on $\mathbb{M}$, respectively. Their fibers are equipped with the epsilon tensors $\epsilon^{\alpha\beta}$ and $\tilde{\epsilon}^{\dot{\alpha}\dot{\beta}}$. The tangent bundle $T\mathbb{M}$ is isomorphic to the direct product bundle $S^+\mathbb{M} \otimes S^-\mathbb{M}$. This isomorphism describes a tensor $(\hat{\sigma}_\mu)^{\dot{\alpha}\alpha}$, while its inverse describes a tensor $(\check{\sigma}^\mu)_{\alpha\dot{\alpha}}$. These invariant tensors are used to convert vector indices to spinor indices and vice versa. Various conventions exist in the literature, the detailed discussion on which we might want to omit.

Let x^μ be Cartesian coordinates for the flat spacetime $\mathbb{M}$. Lightcone coordinates refer to

$$x^{\dot{\alpha}\alpha} = (\hat{\sigma}_\mu)^{\dot{\alpha}\alpha} x^\mu \propto \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}. \quad (2.89)$$

We also employ constant dyads: $o^\alpha \doteq \delta_0^\alpha$, $\iota^\alpha \doteq \delta_1^\alpha$, $\tilde{o}^{\dot{\alpha}} \doteq \delta^{\dot{\alpha}}_0$, $\tilde{\iota}^{\dot{\alpha}} \doteq \delta^{\dot{\alpha}}_1$.

Now consider YM theory on $\mathbb{M}$ for a finite-dimensional gauge group $\mathbf{G}$. The gauge potential describes $A^a_{\alpha\dot{\alpha}}$, where $a, b, c, d, \dots$ are adjoint indices of the gauge algebra $\mathfrak{g} = \text{Lie}(\mathbf{G})$. Typically, one has $\mathbf{G} = \text{SU}(N)$ and $\mathfrak{g} = \mathfrak{su}(N)$, for instance. The lightcone gauge sets $o^\alpha A^a_{\alpha\dot{\alpha}} = A^a_{0\dot{\alpha}} = 0$ for a fixed reference spinor o^α . Let us denote the remaining components of the gauge potential as

$$j^a_{\dot{\alpha}} := \iota^\alpha A^a_{\alpha\dot{\alpha}} = A^a_{1\dot{\alpha}}. \quad (2.90)$$

The SDYM equations, Eq. (2.85), impose vanishing of the ASD field strength $F^a_{\alpha\beta}$. This describes

$$\tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \left(\partial_{\alpha\dot{\alpha}} A^a_{\beta\dot{\beta}} - \partial_{\beta\dot{\beta}} A^a_{\alpha\dot{\alpha}} + f^a_{bc} A^b_{\alpha\dot{\alpha}} A^c_{\beta\dot{\beta}} \right) = 0. \quad (2.91)$$

In the lightcone gauge, the nontrivial contents of Eq. (2.91) are

$$\partial_{0\dot{\alpha}} j^{a\dot{\alpha}} = 0, \quad \partial_{1\dot{\alpha}} j^a_{\dot{\beta}} - \partial_{1\dot{\beta}} j^a_{\dot{\alpha}} + f^a_{bc} j^b_{\dot{\alpha}} j^c_{\dot{\beta}} = 0. \quad (2.92)$$

The first condition in Eq. (2.92) states that $j^{a\dot{\alpha}}$ is “divergence-free,” implying the existence of a $\mathfrak{g}$ -valued scalar field Φ^a such that

$$j^{a\dot{\alpha}} = \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \partial_{0\dot{\beta}} \Phi^a. \quad (2.93)$$

The second condition in Eq. (2.92) states that $j^a_{\dot{\alpha}}$ is a “flat connection,” which is solved by $j^a_{\dot{\alpha}} = (U^{-1} \partial_{1\dot{\alpha}} U)^a$ for a $\mathbf{G}$ -valued field U .

By plugging in Eq. (2.93) to Eq. (2.92), one finds

$$\square \Phi^a = f^a_{bc} \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \partial_{0\dot{\alpha}} \Phi^b \partial_{0\dot{\beta}} \Phi^c, \quad (2.94)$$

where $\square$ is the four-dimensional d’Alembertian operator. Eq. (2.94) is a *cubic* formulation of SDYM theory in terms of a colored *scalar* field Φ^a ; see, e.g., Ref. [333]. This reveals the simplicity of SDYM theory.

Notably, we see that Eq. (2.94) identifies a (degenerate) Poisson structure on the spacetime $\mathbb{M}$:

$$\Pi_{\mathcal{H}} := \frac{1}{2} \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \partial_{0\dot{\alpha}} \wedge \partial_{0\dot{\beta}}. \quad (2.95)$$

The Poisson bracket,⁵

$$\{F, G\}_{\mathcal{H}} := \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \partial_{0\dot{\alpha}} F \partial_{0\dot{\beta}} G, \quad (2.96)$$

satisfies the standard axioms: skew symmetry, Leibniz rule, and Jacobi identity. With this understanding, Eq. (2.94) can be written as

$$\mathbf{SDYM}(\mathfrak{g}) : \quad \square \Phi^a = f^a_{bc} \{\Phi^b, \Phi^c\}_{\mathcal{H}}. \quad (2.97)$$

In this way, Eq. (2.94) describes two Lie algebras: the color Lie algebra $\mathfrak{su}(N)$ and the Poisson Lie algebra $\text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})$.

Now consider replacing the color Lie algebra with the Poisson Lie algebra. This maps Eq. (2.97) to

$$\mathbf{SDGR} : \quad \square \Phi = \{\{\Phi, \Phi\}\}_{\mathcal{H}}, \quad (2.98)$$

where the “double Poisson bracket”⁶ is defined as

$$\{\{F, G\}\}_{\mathcal{H}} := \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \tilde{\epsilon}^{\dot{\gamma}\dot{\delta}} \partial_{0\dot{\alpha}} \partial_{0\dot{\gamma}} F \partial_{0\dot{\beta}} \partial_{0\dot{\delta}} G, \quad (2.99)$$

literally doubling the Poisson bracket in Eq. (2.96). Remarkably, Eq. (2.98) is a formulation of SD gravity due to Plebański [314], known as the *second heavenly equation*. Interested readers may refer to App. A.1 for its derivation. This reveals the simplicity of SD gravity: it admits a *cubic* formulation in terms of a single *scalar* field Φ .

⁵This gives a fiberwise Poisson bracket via a $2 + 2$ splicing of spacetime: see App. A.2.

⁶This is the B_2 bracket in Ref. [336] for Moyal deformation quantization.

The above findings can be summarized in terms of the BAS equation,

$$\mathbf{BAS}(\mathfrak{g}, \tilde{\mathfrak{g}}) : \quad \square \Phi^{a\tilde{a}} = f^a_{bc} \tilde{f}^{\tilde{a}}_{\tilde{b}\tilde{c}} \Phi^{b\tilde{b}} \Phi^{c\tilde{c}}, \quad (2.100)$$

which describe two Lie algebras $\mathfrak{g}$ and $\tilde{\mathfrak{g}}$. The formulation of SDYM theory in Eq. (2.97) implies that SDYM theory describes an instance of BAS theory for $\tilde{\mathfrak{g}} = \text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})$:

$$\mathbf{SDYM}(\mathfrak{g}) \cong \mathbf{BAS}(\mathfrak{g}, \text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})). \quad (2.101)$$

The Plebański formulation of SD gravity in Eq. (2.98) implies that SD gravity describes an instance of SDYM theory that takes the Poisson algebra $\text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})$ as the color algebra:

$$\begin{aligned} \mathbf{SDGR} &\cong \mathbf{BAS}(\text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}}), \text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})), \\ &\cong \mathbf{SDYM}(\text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})). \end{aligned} \quad (2.102)$$

The isomorphism in Eq. (2.101) is a manifestation of the very CK duality of YM theory in the SD sector, identifying the kinematic algebra $\mathfrak{g}_{\mathcal{F}}$ as the Poisson algebra:

$$(\mathfrak{g}_{\mathcal{F}} \text{ in four dimensions, in the SD sector}) \cong \text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}}). \quad (2.103)$$

The correspondences in Eq. (2.102) then explain the double copy relationship between YM theory and gravity in the SD sector. These facts were established by the seminal work [327] by Monteiro and O’Connell; see also Ref. [337].

2.4.3 Mason-Newman Formulation

In component terms, the SDYM equations in Eq. (2.85) describe

$$\frac{1}{2} \epsilon_{ABCD} [D^C, D^D] = i [D_A, D_B]. \quad (2.104)$$

Here, the field strength is understood as an operator $F_{AB} = [D_A, D_B]$ arising from the commutator between covariant derivatives. Also, for this moment, we have temporarily used $A, B, C, D, \dots$ for denoting the *global* Lorentz indices of flat spacetime $\mathbb{M}$, raised and lowered via the flat metric η_{AB} .

Remarkably, Mason and Newman [309] established that SD gravity on a four-manifold $\mathcal{M}$ can be formulated by the following equation:

$$\frac{1}{2} \epsilon_{ABCD} [V^C, V^D] = i [V_A, V_B]. \quad (2.105a)$$

Here, $V_A = V^\mu{}_A \partial_\mu$ are four linearly independent vector fields on $\mathcal{M}$ that preserve a volume form ν :

$$\mathcal{L}_{V_A} \nu = 0. \quad (2.105b)$$

$A, B, C, D, \dots$ are frame indices, raised and lowered via the flat metric η_{AB} .

Specifically, given a volume-preserving frame $V^\mu{}_A$ that satisfies Eq. (2.105a), one constructs another frame $E^\mu{}_A$ via

$$E_A = f^{-1} V_A \quad \text{where} \quad f^2 = \nu(V_0, V_1, V_2, V_3). \quad (2.106)$$

The statement is that this $E^\mu{}_A$ defines a metric $g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu$ whose curvature is SD, where $e^A{}_\mu E^\mu{}_B = \delta^A_B$.

Conversely, Mason and Newman established that an SD spacetime $(\mathcal{M}, g)$ always admits an orthonormal frame E_A and a nonvanishing function f such that $V_A = f E_A$ satisfy Eqs. (2.105a) and (2.105b) for some volume form ν .

Mason and Newman's formulation of SD gravity generalizes an earlier result by Ashtekar, Jacobson, and Smolin [338].⁷ Below, we wish to review the core part of their derivation of the converse statement.

In the complexified setup, spacetime is an oriented complex four-manifold $\mathcal{M}$ equipped with a holomorphic metric g . Let ε be the metric volume form, and let $\mathcal{M} \cong \mathbb{C}^4$ for simplicity. $(\mathcal{M}, g)$ is said to be a SD spacetime if the metric g describes SD Riemannian curvature as $*R_{\mu\nu\rho\sigma} = +i R_{\mu\nu\rho\sigma}$.

Suppose an orthonormal frame E_A in a SD spacetime. Firstly, we recall the definition of the anholonomy coefficients in Eq. (2.52): $[E_A, E_B] = \Omega^C{}_{AB} E_C$. Upon a rescaling $V_A = f E_A$, we find

$$[V_A, V_B] = \left(\Omega^C{}_{AB} + E_A[\log f] \delta^C{}_B - E_B[\log f] \delta^C{}_A \right) f^2 E_C. \quad (2.107)$$

Secondly, by utilizing the freedom of local Lorentz transformations, the ASD part of the spin connection can be made to vanish:

$$\star \gamma_{ABC} = +i \gamma_{ABC}. \quad (2.108a)$$

Eq. (2.108a) implies $\gamma_{ABC} \epsilon^{ABCD}/2 = -\star \gamma^{CD}{}_C = -i \gamma^{CD}{}_C$, or equivalently

$$3\gamma_{[ABC]} = -i \gamma^{KD}{}_K \epsilon_{ABCD}. \quad (2.108b)$$

⁷To this end, Mason and Newman [309] assume that the spacetime $\mathcal{M}$ is topologically a product of a three-manifold and a line. Here, we may work locally or simply take $\mathcal{M} \cong \mathbb{C}^4$.

For the anholonomy coefficients, this self-duality condition translates to

$$\Omega_{CAB} = \gamma_{ABC} - 3\gamma_{[ABC]} = \gamma_{ABC} + i\gamma^{KD}{}_K \epsilon_{ABCD}, \quad (2.108c)$$

where we have used Eq. (2.54). Since $\gamma^K{}_{AK} = \Omega^K{}_{KA}$, Eq. (2.108c) implies that the following combination is SD on the antisymmetric pair of indices $[AB]$:

$$\Omega^C{}_{AB} + \Omega^K{}_{KA} \delta^C{}_B - \Omega^K{}_{KB} \delta^C{}_A. \quad (2.108d)$$

Combining Eqs. (2.107) and (2.108d), one establishes Eq. (2.105a) if there exists a scalar f such that

$$\Omega^K{}_{KA} = E_A[\log f]. \quad (2.109)$$

If Eq. (2.109) holds, the identity stated earlier in Eq. (2.80) implies that

$$\mathcal{L}_{V_A}(f^{-2}\varepsilon) = \mathcal{L}_{E_A}(f^{-1}\varepsilon) = f^{-1}\left(\Omega^C{}_{CA} - E_A[\log f]\right)\varepsilon = 0, \quad (2.110)$$

where the first identity can be seen by the Cartan magic formula. Hence one also establishes Eq. (2.105b) by taking

$$\nu = f^{-2}\varepsilon \implies \nu(V_0, V_1, V_2, V_3) = f^2\varepsilon(E_0, E_1, E_2, E_3) = f^2, \quad (2.111)$$

which explains the recipe in Eq. (2.106).

The condition in Eq. (2.109) is not generally true, but Mason and Newman show that a concrete construction exists in a SD spacetime by utilizing a harmonic function and the Plebański ASD two-forms [315, 339].⁸

Several remarks can be given here. Firstly, note that Eq. (2.105a) follows from Eq. (2.104) by replacing the color Lie bracket with the Lie bracket between vector fields, i.e., the diffeomorphism Lie bracket. As will be made further explicit in Chap. IV, in this sense the Mason-Newman formulation of SD gravity describes that

$$\mathbf{SDGR} \cong \mathbf{SDYM}(\mathbf{diff}(\mathbb{C}^4, \nu)), \quad (2.112)$$

where $\mathbf{diff}(\mathbb{C}^4, \nu)$ denotes the Lie algebra of diffeomorphisms of $\mathbb{C}^4$ that preserve the volume form ν , i.e., the volume-preserving diffeomorphism algebra (also denoted as $\mathbf{sdiff}(\mathbb{C}^4)$).

⁸Note that one will be able to fix $\nu = d^4x$, i.e., $\nu_{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma}$, by a choice of coordinates. In this case, one can take $f = |E|^{-1/2}$, so $V_A = |E|^{-1/2}E_A = \mathfrak{E}_A$ precisely becomes the densitized frame that serves as the “square root” of the LL gothic metric in Sec. 2.1.2: $\mathbf{g}^{\mu\nu} = \mathfrak{E}^\mu{}_A \mathfrak{E}^\nu{}_B \eta^{AB}$.

Secondly, note an interesting “role reversal” between the spacetime and internal indices in establishing the relationship in Eq. (2.112). The frame indices of SD gravity, $A, B, C, D, \dots$, are made to correspond to the flat spacetime indices of SDYM theory on $\mathbb{M}$. It seems that this point could be generally relevant since YM equations crucially employ a flat metric η^{AB} to contract indices: $\eta^{AB} D_A F^a_{BC} = 0$.

Thirdly, it is easily seen that Eq. (2.105a) implies

$$[V^A, [V_A, V_B]] = 0, \quad (2.113a)$$

when combined with the Jacobi identity

$$[V_A, [V_B, V_C]] + [V_B, [V_C, V_A]] + [V_C, [V_A, V_B]] = 0. \quad (2.113b)$$

As will be elaborated in Chap. IV, we can identify Eqs. (2.113a) and (2.113b) as the electric and magnetic equations of volume-preserving diffeomorphism YM theory. Unfortunately, Mason and Newman [309] report that Eq. (2.113a) does not describe GR beyond the SD sector; recall our earlier exploration in Sec. 2.3.

Lastly, with the aid of an auxiliary ASD spinor λ_α , the SDYM and SD gravity equations in Eqs. (2.104) and (2.105a) can be stated as

$$[L_{\dot{\alpha}}, L_{\dot{\beta}}] = 0, \quad (2.114)$$

via $L_{\dot{\alpha}} = \lambda^\alpha D_{\alpha\dot{\alpha}}$ or $L_{\dot{\alpha}} = \lambda^\alpha V_{\alpha\dot{\alpha}}$. Geometrically, Eq. (2.114) implies integrability of (gauge-covariant) translations along SD null planes, known as α -surfaces. This gives the Lax pair (and eventually twistor) formulations of SDYM and SD gravity [310, 313, 340, 341].

2.5 Gravity as a Tapestry of Self-Dual and Anti-Self-Dual Parts

In Sec. 2.4, we have briefly reviewed the simplicity of SD gravity. Plebański’s heavenly equation [314] shows that a single scalar field Φ dictates the full dynamics of SD gravity. This also reveals that a Poisson algebra $\text{diff}(\mathbb{C}^4, \Pi_{\mathcal{H}})$ lies at the heart of perturbative gravity in the SD sector, from which one sees that SD gravity is a double copy of SDYM theory. In Mason and Newman [309]’s version, SD gravity is understood as an instance of SDYM theory for a volume-preserving diffeomorphism algebra.

As we have remarked in a footnote, Newman [258, 328] and Plebański [314, 315] coined the technical term “heaven” for referring to SD spacetimes. This

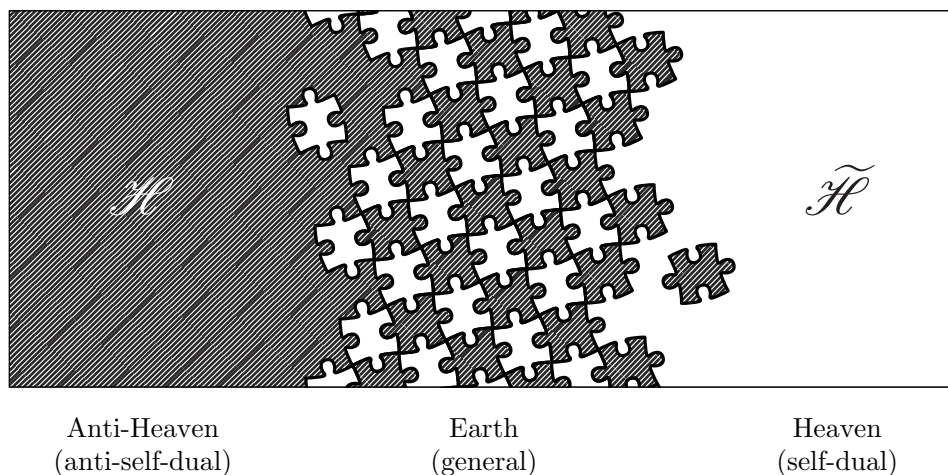


Figure 2.2: *Heaven* refers to SD spacetimes. *Earth* refers to spacetimes with both SD and ASD curvatures nonvanishing. Right-handed gravitons inhabit heaven; left-handed gravitons inhabit anti-heaven; both coexist on earth [342, 343].

terminology is to be taken as acknowledging the simplicity and ideal nature of SD gravity. The nonlinear dynamics of full gravity, however, is much more complicated. Newman [258, 328] and Plebański [314, 315] refer to spacetimes with both SD and ASD curvatures as “earth.”

In Plebański’s world view, a real spacetime is a tapestry of SD and ASD building blocks: an interacting mixture of heaven and anti-heaven, so to speak. The idea of “patching” SD and ASD spacetimes together was the dream of the Plebański school for an era [344–346].⁹ An artist’s impression (in the style of M. C. Escher) is given in Fig. 2.2.

Accordingly, the program of *understanding earth by maximally utilizing the simplicity of heaven* has existed as a reasonable slogan for four-dimensional gravitation. This approach has been systematized and largely advocated by twistor theorists. In this context, two perspectives have been pursued.

- The *googly* agenda [347–350].—In essence, solving the googly problem means to describe ASD modes in the same variables in which SD modes are formulated. Though considered incomplete, a viable approach to this problem is to understand the full theory by perturbing away from the SD sector.
- The *ambitwistor* approach [351–355].—This line of thinking seeks to formulate SD and ASD modes on an equal footing, by employing both “untilded”

⁹The author thanks Maciej Dunajski for this anecdote.

and “tilded” variables. It is also open to describing theories in general dimensions and employing non-spinorial variables.

With this broad remark made, in this final section for reimagining gravity we would like to describe two threads on grasping the full dynamics of GR as a nonlinear interaction of SD and ASD modes. From Secs. 2.5.1 to 2.5.3, we implement the chiral, “googly” philosophy that perturbatively adds ASD modes into SD gravity. From Secs. 2.5.4 to 2.5.6, we follow the “ambitwistor” spirit to treat SD and ASD modes symmetrically. We clarify, however, that our explorations are strictly field-theoretical in spacetime, not in twistor spaces.

2.5.1 Chalmers-Siegel Action

We begin with the YM analog. YM theory in four dimensions describes the Lagrangian four-form

$$L[A] = -\frac{1}{2g^2} \delta_{ab} F^a[A] \wedge *F^b[A], \quad (2.115)$$

where δ_{ab} is a Killing form, and g is the coupling. A well-known identity on the Chern-Simons (CS) three-form reads

$$d\left(\delta_{ab} A^a \wedge dA^b + \frac{1}{3} f_{abc} A^a \wedge A^b \wedge A^c\right) = \delta_{ab} F^a[A] \wedge F^b[A], \quad (2.116)$$

so the Lagrangian four-form in Eq. (2.115) can be modified to

$$L[A] = \frac{i}{g^2} \delta_{ab} F^a[A] \wedge F^{-b}[A], \quad (2.117)$$

without changing the Euler-Lagrange equations. Eq. (2.117) is perturbatively equivalent to Eq. (2.115). Here, $F^{\pm a}[A] = \frac{1}{2}(1 \mp i*)F^a[A]$. Switching to the first-order formulation gives

$$L[A, B] = -iB_a \wedge F^a[A] - \frac{ig^2}{4} \delta^{ab} B_a \wedge B_b, \quad (2.118)$$

where B_a is a coadjoint-valued ASD two-form:

$$*B_a = -iB_a. \quad (2.119)$$

Eq. (2.118) is known to provide the Chalmers-Siegel [356] formulation of YM theory in four dimensions.

In the limit $g \rightarrow 0$, the Chalmers-Siegel Lagrangian four-form in Eq. (2.118) reduces to $B_a \wedge F^a[A]$, which describes SDYM theory: the B -field is a Lagrange multiplier stipulating that the ASD part of $F^a[A]$ vanishes. In this

sense the Chalmers-Siegel formulation understands full YM theory as a perturbation away from SDYM theory. This approach exactly resonates with the perturbative solution to the googly problem in spirit (see, e.g., Ref. [350]).

The EoM of Eq. (2.118) are

$$F^{-a}[A] = -\frac{g^2}{2}B^a, \quad DB_a = 0. \quad (2.120)$$

In the $g \rightarrow 0$ limit, $F^{-a}[A] = 0$, so one can interpret the A -field as describing SD (positive-helicity) gluons, on the background of which the “cotangent” B -field propagates as a linearized ASD (negative-helicity) perturbation. For $g \neq 0$, this ASD perturbation “backreacts” to the A -field.

2.5.2 Chiral Coframe Action (Plebański Action)

For gravity, the Plebański formulation of GR [315] has been widely identified as the gravitational analog of the Chalmers-Siegel formulation; see, e.g., Ref. [310]. Note that this is different from the Plebański formulation of SD gravity (heavenly equation) [314] described in Sec. 2.4.2.

Nonchiral Version Let us first review the nonchiral formulation before reproducing the chiral version of Plebański [315]. The point of departure is the coframe formulation of GR. In four dimensions, Eq. (2.47) describes the Lagrangian four-form

$$L[e, \gamma] = \frac{1}{2\kappa^2} \left(\frac{1}{2} \epsilon_{ABCD} e^C \wedge e^D \right) \wedge R^{AB}[\gamma]. \quad (2.121)$$

The key observation is that the two-form in parenthesis in Eq. (2.121) emerges as a solution to the following “simplicity constraint”:

$$B_{AB} \wedge B_{CD} = B_{[AB} \wedge B_{CD]}. \quad (2.122)$$

With this understanding, the proposed reformulation for Eq. (2.121) is

$$L[B, \gamma, \Psi] = \frac{1}{2\kappa^2} B_{AB} \wedge R^{AB}[\gamma] - \frac{1}{8\kappa^2} \Psi^{ABCD} B_{AB} \wedge B_{CD}, \quad (2.123)$$

where $\Psi^{ABCD} = \Psi^{[AB][CD]} = \Psi^{[CD][AB]}$ is a Lagrange multiplier subject to the condition (assuming vanishing cosmological constant)

$$\epsilon_{ABCD} \Psi^{ABCD} = 0, \quad (2.124)$$

hence imposing $6(6+1)/2-1=20$ constraints. As a result, the 36 components of $B_{AB\mu\nu}$ are reduced down to 16 degrees of freedom, which precisely match with the number of components in the tetrad e^A_μ .

To be precise, the general solution to Eq. (2.122) is [357–359]

$$B_{AB} = \pm \frac{1}{2} \epsilon_{ABCD} e^C \wedge e^D \quad \text{or} \quad \pm e_A \wedge e_B \quad \text{or} \quad 0. \quad (2.125)$$

These three branches of solutions may be referred to as the gravitational, topological, and degenerate sectors, respectively [358]. One may assume that the vacuum expectation value of the metric (see Eq. (2.127)) chooses the positive gravitational branch, $B_{AB} = +\frac{1}{2} \epsilon_{ABCD} e^C \wedge e^D$, as a superselection sector. With this assumption, Eq. (2.123) can be brought back to the tetradic Palatini formulation in Eq. (2.121) by integrating out the Ψ -field.

In the meantime, from the variations of Eq. (2.123) with respect to the B - and γ -fields, one finds

$$R^{AB}[\gamma] = \frac{1}{2} \Psi^{ABCD} B_{CD}, \quad DB_{AB} = 0. \quad (2.126)$$

The first equation tells us that Ψ^{ABCD} encode the components of the Weyl tensor in the orthonormal frame. The second equation encodes the torsion-free condition for the spin connection γ .

We see that an interesting perspective toward gravity has emerged. Firstly, the area element B_{AB} is taken as a fundamental variable for describing Riemannian geometry, instead of lengths or orthonormal frames. Intriguingly, the line element is retrieved via a remarkable formula due to Urbantke [339, 360]:

$$\begin{aligned} g_{\mu\nu} &= -\frac{1}{12|e|} B^A_{B\mu\kappa} B^B_{C\nu\lambda} B^C_{A\rho\sigma} \epsilon^{\kappa\lambda\rho\sigma}, \\ |e| &= \frac{1}{96} \epsilon^{ABCD} B_{AB\mu\nu} B_{CD\rho\sigma} \epsilon^{\mu\nu\rho\sigma}, \end{aligned} \quad (2.127)$$

where we have assumed the positive gravitational branch. For the topological and degenerate branches, Eq. (2.127) gives $g_{\mu\nu} = 0$.

Secondly, the graviton is “carved out” from a topological BF theory by constraints. This illustrates how gravity as a diffeomorphism-invariant theory is close to topological field theories. In a sense, gravitons are being viewed as “impurities” in a topological theory. The B -field could have been a Lagrange multiplier stipulating flatness $R^{AB}[\gamma] = 0$ on the spin connection, but it is

made “defective” due to the Ψ -field. In consequence, γ acquires the freedom to be not perfectly flat. In fact, performing a rescaling $B_{AB} \mapsto \kappa^2 B_{AB}$ brings Eq. (2.123) to

$$L[B, \gamma, \Psi] = B_{AB} \wedge R^{AB}[\gamma] - \frac{\kappa^2}{8} \Psi^{ABCD} B_{AB} \wedge B_{CD}, \quad (2.128)$$

in which case the $\kappa \rightarrow 0$ limit describes the BF theory.

Chiral Version We are now ready to describe the chiral version. In the tetradic Palatini formalism, the Nieh-Yan identity [361] reads

$$d(e_A \wedge T^A) = T_A \wedge T^A - e_A \wedge e_B \wedge R^{AB}, \quad (2.129)$$

where $T^A = De^A = de^A + \gamma^A_B \wedge e^B$ is the torsion two-form. Based on this fact, a simplified alternative for Eq. (2.121) has been proposed as

$$L[e, \gamma] = -\frac{i}{\kappa^2} B_{AB}[e] \wedge R^{AB}[\gamma], \quad (2.130)$$

which is attributed to Holst [362]. Here, $B_{AB}[e]$ denotes the two-form

$$B_{AB}[e] = \frac{1}{2} \left(e_A \wedge e_B + \frac{i}{2} \epsilon_{ABCD} e^C \wedge e^D \right), \quad (2.131)$$

which is ASD on the pair of indices $[AB]$:

$$\star B_{AB} = -i B_{AB}. \quad (2.132)$$

Note that Eq. (2.132) could be compared to Eq. (2.119).

The Holst action is classically equivalent to the tetradic Palatini action in the sense that it gives the same bulk EoM, provided that the coframe e is nondegenerate.¹⁰ However, the two actions are not equivalent off shell, and further subtleties may arise in the presence of fermions or spin sources.

The ASD two-form in Eq. (2.131) is called the Plebański two-form [315, 339]. In the spinor notation, it boils down to

$$B^{\alpha\beta}[e] = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} (e^{\dot{\alpha}\alpha} \wedge e^{\dot{\beta}\beta}). \quad (2.133)$$

With this understanding, Plebański [315] reformulates Eq. (2.130) as

$$L[B, \gamma, \Psi] = -\frac{2i}{\kappa^2} \left(B_{\alpha\beta} \wedge R^{\alpha\beta}[\gamma] - \frac{1}{2} \Psi^{\alpha\beta\gamma\delta} B_{\alpha\beta} \wedge B_{\gamma\delta} \right), \quad (2.134)$$

¹⁰Variation of γ imposes vanishing torsion $T^A[e, \gamma] = 0$ provided e is nondegenerate. As a result, dropping the torsion-squared term in Eq. (2.130) does not alter the classical EoM.

where $B_{\alpha\beta} = B_{(\alpha\beta)}$ is now taken as a fundamental field. The Lagrange multiplier $\Psi^{\alpha\beta\gamma\delta} = \Psi^{(\alpha\beta\gamma\delta)}$ imposes 5 constraints, reducing the 18 components of $B_{(\alpha\beta)[\mu\nu]}$ down to $13 = 16 - 3$. This precisely corresponds to the degrees of freedom in $e^{\dot{\alpha}\alpha}{}_{\mu}$ modulo SD local Lorentz transformations to which Eq. (2.133) is ignorant.

It should also be clarified that γ in Eq. (2.134) refers to an ASD spin connection, and the SD part completely drops out:

$$R_{\alpha}{}^{\beta}[\gamma] = d\gamma_{\alpha}{}^{\beta} + \gamma_{\alpha}{}^{\gamma} \wedge \gamma_{\gamma}{}^{\beta}. \quad (2.135)$$

Finally, by a rescaling $\gamma \mapsto \kappa^2 \gamma$ [310, 311], together with $B \mapsto iB/2$ and $\Psi \mapsto -2i\kappa^2 \Psi$, Eq. (2.134) is brought to

$$L[B, \gamma, \Psi] = B_{\alpha\beta} \wedge \left(d\gamma^{\alpha\beta} + \kappa^2 \gamma^{\alpha\gamma} \wedge \gamma_{\gamma}{}^{\beta} \right) - \frac{1}{2} \Psi^{\alpha\beta\gamma\delta} B_{\alpha\beta} \wedge B_{\gamma\delta}. \quad (2.136)$$

The interpretation of Eq. (2.136) is detailed in Refs. [310, 311], for instance. In the limit $\kappa \rightarrow 0$, the saddle of Eq. (2.136) describes $dB_{\alpha\beta} = 0$ with $B_{\alpha\beta}$ subject to the simplicity constraint $B_{(\alpha\beta} \wedge B_{\gamma\delta)} = 0$. It is not difficult to see that this implies a coframe $e^{\dot{\alpha}\alpha}$ whose ASD spin connection coefficients vanish. Hence the B -field defines a SD metric via the (spinorial version of) Urbantke formula.

The Plebański formulation of GR in Eq. (2.136) will be compared with the Chalmers-Siegel formulation of YM theory in Eq. (2.118). In both cases, the full theory is formulated in a way that perturbs away from the SD sector.

2.5.3 Chiral Frame Action

The naive perturbation theory of the Lagrangian in Eq. (2.136) might be somewhat confusing when taken literally, however. The B -field is subject to the simplicity constraint. It gains a vacuum expectation value, so the ΨBB term produces a tadpole. Note also that, while the variation of Eq. (2.118) directly yields the SDYM equations in Eq. (2.104), $F^a{}_{\alpha\beta}[A] = 0$, the variation of Eq. (2.136) does not directly yield the SD gravity equations in the form of Eq. (2.105a). Hence the map between YM and gravity in the SD sector, stated in Eq. (2.112), is not direct.

In this light, we may want to envision an alternative approach, based on the frame formulation instead of the coframe formulation.

The point of departure is the Holst formulation in Eq. (2.130). This time, we present it as

$$L[e, \gamma^-] = -i e_A \wedge e_B \wedge d\gamma^{-AB} - i\kappa^2 e_A \wedge e_B \wedge \gamma^{-A}{}_C \wedge \gamma^{-CB}, \quad (2.137)$$

where it is emphasized that γ^- is an ASD connection by fiat:

$$\star \gamma^{-AB} = -i \gamma^{-AB}. \quad (2.138)$$

We have also implemented the rescaling $\gamma^- \mapsto \kappa^2 \gamma^-$.

While Plebański has oriented us to view $e \wedge e$ as the B -field, here we wish to view γ^- as the B -field instead. Hence, via integration by parts, we bring Eq. (2.137) to the sum of the following two terms:

$$L_1[e, \gamma^-] = -i \gamma_{AB}^- \wedge d(e^A \wedge e^B), \quad (2.139a)$$

$$L_2[e, \gamma^-] = -i\kappa^2 e_A \wedge e_B \wedge \gamma^{-A}{}_C \wedge \gamma^{-CB}. \quad (2.139b)$$

With $\varepsilon = \frac{1}{4!} \epsilon_{ABCD} e^A \wedge e^B \wedge e^C \wedge e^D$, it holds that

$$d(e^A \wedge e^B) = \frac{1}{2} \epsilon^{ABCD} d\iota_{E_C} \iota_{E_D} \varepsilon, \quad (2.140a)$$

while Cartan calculus shows that

$$\begin{aligned} d\iota_{E_C} \iota_{E_D} \varepsilon &= \mathcal{L}_{E_C} \iota_{E_D} \varepsilon - \iota_{E_C} d\iota_{E_D} \varepsilon, \\ &= \iota_{[E_C, E_D]} \varepsilon + \iota_{E_D} \mathcal{L}_{E_C} \varepsilon - \iota_{E_C} \mathcal{L}_{E_D} \varepsilon, \\ &= \left(\Omega^I{}_{CD} + \Omega^K{}_{KC} \delta^I{}_D - \Omega^K{}_{KD} \delta^I{}_C \right) \iota_{E_I} \varepsilon. \end{aligned} \quad (2.140b)$$

In the last equality, we have recalled Eq. (2.80). Via Eqs. (2.140a) and (2.140b), Eq. (2.139a) is brought to

$$L_1[E, \gamma^-] = \left(\Omega^C{}_{AB} + \Omega^K{}_{KA} \delta^C{}_B - \Omega^K{}_{KB} \delta^C{}_A \right) \gamma^{-AB}{}_C |E|^{-1} d^4x. \quad (2.141a)$$

We observe the same combination that appeared previously in Eq. (2.108d). Meanwhile, Eq. (2.139b) boils down to

$$L_2[E, \gamma^-] = \kappa^2 \left(\gamma_{ABC}^- \gamma^{-ACB} - \gamma^{-B}{}_{AB} \gamma^{-CA}{}_C \right) |E|^{-1} d^4x. \quad (2.141b)$$

The sum of Eqs. (2.141a) and (2.141b) enjoys three kinds of gauge redundancies. The first is diffeomorphisms. The second is SD local Lorentz transformations, which alter E but fix γ^- . The third is ASD local Lorentz transformations, which alter both E and γ^- . Only the first and the second are robust in the decoupling limit $\kappa \rightarrow 0$.

With this understanding, let us introduce a new scalar degree of freedom f , by plugging in $E_A = f^{-1}V_A$ while declaring the redundancy

$$f \sim e^\varsigma f, \quad V_A \sim e^\varsigma V_A, \quad (2.142)$$

for a local scalar parameter ς (cf. Stueckelberg trick [363]). Now our field basis is (V, γ^-, f) . This brings Eqs. (2.141a) and (2.141b) to

$$L_1[V, \gamma^-, f] = \left(v^C{}_\mu [V_A, V_B]^\mu + 2\Delta_{[A} \delta^C{}_{B]} \right) \gamma^{-AB}{}_C |V|^{-1} d^4x, \quad (2.143a)$$

$$L_2[V, \gamma^-, f] = \kappa^2 \left(\gamma_{ABC}^- \gamma^{-ACB} - \gamma^{-B}{}_{AB} \gamma^{-CA}{}_C \right) f |V|^{-1} d^4x, \quad (2.143b)$$

where $v^A{}_\mu V^\mu{}_B = \delta^A{}_B$, and

$$\begin{aligned} \Delta_A[V, f] &= f \left(\Omega^K{}_{KA} - E_A[\log f] \right), \\ &= f^2 v^K{}_\mu [f^{-1}V_K, f^{-1}V_A]^\mu - V_A[\log f]. \end{aligned} \quad (2.144)$$

Note that the computation performed in Eq. (2.110) implies

$$\mathcal{L}_{V_A}(f^{-2}\varepsilon) = \Delta_A[V, f](f^{-2}\varepsilon). \quad (2.145)$$

Now the sum of Eqs. (2.143a) and (2.143b) exhibits four kinds of gauge redundancies. The first kind is diffeomorphisms. The second kind is SD local Lorentz transformations. The third kind is ASD local Lorentz transformations. The fourth kind is the scaling transformation in Eq. (2.142). The first, the second, and the fourth are robust in the decoupling limit $\kappa \rightarrow 0$.

Direct computation verifies that the set of four functionals in Eq. (2.144), $\Delta_A[V, f]$, changes nontrivially under the scaling and local Lorentz transformations. Hence, at the classical level, we might envision an explicitly gauge-fixed action by introducing the term

$$L_{\text{gf}}[V, \gamma^-, f, \rho] = -\rho^A \Delta_A[V, f] f^{-2}\varepsilon = -\rho^A \mathcal{L}_{V_A}(f^{-2}\varepsilon), \quad (2.146)$$

where ρ^A is a Lagrange multiplier, and we have used the dynamical volume form $\varepsilon = \frac{1}{4!} f^4 \epsilon_{ABCD} v^A \wedge v^B \wedge v^C \wedge v^D$ to prevent interference with the diffeomorphism redundancies. Four combinations of the scaling and local Lorentz redundancies will be eliminated by Eq. (2.146). See Ref. [364] for a remark on the existence of this f for off-shell geometries.

The sum of Eqs. (2.143a), (2.143b), and (2.146) is equivalent to

$$\begin{aligned} &L[V, B, f, \tilde{\rho}] \\ &= \frac{1}{2} B^{AB}{}_\mu [V_A, V_B]^\mu + \frac{\kappa^2}{8} f \mathcal{G}^{\mu\nu}{}_{ABCD}[V] B^{AB}{}_\mu B^{CD}{}_\nu - \tilde{\rho}^A \Delta_A[V, f], \end{aligned} \quad (2.147)$$

where we have performed algebraic field redefinitions,

$$\gamma_{ABC}^- \mapsto \frac{1}{2} B_{AB\mu} V^\mu{}_C |V|, \quad \rho^A \mapsto f|V| \tilde{\rho}^A, \quad (2.148)$$

and also defined

$$\mathcal{G}_{A_1 A_2 B_1 B_2}^{\mu\nu}[V] = 4|V| V^{[\mu}{}_{[A_1} \eta_{A_2][B_1} V^{\nu]}{}_{B_2]} . \quad (2.149)$$

Eqs. (2.147) and (2.149) provide a chiral frame formulation of GR, which is to be compared with the nonchiral frame formulation given in Eqs. (2.69) and (2.68). The action due to Eq. (2.147) will enjoy the diffeomorphism redundancy as well as three more redundancies.

In the decoupling limit $\kappa \rightarrow 0$, one will obtain a Lagrangian formulation of the Mason-Newman [309] formulation of SD gravity: the ASD Lagrange multiplier $\star B^{AB}{}_\mu = -i B^{AB}{}_\mu$ imposes Eq. (2.105a), while the other Lagrange multiplier $\tilde{\rho}^A$ imposes the volume-preserving condition in Eq. (2.105b) with $\nu = f^{-2}\varepsilon$. In this manner, Eq. (2.147) describes full gravity by perturbing away from SD gravity as a SDYM theory for volume-preserving diffeomorphisms. It is a Chalmers-Siegel version of teleparallel gravity, so to speak.

Speaking of which, the Chalmers-Siegel action in Eq. (2.118) unpacks to

$$L[A, B] = \frac{1}{2} B_a{}^{AB} F^a{}_{AB}[A] + \frac{g^2}{8} \left(\delta^{ab} \eta_{A[C} \eta_{D]B} \right) B_a{}^{AB} B_b{}^{CD}, \quad (2.150)$$

where we have used $A, B, \dots$ for the global Lorentz indices; suppose we install a trivial frame $\delta^\mu{}_A$ on the flat spacetime to convert $\mu, \nu, \dots$ to $A, B, \dots$. Eq. (2.150) can be compared with Eq. (2.147). The correspondence is only exact in the decoupling limit, since $f \mathcal{G}_{ABCD}^{\mu\nu}[V]$ in Eq. (2.147) is dynamical.¹¹

2.5.4 Covariant Color-Kinematics Duality

We now wish to treat SD and ASD modes symmetrically. There can be many approaches, but our goal is to demonstrate one concrete version.

To begin with, consider YM theory on a d -dimensional flat spacetime. The electric and magnetic equations of YM theory, shown in Eq. (2.84), are

$$D^\mu F^a{}_{\mu\nu} = 0, \quad D_{[\rho} F^a{}_{\mu\nu]} = 0. \quad (2.151)$$

For the moment, let us assume generic d . We will later specialize in $d = 4$.

¹¹It will be interesting to integrate out the B -field in Eq. (2.147) as an auxiliary and understand the discarded term (around Eq. (2.129)) in terms of frame variables.

In the usual treatment, the gauge potential A^a_μ is taken as the fundamental field. By plugging in the definition of the field strength,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^a_{bc} A^b_\mu A^c_\nu, \quad (2.152)$$

to the electric equation in Eq. (2.151), one derives the second-order EoM,

$$\begin{aligned} \square A^{a\mu} + \partial^\mu(\partial \cdot A^a) &= f^a_{bc} \left(A^{b\rho} \partial_\rho A^{c\mu} - A^{c\rho} \partial_\rho A^{b\mu} \right) - f^a_{bc} A^{b\rho} \partial^\mu A^c_\rho \\ &\quad - f^a_{cb} A^{c\mu} (\partial \cdot A^b) + f^a_{bc} f^c_{de} A^b \cdot A^d A^{e\mu}, \end{aligned} \quad (2.153)$$

where $\square := -\eta^{\mu\nu} \partial_\mu \partial_\nu$.

As a classical field theory, pure YM theory describes the dynamics of “color waves.” The second-order equation in Eq. (2.153) describes a *nonlinear wave equation*. At the linearized order, the solutions to Eq. (2.153) take the form

$$A^{a\mu} = g \sum_i \epsilon_i c_i^a e_i^\mu e^{ik_i x} + \mathcal{O}(g^2), \quad (2.154)$$

where $k_{i\mu}$, c_i^a , and e_i^μ are the momentum, color polarization, and kinematic polarization of the i^{th} color plane wave, which are subject to the massless on-shell condition $k_i^2 = 0$ and $k_i \cdot e_i = 0$. Also, we have chosen to employ formal parameters ϵ_i such that $\epsilon_i^2 = 0$. With $k_{ij} := k_i + k_j$, the generic perturbative solution to Eq. (2.153) is then given by

$$A^{a\mu} = g \sum_i \epsilon_i c_i^a e_i^\mu e^{ik_i x} + g^2 \sum_{i,j} \epsilon_i \epsilon_j \frac{c_{ij}^a e_{ij}^\mu}{k_{ij}^2} e^{ik_{ij} x} + \mathcal{O}(g^3), \quad (2.155)$$

which takes the linearized solution in Eq. (2.154) as a “seed” and constructs an on-shell “nonlinear completion” to a desired order in the coupling g . Eq. (2.155) represents the “nonlinear superposition” of color plane waves.

This is the idea of BG recursion [154] (perturbative method [155–157]): solve nonlinear wave equations perturbatively. Diagrammatically, the perturbative solution at each order in g is given as a sum over tree graphs where each vertex exhibits degree 3 or 4. In this way, one sees that Eq. (2.153) describes cubic and quartic interactions between color plane waves (gluons).

Meanwhile, suppose one wants to see how SD and ASD color waves interact in $d = 4$ dimensions. The field basis $A^{a\mu}$ is not a good choice, since it does not split cleanly into SD and ASD parts.

To this end, we *reimagine* YM theory such that the field strength $F^a_{\mu\nu}$ serves as a fundamental field variable. Since $F^a_{\mu\nu}$ is a two-form, it can be cleanly split into SD and ASD parts in $d = 4$ dimensions.

First of all, a second-order EoM should be derived for the field strength. The electric and magnetic equations in Eq. (2.151) can be combined as follows, yielding a *covariant nonlinear wave equation* for YM theory:

$$\begin{aligned}
D^2 F^a_{\mu\nu} &= -D^\rho D_\mu F^a_{\nu\rho} + D^\rho D_\nu F^a_{\mu\rho}, \\
&= -f^a_{bc} F^{b\rho}_{\mu} F^c_{\nu\rho} + f^a_{bc} F^{b\rho}_{\nu} F^c_{\mu\rho}, \\
&= -2f^a_{bc} F^b_{\mu\rho} F^{c\rho}_{\nu} = -f^a_{bc} \left(F^b_{\mu\rho} F^{c\rho}_{\nu} - F^c_{\mu\rho} F^{b\rho}_{\nu} \right).
\end{aligned} \tag{2.156}$$

The first line uses the magnetic equation. The second line uses the electric equation while commuting covariant derivatives. This gymnastics is due to Cheung and Mangan [90].

Second of all, the field strength should be promoted to a fundamental variable. To this end, one can employ the axial gauge. By introducing a reference vector η^μ , one imposes

$$A^a_{\mu} \eta^\mu = 0 \implies A^a_{\mu} = \frac{1}{\eta \cdot \partial} \eta^\rho F^a_{\rho\mu}. \tag{2.157}$$

After this gauge fixing, the A -field is viewed as a nonlocal functional of the field strength: $A^a_{\mu}[F]$.

Cheung and Mangan [90] make an intriguing observation: the YM field strength $F^{a\mu\nu}$ can be viewed as a BAS field $F^{a\tilde{a}}$ that takes the Lorentz algebra as its tilded Lie algebra, i.e.,

$$\tilde{\mathfrak{g}} = \mathfrak{so}(1, d-1). \tag{2.158}$$

That is, the antisymmetric pair of indices $[\mu\nu]$ is identified as an adjoint index $\tilde{a}$ via the generators of the Lorentz algebra, say $(\Sigma_{\tilde{a}})^{\mu\nu}$:

$$(\Sigma_{\tilde{a}})^\mu_{\rho} (\Sigma_{\tilde{b}})^\rho_{\nu} - (\Sigma_{\tilde{b}})^\mu_{\rho} (\Sigma_{\tilde{a}})^\rho_{\nu} = \tilde{f}^{\tilde{c}}_{\tilde{a}\tilde{b}} (\Sigma_{\tilde{c}})^\mu_{\nu}. \tag{2.159}$$

With this realization, the nonlinear wave equation in Eq. (2.156) appears as

$$-D^2 F^{a\tilde{a}} = f^a_{bc} \tilde{f}^{\tilde{c}}_{\tilde{b}\tilde{c}} F^{b\tilde{b}} F^{c\tilde{c}}, \tag{2.160}$$

which very much resembles the BAS equation in Eq. (1.1). In Eq. (2.160), the covariant derivative gauges the untilded Lie algebra only:

$$D_\rho F^{a\tilde{a}} = \partial_\rho F^{a\tilde{a}} + f^a_{cb} A^c_{\rho} F^{b\tilde{a}}. \tag{2.161}$$

Expanding out the covariant derivative in Eq. (2.160) by Eq. (2.161) gives

$$\begin{aligned} \square F^{a\tilde{a}} = & f^a_{bc} \tilde{f}^{\tilde{a}}_{\tilde{b}\tilde{c}} F^{b\tilde{b}} F^{c\tilde{c}} + \left(2f^a_{bc} A^b \cdot \partial F^{c\tilde{a}} + f^a_{bc} \partial \cdot A^b F^{c\tilde{a}} \right) \\ & + f^a_{be} f^e_{cd} (A^b \cdot A^c) F^{d\tilde{a}}, \end{aligned} \quad (2.162)$$

which explicitly spells out how exactly the nonlinear wave equation for $F^{a\tilde{a}}$ is mismatched from the BAS equation in Eq. (1.1). In this sense, one finds an *approximate* notion of CK duality:

$$\mathbf{YM}(\mathfrak{g}) \cong \mathbf{BAS}(\mathfrak{g}, \mathfrak{so}(1, d-1)) \text{ with corrections.} \quad (2.163)$$

In Cheung and Mangan [90]’s version, the A -field and the F -field are treated as a priori independent objects. Then Eq. (2.160) can be interpreted as describing a “color-Lorentz” scalar wave $F^{\alpha\tilde{a}}$ propagating in the background of the A -field while exhibiting cubic self-interaction. Namely, it is pointed out that Eq. (2.160) is isomorphic to a partially gauged BAS theory. This approximate version of CK duality was dubbed covariant CK duality.

For our purposes, it could be simpler to view $F^{a\tilde{a}}$ as the sole dynamical field by taking the A -field as a dependent field via Eq. (2.157). In $d = 4$ dimensions, the axial gauge condition in Eq. (2.157) allows us to split the A -field *linearly* into SD and ASD parts as

$$A^{\pm a\mu} = \frac{1}{\eta \cdot \partial} F^{a\tilde{a}} (\eta \Sigma_a^{\pm})^\mu \implies A^{a\mu} = A^{+a\mu} + A^{-a\mu}, \quad (2.164)$$

where $(\Sigma_a^{\pm})$ are the SD and ASD Lorentz generators, and $(\eta \Sigma_a^{\pm})^\mu = \eta_\rho (\Sigma_a^{\pm})^{\rho\mu}$. More details will be explored in Sec. 2.5.6.

2.5.5 Electric and Magnetic Equations, II

In Sec. 2.2.1, we identified the electric and magnetic equations of pure GR as Ricci flatness and the algebraic Bianchi identity. In terms of the number of derivatives on the potential field, for instance, this identification aligned with the electric and magnetic equations of gauge theory, Eq. (2.151).

In Sec. 2.4.1, this parallel was reviewed and was refashioned into alternative proposals in Secs. 2.4.3 and 2.4.2, whose validity is yet restricted within the SD sector. As illustrated in Fig. 2.3, these proposals reduce or preserve the derivative count (if the gauge potential A and the metric g are declared as carrying derivative count zero).

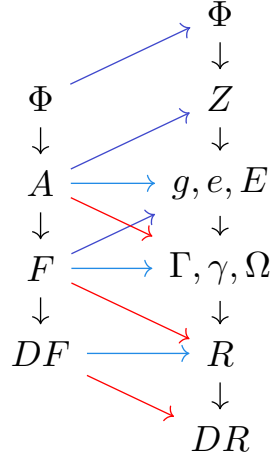


Figure 2.3: Parallels between gauge theory and gravity in the context of double copy, organized in terms of the derivative count. (a) The SD double copy via the second heavenly equation (Sec. 2.4.2) climbs up one level. (b) The Mason-Newman formulation of SD gravity (Sec. 2.4.3) preserves the level. (c) The covariant CK duality or Penrose wave equation (Sec. 2.5.5) step down by one level. For classical double copies, the SD double copy in terms of Tod’s theorem [175, 365] describes case (a), the KS double copy [160] describes case (b), and the Weyl double copy [182, 183] describes case (c). Recall also Fig. 2.1, the structure of electromagnetism and geometrodynamics in outline form.

However, an indisputable parallel between gauge theory and gravity that shall be also seen as classic is the obvious parallel between the connection A and the Levi-Civita connection Γ , or the curvature F and the curvature R . In particular, this parallel is the very insight encoded in Ashtekar variables [61] whose covariant incarnation is Plebański gravity reviewed in Sec. 2.5.2.

In this light, the electric and magnetic equations of gauge theory in Eq. (2.151) shall be paralleled with covariant first-order differential equations on the Riemannian curvature. It can be shown that the Riemannian curvature satisfies

$$\nabla^\rho R^{\mu\nu}{}_{\rho\sigma} = 0, \quad \nabla_{[\kappa} R^{\mu\nu}{}_{\rho\sigma]} = 0, \quad (2.165)$$

in generic d spacetime dimensions. Eq. (2.165) provides our second version of the *electric and magnetic equations of GR*.

The magnetic equation in Eq. (2.165) is the differential Bianchi identity in Eq. (2.8). The electric equation in Eq. (2.165) is implied by the index symmetry in Eq. (2.7), the differential Bianchi identity, and Ricci flatness:

$$\nabla^\rho R^{\mu\nu}{}_{\rho\sigma} = -\nabla^\mu R^{\nu\rho}{}_{\rho\sigma} - \nabla^\nu R^{\rho\mu}{}_{\rho\sigma} = \nabla^\mu \text{Ric}^\nu{}_\sigma - \nabla^\nu \text{Ric}^\mu{}_\sigma = 0. \quad (2.166)$$

Again, a second-order equation on the curvature tensor can be derived by combining the electric and magnetic equations in Eq. (2.165). We implement the following gymnastics:

$$\begin{aligned}
& g^{\kappa\lambda} \nabla_\kappa \nabla_\lambda R^\mu{}_\nu{}^\rho{}_\sigma \\
&= \nabla^\lambda \left(\nabla_\sigma R^\mu{}_\nu{}^\rho{}_\lambda - (\rho \leftrightarrow \sigma) \right), \\
&= \left(R^\mu{}_\kappa{}^\lambda{}_\sigma R^\kappa{}_\nu{}^\rho{}_\lambda - R^\mu{}_\kappa{}^\rho{}_\lambda R^\kappa{}_\nu{}^\lambda{}_\sigma + R^\rho{}_\kappa{}^\lambda{}_\sigma R^\mu{}_\nu{}^\kappa{}_\lambda - R^\mu{}_\nu{}^\rho{}_\kappa R^\kappa{}_\lambda{}^\lambda{}_\sigma \right) - (\rho \leftrightarrow \sigma), \\
&= 2 \left(R^\mu{}_\kappa{}^\lambda{}_\sigma R^\kappa{}_\nu{}^\rho{}_\lambda - R^\mu{}_\kappa{}^\rho{}_\lambda R^\kappa{}_\nu{}^\lambda{}_\sigma \right) + R^\lambda{}_\kappa{}^\rho{}_\sigma R^\mu{}_\nu{}^\kappa{}_\lambda, \\
&= -2 \left(R^\mu{}_\kappa{}^\rho{}_\lambda R^\kappa{}_\nu{}^\lambda{}_\sigma - R^\mu{}_\kappa{}^\lambda{}_\sigma R^\kappa{}_\nu{}^\rho{}_\lambda \right) + R^\mu{}_\nu{}^\kappa{}_\lambda R^\lambda{}_\kappa{}^\rho{}_\sigma.
\end{aligned} \tag{2.167}$$

The first equality uses the magnetic equation in Eq. (2.165). The second equality uses the electric equation in Eq. (2.165). The third equality uses the algebraic Bianchi identity and Ricci flatness.

Eq. (2.167) is known as the Penrose wave equation [366, 367]. By installing an orthonormal frame $E^\mu{}_A$, Eq. (2.167) boils down to

$$\begin{aligned}
D^2 R^A{}_B{}^C{}_D &= -2 \left(R^A{}_E{}^C{}_F R^E{}_B{}^F{}_D - R^A{}_E{}^F{}_D R^E{}_B{}^C{}_F \right) \\
&\quad + R^A{}_B{}^E{}_F R^F{}_E{}^C{}_D.
\end{aligned} \tag{2.168}$$

Again, by identifying the antisymmetrized pair of indices $[AB]$ with a Lorentz Lie algebra index $\tilde{a}$, we view the curvature tensor $R^{ABCD} = R^{[AB][CD]} = R^{[CD][AB]}$ as a symmetric BAS field $R^{\tilde{a}\tilde{b}} = R^{\tilde{b}\tilde{a}}$ that takes the Lorentz Lie algebra as both of its Lie algebras:

$$\mathfrak{g} = \tilde{\mathfrak{g}} = \mathfrak{so}(1, d-1). \tag{2.169}$$

With this understanding, Eq. (2.168) boils down to

$$-D^2 R^{\tilde{a}_1 \tilde{a}_2} = \tilde{f}^{\tilde{a}_1}{}_{\tilde{b}_1 \tilde{c}_1} \tilde{f}^{\tilde{a}_2}{}_{\tilde{b}_2 \tilde{c}_2} R^{\tilde{b}_1 \tilde{b}_2} R^{\tilde{c}_1 \tilde{c}_2} + 2 R^{\tilde{a}_1 \tilde{d}} R_d{}^{\tilde{a}_2}, \tag{2.170}$$

where $\eta^{AB} D_A D_B$ utilizes the *flat* metric, and the covariant derivative describes

$$D_C \phi^{\tilde{a}} = E^\rho{}_C \partial_\rho \phi^{\tilde{a}} + f^{\tilde{a}}{}_{\tilde{c}\tilde{b}} \gamma^{\tilde{c}}{}_C \phi^{\tilde{b}}, \tag{2.171}$$

where $\gamma^{\tilde{a}}{}_C$ is the spin connection coefficient, $\gamma^{AB}{}_C$.

Some crucial clarifications shall be made. First, in Eq. (2.170), all indices are gauged. The tilded color indices are local Lorentz indices coupled to the spin connection as a Lorentz-valued connection $A^{\tilde{a}}{}_\rho$. Second, the covariant

derivative in Eq. (2.171) differs from Eq. (2.161) because it is dressed by the vielbein E^ρ_C .¹² That is, the covariant derivative in Eq. (2.171) does not map to the covariant derivative in Eq. (2.161) by the mere replacement of Lorentz with color. With these understandings, Eq. (2.170) may be seen as a doubly gauged instance of a symmetric BAS theory.¹³

Again, Eq. (2.170) provides an approximate idea for double copy:

$$\mathbf{GR} \cong \mathbf{BAS}(\mathfrak{so}(1, d-1), \mathfrak{so}(1, d-1)) \text{ with corrections.} \quad (2.172)$$

The mismatch, however, is more serious than the YM case in Eq. (2.163). Besides the subtle difference in the covariant derivative and the doubly gauged nature, the term $2R^{\tilde{a}_1\tilde{d}}R_{\tilde{d}}^{\tilde{a}_2}$ in Eq. (2.170) explicitly ruins index factorization. Namely, even with the covariantization, the right-hand side of Eq. (2.170) does not take the form of the right-hand side of the BAS equation in Eq. (1.1).

Despite this mismatch, the Penrose wave equation in the form of Eq. (2.170) offers an interesting perspective on GR. It reimagines gravitational waves as “Lorentz-Lorentz” waves, exhibiting cubic and higher-order self-interactions in terms of Eq. (2.170). It facilitates a suggestive parallel with YM theory as a theory of “color-Lorentz” waves.

2.5.6 Nonlinear Superposition of Self-Dual and Anti-Self-Dual Modes

In Secs. 2.5.4 and 2.5.5, we have reviewed the covariant wave equations for the curvature field in YM theory and GR [90, 366, 367]. We learned that they describe nonlinear waves exhibiting color and Lorentz polarizations:

$$\mathbf{YM}(\mathfrak{g}) : \quad -D^2 F^{a\tilde{a}} = f^a_{bc} \tilde{f}^{\tilde{a}}_{\tilde{b}\tilde{c}} F^{b\tilde{b}} F^{c\tilde{c}}, \quad (2.173a)$$

$$\mathbf{GR} : \quad -D^2 R^{\tilde{a}\tilde{b}} = \tilde{f}^{\tilde{a}}_{\tilde{c}\tilde{e}} \tilde{f}^{\tilde{b}}_{\tilde{d}\tilde{f}} R^{\tilde{c}\tilde{d}} R^{\tilde{e}\tilde{f}} + 2R^{\tilde{a}\tilde{c}} R_{\tilde{c}}^{\tilde{b}}. \quad (2.173b)$$

The purpose of this last subsection is to concretely demonstrate that the above equations uniquely determine the nonlinear superposition between plane waves in YM theory and GR via BG recursion. Crucially, we take the curvature fields as fundamental by employing the axial gauge. We investigate the nonlinear superposition of two plane waves at the leading order in coupling.

¹²When expanded around a flat background, this gives a diffeomorphism connection component; see Chap. IV.

¹³Note that Cheung and Mangan [90]’s covariant CK duality instead concerned Einstein-YM theory, not pure GR. Nevertheless, for our purposes here we study the Penrose wave equation in Eq. (2.170) as the nonlinear wave equation for GR.

Nonlinear Superposition of Yang-Mills Waves We investigate the nonlinear superposition of plane waves in YM theory as color-Lorentz waves.

The perturbative solution to Eq. (2.173a) takes the form

$$F^{a\tilde{a}} = \frac{ig}{\sqrt{2}} \sum_i \epsilon_i c_i^a \varphi_i^{\tilde{a}} e^{ik_i x} - g^2 \sum_{i,j} \epsilon_i \epsilon_j \frac{c_{ij}^a \varphi_{ij}^{\tilde{a}}}{k_{ij}^2} e^{ik_{ij} x} + \mathcal{O}(g^3). \quad (2.174)$$

At the linearized level, $k_{i\mu}$, c_i^a , and $\varphi_i^{\tilde{a}}$, are the momentum, color polarization, and Lorentz polarization of the i^{th} color-Lorentz plane wave. The on-shell condition imposes

$$k_i^2 = 0, \quad k_i^\mu \varphi_{i\mu\nu} = 0, \quad k_{i[\rho} \varphi_{i\rho\nu]} = 0, \quad \varphi_{i\mu}{}^\rho \varphi_{i\rho\nu} = 0. \quad (2.175)$$

The Lorentz polarization $\varphi_i^{\mu\nu}$ recasts the usual vector polarization e_i^μ in a gauge-invariant way. It will also be called the bivector polarization. Here, we work in the convention

$$\varphi_i^{\tilde{a}} (\Sigma_{\tilde{a}})_{\mu\nu} = \varphi_{i\mu\nu} = \sqrt{2} (k_i \wedge e_i)_{\mu\nu}. \quad (2.176)$$

By imposing the axial gauge condition in Eq. (2.157), we pursue a “flipside view” where the color potential $A^{a\mu}$ is a nonlocal functional of the curvature field $F^{\alpha\tilde{a}}$. For simplicity, consider two incident plane waves labeled with 3 and 4. By plugging in Eq. (2.174) to Eq. (2.173a) and examining terms at the order $\mathcal{O}(g^2)$, the perturbative solution is determined up to $\mathcal{O}(g^2)$ as

$$F^{a\tilde{a}} = \frac{ig}{\sqrt{2}} c_3^a \varphi_3^{\tilde{a}} e^{ik_3 x} + \frac{ig}{\sqrt{2}} c_4^a \varphi_4^{\tilde{a}} e^{ik_4 x} - g^2 [c_3, c_4]^a \left([\varphi_3, \varphi_4]^{\tilde{a}} - \varphi_3^{\tilde{a}} \frac{\eta \varphi_4 k_3}{k_4 \eta} + \varphi_4^{\tilde{a}} \frac{\eta \varphi_3 k_4}{k_3 \eta} \right) \frac{1}{k_{34}^2} e^{ik_{34} x} + \mathcal{O}(g^3), \quad (2.177)$$

where $k_{34} := k_3 + k_4$, $[c_3, c_4]^a := f_{bc}^a c_3^b c_4^c$, and $[\varphi_3, \varphi_4]^{\tilde{a}} := \tilde{f}_{\tilde{b}\tilde{c}}^{\tilde{a}} \varphi_3^{\tilde{b}} \varphi_4^{\tilde{c}}$. Accordingly, the color potential $A^{a\mu}$ is determined as

$$\frac{g}{\sqrt{2}} c_3^a \frac{(\eta \varphi_3)^\mu}{k_3 \eta} e^{ik_3 x} + \frac{g}{\sqrt{2}} c_4^a \frac{(\eta \varphi_4)^\mu}{k_4 \eta} e^{ik_4 x} + ig^2 [c_3, c_4]^a \left([\varphi_3, \varphi_4]^{\tilde{a}} - \varphi_3^{\tilde{a}} \frac{\eta \varphi_4 k_3}{k_4 \eta} + \varphi_4^{\tilde{a}} \frac{\eta \varphi_3 k_4}{k_3 \eta} \right) \frac{(\eta \Sigma_{\tilde{a}})^\mu}{k_{34} \eta} \frac{1}{k_{34}^2} e^{ik_{34} x} + \mathcal{O}(g^3), \quad (2.178)$$

where $(\eta \varphi_i)^\mu = \eta_\rho \varphi_{i\rho\mu}$. When listing these solutions, we have replaced the formal parameters ϵ_i with 1 for simplicity.

Eq. (2.177) is an explicit description of how two plane waves in YM theory interact nonlinearly. It describes that two incident plane waves create a

new plane wave at the next order in the coupling, whose color and Lorentz polarizations can be respectively $c_{34}^a = [c_3, c_4]^a$ and

$$\varphi_{34}^{\tilde{a}} = [\varphi_3, \varphi_4]^{\tilde{a}} - \varphi_3^{\tilde{a}} \frac{\eta \varphi_4 k_3}{k_4 \eta} + \varphi_4^{\tilde{a}} \frac{\eta \varphi_3 k_4}{k_3 \eta}, \quad (2.179)$$

up to stripping off the pole $1/k_{34}^2$. The Lie bracket term in Eq. (2.179) arises from the double Lie bracket structure in the right-hand side of Eq. (2.173a), while the reference-dependent terms trace back to the covariantization involved in the left-hand side.

Color-Lorentz Waves in Helicity Basis In $d = 4$ dimensions, we can specialize Eq. (2.179) into two subcases: same-helicity configuration, $(3^+, 4^+)$, and opposite-helicity configuration, $(3^+, 4^-)$. Crucially, the Lorentz polarizations cleanly split into SD and ASD parts, facilitating a systematic examination of the nonlinear superposition in the helicity (more precisely chirality) basis.

For the same-helicity configuration $(3^+, 4^+)$, we see that all contributions in Eq. (2.179) survive, and the resulting $\varphi_{34}^{\tilde{a}}$ also exhibits positive helicity:

$$\varphi_{34}^+ = [\varphi_3^+, \varphi_4^+] - \varphi_3^+ \frac{\eta \varphi_4 k_3}{k_4 \eta} + \varphi_4^+ \frac{\eta \varphi_3 k_4}{k_3 \eta}, \quad \varphi_{34}^- = 0. \quad (2.180)$$

This is because the Lorentz algebra $\tilde{\mathfrak{g}} = \mathfrak{so}(1, 3)$ decomposes into the SD and ASD sectors, each of which is closed by itself.

For the opposite-helicity configuration, we see that the Lorentz interaction term in Eq. (2.179) vanishes: $[\varphi_3^+, \varphi_4^-] = 0$. Again, this is because the SD and ASD sectors of the Lorentz algebra do not talk to each other. As a result, we find that SD and ASD color-Lorentz waves are *invisible* to each other in the sense of the right-hand side of Eq. (2.173a). Yet, the left-hand side effect (contributions from covariantization) survives to give

$$\varphi_{34}^+ = -\varphi_3^+ \frac{\eta \varphi_4^- k_3}{k_4 \eta}, \quad \varphi_{34}^- = \varphi_4^- \frac{\eta \varphi_3^+ k_4}{k_3 \eta}. \quad (2.181)$$

Hence the collision of SD and ASD waves produces both SD and ASD modes.

Overall, we find two types of cubic interaction vertices in the helicity basis at this order: $(+, +, -)$ and $(-, -, +)$, in the all-incoming convention. Surely, this is consistent with a well-known fact about YM theory. Explicitly, one should be able to find the three-point gluon amplitudes by implementing a proper LSZ reduction recipe on the one-point function.

To reiterate, note how the perturbation theory in which the curvature is the foundation of all systematically facilitates viewing YM theory as a nonlinearly interacting system of SD and ASD gluons. In Plebański and Newman's language in Fig.2.2, we just have witnessed how heaven and anti-heaven meet and create earth.

Nonlinear Superposition of Gravitational Waves We investigate the nonlinear superposition of gravitational waves in GR as Lorentz-Lorentz waves.

The perturbative solution to Eq. (2.173b) takes the form

$$R^{\tilde{a}\tilde{b}} = \frac{\kappa}{2} \sum_i \epsilon_i \varphi_i^{\tilde{a}} \varphi_i^{\tilde{b}} e^{ik_i x} + \frac{\kappa^2}{2} \sum_{i,j} \epsilon_i \epsilon_j \frac{\varphi_{ij}^{\tilde{a}\tilde{b}}}{k_{ij}^2} e^{ik_{ij} x} + \mathcal{O}(\kappa^3), \quad (2.182)$$

the linearized part of which describes a superposition of Lorentz-Lorentz plane waves characterized by momentum $k_{i\mu}$ and Lorentz polarization $\varphi_i^{\tilde{a}}$. The on-shell condition imposes $k_i^2 = 0$, $k_i^A \varphi_{iAB} = 0$, $k_{i[C} \varphi_{iAB]} = 0$, and $\varphi_{iA}{}^C \varphi_{iCB} = 0$, where $A, B, \dots$ are flat indices. At the quadratic order (and beyond), our perturbation theory would not manifestly factorize $\varphi_{ij}^{\tilde{a}\tilde{b}}$ into a tensor product of two Lorentz polarizations, in general.

Eq. (2.182) represents the nonlinear superposition of gravitational waves in terms of the curvature tensor. In textbook approaches, gravitational waves and their nonlinear interaction are described in terms of the metric perturbation or the gothic metric perturbation $h^{\mu\nu} = \mathbf{g}^{\mu\nu} - \eta^{\mu\nu}$; see [291], for instance. However, we would like to see if a perturbation theory is viable in which the curvature field is fundamental and is directly governed by the Penrose wave equation in Eq. (2.173b). Again, a charm of this pathway is that a systematic decomposition into SD and ASD modes will be possible in $d = 4$ dimensions. The symmetric field $h^{\mu\nu}$ does not split nicely into SD and ASD parts.

To this end, we devise an *axial gauge prescription for tetradic Palatini gravity*. We have found that the following recipe can be reasonable. First, we must stipulate that the coframe $e^A{}_\mu$ exhibits a nondegenerate vacuum expectation value $\delta^A{}_\mu$, the components of which can simply describe the Kronecker delta. Second, we employ a reference *vector* whose components η^μ in the spacetime frame are constants. For clarity, its index will not be lowered by any means. On the spin connection one-form $\gamma^{\tilde{a}}{}_\mu$ and the *coframe* $e^A{}_\mu$, we impose

$$\gamma^{\tilde{a}}{}_\mu \eta^\mu = 0, \quad e^A{}_\mu \eta^\mu = \delta^A{}_\mu, \quad (2.183)$$

where the vacuum expectation value δ^A_μ is exploited as a reference coframe. These stipulations together determine $\gamma^{\tilde{a}}_\mu$ and the frame E^μ_A as

$$\gamma^{\tilde{a}}_\nu = \frac{1}{\eta \cdot \partial} \left(R^{\tilde{a}\tilde{b}}(\Sigma_{\tilde{b}})_{AB} (\delta^A_\mu \eta^\mu) e^B_\nu \right), \quad (2.184a)$$

$$E^\mu_B = \delta^\mu_B + \frac{1}{\eta \cdot \partial} \left(E^\mu_A \gamma^A_{C\nu} (\delta^C_\mu \eta^\mu) E^\nu_B \right), \quad (2.184b)$$

which follow from the relations

$$\begin{aligned} R^{\tilde{a}\tilde{b}}(\Sigma_{\tilde{b}})_{CD} e^C_\rho e^D_\sigma &= \partial_\rho \gamma^{\tilde{a}}_\sigma - \partial_\sigma \gamma^{\tilde{a}}_\rho + \tilde{f}^{\tilde{a}\tilde{b}\tilde{c}} \gamma^{\tilde{b}}_\rho \gamma^{\tilde{c}}_\sigma, \\ E^\mu_{B,[\rho} e^B_{\sigma]} &= E^\mu_A \gamma^A_{B[\rho} e^B_{\sigma]}, \end{aligned} \quad (2.185)$$

which are Eqs. (2.30) and (2.25). In Eq. (2.184b), δ^μ_B is the inverse of δ^B_μ .

Eq. (2.184) determines the spin connection, the frame, and the coframe as nonlocal functionals of the orthonormal-frame curvature $R^{\tilde{a}\tilde{b}}$, around the flat background δ^A_μ . In this way, a flipside view is provided on tetradic Palatini gravity: curvature as the foundation of all.

By plugging in Eq. (2.184) to Eq. (2.173b), imposing the ansatz in Eq. (2.182), and solving the $\mathcal{O}(\kappa^2)$ part, it can be found that

$$\begin{aligned} \varphi^{\tilde{a}\tilde{b}}_{34} &= \left([\varphi_3, \varphi_4] - \varphi_3 \frac{\eta \varphi_4 k_3}{k_4 \eta} \right)^{\tilde{a}} \left([\varphi_3, \varphi_4] - \varphi_3 \frac{\eta \varphi_4 k_3}{k_4 \eta} \right)^{\tilde{b}} \\ &\quad + \left([\varphi_3, \varphi_4] - \varphi_4 \frac{\eta \varphi_3 k_4}{k_3 \eta} \right)^{\tilde{a}} \left([\varphi_3, \varphi_4] - \varphi_4 \frac{\eta \varphi_3 k_4}{k_3 \eta} \right)^{\tilde{b}} \\ &\quad - [\varphi_3, \varphi_4]^{\tilde{a}} [\varphi_3, \varphi_4]^{\tilde{b}} + \varphi_3 \cdot \varphi_4 \left(\varphi^{\tilde{a}}_3 \varphi^{\tilde{b}}_4 + \varphi^{\tilde{a}}_4 \varphi^{\tilde{b}}_3 \right), \end{aligned} \quad (2.186)$$

where $\varphi_3 \cdot \varphi_4 = \varphi_{3\tilde{a}} \varphi^{\tilde{a}}_4 = \frac{1}{2} \varphi_3^{AB} \varphi_{4AB}$. Again, we assume two incident gravitational waves for simplicity, labeled with 3 and 4.

Again, the squaring of the factor $[\varphi_3, \varphi_4]$ in Eq. (2.186) arises by the double Lie bracket structure in the right-hand side of Eq. (2.173b). We recall the two mismatch factors that made Eq. (2.173b) deviate from the BAS grammar: first, the covariantization of the d'Alembertian (the left-hand side effect), and second, the loss of index factorization by an additional term in the right-hand side. The reference-dependent terms in Eq. (2.186) will be attributed to the former while the term $\varphi_3 \cdot \varphi_4 (\varphi^{\tilde{a}}_3 \varphi^{\tilde{b}}_4 + \varphi^{\tilde{a}}_4 \varphi^{\tilde{b}}_3)$ will be traced back to the latter.

Lorentz-Lorentz Waves in Helicity Basis In $d = 4$ dimensions, we can specialize Eq. (2.186) into two subcases: same-helicity configuration, $(3^+, 4^+)$,

and opposite-helicity configuration, $(3^+, 4^-)$. Again, this analysis explicates the content of the nonlinear superposition of gravity in the chirality basis.

For the same-helicity configuration $(3^+, 4^+)$, all contributions in Eq. (2.186) survive, and the resulting $\varphi_{34}^{\tilde{a}\tilde{b}}$ is also SD on each of its Lorentz indices.

For the opposite-helicity configuration, the decomposition of the Lorentz algebra implies $[\varphi_3, \varphi_4] = 0$ and $\varphi_3 \cdot \varphi_4 = 0$. Hence Eq. (2.186) boils down to

$$\varphi_{34} = \varphi_3^+ \otimes \varphi_3^+ \left(\frac{\eta \varphi_4 k_3}{k_4 \eta} \right)^2 + \varphi_4^- \otimes \varphi_4^- \left(\frac{\eta \varphi_3 k_4}{k_3 \eta} \right)^2, \quad (2.187)$$

which represents a combination of SD and ASD spin-2 modes. Physically, Eq. (2.187) describes that the collision of SD and ASD gravitational waves produces both SD and ASD modes.

Again, we find that $(+, +, -)$ and $(-, -, +)$ interaction vertices are encoded in this computation, in the all-incoming convention. Surely, this is consistent with a well-known fact about GR. Explicitly, one should be able to find the three-point graviton amplitudes via a proper LSZ reduction, which will describe a square of gluon amplitudes.

As a consistency check, it should also be remarked that the curvature field $R^{\tilde{a}\tilde{b}}$ has been subject to algebraic constraints:

$$R^{\tilde{a}\tilde{b}} = R^{\tilde{b}\tilde{a}}, \quad \delta_{\tilde{a}\tilde{b}} R^{\tilde{a}\tilde{b}} = 0, \quad \star_{\tilde{a}\tilde{b}} R^{\tilde{a}\tilde{b}} = 0. \quad (2.188)$$

Here, $\star_{\tilde{b}}$ is the internal Hodge star on the space of antisymmetric rank-two tensors. The second condition is by vanishing Ricci scalar, while the third condition is by algebraic Bianchi. It can be checked that these conditions are dynamically preserved and are also enjoyed by $\varphi_{34}^{\tilde{a}\tilde{b}}$ in Eq. (2.186).

The above computation should provide a concrete and explicit demonstration of the nonlinearity of GR in the chirality basis. This gives a quantitative exploration of the view presented in Fig. 2.2, of Plebański and Newman: four-dimensional GR as a nonlinearly interacting system of SD and ASD modes.

Curvature as the Foundation of All We end with a broad remark. Consider how electromagnetism is taught in our classrooms. The very first concepts are charges, currents, and electromagnetic forces. This leads to the definition of electric and magnetic fields as the physical fields that we directly experience and measure. The scalar and vector potentials are then introduced later, through involving some degree of mathematical sophistication, and are

portrayed as secondary “human” inventions subject to unphysical redundancies. In this way, the field strength F is viewed as more fundamental than the manmade construct A , the gauge potential:¹⁴

$$F \rightsquigarrow A. \quad (2.189)$$

In comparison, consider how GR is taught in our classrooms. The very first concept is the *metric*. The metric is the foundation of all, defining Christoffel symbols (the inertial/gravitational forces—acceleration and frame dragging—experienced in a local laboratory), the Riemann curvature (the tidal forces), and finally the Ricci and Einstein tensors (the sources of gravitation):

$$g \rightsquigarrow \Gamma \rightsquigarrow R. \quad (2.190)$$

Amusingly, this describes the exact *opposite* approach of the pedagogy pursued for electromagnetism. Geometrically, the metric seems to be a fundamental measurable quantity, i.e., length. Physically, however, it is also the potential field for gravitation.

As an experiment, it is interesting to envision an alternative approach to teaching GR that parallels the pedagogy of electromagnetism in Eq. (2.189):

$$R \rightsquigarrow \Gamma \rightsquigarrow g. \quad (2.191)$$

In this flipside view, one starts with stress-energies and the measurement of tidal forces. While taking the curvature as fundamental, the Christoffel symbols are introduced as a secondary construct, from which the metric is eventually reconstructed.¹⁵

Of course, this creative attempt can turn out to be a terrible idea in reality, given the mathematical subtleties and complexities. However, suppose length was somehow not an everyday concept straightforwardly available to human intuition. If length was not an easy, visual, and tangible concept to process, then the only way to discover the metric could be starting from the tidal forces and then inventing secondary constructs in order.

¹⁴Of course, the celebrated debate [368] arises to demonstrate a tension between manifest locality and manifest gauge invariance.

¹⁵If GR was taught in this way, the gravitational Aharonov-Bohm (AB) effect where the proper time gives the quantum phase of a particle [369–373] (see also Refs. [172, 374]) would be perceived as more shocking. The AB effect [368] gives surprise because it seems to necessitate the gauge potential A , seen as a human invention. Length and time, in contrast, are usually not viewed as human inventions.

Notably, the axial gauge formalism established in Eqs. (2.183) and (2.184) has concretely conducted this experiment. There may be more elegant and geometrically more interesting approaches, representing the frame and spin connection with different nonlocal functionals. Some relevant ideas could be Wilson lines, gravitational dressing, “dynamical” lightcone gauge, etc., to record.

2.6 Ending Remarks

In this chapter, we reviewed, derived, or envisioned various formulations of GR in general dimensions and in four dimensions, at the EoM level or at the action level. Certainly, this is not an exhaustive exploration on the subject; see, e.g., Ref. [375]. The fascinating mystery of double copy motivated and guided us throughout this journey as a concrete anchor. We have encountered a vast pool of options for the field space for gravity: a metric or its densitization, a coframe (Lorentz gauge theory), a frame (teleparallel gravity), a single scalar field (Plebański heavenly equation), area elements (Plebański gravity), and the curvature as a gauged BAS field (Penrose wave equation).

Gravitation was once a manifestation of a dynamical metric, but now we are presented with a decentralized view. None of the formulations, however, seem to explain the double copy. It is unclear when the sought-after manifest double copy formulation of GR, if it exists, will make its entrance onto the scene and provide a new invariant center.

One might have to perform field redefinitions that are more aggressively nonlocal.¹⁶ Surely, twistor space [377–380] has been an aspiration. Given that the double copy exists beyond the SD sector and in general d dimensions, the ambitwistor approaches [351–355] can be more relevant. Ambitwistor space is the space of complexified null geodesics in curved spacetime [381]. The ambitwistor string models of Refs. [382, 383] have provided an explanation for the Cachazo-He-Yuan formulae [384, 385], which represent scattering amplitudes of massless particles as moduli space integrals of a factorized integrand. This factorization yields concrete expressions for the KLT [86] version of double copy [386]. For the BCJ [87, 88] version of double copy, however, it has been emphasized that a purely field-theoretic explanation would be desirable. It could be interesting to see if the worldsheet constructions of Refs. [382, 383] have worldline counterparts [387–390]. In this case the kinematic algebra $\mathfrak{g}_?$

¹⁶A related idea is loop space [376].

may be realized as the Poisson algebra of ambitwistor space [391].

The frame formulations of GR apparently do not confirm that $\mathfrak{g}_{\mathcal{F}}$ is the diffeomorphism algebra of spacetime. Still, explorations of the frame formulations could turn out to be meaningful, as diffeomorphism algebras seem to be the best (and perhaps the only) candidates for kinematic algebras regarding the double copy. On a related note, our demonstration of the nonlinear interaction between SD and ASD modes in Sec. 2.5.6 was based on the Lorentz gauge theory picture. In the teleparallel picture, Grant [364] pursued an ambitwistor construction for four-dimensional gravity that generalizes the Mason-Newman construction in Eq. (2.113a), finding an equation of the schematic form

$$[E^A, [E_A, E_B]] \sim \Omega^+ \Omega^-, \quad (2.192)$$

where $\Omega^\pm$ are the SD and ASD components of the anholonomy Ω (by applying the internal Hodge star on its naturally antisymmetric pair of flat indices). Certainly, $\Omega^\pm$ encode the SD and ASD components of the “diffeomorphism field strength.” More explorations will be given in Chap. IV.

GENERALIZED SYMMETRIES IN DYNAMICAL GRAVITY

We explore generalized symmetry in the context of nonlinear dynamical gravity. Our basic strategy is to transcribe known results from YM theory directly to gravity via the tetrad formalism, which recasts general relativity as a gauge theory of the local Lorentz group. By analogy, we deduce that gravity exhibits a one-form symmetry implemented by an operator U_α labeled by a center element α of the Lorentz group and associated with a certain area measured in Planck units. The corresponding charged line operator W_ρ is the holonomy in a spin representation ρ , which is the gravitational analog of a Wilson loop. The topological linking of U_α and W_ρ has an elegant physical interpretation in classical gravitation: the former materializes a tetradic version of a cosmic string whose quantized spin precession angle is measured by the latter. Notably, our conclusions imply that the standard model exhibits a new symmetry of nature at scales below the lightest neutrino mass. More generally, the absence of global symmetries in quantum gravity suggests that the gravitational one-form symmetry is either gauged or explicitly broken; the latter mandates the existence of fermions. Finally, we comment on various generalizations, such as magnetic higher-form or higher-group gravitational symmetries.

Contents of this chapter are adapted from Ref. [186], C. Cheung, M. Derda, J.-H. Kim, V. Nevoa, I. Rothstein, and N. Shah, “Generalized symmetry in dynamical gravity,” *JHEP* **10** (2024) 007, [arXiv:2403.01837 \[hep-th\]](#).

3.1 Introduction

Symmetry has long been a vital tool for investigating complex physical systems, particularly at strong coupling. Historically, most efforts in this expansive subject have focused on conventional symmetries, which act on *local* operators. The standard model of physics exhibits numerous exact and approximate symmetries of this type, for example relating to charge in electromagnetism and chiral symmetry in the strong interactions.

In the past decade, however, the fundamental concept of symmetry has broadened considerably [392–401]. As described in the seminal work [76], it is

now understood that the traditional formulation of symmetry is actually the tip of a colossal iceberg. Rather, there exists a rich patchwork of so-called higher-form symmetries whose distinguishing feature is that they act intrinsically on extended objects described by *nonlocal* operators supported on lines, surfaces, and membranes. Since higher-form symmetries act trivially on local operators, their physical implications are sometimes quite subtle to diagnose. From this point of view, the standard symmetries found in most QFT textbooks are brusquely relegated to the special case of zero-form symmetry.

The growing body of work on generalized symmetries has revealed new perspectives on a broad spectrum of assorted phenomena in QFT, including phase transitions [402–407], anomalies [408–419], and symmetry breaking [420–424]. Recent work has even explored new opportunities for physics beyond the standard model, for example in the context of flavor physics [425], neutrinos [426], and axions [427–436]. Such efforts are a welcome development, as they attempt to draw an explicit connection between highly formal developments in mathematical physics and high-energy physics of actual experimental relevance. That said, the constraints imposed by generalized symmetry on particle physics models tend to be explicable via more conventional means. This is perhaps not so surprising—these models are easily embedded within renormalizable theories in which all is calculable and there are no surprises to be had or which require explanation.

Gravity, on the other hand, is another story. Far less is understood about its putative UV completion. Consequently, the only truly theory-agnostic approach is to retreat to safely low energies, where gravitational dynamics are described universally by an EFT of gravitons on a fixed background, augmented by possible higher-derivative corrections. For example, see Refs. [66, 437] for a review of this perspective. The EFT of gravity is clearly a natural target for understanding generalized symmetry in a refreshingly different context. There has, however, been relatively little effort in this vein.¹ Some notable exceptions include interesting recent work studying the higher-form symmetries associated with parity [438] and topology change [439], as well as generalizations of continuous higher-form symmetries to *linearized gravity* [440–442].

¹The bulk of work that makes reference to both gravity and generalized symmetries has focused on the implications of swampland conjectures. In this picture, one posits a QFT that exhibits certain generalized global symmetries. The conjectured absence of global symmetries in a theory of quantum gravity then imposes constraints on the theory in order to explicitly break or gauge these symmetries.

In this work, we extend the now well-established insights of higher-form symmetry in gauge theory to the effective theory of *nonlinear gravity* in four-dimensional spacetime. Our key ingredient is the well-known fact that gravity can itself be recast in gauge theoretic language. As history would have it, this perspective carries dual meanings. On the one hand, gravity is a theory of diffeomorphisms, nonlinearly realized by a self-interacting, massless spin-two field. Since diffeomorphisms are a redundancy, they are on occasion referred to as a gauge symmetry, though colloquially and not in the strict technical sense. On the other hand, it is well-known that gravity can also be described by a bona fide gauge theory of local Lorentz transformations, which is the so-called Palatini formalism for the tetrad and spin connection. Formally, these descriptions are equivalent since gauge symmetry is, after all, pure redundancy and no redundancy is more valid than any other.² As we will see, tetradic Palatini gravity is perfectly suited to our purposes because we can work in lockstep analogy with the familiar approach taken in gauge theory. When discussing gravity, we work in the phenomenologically relevant case of four spacetime dimensions. Our conclusions are as follows.

First and foremost, our central claim is that tetradic Palatini gravity exhibits an electric one-form symmetry described by the center subgroup $Z(\mathbf{G})$ of the Lorentz group $\mathbf{G}$. This one-form symmetry depends crucially on the signature and global structure of $\mathbf{G}$. For example, in Euclidean signature the center is nontrivial when we consider $Z(\mathbf{SO}(4)) = \mathbb{Z}_2$ or $Z(\mathbf{Spin}(4)) = \mathbb{Z}_2 \times \mathbb{Z}_2$, while in Lorentzian signature the center is nontrivial for $Z(\mathbf{SL}(2, \mathbb{C})) = \mathbb{Z}_2$. In all cases, these center subgroups have a zero-form symmetry action as various parities on Lorentz vector and spinor indices.

Second, we show how the one-form symmetry of gravity is implemented by a topological symmetry operator U_α . This object is constructed explicitly in terms of the local degrees of freedom as the exponential of a certain area operator for a closed surface measured in Planck units and labeled by an element of the center α . The symmetry operator U_α acts on a line operator W_ρ known as the spin holonomy, which is simply the Wilson loop for the spin connection computed in a spin representation ρ along a chosen contour. The symmetry transformation precisely takes the form of a local Lorentz transformation, but

²At low energies, metric gravity and tetradic Palatini gravity are perturbatively equivalent classically and quantum mechanically since they both reproduce a local EFT of a massless spin-two particle. As is well-known, the dynamics of such a theory is uniquely fixed, up to unknown Wilson coefficients.

multivalued such that a nontrivial winding is exhibited, precluding it from being a genuine gauge transformation. By using this multivalued local Lorentz transformation, we show that W_ρ transforms by a center-valued phase that depends only on the topological linking of the surface and curve which define U_α and W_ρ , respectively. In turn, we prove the Ward identity for U_α and W_ρ using both covariant and canonical approaches. This proof is valid to all orders in perturbation theory, at least within the context of the EFT description of gravity³ where the topology and dimension of spacetime are preserved.

Thirdly, we show that the interplay of U_α and W_ρ has a remarkably simple interpretation in terms of *classical gravitation*. The symmetry operator U_α creates a vortex geometry in spacetime that materializes a tetradic version of a cosmic string, serving as a certain gravitational analog of the Dirac string. The strength of this vortex is quantized so as to induce a π spin precession angle which is directly measured by the spin holonomy W_ρ as the center-valued linking number. This linking number can be computed in various backgrounds with or without matter. The topological nature of U_α implies that its linking with W_ρ arises purely from contributions at leading order in the SF expansion, where U_α is treated as a nondynamical background. Furthermore, this implies that higher-order SF corrections vanish, so evidently the classical spin precession angle is not quantum corrected at any perturbative order.

Last but not least, we discuss the breaking of the gravitational one-form symmetry. As expected, explicit breaking requires a local operator in the representation ρ that renders the spin holonomy W_ρ “endable,” thus unspooling its linking with U_α . Physically, this corresponds to the screening of the spin holonomy by spinning particles. Interestingly, the spin holonomy in the vector representation is automatically screened in pure gravity by orbital angular momentum. This mirrors the phenomenon in gauge theory where adjoint Wilson lines are screened by the gluon field itself. On the other hand, holonomies in the spinor representation are endable only by local fermionic operators. If no such operators exist, then the one-form gravitational symmetry is exact. Remarkably, this implies the emergence of a hidden symmetry of the real world: below the lightest neutrino mass, there is a gravitational one-form symmetry under which spinor holonomies are charged. More generally, in a theory

³As is well-known, quantum corrections are perfectly well-defined even within a low-energy EFT, provided one enforces systematic power counting. In the EFT of gravity, most quantum corrections are ultraviolet sensitive and thus absorbed into incalculable counterterms. However, there also exist calculable long-distance quantum corrections [437].

in which the gravitational one-form symmetry is not gauged, the conjectured absence of exact global symmetries in quantum gravity directly implies the existence of fermions.

3.2 The Symmetry Triangle

The modern view states that having a p -form symmetry amounts to having a topological operator supported on closed oriented codimension $p+1$ submanifolds [76]; see Refs. [77–84] for reviews. Concretely, establishing a p -form symmetry means to identify the following:

-
- Charged operators, $W(\mathcal{C}_p)$,
 - Symmetry operator, $U(\mathcal{S}_{d-p-1})$,
 - Ward identity, $\langle W(\mathcal{C}_p) U(\mathcal{S}_{d-p-1}) \rangle = (\text{charge})^{\text{link}(\mathcal{C}_p, \mathcal{S}_{d-p-1})} \langle W(\mathcal{C}_p) \rangle$.
-

Here, $\text{link}(\mathcal{C}_p, \mathcal{S}_{d-p-1})$ denotes the linking number between two closed oriented submanifolds $\mathcal{C}_p$ and $\mathcal{S}_{d-p-1}$, while d is the spacetime dimension.

A useful physical insight is that the symmetry operator $U(\mathcal{S}_{d-p-1})$ serves as the creation operator for a vortex [443–447] configuration, on the background of which the charged operators $W(\mathcal{C}_p)$ measure the degree of “twist” or “swirl”. Notably, this geometrical identification establishes the core part of the Ward identity via the path integral formalism. (In this sense, it may be said that the problem of establishing higher-form symmetries practically boils down to the task of finding vortex configurations in the theory.) This relationship between vortex geometry, charged operators, and symmetry operator is schematically visualized in Fig. 3.1: a “Symmetry Triangle.” Our goal is to demonstrate this template as a recurring structure and apply it to dynamical gravity.

3.2.1 An Abelian Primer

As a prototype demonstration, let us consider abelian p -form gauge theories defined on a Lorentzian d -dimensional background spacetime $\mathcal{M}$. In the first-order formulation, the Lagrangian d -form can be taken as

$$L[A^{(p)}, B^{(d-p-1)}] = B^{(d-p-1)} \wedge dA^{(p)} + f[B^{(d-p-1)}], \quad (3.1)$$

where $f[B^{(d-p-1)}]$ is a d -form functional of the auxiliary $(d-p-1)$ -form field $B^{(d-p-1)}$. The action is the integration of the Lagrangian d -form over $\mathcal{M}$.

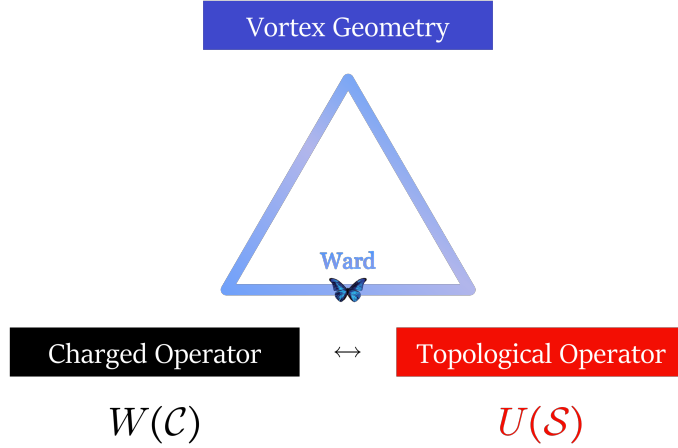


Figure 3.1: The “Symmetry Triangle.” The linking between the charged operator $W(\mathcal{C})$ and the topological symmetry operator $U(\mathcal{S})$ establishes a Ward identity. Physically, $U(\mathcal{S})$ is the creation operator for a vortex configuration whose topological nontriviality is measured by $W(\mathcal{C})$.

The gauge transformations are

$$A^{(p)} \mapsto A^{(p)} + d\xi^{(p-1)}, \quad B^{(d-p-1)} \mapsto B^{(d-p-1)}, \quad (3.2)$$

for a single-valued $(p-1)$ -form parameter $\xi^{(p-1)} \in \Omega^{(p-1)}(\mathcal{M})$. Eq. (3.2) leaves the Lagrangian d -form in Eq. (3.1) invariant.

Now consider the transformation

$$A^{(p)} \mapsto A^{(p)} + 2\pi n \alpha^{(p)}, \quad B^{(d-p-1)} \mapsto B^{(d-p-1)}, \quad (3.3)$$

where $n \in \mathbb{R}$ takes any real value. This assumes a closed oriented codimension $p+1$ submanifold $\mathcal{S}_{d-p-1} \subset \mathcal{M}$, so the p -form $\alpha^{(p)}$ represents the integral de Rham cohomology element $[\alpha^{(p)}] \in H^p(\mathcal{M} \setminus \mathcal{S}_{d-p-1}, \mathbb{Z})$ such that

$$\oint_{\mathcal{C}_p} \alpha^{(p)} = \text{link}(\mathcal{C}_p, \mathcal{S}_{d-p-1}), \quad (3.4)$$

for any p -dimensional submanifold $\mathcal{C}_p \subset \mathcal{M}$ that is closed, oriented, and does not intersect or overlap with $\mathcal{S}_{d-p-1}$.

Note that there are two ways to think about the representative $\alpha^{(p)}$. Mathematicians would like to view it as a closed but not exact p -form defined on the restricted domain $\mathcal{M} \setminus \mathcal{S}_{d-p-1}$. Physicists, on the other hand, will prefer to *formally* view it as a p -form on the entire space $\mathcal{M}$ whose exterior derivative produces a “Dirac delta form” supported on $\mathcal{S}_{d-p-1}$ [80–82, 448, 449],

$$d\alpha^{(p)} = \delta(\mathcal{S}_{d-p-1}). \quad (3.5)$$

Namely, $\alpha^{(p)}$ is a solenoidal field generated by a distributional current. For any oriented p -dimensional submanifold $\mathcal{N}_p \subset \mathcal{M}$, the Dirac delta form is defined as the $(d-p)$ -form such that, for any test p -form field $\varphi^{(p)}$,

$$\int_{\mathcal{N}_p} \varphi^{(p)} = \int_{\mathcal{M}} \varphi^{(p)} \wedge \delta(\mathcal{N}_p). \quad (3.6)$$

These two views are equivalent; see, e.g., Nakahara [30], Secs. 1.9 and 10.5.

It can be seen that Eq. (3.3) is a transformation that creates a codimension $p+1$ vortex configuration. This vortex describes a thin flux tube of strength n such that the Wilson loops,

$$W_m(\mathcal{C}_p) = \exp\left(im \oint_{\mathcal{C}_p} A^{(p)}\right), \quad (3.7)$$

gain AB [368] phases as

$$W_m(\mathcal{C}_p) \mapsto W'_m(\mathcal{C}_p) = W_m(\mathcal{C}_p) \exp(2\pi imn)^{\text{link}(\mathcal{C}_p, \mathcal{S}_{d-p-1})}, \quad (3.8)$$

where $m \in \mathbb{Z}$ is a $\text{U}(1)$ representation (electric charge). As is well known, the AB phase is invisible when the strength n of the flux tube is quantized as $n \in \mathbb{Z}$.

The above observation boils down to the following path integral identity:

$$\begin{aligned} & \int \mathcal{D}A \mathcal{D}B \exp\left(i \int_{\mathcal{M}} B^{(d-p-1)} \wedge d(A^{(p)} - 2\pi n \alpha^{(p)}) + f[B^{(d-p-1)}]\right) W_m(\mathcal{C}_p) \\ &= \int \mathcal{D}A \mathcal{D}B \exp\left(i \int_{\mathcal{M}} B^{(d-p-1)} \wedge dA^{(p)} + f[B^{(d-p-1)}]\right) W'_m(\mathcal{C}_p). \end{aligned} \quad (3.9)$$

This is established by using Eq. (3.3) as a change of path integration variables.

By definition, the right-hand side of Eq. (3.9) is the expectation value of the transformed Wilson loop, $W'_m(\mathcal{C}_p)$. The left-hand side, on the other hand, describes the expectation value of $W_m(\mathcal{C}_p)$ put together with the operator

$$U_n(\mathcal{S}_{d-p-1}) = \exp\left(-2\pi in \int_{\mathcal{M}} B^{(d-p-1)} \wedge d\alpha^{(p)}\right). \quad (3.10)$$

This operator is supported on the submanifold $\mathcal{S}_{d-p-1}$. In view of Eqs. (3.5) and (3.6), Eq. (3.10) boils down to

$$U_n(\mathcal{S}_{d-p-1}) = \exp\left(-2\pi in \int_{\mathcal{S}_{d-p-1}} B^{(d-p-1)}\right). \quad (3.11)$$

The Lagrangian d -form in Eq. (3.1) describes that the saddles of the theory satisfy $dB^{(d-p-1)} = 0$. Hence Eq. (3.11) is a topological operator.

In conclusion, the path integral identity in Eq. (3.9) establishes the p -form symmetry of p -form abelian gauge theories in terms of the Ward identity

$$\langle W_m(\mathcal{C}_p) U_n(\mathcal{S}_{d-p-1}) \rangle = \exp(2\pi i m n)^{\text{link}(\mathcal{C}_p, \mathcal{S}_{d-p-1})} \langle W_m(\mathcal{C}_p) \rangle. \quad (3.12)$$

The charged operators are the Wilson loops $W_m(\mathcal{C}_p)$ in Eq. (3.7), which transform as Eq. (3.8). The symmetry operator $U_n(\mathcal{S}_{d-p-1})$ is explicitly constructed in Eq. (3.11), which is topological under smooth deformations of its support.

We see that Eq. (3.3), the transformation for creating a codimension $p+1$ vortex, is the very p -form symmetry transformation of abelian p -form gauge theories. Note how the symmetry operator $U_n(\mathcal{S}_{d-p-1})$ is systematically deduced as the operator that induces Eq. (3.3) when inserted inside the path integral. That is, it is the operator that precisely gets absorbed into the exponentiated action when using Eq. (3.3) as a change of path integration variables.

A few remarks are in order. First, the Ward identity in Eq. (3.12) admits another physical interpretation as measuring the charge of $W_m(\mathcal{C}_p)$ by a Gaussian pillbox $\mathcal{S}_{d-p-1}$; Eq. (3.11) exponentiates the charge as the Gaussian flux integral. In nonabelian gauge theories, however, the AB phase interpretation turns out to be more robust than the Gaussian pillbox intuition.

Second, note how the “ BdA ” structure of Eq. (3.1) provides a covariant reincarnation of the canonical formalism (in which “ pdx ” gives rise to $[x, p] = i$) such that the charged and symmetry operators develop a conjugate-type relationship as functionals of A and B , serving as the very basis of Eq. (3.12).

Third, an explicit formula for the Dirac delta form in Eq. (3.6) reads

$$\delta(\mathcal{N}_p)_{\mu_1 \dots \mu_{d-p}} = \int d^p \sigma \, \delta^{(d)}(x - X(\sigma)) \frac{\partial X^{\lambda_1}}{\partial \sigma^1} \cdots \frac{\partial X^{\lambda_p}}{\partial \sigma^p} \epsilon_{\lambda_1 \dots \lambda_p \mu_1 \dots \mu_{d-p}}, \quad (3.13)$$

where $\delta^{(d)}(x - X(\sigma))$ is the ordinary Dirac delta function, $X^\mu(\sigma)$ parametrizes $\mathcal{N}_k$, and $\epsilon_{\mu_1 \dots \mu_d}$ is the permutation symbol taking values from $\{+1, -1, 0\}$. The ordinary Dirac delta is reproduced for $p=0$ as an invariant top form. The Dirac delta form of a top-dimensional submanifold is its characteristic function.

Mathematically, $\delta(\mathcal{N}_p)$ is the distributional Poincaré dual of the embedded oriented submanifold $\mathcal{N}_p$: a singular current representative of the Poincaré dual

class [82, 448–452]. This facilitates a consistent definition of the linking and intersection numbers between various oriented submanifolds and a derivation of their relations, with one’s favorite sign conventions. A notable identity is $d\delta(\mathcal{N}_p) = (-1)^p \delta(\partial\mathcal{N}_p)$, implied by Stokes’ theorem.

Last but not least, note that the symmetry transformation in Eq. (3.3) can be interpreted as a *multivalued gauge transformation*. Locally, it looks like the small gauge transformation in Eq. (3.2) since a single-valued $(p-1)$ -form $\xi^{(p-1)}$ such that $\alpha^{(p)} = d\xi^{(p-1)}$ can be constructed within each small patch in view of Poincarè lemma. However, the global existence of such a single-valued $(p-1)$ -form gauge parameter is obstructed, which is precisely the content of the de Rham cohomology $H^p(\mathcal{M} \setminus \mathcal{S}_{d-p-1}, \mathbb{Z})$. If one insists on taking $\alpha^{(p)} = d\xi^{(p-1)}$ for some $\xi^{(p-1)}$, one inevitably has to take $\xi^{(p-1)}$ as multivalued.

3.2.2 Multivalued Gauge Transformation

In gauge theories, one can argue that any higher-form symmetry transformation must locally look like a gauge transformation. This is because it has to preserve all local, i.e., pointlike gauge-invariant operators. Meanwhile, to describe a genuine global symmetry (not redundancy), the higher-form symmetry transformation must also be globally nontrivial at the same time.

A multivalued gauge transformation is a field redefinition that takes the form of a gauge transformation when restricted to each small patch in spacetime. When viewed as a whole, however, it glues such local gauge transformations together into a nontrivial configuration.

**Higher-form symmetry transformations are
multivalued gauge transformations.**

A comparison will be drawn between multivalued gauge transformations and large gauge transformations (in the sense of Ref. [31], not Ref. [453]). Crucially, the latter demands that the gauge parameter is single-valued. The former is a symmetry transformation, while the latter is a gauge transformation.

Take Maxwell theory, for instance. The Dirac string is a magnetic flux tube or equivalently a thin solenoid, which describes a codimension two surface in the spacetime view [454–456]. Its flux is quantized when generated by a single-valued $U(1)$ gauge transformation exhibiting a nontrivial winding [30].

In contrast, a multivalued $U(1)$ gauge transformation generates a Dirac

string whose flux is improperly quantized. This serves as the electric one-form symmetry transformation of Maxwell theory. The symmetry operator creates an improperly quantized Dirac string as the vortex configuration. The charged operators are the Wilson loops, measuring the AB phases around this thin solenoid. An explicit demonstration of these points will appear in Sec. 3.4.

3.3 Universal Template for One-Form Symmetry

Gathering the insights from Sec. 3.2, this section shows how any nonabelian gauge theory of a finite-dimensional (matrix) Lie group $\mathbf{G}$, whose Lagrangian top form admits a first-order formulation in the form

$$L[A, B] = \frac{1}{g^2} \left(B_a \wedge F^a[A] + f[B] \right), \quad (3.14)$$

exhibits a one-form symmetry valued in the center $Z(\mathbf{G})$ of the gauge group. In Eq. (3.14), $F^a[A]$ denotes the field strength of the nonabelian gauge connection,

$$F^a[A] = dA^a + \frac{1}{2} f^a_{bc} A^b \wedge A^c, \quad (3.15)$$

while $f[B]$ is a gauge-invariant algebraic d -form functional of the auxiliary, coadjoint-valued $(d-2)$ -form field B_a . We assume that the theory is defined on a d -dimensional spacetime manifold $\mathcal{M}$. We use $a, b, c, d, \dots$ for adjoint indices and $i, j, k, l, \dots$ for fundamental indices.

The gauge transformations are⁴

$$A \mapsto \text{Ad}_{\Omega^{-1}} A + \Omega^{-1} d\Omega, \quad B \mapsto \text{Ad}_{\Omega^{-1}}^* B, \quad (3.16)$$

where Ω is a single-valued zero-form parameter valued in the gauge group $\mathbf{G}$. Eq. (3.16) leaves the Lagrangian d -form in Eq. (3.14) invariant.

3.3.1 Multivalued Gauge Transformation

Recalling our earlier discussion via Eqs. (3.2) and (3.3), we expect that the one-form symmetry transformation will be obtained by promoting g in Eq. (3.16) to a multivalued parameter $\Omega_{\mathcal{S}}$ that exhibits a nontrivial winding around a closed oriented codimension two submanifold $\mathcal{S} \subset \mathcal{M}$:

$$A \mapsto \text{Ad}_{\Omega_{\mathcal{S}}^{-1}} A + \Omega_{\mathcal{S}}^{-1} d\Omega_{\mathcal{S}}, \quad B \mapsto \text{Ad}_{\Omega_{\mathcal{S}}^{-1}}^* B. \quad (3.17)$$

⁴Of course, Ω should properly approach the identity at the infinities if $\mathcal{M}$ is noncompact.

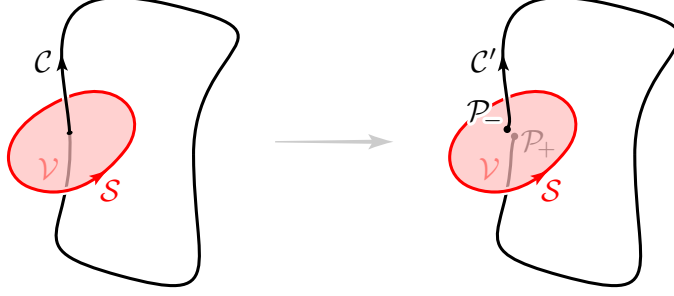


Figure 3.2: Infinitesimal opening of the closed contour $\mathcal{C}$ at the intersection point $\mathcal{C} \cap \mathcal{V}$.

A detailed elaboration on the geometry and definition of $\Omega_{\mathcal{S}}$ is postponed to Sec. 3.4. For the current moment, it suffices to note that the multivaluedness of $\Omega_{\mathcal{S}}$ can be characterized by the discontinuity exhibited across a branch sheet $\mathcal{V}$ such that $\mathcal{S} = \partial\mathcal{V}$. This means to stipulate the condition

$$\lim_{\mathcal{P}_+ \rightarrow \mathcal{P}_-} \Omega_{\mathcal{S}}(\mathcal{P}_+) \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) = \alpha, \quad (3.18)$$

where $\mathcal{P}_+$ and $\mathcal{P}_-$ are points infinitesimally displaced away from the same point on the branch sheet $\mathcal{V}$, but in opposite directions: see Fig. 3.2.

3.3.2 Charged Operator

Next, we identify the charged operator as the Wilson loop $W_{\rho}(\mathcal{C})$ for a representation ρ of the gauge group $\mathbf{G}$, defined by the path-ordered integral formula⁵

$$W_{\rho}(\mathcal{C}) = \text{tr}_{\rho} \bar{\text{Pexp}} \left(\oint_{\mathcal{C}} A \right). \quad (3.19)$$

Here, $\mathcal{C}$ is a closed oriented one-dimensional submanifold in $\mathcal{M}$.

To show that $W_{\rho}(\mathcal{C})$ is charged under the proposed symmetry transformation in Eq. (3.17), it suffices to first consider the case when $\mathcal{C}$ links with $\mathcal{S}$ once. With this assumption, we consider an auxiliary open contour $\mathcal{C}'$ that arises by cutting the loop $\mathcal{C}$ by a branch sheet $\mathcal{V}$, as is illustrated in Fig. 3.2. Let $\partial\mathcal{C}' = \mathcal{P}_+ - \mathcal{P}_-$, where $\mathcal{P}_+$ and $\mathcal{P}_-$ are infinitesimally separated away from the exact intersection point $\mathcal{P} = \mathcal{C} \cap \mathcal{V}$.

Consider the Wilson *line* about the open contour $\mathcal{C}'$, which outputs a group element: $W(\mathcal{C}') = \bar{\text{Pexp}} \left(\int_{\mathcal{C}'} A \right)$. Since Wilson lines exhibit the bilocal trans-

⁵For technical reasons, within this chapter we let Wilson lines solve the right-multiplication transport equation, unlike in Chap. IV where the left-multiplication convention is taken. Hence the path ordering in Eq. (3.19) places earlier points on the left and later points to the right as indicated by an overbar.

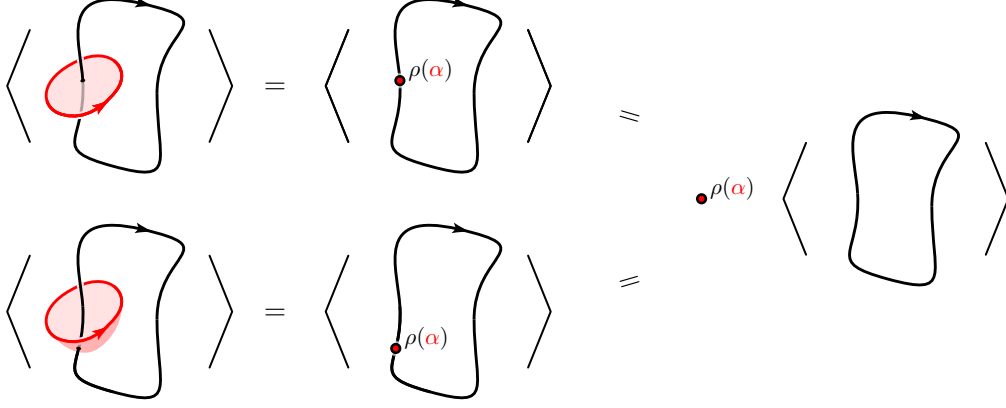


Figure 3.3: The one-form symmetry operator inserts the group element α inside the trace of the Wilson loop. The location of this insertion can be arbitrary if α belongs to the center of the gauge group: $\alpha \in Z(\mathbf{G})$.

formation behavior, Eq. (3.17) transforms $W(\mathcal{C}')$ as

$$W(\mathcal{C}') \mapsto \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \Omega_{\mathcal{S}}(\mathcal{P}_+). \quad (3.20)$$

It follows that the Wilson loop $W_{\rho}(\mathcal{C})$ transforms under Eq. (3.17) as

$$\begin{aligned} W_{\rho}(\mathcal{C}) &\mapsto \text{tr}_{\rho} \lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \left(\Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \Omega_{\mathcal{S}}(\mathcal{P}_+) \right), \\ &= \text{tr}_{\rho} \lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \left(\Omega_{\mathcal{S}}(\mathcal{P}_+) \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \right), \\ &= \text{tr}_{\rho} \left(\lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \alpha W(\mathcal{C}') \right), \end{aligned} \quad (3.21)$$

which shows that the effect of the multivalued gauge transformation is the insertion of the group element α into the Wilson loop at the intersection point $\mathcal{P} = \mathcal{C} \cap \mathcal{V}$. See Fig. 3.3.

At this point, we remind ourselves that the branch sheet $\mathcal{V}$ has been an utterly arbitrary construct. The transformation behavior of the Wilson loop should not depend on one's arbitrary choice of a branch sheet.

Physically, the surface $\mathcal{S}$ will be the support of the symmetry operator. Crucially, the symmetry operator needs to be *topological*. As illustrated in Fig. 3.4, the branch sheet choice precisely corresponds to one's prescription of collapsing the symmetry operator down to a point on the Wilson loop. If this prescription affects the outcome of charge measurement, the very premise that the symmetry operator is topological gets violated.

It is also well-known that any higher-form symmetry algebra must be abelian, since there does not exist a canonical ordering between submanifolds whose

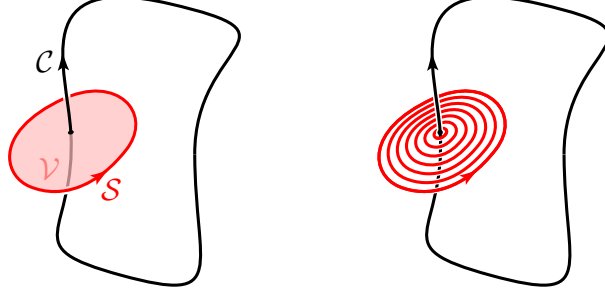


Figure 3.4: *Left:* The linking number between a one-dimensional loop $\mathcal{C}$ and an exact codimension two surface $\mathcal{S} = \partial\mathcal{V}$ encodes the number of intersections between $\mathcal{C}$ and a branch sheet $\mathcal{V}$. *Right:* The branch sheet $\mathcal{V}$ defines a homotopy for shrinking the surface $\mathcal{S}$ to a point on $\mathcal{C}$. Physically, this precisely represents the operation of measuring the charge.

codimension is greater than one. It is not difficult to see that the above “branch sheet independence” is a consistency condition needed for this abelianity of higher-form symmetry algebra as well.

In conclusion, we must stipulate that the transformed Wilson loop is ignorant of the choice of a branch sheet $\mathcal{V}$. As illustrated in Fig. 3.3, this means that the group element α should commute with the rest of the Wilson line regardless of the location of its insertion. This is viable if α belongs to the center $Z(\mathbf{G})$ of the gauge group:

$$\lim_{\mathcal{P}_+ \rightarrow \mathcal{P}_-} \Omega_{\mathcal{S}}(\mathcal{P}_+) \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) = \alpha \in Z(\mathbf{G}). \quad (3.22)$$

The implication is that the multivaluedness of $\Omega_{\mathcal{S}}$ must be valued in $Z(\mathbf{G})$.

Finally, provided this center stipulation, Eq. (3.21) boils down to $W_{\rho}(\mathcal{C}) \mapsto \rho(\alpha) W_{\rho}(\mathcal{C})$, where $\rho(\alpha)$ denotes the representation of the center element α as a complex phase. This establishes that Wilson loops do transform under the multivalued gauge transformation in Eq. (3.17) by the factor $\rho(\alpha)$, for any branch sheet choice. Recalling the fact that we have assumed unit linking between $\mathcal{C}$ and $\mathcal{S}$ for simplicity, we conclude that, for generic $\mathcal{C}$ and $\mathcal{S}$ that do not intersect or overlap,

$$W_{\rho}(\mathcal{C}) \mapsto \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} W_{\rho}(\mathcal{C}). \quad (3.23)$$

3.3.3 Ward Identity

To recapitulate, we have established that Wilson loops are charged under the multivalued gauge transformation in Eq. (3.17) as in Eq. (3.23), provided the

center condition in Eq. (3.22). This result boils down to the following path integral identity:

$$\begin{aligned} & \int \mathcal{D}A \mathcal{D}B \exp \left(\frac{i}{g^2} \int_{\mathcal{M}} B_a \wedge \left(F^a[A] - (\Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}})^a \right) + f[B] \right) W_{\rho}(\mathcal{C}) \\ &= \int \mathcal{D}A \mathcal{D}B \exp \left(\frac{i}{g^2} \int_{\mathcal{M}} B_a \wedge F^a[A] + f[B] \right) \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} W_{\rho}(\mathcal{C}). \end{aligned} \quad (3.24)$$

Recall Eq. (3.9). Again, Eq. (3.24) is established by using Eq. (3.17) as a change of path integration variables. Note that Eqs. (3.15) and (3.17) imply

$$F[\text{Ad}_{\Omega_{\mathcal{S}}^{-1}} A + \Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}}] = \text{Ad}_{\Omega_{\mathcal{S}}^{-1}} F[A] + \Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}}, \quad (3.25)$$

where $dd\Omega_{\mathcal{S}}$ should not be equated to zero due to the multivaluedness of $\Omega_{\mathcal{S}}$; it will eventually produce the Dirac delta form $\delta(\mathcal{S})$.

By definition, the right-hand side of Eq. (3.24) is the expectation value of the transformed Wilson loop. The left-hand side, on the other hand, describes the expectation value of $W_{\rho}(\mathcal{C})$ put together with the operator

$$U_{\alpha}(\mathcal{S}) = \exp \left(-\frac{i}{g^2} \int_{\mathcal{M}} B_a \wedge (\Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}})^a \right). \quad (3.26)$$

The invariant geometrical data characterizing this operator are $\mathcal{S}$, the Dirac delta support, and α , the multivaluedness of $\Omega_{\mathcal{S}}$. Hence it is named $U_{\alpha}(\mathcal{S})$.

To conclude, the path integral identity in Eq. (3.24) establishes the one-form symmetry of nonabelian gauge theories in terms of the Ward identity

$$\langle W_{\rho}(\mathcal{C}) U_{\alpha}(\mathcal{S}) \rangle = \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} \langle W_{\rho}(\mathcal{C}) \rangle. \quad (3.27)$$

The charged operators are the Wilson loops $W_{\rho}(\mathcal{C})$ defined in Eq. (3.19), which transform as Eq. (3.23). The symmetry operator is stipulated to be the operator $U_{\alpha}(\mathcal{S})$ defined in Eq. (3.26). This operator is topological because its action on $W_{\rho}(\mathcal{C})$ does not change under arbitrary smooth deformations of its support $\mathcal{S}$, provided that the symmetry group is an abelian subgroup of the gauge group.

Again, note how the symmetry operator $U_{\alpha}(\mathcal{S})$ is systematically deduced as the operator that induces the multivalued gauge transformation Eq. (3.17) when inserted inside the path integral. It also follows that $U_{\alpha}(\mathcal{S})$ is the creation operator for a codimension two vortex configuration that induces AB phases for colored particles (as Wilson loops) around $\mathcal{S}$. It should also be clear that

$$\langle U_{\alpha}(\mathcal{S}) \mathcal{O} \rangle = \langle \mathcal{O}_{\Omega_{\mathcal{S}}} \rangle \quad (3.28)$$

for any operator $\mathcal{O}$ in the theory, if $\mathcal{O}_{\Omega_{\mathcal{S}}}$ denotes the image of $\mathcal{O}$ under the multivalued gauge transformation in Eq. (3.17). For gauge-invariant local operators, $\mathcal{O}_{\Omega_{\mathcal{S}}} = \mathcal{O}$.

3.3.4 Symmetry Operator

Note that Eq. (3.26) is a concrete formula that expresses the one-form symmetry operator $U_{\alpha}(\mathcal{S})$ in terms of the local degrees of freedom of the theory. A further explicit formula can be obtained by choosing a specific representative for the multivalued gauge parameter $\Omega_{\mathcal{S}}$. As will be discussed in detail in Sec. 3.4, there are many choices one can make. One nice choice is

$$\Omega_{\mathcal{S}} = \exp(\lambda \phi_{\mathcal{S}}) \quad \text{where} \quad e^{2\pi\lambda} = \alpha \in Z(\mathbf{G}), \quad (3.29)$$

where $\phi_{\mathcal{S}}$ is a multivalued function that exhibits a unit winding from 0 to 2π around $\mathcal{S}$, and λ^a is an element of the gauge algebra $\mathfrak{g} = \text{Lie}(\mathbf{G})$ that exponentiates to the center element α . We take λ^a as a constant. With this choice, it follows that

$$dd\phi_{\mathcal{S}} = 2\pi\delta(\mathcal{S}) \implies (\Omega_{\mathcal{S}}^{-1}dd\Omega_{\mathcal{S}})^a = 2\pi\lambda^a\delta(\mathcal{S}), \quad (3.30)$$

so Eq. (3.26) boils down to

$$U_{\alpha}(\mathcal{S}) = \exp\left(-\frac{2\pi i}{g^2} \int_{\mathcal{S}} B_a \lambda^a\right), \quad (3.31)$$

manifesting the surface support on $\mathcal{S}$.

Eq. (3.31) is one possible specific realization of the symmetry operator. This formula, Eq. (3.31), has appeared in the literature in several contexts. In Eq. (46) of Ref. [443] and Eq. (30) of Ref. [444], it was introduced as a vortex creation operator. As a one-form symmetry operator, Ref. [80] provided its realization in a particular setup where temporal winding is utilized. See Ref. [81] for a more sophisticated realization; Eq. (3.31) is also a special case of the family of symmetry operators constructed in Ref. [457]. For a lattice counterpart, see Ref. [458], for instance.

One may worry that Eq. (3.31) utilizes a color reference λ^a on its right-hand side, which is not unique for a given α on the left-hand side. Even worse, λ^a seems to specify an arbitrary direction in color space that naively violates gauge invariance. However, this dependence on λ^a is spurious because the properties of $U_{\alpha}(\mathcal{S})$ are dictated entirely by its action on Wilson loops via

the Ward identity in Eq. (3.27), which depends only on α . Hence any distinct choices of λ^a with the same twist α are regarded as physically equivalent.⁶

It remains to elaborate on the interpretation of the vortex geometry which $U_\alpha(\mathcal{S})$ creates: a colored Dirac string. We wish to do so while revealing more details on the multivalued gauge transformation and its mathematical definition. Yet, readers may skip the following and directly jump to Sec. 3.5.

3.4 More on Multivalued Gauge Transformations

Let us show how the multivalued gauge transformation in abelian and non-abelian gauge theories, or equivalently the geometry of the Dirac strings, is described and defined by the physicists and the mathematicians. See Ref. [30], Secs. 1.9 and 10.5, and also Refs. [456, 460], for instance. For simplicity and concreteness, we take $\mathcal{M}$ as the four-dimensional space of coordinates (t, x, y, z) and take the support $\mathcal{S}$ of the Dirac strings as the t - z plane, for a moment.

3.4.1 Physicists' Approach

To a physicist, a Dirac string in an abelian gauge theory means the following gauge potential:

$$A = n d\phi_{\mathcal{S}} = n \frac{x dy - y dx}{x^2 + y^2}. \quad (3.32)$$

Here, $\phi_{\mathcal{S}}(x, y)$ is the azimuthal angle: a *multivalued function* of x and y with periodicity 2π . By formally treating Eq. (3.32) as a one-form field on the entire space $\mathcal{M}$, the physicist computes its curl as

$$F = dA = n dd\phi_{\mathcal{S}} = 2\pi n \delta(x)\delta(y) dx \wedge dy. \quad (3.33)$$

This describes the magnetic field of an infinitely thin solenoid, placed along the z -axis. The total magnetic flux flowing inside the solenoid is $2\pi n$. This can be either quantized or not quantized: $n \in \mathbb{R}$. The AB phase of a particle with electric charge $m \in \mathbb{Z}$ is $e^{2\pi i m n}$, which can be obtained by computing either $\oint A$ or $\int F$ on account of the Stokes theorem.

⁶This is similar to what occurs in the case of instantons in gauge theory [459]. These pure gauge configurations necessarily specify some explicit path in color space, and hence appear naively color breaking. However, only the topological winding number is the invariant label on these configurations. If desired, one could implement a sum over group orbit for λ^a , for instance.

Clearly, Eq. (3.32) is generated by applying a gauge transformation with the *multivalued parameter* $n\phi_S$ to the trivial configuration $A = 0$. This is why the field strength F is almost zero everywhere, except on a measure-zero support.

The nonabelian generalization is straightforward. A *colored Dirac string* is the following field configuration in a nonabelian gauge theory:

$$A^a = \lambda^a d\phi_S = \lambda^a \frac{x dy - y dx}{x^2 + y^2}. \quad (3.34)$$

Here, λ^a is a constant color factor: a fixed element in the gauge algebra $\mathfrak{g} = \text{Lie}(\mathbf{G})$. The field strength easily computes to

$$F^a = \lambda^a dd\phi_S = 2\pi\lambda^a \delta(x)\delta(y) dx \wedge dy, \quad (3.35)$$

which is simply the colored version of Eq. (3.33). Clearly, this describes the nonabelian magnetic field of an infinitely thin solenoid, placed along the z -axis. The total color flux flowing inside the solenoid is $2\pi\lambda^a$. It describes a color vortex of codimension two.

The analog of the AB phase is the Wilson loop for a representation ρ of the gauge group $\mathbf{G}$, defined in Eq. (3.19). For instance, let $\mathcal{C}$ be the circular contour of constant radius R , parametrized as $(x, y, z) = (R \cos \sigma, R \sin \sigma, 0)$ where σ runs from 0 to 2π . By plugging in Eq. (3.34) to Eq. (3.19), one obtains $W_\rho(\mathcal{C}) = \text{tr}_\rho(e^{2\pi\lambda})$.

As per the conclusion in Sec. 3.3.2, we consider the physically relevant case of a *center* vortex [443–447]. To this end, we stipulate the condition $e^{2\pi\lambda} = \alpha \in Z(\mathbf{G})$ in Eq. (3.29). Concretely, suppose $\mathbf{G} = \text{SU}(N)$, so $Z(\mathbf{G}) = \mathbb{Z}_N$. In this case, we can take [80]

$$\lambda^a = \frac{k}{N} \tau^a, \quad \tau^i_j = i \text{diag}(1, 1, \dots, 1, -N+1), \quad (3.36)$$

where $k \in \mathbb{Z}$. With Eq. (3.36), the Wilson loop for the fundamental representation, for instance, evaluates to

$$W_{\text{fund}}(\mathcal{C}) = e^{2\pi i k/N} N. \quad (3.37)$$

Note that shifting k by an integer multiple of N does not change this result. Physically, this means that two solenoids of different “strengths” can be indistinguishable in terms of the AB phase. In fact, it will become self-evident that such a quantized shift of color flux corresponds to a *single-valued* gauge

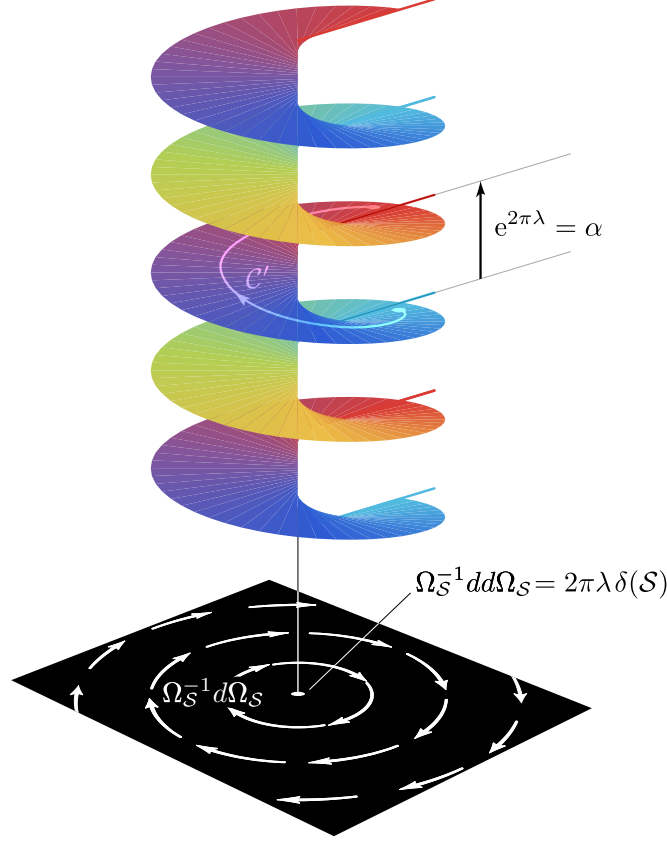


Figure 3.5: The twist of a multivalued gauge parameter $\Omega_{\mathcal{S}}$ is characterized by the discontinuity α exhibited across a branch cut. Physicists approach this twist as the localization of the curl $\Omega_{\mathcal{S}}^{-1}dd\Omega_{\mathcal{S}}$ of the flat connection $\Omega_{\mathcal{S}}^{-1}d\Omega_{\mathcal{S}}$ along $\mathcal{S}$.

transformation (a genuine gauge transformation describing physical equivalence) that is not continuously connected to the identity, i.e., a large gauge transformation in the sense of Ref. [31].

Clearly, Eqs. (3.34) and (3.35) can be written as

$$A = \Omega_{\mathcal{S}}^{-1}d\Omega_{\mathcal{S}}, \quad F = \Omega_{\mathcal{S}}^{-1}dd\Omega_{\mathcal{S}}, \quad \Omega_{\mathcal{S}} = \exp(\lambda\phi_{\mathcal{S}}). \quad (3.38)$$

This shows that the colored Dirac string is created by acting on a multivalued gauge transformation to the trivial configuration $A^a = 0$, by the parameter $\Omega_{\mathcal{S}}$.

The computation of Wilson loops, demonstrated above, can be revisited from this angle. Consider a Wilson *line* $W(\mathcal{C}')$ about a contour $\mathcal{C}'$, obtained as an infinitesimal opening of $\mathcal{C}$ by a branch cut: $(x, y, z) = (R \cos \sigma, R \sin \sigma, 0)$ for $\sigma \in [\epsilon, 2\pi - \epsilon]$; this repeats the approach in Eq. (3.21). In the trivial configuration $A^a = 0$, $W(\mathcal{C}')$ simply computes to the identity element, $\mathbb{1}$. The bilocal transformation behavior of Wilson lines implies that the multivalued

gauge transformation by $\Omega_{\mathcal{S}} = \exp(\lambda\phi_{\mathcal{S}})$ transforms it to

$$W(\mathcal{C}') = \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) \Omega_{\mathcal{S}}(\mathcal{P}_+) = e^{2(\pi-\epsilon)\lambda}, \quad (3.39)$$

if $\mathcal{P}_{\pm}$ are the endpoints of $\mathcal{C}'$ such that $\partial\mathcal{C}' = \mathcal{P}_+ - \mathcal{P}_-$. In the limit $\epsilon \rightarrow 0$, the contour $\mathcal{C}'$ approaches the loop $\mathcal{C}$, so

$$W(\mathcal{C}) = \lim_{\mathcal{C}' \rightarrow \mathcal{C}} \text{tr}_{\rho} W(\mathcal{C}') = \text{tr}_{\rho}(e^{2\pi\lambda}) = \rho(\alpha) \dim \rho, \quad (3.40)$$

reproducing the earlier result. This reinterprets the AB phase in terms of the discontinuity α of the multivalued gauge parameter $\Omega_{\mathcal{S}}$ exhibited at a branch cut. That is, the multivaluedness of $\Omega_{\mathcal{S}}$ is measured by the jump entailed in “gluing” its values across a branch cut.

3.4.2 Mathematician’s Approach

To a mathematician, the Dirac string gauge potential in Eq. (3.32) is perhaps best formulated by not mentioning the multivalued function $\phi_{\mathcal{S}}$. It is rather directly understood as

$$A = 2\pi n \alpha_{\mathcal{S}}, \quad (3.41)$$

with the aid of a representative for an integral de Rham cohomology element:

$$\alpha_{\mathcal{S}} = \frac{x dy - y dx}{x^2 + y^2}, \quad [\alpha_{\mathcal{S}}] = 1 \in H^1(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z}) \cong \mathbb{Z}. \quad (3.42)$$

It should be clear that this exactly is the formalization given around Eq. (3.4). In particular, for any closed loop $\mathcal{C}$ in $\mathcal{M} \setminus \mathcal{S}$, Eq. (3.42) states that

$$\oint_{\mathcal{C}} \alpha_{\mathcal{S}} = \text{link}(\mathcal{C}, \mathcal{S}). \quad (3.43)$$

However, suppose one insists that a satisfactory formulation should also be given to the multivalued function $\phi_{\mathcal{S}}$. There can be many ways to achieve so, but our favorite version might be the following.

In essence, $\phi_{\mathcal{S}}$ is a circle-valued function defined over the punctured domain: $\phi_{\mathcal{S}} \in C^{\infty}(\mathcal{M} \setminus \mathcal{S}, 2\pi\mathbb{R}/\mathbb{Z})$. To characterize its winding, mathematicians study how $\phi_{\mathcal{S}}$ arises by gluing local data.

To this end, suppose an open cover $\{U_i\}$ of $\mathcal{M} \setminus \mathcal{S}$. For each patch U_i , let $\phi_i \in C^{\infty}(U_i, \mathbb{R})$ be the restriction of the azimuthal angle as a single-valued function via a branch choice. It follows that each overlapping region $U_i \cap U_j$

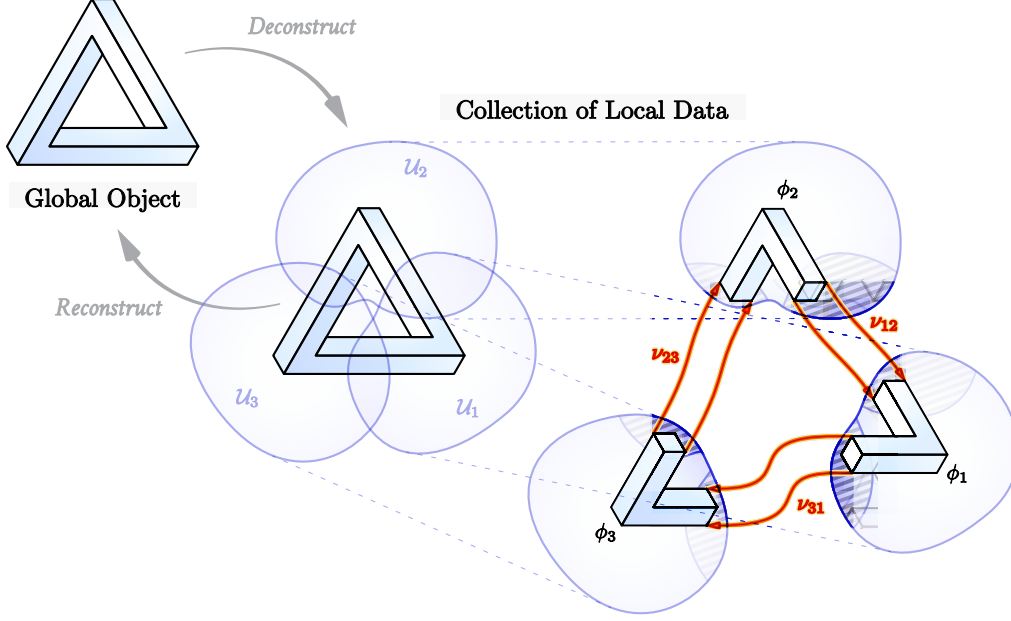


Figure 3.6: Čech cohomology studies whether a collection of local data can be consistently glued into a global object.

is equipped with an integer ν_{ij} via $\phi_i - \phi_j = 2\pi\nu_{ij}$. It also follows that $\nu_{ij} + \nu_{jk} + \nu_{ki} = 0$ on each triple overlap $U_i \cap U_j \cap U_k$. Note that $\nu_{ij} = -\nu_{ji}$.

A collection of data from patch overlaps such as $\{\nu_{ij}\}$ is referred to as a Čech 1-cochain: $\{\nu_{ij}\} \in \check{C}^1(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z})$. The condition $(\delta\nu)_{ijk} = \nu_{ij} + \nu_{jk} + \nu_{ki} = 0$ is known as Čech cocycle condition, where δ is the Čech coboundary operator. Hence it is said that $\{\nu_{ij}\}$ defines not only a Čech 1-cochain but a Čech 1-cocycle: $\{\nu_{ij}\} \in \check{Z}^1(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z})$. See, e.g., Refs. [451, 461].

Ref. [462], Sec. 33.9 provides some helpful visual intuitions for Čech cohomology. See also Fig. 3.6, which recalls Ref. [463].

Surely, there is an arbitrariness in picking ϕ_i for each U_i , which is inherited to ν_{ij} : $\{\nu_{ij}\} \sim \{\nu'_{ij}\}$ if there exists a collection of integers $\{\eta_i\} \in \check{C}^0(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z})$ assigned to each patch such that $\nu'_{ij} - \nu_{ij} = (\delta\eta)_{ij} = \eta_i - \eta_j$.

Invariantly, however, it turns out that $\{\nu_{ij}\}$ is a Čech 1-cocycle that is not a Čech coboundary of any 0-cocycle. It represents a nontrivial element of a Čech cohomology,

$$[\{\nu_{ij}\}] = 1 \in \check{H}^1(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z}) \cong \mathbb{Z}. \quad (3.44)$$

We should talk in terms of explicit examples. Suppose $\mathcal{M} \setminus \mathcal{S}$ is covered by

four patches,

$$\begin{aligned}
U_1 &= \{ (r \cos \sigma_1, r \sin \sigma_1, z) \in \mathbb{R}^3 \mid r > 0, \sigma_1 \in (-\epsilon, \pi/2 + \epsilon), z \in \mathbb{R} \}, \\
U_2 &= \{ (r \cos \sigma_2, r \sin \sigma_2, z) \in \mathbb{R}^3 \mid r > 0, \sigma_2 \in (\pi/2 - \epsilon, \pi + \epsilon), z \in \mathbb{R} \}, \\
U_3 &= \{ (r \cos \sigma_3, r \sin \sigma_3, z) \in \mathbb{R}^3 \mid r > 0, \sigma_3 \in (\pi - \epsilon, 3\pi/2 + \epsilon), z \in \mathbb{R} \}, \\
U_4 &= \{ (r \cos \sigma_4, r \sin \sigma_4, z) \in \mathbb{R}^3 \mid r > 0, \sigma_4 \in (3\pi/2 - \epsilon, 2\pi + \epsilon), z \in \mathbb{R} \},
\end{aligned} \tag{3.45}$$

where $\epsilon > 0$ is infinitesimal. In this case, the collection of integers on patch overlaps, $\{\nu_{ij}\} \in \check{Z}^1(\mathcal{M} \setminus \mathcal{S}, \mathbb{Z})$, can be given as

$$(\nu_{12}, \nu_{23}, \nu_{34}, \nu_{41}) = (0, 0, 0, 1), (0, -1, 0, 2), (-3, 7, -2, -1), \dots \tag{3.46}$$

The first case can arise by choosing $\phi_i = \sigma_i$. The second case can arise by choosing $\phi_1 = \sigma_1 - 2\pi$, $\phi_2 = \sigma_2 - 2\pi$, $\phi_3 = \sigma_3$, and $\phi_4 = \sigma_4$. The third case can arise by choosing $\phi_1 = \sigma_1 + 4\pi$, $\phi_2 = \sigma_2 + 10\pi$, $\phi_3 = \sigma_3 - 4\pi$, and $\phi_4 = \sigma_4$. Invariantly, we find $\nu_{12} + \nu_{23} + \nu_{34} + \nu_{41} = 1$.

The equivalence between $\{\nu_{ij}\}$ in Eq. (3.46) can also be demonstrated explicitly. For example, take the difference between the second and third realizations: $(3, -8, 2, 3)$. It can be seen that this describes the Čech coboundary of $(\eta_1, \eta_2, \eta_3, \eta_4) = (5, 2, 10, 8)$, for instance. This explicitly constructs a “gauge transformation”: $(\delta\eta_{12}, \delta\eta_{23}, \delta\eta_{34}, \delta\eta_{41}) = (3, -8, 2, 3)$.

It remains to provide the formalization for the nonabelian case. Earlier, we have introduced $\Omega_{\mathcal{S}}$ as a multivalued gauge parameter that exhibits a branch cut discontinuity $\alpha \in Z(\mathbf{G})$. In essence, this is a $\mathbf{G}/Z(\mathbf{G})$ -valued map, $\Omega_{\mathcal{S}} \in C^\infty(\mathcal{M} \setminus \mathcal{S}, \mathbf{G}/Z(\mathbf{G}))$, locally lifted to $\mathbf{G}$ with $Z(\mathbf{G})$ -valued transition jumps such that a net monodromy α is exhibited around $\mathcal{S}$.

In the Čech picture, this global object $\Omega_{\mathcal{S}}$ is deconstructed into local data $\{\Omega_i\} \in \check{C}^0(\mathcal{M} \setminus \mathcal{S}, \mathbf{G})$ by again supposing an open cover $\{U_i\}$ of $\mathcal{M} \setminus \mathcal{S}$. For each patch U_i , we choose a group-valued function $\Omega_i \in C^\infty(U_i, \mathbf{G})$. On each overlapping region $U_i \cap U_j$, we stipulate that

$$\Omega_i \Omega_j^{-1} = z_{ij} \in Z(\mathbf{G}), \tag{3.47}$$

where $\{z_{ij}\} \in \check{Z}^1(\mathcal{M} \setminus \mathcal{S}, Z(\mathbf{G}))$ is a center-valued Čech 1-cocycle. This means that, on each triple overlap $U_i \cap U_j \cap U_k$, we impose

$$z_{ij} z_{jk} z_{ki} = \mathbb{1}. \tag{3.48}$$

The topological nontriviality of $\{\Omega_i\}$ arises by stipulating that $\{z_{ij}\}$ represents a nontrivial Čech cohomology element,

$$[\{z_{ij}\}] = \alpha \in \check{H}^1(\mathcal{M} \setminus \mathcal{S}, Z(\mathbf{G})) \cong Z(\mathbf{G}). \quad (3.49)$$

This characterizes the multivaluedness of $\Omega_{\mathcal{S}}$ as an obstruction to a single-valued $\mathbf{G}$ -lift by $\alpha \in Z(\mathbf{G})$.

A few remarks are in order. Firstly, note that the adjoint action of $\Omega_{\mathcal{S}}$ is globally defined as a single-valued operation. This can be seen from Eq. (3.47): $\text{Ad}_{\Omega_i} = \text{Ad}_{\Omega_j}$.

Secondly, it should be clear that this construction describes a generic multivalued gauge transformation parameter $\Omega_{\mathcal{S}}$, so the earlier $\Omega_{\mathcal{S}} = \exp(\lambda\phi_{\mathcal{S}})$ with $e^{2\pi\lambda} = \alpha$ from Eq. (3.38) has provided just one simple realization of $\Omega_{\mathcal{S}}$ among many possibilities; note how the color reference λ^a appears *nowhere* in the above definition of $\Omega_{\mathcal{S}}$. Explicitly, recall the covering of $\mathcal{M} \setminus \mathcal{S}$ in Eq. (3.45). Assuming $\mathbf{G} = \text{SU}(3)$ for concreteness, the center-valued Čech 1-cocycle can be chosen as $(z_{12}, z_{23}, z_{34}, z_{41}) = (\mathbb{1}, e^{-2\pi i/3} \mathbb{1}, \mathbb{1}, e^{4\pi i/3} \mathbb{1})$, so

$$[\{z_{ij}\}] = e^{2\pi i/3} \mathbb{1} \in Z(\mathbf{G}) = \mathbb{Z}_3. \quad (3.50)$$

A possible local deconstruction of $\Omega_{\mathcal{S}}$ can be

$$\Omega_1 = g(\sigma_1), \quad \Omega_2 = g(\sigma_2), \quad \Omega_3 = e^{2\pi i/3} g(\sigma_3), \quad \Omega_4 = e^{2\pi i/3} g(\sigma_4), \quad (3.51)$$

where $g(\sigma)$ is given by

$$\begin{pmatrix} e^{i\sigma/3} \cos^2 \sigma + e^{-2i\sigma/3} \sin^2 \sigma & 0 & (e^{i\sigma/3} - e^{-2i\sigma/3}) \sin \sigma \cos \sigma \\ 0 & e^{i\sigma/3} & 0 \\ (e^{i\sigma/3} - e^{-2i\sigma/3}) \sin \sigma \cos \sigma & 0 & e^{-2i\sigma/3} \cos^2 \sigma + e^{i\sigma/3} \sin^2 \sigma \end{pmatrix}, \quad (3.52)$$

so $g(2\pi) = e^{2\pi i/3}$. Note how this example describes a complicated instance of $\Omega_{\mathcal{S}}$ that encodes a continuously (up to the center jumps) varying direction in the color space, instead of using a fixed color direction λ^a as in Eq. (3.38).

Last but not least, the Čech picture ensures that the composition $\Omega_{\mathcal{S}} \tilde{\Omega}_{\mathcal{S}}$ of two multivalued gauge parameters is well-defined. Let $\{\Omega_i\} \in \check{C}^0(\mathcal{M} \setminus \mathcal{S}, \mathbf{G})$ and $\{\tilde{\Omega}_i\} \in \check{C}^0(\mathcal{M} \setminus \mathcal{S}, \mathbf{G})$ be local deconstructions of $\Omega_{\mathcal{S}}$ and $\tilde{\Omega}_{\mathcal{S}}$, respectively. It follows that $\{\Omega_i \tilde{\Omega}_i\} \in \check{C}^0(\mathcal{M} \setminus \mathcal{S}, \mathbf{G})$ also qualifies as a local deconstruction of a multivalued gauge parameter. Eq. (3.47) implies that $(\Omega_i \tilde{\Omega}_i)(\Omega_j \tilde{\Omega}_j)^{-1} = \Omega_i \tilde{z}_{ij} \Omega_i^{-1} = z_{ij} \tilde{z}_{ij}$, so $\{z_{ij} \tilde{z}_{ij}\}$ can be shown to define a center-valued 1-cocycle

that represents $[\{z_{ij}\tilde{z}_{ij}\}] = \omega\tilde{\omega} \in \check{H}^1(\mathcal{M}\backslash\mathcal{S}, Z(\mathbb{G}))$. The composition $\Omega_{\mathcal{S}}\tilde{\Omega}_{\mathcal{S}}$ is defined as the multivalued function whose local deconstruction is $\{\Omega_i\tilde{\Omega}_i\}$.

For instance, suppose $\Omega_{\mathcal{S}} = \exp(\lambda\phi_{\mathcal{S}})$ and $\tilde{\Omega}_{\mathcal{S}} = \exp(\tilde{\lambda}\phi_{\mathcal{S}})$. The composition of these two gauge parameters is $\Omega_{\mathcal{S}}\tilde{\Omega}_{\mathcal{S}} = \exp(\lambda\phi_{\mathcal{S}})\exp(\tilde{\lambda}\phi_{\mathcal{S}})$, whose multivaluedness does lie within $Z(\mathbb{G})$.

On a related note, the group multiplication rule for the symmetry operator,

$$U_{\alpha}(\mathcal{S})U_{\tilde{\alpha}}(\mathcal{S}) = U_{\alpha\tilde{\alpha}}(\mathcal{S}), \quad (3.53)$$

is required axiomatically. Since our symmetry operator implements a multivalued gauge transformation, the composition of two such transformations automatically yields a third, so we know a priori that the group composition law is valid.⁷

3.4.3 Explicit Examples

Wiggly String We end by providing more explicit examples of the multivalued gauge transformation. Our last example demonstrated that the direction of the color factor can vary. The following example instead fixes the color direction but deforms the shape of the Dirac string to a “wiggly” shape:

$$\Omega_{\mathcal{S}} = \exp\left(\frac{k}{N}\tau\phi_{\mathcal{S}}\right) \quad \text{where} \quad \tan\phi_{\mathcal{S}} = \frac{y-Y(z)}{x-X(z)}. \quad (3.54)$$

Here, the string is realized along $\{(X(z), Y(z), z) | z \in \mathbb{R}\}$. Indeed, a simple calculation verifies that the field strength $F = \Omega^{-1}d\Omega$, arising by applying the multivalued gauge transformation due to Eq. (3.54) to the flat background, is proportional to

$$\delta(x-X(z))\delta(y-Y(z))d(x-X(z))\wedge d(y-Y(z)), \quad (3.55)$$

precisely detecting the wiggly support. It is obvious that the AB phase for an electric charge looping around the string is not modified by these wiggles, demonstrating the topological nature of the Dirac string.

⁷Establishing this at the level of the representatives could be yet subtle. Suppose $U_{\alpha}(\mathcal{S})$ and $U_{\tilde{\alpha}}(\mathcal{S})$ realized with color references λ and $\tilde{\lambda}$ in terms of Eq. (3.31). In the path integral formalism, they naively merge into $\exp(\int_{\mathcal{S}} B_a(\lambda+\tilde{\lambda})^a)$. Confusingly, the multivaluedness of $\exp((\lambda+\tilde{\lambda})\phi_{\mathcal{S}})$ is not guaranteed to lie within $Z(\mathbb{G})$: irrational period. To this end, one can realize α and $\tilde{\alpha}$ as $\exp(2\pi\text{Ad}_{\hat{\Omega}^{-1}}\lambda)$ and $\exp(2\pi\lambda)$.

Gauge Theory at Finite Temperature Another interesting example is gauge theory at finite temperature, described by a compact product manifold with compactified Euclidean time:

$$\mathcal{M} = \mathcal{M}_3 \times \mathbb{S}^1, \quad t \sim t + \beta. \quad (3.56)$$

Here, $\mathcal{M}_3$ is a three-manifold. In addition to the trivial vacuum, there is an infinite set of gauge equivalence classes for the background gauge field. For example, consider

$$A^a = p \frac{2\pi\tau^a}{\beta} dt \quad \text{where } p \in \mathbb{Z}. \quad (3.57)$$

In this background, Wilson loops winding about the thermal circle are trivial. Meanwhile, consider the following multivalued gauge transformation:

$$\Omega = \exp\left(\frac{2\pi k\tau}{N} \frac{t}{\beta}\right). \quad (3.58)$$

Via Eq. (3.17), Eq. (3.58) maps the background in Eq. (3.57) to

$$A^a = \left(p + \frac{k}{N}\right) \frac{2\pi\tau^a}{\beta} dt, \quad (3.59)$$

so the Wilson loops gain a nontrivial phase factor of $e^{2\pi i k/N}$ per each thermal circle. Hence, the multivalued gauge transformation has induced a monodromy in the time direction.

Note that Eqs. (3.57) and (3.59) can be obtained by identifying the ends of a flat gauge field configuration in $\mathcal{M}_3$ times an interval with a twisted boundary condition. An implementation of this construction in Lorentzian signature can be found in Ref. [80].

This example exhibits several interesting features, originating from the topological nontriviality of the spacetime $\mathcal{M}$. A multivalued gauge transformation is constructed without addressing the singular support $\mathcal{S}$. The Wilson loops do not generally admit a coboundary, so they can be closed but not exact.

Compact Support Lastly, we wish to provide an example that explicitly demonstrates that the support $\mathcal{S}$ of the Dirac string can be compact. This is meaningful because the multivalued gauge transformation for the one-form symmetry takes $\mathcal{S}$ as a closed surface. We take $\mathcal{M} \cong \mathbb{R}^4$, coordinatized by (t, x, y, z) .

We employ what is known as toroidal coordinates (η, σ, ϕ) , which foliate the three-dimensional space of (x, y, z) according to a “reference ring” on the x - y plane. For our purposes, we take its radius to be a function of time, $a(t)$:

$$(x, y) = \frac{a(t) \sinh \eta}{\cosh \eta - \cos \sigma} (\cos \phi, \sin \phi), \quad z = \frac{a(t) \sin \sigma}{\cosh \eta - \cos \sigma}. \quad (3.60)$$

While ϕ is the azimuthal angle, surfaces of fixed $\eta \geq 0$ label concentric two-tori which enclose the reference ring, while surfaces of fixed $-\pi \leq \sigma < \pi$ label two-spheres which intersect the reference ring. Also, spatial infinity corresponds to $\sigma = 0$. Let us choose the branch cut for σ such that the discontinuity at $\sigma = \pm\pi$ develops on the disc enclosed by the reference ring.

Crucially, σ describes an angular coordinate that winds like a solenoid about the reference ring. With this understanding, consider

$$\Omega_{\mathcal{S}} = \exp \left(\frac{k}{N} \tau \sigma \right). \quad (3.61)$$

It can be seen that Eq. (3.61) is the multivalued gauge parameter that creates a Dirac string along $\mathcal{S} = \{ (t, x, y, z) \in \mathbb{R}^4 \mid x^2 + y^2 = a^2(t), z = 0 \}$. Clearly, it induces a rephasing for any Wilson loop that wraps the reference ring. In the spacetime view, the branch cut for Eq. (3.61) resides on a volume $\mathcal{V}$ corresponding to the region swept out by the disc enclosed by the reference ring. Furthermore, $\Omega_{\mathcal{S}}$ nicely approaches the identity at spatial infinity.

By choreographing the function $a(t)$ to smoothly increase from and decrease to zero within a time interval $t \in [t_1, t_2]$, the surface $\mathcal{S}$ is made to take finite support in both time and space, parameterized by $t \in [t_1, t_2]$ and $\phi \in [0, 2\pi[$ as $(t, a(t) \cos \phi, a(t) \sin \phi, 0)$.

Existence Argument In general, one wants to construct the multivalued gauge parameter $\Omega_{\mathcal{S}}$ for an arbitrarily shaped codimension two surface $\mathcal{S}$ in an arbitrary manifold $\mathcal{M}$, with or without boundaries. How does one know that such an $\Omega_{\mathcal{S}}$ always exists, given some choice of $\mathcal{S}$ and $\mathcal{M}$?

It can be argued that an $\Omega_{\mathcal{S}}$ can always be found for a given $\mathcal{S}$ and $\mathcal{M}$, on account of an analogous question in *classical magnetostatics*. Deducing $\Omega_{\mathcal{S}}$ from $\mathcal{S}$ and $\mathcal{M}$ is mathematically identical to deducing the static magnetic field and potential of an electric current loop. Imagine we are experimentalists who construct a loop of electrical line current $\mathbf{J}$, built to specification according to some arbitrary contour. On account of Ampère’s law, $\nabla \times \mathbf{H} = \mathbf{J}$, we

can deduce the magnetic field $\mathbf{H}$ by simply “measuring” it. In regions away from the current, we can then reconstruct a magnetic scalar potential via $\mathbf{H} = -\nabla\Psi$. If $\mathcal{M}$ is not closed, we make its boundary $\partial\mathcal{M}$ superconducting to enforce the boundary condition $\mathbf{H}_\perp=0$, in which case Ψ will be set to a constant over $\partial\mathcal{M}$ that we fix to zero. Finally, Ω_S arises by exponentiating Ψ while $\mathbf{H}$ and $\mathbf{J}$ correspond to $\Omega_S^{-1}d\Omega_S$ and $\Omega_S^{-1}dd\Omega_S$, respectively.

3.5 Application to Gravity

Do gravitational higher-form symmetries exist? Suppose gravity admits a higher-form symmetry. On account of the symmetry triangle in Fig. 3.1, the symmetry operator will generate a vortex configuration. One can argue that this vortex must *gravitate topologically*, since otherwise the symmetry operator cannot be topological. For instance, the vortex must not produce a Newtonian $1/r$ gravitational field. One should not be able to deduce the presence of the vortex by measurements bound to each local laboratory. The vortex will be detected only through global interferometer-type experiments.

This provides an intriguing top-down angle unique to gravity. To relativists, it has been known that string defects in gravity fall into two categories:

- *Cosmic string* [464, 465]. A concentration of stress-energy, characterized by the tension μ (energy per unit length). Detected by measuring the conical deficit angle $\Delta\theta = 8\pi G\mu$ in terms of scalar particle geodesics. Can be obtained via removing a “wedge” from spacetime and performing a gluing. Entails a distributional Ricci curvature and is sourced by the Nambu-Goto worldsheet. See Refs. [466–469] and also Refs. [470–473].
- *Misner string* [474–478]. A thin flux tube of NUT charge, characterized by the total gravitomagnetic flux N . Serves precisely as the gravitational counterpart of the Dirac string when the NUT charge is seen as the gravitomagnetic charge [331, 479]. Detected by measuring the time monodromy $\Delta t = 8\pi GN$ as a pure frame dragging effect: Sagnac interferometry [480, 481] as gravitational AB effect [482–484]. Involves no conical deficit angle and thus is tensionless. Describes the very string defect of the Taub-NUT solution. See also Refs. [163, 172, 290, 485–493].

A top-down angle for relativists would be constructing a one-form symmetry that corresponds to each of these topologically gravitating defects.

Meanwhile, we see that the universal template established in Eq. (3.3) rather immediately implies the existence of a one-form symmetry in tetradic Palatini gravity, valued in the center of the local Lorentz group. Amusingly, this identifies a string defect that is subtly different from the standard cosmic string:

- *Tetradic Cosmic String.* A concentration of stress-energies that creates quantized spin precession angles. Detected by interferometry with fermions, but *cannot* be detected by scalar particle geodesics. Hence differs from the standard cosmic string.

As a related, more elementary object, we will also find the following:

- *Chiral Tetradic Cosmic String.* A concentration of a chiral (SD or ASD) stress-energy that creates quantized spin precession angles. Namely, a combined concentration of both electric and magnetic stress-energies in equal magnitudes. Detected by interferometry with spinning particles, including fermions, but *cannot* be detected by scalar particle geodesics. Entails a distributional Ricci curvature as well as a distributional violation of the algebraic Bianchi identity.

Below, we investigate the gravitational one-form symmetry whose vortex configuration is the tetradic cosmic string, although investigating other proposals such as the Misner string will be an intriguing research direction (see Sec. 4.6).

3.5.1 Gravity as a Lorentz Gauge Theory

To recapitulate, the universal construction in Sec. 3.3 establishes the existence of a one-form symmetry in nonabelian gauge theories of a finite-dimensional matrix Lie group $\mathbf{G}$ by deriving the Ward identity, in which the Lagrangian top form, the charged operator, and the symmetry operator are

$$L[A, B] = \frac{1}{g^2} \left(B_a \wedge F^a[A] + f[B] \right), \quad (3.62a)$$

$$W_\rho(\mathcal{C}) = \text{tr}_\rho \bar{\text{Pexp}} \left(\oint_{\mathcal{C}} A \right), \quad (3.62b)$$

$$U_\alpha(\mathcal{S}) = \exp \left(-\frac{i}{g^2} \int_{\mathcal{M}} B_a \wedge (\Omega_{\mathcal{S}}^{-1} d\Omega_{\mathcal{S}})^a \right), \quad (3.62c)$$

respectively. The center element $\alpha \in Z(\mathbf{G})$ and the closed oriented submanifold support $\mathcal{S} \subset \mathcal{M}$ of codimension two provide the invariant characterization of the multivalued gauge parameter $\Omega_{\mathcal{S}}$.

For instance, take YM theory defined on a background d -dimensional pseudo-Riemannian manifold $\mathcal{M}$. In the first-order formulation, the Lagrangian d -form reads

$$L[A, B] = \frac{1}{g^2} B_a \wedge F^a[A] - \frac{1}{2g^2} B_a \wedge *B^a, \quad (3.63)$$

where $*$ denotes the Hodge dual acting on spacetime indices. The universal construction of Sec. 3.3 immediately proves the electric one-form symmetry, identifies the Wilson loops as the charged operators, and constructs the symmetry operator explicitly in terms of the theory's local degrees of freedom.

Now consider gravity in four spacetime dimensions. The tetradic Palatini formulation of GR describes a gauge theory of a Lorentz group $\mathbf{G}$, i.e., a Lie group whose algebra $\mathfrak{g} = \text{Lie}(\mathbf{G})$ is either $\mathfrak{so}(1, 3)$ or $\mathfrak{so}(4)$ depending on the signature. To be concrete, we may presume $\mathfrak{g} = \mathfrak{so}(1, 3)$ for the moment.

The Lagrangian four-form reads

$$L[\gamma, e] = \frac{1}{4\kappa^2} \epsilon_{ABCD} e^A \wedge e^B \wedge R^{CD}[\gamma] - \frac{\Lambda}{24\kappa^2} \epsilon_{ABCD} e^A \wedge e^B \wedge e^C \wedge e^D, \quad (3.64)$$

where the internal indices $A, B, \dots \in \{0, 1, 2, 3\}$ are the local Lorentz indices, and ϵ_{ABCD} is the permutation symbol that is nondynamical. We have assumed a generic cosmological constant Λ and identified the gravitational coupling as $\kappa = \sqrt{8\pi G_{\text{N}}}$.

The coframe one-form e^A provides local trivializations of the cotangent bundle $T^*\mathcal{M}$. It defines the spacetime metric as a *composite* operator,

$$g_{\mu\nu}[e] = \eta_{AB} e^A{}_\mu e^B{}_\nu, \quad (3.65)$$

via an invariant constant internal metric η_{AB} that facilitates the raising and lowering of the internal indices.

The spin connection one-form $\gamma^{AB} = -\gamma^{BA}$ refers to a Lorentz-valued one-form that defines a connection on the tangent bundle $T\mathcal{M}$. Its covariant exterior derivative D preserves the constant internal metric η_{AB} . Its curvature two-form $R^A{}_B[\gamma]$ is defined as

$$R^A{}_B[\gamma] = d\gamma^A{}_B + \gamma^A{}_C \wedge \gamma^C{}_B, \quad (3.66a)$$

which precisely reproduces Eq. (3.15). In fact, by identifying the antisymmetric pair of fundamental indices $[AB]$ with an adjoint index a , Eq. (3.66a) can be

written as

$$R^a[\gamma] = d\gamma^a + \frac{1}{2} f^a{}_{bc} \gamma^b \wedge \gamma^c, \quad (3.66b)$$

if $f^a{}_{bc}$ denotes the structure constants of the Lorentz algebra $\mathfrak{g}$.

The Lagrangian four-form in Eq. (3.64) is quadratic in the spin connection $\gamma^A{}_B$. Its EoM give $D(e^A \wedge e^B) = 0$, which is algebraically⁸ equivalent to the vanishing of the torsion two-form $De^A = de^A + \gamma^A{}_B \wedge e^B$ provided e^A is nondegenerate. This analysis shows that $\gamma^A{}_B = \gamma^A{}_{BC} e^C$ can be viewed as an auxiliary field that can be eliminated at tree level by plugging its classical solution in terms of e^A which the torsion-free condition stipulates. This means to equate $\gamma^A{}_{BC}$ with the connection coefficients of the Levi-Civita connection of the metric in Eq. (3.65) in the local Lorentz frame. It is not difficult to show that the resulting action is equated with the Einstein-Hilbert action for the metric in Eq. (3.65). Variation with respect to the coframe e^A precisely reproduces Einstein's equations with the cosmological constant Λ .

In conclusion, tetradic Palatini gravity is classically equivalent to the metric theory of GR, provided there are no sources that couple directly to the spin connection. As noted earlier, quantum equivalence also follows since the EFT of a massless spin-two particle is unique, modulo Wilson coefficients.

As emphasized earlier, we work in an EFT description of gravity which describes gravitons propagating over a fixed background. Hence, throughout our analysis the coframe and metric are implicitly expanded as fluctuations about some choice of background values $\bar{e}^A{}_\mu$ and $\bar{g}_{\mu\nu}$, respectively, though it will usually be simpler to manipulate the full field variables rather than their fluctuations.⁹

In consideration of Eq. (3.66b), we identify the two-form composite operator

$$B_{AB}[e] = \frac{1}{2} \epsilon_{ABCD} e^C \wedge e^D, \quad (3.67)$$

⁸The torsion is non-propagating: any coupling of the spin connection to an external source generates nonzero torsion precisely only on the support of that source.

⁹In any sensible EFT, the background spacetime is nondegenerate and thus the background tetrad $\bar{e}^A{}_\mu$ must be nonzero. This means the vacuum breaks diffeomorphism invariance and local Lorentz symmetry down to the diagonal, which naively hinders our analysis. However, our calculation of the Ward identity for the symmetry and line operators utilizes the full tetrad field $e^A{}_\mu$, which transforms covariantly, so there is no additional complication. The very same phenomenon occurs in gauge theory, where expanding about a background gauge field $\bar{A}^a{}_\mu$ technically breaks Lorentz invariance and color symmetry down to the diagonal, but of course with no effect on the Ward identities in the theory.

which we may want to refer to as the Plebański two-form [315, 339]. This is a two-form valued in the coadjoint of the local Lorentz algebra $\mathfrak{g}$. Eq. (3.67) facilitates a rewriting of Eq. (3.64) as

$$L[\gamma, e] = \frac{1}{\kappa^2} \left(B_a[e] \wedge R^a[\gamma] - \frac{\Lambda}{6} B_a[e] \wedge \star B^a[e] \right), \quad (3.68)$$

where $a, b, \dots \in \{1, 2, 3, 4, 5, 6\}$ are the adjoint indices. Here, $\star$ denotes the Hodge star in the *internal* space. When the adjoint indices are all converted to fundamental indices by the generators $(t_a)^{AB} = -(t_a)^{BA}$ of $\mathfrak{g}$, we have

$$B_{AB}[e] = \frac{1}{2} \epsilon_{ABCD} e^C \wedge e^D, \quad \star B^{AB}[e] = e^A \wedge e^B. \quad (3.69)$$

Note that the internal Hodge dual $\star$ should be distinguished from the spacetime Hodge dual $*$ in Eq. (3.63).

When presented as in Eq. (3.68), it is evident that tetradic Palatini gravity conforms to the “ BF ” grammar of the universal template in Sec. 3.3: compare it with Eq. (3.14).¹⁰ The only difference is that the B -field is only *composite* for gravity, via the formula in Eq. (3.67). Importantly, we will see shortly that this does not raise any issue in the path integral formalism since the B -field exhibits the correct behavior under the multivalued gauge transformation regardless of its compositeness. Hence the universal construction of Sec. 3.3 precisely applies to tetradic Palatini gravity up to a straightforward refinement.

The implication is that dynamical gravity exhibits a one-form symmetry valued in the center $Z(\mathbf{G})$ ¹¹ as a gauge theory of the local Lorentz group $\mathbf{G}$. The summary in Eq. (3.62) applies as

$$L[\gamma, e] = \frac{1}{\kappa^2} \left(B_a[e] \wedge R^a[\gamma] - \frac{\Lambda}{6} B_a[e] \wedge \star B^a[e] \right), \quad (3.70a)$$

$$W_\rho(\mathcal{C}) = \text{tr}_\rho \bar{\text{P}} \exp \left(\oint_{\mathcal{C}} \gamma \right), \quad (3.70b)$$

¹⁰It should be clear that the statement here by no means implies that gravity is a topological BF theory, even when the cosmological constant term is neglected. In Eq. (3.68), the B -field refers to the composite operator defined in Eq. (3.67) and does not constrain all components of $R^{AB}[\gamma]$, which is precisely why Einstein’s equations do not kill all Riemann components. The Plebański formulation [315] could only promote B_a as fundamental field variable by imposing a constraint on it (the simplicity constraint) so that not all components of B_a are independent Lagrange multipliers.

¹¹To be precise, this statement is up to self-screening: see the discussion in Sec. 3.6. The symmetry group can technically be smaller than $Z(\mathbf{G})$ for tetradic theories in Euclidean or complexified signatures, while, for the Plebański formulation where the (co)adjoint-valued B -field is fundamental, no such subtlety arises in any signature. Below, we assume that this point on self-screening is understood.

$$U_\alpha(\mathcal{S}) = \exp\left(-\frac{i}{\kappa^2} \int_{\mathcal{M}} B_a[e] \wedge (\Omega_S^{-1} dd\Omega_S)^a\right), \quad (3.70c)$$

where $W_\rho(\mathcal{C})$ is a functional of γ and $U_\alpha(\mathcal{S})$ is a functional of e .

This one-form symmetry will be preserved by the inclusion of higher-curvature corrections for the EFT of gravity. In any case, the B -field will always describe the conjugate field variable of the curvature in the EFT.¹²

It remains to elaborate on the viable options for the symmetry group $Z(\mathbf{G})$. Note that actions and Lagrangian top forms do not specify the global structure of the gauge group $\mathbf{G}$ but only specify the gauge algebra $\mathfrak{g}$.

In Lorentzian signature, the viable options for the gauge group $\mathbf{G}$ are¹³

$$\text{Spin}(1, 3) \cong \text{SL}(2, \mathbb{C}), \quad \text{SO}^+(3, 1) \cong \frac{\text{Spin}(3, 1)}{\mathbb{Z}_2}. \quad (3.71)$$

The latter is the restricted Lorentz group. The former is the double cover of the latter. The corresponding center subgroups are, respectively,

$$\mathbb{Z}_2, \quad \mathbb{1}. \quad (3.72)$$

These are the allowed one-form symmetries of gravity in Lorentzian signature. Note that the zero-form symmetry associated with the former corresponds to *net* parity on chiral and antichiral spinor indices, also known as fermion parity.

For completeness of our exposition, we also discuss Euclidean signature, in which case $\mathfrak{g} = \mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2) \cong \mathfrak{spin}(4)$. Hence we can choose the gauge group $\mathbf{G}$ (which we may refer to as the “local Lorentz group” by abuse of terminology) to be either¹⁴

$$\begin{aligned} \text{Spin}(4) &\cong \text{SU}(2) \times \text{SU}(2), \\ \text{SO}(4) &\cong \frac{\text{Spin}(4)}{\mathbb{Z}_2}, \\ \frac{\text{SO}(4)}{\mathbb{Z}_2} &\cong \frac{\text{Spin}(4)}{\mathbb{Z}_2 \times \mathbb{Z}_2}, \end{aligned} \quad (3.73)$$

¹²For instance, utilizing a Lagrangian such as $L[\gamma, e, r, B] = \kappa^{-2} (B_a \wedge (R^a[\gamma] - r^a) + f[e, r])$ would incorporate the cases with algebraic higher-curvature operators.

¹³For simplicity, we restrict our attention to connected gauge groups.

¹⁴A priori, one can also consider semispin groups such as $\text{SemiSpin}(4) \cong \text{SU}(2) \times \text{SO}(3)$, which is yet a nonstandard quotient. Note also that, in the cases of such semispin groups or $\mathbf{G} = \text{PSO}(4) = \text{SO}(4)/\mathbb{Z}_2$, the local Lagrangian will be realized in the Plebański formulation (reviewed in Secs. 2.5.2 and 3.A.3) for not invoking fields in the vector representation such as the coframe e^A .

whose center subgroups are given by

$$\mathbb{Z}_2 \times \mathbb{Z}_2, \quad \mathbb{Z}_2, \quad \mathbb{1}, \quad (3.74)$$

respectively. The one-form charges will be valued in either of these center subgroups. Note that the zero-form symmetry associated with the center $\mathbb{Z}_2 \times \mathbb{Z}_2$ of $\text{Spin}(4) \cong \text{SU}(2) \times \text{SU}(2)$ acts as parity on chiral and antichiral spinor indices, while that of the center $\mathbb{Z}_2$ of $\text{SO}(4)$ acts as parity on *vector* indices.¹⁵

Lastly, as a formal consideration one may be interested in the case of the complexified gauge algebra $\mathfrak{g} = \mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$. The viable center subgroups are again $\mathbb{Z}_2 \times \mathbb{Z}_2$ and its quotients as in Eq. (3.74).

Below, we explicitly spell out how the general construction in Sec. 3.3 applies to the tetradic Palatini formalism to establish the one-form symmetry of gravity. Without loss of generality, we assume the context of Lorentzian signature while being agnostic about the local Lorentz group $\mathbf{G}$. Concretely, we show that the dynamical theory of a coframe e^A and a spin connection $\gamma^{AB} = -\gamma^{BA}$, defined on a four-manifold $\mathcal{M}$ by the Lagrangian four-form

$$L[\gamma, e] = \frac{1}{\kappa^2} \left(B_a[e] \wedge R^a[\gamma] + f[e] \right), \quad (3.75)$$

exhibits a one-form symmetry valued in the center $Z(\mathbf{G})$. The definitions in Eqs. (3.67) and (3.66b) state that $B_a[e]$ and $R^a[\gamma]$ respectively transform in the coadjoint and adjoint of the local Lorentz group, while $f[e]$ in Eq. (3.75) is an algebraic four-form functional invariant under any local Lorentz transformations.

3.5.2 Multivalued Local Lorentz Transformation

The gauge transformations of tetradic Palatini gravity are

$$\gamma^A{}_B \mapsto (\Omega^{-1})^A{}_C \gamma^C{}_D \Omega^D{}_B + (\Omega^{-1})^A{}_C d\Omega^C{}_B, \quad e^A \mapsto (\Omega^{-1})^A{}_B e^B, \quad (3.76)$$

where Ω is a single-valued zero-form parameter valued in the local Lorentz group $\mathbf{G}$. Eq. (3.76) leaves the Lagrangian four-form in Eq. (3.75) invariant.

The symmetry transformation is a multivalued local Lorentz transformation. For a multivalued parameter Ω_S that exhibits a nontrivial winding around a

¹⁵It is instructive to compare the cases when the gravitational one-form symmetry group is $\mathbb{Z}_2$. In the Lorentzian signature, the local Lorentz group should be $\mathbf{G} = \text{Spin}(1, 3)$, in which case vector holonomies are *not charged*. In the Euclidean signature, we have $\mathbf{G} = \text{SO}(4)$ (for the standard option) so fermions do not exist while vector holonomies are *charged*.

closed oriented two-dimensional surface $\mathcal{S} \subset \mathcal{M}$, it is

$$\begin{aligned}\gamma^A{}_B &\mapsto (\Omega_{\mathcal{S}}^{-1})^A{}_C \gamma^C{}_D (\Omega_{\mathcal{S}})^D{}_B + (\Omega_{\mathcal{S}}^{-1})^A{}_C d(\Omega_{\mathcal{S}})^C{}_B, \\ e^A &\mapsto (\Omega_{\mathcal{S}}^{-1})^A{}_B e^B.\end{aligned}\tag{3.77}$$

Just as in Eq. (3.22), $\Omega_{\mathcal{S}}$ is stipulated to satisfy the center jump condition

$$\lim_{\mathcal{P}_+ \rightarrow \mathcal{P}_-} \Omega_{\mathcal{S}}(\mathcal{P}_+) \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) = \alpha \in Z(\mathbf{G}),\tag{3.78}$$

across a branch sheet $\mathcal{V}$ such that $\mathcal{S} = \partial\mathcal{V}$. The relevant illustration has been given in Fig. 3.2, clarifying the meaning of the points $\mathcal{P}_+$ and $\mathcal{P}_-$ in Eq. (3.78).

3.5.3 Spin Holonomy as Charged Operator

The charged operator is the spin holonomy $W_{\rho}(\mathcal{C})$ for a representation ρ of the local Lorentz group $\mathbf{G}$, i.e., the Wilson loop for the spin connection about a closed oriented one-dimensional contour in $\mathcal{M}$:

$$W_{\rho}(\mathcal{C}) = \text{tr}_{\rho} \bar{\text{Pexp}} \left(\oint_{\mathcal{C}} \gamma \right).\tag{3.79}$$

To show that $W_{\rho}(\mathcal{C})$ is charged under the symmetry transformation stated in Eq. (3.77), it suffices to first consider the case when $\mathcal{C}$ links with $\mathcal{S}$ once. With this assumption, we consider an auxiliary open contour $\mathcal{C}'$ defined previously in Fig. 3.2. The Wilson line about this open contour is the parallel propagator with respect to the spin connection γ . Its bilocal transformation behavior implies

$$W(\mathcal{C}') \mapsto \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \Omega_{\mathcal{S}}(\mathcal{P}_+),\tag{3.80}$$

so the spin holonomy in Eq. (3.79) transforms as

$$\begin{aligned}W_{\rho}(\mathcal{C}) &\mapsto \text{tr}_{\rho} \lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \left(\Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \Omega_{\mathcal{S}}(\mathcal{P}_+) \right), \\ &= \text{tr}_{\rho} \lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \left(\Omega_{\mathcal{S}}(\mathcal{P}_+) \Omega_{\mathcal{S}}^{-1}(\mathcal{P}_-) W(\mathcal{C}') \right), \\ &= \text{tr}_{\rho} \left(\lim_{\mathcal{P}_{\pm} \rightarrow \mathcal{P}} \alpha W(\mathcal{C}') \right) = \rho(\alpha) W_{\rho}(\mathcal{C}),\end{aligned}\tag{3.81}$$

where $\rho(\alpha)$ denotes the representation of the center element α as a complex phase. Notably, this computation is independent of the branch sheet choice by virtue of the fact that α commutes with any element in the local Lorentz group $\mathbf{G}$. Finally, for a generic contour $\mathcal{C}$, it follows that

$$W_{\rho}(\mathcal{C}) \mapsto \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} W_{\rho}(\mathcal{C}).\tag{3.82}$$

3.5.4 Ward Identity

The above computation boils down to the following path integral identity:

$$\begin{aligned} & \int \mathcal{D}\gamma \mathcal{D}e \exp\left(\frac{i}{\kappa^2} \int_{\mathcal{M}} B_a[e] \wedge \left(R^a[\gamma] - (\Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}})^a\right) + f[e]\right) W_{\rho}(\mathcal{C}) \\ &= \int \mathcal{D}\gamma \mathcal{D}e \exp\left(\frac{i}{\kappa^2} \int_{\mathcal{M}} B_a[e] \wedge R^a[\gamma] + f[e]\right) \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} W_{\rho}(\mathcal{C}). \end{aligned} \quad (3.83)$$

This follows by using Eq. (3.77) as a change of path integration variables. Note that Eqs. (3.66) and (3.77) imply

$$\begin{aligned} & R^A{}_B[\Omega_{\mathcal{S}}^{-1} \gamma \Omega_{\mathcal{S}} + \Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}}] \\ &= (\Omega_{\mathcal{S}}^{-1})^A{}_C R^C{}_D[\gamma] (\Omega_{\mathcal{S}})^D{}_B + (\Omega_{\mathcal{S}}^{-1})^A{}_C dd(\Omega_{\mathcal{S}})^C{}_B, \end{aligned} \quad (3.84)$$

where $dd\Omega_{\mathcal{S}}$ should not be equated to zero due to the multivaluedness of $\Omega_{\mathcal{S}}$.

By definition, the right-hand side of Eq. (3.83) is the expectation value of the transformed spin holonomy. The left-hand side, on the other hand, describes the expectation value of $W_{\rho}(\mathcal{C})$ put together with the operator

$$U_{\alpha}(\mathcal{S}) = \exp\left(-\frac{i}{\kappa^2} \int_{\mathcal{M}} B_a[e] \wedge (\Omega_{\mathcal{S}}^{-1} dd\Omega_{\mathcal{S}})^a\right). \quad (3.85)$$

As explicated in detail in Sec. 3.4, the invariant geometrical data characterizing this operator are $\mathcal{S}$, the Dirac delta support, and α , the multivaluedness of $\Omega_{\mathcal{S}}$. Hence it is named $U_{\alpha}(\mathcal{S})$.

To conclude, the path integral identity in Eq. (3.83) establishes the one-form symmetry of nonabelian gauge theories in terms of the Ward identity

$$\langle W_{\rho}(\mathcal{C}) U_{\alpha}(\mathcal{S}) \rangle = \rho(\alpha)^{\text{link}(\mathcal{C}, \mathcal{S})} \langle W_{\rho}(\mathcal{C}) \rangle. \quad (3.86)$$

The charged operators are the Wilson loops $W_{\rho}(\mathcal{C})$ defined in Eq. (3.79), which transform as Eq. (3.82). The symmetry operator is stipulated to be the operator $U_{\alpha}(\mathcal{S})$ defined in Eq. (3.85). This operator is topological because its action on $W_{\rho}(\mathcal{C})$ does not change under arbitrary smooth deformations of its support $\mathcal{S}$, provided that the symmetry group is an abelian subgroup of the local Lorentz group.

Again, the symmetry operator $U_{\alpha}(\mathcal{S})$ is systematically deduced as the operator that induces the multivalued local Lorentz transformation in Eq. (3.77) when inserted inside the path integral. It should also be clear that

$$\langle U_{\alpha}(\mathcal{S}) \mathcal{O} \rangle = \langle \mathcal{O}_{\Omega_{\mathcal{S}}} \rangle \quad (3.87)$$

for any operator $\mathcal{O}$ in the theory, if $\mathcal{O}_{\Omega_S}$ denotes the image of $\mathcal{O}$ under the multivalued local Lorentz transformation in Eq. (3.77). For local operators valued in the singlet of the local Lorentz group, $\mathcal{O}_{\Omega_S} = \mathcal{O}$. It is worth emphasizing that the metric $g_{\mu\nu}[e] = \eta_{AB} e^A_\mu e^B_\nu$ in Eq. (3.65) is one such singlet operator.

As will be elaborated on in Sec. 3.5.6, it follows that $U_\alpha(\mathcal{S})$ is the creation operator for a two-dimensional vortex configuration, i.e., a string defect, that induces spin precession angles for spinning particles (as spin holonomies) around $\mathcal{S}$. This is the gravitational AB effect.

Before moving on, we may need a space to take stock on some subtle aspects of the gravitational one-form symmetry.

Firstly, we may have to take stock on the thorny issue of diffeomorphism invariance. Any local or quasilocal object such as a point, curve, or surface in spacetime is famously not diffeomorphism invariant in the presence of dynamical gravitation, and the charged operator $W_\rho(\mathcal{C})$ defined in Eq. (3.79) is no exception. Although it does not carry dangling indices, it does specify a particular collection of spacetime points in the same way as a local operator $\mathcal{O}(x)$, e.g., the Ricci scalar $\text{Ric}(x)$, picks a point $x \in \mathcal{M}$. In fact, this annoyance has also been present in any investigation of Maxwell or YM theory Wilson loops in the presence of dynamical gravitation, which is central to many well-established discussions of swampland conjectures. A related discourse could be found in Refs. [494, 495], known as gravitational dressing.

However, we see that the Ward identity in Eq. (3.86) is a concrete result in tetradic Palatini gravity. Most importantly, the linking number $\text{link}(\mathcal{C}, \mathcal{S})$ is clearly diffeomorphism invariant. The linking between $\mathcal{C}$ and $\mathcal{S}$ is topological information invariant under coordinate transformations, encoded in their relative relationship. We do not work in the topology-changing regime of gravity, so there is no need to worry about gravitational fluctuations unlinking $\mathcal{C}$ and $\mathcal{S}$ or changing $\text{link}(\mathcal{C}, \mathcal{S})$. The coordinate representations of the submanifolds $\mathcal{C}$ and $\mathcal{S}$ will vary, but $\text{link}(\mathcal{C}, \mathcal{S})$ is always well-defined as the same value. With this understanding, Eq. (3.86) stands as a perfectly diffeomorphism-covariant¹⁶ identity, where the covariances of the left-hand side and the right-hand side match. In this context, diffeomorphisms pose no threat to the well-definedness of Eq. (3.86).

Secondly, the one-form symmetry of gravity will still be present in pure

¹⁶Again, even a scalar operator such as $\text{Ric}(x)$ is diffeomorphism covariant, not invariant.

tetrad gravity. While our derivations have made elaborate use of the first-order (Palatini) formalism, our final conclusions on the one-form symmetry will remain valid also after integrating out the spin connection.

Certainly, it is generally said that global symmetries shall be viewed as independent of one's choice of formalism. A much subtler question along this line is whether there is a one-form symmetry in the pure-metric or metric-affine theories of gravity, where there is no tetrad, spin connection, or local Lorentz symmetry to speak of. In particular, there will be no such thing as the one-form symmetry for spinor holonomy, because of the simple reason that one cannot even write it down since there is no tetrad field to characterize the gravitational interactions of fermions.

On the one hand, it seems that the various formulations of gravity secretly had a subtle effect on changing the declaration of admissible operators/observables. An analogous situation is studying YM theory with a stipulation that one can only ever use adjoint indices, thus precluding the existence of the very fundamental Wilson loops which are charged under the one-form symmetry. Conversely, it seems that declaring a set of symmetries constrains one's theory such that certain formulations might become not available.

On the other hand, one can also think of integrating out field variables subject to the local Lorentz symmetry. From this perspective, the vector holonomy and its one-form symmetry should presumably have a description in the metric-affine formulation since the spin structure should not be necessary.

3.5.5 Exponentiated Area as Symmetry Operator

Eq. (3.85) is a concrete formula expressing the gravitational one-form symmetry operator $U_\alpha(\mathcal{S})$ in terms of the local degrees of freedom of the theory. A further explicit formula can be obtained by choosing a specific representative for the multivalued gauge parameter $\Omega_{\mathcal{S}}$. As shown in Sec. 3.4, there are many choices one can make. One nice choice is

$$\Omega_{\mathcal{S}} = \exp(\lambda \phi_{\mathcal{S}}) \quad \text{where} \quad e^{2\pi\lambda} = \alpha \in Z(\mathbf{G}), \quad (3.88)$$

where $\phi_{\mathcal{S}}$ is a multivalued function that exhibits a unit winding from 0 to 2π around $\mathcal{S}$, and λ^a is an element of the local Lorentz algebra $\mathfrak{g} = \text{Lie}(\mathbf{G})$ that exponentiates to the center element α . We take λ^a as a constant. With this

choice, Eq. (3.85) boils down to

$$U_\alpha(\mathcal{S}) = \exp\left(-\frac{2\pi i}{\kappa^2} \int_{\mathcal{S}} B_a[e] \lambda^a\right), \quad (3.89)$$

manifesting the surface support on $\mathcal{S}$. Again, Eq. (3.89) should read as one possible realization for $U_\alpha(\mathcal{S})$, and our previous discussions in Sec. 3.3.4 on the spuriously breaking nature of λ^a applies in the same way.

Intriguingly, this surface operator $U_\alpha(\mathcal{S})$ literally computes a certain area-like quantity associated with $\mathcal{S}$. By plugging in Eq. (3.67) and unpacking the gravitational coupling as $\kappa = (8\pi G_N)^{1/2}$, we find that Eq. (3.89) describes

$$U_\alpha(\mathcal{S}) = \exp\left(-\frac{i}{4G_N} \int_{\mathcal{S}} \frac{1}{2} \star \lambda_{AB} (e^A \wedge e^B)\right), \quad (3.90)$$

where $\star \lambda_{AB} = \frac{1}{2} \epsilon_{ABCD} \lambda^{CD}$ is the Hodge dual of λ^{AB} . Here we recognize $e^A \wedge e^B$ as the infinitesimal area element in the orthonormal frame, so the exponent computes the area smeared with a reference $\star \lambda_{AB}$, measured in Planck units; $1/2$ is the canonical normalization factor for contracting antisymmetric tensors.

Additionally, it is curious that the exponent in Eq. (3.90) is tantalizingly similar to the area operator in loop quantum gravity which leads to the quantization of tetrahedral volume [496, 497]. The only crucial difference is that here the area is “dotted” with a Lorentz generator λ^{AB} that exponentiates to the center, so in a sense it serves as a discrete and topological reincarnation of the area operator in loop quantum gravity. On the other hand, $U_\alpha(\mathcal{S})$ is also reminiscent of the Bekenstein-Hawking entropy formula [498, 499]: $(\text{Area})/4G_N$. Of course, these could easily be accidents of dimensional analysis on account of the $1/G_N$ normalization in the exponent. But in any case, it would be fascinating to see if any of these superficial similarities carry deeper significance.

It is also useful to define the *chiral surface operators* as

$$\tilde{U}(\mathcal{S}) = \exp\left(\frac{1}{4G_N} \int_{\mathcal{S}} \epsilon_{\alpha\beta} \tilde{\lambda}_{\dot{\alpha}\dot{\beta}} (e^{\dot{\alpha}\alpha} \wedge e^{\dot{\beta}\beta})\right), \quad (e^{2\pi\tilde{\lambda}})^{\dot{\alpha}}_{\dot{\beta}} = -\delta^{\dot{\alpha}}_{\dot{\beta}}, \quad (3.91a)$$

$$U(\mathcal{S}) = \exp\left(-\frac{1}{4G_N} \int_{\mathcal{S}} \lambda_{\alpha\beta} \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} (e^{\dot{\alpha}\alpha} \wedge e^{\dot{\beta}\beta})\right), \quad (e^{2\pi\lambda})_{\alpha}^{\beta} = -\delta_{\alpha}^{\beta}, \quad (3.91b)$$

which point to an intriguing idea of measuring the chiral areas in Planck units in terms of the chiral Plebański two-forms [315, 339].

In Eq. (3.91), $\tilde{\lambda}_{\dot{\alpha}\dot{\beta}}$ and $\lambda_{\alpha\beta}$ belong to the SD and ASD sectors of the local Lorentz algebra $\mathfrak{g}$, respectively. Since the two sectors commute, any element

of $\mathfrak{g}$ that exponentiates to a center element splits into SD and ASD generators that *separately* exponentiate to center elements. Therefore, it follows that the chiral surface operators in Eq. (3.91) serve as primordial building blocks for any symmetry operator. The details of this decomposition yet vary depending on the signature.

For example, suppose the complexified setup with the largest group $\mathbf{G} = \mathbf{Spin}(4, \mathbb{C}) \cong \mathbf{SL}(2, \mathbb{C}) \times \mathbf{SL}(2, \mathbb{C})$. Then each of $\tilde{U}(\mathcal{S})$ and $U(\mathcal{S})$ in Eq. (3.91) is separately a one-form symmetry operator, implementing $\mathbb{Z}_2$ flips of the holonomies of the chiral spin connections $\tilde{\gamma}^{\dot{\alpha}}_{\dot{\beta}}$ and γ_{α}^{β} . Note that, as zero-form symmetries, each of these $\mathbb{Z}_2$ acts as parity in the numbers of SD (dotted) and ASD (undotted) spinor indices, respectively.

For another example, suppose Euclidean signature with $\mathbf{G} = \mathbf{Spin}(4)$. This case is similar to the above complexified case. The chiral surface operators $\tilde{U}(\mathcal{S})$ and $U(\mathcal{S})$ are precisely the one-form symmetry operators corresponding to each $\mathbb{Z}_2$ factor of the center subgroup $Z(\mathbf{G}) = \mathbb{Z}_2 \times \mathbb{Z}_2$. Again, as zero-form symmetries, each factor of this center acts as parity in the numbers of dotted and undotted spinor indices, respectively.

Finally, in the Lorentzian signature without complexification, the one-form symmetry is only nontrivial if $\mathbf{G} = \mathbf{Spin}(1, 3) = \mathbf{SL}(2, \mathbb{C})$, in which case the center is $Z(\mathbf{G}) = \mathbb{Z}_2$. In this case, only the *real* combination of the chiral operators, $\tilde{U}(\mathcal{S})U(\mathcal{S})$, can qualify as a one-form symmetry operator. Note that as a zero-form symmetry this acts as *net* parity on spinor indices, drawing no distinction between the dotted and undotted ones. For instance, it acts *trivially* on vectors but flips sign for fermions.

3.5.6 Tetradic Cosmic String as a Local Lorentz Vortex

Finally, we explore the top corner of the symmetry triangle in Fig. 3.1: vortex geometry. Our goal is to determine the physical interpretation of the vortex configuration which the symmetry operator $U_{\alpha}(\mathcal{S})$ in Eq. (3.90) generates in GR. The chiral surface operators in Eq. (3.91) will also be considered in turn.

Recall how Sec. 3.4 demonstrated that the color vortex in gauge theory is the Dirac string. First, a flat configuration $\bar{A}^a$ was assumed where $F^a[\bar{A}] = 0$. Second, a multivalued gauge transformation $\Omega_{\mathcal{S}} = \exp(\lambda \phi_{\mathcal{S}})$ derived a new configuration $A = \text{Ad}_{\Omega_{\mathcal{S}}^{-1}} \bar{A} + \Omega_{\mathcal{S}}^{-1} d\Omega_{\mathcal{S}}$ where $F^a[A] = 2\pi\lambda^a \delta(\mathcal{S})$. Third, this distributional field strength was identified as implementing a thin tube of color

flux. Concrete demonstrations of this fact were provided by assuming various shapes for $\mathcal{S}$. The background $\bar{A}^a$ could have been chosen arbitrarily, but making it flat helps focus on the geometry and physics of the string itself.

To repeat this analysis for GR, we suppose a flat spacetime described by a metric $\bar{g}_{\mu\nu} = \eta_{AB} \bar{e}^A{}_\mu \bar{e}^B{}_\nu$ and a spin connection $\bar{\gamma}^A{}_B$ such that $R^A{}_B[\bar{\gamma}] = 0$. For instance, we may take $\bar{e}^A{}_\mu = \delta^A{}_\mu$ and $\bar{\gamma}^A{}_B = 0$. Next, we apply a multivalued local Lorentz transformation represented as $\Omega_{\mathcal{S}} = \exp(\lambda \phi_{\mathcal{S}})$. The resulting geometry describes

$$e^A = \exp(-\lambda \phi_{\mathcal{S}})^A{}_B \bar{e}^B, \quad R^A{}_B = 2\pi \lambda^A{}_B \delta(\mathcal{S}). \quad (3.92)$$

By analogy, Eq. (3.92) realizes a “local Lorentz vortex.” Notably, it encodes a curvature singularity distributed over the two-dimensional surface $\mathcal{S}$, reminiscent of the cosmic string. To determine the precise physical interpretation of this string geometry, we consider the following facts.

Invisibility to Scalar Particles First and foremost, the metric is invariant under any local Lorentz transformation, whether multivalued or not. The coframe given in Eq. (3.92) implies

$$g_{\mu\nu} = \bar{g}_{\mu\nu}, \quad (3.93)$$

so the metric remains identically flat. Physically, this implies that our string geometry is *completely invisible to scalar particles*. The geodesics of scalar particles are left the same, in both local and global senses. Locally, the geodesic equation uses the Christoffel symbols $\Gamma^\mu{}_{\rho\sigma}[g]$ defined by the metric. Globally, the scalar particle geodesics will detect *no angle deficit* since no multivaluedness has been introduced regarding the points of the base spacetime manifold. The conclusion is that our string geometry should somehow differ from the standard cosmic string.

Matter Content Secondly, we reverse engineer the matter source that generates our string geometry. Certainly, this analysis will provide an explicit physical characterization. With vanishing cosmological constant, Einstein’s equations describe

$$-\frac{1}{\kappa^2} \star R_{AB} \wedge e^B = \frac{1}{3!} T^\mu{}_A \varepsilon_{\mu\nu\rho\sigma} dx^\nu \wedge dx^\rho \wedge dx^\sigma. \quad (3.94)$$

By plugging in Eq. (3.92) to Eq. (3.94), the stress-energy tensor T^μ_A is determined as

$$T^\mu{}_\kappa = \frac{1}{8G_N} \varepsilon^{\mu\nu\rho\sigma} \star \lambda_{\nu\kappa} \delta(\mathcal{S})_{\rho\sigma}, \quad (3.95)$$

which acts as a source for the coframe in the first-order formulation. Note that the formula given in Eq. (3.13) expresses this $T^\mu{}_\kappa$ as

$$\frac{1}{4G_N} \frac{1}{|e|} \star \lambda_{\nu\kappa} \int d^2\sigma \delta^{(4)}(x - X(\sigma^1, \sigma^2)) \left(\frac{\partial X^\mu}{\partial \sigma^1} \frac{\partial X^\nu}{\partial \sigma^2} - \frac{\partial X^\nu}{\partial \sigma^1} \frac{\partial X^\mu}{\partial \sigma^2} \right), \quad (3.96)$$

when the surface $\mathcal{S}$ is parameterized as $X^\mu(\sigma^1, \sigma^2)$ in terms of worldsheet coordinates (σ^1, σ^2) . We see that the stress-energy tensor is localized along a membrane $\mathcal{S}$ and describes a formula reminiscent of the cosmic string, though it is not literally identical to the Nambu-Goto string.

Interestingly, we also see that the algebraic Bianchi identity, $R_{AB} \wedge e^B = 0$, *fails* for our string geometry, indicating the existence of *magnetic* stress-energy.¹⁷ The dual stress-energy tensor $T^{\star\mu}{}_A$ is defined as [294, 492]

$$-\frac{1}{\kappa^2} R_{AB} \wedge e^B = \frac{1}{3!} T^{\star\mu}{}_A \varepsilon_{\mu\nu\rho\sigma} dx^\nu \wedge dx^\rho \wedge dx^\sigma. \quad (3.97)$$

By plugging in $R^A{}_B = \lambda^A{}_B \delta(\mathcal{S})$, we obtain

$$T^{\star\mu}{}_\kappa = \frac{1}{8G_N} \varepsilon^{\mu\nu\rho\sigma} \lambda_{\nu\kappa} \delta(\mathcal{S})_{\rho\sigma}. \quad (3.98)$$

It follows that Eq. (3.98) describes a line distribution of NUT charge, just as Eq. (3.95) describes a line distribution of mass.

The conclusion here is that the geometry in Eq. (3.92) represents a cousin of cosmic string, sourced by a *combined concentration of both electric and magnetic matter* localized on a two-dimensional surface $\mathcal{S}$.

No Misner String Component Lastly, we shall clarify if a Misner string component is involved as well. The answer turns out to be no. First, the Misner string describes a stack of gravitomagnetic *dipoles* [475, 476], and as such it can terminate on gravitomagnetic monopoles (NUT charges). However, the magnetic stress-energy found in Eq. (3.98) itself is a line density of distributed gravitomagnetic *monopole*. Second, the Misner string could also be

¹⁷This failure of algebraic Bianchi in the presence of magnetic matter is analogous to the failure of the Bianchi identity in Maxwell theory in the presence of magnetic charges.

characterized as a localized source of torsion, as it induces a time monodromy [172, 474, 486]. However, it is easy to see that the torsion transforms covariantly and linearly even under the multivalued local Lorentz transformation, so if the initial background exhibited vanishing torsion, it stays identically zero after the insertion of the symmetry operator as well. Concretely, one can establish that our string geometry has vanishing torsion by plugging in the transformed tetrad and spin connection into $De^A = de^A + \gamma^A_B \wedge e^B$, which also reveals that the failure of algebraic Bianchi is solely due to the multivaluedness of the tetrad, $dde^A \neq 0$, *not* by a nonzero torsion. The absence of torsion is also clear if one recalls the fact that torsion is induced by local source for the spin connection in the tetradic Palatini formalism, while the symmetry operator only couples to the coframe. For these reasons, we conclude that the geometry in Eq. (3.92) is of a purely cosmic kind, with no Misner string component.

Tetradic Cosmic String Let us name the geometry due to Eq. (3.92) a *tetradic cosmic string*. Our analysis above shows that the tetradic cosmic string differs from the standard cosmic string in terms of its invisibility to scalar particle geodesics, involves both electric and magnetic stress-energies, and carries no Misner string component. It seems that the tetradic cosmic string cannot be realized in non-tetradic theories of gravity such as pure-metric or metric-affine gravity. The tetradic cosmic string will be detected by the spin holonomy in Eq. (3.79), i.e., interferometer experiments with spinning particles to speak physically. See Fig. 3.7 for a visualization of the geometry of a tetradic cosmic string.

Concrete Construction and Relation to Standard Cosmic String To be further explicit about these points, we demonstrate below how a tetradic cosmic string can be derived by modifying the geometry of the standard cosmic string such that scalar particle geodesics get globally straightened out. We wish to assume the complexified setup with $G = \text{Spin}(4, \mathbb{C})$.

We begin by reviewing the geometry of the standard cosmic string with tension μ . In cylindrical coordinates, a straight cosmic string inserted along the z -axis describes the line element

$$ds^2 = -dt^2 + ds^2 + (1 - 4G_N\mu)^2 s^2 d\phi^2 + dz^2, \quad \phi \sim \phi + 2\pi. \quad (3.99)$$

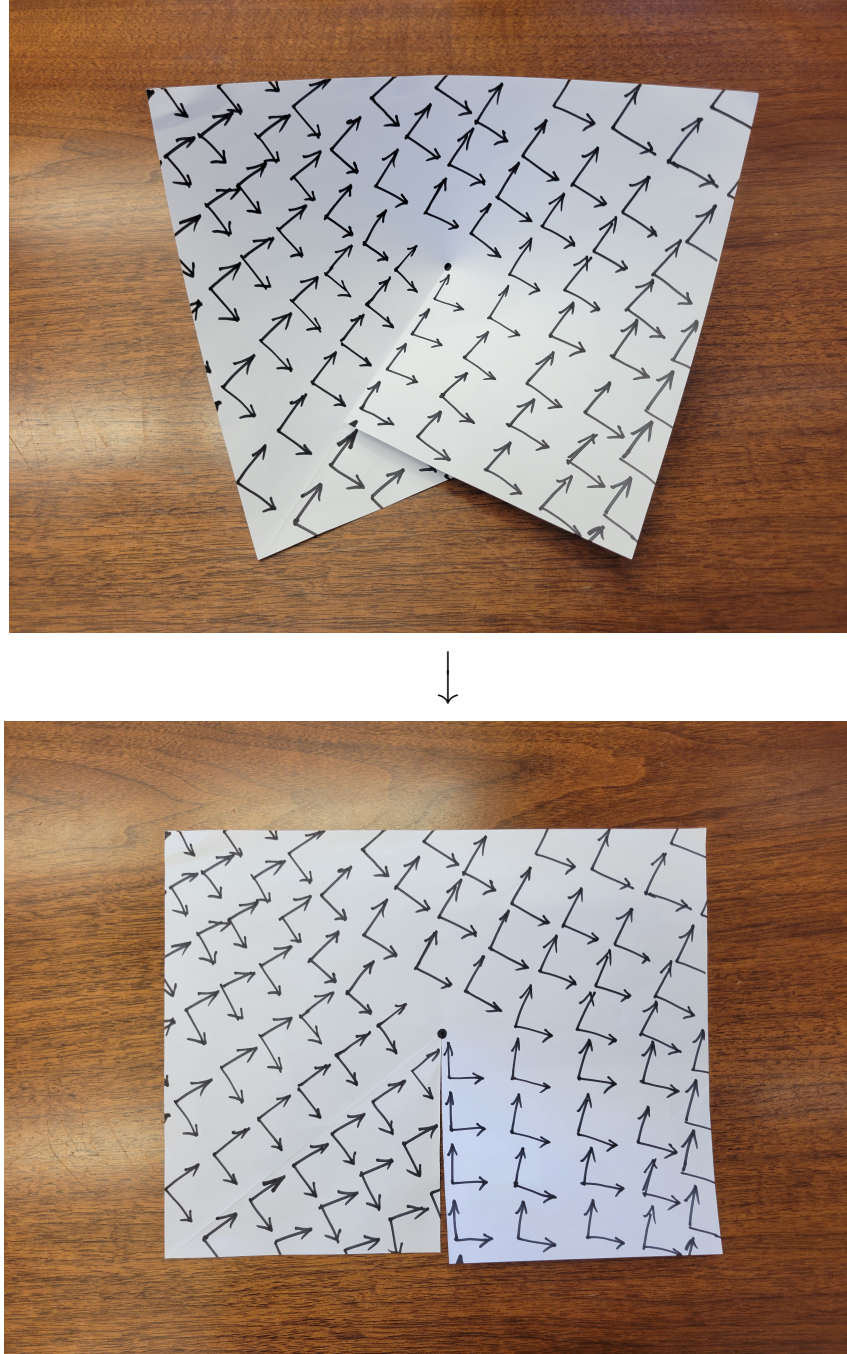


Figure 3.7: Construction of a tetradic cosmic string from the standard cosmic string. (*Upper panel*) A single-valued tetrad is installed on the spacetime of a standard cosmic string. (*Lower panel*) Unwrapping the spacetime by adding a wedge produces a multivalued tetrad on a spacetime without an angle deficit, realizing a tetradic cosmic string. The required center periodicity condition will be achieved by tuning the original string tension (figure depicts improper quantization for clarity of illustration). In this procedure, note the interplay between the multivalued geometries of the base and the fibration, manipulated by diffeomorphisms and local Lorentz transformations, respectively.

The angle deficit is detected as

$$2\pi - \Delta\theta = \oint (1 - 4G_N\mu) d\phi \implies \Delta\theta = 8\pi G_N\mu. \quad (3.100)$$

A complexified orthonormal coframe for the line element in Eq. (3.99) can be given as

$$\begin{aligned} e^0 &= \cos\phi dt - i \sin\phi dz, \\ e^1 &= \cos\phi ds - (1 - 4G_N\mu) \sin\phi s d\phi, \\ e^2 &= \sin\phi ds + (1 - 4G_N\mu) \cos\phi s d\phi, \\ e^3 &= -i \sin\phi dt + \cos\phi dz, \end{aligned} \quad (3.101)$$

which is *single-valued* over the domain $t \in \mathbb{R}$, $s \geq 0$, $\phi \in [0, 2\pi[$, $z \in \mathbb{R}$.

Now consider plugging in $\phi = \psi/(1 - 4G_N\mu)$ and *changing* the periodicity condition as $\phi \sim \phi + 2\pi/(1 - 4G_N\mu)$:

$$ds^2 = -dt^2 + ds^2 + s^2 d\psi^2 + dz^2, \quad \psi \sim \psi + 2\pi. \quad (3.102)$$

Crucially, this derives a *new* geometry that globally differs from the previous one. The line element in Eq. (3.102) describes *no angle deficit* with the new periodicity condition:

$$2\pi - \Delta\theta = \oint (1 - 4G_N\mu) d\phi = \oint d\psi \implies \Delta\theta = 0. \quad (3.103)$$

In the meantime, the orthonormal coframe in Eq. (3.101) is brought to

$$\begin{aligned} e^0 &= \cos\left(\frac{\psi}{1 - 4G_N\mu}\right) dt - i \sin\left(\frac{\psi}{1 - 4G_N\mu}\right) dz, \\ e^1 &= \cos\left(\frac{\psi}{1 - 4G_N\mu}\right) ds - \sin\left(\frac{\psi}{1 - 4G_N\mu}\right) s d\psi, \\ e^2 &= \sin\left(\frac{\psi}{1 - 4G_N\mu}\right) ds + \cos\left(\frac{\psi}{1 - 4G_N\mu}\right) s d\psi, \\ e^3 &= -i \sin\left(\frac{\psi}{1 - 4G_N\mu}\right) dt + \cos\left(\frac{\psi}{1 - 4G_N\mu}\right) dz, \end{aligned} \quad (3.104)$$

which is *multivalued* over the domain $t \in \mathbb{R}$, $s \geq 0$, $\psi \in [0, 2\pi[$, $z \in \mathbb{R}$ unless $(1 - 4G_N\mu)^{-1} \in \mathbb{Z}$. Concretely, let us take

$$\frac{1}{1 - 4G_N\mu} = k + \frac{1}{2} \iff \mu = \frac{1}{4G_N} \frac{2k - 1}{2k + 1}, \quad (3.105)$$

where $k \in \mathbb{Z}$.

Clearly, Eq. (3.104) arises by applying a multivalued local Lorentz transformation on the background coframe $(\bar{e}^0, \bar{e}^1, \bar{e}^2, \bar{e}^3) = (dt, ds, s d\psi, dz)$:

$$\Omega_{\mathcal{S}} = \exp(\lambda\psi), \quad \lambda^A{}_B = \left(k + \frac{1}{2}\right) \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}. \quad (3.106)$$

As per Eq. (3.78), the multivaluedness is characterized by the discontinuity,

$$\alpha = \lim_{\epsilon \rightarrow 0} \left(\Omega_{\mathcal{S}}(\psi = 2\pi - \epsilon) \Omega_{\mathcal{S}}^{-1}(\psi = \epsilon) \right) = e^{2\pi\lambda} = (-1)^{2k+1} \mathbb{1}. \quad (3.107)$$

This shows how a tetradic cosmic string can be concretely constructed by modifying the geometry of the standard cosmic string whose tension μ is made to take specific values as stipulated in Eq. (3.105).

Chiral Tetradic Cosmic String It can also be seen that the Lorentz generator $\lambda^A{}_B$ in Eq. (3.106) is SD: $\star\lambda^{AB} = +i\lambda^{AB}$. Therefore, the above example has constructed the geometry created by a SD surface operator, $\tilde{U}(\mathcal{S})$, in Eq. (3.91a). Hence more precisely, it will be called a *SD tetradic cosmic string*. In general, we refer to the vortex configurations generated by the chiral surface operators in Eq. (3.91) as *chiral tetradic cosmic strings*.

For chiral tetradic cosmic strings, the electric and magnetic stress-energy tensors in Eqs. (3.95) and (3.98) satisfy the relation

$$T^\mu{}_\kappa = \pm i T^{\star\mu}{}_\kappa. \quad (3.108)$$

Hence they are concentrations of *chiral matter*: combinations of electric and magnetic stress-energies in “equal magnitudes.”

It is straightforward to compute the holonomies for various spin representations in the background of Eq. (3.104):

$$\begin{aligned} \text{tr}_{\text{vector}} \text{Pexp} \left(\lambda \int_0^{2\pi} d\psi \right) &= 4(-1)^{2k+1} = -4, \\ \text{tr}_{\widetilde{\text{spinor}}} \text{Pexp} \left(\lambda \int_0^{2\pi} d\psi \right) &= 2(-1)^{2k+1} = -2, \\ \text{tr}_{\text{spinor}} \text{Pexp} \left(\lambda \int_0^{2\pi} d\psi \right) &= 2. \end{aligned} \quad (3.109)$$

Provided Eq. (3.105) with $k \in \mathbb{Z}$, we see that the vector and SD spinor holonomies flip their sign while the ASD spinor holonomy is left the same.

With the imposition of the Lorentzian reality condition, one considers the real combination $\tilde{U}(\mathcal{S})U(\mathcal{S})$ as discussed in Sec. 3.5.5. The SD tetradic cosmic string described in Eq. (3.104) can be superimposed with its ASD counterpart (“complex conjugate”) in this sense, in which case the coframe is¹⁸

$$\begin{aligned} e^0 &= dt, \\ e^1 &= \cos\left((2k+1)\psi\right)ds - \sin\left((2k+1)\psi\right)s d\psi, \\ e^2 &= \sin\left((2k+1)\psi\right)ds + \cos\left((2k+1)\psi\right)s d\psi, \\ e^3 &= dz. \end{aligned} \tag{3.110}$$

This arises by the local Lorentz transformation

$$\Omega_{\mathcal{S}} = \exp(\lambda\psi), \quad \lambda^A{}_B = (2k+1) \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{3.111}$$

which is single-valued in the vector representation. The computation of the holonomies in this background gives

$$\begin{aligned} \text{tr}_{\text{vector}} \text{Pexp}\left(\lambda \int_0^{2\pi} d\psi\right) &= 4(-1)^{4k+2} = 4, \\ \text{tr}_{\widetilde{\text{spinor}}} \text{Pexp}\left(\lambda \int_0^{2\pi} d\psi\right) &= 2(-1)^{2k+1} = -2, \\ \text{tr}_{\text{spinor}} \text{Pexp}\left(\lambda \int_0^{2\pi} d\psi\right) &= 2(-1)^{2k+1} = -2, \end{aligned} \tag{3.112}$$

so the vector holonomy is not transformed.

3.5.7 Linking and Spin Precession Angle

In Sec. 3.5.6, we have established that the symmetry operator $U_{\alpha}(\mathcal{S})$ creates a “tetradic cosmic string” that is invisible to scalar particle trajectories but affects the spin holonomies $W_{\rho}(\mathcal{C})$ linking with it.

What is the physical interpretation of this linking between $U_{\alpha}(\mathcal{S})$ and $W_{\rho}(\mathcal{C})$, leading to the Ward identity in Eq. (3.86)? Notably, this too has a simple physical interpretation in terms of classical gravitation: *spin precession angles* as a gravitational AB effect.

¹⁸In principle, a real tetradic cosmic string will be able to involve both electric and magnetic stress-energies depending on the relative orientation between the surface support $\mathcal{S}$ and the Lorentz generator λ . For our specific example here, the magnetic stress-energy happens to vanish.

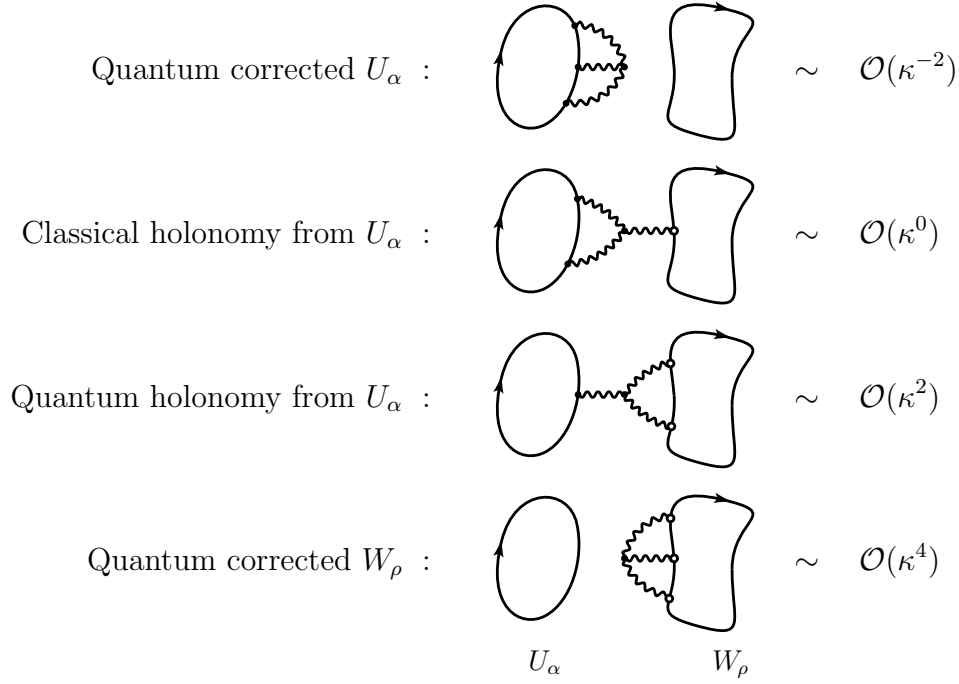


Figure 3.8: A schematic depiction of Feynman diagrams contributing to the Ward identity at various perturbative orders in the gravitational coupling κ . The wavy lines denote gravitons, while the loops of solid lines on the left and right depict the symmetry operator U_α and line operator W_ρ , respectively. The first and fourth rows describe quantum corrections to U_α and W_ρ separately. The second row is a contribution to the classical holonomy measured by W_ρ , treating U_α as a source for the background spacetime. The third row is the one-loop quantum correction to this quantity in that background. Since the linking number is dimensionless and topological, it arises purely from the classical holonomy.

It is easy to see that spin holonomies arise from the parallel transport of spinning particles around a loop. For example, take the example calculation in Eq. (3.109). It describes that vector particles and SD fermions exhibit spin precession angles continuously accumulating from 0 to $(2k+1)\pi$ while traveling along a loop that encloses the tetradic cosmic string. This provides the physical interpretation of the Ward identity: the SD surface operator $\tilde{U}(\mathcal{S})$ induces quantized spin precession angles for vector particles and SD fermions.

For the example calculation in Eq. (3.112), one learns that fermions exhibit spin precession angles continuously accumulating from 0 to $(2k+1)\pi$. In the meantime, vector particles exhibit the total spin precession angle $(4k+2)\pi$, bringing it back to its initial state.

One may wonder how this *classical* computation in the background geome-

tries of tetradic cosmic strings captures the full content of the *quantum* Ward identity. Of course, the exact path integral derivation of the Ward identity in Sec. 3.5.4 clarifies that there is no mystery at the mathematical level. However, one may still want to revisit the path integral derivation from the perturbation theory point of view, in which case the linking number is computed by an infinite set of perturbative diagrams. As sketched in Fig. 3.8, a careful power counting argument with the formulae in Eqs. (3.79), (3.89), and (3.68) shows that the linking number receives contributions solely from the diagrams computing the classical holonomy in the background of a $U_\alpha(\mathcal{S})$ insertion. Note that $U_\alpha(\mathcal{S})$ couples gravitons to the surface $\mathcal{S}$ via the inverse factor $1/\kappa^2$. Similarly, one can also argue that the spin precession angles are not renormalized.

Another remark is that the computation of spin precession angles under the insertion of a tetradic cosmic string can be demonstrated in general backgrounds, such as the spacetime of an Anti-de-Sitter-Schwarzschild black hole. First, identify the background coframe. Second, apply a multivalued local Lorentz transformation. Third, compute the spin holonomies directly from the transformed spin connection by using the path-ordered exponential formula. By one's choice of the background spin connection and the Lorentz generator λ , one will be able to demonstrate a rather nontrivial calculation.

3.6 Physical Implications

As is well-known, global higher-form symmetries can be broken, either explicitly or spontaneously. In analogy with YM theory, the one-form symmetry of gravity is explicitly broken in the presence of matter fields that transform nontrivially under the Lorentz group. For example, if the theory includes a local operator in the spin representation ρ , then it is possible to define a Lorentz invariant line holonomy $W_\rho(\mathcal{C})$ for a contour $\mathcal{C}$ that is not closed, but rather terminates on this local operator. The spin holonomy is then “endable,” so it can be unlinked topologically from the symmetry operator $U_\alpha(\mathcal{S})$, and the one-form symmetry is explicitly broken. The physical interpretation of this phenomenon is that the spin holonomy is screened by spinning particles.

Interestingly, this implies that the gravitational one-form symmetry is explicitly broken by particles with spin. For example, if the local Lorentz group is $\mathbf{G} = \mathbf{Spin}(4)$, then a holonomy in the spinor representation can only end on a local operator $\mathcal{O}_\alpha$ with a free spinor index. Hence, the corresponding holonomy is endable if there exist fermions in the spectrum. The case of $\mathbf{G} = \mathbf{SO}(4)$ is

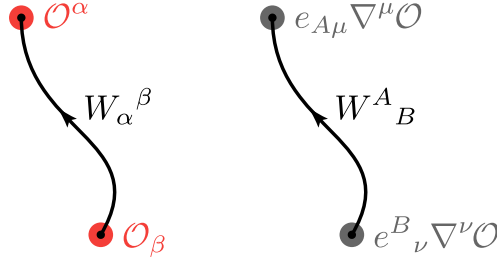


Figure 3.9: Spinor holonomy is screened by fermions. Vector holonomy is screened by orbital angular momentum.

more subtle, however. A holonomy in the vector representation of the Lorentz group must end on an operator with a free Lorentz vector index, which can be any operator carrying orbital angular momentum dotted with a tetrad such as $e^A_\mu \nabla^\mu \mathcal{O}$.¹⁹ Consequently, the vector holonomy is analogous to the adjoint Wilson loop of gauge theory, which is automatically screened by dynamical gluons. See Fig. 3.9 for a depiction of this phenomenon.

The above logic has implications for the standard model of physics. Since fermions exist, we know that the local Lorentz group is $G = \text{SL}(2, \mathbb{C})$, which implies that there is a $\mathbb{Z}_2$ one-form gravitational symmetry under which spinor holonomies are charged. Presuming the lightest neutrino is not massless, this one-form symmetry is unbroken below that scale. This implies a new, albeit subtle, exact symmetry in the known laws of physics.

Notably, the explicit breaking of higher-form symmetry is strongly suggested by the so-called swampland conjectures. In particular, there is strong evidence that all global symmetries are necessarily broken at some scale in a consistent theory of quantum gravity [70, 71, 73, 74]. A well-known avatar of this is the weak gravity conjecture [75, 500], which states that a $U(1)$ gauge theory must exhibit a state whose charge exceeds its mass in Planck units. The weak gravity conjecture quantitatively forbids the strict global symmetry limit of vanishing charge in a $U(1)$ gauge theory.

¹⁹A more subtle question arises in formulations of gravity with differences in field content. For example, in the Plebański theory [315] described in Sec. 2.5.2, the associated two-form field B is a fundamental degree of freedom. The tetrad is sculpted from B through a constraint [358, 359], while the metric can be expressed in terms of B in closed form [339, 360]. Consequently, there is no field in the pure gravity sector that carries a single vector index, so it is unclear whether the vector spin holonomy is endable in this case. This illustrates a situation where the one-form symmetry may be capable of distinguishing between theories that only differ nonperturbatively.

The swampland conjectures imply that any higher-form symmetry should be either gauged or explicitly broken. In the case of gauge theory coupled to gravity, the latter scenario requires the existence of a tower of charged states, as described by the so-called completeness conjectures [85]. Applying the same logic to dynamical gravity, we expect that something similar applies to the gravitational one-form symmetry. One option is that this symmetry is gauged, for example as would occur if the Lorentz group is $\mathrm{SO}^+(3, 1) = \mathrm{SL}(2, \mathbb{C})/\mathbb{Z}_2$. Alternatively, if the Lorentz group is $\mathrm{SL}(2, \mathbb{C})$, then the one-form symmetry is not gauged and must be explicitly broken, thus implying the existence of fermions in the spectrum.

Finally, let us speculate briefly on the possibility of phases in gravity. Taking inspiration from gauge theory, it is natural to wonder whether the expectation value of the spin holonomy is an order parameter for symmetry breaking. In the case of gauge theory, it is well-known that an area versus perimeter law scaling of the Wilson loop expectation value is a diagnostic of whether or not a theory is confining. However, the analogous construction in gravity is far murkier. In particular, the diffeomorphism invariance of dynamical gravity may suggest that the contour of the spin holonomy should be defined relationally with respect to some invariant boundary data. So to be an order parameter, the spin holonomy may presumably be computed for a contour that circumscribes the boundary.

Even ignoring these subtleties, the fact that the EFT description of gravity is intrinsically weakly coupled suggests that confinement is not in play. More generally it is very unclear whether a low-energy EFT of gravity on a fixed background could even access different phases, or what that would even mean. One speculation is that this might have something to do with degenerate configurations of the metric and their corresponding domain walls [501–504]. Another possibility is that a putative gravitational phase diagram might delineate various choices of compactification or of asymptotic behavior of the metric. Indeed, it is easy to see that the spin holonomy is highly sensitive to the cosmological constant. For these reasons, it would be interesting to explicitly compute the expectation value of the spin holonomy in various examples. A number of existing works have calculated the spin holonomy in various contexts [374, 505–510].

3.7 Future Directions

In this work, we have initiated an exploration of generalized symmetry in the context of dynamical gravity. Taking our cues from the one-form symmetries of YM theory, we have considered gravity in the tetradic Palatini formalism, which is a gauge theory of the local Lorentz group. We have argued that the gravitational one-form symmetry is defined by the center of the Lorentz group. The object which is charged under this symmetry is the spin holonomy $W_\rho(\mathcal{C})$. The one-form symmetry transformation is implemented by an operator $U_\alpha(\mathcal{S})$, which has dual interpretations, both as a twisted Lorentz transformation but also as a chiral cosmic string defect carrying both electric and magnetic gravitational charge. The topological linking of the line and symmetry operators corresponds to the measurement of a quantized spin precession angle by the spin holonomy as a gravitational AB effect. In the standard model, this implies the existence of a new symmetry below the mass of the lightest neutrino. The present work leaves numerous avenues for future exploration, which we now describe.

First and foremost are a number of very simple extensions of this work which should be relatively straightforward. These include the question of generalization to higher spacetime dimensions, which offers a richer spectrum of one-form symmetry groups. For example, the center group is maximally $Z(\text{Spin}(4)) = \mathbb{Z}_2 \times \mathbb{Z}_2$ in four dimensions, but this grows to $Z(\text{Spin}(6)) = \mathbb{Z}_4$ in six dimensions. Another concrete direction is the inclusion of gravitational higher-curvature corrections. As mentioned earlier, these contributions will clearly preserve the one-form symmetry, whose corresponding symmetry operator will be equal to the surface integral of the canonical conjugate of the curvature in the EFT.

Secondly, while the present work has focused on a gravitational one-form symmetry of electric type, it is natural to ask whether one can derive a magnetic counterpart. For gauge theories, the electric and magnetic symmetries are straightforwardly related via Hodge duality [79]. In contrast, constructing the magnetic counterpart of our gravitational one-form symmetry stands as an open question. One might possibly be considering the gravitational analog of the 't Hooft line and attempting to identify the conjugate field for the (internal) Hodge dual of the curvature two-form, while an alternative pathway aiming to dualize the Plebański two-form might also be a speculation.

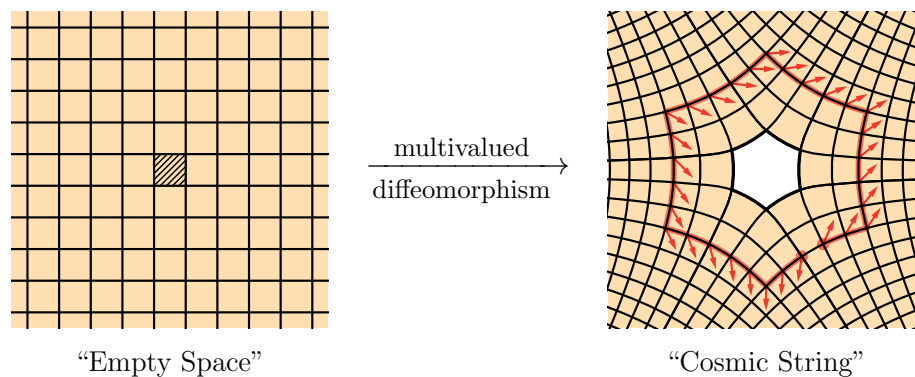


Figure 3.10: A diffeomorphism with a multivalued Jacobian creates a cosmic string from empty space, analogous to disinclinations in lattice systems.

A third topic of future study is higher-group symmetry, which describes a certain nonabelian structure built from the fusion of multiple higher-form symmetries [511–514]. A well-studied example of this is axion-YM theory, which exhibits a two-form symmetry for the axion, together with the one-form symmetry of the gauge theory [429]. These symmetries fuse to yield a two-group structure, which in fact bounds the scale of axion strings from below by the lightest particle in the fundamental of color. Acquainted with this remarkable fact, it would be interesting to investigate if gravity can also exhibit a higher-group symmetry. It seems quite likely that a similar two-group symmetry will appear in gravity coupled to an axion, in which case we should expect that the axion string scale is bounded by the mass of the lightest fermion.

Fourthly, while our approach of formulating gravity into the “ BF ” grammar yields a simple route to the one-form symmetry, there is the question of how this framework is explicitly realized in other physically equivalent formulations of gravity [515] in the literature. Another open question is the fate of the one-form symmetry in gravitational formulations with subtly different field content, as referred to in a footnote in Sec. 3.6.

A fifth area of study concerns the question of what topological symmetry can teach us about classical gravitation. As a theory of spacetime geometry, gravity boasts a rich array of classical vacuum solutions, each showcasing distinctive *singularity structures* that are themselves a focal point of study [11]. In particular, it could be very illuminating to initiate a systematic analysis and

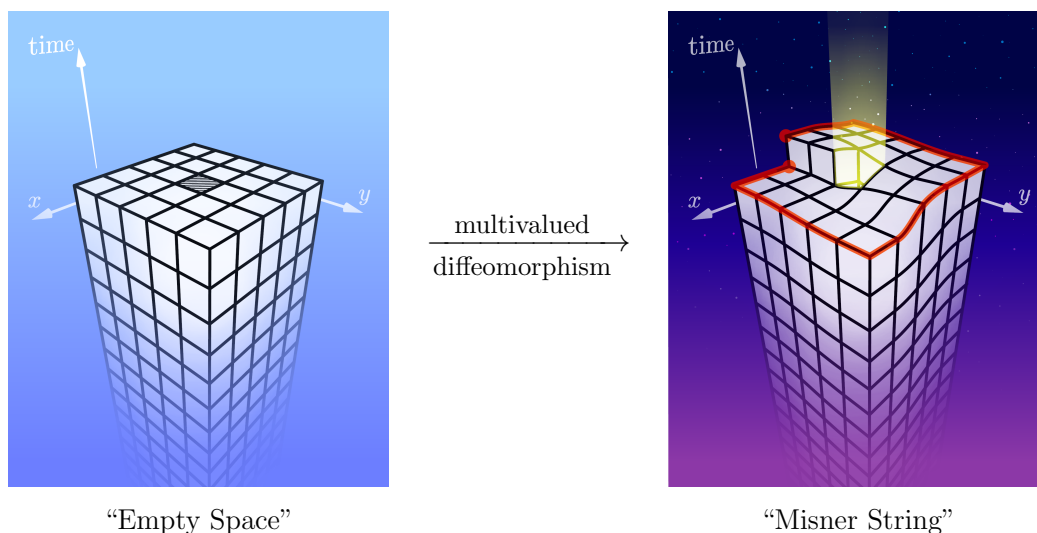


Figure 3.11: A multivalued diffeomorphism creates a Misner string from empty space, whose time monodromy is analogous to the Burgers vector of dislocations in lattice systems.

classification of gravitational singularities from the perspective of topological operators and their algebra. For example, even in the absence of spin holonomy, the one-form symmetry operator can link with “holes” in spacetime. It would be interesting to study whether there is physical information encoded in such a linking, for example regarding the singularity of a black hole. Moreover, it is conceivable that known spacetime singularities in the literature have an alternative interpretation as symmetry operators, like we discovered for the chiral cosmic string.

A sixth question is whether there exists a gravitational one-form symmetry that arises by the geometry of *multivalued diffeomorphisms*. While our construction has viewed gravity as a gauge theory of local Lorentz group, the more fundamental redundancy endemic to dynamical gravitation is diffeomorphisms. As described below as well, we anticipate that this line of inquiry will likely realize the Misner string proposal suggested in Sec. 3.5. Difficulties are apparent, however, since the diffeomorphism group is not finite, does not admit a nice center, and even does not describe *internal* transformations in the strict sense of gauge theories.

Last but not least, it would be interesting to see if the theoretical framework developed in this chapter can be applied to lattice systems such as crystals with

impurities [516–518]. Indeed, as described in [460, 519, 520], the physics of lattice systems has a formulation that is strikingly reminiscent of gravity. In this picture, lattice disinclinations are analogous to cosmic strings, as depicted in Fig. 3.10. Furthermore, the analog of “translation holonomy” is manifested by the Burgers vector, which measures the net drift in the “lattice frame” per round trip about a defect. Hence, as depicted in Fig. 3.11, we observe that lattice dislocations are analogous to the Misner string—a Dirac string of time monodromy flux endable on Taub-NUT charges [172, 294, 474–476, 486, 521]. These two types of lattice defects are described in terms of *multivalued* coordinate transformations [460, 519, 520], which are the diffeomorphism analogs of the twisted gauge transformations that played such a crucial role in our construction of a gravitational one-form symmetry. It would be interesting if this convergence between gravity and lattice systems could cross-pollinate new insights across these fields.

3.A Canonical Formalism

The one-form symmetry of gauge theories can also be understood from the complementary point of view of the Hamiltonian formalism. To this end, we suppose a product spacetime $\mathcal{M} = \mathcal{M}_3 \times \mathbb{R}$ equipped with coordinates $x^\mu = (x^i, x^4)$. We perform a 3 + 1 decomposition in which x^i denotes coordinates of the spatial three-manifold $\mathcal{M}_3$ and $x^4 = t$ defines equal-time slices.

3.A.1 Phase Space

We suppose the first-order template Lagrangian top form in Eq. (3.14). Writing out all indices explicitly, we obtain

$$\frac{1}{g^2} \left[\frac{1}{2} B_{a\mu\nu} \left(\partial_\rho A^a{}_\sigma + \frac{1}{2} f^a{}_{bc} A^b{}_\rho A^c{}_\sigma \right) \epsilon^{\mu\nu\rho\sigma} \right] d^4x + f[B], \quad (3.113)$$

where $\epsilon^{\mu\nu\rho\sigma}$ denotes the permutation symbol. Carrying out the 3 + 1 decomposition and integrating out B_{ai4} , we immediately see that the dynamical coordinates on the phase space are $A^a{}_i$ together with their canonical conjugates,

$$E^i{}_a = \frac{1}{2} B_{ajk} \epsilon^{ijk}. \quad (3.114)$$

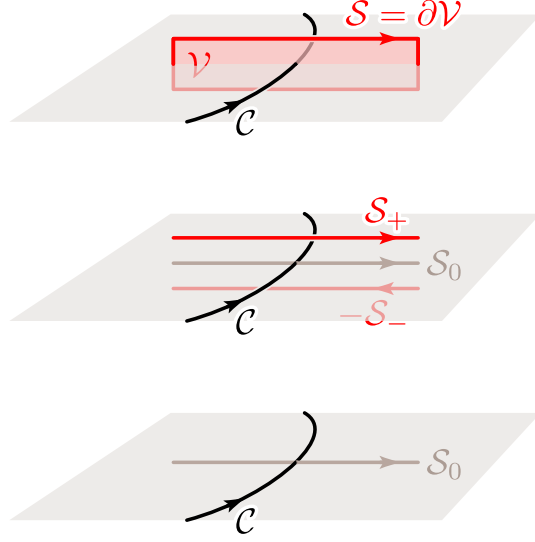


Figure 3.12: Pancaking the symmetry operator on a purely spatial line operator $\mathcal{C}$. The plane depicts an equal-time slice while time flows upwards. *Top:* The equal-time slice bisects the volume $\mathcal{V}$ by a surface $\mathcal{S}_0$. *Middle:* The surface $\mathcal{S} = \partial\mathcal{V}$ splits into surfaces $\mathcal{S}_+$ and $-\mathcal{S}_-$ at the infinitesimal future and past. *Bottom:* $\mathcal{S}_+$ and $\mathcal{S}_-$ both project down to the surface $\mathcal{S}_0$.

As a result, we identify the canonical commutation relations,

$$\begin{aligned} [A^a_i(x), A^b_j(x')] &= 0, \\ [E^i_a(x), A^b_j(x')] &= ig^2 \delta^i_j \delta^b_a \delta^{(3)}(x - x'), \\ [E^i_a(x), E^j_b(x')] &= 0, \end{aligned} \quad (3.115)$$

where $x, x' \in \mathcal{M}_3$ are points in the spatial manifold. The phase space is also equipped with the Gauss constraint and a Hamiltonian, the details of which are not important for our purposes.

3.A.2 Ward Identity

In the language of the path integral, a one-form symmetry transformation is implemented through the insertion of a symmetry operator which wraps the line operator. In the operator formalism, however, this corresponds to a conjugation of the latter by the former. To see how this works in detail, consider a line operator $W_\rho(\mathcal{C})$, where $\mathcal{C}$ is restricted to an equal-time slice, say at $t=0$. As before, we take the symmetry operator $U_\alpha(\mathcal{S})$ to be supported on an exact surface $\mathcal{S}$ with an associated coboundary $\mathcal{V}$, so $\mathcal{S} = \partial\mathcal{V}$.

The geometric set-up is depicted in Fig. 3.12. We assume that the surface $\mathcal{S}$ *links* once with the purely spatial loop $\mathcal{C}$. Consequently, the coboundary

$\mathcal{V}$ is *intersected* by the loop $\mathcal{C}$ and *bisected* by the spatial slice. Now imagine continuously squashing or pancaking the coboundary $\mathcal{V}$ along the time direction such that its two-dimensional boundary $\mathcal{S}$ infinitesimally hugs the spatial slice. In this limit, $\mathcal{S} = \mathcal{S}_+ \cup (-\mathcal{S}_-)$ is the union of two disjoint discs $\mathcal{S}_+$ and $-\mathcal{S}_-$ at the infinitesimal future and past across $t=0$. Once $\mathcal{V}$ has completely collapsed into the spatial slice at $t=0$, both discs $\mathcal{S}_\pm$ approach the same surface, which we denote by $\mathcal{S}_0$. From Fig. 3.12, we see that this $\mathcal{S}_0$ will be the intersection between $\mathcal{V}$ and $t=0$. As a result, the intersection of $\mathcal{C}$ and $\mathcal{V}$ in the four-manifold $\mathcal{M}$ is equivalent to intersection of $\mathcal{C}$ and $\mathcal{S}_0$ in the three-manifold $\mathcal{M}_3$ as the slice $t=0$. Therefore, we have in general

$$\text{link}(\mathcal{C}, \mathcal{S}) = \text{int}(\mathcal{C}, \mathcal{V}) = \text{int}_3(\mathcal{C}, \mathcal{S}_0), \quad (3.116)$$

where int_3 denotes intersection number in $\mathcal{M}_3$.

Now, we can describe how this pancaking procedure boils down the Ward identity to an equal-time operator equation. Since $\mathcal{S} = \mathcal{S}_+ \cup (-\mathcal{S}_-)$, we see that the symmetry operator factorizes into $U_\alpha(\mathcal{S}) = U_\alpha(\mathcal{S}_+) U_\alpha(-\mathcal{S}_-)$.²⁰ In turn, the left-hand side of the Ward identity in Eq. (3.27) becomes the time-ordered expression $U_\alpha(\mathcal{S}_+) W_\rho(\mathcal{C}) U_\alpha(-\mathcal{S}_-)$, which in the process of pancaking limits to the equal-time operator product $U_\alpha(\mathcal{S}_0) W_\rho(\mathcal{C}) U_\alpha^{-1}(\mathcal{S}_0)$. Using Eq. (3.116), we then find that the Ward identity translates to

$$U_\alpha(\mathcal{S}_0) W_\rho(\mathcal{C}) U_\alpha^{-1}(\mathcal{S}_0) = \rho(\alpha)^{\text{int}_3(\mathcal{C}, \mathcal{S}_0)} W_\rho(\mathcal{C}), \quad (3.117)$$

which is an equal-time operator equation.

Eq. (3.117) is the avatar of the Ward identity in the operator formalism, where all the relevant geometric objects and operations, including the disc $\mathcal{S}_0$ and the closed contour $\mathcal{C}$, are defined within the three-manifold $\mathcal{M}_3$. The key insight here is that time ordering in the path integral formalism turns into operator ordering in the operator formalism.

Finally, it is straightforward to explicitly evaluate Eq. (3.117). Applying the $3+1$ decomposition and using Eq. (3.114), the Hamiltonian formalism avatar of the symmetry operator, supported on a disc $\mathcal{S}_0 \subset \mathcal{M}_3$ in three dimensions, is given as

$$U_\alpha(\mathcal{S}_0) = \exp\left(\frac{-i\pi}{g^2} \int_{\mathcal{S}_0} dx^j \wedge dx^k \epsilon_{ijk} E^i_a \lambda^a\right), \quad e^{2\pi\lambda} = \alpha. \quad (3.118)$$

²⁰Note that here we have allowed nonclosed surfaces for the support of symmetry operators, which might be a slight abuse of notation.

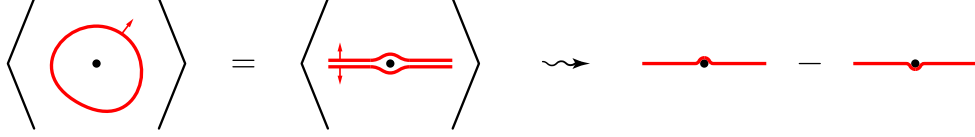


Figure 3.13: Zero-form version of the pancaking procedure, which the reader may be familiar with. Wrapping an operator with the symmetry operator translates to an equal-time conjugation, where operator ordering traces back to time ordering.

To compute $U_\alpha(\mathcal{S}_0) W_\rho(\mathcal{C}) U_\alpha^{-1}(\mathcal{S}_0)$, let us first deduce how the conjugation acts on the phase space. While $E^i_a(x)$ is left invariant because $[E^i_a(x), E^j_b(x')] = 0$, the spatial gauge connection $A^a_k(x)$ has a nonvanishing commutator with the exponent of Eq. (3.118),

$$\begin{aligned} & \frac{-i\pi}{g^2} \int_{\mathcal{S}_0} dx'^i \wedge dx'^j \lambda^b(x') \epsilon_{ijl} [E^l_b(x'), A^a_k(x)] \\ & = 2\pi \int d\sigma_1 d\sigma_2 \frac{\partial X^i}{\partial \sigma_1} \frac{\partial X^j}{\partial \sigma_2} \lambda^a(X) \epsilon_{ijk} \delta^{(3)}(x - X), \end{aligned} \quad (3.119)$$

where $(\sigma_1, \sigma_2) \mapsto X^i(\sigma_1, \sigma_2)$ is a parameterization of the surface $\mathcal{S}_0$. According to Eq. (3.13), this describes the components of the one-form $2\pi\lambda^a \delta_3(\mathcal{S}_0)$, where $\delta_3(\mathcal{S}_0)$ is the Dirac delta one-form of $\mathcal{S}_0$ defined in the spatial three-manifold $\mathcal{M}_3$. Therefore, we can summarize the action of the symmetry transformation on the phase space variables as the following:

$$\begin{aligned} U_\alpha(\mathcal{S}_0) A^a_i U_\alpha^{-1}(\mathcal{S}_0) &= (A^a + 2\pi\lambda^a \delta_3(\mathcal{S}_0))_i, \\ U_\alpha(\mathcal{S}_0) E^i_a U_\alpha^{-1}(\mathcal{S}_0) &= E^i_a. \end{aligned} \quad (3.120)$$

As a result, we find that the Wilson loop transforms as

$$U_\alpha(\mathcal{S}_0) W_\rho(\mathcal{C}) U_\alpha^{-1}(\mathcal{S}_0) = \rho(\alpha)^{\text{int}_3(\mathcal{C}, \mathcal{S}_0)} W_\rho(\mathcal{C}), \quad (3.121)$$

which proves the Hamiltonian counterpart of the Ward identity. We have used the fact that the three-dimensional intersection number is $\text{int}_3(\mathcal{C}, \mathcal{S}_0) = \oint_{\mathcal{C}} \delta_3(\mathcal{S}_0)$.

3.A.3 Plebański Gravity

The above canonical analysis for the Ward identity straightforwardly applies to gravity with the same formulae in Eqs. (3.114), (3.115), and (3.118) (up to the $g \mapsto \kappa$ replacement), by utilizing the Plebański formulation [315] of

gravity in which the B -field is promoted to a fundamental field variable;²¹ see Sec. 2.5.2. The resulting Hamiltonian formulation of gravity essentially describes a nonchiral version of the Ashtekar [61, 522] formalism. Namely, it is known that implementing the $3+1$ decomposition and integrating out the nondynamical degrees of freedom from the Plebański Lagrangian results in a well-posed constrained Hamiltonian system in the phase space (γ^a_i, E^i_a) coordinatized in the Darboux basis [358, 359, 523, 524]. The constraints include the Gauss constraints for local Lorentz transformations as well as the Hamiltonian and diffeomorphism constraints familiar from the ADM [187] analysis.

²¹While the path originally taken by Ashtekar for deriving canonical gravity starts from the tetradic Palatini formulation (see Ref. [522], for instance), the Plebański formulation offers a more straightforward path because it manifests the symplectic structure in the “ BF ” form with the B -field non-composite.

DOUBLE COPY AND DIFFEOMORPHISM GAUGE THEORIES

We investigate the idea of diffeomorphism gauge theory as a strict implication of EoM level CK duality. In field theory, diffeomorphisms are external transformations shifting the base spacetime points and are considered disparate from the internal, fiberwise transformations of gauge theories such as YM theory. However, an EoM level formulation of manifest CK duality is known to necessitate taking diffeomorphisms like the internal color algebra of a gauge theory, via the so-called color-to-kinematics replacement (CKR). It is shown that CK duality systematically establishes a consistent definition of diffeomorphism gauge transformation, diffeomorphism gauge connection, diffeomorphism covariant derivative, diffeomorphism field strength, and diffeomorphism Wilson line. Notably, such a diffeomorphism gauge theory can admit a simple and consistent geometrical construction as a field theory of a dynamical vielbein. In particular, this implies that classical Born-Infeld theory can be embedded in a teleparallel theory. For gravity, we note that diffeomorphism gauge theory formulations are viable in three dimensions or in the SD sector in four dimensions. For a classical solution of diffeomorphism gauge theories, we reinterpret the Misner string as a topological configuration generated by a multivalued diffeomorphism gauge transformation and view it as the counterpart of the Dirac string.

4.1 Introduction

YM theory is a gauge theory. Gravity is a gauge theory. What these statements precisely mean, however, is utterly different.

By the late twentieth century, gauge theory acquired a systematic geometric interpretation, establishing a paradigm in mathematical physics. Yang and Mills had introduced the nonabelian gauge principle as a local version of internal symmetry [24]. Wu and Yang later studied the global aspects and emphasized the role of principal bundles [25]. As per the Wu-Yang dictionary [25], the gauge potential became a connection on a principal bundle; the field

strength became its curvature; gauge transformations became changes in local trivializations; and Wilson loops, AB phases, monopoles, and instantons became manifestations of holonomy and global topology [24–32, 39].

The success of this geometric ideal of Wu, Yang, and Mills was multifold. To physicists, it gave a common language to the three fundamental forces of our nature: *electromagnetism, the weak, and the strong*. It also provided systematic tools for investigating topics such as monopoles, instantons, and anomalies [29–32]. To mathematicians, it gave the realization that physics can be used as a tool for studying the topology of manifolds and moduli spaces of bundles [33–38].

A gauge theory, in this narrower context, is a theory on a fixed base manifold $\mathcal{M}$, equipped with an internal fiber modeled on a group G . The gauge transformations are vertical: they act within each fiber while leaving the base point $x \in \mathcal{M}$ completely fixed. This fiberwise, *internal* character is necessary.

YM theory is a gauge theory in the narrower, strict mathematical sense of Wu, Yang, and Mills, based on internal local transformations.

Interestingly, *gravity* has been both the most tempting and the most resistant candidate for inclusion in this class. On the one hand, GR is already inherently geometric. It features the Levi-Civita connection and the Riemannian curvature as well as parallel transport, holonomy, and topological invariants. In particular, the Riemannian curvature looks like a field strength of a gauge theory. This proximity makes it natural to ask whether gravity, too, should be a gauge theory in the same mathematical sense of Wu, Yang, and Mills.

Crucially, the local symmetry of GR describes diffeomorphisms: the freedom to choose coordinates. Diffeomorphisms move spacetime points and pull back fields. They act on the base manifold $\mathcal{M}$ itself. They describe *external* transformations to say, not internal. In YM theory, spacetime is the fixed arena on top of which the physics of gluons take place. In GR, the spacetime itself is dynamical. Diffeomorphisms are not gauge transformations in the YM sense.

GR is sometimes said to be a gauge theory, but this is only in the broader sense where the word “gauge” is used as a shorthand for “redundancy.” The redundancies of GR are diffeomorphisms, which are external local transformations.

A related point is that GR employs a connection on the tangent bundle $T\mathcal{M}$ of the spacetime $\mathcal{M}$, and $T\mathcal{M}$ is not any bundle (like the fiber bundles in YM theory) but is intimately tied to $\mathcal{M}$. Elements of each tangent fiber describe actual directions in spacetime, as a vector describes a pair of infinitesimally neighboring points on the very base space $\mathcal{M}$. This is the soldering structure, which can be represented as a coframe (vielbein). Giving a gauge theory style interpretation to the soldering structure could be subtle and tricky [28, 39, 53].

Historically, there had been aspirations toward reformulating GR as a gauge theory [41–61], envisioning a unified geometrization of the four fundamental forces. Early attempts began with gauging spacetime symmetries. Utiyama [41] gauged the Lorentz group and obtained the spin connection, but the vielbein had to be supplied as an additional structure. To this end, Kibble [42] considered gauging the Poincaré group so that the vielbein may be recast as a kind of a “translation gauge connection,” arriving at Einstein-Cartan theory. Other notable or adjacent results include the Poincaré gauge theory formulation of three-dimensional gravity [525, 526], MacDowell–Mansouri formulation and its descendants [58–60], Ashtekar variables [61], and Plebański or BF gravity [315, 339, 357–360, 523, 524]. Despite these developments, the gauge-theoretic formulation of GR still appears as an open problem, and the practical outcomes are unclear.

Notably, we remark that modern explorations through scattering amplitudes bring a new perspective into this discourse. In the scattering amplitudes program, gauge redundancies may be portrayed as pure evil, hindering efficient computations and obscuring physical structures. Hence one seeks for more efficient representations of scattering amplitudes instead of sticking to the Feynman diagram representations arising from conventional Lagrangians.

Surprisingly, deep computational explorations into gluon amplitudes have concluded that the entire nonlinear content of YM theory might be encapsulated in a single, universal *cubic* interaction vertex, given as the product of two structure constants. The first are the very structure constants of the color gauge algebra, such as $\mathfrak{g} = \mathfrak{su}(N)$. The second, however, are the structure constants of a mysterious Lie algebra $\mathfrak{g}_\varphi$. This mystery Lie algebra $\mathfrak{g}_\varphi$ is known as the kinematic algebra of YM theory, as it would have to take the kinematic data (momenta and polarizations) as “adjoint indices.” The fact that YM amplitudes admit such remarkable cubic representations is referred



Figure 4.1: “What is the kinematic algebra of YM theory?”

to as CK duality [87, 88]. When extrapolated to a field theory statement [90, 527, 528], CK duality provides striking evidence for a cubic reformulation of YM theory, although the Feynman rules of the conventional YM Lagrangian describe a quartic perturbation theory. At this moment, CK duality stands well-established at the level of on-shell scattering amplitudes. See [89] for a comprehensive review.

Further surprisingly, deep computational explorations into graviton amplitudes have concluded that the entire nonlinear content of GR might be encapsulated in a single, universal cubic interaction vertex that simply squares the structure constants of the kinematic algebra $\mathfrak{g}_?$ of YM theory.¹ This is the celebrated double copy correspondence between YM theory and GR [87–89], which has a string-theoretic precursor as well [86]. Again, the double copy is well-established at the level of on-shell scattering amplitudes. When extrapolated to a field theory statement [90, 527, 528], it gives striking evidence for the conjecture that GR could be reformulated as a YM theory for a choice of a mysterious gauge algebra $\mathfrak{g}_?$, at least at the EoM level:

$$\mathbf{GR} \cong \mathbf{YM}(\mathfrak{g}_?). \quad (4.1)$$

At this moment, this conjecture stands verified only within the SD sector [309, 327], while its validity beyond the SD sector has been inconclusive.

CK duality and double copy are not properties that are exclusively enjoyed by YM theory or GR; as well-reviewed in Ref. [89], there exists a web of theories. For example, nonlinear sigma model (NLSM) (as principal chiral model

¹Technically, the precise double copy of YM theory is “fat gravity,” containing not only the graviton but also a dilaton and a Kalb-Ramond B -field (the Neveu-Schwarz-Neveu-Schwarz (NS-NS) sector of type II supergravity). Yet, one can consistently truncate such extra modes.

for a Lie group) is known to exhibit CK duality, whose field-theoretic explanation was established by Refs. [90, 527, 528]. In particular, Cheung and Mangan [90] explicitly identified the kinematic algebra of NLSM at the EoM level as the Lie algebra of volume-preserving diffeomorphisms on spacetime, $\mathbf{sdiff}(\mathcal{M})$. The double copy of NLSM is a special instance of a Galileon [529–531] theory, or “special Galileon” (SG) in short, which arises by doubling the volume-preserving diffeomorphism algebra [90, 157, 527, 528, 532].

It is also known that, despite the mysterious status of the kinematic algebra $\mathbf{g}_?$ of YM theory, one can replace its color algebra with the volume-preserving diffeomorphism algebra to obtain a theory with two kinematic algebras, which turns out to be BI theory [89, 90]:

$$\mathbf{BI} \cong \mathbf{YM}(\mathbf{sdiff}(\mathcal{M})). \quad (4.2)$$

This action of replacing a color² (meaning finite-dimensional) Lie algebra with a kinematic Lie algebra is known as CKR and is implemented in Ref. [90]. The correspondence in Eq. (4.2) precisely conforms to the grammar of Eq. (4.1), and BI theory serves as a concrete testing ground for enhancing our understandings on kinematic algebras and double copy.

The implication of the double copy correspondence in Eq. (4.2) is the following. The YM equations describe the vanishing of the covariant divergence of the nonabelian field strength $F^a_{\alpha\beta}$, whose explicit formula reads

$$F^a_{\alpha\beta} = \partial_\alpha A^a_\beta - \partial_\beta A^a_\alpha + f^a_{bc} A^b_\alpha A^c_\beta. \quad (4.3)$$

Here, $a, b, c, \dots$ are the adjoint indices of the color Lie algebra, say $\mathbf{g} = \mathbf{su}(N)$, while greek indices are spacetime indices. Also, f^a_{bc} denote the structure constants of $\mathbf{g}$, so $f^a_{bc} A^b_\alpha A^c_\beta$ equals the Lie bracket $[A_\alpha, A_\beta]^a$. Cheung and Mangan [90] showed that the scattering amplitudes of BI theory are precisely reproduced by applying the framework of BG recursion [154–157] to the YM equations with the color algebra replaced with the volume-preserving diffeomorphism algebra $\mathbf{sdiff}(\mathcal{M})$. What this means is to impose vanishing covariant divergence on the “diffeomorphism field strength,” $F^\mu_{\alpha\beta}$. In their Eq. (3.28), Cheung and Mangan [90] explicitly write down the definition of this $F^\mu_{\alpha\beta}$ as

$$F^\mu_{\alpha\beta} = \partial_\alpha A^\mu_\beta - \partial_\beta A^\mu_\alpha + \left(A^\rho_\alpha \partial_\rho A^\mu_\beta - A^\rho_\beta \partial_\rho A^\mu_\alpha \right). \quad (4.4)$$

²Strictly speaking, “color” in the context of CK duality sometimes refers to a *flavor* index, not necessarily gauged. This is a common abuse of terminology.

Here, A^μ_α is a “diffeomorphism gauge connection,” and the bracketed terms in Eq. (4.4) evaluate $[A_\alpha, A_\beta]^\mu$, the Lie bracket of $\mathfrak{sdiff}(\mathcal{M})$ that computes the commutator between vector fields as differential operators: $[A^\mu_\alpha \partial_\mu, A^\nu_\beta \partial_\nu] = [A_\alpha, A_\beta]^\mu \partial_\mu$. As Cheung and Mangan [90] explicitly state in their Eqs. (3.27)-(3.29) the CKR means to replace the color index a with the vector index μ while replacing the color Lie bracket with the Lie bracket between vector fields. Note that the elements of the diffeomorphism algebra are vector fields. The volume-preserving condition simply puts an additional restriction $\partial_\mu A^\mu_\alpha = 0$.

To reiterate, this “diffeomorphism YM theory” formulation of BI theory is an inevitable logical consequence provided the existence of manifest CK duality at the EoM level and is also concretely verified in terms of explicit scattering amplitudes calculations, as shown by Cheung and Mangan [90].

If one questions the mathematical basis of this construction, however, the very phrase “diffeomorphism YM theory” appears seriously wrong in the first place. What does it mean to take *diffeomorphisms as the internal symmetry of a YM theory*? It directly violates the careful geometrical distinction made at the beginning of this introduction, where it is literally written that “Diffeomorphisms are not gauge transformations in the YM sense.” To reiterate, the CKR wishes to take diffeomorphisms, which are *external* transformations, as the *internal* gauge transformations of a YM theory. From this point of view, the CKR seems to be a fundamentally forbidden operation.

Clearly, physics tells otherwise. The idea of “diffeomorphism YM theory” is valid in concrete senses as clarified above, and it is indeed even the very gist of CK duality. The kinematic algebra describes spacetime transformations because it is kinematic, and CK duality literally likens it to a color algebra.

The purpose of this work is to show that these points of the mathematicians and the physicists can be reconciled into a single consistent picture, and there is no strict illegality in thinking about a “diffeomorphism YM theory,” yet with some crucial clarifications. Arguably, the insights gained here would show us some expectations onto how the unsolved problem of finding the kinematic algebra $\mathfrak{g}_?$ of GR could unfold.

We begin by showing that the CKR provides a working definition of diffeomorphism gauge theory. We do so by concrete demonstrations. We find that CKR mechanically constructs diffeomorphism gauge transformation, diffeomorphism gauge connection, diffeomorphism covariant derivative, diffeomor-

phism field strength, and even diffeomorphism Wilson line.

In later parts, we discover a consistent and remarkably simple geometrical picture that justifies diffeomorphism gauge theory within standard concepts of differential geometry or GR in the textbook regime: the diffeomorphism gauge connection $A^\mu{}_\alpha$ is nothing but the vielbein perturbation. Based on this fact, we establish a novel geometric interpretation of Cheung and Mangan [90]’s formulation of BI theory as a teleparallel theory. For GR, we remark that three-dimensional gravity and four-dimensional SD gravity admit diffeomorphism gauge theory formulations by the constructions due to Refs. [309, 533, 534]. Lastly, we revisit the Misner string [474–478] geometry as a classical solution of any diffeomorphism gauge theories and view it as the classical double copy of the Dirac string in ordinary gauge theories.

4.2 Color-Kinematics Duality from Equations of Motion

For a precise formulation of ideas, we begin by a review of the EoM level manifest CK duality [90].³

4.2.1 Bi-Adjoint Scalar Theory as a Template

A BAS field is a scalar field $\Phi^{a\tilde{a}}$ carrying two adjoint indices $a = 1, \dots, \dim \mathfrak{g}$ and $\tilde{a} = 1, \dots, \dim \tilde{\mathfrak{g}}$ for two Lie algebras $\mathfrak{g}$ and $\tilde{\mathfrak{g}}$. By definition, their structure constants $f^a{}_{bc}$ and $\tilde{f}^{\tilde{a}}{}_{\tilde{b}\tilde{c}}$ must satisfy the Jacobi identity.

Suppose a field theory is found to exhibit tree-level CK duality at the scattering amplitudes level. Cheung and Mangan [90] concretely established that this tree-level CK duality can be manifested and proved to all multiplicities if the perturbative EoM of the theory can be formulated in the form of BAS equations for a choice of the untilded and tilded Lie algebras $\mathfrak{g}$ and $\tilde{\mathfrak{g}}$,

$$\mathbf{BAS}(\mathfrak{g}, \tilde{\mathfrak{g}}) : \quad \square \Phi^{a\tilde{a}} = \frac{1}{2} f^a{}_{bc} \tilde{f}^{\tilde{a}}{}_{\tilde{b}\tilde{c}} \Phi^{b\tilde{b}} \Phi^{c\tilde{c}}, \quad (4.5)$$

provided that the relevant field redefinition is invertible on external scattering states (asymptotically free states).

Cheung and Mangan [90] showed that the EoM of NLSM and SG can be formulated in the form of BAS equations with the following choices of Lie

³The broader program has concerned manifesting CK duality at the EoM level [90, 327], at the Lagrangian level [527, 528, 535, 536], or at the fully off-shell level [533, 537]. Here, we restrict our scope to the EoM level manifestation of Ref. [90], which makes the most conservative claim.

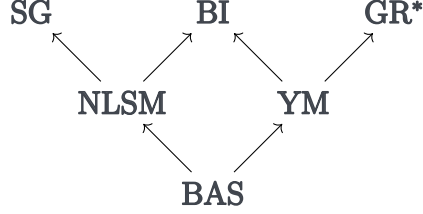


Figure 4.2: A web of field theories via CK duality and double copy. Each arrow implements a CKR. The arrows pointing to the left replace $\mathfrak{su}(N)$ with $\mathfrak{sdiff}(\mathbb{R}^d)$. The arrows pointing to the right replace $\mathfrak{su}(N)$ with $\mathfrak{g}_?$, the mystery Lie algebra. Note how this web is structured in terms of spin (helicities 0, 1, 2). It is remarkable that such an algebraic organization arises between forces in our nature.

algebras, thus exhibiting manifest CK duality. For NLSM, the algebras are

$$\mathbf{NLSM}(\mathfrak{g}) \cong \mathbf{BAS}(\mathfrak{g}, \mathfrak{sdiff}(\mathcal{M})), \quad (4.6)$$

where the color algebra is typically $\mathfrak{g} = \mathfrak{su}(N)$. For SG, the algebras are

$$\mathbf{SG} \cong \mathbf{NLSM}(\mathfrak{sdiff}(\mathcal{M})) \cong \mathbf{BAS}(\mathfrak{sdiff}(\mathcal{M}), \mathfrak{sdiff}(\mathcal{M})). \quad (4.7)$$

We will review the derivation of Eqs. (4.6) and (4.7) shortly. Here, $\mathfrak{sdiff}(\mathcal{M})$ denotes the volume-preserving diffeomorphism algebra of the spacetime, $\mathcal{M}$.

Similarly, the CK duality of YM theory [87] is manifested at the EoM level if one shows that

$$\mathbf{YM}(\mathfrak{g}) \cong \mathbf{BAS}(\mathfrak{g}, \mathfrak{g}_?), \quad (4.8)$$

for a choice of a Lie algebra $\mathfrak{g}_?$. The double copy property of gravity [88] is manifested at the EoM level if one shows that

$$\mathbf{GR}^* \cong \mathbf{YM}(\mathfrak{g}_?) \cong \mathbf{BAS}(\mathfrak{g}_?, \mathfrak{g}_?), \quad (4.9)$$

for a choice of the same Lie algebra $\mathfrak{g}_?$ as in Eq. (4.8). Here, $\mathbf{GR}^*$ denotes the NS-NS sector of type II supergravity, sometimes called “fat gravity.” As remarked earlier in a footnote, this theory contains not only the graviton but also a dilaton and a Kalb-Ramond B -field, while a consistent truncation into GR is viable.

Note that these are “if” type statements. Strictly speaking, there might be no guarantee that CK duality or double copy can always be manifested at the EoM level. Still, Eqs. (4.8) and (4.9) may be taken as reasonable conjectures, which are more conservative than Lagrangian or off-shell level claims, in fact.

The reformulation statements in Eqs. (4.6), (4.7), (4.8), and (4.9) are summarized in Fig. 4.2, which show a web of massless field theories. To reiterate and clarify, these reformulations are at the perturbative EoM level and presume invertibility on external scattering states, the meaning of which will become clear soon.

BI theory holds an interesting position in this web:

$$\mathbf{NLSM}(\mathfrak{g}_{\mathcal{P}}) \cong \mathbf{BI} \cong \mathbf{YM}(\mathbf{sdiff}(\mathcal{M})) \quad (4.10)$$

The statement here is the double copy property of BI theory, which utilizes two different kinematic algebras, is manifested at the EoM level if one shows either one of the proposed equivalences in Eq. (4.10). Examining this unique “double duty” of BI theory might be holding a key toward identifying $\mathfrak{g}_{\mathcal{P}}$.

The equivalence $\mathbf{BI} \cong \mathbf{YM}(\mathbf{sdiff}(\mathcal{M}))$ was established at the scattering amplitudes level by Cheung and Mangan [90]. Showing this equivalence at the EoM level, however, remains as an open problem. Concretely, it demands identifying a field redefinition from an abelian gauge connection a_α to a diffeomorphism gauge connection A^μ_α ; we will revisit this point in Sec. 4.7.

4.2.2 Nonlinear Sigma Model

Defining Formulation In the present context, $\mathbf{NLSM}(\mathfrak{g})$ refers to the principal chiral model $U : \mathcal{M} \rightarrow \mathbf{G}$ on flat spacetime $(\mathcal{M}, \eta)$ for a finite-dimensional matrix Lie group $\mathbf{G}$ whose Lie algebra is $\text{Lie}(\mathbf{G}) = \mathfrak{g}$. Here, η is a Lorentzian-signature flat metric given to the spacetime manifold $\mathcal{M}$ with $\dim \mathcal{M} = d$, while U describes a section of a principal $\mathbf{G}$ -bundle over $\mathcal{M}$ which will be taken as trivial. To define this as a perturbative field theory, we presume a local coordinatization $u : \mathfrak{g} \rightarrow \mathbf{G}$ of the target manifold $\mathbf{G}$ centered around the vacuum expectation value $U = \mathbb{1}$, while recalling that each tangent fiber of a Lie group $\mathbf{G}$ is isomorphic to its Lie algebra $\mathfrak{g}$. In this way we obtain the pion field $\pi : \mathcal{M} \rightarrow \mathfrak{g}$ as the composition $\pi = u^{-1} \circ U$ constructed locally.

The Lagrangian d -form is

$$L[\pi] = -\frac{1}{2\kappa^2} \delta_{ab} j^a[\pi] \wedge *j^b[\pi], \quad (4.11)$$

where δ_{ab} is a Killing form on $\mathfrak{g}$, $a, b, c, d, \dots$ are the adjoint indices, and $\kappa = 1/f_\pi$ is the coupling. The left-invariant Maurer-Cartan form is

$$j[\pi] = (u(\pi))^{-1} du(\pi). \quad (4.12)$$

From Eq. (4.12), it is clear that the pion field satisfies the following equation off shell:

$$dj^a[\pi] + \frac{1}{2} f^a_{bc} j^b[\pi] \wedge j^c[\pi] = 0, \quad (4.13a)$$

Variation of Eq. (4.11) shows that the pion field satisfies the following equation on shell:

$$d*j^a[\pi] = 0. \quad (4.13b)$$

Eq. (4.13b) implies that $j^a[\pi]$ is a conserved current: the chiral current.

The arbitrariness in the coordinatization $u(\pi)$ describes a prime category of field redefinitions. The only condition we demand on u is that it is “proper,” in the sense that the chiral current in Eq. (4.12) admits a well-behaved perturbative expansion

$$j^a[\pi] = d\pi^a + \mathcal{O}(\pi^2). \quad (4.14)$$

Provided Eq. (4.14), any coordinatization of $\mathbf{G}$ is allowed. For example, take $u = \exp(\pi)$: normal coordinates via the exponential map. In this case, the chiral current exhibits the expansion

$$j = e^{-\pi} d e^{\pi} = d\pi - \frac{1}{2!} [\pi, d\pi] + \frac{1}{3!} [\pi, [\pi, d\pi]] + \mathcal{O}(\pi^4). \quad (4.15a)$$

For another example, take $u = (\mathbb{1} + \pi/2)(\mathbb{1} - \pi/2)^{-1}$, which is known as Cayley basis. In this case, the chiral current exhibits the expansion

$$j = d\pi - \frac{1}{2} [\pi, d\pi] + \frac{1}{4} \left(\pi^2 d\pi - \pi d\pi \pi + d\pi \pi^2 \right) + \mathcal{O}(\pi^4). \quad (4.15b)$$

From Eqs. (4.15a) and (4.15b), we see that both coordinatizations are proper.

Plugging in either of Eqs. (4.15a) and (4.15b) to Eq. (4.12) determines the EoM of the pion field explicitly. As a result, one obtains a perturbation theory with an infinite number of interaction vertices. However, since both of Eqs. (4.15a) and (4.15b) exhibit the same leading behavior stipulated in Eq. (4.14) at the linearized level, the S-matrix equivalence theorem [276–280] applies and ensures that the scattering amplitudes are the same.

New Formulation An insight conceived by Cheung and Mangan [90] was that one may have to focus on objects that are invariant under the coordinate choice $u(\pi)$ on the target group manifold to manifest CK duality, a property observed in the field basis invariant S-matrix. Again, coordinate transformations in the target space are a prime example of field redefinitions.

Motivated by this argument (and in connection with an earlier creative exploration [538]), Cheung and Mangan [90] envisioned taking the chiral current as a fundamental field variable j . This is viable by imposing the flatness condition as a constraint,

$$\partial_\mu j^a{}_\nu - \partial_\nu j^a{}_\mu + f^a{}_{bc} j^b{}_\mu j^c{}_\nu = 0, \quad (4.16a)$$

along with the conservation equation,

$$\partial_\mu j^{a\mu} = 0. \quad (4.16b)$$

Combining Eqs. (4.16a) and (4.16b) gives a second-order equation,

$$\square j^{a\mu} = f^a{}_{bc} j^{b\rho} \partial_\rho j^{c\mu} = \frac{1}{2} f^a{}_{bc} [j^b, j^c]^\mu, \quad (4.17)$$

where $[j^b, j^c]^\mu = j^{b\rho} \partial_\rho j^{c\mu} - j^{c\rho} \partial_\rho j^{b\mu}$ describes the diffeomorphism Lie bracket. Crucially, Eq. (4.17) takes the form of BAS equations in Eq. (4.5) for the choice $\tilde{\mathbf{g}} = \text{diff}(\mathcal{M})$. More precisely, the divergence-free condition in Eq. (4.16b) implies a restriction to the volume-preserving subalgebra $\tilde{\mathbf{g}} = \text{sdiff}(\mathcal{M})$. That is, the index-raised chiral current $j^{a\mu}$ can be seen as a BAS field.

Let $\mathbf{BAS}(\mathfrak{g}, \text{sdiff}(\mathcal{M}))$ denote the classical field theory of the field $j^{a\mu}$ whose on-shell configurations satisfy Eqs. (4.16a) and (4.16b). The claim in Eq. (4.6) is that this new formulation is equivalent to $\mathbf{NLSM}(\mathfrak{g})$ as a perturbative classical field theory in specific senses.

Equivalence Firstly, $\mathbf{NLSM}(\mathfrak{g})$ is embedded in $\mathbf{BAS}(\mathfrak{g}, \text{sdiff}(\mathcal{M}))$ by the map $\pi \mapsto j[\pi]$ given in Eq. (4.12), for any proper target-space coordinatization $u(\pi)$. Each on-shell configuration π^a in $\mathbf{NLSM}(\mathfrak{g})$, satisfying Eq. (4.13), is mapped to an on-shell configuration in $\mathbf{BAS}(\mathfrak{g}, \text{sdiff}(\mathcal{M}))$, satisfying Eq. (4.16).

Secondly, consider

$$U[j](x) = \text{Pexp} \left(\int_{\mathcal{C}_x} j^a{}_\mu dx^\mu \right), \quad (4.18)$$

which computes the Wilson line along a semi-infinite contour $\mathcal{C}_x$ that ends at the bulk point x . Then consider the map $j \mapsto u^{-1}(U[j])$, where $u^{-1} : \mathbf{G} \rightarrow \mathfrak{g}$ is the inverse of $u : \mathfrak{g} \rightarrow \mathbf{G}$ as a map. It follows that $\mathbf{BAS}(\mathfrak{g}, \text{sdiff}(\mathcal{M}))$ is embedded in $\mathbf{NLSM}(\mathfrak{g})$ by this map for any choice of $\mathcal{C}_x$. Each on-shell configuration $j^a{}_\mu$ in $\mathbf{BAS}(\mathfrak{g}, \text{sdiff}(\mathcal{M}))$, satisfying Eq. (4.16), is mapped to an on-shell configuration in $\mathbf{NLSM}(\mathfrak{g})$, satisfying Eq. (4.13). Especially, the flatness condition in Eq. (4.16a) ensures that Eq. (4.18) evaluates to the same value for each fixed $x \in \mathcal{M}$, regardless of one's choice of the contour $\mathcal{C}_x$.

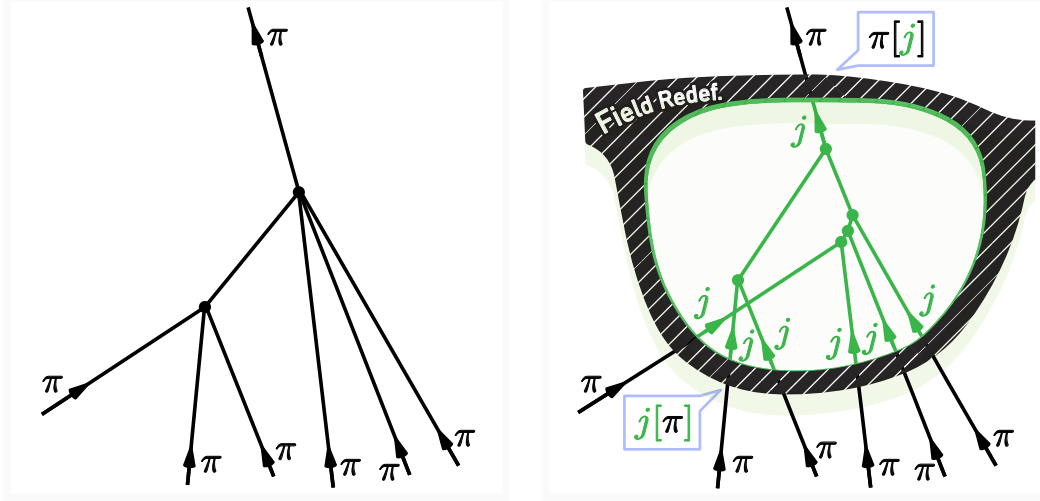


Figure 4.3: Field redefinitions change one’s description of perturbative physics in the bulk. In one frame, one sees π -particles interacting via vertices of various valence. In another frame, one sees j -particles interacting via a universal cubic vertex. To obtain the same scattering amplitude, one simply needs to know how a single, *free* π -particle turns into a free j -particle and vice versa. That is, it suffices to perform the conversion between the π - and j -frames at the boundary. When the amplitude is obtained via computing a one-point function by BG recursion, $j[\pi]$ is needed for the leaf legs while $\pi[j]$ is needed for the one last conversion for the root leg.

Lastly, the above two maps $\pi \mapsto j[\pi]$ and $j \mapsto u^{-1}(U[j])$ are invertible and are well-behaved in the perturbative regime. At the linearized level, in particular, the former gives $j^a_\mu[\pi] = \partial_\mu \pi^a + \mathcal{O}(\pi^2)$ as stipulated in Eq. (4.14) while the latter gives, e.g., $\pi^a[j] = (q \cdot \partial)^{-1} j^a_\mu q^\mu + \mathcal{O}(j^2)$ when $\mathcal{C}_x$ is realized as a straight semi-infinite line extending along the direction of a constant reference vector q^μ . It is easily checked that these are inverse to each other:

$$j^a_\mu = \partial_\mu \pi^a + \mathcal{O}(\pi^2) \quad \Longleftrightarrow \quad \pi^a = \frac{1}{q \cdot \partial} j^a_\mu q^\mu + \mathcal{O}(j^2). \quad (4.19)$$

The linearized-level invertibility in Eq. (4.19) implies that there exists a prescription for converting a free π -state to a unique free j -state and vice versa. In **NLSM**($\mathfrak{g}$), a linearized on-shell configuration is given by $\pi^a = c^a e^{ikx}$ for constant c^a and $k^2 = 0$. Eq. (4.19) embeds this to **BAS**($\mathfrak{g}, \text{sdiff}(\mathcal{M})$) by $j^a_\mu = \partial_\mu \pi^a = ik_\mu c^a e^{ikx}$, which indeed solves the linear part of Eq. (4.16). Conversely, in **BAS**($\mathfrak{g}, \text{sdiff}(\mathcal{M})$), a linearized on-shell state is given by $j^{a\mu} = w^{a\mu} e^{ikx}$ for $k^2 = 0$ subject to $w^{a\mu} k_\mu = 0$ and $w^{a[\mu} k^{\nu]} = 0$, so $j^{a\mu} = ic^a k^\mu e^{ikx}$ for some c^a . Eq. (4.19) embeds this to **NLSM**($\mathfrak{g}$) by $\pi^a = (iq \cdot k)^{-1} (ic^a k_\mu e^{ikx}) q^\mu = c^a e^{ikx}$, which indeed solves the linear part of Eq. (4.13).

In sum, an invertible field redefinition is constructed between $\mathbf{NLSM}(\mathfrak{g})$ and $\mathbf{BAS}(\mathfrak{g}, \mathbf{sdiff}(\mathcal{M}))$ that is well-behaved in perturbation theory and thus on free on-shell states. This establishes the proposed perturbative equivalence $\mathbf{NLSM}(\mathfrak{g}) \cong \mathbf{BAS}(\mathfrak{g}, \mathbf{sdiff}(\mathcal{M}))$ stated in Eq. (4.6).

To recapitulate, Cheung and Mangan [90]’s construction can be taken as providing the *definition* of EoM level manifest CK duality:

A classical field theory exhibits *EoM level manifest CK duality* if its EoM can be reformulated as BAS equations by a field redefinition that is invertible on asymptotic scattering states.

To show that this EoM level manifest CK duality implies the manifest CK duality of scattering amplitudes, Cheung and Mangan [90] utilize the framework of BG recursion [154] (perturbative method [155–157]). Details are to be found in Ref. [90], but nevertheless a quick review is provided in Sec. 4.A.

Note also that it is unnecessary to invoke Lagrangian formulations in this discussion, although the Freedman-Townsend [539] formulation of NLSM may provide tempting pathways: see Sec. 4.B.

4.2.3 Born-Infeld Theory

Given the identification of the kinematic algebra of NLSM as $\mathbf{sdiff}(\mathcal{M})$, and also the amplitudes-level fact that BI theory arises by combining the NLSM and YM kinematic algebras (recall Fig. 4.2 and Eq. (4.10)), Cheung and Mangan [90] established a novel formulation of BI theory at the EoM level in their Sec. 3.4.3. For the reader’s sake, let us reproduce their equations below up to minor change in notation.

In their (3.27), Cheung and Mangan state the NLSM CKR rule applied on YM gauge connection and field strength:

$$\mathfrak{g} \mapsto \mathbf{sdiff}(\mathcal{M}) : \quad A^a_{\alpha} \mapsto A^{\mu}_{\alpha}, \quad F^a_{\alpha\beta} \mapsto F^{\mu}_{\alpha\beta}. \quad (4.20a)$$

This means to take the gauge algebra $\mathfrak{g}$ of YM theory as the Lie algebra $\mathbf{sdiff}(\mathcal{M})$ of volume-preserving diffeomorphisms. In their (3.28), Cheung and Mangan expand out the precise mechanics of this replacement for the definition of field strength:

$$\begin{aligned} F^a_{\alpha\beta} &= \partial_{\alpha} A^a_{\beta} - \partial_{\beta} A^a_{\alpha} + f^a_{bc} A^b_{\alpha} A^c_{\beta} \\ &\mapsto F^{\mu}_{\alpha\beta} = \partial_{\alpha} A^{\mu}_{\beta} - \partial_{\beta} A^{\mu}_{\alpha} + A^{\rho}_{\alpha} \partial_{\rho} A^{\mu}_{\beta} - A^{\rho}_{\beta} \partial_{\rho} A^{\mu}_{\alpha}. \end{aligned} \quad (4.20b)$$

This clearly replaces $f^a_{bc} A^b_\alpha A^c_\beta = [A_\alpha, A_\beta]^a$ with $[A_\alpha, A_\beta]^\mu = A^\rho_\alpha \partial_\rho A^\mu_\beta - A^\rho_\beta \partial_\rho A^\mu_\alpha$. Lastly, in their (3.29), Cheung and Mangan explicate how the YM equations are mapped by this replacement. For the source-free case, it is⁴

$$\begin{aligned} D^\alpha F^a_{\alpha\beta} &= \partial^\alpha F^a_{\alpha\beta} + f^a_{bc} A^{b\alpha} F^c_{\alpha\beta} = 0 \\ \mapsto D^\alpha F^\mu_{\alpha\beta} &= \partial^\alpha F^\mu_{\alpha\beta} + A^{\rho\alpha} \partial_\rho F^\mu_{\alpha\beta} - F^\rho_{\alpha\beta} \partial_\rho A^{\mu\alpha} = 0. \end{aligned} \quad (4.20c)$$

Cheung and Mangan then state that their “Eqs. (3.28) and (3.29) constitute a new first-order formulation of BI theory.” Furthermore, they report that they verified “up to six-point scattering that the resulting BI amplitudes agree with known expressions.”

For NLSM, the CKR mapped a global Lie algebra index (flavor index) to a kinematic index. For BI theory, the CKR maps the gauged index (color index) of YM theory to a kinematic index. As a result, one inevitably encounters the notion of “diffeomorphism gauge transformation” and “diffeomorphism gauge covariance,” which seem nonsensical at first sight. What does it mean to take diffeomorphisms as the gauge algebra of a YM theory? What does “**YM**(sdiff($\mathcal{M}$))” mean? Crucially, the EoM level CK duality strictly necessitates such concepts, as is asserted by Cheung and Mangan [90] and as should be clear from our review and exposition above. To our understanding, however, this point has not been properly clarified nor addressed in the literature.

With this remark made, we proceed to Secs. 4.3 and 4.4 to define diffeomorphism gauge theory as a strict implication of EoM level CK duality.

4.3 Ordinary Gauge Theory

To begin with, we review basic concepts in ordinary gauge theory. By an ordinary gauge theory, we mean a gauge theory whose gauge group is a finite-dimensional matrix Lie group $\mathbf{G}$. For concreteness, we may take $\mathbf{G} = \mathrm{SU}(N)$. The gauge algebra $\mathfrak{g} = \mathfrak{su}(N)$ describes fundamental indices $i, j, k, l, \dots$ and adjoint indices $a, b, c, d, \dots$. For $\psi \in \mathbb{C}^N$ and $u, v \in \mathfrak{su}(N)$, we denote

$$(u\psi)^i = u^i_j \psi^j = u^a (t_a)^i_j \psi^j, \quad [u, v]^a = f^a_{bc} u^b v^c, \quad (4.21)$$

where $(t_a)^i_j$ and f^a_{bc} are the generators and structure constants. When working in the fundamental representation, we may sometimes make indices implicit

⁴The source term in their Eq. (3.29) is essentially used for specifying the external state prescription. It states $A^\mu_\alpha = \partial^\mu a_\alpha + \mathcal{O}(a^2)$ if a_α is the conventional BI photon. Certainly, the linearized on-shell state is $A^\mu_\alpha \sim p^\mu e_\alpha e^{ipx}$ when $a_\alpha \sim e_\alpha e^{ipx}$. As mentioned earlier, the exact field redefinition $A[a]$ is unknown for this moment (as well as its inverse $a[A]$).

to avoid clutter. We assume d -dimensional flat spacetime and use greek letters for spacetime indices.

4.3.1 Gauge Covariance

In ordinary gauge theories such as YM theory, local operators are subject to local $\mathbf{G}$ -transformations, geometrically formulated as transformations acting on the *fibers* of a bundle over spacetime. In this sense, gauge transformations are said to be *internal* transformations: the points in the base spacetime manifold are not altered at all. It is important to understand that gauge transformations define equivalences, or redundancies, under a gauge orbit, so physically meaningful quantities are gauge-invariant objects.

To describe gauge transformations in an explicit manner, let $g(x)$ assign a group element in $\mathbf{G}$ for each spacetime point x . Gauge-covariant operators, such as those valued in the singlet, fundamental, or adjoint, exhibit the following transformation behaviors:

$$\begin{aligned} \text{SINGLET :} \quad \phi(x) &\mapsto \phi(x), \\ \text{FUND. :} \quad \phi^i(x) &\mapsto g^i_j(x) \phi^j(x), \\ \text{ADJOINT :} \quad \phi^a(x) &\mapsto g^a_b(x) \phi^b(x). \end{aligned} \tag{4.22}$$

Being valued in the singlet is a synonym for being gauge invariant.

Suppose a non-singlet field. Due to the fact that the parameter $g(x)$ vary over spacetime, the field values at different points transform differently. As a result, their difference is not physically meaningful. The implication is that the partial derivative of a non-singlet field is not a gauge-covariant object.

This provides the very motivation for introducing the gauge connection A , which defines the gauge-covariant derivative D . For each gauge-covariant operator, the covariant derivative returns an operator with one more spacetime index that exhibits the *same* gauge-covariant transformation behavior:

$$\begin{aligned} \text{SINGLET :} \quad D_\alpha \phi(x) &= \partial_\alpha \phi(x), \\ \text{FUND. :} \quad D_\alpha \phi^i(x) &= \partial_\alpha \phi^i(x) + A^i_{j\alpha}(x) \phi^j(x), \\ \text{ADJOINT :} \quad D_\alpha \phi^a(x) &= \partial_\alpha \phi^a(x) + [A_\alpha(x), \phi(x)]^a. \end{aligned} \tag{4.23}$$

This is easily verified by direct computation, provided that the connection transforms as

$$A_\alpha(x) \mapsto g(x) \partial_\alpha g^{-1}(x) + g(x) A_\alpha(x) g^{-1}(x). \tag{4.24}$$

For reference, we also record the infinitesimal version of the gauge transformations in Eq. (4.23):

$$\begin{aligned} \text{SINGLET :} \quad \phi(x) &\mapsto \phi(x), \\ \text{FUND. :} \quad \phi^i(x) &\mapsto \phi^i(x) + (\varepsilon(x)\phi(x))^i, \\ \text{ADJOINT :} \quad \phi^a(x) &\mapsto \phi^a(x) + [\varepsilon(x), \phi(x)]^a. \end{aligned} \quad (4.25)$$

Here, we have taken $g(x) = \exp(\varepsilon(x))$, in which case Eq. (4.24) becomes

$$A_\alpha^a(x) \mapsto A_\alpha^a(x) - D_\alpha \varepsilon^a(x). \quad (4.26)$$

The field strength F is the curvature associated with the gauge connection A . This describes the commutator between two covariant derivatives, say D_α and D_β . Explicitly, it is given as

$$F_{\alpha\beta}^a(x) = \partial_\alpha A_\beta^a(x) - \partial_\beta A_\alpha^a(x) + [A_\alpha(x), A_\beta(x)]^a. \quad (4.27)$$

The field strength is an adjoint-valued two-form. Hence its gauge-covariant derivative is

$$D_\gamma F_{\alpha\beta}^a(x) = \partial_\gamma F_{\alpha\beta}^a(x) + f_{bc}^a A_\gamma^b(x) F_{\alpha\beta}^c(x). \quad (4.28)$$

It is easy to establish either from Eqs. (4.24) or (4.26) that the field strength transforms covariantly. Note also the Bianchi identity:

$$D_{[\gamma} F_{\alpha\beta]}^a(x) = D_\gamma F_{\alpha\beta}^a(x) + D_\alpha F_{\beta\gamma}^a(x) + D_\beta F_{\gamma\alpha}^a(x) = 0, \quad (4.29)$$

which holds by virtue of the Jacobi identity of the gauge algebra $\mathfrak{g}$.

4.3.2 Wilson Line

Geometrically, the gauge connection A defines parallel transport along contours in spacetime. For a contour $\mathcal{C}$ that stretches from point $\mathcal{P}$ to point $\mathcal{P}'$, the parallel transport establishes an isomorphism between the gauge fibers at $\mathcal{P}$ and $\mathcal{P}'$. This isomorphism describes a group element in $\mathbf{G}$ assigned for the contour $\mathcal{C}$, which is the Wilson line $W_{\mathcal{C}}$.

To be concrete, suppose a finite contour $\mathcal{C}$ parameterized as $s \mapsto x^\mu + y^\mu(s)$, where $y^\mu(0) = 0$ and $s \in [0, 1]$. The point $\mathcal{P}$ has coordinates x^μ , while the point $\mathcal{P}'$ has coordinates $x^\mu + y^\mu(1)$. The parallel transport equation for this contour is given by

$$\frac{d}{ds} \Psi^i(s) + A_{j\alpha}^i(x + y(s)) \dot{y}^\alpha(s) \Psi^j(s) = 0, \quad (4.30)$$

where $\Psi^i(s)$ is a fundamental-valued variable restricted on the contour $\mathcal{C}$. As is well-known, the solution to this equation is available in terms of a path-ordered exponential. For

$$W_{\mathcal{C}} = \text{P exp} \left(- \int_0^1 ds A_{\alpha}(x + y(s)) \dot{y}^{\alpha}(s) \right), \quad (4.31)$$

it follows that

$$\Psi^i(1) = (W_{\mathcal{C}})^i_j \Psi^j(0). \quad (4.32)$$

Eq. (4.31) is the definition of the Wilson line $W_{\mathcal{C}}$ about the contour $\mathcal{C}$.

In Eq. (4.32), $\Psi^i(0)$ and $\Psi^i(1)$ are covariant objects located at the points $\mathcal{P}$ and $\mathcal{P}'$ (x^{μ} and $x^{\mu} + y^{\mu}(1)$), respectively. In particular, they transform with the gauge parameters at $\mathcal{P}$ and $\mathcal{P}'$. This establishes that the gauge transformation behavior of the Wilson line is

$$W_{\mathcal{C}} \mapsto g(\mathcal{P}') W_{\mathcal{C}} g^{-1}(\mathcal{P}). \quad (4.33)$$

Namely, the Wilson line transforms bilocally.

When $\mathcal{C}$ is a loop, a singlet is formed by taking a trace of $W_{\mathcal{C}}$ for a chosen representation. This singlet is the Wilson *loop*, which should be distinguished from the open Wilson *line* described above. The former is a number, whereas the latter is a matrix (group element). The focus of this chapter is put on Wilson lines rather than Wilson loops.

An implication of the bilocal behavior in Eq. (4.33) is that a gauge-covariant operator at $\mathcal{P}$ can be alternatively covariantized at $\mathcal{P}'$ via the Wilson line $W_{\mathcal{C}}$. For a fundamental-valued operator, one constructs

$$\text{FUND. : } W_{\mathcal{C}} \phi(\mathcal{P}) \mapsto g(\mathcal{P}') \left(W_{\mathcal{C}} \phi(\mathcal{P}) \right). \quad (4.34)$$

For adjoint-valued operators, one constructs

$$\text{ADJOINT : } W_{\mathcal{C}} \phi(\mathcal{P}) W_{\mathcal{C}}^{-1} \mapsto g(\mathcal{P}') \left(W_{\mathcal{C}} \phi(\mathcal{P}) W_{\mathcal{C}}^{-1} \right) g^{-1}(\mathcal{P}'). \quad (4.35)$$

Finally, it is instructive to note that another explicit formula is available for the Wilson line in Eq. (4.31), known as Magnus series [540]. The Magnus series computes the log of the path-ordered exponential. For $W_{\mathcal{C}} = \exp(\Omega_{\mathcal{C}})$,

it follows that

$$\begin{aligned}\Omega_{\mathcal{C}}^a = & - \int_0^1 ds_1 A^a_{\alpha}(x + y(s_1)) \dot{y}^{\alpha}(s_1) \\ & + \frac{1}{2} \int_0^1 ds_1 \int_0^{s_1} ds_2 [A_{\alpha}(x + y(s_1)), A_{\beta}(x + y(s_2))]^a \dot{y}^{\alpha}(s_1) \dot{y}^{\beta}(s_2) \\ & + \cdots ,\end{aligned}$$

where higher-order terms describe integrals involving more Lie brackets.

The insight to be gained from this alternative representation is that Eq. (4.36) manifests the fact that the Wilson line defines a group element. This is because each term in Eq. (4.36) is manifestly a Lie algebra element, constructed in terms of nested Lie brackets. Hence $\Omega_{\mathcal{C}} \in \mathfrak{g}$, which manifests that $W_{\mathcal{C}} = \exp(\Omega_{\mathcal{C}}) \in \mathbf{G}$.

It should be noted that the derivation of such nested Lie brackets counts on the fact that the commutator of adjoint actions is the adjoint action of the commutator. This is implied by the Jacobi identity of the gauge algebra $\mathfrak{g}$: $[u, [v, w]] - [v, [u, w]] = [[u, v], w]$.

Note also that the sandwiched operator in Eq. (4.35) can now be described as a sum of integrals of nested Lie brackets rather than matrix products:

$$\begin{aligned}(\text{Ad}_{W_{\mathcal{C}}} \phi(\mathcal{P}))^a &= (e^{\text{ad}_{\Omega_{\mathcal{C}}}} \phi(x))^a , \\ &= - \int_0^1 ds_1 [A_{\alpha}(x + y(s_1)), \phi(x)]^a \dot{y}^{\alpha}(s_1) + \cdots .\end{aligned}\tag{4.36}$$

This expression manifests that the structure constants f^a_{bc} are the only necessary algebraic information needed to compute the adjoint-valued sandwiched operator, crucially.

4.3.3 Covariant Taylor Expansion

In Eqs. (4.34) and (4.35), we have seen that the point of covariantization can be shifted by attaching Wilson lines. For instance, $W_{\mathcal{C}} \phi(\mathcal{P})$ in Eq. (4.34), which defines a nonlocal operator, transforms like an operator at the point $\mathcal{P}'$. In this light, one might wonder if it is possible to explicitly express $W_{\mathcal{C}} \phi(\mathcal{P})$ as an infinite sum of local operators at $\mathcal{P}'$.

This is indeed possible, assuming that $\mathcal{P}'$ is sufficiently close to $\mathcal{P}$. To understand this explicitly, let us consider a straight-line contour $s \mapsto x^{\mu} + q^{\mu} s$, where q^{μ} describes a fixed vector. The Wilson line for a part of this contour

will be denoted as

$$W(x + qs_2, x + qs_1) = \text{Pexp} \left(- \int_{s_1}^{s_2} ds A_\alpha(x + qs) q^\alpha \right). \quad (4.37)$$

For any s , it holds that

$$W(x + qs_2, x + qs) W(x + qs, x + qs_1) = W(x + qs_2, x + qs_1). \quad (4.38)$$

Also, due to the transport equation in Eq. (4.30), it follows that

$$\dot{W}(x + qs, x + qs_1) = -A_\alpha(x + qs) q^\alpha W(s, s_1). \quad (4.39)$$

Suppose a fundamental-valued field $\phi^i(x)$ in spacetime and its restriction on the straight-line contour, $\phi(x + qs)$. Direct computation shows that

$$\frac{d}{ds} \left(W(x, x + qs) \phi(x + qs) \right) = W(x, x + qs) q^\alpha D_\alpha \phi(x + qs), \quad (4.40)$$

where $D_\alpha \phi(x + qs) = \partial_\alpha \phi(x + qs) + A_\alpha(x + qs) \phi(x + qs)$. Moreover, it also follows that

$$\frac{d^\ell}{ds^\ell} \left(W(x, x + qs) \phi(x + qs) \right) = W(x, x + qs) (q \cdot D)^\ell \phi(x + qs), \quad (4.41)$$

for any positive integer ℓ . As a result, the Taylor expansion of $W(x, x + qs) \phi(x + qs)$ as a one-variable function of s implies that

$$W(x, x + q) \phi(x + q) = \sum_{\ell=0}^{\infty} \frac{1}{\ell!} (q \cdot D)^\ell \phi(x). \quad (4.42)$$

Clearly, Eq. (4.42) shows that a nonlocal operator formed by attaching a Wilson line is expressible in terms of an infinite sum of local operators.

The identity in Eq. (4.42) could be referred to as the covariant Taylor expansion. It generalizes the usual Taylor expansion in spacetime by replacing the partial derivative with the covariant derivative while matching the covariances at two different points via a Wilson line. The similar identity holds for adjoint-valued fields as well.

Another insight that can be gained from Eq. (4.42) is that the covariant derivative has facilitated the comparison between field values at different points via the Wilson line:

$$q^\alpha D_\alpha \phi(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(W(x, x + \epsilon q) \phi(x + \epsilon q) - \phi(x) \right). \quad (4.43)$$

This faithfully expresses the very geometrical idea of gauge connection and covariant derivative. In fact, we shall remark that it is possible and may be desirable to view Wilson lines (holonomies, parallel propagators) as fundamental and take Eq. (4.43) as the definition of the covariant derivative derived from the Wilson lines.

4.4 Diffeomorphism Gauge Theory via Color-Kinematics Duality

In this section, we take the diffeomorphism algebra in the sense that is implied by CK duality. The strategy here is to construct exact counterparts of the well-established constructions of ordinary gauge theory in Sec. 4.3 by mechanically implementing the CKR:

$$\mathfrak{g} = \mathfrak{su}(N) \longrightarrow \mathfrak{g} = \mathfrak{diff}(\mathbb{R}^d). \quad (4.44)$$

This systematically deduces and defines what a diffeomorphism gauge theory is. For instance, every Lie bracket of $\mathfrak{su}(N)$ will be replaced with the Lie bracket between vector fields.

To elaborate, the diffeomorphism Lie algebra in d dimensions, $\mathfrak{diff}(\mathbb{R}^d)$, is identified with the space $\Gamma(T\mathbb{R}^d)$ of vector fields on $\mathbb{R}^d$. For a scalar field $\psi \in C^\infty(\mathbb{R}^d)$ and vector fields $u, v \in \mathfrak{diff}(\mathbb{R}^d)$, we denote

$$\begin{aligned} u\psi(x) &= u^\mu(x) \partial_\mu \psi(x), \\ [u(x), v(x)]^\mu &= u^\nu(x) \partial_\nu v^\mu(x) - v^\nu(x) \partial_\nu u^\mu(x), \end{aligned} \quad (4.45)$$

which defines the fundamental and adjoint actions in parallel with Eq. (4.21). Note that the Lie bracket between vector fields, $[u(x), v(x)]^\mu$, is literally derived by taking the commutator between $u^\mu(x) \partial_\mu$ and $v^\mu(x) \partial_\mu$ as *differential operators*. The mathematical basis of this view is elaborated in Sec. 4.4.4. The diffeomorphism Lie group is denoted as $\text{Diff}(\mathbb{R}^d)$.

4.4.1 Gauge Covariance

In gravitational theories such as general relativity, there exists the freedom to choose coordinates in spacetime. This introduces the notion of *general covariance*, which describes how local fields change under coordinate transformations. It is important to understand that coordinate transformations in generally covariant theories define equivalencies, or redundancies, under a gauge orbit, on which physically meaningful quantities shall be invariant.

However, at the technical, precise mathematical level, the gauge transformations for general covariance work differently than the gauge transformations described in Sec. 4.3, i.e., those in the YM theory, for instance. Crucially, diffeomorphisms alter points in the *base* spacetime manifold themselves. Namely, they are *external* transformations, not internal.

Despite this crucial difference, however, we push the idea of having the diffeomorphism algebra as a gauge algebra in the YM sense, as a necessary implication of CK duality.

We start with the transformation behavior of fields. Based on Eq. (4.25), the CKR in Eq. (4.44) stipulates that the singlet, fundamental, and adjoint transformation behaviors are defined as

$$\begin{aligned}
\text{SINGLET :} \quad \phi(x) &\mapsto \phi(x), \\
\text{FUND. :} \quad \phi(x) &\mapsto \phi(x) + \xi^\mu(x) \partial_\mu \phi(x), \\
\text{ADJOINT :} \quad \phi^\mu(x) &\mapsto \phi^\mu(x) + \xi^\nu(x) \partial_\nu \phi^\mu(x) - \phi^\nu(x) \partial_\nu \xi^\mu(x),
\end{aligned} \tag{4.46}$$

provided the fundamental and adjoint actions in Eq. (4.45). Here, $\xi = \xi^\mu(x) \partial_\mu \in \mathbf{diff}(\mathbb{R}^d)$ is a diffeomorphism Lie algebra element serving as an infinitesimal diffeomorphism gauge transformation parameter, which is the counterpart of $\varepsilon(x) \in \mathfrak{su}(N)$ in Eq. (4.25).

Suppose a non-singlet field, such as the fundamental field in Eq. (4.46). Due to the fact that the diffeomorphism parameter $\xi^\mu(x)$ vary over spacetime, the field values at different points transform differently. As a result, their difference is not meaningful. The implication is that the partial derivative of a non-singlet field is not a gauge-covariant object. Explicitly, the partial derivative of the fundamental field, $\partial_\alpha \phi(x)$, transforms to

$$\begin{aligned}
&\partial_\alpha \left(\phi(x) + \xi^\mu(x) \partial_\mu \phi(x) \right) \\
&= \partial_\alpha \phi(x) + \xi^\mu(x) \partial_\alpha \partial_\mu \phi(x) + \partial_\alpha \xi^\mu(x) \partial_\mu \phi(x),
\end{aligned} \tag{4.47}$$

the behavior of which is neither singlet, fundamental, or adjoint according to Eq. (4.46).

In this light, we are invited to the idea of introducing a diffeomorphism gauge connection A , defining the diffeomorphism-covariant derivative D . To

this end, one mechanically applies the CKR rule. From Eq. (4.23), one finds

$$\begin{aligned}
 \text{SINGLET :} \quad D_\alpha \phi(x) &= \partial_\alpha \phi(x), \\
 \text{FUND. :} \quad D_\alpha \phi(x) &= \partial_\alpha \phi(x) + A^\mu{}_\alpha(x) \partial_\mu \phi(x), \\
 \text{ADJOINT :} \quad D_\alpha \phi^\mu(x) &= \partial_\alpha \phi^\mu(x) + A^\nu{}_\alpha(x) \partial_\nu \phi^\mu(x) - \phi^\nu(x) \partial_\nu A^\mu{}_\alpha(x),
 \end{aligned} \tag{4.48}$$

while Eq. (4.26) stipulates that the diffeomorphism gauge connection transforms as

$$A^\mu{}_\alpha(x) \mapsto A^\mu{}_\alpha(x) - D_\alpha \xi^\mu(x). \tag{4.49}$$

Recall the description given around Eq. (4.23). Again, for each diffeomorphism-covariant operator, the diffeomorphism-covariant derivative returns an operator with one more spacetime index that exhibits the *same* diffeomorphism-covariant transformation behavior. That is, the additional spacetime index is diffeomorphism-*neutral*.

Let us verify this claim in a direct, explicit fashion. For the singlet, Eq. (4.48) stipulates that its covariant derivative $D_\alpha \phi(x)$ is simply the partial derivative $\partial_\alpha \phi(x)$. By Eq. (4.46), this transforms as $\partial_\alpha \phi(x) \mapsto \partial_\alpha \phi(x)$: it is invariant. Therefore, $D_\alpha \phi(x)$ is still a singlet, where the index α is neutral under the diffeomorphism gauge transformations.

Next, consider a diffeomorphism-fundamental field $\phi(x)$, whose infinitesimal gauge transformation is $\delta_\xi \phi(x) = \xi^\mu(x) \partial_\mu \phi(x)$ according to Eq. (4.49). Using Eqs. (4.47) and (4.49), the infinitesimal variation of its covariant derivative $D_\alpha \phi(x)$ is found as the sum of the following three groups of terms:

$$\begin{aligned}
 &\left(\partial_\alpha \xi^\mu \partial_\mu \phi + \xi^\mu \partial_\alpha \partial_\mu \phi \right), \\
 &A^\mu{}_\alpha \left(\partial_\mu \xi^\nu \partial_\nu \phi + \xi^\nu \partial_\mu \partial_\nu \phi \right), \\
 &\left(-\partial_\alpha \xi^\mu - A^\nu{}_\alpha \partial_\nu \xi^\mu + \xi^\nu \partial_\nu A^\mu{}_\alpha \right) \partial_\mu \phi.
 \end{aligned} \tag{4.50}$$

Here, we have temporarily omitted the argument “ (x) ” to avoid clutter and employed a color coding to show cancellations/summations of terms efficiently. As a result, one finds

$$\delta_\xi(D_\alpha \phi) = \xi^\mu \partial_\mu (D_\alpha \phi), \tag{4.51}$$

establishing that $D_\alpha \phi(x)$ transforms exactly like a diffeomorphism-fundamental field à la Eq. (4.46).

Finally, consider a diffeomorphism-adjoint field $\phi^\mu(x)$, whose infinitesimal gauge transformation is given by $\delta_\xi \phi^\mu(x) = [\xi(x), \phi(x)]^\mu = \xi^\nu(x) \partial_\nu \phi^\mu(x) - \phi^\nu(x) \partial_\nu \xi^\mu(x)$ according to Eq. (4.49). The infinitesimal variation of its covariant derivative $D_\alpha \phi^\mu(x)$, given in Eq. (4.48), evaluates to the sum of five groups of terms:

$$\begin{aligned} & \partial_\alpha \left([\xi, \phi]^\mu \right), \quad A^\rho{}_\alpha \partial_\rho \left([\xi, \phi]^\mu \right), \quad \left(-\partial_\alpha \xi^\rho - [A_\alpha, \xi]^\rho \right) \partial_\rho \phi^\mu, \\ & - \left([\xi, \phi]^\rho \right) \partial_\rho A^\mu{}_\alpha, \quad -\phi^\rho \partial_\rho \left(-\partial_\alpha \xi^\mu - [A_\alpha, \xi]^\mu \right). \end{aligned} \quad (4.52)$$

Expanding each group, we find

$$\begin{aligned} & \left(\partial_\alpha \xi^\nu \partial_\nu \phi^\mu + \xi^\nu \partial_\alpha \partial_\nu \phi^\mu - \partial_\alpha \phi^\nu \partial_\nu \xi^\mu - \phi^\nu \partial_\alpha \partial_\nu \xi^\mu \right), \\ & A^\rho{}_\alpha \left(\partial_\rho \xi^\nu \partial_\nu \phi^\mu + \xi^\nu \partial_\rho \partial_\nu \phi^\mu - \partial_\rho \phi^\nu \partial_\nu \xi^\mu - \phi^\nu \partial_\rho \partial_\nu \xi^\mu \right), \\ & \left(-\partial_\alpha \xi^\rho - A^\nu{}_\alpha \partial_\nu \xi^\rho + \xi^\nu \partial_\nu A^\rho{}_\alpha \right) \partial_\rho \phi^\mu, \\ & - \left(\xi^\nu \partial_\nu \phi^\rho - \phi^\nu \partial_\nu \xi^\rho \right) \partial_\rho A^\mu{}_\alpha, \\ & \phi^\rho \left(\partial_\rho \partial_\alpha \xi^\mu + \partial_\rho A^\nu{}_\alpha \partial_\nu \xi^\mu + A^\nu{}_\alpha \partial_\rho \partial_\nu \xi^\mu - \partial_\rho \xi^\nu \partial_\nu A^\mu{}_\alpha - \xi^\nu \partial_\rho \partial_\nu A^\mu{}_\alpha \right). \end{aligned} \quad (4.53)$$

As a result, it follows that

$$\begin{aligned} \delta_\xi(D_\alpha \phi^\mu) &= \xi^\rho \partial_\rho \left(\partial_\alpha \phi^\mu + A^\nu{}_\alpha \partial_\nu \phi^\mu(x) - \phi^\nu \partial_\nu A^\mu{}_\alpha \right) \\ &\quad - \left(\partial_\alpha \phi^\rho + A^\rho{}_\alpha \partial_\rho \phi^\rho - \phi^\rho(x) \partial_\rho A^\mu{}_\alpha \right) \partial_\nu \xi^\mu, \end{aligned} \quad (4.54)$$

which simplifies to

$$\delta_\xi(D_\alpha \phi^\mu) = \xi^\rho \partial_\rho (D_\alpha \phi^\mu) - (D_\alpha \phi^\nu) \partial_\nu \xi^\mu = [\xi, D_\alpha \phi]^\mu. \quad (4.55)$$

Therefore, $D_\alpha \phi^\mu(x)$ transforms exactly like a diffeomorphism-adjoint field à la Eq. (4.46).

These calculations explicitly establish how the CKR rule concretely realizes the notion of a diffeomorphism-covariant connection and diffeomorphism-covariant derivative. A crucial feature to be reiterated is that the covariant derivative does not change the transformation behavior. If $\phi(x)$ is a fundamental, its covariant descendants, $D_\alpha \phi(x)$, $D_\alpha D_\beta \phi(x)$, $D_\alpha D_\beta D_\gamma \phi(x)$, $\dots$, are all fundamentals as well. If $\phi^\mu(x)$ is an adjoint, its covariant descendants, $D_\alpha \phi^\mu(x)$, $D_\alpha D_\beta \phi^\mu(x)$, $D_\alpha D_\beta D_\gamma \phi^\mu(x)$, $\dots$, are all adjoints as well.

Consequently, we discover that there naturally arise two types of space-time indices: diffeomorphism-neutral and diffeomorphism-charged. The astute reader will point out that we have been consistently using early greek

letters $\alpha, \beta, \gamma, \delta, \dots$ and late greek letters $\mu, \nu, \rho, \sigma, \dots$ for the former and the latter, respectively. A simple underlying geometric picture will be disclosed in Sec. 4.5.

The diffeomorphism field strength is defined by applying CKR on Eq. (4.27):

$$\begin{aligned} F^\mu_{\alpha\beta}(x) &= \partial_\alpha A^\mu_\beta(x) - \partial_\beta A^\mu_\alpha(x) \\ &\quad + A^\nu_\alpha(x) \partial_\nu A^\mu_\beta(x) - A^\nu_\beta(x) \partial_\nu A^\mu_\alpha(x). \end{aligned} \quad (4.56)$$

Again, this shall arise from the commutator between two diffeomorphism-covariant derivatives, as is explicitly verified below.

First of all, suppose a fundamental field $\phi(x)$. Computation shows that

$$\begin{aligned} D_\alpha D_\beta \phi &= \partial_\alpha \left(\partial_\beta \phi + A^\nu_\beta \partial_\nu \phi \right) + A^\mu_\alpha \partial_\mu \left(\partial_\beta \phi + A^\nu_\beta \partial_\nu \phi \right), \\ &= \left(\partial_\alpha \partial_\beta \phi + \partial_\alpha A^\nu_\beta \partial_\nu \phi + A^\nu_\beta \partial_\alpha \partial_\nu \phi \right) \\ &\quad + A^\mu_\alpha \left(\partial_\mu \partial_\beta \phi + \partial_\mu A^\nu_\beta \partial_\nu \phi + A^\nu_\beta \partial_\mu \partial_\nu \phi \right), \end{aligned} \quad (4.57)$$

where the sum of the terms marked in red does not survive after antisymmetrization in the indices α, β :

$$[D_\alpha, D_\beta] \phi = \left(\partial_{[\alpha} A^\nu_{\beta]} + A^\rho_{[\alpha} \partial_\rho A^\nu_{\beta]} \right) \partial_\nu \phi = F^\nu_{\alpha\beta} \partial_\nu \phi = F_{\alpha\beta} \phi. \quad (4.58)$$

Next, suppose an adjoint field $\phi^\mu(x)$. Computation shows that

$$\begin{aligned} D_\alpha D_\beta \phi^\mu &= \partial_\alpha \left(\partial_\beta \phi^\mu + [A_\beta, \phi]^\mu \right) + [A_\alpha, \partial_\beta \phi]^\mu + [A_\alpha, [A_\beta, \phi]]^\mu, \\ &= \left(\partial_\alpha \partial_\beta \phi^\mu + [\partial_\alpha A_\beta, \phi]^\mu + [A_\beta, \partial_\alpha \phi]^\mu \right) + [A_\alpha, \partial_\beta \phi]^\mu + [A_\alpha, [A_\beta, \phi]]^\mu, \end{aligned} \quad (4.59)$$

where the sum of the terms marked in red does not survive after antisymmetrization in the indices α, β :

$$\begin{aligned} [D_\alpha, D_\beta] \phi^\mu &= [\partial_{[\alpha} A_{\beta]}, \phi]^\mu + [A_\alpha, [A_\beta, \phi]]^\mu - [A_\beta, [A_\alpha, \phi]]^\mu, \\ &= [\partial_{[\alpha} A_{\beta]}, \phi]^\mu + [[A_\alpha, A_\beta], \phi]^\mu, \\ &= [F_{\alpha\beta}, \phi]^\mu = F^\nu_{\alpha\beta} \partial_\nu \phi^\mu - \phi^\nu \partial_\nu F^\mu_{\alpha\beta}. \end{aligned} \quad (4.60)$$

Importantly, we have used the Jacobi identity of the diffeomorphism Lie bracket, in the form of $[A_\alpha, [A_\beta, \phi]]^\mu - [A_\beta, [A_\alpha, \phi]]^\mu = [[A_\alpha, A_\beta], \phi]^\mu$.

These calculations show that the diffeomorphism field strength in Eq. (4.56) is covariantly constructed from the commutator between two covariant derivatives, so it serves as a diffeomorphism-covariant operator. To verify this explicitly, we apply the transformation law for the gauge connection stipulated

in Eq. (4.49), $\delta_\xi A^\mu{}_\alpha = -D_\alpha \xi^\mu$, to Eq. (4.56):

$$\begin{aligned}
\delta_\xi F^\mu{}_{\alpha\beta} &= \partial_{[\alpha}(-D_{\beta]}\xi^\mu) + [-D_\alpha \xi, A_{\beta}]^\mu + [A_\alpha, -D_\beta \xi]^\mu, \\
&= -\partial_{[\alpha}(D_{\beta]}\xi^\mu) - [A_{[\alpha}, D_{\beta]}\xi]^\mu, \\
&= -D_{[\alpha}D_{\beta]}\xi^\mu = -[F_{\alpha\beta}, \xi]^\mu = [\xi, F_{\alpha\beta}]^\mu, \\
&= \xi^\nu \partial_\nu F^\mu{}_{\alpha\beta} - F^\nu{}_{\alpha\beta} \partial_\nu \xi^\mu.
\end{aligned} \tag{4.61}$$

Eq. (4.61) shows that the diffeomorphism field strength transforms as an adjoint in the precise sense of Eq. (4.46).

The diffeomorphism field strength $F^\mu{}_{\alpha\beta}$ is a diffeomorphism adjoint. Hence its diffeomorphism covariant derivative is defined as

$$D_\gamma F^\mu{}_{\alpha\beta}(x) = \partial_\gamma F^\mu{}_{\alpha\beta}(x) + A^\rho{}_\gamma(x) \partial_\rho F^\mu{}_{\alpha\beta}(x) - F^\mu{}_{\alpha\beta}(x) \partial_\rho A^\rho{}_\gamma(x), \tag{4.62}$$

which arises by applying the CKR on Eq. (4.28).(x)

Finally, the CKR yields the diffeomorphism Bianchi identity from Eq. (4.29):

$$D_{[\gamma} F^\mu{}_{\alpha\beta]}(x) = D_\gamma F^\mu{}_{\alpha\beta}(x) + D_\alpha F^\mu{}_{\beta\gamma}(x) + D_\beta F^\mu{}_{\gamma\alpha}(x) = 0, \tag{4.63}$$

Again, this is satisfied by the Jacobi identity of diffeomorphism Lie algebra. The essential cancellation describes $[A_\gamma, [A_\alpha, A_\beta]]^\mu + [A_\alpha, [A_\beta, A_\gamma]]^\mu + [A_\beta, [A_\gamma, A_\alpha]]^\mu = 0$.

These explicit and mechanical computations should convince the reader that the idea a consistent working notion of “diffeomorphism gauge theory” is not inconceivable from the field theory perspective.

4.4.2 Diffeomorphism Wilson Line

Amusingly, the CKR allows us to construct the “diffeomorphism Wilson line.”

Take the path-ordered exponential formula in Eq. (4.31), given in the fundamental representation. The CKR maps it to

$$W_C = \text{Pexp} \left(- \int_0^1 ds A^\mu{}_\alpha(x + y(s)) \dot{y}^\alpha(s) \partial_\mu \right), \tag{4.64}$$

where we remind ourselves that ∂_μ denotes $\partial/\partial x^\mu$. Again, $s \mapsto y^\alpha(s)$ is a finite contour such that $y^\alpha(0) = 0$ and $s \in [0, 1]$. We propose that Eq. (4.64) defines the counterpart of Wilson line in diffeomorphism gauge theory. As will be further elaborated on in Sec. 4.4.4, we understand Eq. (4.64) as a *differential*

operator. That is, Eq. (4.64) states that

$$\begin{aligned}
W_{\mathcal{C}}\phi(x) &= \phi(x) - \int_0^1 ds_1 A^\mu{}_\alpha(x+y(s_1)) \dot{y}^\alpha(s_1) \partial_\mu \phi(x) \\
&+ \frac{1}{2} \int_0^1 ds_1 \int_0^{s_1} ds_2 A^\mu{}_\alpha(x+y(s_1)) \partial_\mu A^\nu{}_\beta(x+y(s_2)) \partial_\nu \phi(x) \dot{y}^\alpha(s_1) \dot{y}^\beta(s_2) \\
&+ \frac{1}{2} \int_0^1 ds_1 \int_0^{s_1} ds_2 A^\mu{}_\alpha(x+y(s_1)) A^\nu{}_\beta(x+y(s_2)) \partial_\mu \partial_\nu \phi(x) \dot{y}^\alpha(s_1) \dot{y}^\beta(s_2) \\
&+ \dots,
\end{aligned} \tag{4.65}$$

for any test function $\phi(x)$ valued in the fundamental.

From Eq. (4.65), one can only deduce that $W_{\mathcal{C}}$ is a generic differential operator, collecting arbitrarily high orders of derivatives: ∂_μ , $\partial_\mu \partial_\nu$, $\partial_\mu \partial_\nu \partial_\rho$, $\dots$. Crucially, however, a closer look reveals that it specifically describes the transformation of test functions (pullback) induced by a diffeomorphism on spacetime, establishing that it is indeed valued in the diffeomorphism group: $W_{\mathcal{C}} \in \mathbf{G} = \text{Diff}(\mathbb{R}^d)$.

Again, this will be established by exploiting the Magnus series [540]. Applying the CKR to Eq. (4.36), we find that

$$W_{\mathcal{C}} = \exp\left(\Omega_{\mathcal{C}}^\mu(x) \partial_\mu\right), \tag{4.66}$$

where

$$\begin{aligned}
\Omega_{\mathcal{C}}^\mu(x) &= - \int_0^1 ds_1 A^\mu{}_\alpha(x+y(s_1)) \dot{y}^\alpha(s_1) \\
&+ \frac{1}{2} \int_0^1 ds_1 \int_0^{s_1} ds_2 [A_\alpha(x+y(s_1)), A_\beta(x+y(s_2))]^\mu \dot{y}^\alpha(s_1) \dot{y}^\beta(s_2) + \dots.
\end{aligned} \tag{4.67}$$

Again, the derivation of this formula makes use of the Jacobi identity of the diffeomorphism Lie algebra, from which the nested bracket structure arises. To clarify, note that the bracket $[A_\alpha(x+y(s_1)), A_\beta(x+y(s_2))]^\mu$ in Eq. (4.67) is, with $\partial_\nu = \partial/\partial x^\nu$,

$$A^\nu{}_\alpha(x+y(s_1)) \partial_\nu A^\mu{}_\beta(x+y(s_2)) - A^\nu{}_\beta(x+y(s_2)) \partial_\nu A^\mu{}_\alpha(x+y(s_1)). \tag{4.68}$$

The formula in Eq. (4.67) describes that a choice of a contour $\mathcal{C}$ defines a vector field $\Omega_{\mathcal{C}}^\mu(x)$ on spacetime. The formula in Eq. (4.66) manifests that the Wilson line $W_{\mathcal{C}}$ describes the unit-time flow under this vector field $\Omega_{\mathcal{C}}^\mu(x)$ in spacetime. The crucial point here is that $W_{\mathcal{C}}$ is not just any differential operator but is the exponential of a *first-order* derivative, as Eq. (4.66) precisely shows.

In sum, the diffeomorphism Wilson line defined in Eq. (4.64) can be understood as an element of the diffeomorphism Lie group assigned for each contour $\mathcal{C}$ in spacetime: $W_{\mathcal{C}} \in \mathbf{G} = \text{Diff}(\mathbb{R}^d)$. This is because it describes the exponentiation of an element of the diffeomorphism Lie algebra as a vector field $\Omega_{\mathcal{C}} \in \mathbf{g} = \text{diff}(\mathbb{R}^d)$.

Another way to establish this fact is to study the transport equation. From the path-ordered exponential formula described in Eq. (4.64), it follows that

$$\begin{aligned} & \frac{\partial}{\partial s} \left[\text{Pexp} \left(- \int_0^s ds' A^\mu{}_\alpha(x+y(s')) \dot{y}^\alpha(s') \partial_\mu \right) \phi(x) \right] \\ &= -A^\mu{}_\alpha(x+y(s)) \partial_\mu \left[\text{Pexp} \left(- \int_0^s ds' A^\mu{}_\alpha(x+y(s')) \dot{y}^\alpha(s') \partial_\mu \right) \phi(x) \right], \end{aligned} \quad (4.69)$$

for $\phi(x)$ fundamental. This implies that the solution to the transport equation in the fundamental representation,

$$\frac{\partial}{\partial s} \Psi(x; s) + A^\mu{}_\alpha(x+y(s)) \dot{y}^\alpha(s) \partial_\mu \Psi(x; s) = 0, \quad (4.70)$$

with initial condition $\Psi(x; 0) = \phi(x)$, is given by $\Psi(x; 1) = W_{\mathcal{C}} \phi(x)$. Namely,

$$\Psi(x; 1) = W_{\mathcal{C}} \Psi(x; 0). \quad (4.71)$$

Again, here $W_{\mathcal{C}}$ acts on the function $\Psi(x; 0)$ of x^μ as a differential operator.

To establish the bilocal transformation behavior of the diffeomorphism Wilson line, one can study the transport equation in Eq. (4.70) in the fundamental. Recalling Eq. (4.46), we compute three terms:

$$\frac{\partial}{\partial s} \left(\xi^\mu(x+y(s)) \partial_\mu \Psi(x; s) \right), \quad (4.72a)$$

$$- \left(\partial_\alpha \xi^\mu(x+y(s)) + [A_\alpha(x+y(s)), \xi(x+y(s))]^\mu \right) \dot{y}^\alpha(s) \partial_\mu \Psi(x; s), \quad (4.72b)$$

$$A^\mu{}_\alpha(x+y(s)) \dot{y}^\alpha(s) \partial_\mu \left(\xi^\rho(x+y(s)) \partial_\rho \Psi(x; s) \right). \quad (4.72c)$$

The first term evaluates to

$$\partial_\alpha \xi^\mu(x+y(s)) \dot{y}^\alpha(s) \partial_\mu \Psi(x; s) + \xi^\mu(x+y(s)) \partial_\mu \frac{\partial}{\partial s} \Psi(x; s), \quad (4.73a)$$

while the sum of the second and third terms evaluates to

$$\begin{aligned} & - \partial_\alpha \xi^\mu(x+y(s)) \dot{y}^\alpha(s) \partial_\mu \Psi(x; s) \\ & + \xi^\rho(x+y(s)) \partial_\rho A^\mu{}_\alpha(x+y(s)) \dot{y}^\alpha(s) \partial_\mu \Psi(x; s) \\ & + A^\mu{}_\alpha(x+y(s)) \dot{y}^\alpha(s) \xi^\rho(x+y(s)) \partial_\rho \partial_\mu \Psi(x; s). \end{aligned} \quad (4.73b)$$

As a result, the sum of Eqs. (4.73a) and (4.73b) is

$$\xi^\rho(x+y(s))\partial_\rho\left(\partial_\mu\Psi(x;s)+A^\mu{}_\alpha(x+y(s))\dot{y}^\alpha(s)\partial_\mu\Psi(x;s)\right). \quad (4.74)$$

This shows that the transport equation in Eq. (4.70) is diffeomorphism gauge covariant at position $x+y(s)$, if $\Psi(x;s)$ is diffeomorphism gauge covariant at position $x+y(s)$. With this observation, one argues that $\Psi(x;1)=W_{\mathcal{C}}\Psi(x;0)$ in Eq. (4.71) is diffeomorphism gauge covariant at position $x+y(1)$ if $\Psi(x;0)$ is diffeomorphism gauge covariant at position x . Namely, if $\mathcal{C}$ is a contour that starts at $\mathcal{P}$ and ends at $\mathcal{P}'$, $W_{\mathcal{C}}\phi(\mathcal{P})$ transforms like a local operator at $\mathcal{P}'$ if $\phi(\mathcal{P})$ is a local operator at $\mathcal{P}$ valued in the fundamental or adjoint.

4.4.3 Covariant Taylor Expansion

Again, one can express $W_{\mathcal{C}}\phi(\mathcal{P})$ as an infinite sum of local operators at $\mathcal{P}'$ provided $\mathcal{P}'$ is sufficiently close to $\mathcal{P}$. To understand this explicitly, let us consider a straight-line contour $s \mapsto x^\mu + q^\mu s$, where q^μ describes a fixed vector. The diffeomorphism Wilson line for a part of this contour will be denoted as

$$W(x+qs_2, x+qs_1) = \text{Pexp}\left(-\int_{s_1}^{s_2} ds A^\mu{}_\alpha(x+qs) q^\alpha \partial_\mu\right). \quad (4.75)$$

For any s , the identities in Eqs. (4.38) and (4.39) hold, with the understanding that product of diffeomorphism Wilson lines describes composition of differential operators.

For a fundamental-valued field $\phi(x)$, computation shows that the same identities as in Eqs. (4.40) and (4.41) hold, but with $D_\alpha\phi(x+qs) = \partial_\alpha\phi(x+qs) + A^\mu{}_\alpha(x+qs)\partial_\mu\phi(x+qs)$. Therefore the covariant Taylor expansion in Eq. (4.42) holds for the diffeomorphism gauge connection as well, which also generalizes to adjoint-valued fields.

The insight that can be gained from this fact is that the diffeomorphism-covariant derivative has facilitated the comparison between field values at different points via the diffeomorphism Wilson line. It is a nice exercise to verify that, for a fundamental-valued scalar $\phi(x)$ and an adjoint-valued $\phi^\mu(x)$,

$$\text{FUND. : } q^\alpha D_\alpha\phi(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(W(x, x+\epsilon q) \phi(x+\epsilon q) - \phi(x) \right), \quad (4.76)$$

$$\text{ADJOINT : } q^\alpha D_\alpha\phi^\mu(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(e^{[\Omega^\rho(x, x+\epsilon q) \partial_\rho, \cdot]} \phi^\mu(x+\epsilon q) - \phi^\mu(x) \right),$$

where the second line will compute the adjoint action with the help of the Magnus representation in Eq. (4.66). Notably, this faithfully expresses the very geometrical idea of a gauge connection and a gauge-covariant derivative, in the strict gauge theory sense. The diffeomorphism Wilson line provides a law of parallel transport so that objects at different points can be compared.

4.4.4 A Mathematical Perspective

In the above exposition, we regarded vector fields as linear differential operators. In pursuit of this differential operator approach, we computed and defined W_C as a differential operator and then identified it rather directly as an element of the diffeomorphism group as $W_C \in \text{Diff}(\mathbb{R}^d)$, although a diffeomorphism on $\mathbb{R}^d$ is usually thought of as a map $\mathbb{R}^d \rightarrow \mathbb{R}^d$.

We shall remark that this identification is based on the differential operator representation of diffeomorphisms [541] by the pullback, which traces back to an algebraic perspective on geometry [541–547]. Below, we elaborate on this mathematical viewpoint briefly.

The idea is that diffeomorphisms can be thought of as maps that preserve the algebraic structure of smooth functions under pointwise addition and multiplication: automorphisms of the commutative ring $C^\infty(\mathbb{R}^d)$ of smooth functions. Namely, a differential operator $U : C^\infty(\mathbb{R}^d) \rightarrow C^\infty(\mathbb{R}^d)$ is a diffeomorphism $U \in \text{Diff}(\mathbb{R}^d)$ iff

$$U[\psi_1 + \psi_2] = U[\psi_1] + U[\psi_2], \quad U[\psi_1\psi_2] = U[\psi_1]U[\psi_2] \quad (4.77)$$

for all $\psi_1, \psi_2 \in C^\infty(\mathbb{R}^d)$. In this precise mathematical sense, the differential operator in Eq. (4.64) has defined a diffeomorphism $W_C \in \text{Diff}(\mathbb{R}^d)$.

Similarly, vector fields on $\mathbb{R}^d$, as infinitesimal diffeomorphisms, are derivations on the commutative algebra $C^\infty(\mathbb{R}^d)$: they are differential operators that preserve the pointwise addition and multiplication. That is, a differential operator $v : C^\infty(\mathbb{R}^d) \rightarrow C^\infty(\mathbb{R}^d)$ is a diffeomorphism generator $v \in \text{diff}(\mathbb{R}^d)$ iff

$$v[\psi_1 + \psi_2] = v[\psi_1] + v[\psi_2], \quad v[\psi_1\psi_2] = v[\psi_1]\psi_2 + \psi_1 v[\psi_2]. \quad (4.78)$$

This definition implies that v is a linear differential operator, $v = v^\mu \partial_\mu$. The Lie bracket between vector fields is simply the commutator between linear differential operators.

In ordinary gauge theory, the gauge connection provides the isomorphism between color spaces (finite-dimensional fibers) at two different points. Impor-

tantly, this law of parallel transport preserves the color Lie algebra structure. In diffeomorphism gauge theory, the diffeomorphism gauge connection may be thought of as providing the isomorphism between the infinite-dimensional algebra $C^\infty(\mathbb{R}^d)$ of smooth functions at two different points.

These mathematical formalizations also conform to those of noncommutative and Poisson gauge theories [542–550]. For instance, the diffeomorphism Wilson line in Eq. (4.64) can be deduced as a generalization of the Poisson Wilson line from, e.g., Refs. [551, 552].

These mathematical justifications, however, may not be conceived as providing the most clear or practically useful picture for physicists. Notably, a simple geometrical interpretation is found below.

4.5 Interpretation as Teleparallel Theory

4.5.1 Frame and Anholonomy

As we have discussed in Sec. 2.3, a *teleparallel theory* refers to a field theory of a dynamical frame field [296–307]. Strictly speaking, the name “teleparallel theory” suits theories that describe geometries of absolute parallelism in which the frame (diffeomorphism neutral) index is *global*, as is discussed above Eq. (2.81). In the case of GR, we have seen in Eq. (2.73) that the frame index is, in fact, *gauged* due to local Lorentz symmetry. Thus, the frame formulation of GR in Eqs. (2.69) or (2.77) is not a teleparallel field theory in the strict sense.

A *genuine teleparallel theory* is a field theory of a dynamical frame E^μ_α whose diffeomorphism neutral index α is global, i.e., not charged under any local symmetries. It governs a dynamical geometry of absolute parallelism.

Let us study a genuine teleparallel theory. For clarity, we use $\alpha, \beta, \gamma, \delta, \dots$ for the global diffeomorphism neutral index (since $A, B, C, D, \dots$ have been used and reserved as the local Lorentz index in our discussions on GR). With this clarification in mind, we recall the elements of frame geometry from Sec. 2.3. The frame’s degree of anholonomy, i.e., noncoordinateness, is captured by the Lie bracket $[E_\alpha, E_\beta]$, which we propose to denote as $F_{\alpha\beta}$:

$$F^\mu_{\alpha\beta} = [E_\alpha, E_\beta]^\mu = E^\rho_\alpha \partial_\rho E^\mu_\beta - E^\rho_\beta \partial_\rho E^\mu_\alpha. \quad (4.79)$$

For example, the anholonomy coefficients defined in Eq. (2.52) are $\Omega^\gamma_{\alpha\beta} = e^\gamma_\mu F^\mu_{\alpha\beta}$, if the frame E^μ_α is invertible to a coframe e^α_μ . The identity derived

in Eq. (2.50) describes $-E^\mu{}_\gamma de^\gamma = \frac{1}{2} F^\mu{}_{\alpha\beta} e^\alpha \wedge e^\beta$. Also notable is

$$[E_\gamma, F_{\alpha\beta}]^\mu = E^\rho{}_\gamma \partial_\rho F^\mu{}_{\alpha\beta} - F^\rho{}_{\alpha\beta} \partial_\rho E^\mu{}_\gamma, \quad (4.80)$$

so the Bianchi identity is

$$[E_\gamma, F_{\alpha\beta}]^\mu + [E_\alpha, F_{\beta\gamma}]^\mu + [E_\beta, F_{\gamma\alpha}]^\mu = 0. \quad (4.81)$$

Suppose the frame $E^\mu{}_\alpha$ takes the Kronecker delta $\delta^\mu{}_\alpha$ as its vacuum expectation value. The *frame perturbation* $A^\mu{}_\alpha$ is defined as

$$E^\mu{}_\alpha = \delta^\mu{}_\alpha + A^\mu{}_\alpha. \quad (4.82)$$

The background $\delta^\mu{}_\alpha$ will spontaneously break diffeomorphisms. Let us learn about how the “background independent” formulae of teleparallel geometry boils down when expanded around this vacuum expectation value. By plugging in Eq. (4.82) to Eq. (4.79), we learn that the anholonomy boils down to

$$F^\mu{}_{\alpha\beta} = \partial_\alpha A^\mu{}_\beta - \partial_\beta A^\mu{}_\alpha + A^\rho{}_\alpha \partial_\rho A^\mu{}_\beta - A^\rho{}_\beta \partial_\rho A^\mu{}_\alpha, \quad (4.83)$$

which is nothing but the diffeomorphism field strength in Eq. (4.56).

The vacuum expectation value of the anholonomy $F^\mu{}_{\alpha\beta}$ is zero. With this in mind, the expansion of Eq. (4.80) around the Kronecker vacuum is

$$[E_\gamma, F_{\alpha\beta}]^\mu = \partial_\gamma F^\mu{}_{\alpha\beta} + A^\rho{}_\gamma \partial_\rho F^\mu{}_{\alpha\beta} - F^\rho{}_{\alpha\beta} \partial_\rho A^\mu{}_\gamma = D_\gamma F^\mu{}_{\alpha\beta}, \quad (4.84)$$

which is nothing but the diffeomorphism covariant derivative of the diffeomorphism field strength in Eq. (4.28). Eventually, this leads to the realization that the diffeomorphism covariant derivative is nothing but the Lie derivative with respect to the frame vector field, on various tensor fields:

$$D_\alpha = \mathcal{L}_{E_\alpha}. \quad (4.85)$$

In particular, Eq. (4.46) is reproduced by literally performing a spacetime diffeomorphism while taking $\phi(x)$ and $\phi^\mu(x)$ as scalar and vector fields, while Eq. (4.49) arises by $E_\alpha \mapsto E_\alpha + \mathcal{L}_\xi E_\alpha$ realized around the Kronecker vacuum.

In sum, we realize that teleparallel theory provides an alternative definition of diffeomorphism gauge theory that is equivalent to the definition in Sec. 4.4 by the CKR, provided the crucial identification that⁵

The frame perturbation around the Kronecker vacuum is
the diffeomorphism gauge connection.

⁵To our best understanding, this realization was first made in Ref. [391], App.D. Later, it was also realized in Ref. [534]. A historical precursor will be Mason and Newman [309].

The diffeomorphism covariant derivative is the Lie derivative by the frame.

Lastly, consider a genuine teleparallel theory provided with a constant metric $\eta^{\alpha\beta}$ as an invariant global structure. Suppose its classical EoM are given by

$$\eta^{\gamma\alpha}[E_\gamma, F_{\alpha\beta}]^\mu = \eta^{\gamma\alpha}[E_\gamma, [E_\alpha, E_\beta]]^\mu = 0. \quad (4.86)$$

Evidently, Eq. (4.86) is a background-independent equation. Around the Kronecker vacuum, plugging in Eq. (4.82) to Eq. (4.86) gives

$$\eta^{\gamma\alpha} \left(\partial_\gamma F^\mu_{\alpha\beta} + A^\rho_\gamma \partial_\rho F^\mu_{\alpha\beta} - F^\rho_{\alpha\beta} \partial_\rho A^\mu_\gamma \right) = 0, \quad (4.87)$$

while the vacuum expectation value of the anholonomy $F^\mu_{\alpha\beta}$ is zero. Eq. (4.87) is nothing but the diffeomorphism YM equations of Cheung and Mangan [90], reproduced in Eq. (4.20c).

In conclusion, we realize that Cheung and Mangan [90]’s formulation of BI theory, reproduced in Eq. (4.20), admits a remarkable geometric interpretation. The diffeomorphism gauge connection introduced in Eq. (4.20a) is the vielbein perturbation around the Kronecker vacuum, $A^\mu_\alpha = E^\mu_\alpha - \delta^\mu_\alpha$. The diffeomorphism field strength in Eq. (4.20b) admits a background-independent rewriting as the anholonomy $F^\mu_{\alpha\beta} = [E_\alpha, E_\beta]^\mu$. The diffeomorphism YM equations in Eq. (4.20c) admit a background-independent presentation as $\eta^{\gamma\alpha}[E_\gamma, F_{\alpha\beta}] = 0$. The diffeomorphism covariant derivative means the Lie derivative: $D_\alpha = \mathcal{L}_{E_\alpha}$.

A few remarks are in order. First and foremost, not all diffeomorphism-invariant theories of the dynamical vielbein E would turn out to be obtainable via applying a CKR on ordinary gauge theories. That is, a teleparallel theory is not guaranteed to be a diffeomorphism gauge theory of the strict sense. This point is best demonstrated by examples; we do so in Secs. 4.5.3 to 4.5.6. While the equations that we have explored just now are certainly obtainable from the CKR, there are also objects in teleparallel geometry that do not straightforwardly admit their counterparts in ordinary gauge theory. For example, take the anholonomy *coefficients* $\Omega^\alpha_{\beta\gamma}$, which involve the inverse $e^\alpha_\mu = \delta^\alpha_\mu - \delta^\alpha_\nu A^\nu_\beta \delta^\beta_\mu + \mathcal{O}(A^2)$. For another example, take the determinant $|E| = 1 + \delta^\alpha_\mu A^\mu_\alpha + \mathcal{O}(A^2)$ in general dimensions. The appearance of bare connection A may cause problems when these objects are used in the Lagrangian, for instance.

Secondly, it is known that teleparallel geometry could admit an analogy with lattice systems at the level of mathematical frameworks [460, 516–520],

such as crystals with impurities. In this case one can import intuitions and ideas such as Burgers vector and dislocation defects, as will be briefly touched upon in Sec. 4.6. In this context a diffeomorphism-invariant dressing may be possible, in connection with the diffeomorphism Wilson line in Sec. 4.4.2. The rough idea is to flow along the lattice frame to find an anchor on the infinities. While typical scalar fields are diffeomorphism fundamental, a dressed field will be diffeomorphism singlet, exhibiting the invariant behavior envisioned in Eq. (4.46).

Thirdly, it should be noted that our diffeomorphism gauge theory program for the double copy seeks to represent an approach different from previous constructions such as Refs. [50, 553]’s bundle construction or double field theory [554–560] (cf. generalized geometry [561–564]). In particular, our approach does not employ any auxiliary or extra space other than the spacetime manifold $\mathcal{M}$. In this picture, there is no such thing as two different species of coordinates x^μ , $\tilde{x}^{\tilde{\mu}}$. Ref. [90] writes down “ $A^{\tilde{\mu}}_\mu$ ” as a notation, but both indices there describe the same spacetime kind of indices.

4.5.2 Cofundamental, Coadjoint, and Densities

The realization in Eq. (4.85) facilitates a possible top-down definition of cofundamental, coadjoint, or densities in diffeomorphism gauge theory.⁶ There might be other versions from representation theory points of view, but the identifications below seem to provide one consistent version.

A cofundamental field $\psi(x)$ is a scalar density. It is a field that exhibits the infinitesimal transformation behavior

$$\text{CoFUND. : } \psi(x) \mapsto \psi(x) + \partial_\mu(\xi^\mu(x)\psi(x)), \quad (4.88)$$

which can also be stated as

$$\psi(x)d^d x \mapsto \psi(x)d^d x + \mathcal{L}_\xi(\psi(x)d^d x). \quad (4.89)$$

A coadjoint field $\psi_\mu(x)$ is a one-form density. It is a field that exhibits the infinitesimal transformation behavior

$$\text{CoADJOINT : } \psi_\mu(x) \mapsto \psi_\mu(x) + \partial_\rho(\xi^\rho(x)\psi_\mu(x)) + \psi_\rho(x)\partial_\mu\xi^\rho(x). \quad (4.90)$$

⁶It also facilitates writing down noninfinitesimal diffeomorphism gauge transformations exactly, which we leave as an exercise.

In Eqs. (4.25) and (4.46), we see that the infinitesimal transformation $\delta_\xi \phi^\mu = [\xi, \phi]^\mu = (\text{ad}_\xi \phi)^\mu$ corresponds to $\delta_\varepsilon \phi^a = [\varepsilon, \phi]^a = (\text{ad}_\varepsilon \phi)^a$ in ordinary gauge theory, which identifies the adjoint action ad_ξ for the diffeomorphism algebra. Similarly, comparing Eq. (4.90) with the infinitesimal transformation $\delta_\varepsilon \psi_a = (\text{ad}_\varepsilon^* \psi)_a$ of ordinary gauge theory identifies the coadjoint action for the diffeomorphism algebra as

$$(\text{ad}_\xi^* \psi)_\mu = \partial_\rho (\xi^\rho \psi_\mu) + \psi_\rho \partial_\mu \xi^\rho. \quad (4.91)$$

Suppose $\phi(x)$ is fundamental and $\psi(x)$ is cofundamental. A diffeomorphism-invariant pairing exists between the two:

$$\int d^d x \, \psi(x) \phi(x) \mapsto \int d^d x \, \psi(x) \phi(x). \quad (4.92)$$

Similarly, suppose $\phi^\mu(x)$ is adjoint and $\psi_\mu(x)$ is coadjoint. A diffeomorphism-invariant pairing exists between the two:

$$\int d^d x \, \psi_\mu(x) \phi^\mu(x) \mapsto \int d^d x \, \psi_\mu(x) \phi^\mu(x). \quad (4.93)$$

To show this, consider first how $\psi_\mu(x) \phi^\mu(x)$ transforms. Eqs. (4.46) and (4.90) imply that it changes by

$$\psi_\mu \left(\xi^\rho \partial_\rho \phi^\mu - \phi^\rho \partial_\rho \xi^\mu \right) + \left(\xi^\rho \partial_\rho \psi_\mu + \partial_\rho \xi^\rho \psi_\mu + \psi_\rho \partial_\mu \xi^\rho \right) \phi^\mu, \quad (4.94)$$

which simplifies to $\partial_\rho (\xi^\rho \psi_\mu \phi^\mu)$. This is a total derivative.

It should be clear that one can define various tensor or tensor density fields in this way, with any number of indices.

It will be interesting to reassess these density representations in the axiomatic point of view demonstrated in Sec. 4.4. Also, note that the distinction between fundamental and cofundamental vanishes when the diffeomorphisms are restricted to be volume-preserving.

4.5.3 Case Study I: Diffeomorphism *BF* theory

Consider the action

$$S[E, B] = \int B^{\alpha\beta}{}_\mu F^\mu{}_{\alpha\beta}[E] d^d x. \quad (4.95)$$

Here, $F^\mu{}_{\alpha\beta}[E]$ denotes the anholonomy $[E_\alpha, E_\beta]^\mu$ as a functional of $E^\mu{}_\alpha$.

Eq. (4.95) is invariant under the infinitesimal local transformation

$$\begin{aligned} E^\mu{}_\alpha &\mapsto E^\mu{}_\alpha + \xi^\rho \partial_\rho E^\mu{}_\alpha - E^\rho{}_\alpha \partial_\rho \xi^\mu, \\ B^{\alpha\beta}{}_\mu &\mapsto B^{\alpha\beta}{}_\mu + \partial_\rho (\xi^\rho B^{\alpha\beta}{}_\mu) + B^{\alpha\beta}{}_\rho \partial_\mu \xi^\rho, \end{aligned} \quad (4.96)$$

whose interpretation is an infinitesimal diffeomorphism. We see that the E -field is a diffeomorphism adjoint while the B -field is a diffeomorphism coadjoint, while carrying diffeomorphism-neutral indices.

Note that we have already encountered Eq. (4.96) previously in Eq. (2.70), in fact. Eq. (4.95) is exactly the topological limit of GR proposed in Sec. 2.3.1.

There is no local redundancy acting on the frame index α of $E^\mu{}_\alpha$, so we identify Eq. (4.95) as defining a genuine teleparallel field theory.

We stipulate that the frame $E^\mu{}_\alpha$ gains the Kronecker vacuum expectation value, $\delta^\mu{}_\alpha$. By plugging in Eq. (4.82), the action in Eq. (4.95) boils down to

$$S[A, B] = \int B^{\alpha\beta}{}_\mu F^\mu{}_{\alpha\beta}[A] d^d x, \quad (4.97)$$

where $F^\mu{}_{\alpha\beta}[A]$ denotes the diffeomorphism field strength as a functional of $E^\mu{}_\alpha$. It follows that Eq. (4.97) is obtainable by applying a CKR on an ordinary gauge theory,

$$S[A, B] = \int B^{\alpha\beta}{}_a F^a{}_{\alpha\beta}[A] d^d x, \quad (4.98)$$

which is BF theory. In other words, Eq. (4.98) is the image of Eq. (4.97) under a kinematics-to-color replacement (KCR).

Here, we assumed that the KCR maps a diffeomorphism coadjoint field (with the definition in Eq. (4.90)) in diffeomorphism gauge theory to a coadjoint field in ordinary gauge theory. The rationale is the invariance of the pairing in Eq. (4.93).⁷

In sum, Eq. (4.95) admits a KCR into the ordinary BF theory after acquiring the Kronecker vacuum. In the precise sense of CK duality, the theory which Eq. (4.97) defines can be called diffeomorphism BF theory:

$$\text{KCR} : \quad \mathbf{BF}(\text{diff}(\mathcal{M})) \mapsto \mathbf{BF}(\mathbf{g}). \quad (4.99)$$

The KCR is readily viable at the EoM level. The EoM of Eq. (4.97) impose flatness on the diffeomorphism gauge connection,

$$F^\mu{}_{\alpha\beta}[A] = 0, \quad (4.100a)$$

⁷If considered necessary, the volume-preserving condition can be adopted to take $d^d x$ invariant.

as well as a diffeomorphism-covariant equation on the B -field,

$$0 = \partial_\alpha B^{\alpha\beta}{}_\mu + \partial_\rho (A^\rho{}_\alpha B^{\alpha\beta}{}_\mu) + B^{\alpha\beta}{}_\mu \partial_\rho A^\mu{}_\alpha = D_\alpha B^{\alpha\beta}{}_\mu. \quad (4.100b)$$

Here, we have used Eq. (4.91) to define the diffeomorphism covariant derivative for coadjoint fields; one may repeat the exercise in Sec. 4.4 for justification. With this understanding, Eq. (4.100) admits a KCR image (“single copy”) as

$$F^a{}_{\alpha\beta}[A] = 0, \quad D_\alpha B^{\alpha\beta}{}_a = 0. \quad (4.101)$$

Certainly, Eq. (4.101) gives the EoM of the ordinary BF theory in Eq. (4.98).

4.5.4 Case Study II: Diffeomorphism Chern-Simons Theory

Another concrete instance of a diffeomorphism gauge theory is CS theory with its color algebra replaced with the volume-preserving diffeomorphism algebra in three dimensions, $\mathbf{sdiff}(\mathcal{M}_3)$:

$$\text{CKR} : \quad \mathbf{CS}(\mathfrak{g}) \quad \mapsto \quad \mathbf{CS}(\mathbf{sdiff}(\mathcal{M}_3)). \quad (4.102)$$

The double copy of CS theory was investigated in Ref. [533] with a construction of a Lagrangian. A notable clarification has been given by a recent work [534] at the Lagrangian level, showing that the double copy of CS theory is on-shell equivalent to three-dimensional Einstein gravity:

$$\mathbf{CS}(\mathbf{sdiff}(\mathcal{M}_3)) \cong \mathbf{GR}_3, \quad (4.103)$$

Here, we would like to briefly illuminate a key aspect of Ref. [534]’s construction from our perspective, without reviewing every detail.

Ordinary CS theory is defined by the action that integrates the CS three-form on a three-manifold $\mathcal{M}_3$, written in Eq. (2.116). Crucially, the CS three-form requires the Killing form δ_{ab} on the gauge algebra $\mathfrak{g}$.

In Sec. 4.5.3, we have seen that diffeomorphism BF theory admits a Lagrangian formulation, facilitating the double copy at both EoM and Lagrangian levels. As is well-known, BF and CS Lagrangians are closely related; the former provides the “cotangent” description of the latter. Crucially, however, the CS Lagrangian requires the Killing form δ_{ab} unlike the BF Lagrangian. This raises an immediate problem for the Lagrangian-level double copy: one has to find a diffeomorphism-invariant Killing form.

Interestingly, Ref. [534] shows that a Killing form exists for the volume-preserving diffeomorphism algebra in three dimensions, which is yet nonlocal.

The authors identify an expression that can be compared to the CS three-form with the aid of a nonlocal operator. They then derive the Euler-Lagrangian equations from their proposed $\mathbf{CS}(\text{sdiff}(\mathcal{M}_3))$ action and finds $F^\mu_{\alpha\beta}[A] = 0$, the flatness condition on the diffeomorphism gauge connection. By KCR, this corresponds to $F^a_{\alpha\beta}[A] = 0$, the EoM of CS theory.

4.5.5 Case Study III: Diffeomorphism Yang-Mills Theory

Diffeomorphism YM theory can be defined as a classical teleparallel field theory around the Kronecker vacuum whose EoM are $\eta^{\gamma\alpha}[E_\gamma, [E_\alpha, E_\beta]] = 0$, as discussed several times throughout our discussion. Note that this theory requires the flat frame-indexed metric $\eta^{\gamma\alpha}$ as a background structure.

It seems unclear whether diffeomorphism YM theory could admit a Lagrangian formulation. It will be interesting to see if Ref. [534]’s Killing form can be utilized. As a related subject, it could also be interesting to think about the Pontryagin theory in four dimensions, $(F \wedge F)$.

Ref. [90] established that the volume-preserving diffeomorphism YM equations provide a reformulation of BI theory. Here, we realized that they can be written in terms of the E -variable as $\eta^{\gamma\alpha}[E_\gamma, [E_\alpha, E_\beta]]^\mu = 0$ with $\partial_\mu E^\mu_\alpha = 0$, yet with the crucial clarification that the Kronecker background δ^μ_α has to be imposed. Hence it can be said that BI theory, at the classical EoM level, admits a formulation as a teleparallel theory.

A physical lesson that can be elicited from this geometric rewriting is that expanding around any background of the form $\bar{E}^\mu_\alpha(x) = X^\mu_{,\nu}(x)\delta^\nu_\alpha$ will also yield the same BI scattering amplitudes, if $x^\mu \mapsto X^\mu(x)$ gives a well-behaved diffeomorphism that is trivialized at the infinities at a sufficiently fast speed. This certainly changes many formulae (e.g., the diffeomorphism field strength) and the EoM because the background $\bar{E}^\mu_\alpha(x)$ is position-dependent. Verifying if the same tree amplitudes are obtained as in the constant Kronecker vacuum provides a nontrivial check and test of the teleparallel formulation of BI theory.

4.5.6 Case Study IV: Frame formulation of General Relativity

Diffeomorphism YM theories are genuine teleparallel theories, and there is no local symmetry on the frame index. In contrast, the so-called teleparallel equivalent of GR [304–307] is not a genuine teleparallel theory due to the local Lorentz symmetry. This point applies to both action and EoM level formulations; recall the discussion on the Mason-Newman [309] idea in Eq. (2.60).

An interesting perspective arises by the first-order action in Eq. (2.69), which understands GR as a perturbation away from diffeomorphism BF theory (suppose a rescaling $B \mapsto \kappa^2 B$). Local Lorentz symmetry is not enjoyed in the topological limit but emerges in the full theory.

Can the frame formulation of GR still be viewed as a diffeomorphism gauge theory, which is not necessarily YM? Unfortunately, neither of the actions in Eqs. (2.69) and (2.77) seems to allow for a KCR. The anholonomy coefficients Ω^A_{BC} involves the coframe e^A_μ inside which bare connection coefficients A^μ_A are inserted. The dynamical combination $\mathcal{G}^{\mu\nu}_{ABCD}[E]$ in Eq. (2.69) also suffers from this bare connection problem around the Kronecker vacuum. Hence, in general dimensions, formulating GR as a genuine diffeomorphism gauge theory still seems to be an open question.

4.5.7 General Relativity as a Diffeomorphism Gauge Theory

From the above case study, a lesson emerges that there exist different classes of “teleparallel theories.”

- *Teleparallel theory*: A field theory of a dynamical frame.
- *Genuine teleparallel theory*: A field theory of a dynamical frame whose diffeomorphism-neutral index is not subject to any local symmetry. Governs geometries of absolute parallelism.
- *Nominal teleparallel theory*: A teleparallel theory that is not genuine.
- *Diffeomorphism gauge theory*: A diffeomorphism-invariant teleparallel theory that is mapped to an ordinary gauge theory by a choice of a KCR rule when expanded around a choice of a vacuum.

According to this taxonomy, the theories in Secs. 4.5.3 to 4.5.5 describe diffeomorphism gauge theories at the EoM or Lagrangian levels, while the teleparallel equivalent of GR is a nominal teleparallel theory that seems to be also not diffeomorphism gauge theory in general dimensions. It shall be remarked that, although explorations on teleparallel gravity may generally turn out to be fruitful for envisioning a manifest double copy formulation of GR or understanding CK duality, emphasis must be put on the viability of KCR as a crucial criterion, eventually.

A few instances have been reported in three and four dimensions, however, such that GR can be exactly reformulated as a diffeomorphism gauge theory in the precise sense (classical level, on-shell equivalence).

- As noted earlier in Sec.4.5.4, three-dimensional GR is shown to be on-shell equivalent to volume-preserving Chern-Simons theory: $\mathbf{GR}_3 \cong \mathbf{CS}(\text{sdiff}(\mathcal{M}_3))$. This refers to the result of Ref. [534], which works in terms of Lagrangians.
- In four dimensions, the SD sector of GR is known to admit a formulation as a volume-preserving SDYM theory at the EoM level: $\mathbf{SDGR} \cong \mathbf{SDYM}(\text{sdiff}(\mathcal{M}_4))$. This refers to the result of Mason and Newman [309], reviewed in Sec. 2.4.3.⁸ In this case the diffeomorphism gauge connection is the perturbation of a *rescaled* tetrad V around the Kronecker vacuum.

The three-dimensional GR construction of Ref. [534] depends on several crucial features of three dimensions: first, the nonlocal Killing form construction, second, an interesting identity that rewrites the quadratic part of the determinant $|E|$ via a nonlocal operator, third, the cubic nature of $|E|$ around the Kronecker background, and fourth, the spin connection can be killed in the dreibein-Palatini formalism giving rise to a genuinely teleparallel description, owing to the celebrated fact that Einstein gravity describes flat Riemannian geometry on shell [565, 566]. It will be interesting to investigate if these structures could be properly generalized in $d > 3$ and especially in $d = 4$.⁹

Finally, it is interesting to note that the LL formulation of gravity, reviewed in Sec. 2.1.2, points to suggestive formulae in the context of the current discussion.¹⁰ Suppose the de Donder gauge condition in Eq. (2.17), $\partial_\mu \mathbf{g}^{\mu\alpha} = 0$, and the expansion $\mathbf{g}^{\mu\alpha} = \eta^{\mu\alpha} + h^{\mu\alpha}$ around the flat configuration. A divergence of the H -tensor in Eq. (2.13) gives

$$\begin{aligned} \partial_\mu H^{\mu\nu\alpha\beta} &= \mathbf{g}^{\mu\alpha} \partial_\mu \mathbf{g}^{\nu\beta} - \mathbf{g}^{\mu\beta} \partial_\mu \mathbf{g}^{\nu\alpha}, \\ &= \partial^\alpha h^{\nu\beta} - \partial^\beta h^{\nu\alpha} + h^{\mu\alpha} \partial_\mu h^{\nu\beta} - h^{\mu\beta} \partial_\mu h^{\nu\alpha}, \end{aligned} \quad (4.104)$$

where $\partial^\alpha = \eta^{\mu\alpha} \partial_\mu$. It is tempting to denote this as $\partial_\mu H^{\mu\nu\alpha\beta} = F^{\nu\alpha\beta}$: the anholonomy of the gothic metric as a frame $\mathbf{g}^\alpha = \mathbf{g}^{\mu\alpha} \partial_\mu$. Then the diffeomorphism gauge-covariant derivative, dressing the index ν , is

$$\begin{aligned} D_\alpha F^{\nu\alpha\beta} &= \partial_\alpha F^{\nu\alpha\beta} + h^\rho{}_\alpha \partial_\rho F^{\nu\alpha\beta} - F^{\rho\alpha\beta} \partial_\rho h^\nu{}_\alpha, \\ &= \tau_{\text{LL}}^{\nu\beta} + h^\rho{}_\alpha \partial_\rho F^{\nu\alpha\beta} - F^{\rho\alpha\beta} \partial_\rho h^\nu{}_\alpha, \end{aligned} \quad (4.105)$$

⁸As is reviewed in Sec. 2.4.2, the Plebański second heavenly equation is also a notable formulation of SD gravity, in which case the kinematic algebra is a Poisson algebra. It seems that the Poisson gauge theory (cf. Sec. 4.7) interpretation is yet subtle; see App. A.2.

⁹The determinant theory $|E|$ yields a quartic Lagrangian. Coincidentally or not, the YM and Pontryagin Lagrangians are also quartic.

¹⁰This observation is based on discussions with Clifford Cheung.

where the second line imposes the EoM in Eq. (2.16), and $h^\rho{}_\alpha = h^{\rho\sigma}\eta_{\sigma\alpha}$. Again, Eq. (4.105) fails to be fully organized in the form of YM equations, but it is suggestive that an interpretation of the gothic metric geometry may be possible in terms of anholonomy or teleparallel structures. In fact, on account of the index symmetry $\mathbf{g}^{\mu\nu} = \mathbf{g}^{\nu\mu}$, it would be more relevant to apply the double copy grammar of BAS theory rather than the format of YM theory, where one envisions a repeated insertion of the same kinematic algebra.

4.6 Misner String as a Diffeomorphism Vortex

So far, we have been studying diffeomorphism gauge theories as classical field theories, mainly at the level of EoM. To be comprehensive, we wish to describe *classical solutions* of diffeomorphism gauge theories to some explicit degree. Generally speaking, different classical solutions will arise as each theory exhibits unique dynamics. For instance, the on-shell configurations of the diffeomorphism BF theory in Sec. 4.5.3 are *holonomic* frames, i.e., coordinate vector fields. In contrast, the on-shell configurations of the diffeomorphism YM theory in Sec. 4.5.5 are not necessarily holonomic: the anholonomy dynamically evolves as the field strength. There are propagating degrees of freedom, and the dynamics is far more complex.

The subject of finding and studying classical solutions of diffeomorphism gauge theories seems to be an interesting program. It will be exciting if illuminations can be drawn on the classical double copy [160, 181–185] program, either in general dimensions, in three-dimensions, or for SD gravity. The classical double copy program investigates correspondences and maps between classical solutions of theories in the double-copy web in Fig. 4.2. In particular, the KS classical double copy [160] has been mapping stationary solutions in GR to *Maxwell* theory solutions, which are then trivially embedded to YM theory via a constant color factor. Naively, one could think of pursuing the same to embed Maxwell theory solutions into a diffeomorphism YM theory, thinking of a “constant” diffeomorphism generator. However, this can be subtle because the diffeomorphism algebra involves derivatives. It will be interesting if this nontriviality can lead to insightful illuminations on the nature of classical double copy.

In this work, we do not intend to carry out a comprehensive investigation into classical solutions. Yet, we provide a simple yet concrete example that defines a classical solution in *any* diffeomorphism gauge theories: the Misner

string [474–478], reinterpreted in the teleparallel viewpoint.

As we have remarked in Sec. 3.5, the Misner string has been known as the gravitational analog of the Dirac string, in which case the NUT charge is seen as the gravitomagnetic charge [331, 479]. Notably, the teleparallel geometry interpretation we have established in this chapter facilitates embedding the Misner string to diffeomorphism gauge theories. As a result, we view the Misner string as a *diffeomorphism vortex*, which exists in any diffeomorphism gauge theories as a topological configuration. Surely, the adequate mathematical framework is *multivalued* diffeomorphism gauge transformation, which we have established in detail in Sec. 3.4.

4.6.1 Physicists' Approach

Let us begin by reviewing the Misner string in GR. For simplicity, we assume $d = 4$ spacetime dimensions for the moment.

Metric Description The metric of a typical straight Misner string reads [474–478]

$$ds^2 = -\left(dt + 4G_N n \frac{x dy - y dx}{x^2 + y^2}\right)^2 + dx^2 + dy^2 + dz^2, \quad (4.106)$$

where n is the total NUT flux. Eq. (4.106) describes the image of flat spacetime under a *multivalued diffeomorphism*,

$$t \mapsto t + 4G_N n \phi_S. \quad (4.107)$$

Here, $\phi_S(x, y)$ is the azimuthal angle: a *multivalued function* of x and y with periodicity 2π , such that $\tan \phi_S(x, y) = y/x$. Eq. (4.106) describes an infinite Misner string inserted along the z -axis. By traveling around it, one can meet oneself $8\pi G_N n$ seconds ago, $16\pi G_N n$ seconds ago, $24\pi G_N n$ seconds ago, and so on: closed timelike curves. This is the signature of *multivalued time*:

$$\Delta t = \oint \left(dt + 4G_N n \frac{x dy - y dx}{x^2 + y^2} \right) = 8\pi G_N n. \quad (4.108)$$

In the spacetime view, this Misner string is inserted along the t - z plane, which we denote as $\mathcal{S}$.

The Misner string is a thin tube of gravitomagnetic flux, or equivalently a thin solenoid of mass current according to Bonnor and Sackfield [475, 476]. We

have explained the key facts about the Misner string in Sec. 3.4, and relevant references can be found there. Eq. (4.108) is how the flux n of the Misner string is measured. Physically, this should describe an interferometry experiment that serves as the gravitational AB effect.

Distributional Torsion Under a coordinate transformation from x'^μ to x^μ , the Christoffel symbols transform as

$$\Gamma^\mu_{\nu\rho} = \frac{\partial x^\mu}{\partial x'^\kappa} \Gamma'^\kappa_{\lambda\zeta} \frac{\partial x'^\lambda}{\partial x^\nu} \frac{\partial x'^\zeta}{\partial x^\rho} + \frac{\partial x^\mu}{\partial x'^\lambda} \frac{\partial}{\partial x^\rho} \left(\frac{\partial x^\lambda}{\partial x^\nu} \right). \quad (4.109)$$

Note that the ordering of the two derivatives in the inhomogeneous term is uniquely determined by demanding that $\nabla_\rho V^\mu = \partial_\rho V^\mu + \Gamma^\mu_{\nu\rho} V^\nu$ transforms tensorially with the aid of a test vector field V^μ , for instance. The torsion $T^\mu_{\nu\rho} = \Gamma^\mu_{\rho\nu} - \Gamma^\mu_{\nu\rho}$ transforms as

$$T^\mu_{\nu\rho} = \frac{\partial x^\mu}{\partial x'^\kappa} T'^\kappa_{\lambda\sigma} \frac{\partial x'^\lambda}{\partial x^\nu} \frac{\partial x'^\sigma}{\partial x^\rho} + \frac{\partial x^\mu}{\partial x'^\lambda} \left[\frac{\partial}{\partial x^\nu}, \frac{\partial}{\partial x^\rho} \right] x'^\lambda. \quad (4.110)$$

The inhomogeneous part of Eq. (4.110) vanishes for single-valued coordinate transformations. However, for our case, the multivalued coordinate transformation $t' = t + 4G_N n \phi_S(x, y)$ gives

$$T^t_{xy} = 4G_N \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right] \phi_S(x, y) = 8\pi G_N n \delta(x) \delta(y). \quad (4.111)$$

It can be seen that this gives the only nonvanishing components of the torsion tensor. We conclude that the Misner string spacetime in Eq. (4.106) describes a distributional torsion supported along the surface $\mathcal{S}$. This torsion flux tube is a diffeomorphism vortex, to say, inducing the time monodromy in Eq. (4.108).

Similarly, one can also examine if Riemannian curvature gets generated by a multivalued coordinate transformation:

$$R^\mu_{\nu\rho\sigma} = \frac{\partial x^\mu}{\partial x'^\kappa} R'^\kappa_{\lambda\zeta\xi} \frac{\partial x'^\lambda}{\partial x^\nu} \frac{\partial x'^\zeta}{\partial x^\rho} \frac{\partial x'^\xi}{\partial x^\sigma} + \frac{\partial x^\mu}{\partial x'^\lambda} \left[\frac{\partial}{\partial x^\rho}, \frac{\partial}{\partial x^\sigma} \right] \frac{\partial x'^\lambda}{\partial x^\nu}. \quad (4.112)$$

For our case, the Jacobian is single-valued, so the Misner string does not involve a Riemannian curvature even in the distributional sense.

See also Figs. 3.10 and 3.11. An insightful and inspiring reference relevant to the current discussion, especially around Eqs. (4.110) and (4.112), is Ref. [460].

Coframe Description We can now switch to the tetrad formalism. The coframe for the line element in Eq. (4.106) is

$$e^0 = dt + 4G_N n \frac{x dy - y dx}{x^2 + y^2}, \quad e^1 = dx, \quad e^2 = dy, \quad e^3 = dz, \quad (4.113)$$

which is certainly the image of the Kronecker coframe under the multivalued coordinate transformation $t' = t + 4G_N n \phi_S(x, y)$.

By formally treating Eq. (4.113) as a one-form field on the space $\mathbb{R}^4$ of (t, x, y, z) , a physicist computes its curl as

$$de^0 = 4G_N n dd\phi_S = 8\pi G_N n \delta(x)\delta(y) dx \wedge dy. \quad (4.114)$$

Interestingly, a suggestive parallel arises between Eqs. (3.32) and (4.113) as well as Eqs. (3.33) and (4.114). The Misner string is indeed the gravitational analog of the Dirac string. Eq. (4.114), which captures the torsion computed in Eq. (4.111), will be viewed as describing a “gravitomagnetic field” or “gravitational field strength” in the same sense as F in Eq. (3.33) describes a magnetic field or field strength.

Frame Description To be more precise, we have learned in Eq. (4.5) the anholonomy $F_{\alpha\beta} = [E_\alpha, E_\beta]$ in teleparallel gravity represents a field strength of diffeomorphisms in a certain accurate sense. Thus, we consider the frame field that is dual to the coframe field in Eq. (4.113):

$$\begin{aligned} E_0 &= \frac{\partial}{\partial t}, & E_1 &= \frac{\partial}{\partial x} + 4G_N n \frac{y}{x^2 + y^2} \frac{\partial}{\partial t}, \\ E_3 &= \frac{\partial}{\partial z}, & E_2 &= \frac{\partial}{\partial y} - 4G_N n \frac{x}{x^2 + y^2} \frac{\partial}{\partial t}. \end{aligned} \quad (4.115)$$

Around the Kronecker vacuum, Eq. (4.115) describes

$$\begin{aligned} A_0 &= 0, & A_1 &= +4G_N n \frac{y}{x^2 + y^2} \frac{\partial}{\partial t}, \\ A_3 &= 0, & A_2 &= -4G_N n \frac{x}{x^2 + y^2} \frac{\partial}{\partial t}. \end{aligned} \quad (4.116)$$

The nonvanishing component of the diffeomorphism field strength arises by

$$F^t_{12} = \frac{\partial}{\partial x} A^t_2 - \frac{\partial}{\partial y} A^t_1 = -8\pi G_N n \delta(x)\delta(y), \quad (4.117)$$

while the nonabelian part, $[A_1, A_2]$, vanishes. Eq. (4.117) is a direct characterization of the time monodromy, representing flux measurement by a tiny Ampèrian square loop on the x - y plane around the origin, for instance.

To clarify, we now liberate the Misner string from the GR context and view Eqs. (4.115) and (4.116) as describing a classical solution in a teleparallel or diffeomorphism gauge theory;¹¹ accordingly, we put $8\pi G_N = 1$. Since this is a topological configuration generated by a multivalued *gauge* transformation, it is (locally) on-shell in any theory regardless of the dynamics specified.

Consider the abelian Dirac string in Eq. (3.32). Its trivial embedding in YM theory is given in Eq. (3.34), which uses a constant color factor λ^a . This constant color factor trivializes the nonabelian part of the field strength F^a , which happens to be a common (and even characteristic) situation in classical double copy [160]. Interestingly, the very same kind of trivialization happens in Eq. (4.117), which arises by applying the replacement $\lambda^a \mapsto -n\delta^\mu_t$ on Eq. (3.35). Although the meaning of “constant diffeomorphism generator” would be unclear in general, it appears that this effective constancy is achieved by a collaboration between the direction of the reference δ^μ_t and the orientation of the surface support inducing monodromy for contours on the x - y plane.

With these understandings, we may identify the Misner string in Eq. (4.116), as a classical solution in any diffeomorphism gauge theory, as the classical double copy of the colored Dirac string in Eq. (3.34). In particular, this identifies that the colored Dirac string in YM theory classically double copies to the Misner string in the Cheung and Mangan [90]’s formulation of BI theory.

4.6.2 Mathematician’s Approach

Recalling Sec. 3.4.2, we also briefly remark on a description of the above Misner string geometry that avoids using the Dirac delta. In this case, the surface $\mathcal{S}$ will be excised from the spacetime $\mathcal{M}$, so one studies the manifold $\mathcal{M} \setminus \mathcal{S}$.

To begin with, the multivalued diffeomorphism may be formalized in the basic framework of transition functions. Recall that, in manifold theory, a transition function relates the coordinates of one chart to the coordinates of another. Let (U_i, φ_i) and (U_j, φ_j) be two overlapping charts, where $\varphi_i : U_i \rightarrow \mathbb{R}^d$ and $\varphi_j : U_j \rightarrow \mathbb{R}^d$. The transition function describes the map $\tau_{ij} = \varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \rightarrow \varphi_i(U_i \cap U_j)$.

Let $\mathcal{M}$ be a d -dimensional manifold and $\mathcal{S}$ be a codimension two submanifold in $\mathcal{M}$. Suppose an open cover $\{U_i\}$ of $\mathcal{M} \setminus \mathcal{S}$. For each patch U_i , we define

¹¹In this context, the Misner string geometry is defined by the frame in Eq. (4.115) or perhaps even just by the multivalued diffeomorphism in Eq. (4.107), and the metric in Eq. (4.106) is unnecessary.

coordinates by $\varphi_i : U_i \rightarrow \mathbb{R}^d$. This essentially means to choose a nice set of d elements from $C^\infty(U_i, \mathbb{R}^d)$. On each overlapping region $U_i \cap U_j$, we demand that φ_i and φ_j are related by a transition function τ_{ij} as follows:

$$\varphi_i \circ \varphi_j^{-1} = \tau_{ij}. \quad (4.118)$$

Note that τ_{ij} maps a region of $\mathbb{R}^d$ to a region of $\mathbb{R}^d$. We choose the transition functions such that, on each triple overlap $U_i \cap U_j \cap U_k$,

$$\tau_{ij} \circ \tau_{jk} \circ \tau_{ki} = \mathbb{1}. \quad (4.119)$$

The topological nontriviality of the manifold $\mathcal{M} \setminus \mathcal{S}$, seen as the atlas $\{(U_i, \varphi_i)\}$, will arise by stipulating that the collection $\{\tau_{ij}\}$ of our transition functions represents a nontrivial Čech cohomology element,¹²

$$[\{\tau_{ij}\}] = \alpha \in \check{H}^1(\mathcal{M} \setminus \mathcal{S}, \text{Diff}(\mathbb{R}^d)) \cong \text{Diff}(\mathbb{R}^d). \quad (4.120)$$

This characterizes the multivaluedness of the diffeomorphism entailed in the Misner string as an obstruction to a single-valued lift. Namely, each time one wraps around $\mathcal{S}$, one's coordinates are transformed by α . Note also the parallel between Eqs. (3.47) to (3.49) and Eqs. (4.118) to (4.120).

We should talk in terms of an explicit example. Concretely, suppose $\mathcal{M} = \mathbb{R}^4$ be the space of (t, x, y, z) and take $\mathcal{S}$ to be the t - z plane. Install four patches on $\mathcal{M} \setminus \mathcal{S}$ by taking the product of $\mathbb{R} = \{t | t \in \mathbb{R}\}$ and the four patches in Eq. (3.45). Choose the transition functions as

$$\begin{aligned} \tau_{12} : (t, x, y, z) &\mapsto (t - 3\Delta t, x, y, z), \\ \tau_{23} : (t, x, y, z) &\mapsto (t + 7\Delta t, x, y, z), \\ \tau_{34} : (t, x, y, z) &\mapsto (t - 2\Delta t, x, y, z), \\ \tau_{41} : (t, x, y, z) &\mapsto (t - \Delta t, x, y, z), \end{aligned} \quad (4.121)$$

where $\Delta t = 8\pi G_N n$. Physically, Eq. (4.121) means that four *time zones* are installed on a spacetime. When hopping from zone 3 to zone 2, the reading of the clock jumps by $+7\Delta t$: $t_2 = t_3 + 7\Delta t$. When hopping from zone 2 to zone 1,

¹²Here we have posited the maximally general $\text{Diff}(\mathbb{R}^d)$ while sweeping subtleties under the rug. One expects that an appropriate restriction to a subgroup would be adequate. For instance, our example in Eq. (4.107) will concern the group of constant translations. In the teleparallel theory context, the notion of absolute parallelism could facilitate a nice diffeomorphism-invariant incarnation of “constant translations,” ensuring consistency in identifying α .

the reading of the clock jumps by $-3\Delta t$: $t_1 = t_2 - 3\Delta t$. In this way, one travels through the four time zones and find that one's clock is shifted by the total of $+7\Delta t - 3\Delta t - \Delta t - 2\Delta t = \Delta t$. This is exactly the content of the multivalued diffeomorphism in Eq. (4.107). Also, it should be clear that different choices for the clock jumps can be made while keeping the total jump Δt the same. This is what is implied by the equivalence class in Eq. (4.120). The choice in Eq. (4.121) corresponds to the last example in Eq. (3.46), $(-3, 7, -2, -1)$, which is listed with other choices such as $(0, 0, 0, 1)$ and $(0, -1, 0, 2)$.

This provides a mathematical formalization for a multivalued diffeomorphism. The Misner string as a frame configuration will arise by preparing a single-valued set of vector fields on $\mathcal{M} \setminus \mathcal{S}$ and then applying the pushforward of this multivalued diffeomorphism.

4.6.3 Measurement of Torsion Flux

We end with a brief physical discussion on nonlocal functionals in the teleparallel theory that can characterize the torsion flux of Misner string.

Translation Holonomy Consider a Misner string in a generic d -dimensional spacetime $\mathcal{M}$, supported on a codimension two submanifold $\mathcal{S}$. As a generalization of Eq. (4.113), this Misner string will be described by the coframe

$$e^\alpha = \delta^\alpha_\mu dx^\mu + \frac{n^\alpha}{2\pi} d\phi_{\mathcal{S}}, \quad (4.122)$$

where n^α is a constant factor characterizing the total flux of torsion.

We may take insights from GR, where it has been pointed out that is that the action of a scalar particle gives a version of gravitational AB phase. See, e.g., Refs. [172, 290, 483, 485–491]. The time monodromy in Eq. (4.108) also admits an interpretation as Sagnac interferometry [480–484]. The first-order action of a scalar particle is given by a line integral of the one-form $m_\alpha e^\alpha_\mu dx^\mu$ plus a Lagrange multiplier term imposing the mass-shell constraint on the momentum m_α . Inspired by this fact, one integrates the one-form $m_\alpha e^\alpha_\mu dx^\mu$ over a loop $\mathcal{C}$ in the background in Eq. (4.122) and finds

$$\oint_{\mathcal{C}} m_\alpha e^\alpha = m_\alpha n^\alpha \text{link}(\mathcal{C}, \mathcal{S}). \quad (4.123)$$

Although this calculation describes the basis of the argument for NUT charge quantization in, e.g., Refs. [172, 474, 485, 488, 489, 492], a caveat is that m_α is treated as being constant and nondynamical.

A related idea that has also existed for a long time is “translation holonomy,” which naturally arises in the Poincaré gauge theory [41–49, 51–53] context. See, e.g., Refs. [567–569]. In this case one considers the combination $\mathbf{P}_\alpha e^\alpha + \frac{1}{2}\mathbf{S}_{\alpha\beta}\gamma^{\alpha\beta}$, where $\mathbf{P}_\alpha$ and $\mathbf{S}_{\alpha\beta}$ are the generators of the Poincaré group, and constructs a “Wilson loop” valued in the Poincaré group. When the translation mode is only activated, one obtains $\mathbf{P}_\alpha \oint_{\mathcal{C}} e^\alpha$. This essentially amounts to Eq. (4.123) if one replaces $\mathbf{P}_\alpha$ with the constant frame momentum m_α .

Although insightful, these interpretations can be subtle in GR because the frame index is gauged. Also, the translation algebra is abelian while gravitational interactions are nonlinear. Crucially, it seems unclear what would be the trace for the translation group, which may be needed to achieve a Wilson *loop*. In genuine teleparallel theories, however, we may safely take $\oint_{\mathcal{C}} e^\alpha$ as a valid nonlocal operator since the frame index α is global.

It is interesting to note that $\oint_{\mathcal{C}} e^\alpha$ resembles the Burgers vector in lattice systems, which characterizes disinclination defects [460, 516–520, 570]. In this case the index α describes directions in the so-called *lattice frame* and thus is considered global.

Diffeomorphism Wilson Line In the meantime, the very object that would directly characterize the multivalued diffeomorphism is arguably the diffeomorphism Wilson line defined in Sec. 4.4.2. Concretely, take the example configuration in Eq. (4.116). With the Magnus representation in Eq. (4.66), the logarithm of the diffeomorphism Wilson line $W_{\mathcal{C}} = \exp(\Omega_{\mathcal{C}}^\mu(x)\partial_\mu)$ for a closed contour $\mathcal{C}$ that wraps around the Misner string once is¹³

$$\begin{aligned}\Omega_{\mathcal{C}}^\mu(x) &= -\int A^\mu_1(x+y(s))dy^1(s) + A^\mu_2(x+y(s))dy^2(s), \\ &= -n\delta^\mu_t,\end{aligned}\tag{4.124}$$

regardless of one’s choice of the base point x . Certainly, this gives the direct computation of the diffeomorphism group element α in Eq. (4.120).

This proposal, however, may be considered less robust than the translation holonomy. To have a genuine gauge-invariant Wilson *loop*, one will have to take a trace. It is unclear what would be the trace for the diffeomorphism group.

¹³Note that the plus sign, $+n\delta^\mu_t$, will be obtained in the convention of Chap. III.

In sum, we have established the Misner string as a classical solution in any teleparallel or diffeomorphism gauge theory, which arises by applying a multivalued diffeomorphism. In a certain sense, this Misner string serves as the classical double copy of the Dirac string.

Note that it could be natural to ask if this exploration on the Misner string gives rise to a one-form symmetry in (genuine) teleparallel theories, as per the symmetry triangle in Fig. 3.1. Namely, a one-form symmetry that arises by multivalued diffeomorphisms.¹⁴ We had briefly remarked about this possibility in Chap. III, and if it is indeed viable, it will describe an interesting interaction between the two ideas, generalized symmetries and CK duality. In this context, a question one can ask is “Do symmetries double copy?”

4.7 Poisson Gauge Theory

We end with a brief remark on Poisson gauge theories, which arises by replacing the gauge algebra of an ordinary gauge theory with a Poisson algebra. This is implemented in, e.g., Ref. [158].

In a Poisson gauge theory, the gauge potential A_α carries just one index. The field strength is given by $F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha + \{A_\alpha, A_\beta\}$, where the nonlinear term is the Poisson bracket between A_α and A_β . Infinitesimal Poisson gauge transformations act on the Poisson gauge potential as $A_\alpha \mapsto A_\alpha - D_\alpha \xi$, where $D_\alpha \xi = \partial_\alpha \xi + \{A_\alpha, \xi\}$. On the Poisson field strength, we have $F_{\alpha\beta} \mapsto F_{\alpha\beta} + \{\xi, F_{\alpha\beta}\}$. It is left as an exercise to define Poisson gauge theory via CKR and explicitly establish proper gauge transformation behaviors as in Sec. 4.4, while also constructing Poisson Wilson line¹⁵ and its Magnus formula.

¹⁴Investigations into this direction are beyond the scope of the present work, but several speculative remarks could be made. First, recall that the explicit breaking of the one-form symmetry due to multivalued local Lorentz symmetry has mandated the existence of fermions as spinning particles (Chap. III). Similarly, the explicit breaking of the one-form symmetry due to multivalued diffeomorphisms, if exists, may mandate the existence of massive particles. In this case we encounter a reincarnation of the Poincaré charges (mass, spin) in GR via the higher-form (topological surface operator) framework. Second, the Misner string is famously endable on NUT charges. Hence, if a magnetic picture exists for this hypothetical one-form symmetry, then the explicit breaking of it might mandate the existence of a NUT-charged object in our universe. Indeed, we will see a concrete configuration representing a finite Misner string ending on opposite NUT charges in Chap. VII. Third, it will again be interesting to note similarities with lattice systems (cf. Ref. [571]). Lastly, it will also be natural to consider multivalued volume-preserving diffeomorphisms for SD gravity in the frameworks of Secs. 2.4.3 or 2.5.3. Note that this seems to be what is envisioned or suggested in Ref. [198], although the symmetry transformations were not explicitly discussed.

¹⁵Note that, in noncommutative gauge theories, a gauge-invariant Wilson *loop* is viable;

Although similar in spirit of CK duality, the status of Poisson gauge theory seems more established than diffeomorphism gauge theory; see, e.g., Ref. [572–576]. Part of this status may owe to the fact that Poisson gauge theory can be obtained as the semi-commutative limit of noncommutative gauge theory, which is a well-studied subject [548]. Note also that diffeomorphism gauge theories on $\mathcal{M}$ are automatically Poisson gauge theories on $T^*\mathcal{M}$ by implementing the diffeomorphism generator as the cotangent “momentum” [391, 575].

The semi-commutative limit of the Seiberg-Witten map [548] provides a field redefinition from an ordinary abelian gauge connection a_α to a Poisson gauge connection A_α , which is based on the Moser lemma [572, 573, 577]. It will be interesting to investigate if the semi-commutative limit of Seiberg-Witten map (or its adequate volume-preserving diffeomorphism generalization, if exists) can be useful in explicitly constructing the field redefinition from the BI photon field to the manifest CK dual field basis, explaining the equivalence $\mathbf{BI} \cong \mathbf{YM}(\mathbf{sdiff}(\mathcal{M}))$ in Eq. (4.10) in two (or $d > 2$) dimensions. One lesson that can be gained from this investigation will be the fate of gauge symmetries in the problem of double copy. A key philosophy and feature of the Seiberg-Witten map is that gauge orbits match between the two sides. One of the perennial questions of double copy has been that whether one has to pursue an explicitly gauge-fixed description to manifest double copy or aim to formulate everything in terms of gauge-covariant or gauge-invariant objects (cf. the desire for field basis independence during the NLSM discourse in Sec. 4.2.2).

4.A From Equations of Motion to Amplitudes

To see why formulating a theory as a BAS theory at the EoM level manifests CK duality or double copy, consider the derivation of the scattering amplitudes of BAS theory via BG recursion [154] (perturbative method [155–157]), which performs LSZ reduction on a one-point function evaluated on a background [90]. This represents BAS amplitudes as a sum over cubic rooted tree graphs Γ with momentum labels:

$$\mathcal{A} = \sum_{\Gamma} \frac{C(\Gamma) \tilde{C}(\Gamma)}{D(\Gamma)}. \quad (4.125)$$

Here, $C(\Gamma)$, $\tilde{C}(\Gamma)$, and $D(\Gamma)$ are graph functions. $D(\Gamma)$ encodes the propagators, while $C(\Gamma)$ and $\tilde{C}(\Gamma)$ are color factors associated with $\mathbf{g}$ and $\tilde{\mathbf{g}}$, respectively, see, e.g., Refs. [551, 552].

tively. By the very structure of BAS equations, $C(\Gamma)$ and $\tilde{C}(\Gamma)$ describe the same tensor structure. Hence, for each triplet of cubic graphs $(\Gamma_i, \Gamma_j, \Gamma_k)$ whose color factors are related by the Jacobi identity $C(\Gamma_i) + C(\Gamma_j) + C(\Gamma_k) = 0$, the same relation holds for the tilded color factors as $\tilde{C}(\Gamma_i) + \tilde{C}(\Gamma_j) + \tilde{C}(\Gamma_k) = 0$ and vice versa. As a result, the BCJ definition of CK duality [87–89] is satisfied if a choice for the Lie algebra $\tilde{\mathfrak{g}}$ exists such that $\tilde{C}(\Gamma)$ can define a tree graph function of the external momenta (and also polarizations for spinning theories). For NLSM, one takes $\tilde{\mathfrak{g}} = \mathbf{sdiff}(\mathcal{M})$. Subtleties may lie in converting the external scattering states back and forth between **NLSM**($\mathfrak{g}$) and **BAS**($\mathfrak{g}, \mathbf{sdiff}(\mathcal{M})$), but it can be shown that the invertibility on free on-shell configurations in Eq. (4.19) ensures obtaining the same amplitude. The embedding $j^a{}_\mu = \partial_\mu \pi^a$ is relevant for the leaf-leg conversion of π -particles to j -particles; the inverse mapping $\pi^a = (q \cdot \partial)^{-1} j^a{}_\mu q^\mu$ is relevant for the root-leg conversion of a j -particle to a π -particle. Details are to be found in Ref. [90].

4.B Lagrangian Formulations

BAS theory admits a Lagrangian formulation by the Lagrangian d -form

$$L[\Phi] = \frac{1}{\kappa^2} \delta_{ad} \delta_{\bar{a}\bar{d}} \left(-\frac{1}{2} \Phi^{a\bar{a}} \square \Phi^{d\bar{d}} + \frac{1}{6} f^a{}_{bc} \tilde{f}^{\bar{a}}{}_{\bar{b}\bar{c}} \Phi^{b\bar{b}} \Phi^{c\bar{c}} \Phi^{d\bar{d}} \right) d^d x, \quad (4.126)$$

if κ denotes the coupling.

Although not necessary, it is natural to ask if the chiral current formulation of NLSM, **BAS**($\mathfrak{g}, \mathbf{sdiff}(\mathcal{M})$), admits a Lagrangian formulation, and whether manifest CK duality is exhibited in that description.¹⁶ This will primarily refer to the Freedman-Townsend [539] formulation of NLSM. The idea is to promote the chiral current in Eq. (4.11) to a fundamental field variable by imposing the flatness condition as a constraint:

$$L[j, B] = \left\langle B_a, dj^a + \frac{1}{2} f^a{}_{bc} j^b \wedge j^c \right\rangle - \frac{1}{2\kappa^2} \delta_{ab} \langle j^a, j^b \rangle. \quad (4.127)$$

Here, $\langle \alpha, \beta \rangle := \alpha \wedge * \beta$. The coadjoint two-form B_a is a Lagrange multiplier.

The Freedman-Townsend formulation, however, does not seem to establish CK duality at the Lagrangian level: Eq. (4.127) does not arise by implementing a CKR on Eq. (4.126). To this end, one might be offered with a suggestion that the volume-preserving condition of the j -field may be imposed directly

¹⁶This exploration is based on discussions with Clifford Cheung and Nabha Shah.

by an explicit parametrization,

$$j^a = d^\dagger Z^a \iff j^{a\mu} = \partial_\rho Z^{a\mu\rho}, \quad (4.128)$$

for an adjoint-valued two-form field Z^a . Here, $d^\dagger$ is the Hodge codifferential operator. By analogy with fluid dynamics, a proper name for this Z -field seems to be *streamfunction*.¹⁷ By plugging in Eq. (4.128) to Eq. (4.127) while performing a gauge-fixing $dZ^a = 0$, one finds¹⁸

$$L[Z, B] = \langle B_a, \square Z^a \rangle - \frac{1}{2} \langle B_a, f^a_{bc} [Z^b, Z^c] \rangle - \frac{1}{2\kappa^2} \delta_{ab} \langle Z^a, \square Z^b \rangle, \quad (4.129)$$

where $\square$ describes the Hodge-de Rham Laplacian $d^\dagger d + d d^\dagger$. We have also defined, for any two-forms $Z_1, Z_2 \in \Omega^2(\mathcal{M})$,

$$[Z_1, Z_2] := -d^\dagger Z_1 \wedge d^\dagger Z_2. \quad (4.130)$$

Explicitly, $[Z_1, Z_2]^{\mu\nu} = -(\partial_\rho Z_1^{\mu\rho})(\partial_\sigma Z_2^{\nu\sigma}) + (\partial_\rho Z_2^{\mu\rho})(\partial_\sigma Z_1^{\nu\sigma})$.

As the notation has suggested, Eq. (4.130) defines a Lie bracket on the space $\Omega^2(\mathcal{M})$ of two-forms on $\mathcal{M}$. It is not difficult to verify this directly. In fact, $\Omega^2(\mathcal{M})$ equipped with this Lie bracket realizes a two-form version of the volume-preserving diffeomorphism algebra $\mathbf{sdiff}(\mathcal{M})$ (yet up to the equivalence $Z^a \sim Z^a + d^\dagger \lambda^a$).

Eq. (4.129) is an interesting action, formulated solely in terms of *two-forms*.¹⁹ It also beautifully exhibits the invariant tensors of the Lie algebra, δ^a_b , f^a_{bc} , and δ_{ab} . Diagrammatically, the B -field and the Z -field (as species) serve as “up and down indices” in the Feynman rules.

It is tempting to view Eq. (4.129) as arising from performing the CKR on the following theory via replacing $\tilde{\mathbf{g}}$ with the two-form version of $\mathbf{sdiff}(\mathcal{M})$:

$$\begin{aligned} L[\Phi, \Psi] & \\ &= \langle \Psi_{a\bar{a}}, \square \Phi^{a\bar{a}} \rangle - \frac{1}{2} \langle \Psi_{a\bar{a}}, f^a_{bc} \tilde{f}^{\bar{a}}_{\bar{b}\bar{c}} \Phi^{b\bar{b}} \Phi^{c\bar{c}} \rangle - \frac{1}{2\kappa^2} \delta_{ab} \delta_{\bar{a}\bar{b}} \langle \Phi^{a\bar{a}}, \square \Phi^{b\bar{b}} \rangle. \end{aligned} \quad (4.131)$$

It can be said that Eq. (4.131) embeds BAS theory as a subsector (cf. the EoM for $\delta\Psi_{a\bar{a}}$), but it is unclear whether such a two-field setup could be genuinely relevant for BCJ duality.

¹⁷The j -field could be interpreted as the velocity field for an incompressible fluid (with flavors [538]). The streamfunction is a useful tool in describing two-dimensional incompressible flows, whose contours represent streamlines. The Z -field here is essentially a generalization of this concept in d dimensions.

¹⁸Eq. (4.129) is classically equivalent to Eq. (4.127). In fact, a careful consideration of the one-loop determinant shows that they are also one-loop equivalent.

¹⁹Cf. Plebański gravity in Sec. 2.5.2.

SIMPLICITY OF SELF-DUAL BLACK HOLE

In his classic monograph [262], Chandrasekhar described black holes as “the most perfect macroscopic objects” in the universe. With the first direct detections of gravitational waves [9, 10] and the iconic horizon-scale image produced by the Event Horizon Telescope [14], these theoretical objects have now turned into precision experimental probes of strong-field gravity. It is in this sense that the black hole may be viewed as “the atom of the twenty-first century” [578]: simple enough to be universal, yet rich enough to reveal profound structures.

In the meantime, recent theoretical studies [163, 164, 175, 260, 261, 263–270] have provided deeper investigations into *SD black holes*, realizing their exceptional simplicity. This line of study may be traced back to the classic works [272, 579–582] of Gibbons, Hawking, and Ruback on gravitational instantons and hidden symmetries. The SD Taub-NUT solution, which is an instance of a Gibbons-Hawking gravitational instanton [579, 580], has been investigated as a prime example of a SD black hole (see, e.g., Refs. [175, 261, 264] for justifications of this terminology). In light of its associated hidden symmetry, it has been suggested that the SD Taub-NUT solution is “the analog of the hydrogen atom for black holes” [266]. Moreover, the classical integrability of SD gravity [313] renders its perturbation theory tractable [264]; for instance, it has been shown that the SD Taub-NUT solution exhibits a graviton helicity selection rule up to multiplicity two [261, 264].

Certainly, SD black holes could be viewed as purely mathematical: the self-duality condition inevitably makes their gravitational fields complex-valued. See the discussions on complex spacetimes in Refs. [312–315, 328–330], for instance. One may at least regard them as formal idealizations leading to theoretical curiosities, while keeping in mind that the ultimate desire will be to restore the missing ASD part either perturbatively or nonperturbatively (recall Secs. 2.4 and 2.5).

To give the reader at least a concrete glimpse of why the SD black hole has been compared to the hydrogen atom [266] and why they are of an active theoretical interest [163, 164, 175, 260, 261, 263–270], this chapter aims to

analyze a simpler yet analogous system in electromagnetism: the SD dyon. As is established by Ref. [163] and will be reproduced in Chap. VI of this thesis, the SD dyon has been shown to describe the exact gauge theory counterpart of the SD Taub-NUT solution via the Kerr-Schild double copy. See also works [261, 263, 264] and discussions therein. The SD dyon has served as a simplified testing ground for questions related to SD black holes [263, 583].

Our conclusion in this chapter will be that the motion of a charged test particle in the SD dyon background is *maximally superintegrable*. Note that later parts of this thesis will extend this superintegrability to SD black hole backgrounds and to the all-orders-in-spin dynamics of spinning probe particles.

5.1 Dyon Motion in Dyon Background

Consider flat spacetime with Cartesian coordinates $x^\mu = (t, x^i)$. A dyon is a particle that carries both electric and magnetic charges, say Q and Q^* . The electromagnetic field strength of a static dyon at $\vec{x} = 0$ is given by

$$\vec{E} = \frac{Q}{4\pi} \frac{\hat{r}}{r^2}, \quad \vec{B} = \frac{Q^*}{4\pi} \frac{\hat{r}}{r^2}, \quad (5.1)$$

where $r := |\vec{x}|$ and $\hat{r} := \vec{x}/r$. The reader may consult to [456] for more information on the physics of dyons.

We investigate the special-relativistic dynamics of a charged massive test particle in this background. For full generality, let q and q^* be its electric and magnetic charges, respectively. In a constant einbein parameterization, the classical EoM read

$$m \frac{dt}{d\tau} = p^0, \quad m \frac{dp^0}{d\tau} = \vec{p} \cdot (q\vec{E} + q^*\vec{B}), \quad (5.2a)$$

$$m \frac{d\vec{x}}{d\tau} = \vec{p}, \quad m \frac{d\vec{p}}{d\tau} = p^0(q\vec{E} + q^*\vec{B}) + \vec{p} \times (q\vec{B} - q^*\vec{E}), \quad (5.2b)$$

where m is the rest mass. Plugging in Eq. (5.1), these boil down to

$$\frac{dp^0}{d\tau} = -\frac{d}{d\tau} \left(\frac{\kappa}{r} \right), \quad m \frac{d\vec{p}}{d\tau} = \kappa p^0 \frac{\hat{r}}{r^2} + \tilde{\kappa} \vec{p} \times \frac{\hat{r}}{r^2}, \quad (5.3)$$

where we have denoted

$$\kappa := \frac{qQ + q^*Q^*}{4\pi}, \quad \tilde{\kappa} := \frac{qQ^* - q^*Q}{4\pi}. \quad (5.4)$$

The EoM preserve the mass-shell condition,

$$(p^0)^2 = \vec{p}^2 + m^2. \quad (5.5)$$

As the background in Eq. (5.1) is static and spherically symmetric, we expect the conservation of energy and angular momentum. Indeed, the conserved energy can be immediately identified from the first equation in Eq. (5.3):

$$\mathbb{E} = p^0 + \frac{\kappa}{r}. \quad (5.6)$$

Meanwhile, the conserved angular momentum is given by

$$\vec{\mathbb{J}} = \vec{x} \times \vec{p} - \tilde{\kappa} \hat{r}, \quad (5.7)$$

which follows from using the second equation in Eq. (5.3) and also an identity

$$\frac{1}{r} \hat{r} \times (\vec{p} \times \hat{r}) = \frac{1}{r} \left(\vec{p} - \hat{r} \hat{r} \cdot \vec{p} \right) = \vec{p} \cdot \vec{\partial} \hat{r}. \quad (5.8)$$

Famously, the term $\tilde{\kappa} \hat{r}$ in Eq. (5.7) describes the field angular momentum due to the interaction between the dyon and the test particle.

Consequently, we find three independent conserved quantities in involution: $\mathbb{E}$, $\vec{\mathbb{J}}^2$, and a component of $\vec{\mathbb{J}}$ (say $\mathbb{J}_z$). As the particle has exactly three physical degrees of freedom, the system is Liouville integrable. Hence the EoM are solvable by quadratures.

Let us elaborate on the concrete steps of solving the EoM. First of all, it follows from Eqs. (5.5) and (5.7) that

$$\frac{m}{2} \left(\frac{dr}{d\tau} \right)^2 + \frac{\vec{\mathbb{J}}^2 - \gamma}{2mr^2} + \frac{\mathbb{E}\kappa}{mr} = \frac{\mathbb{E}^2 - m^2}{2m}, \quad (5.9)$$

where we have denoted

$$\gamma := \kappa^2 + \tilde{\kappa}^2 = (Q^2 + Q^{*2})(q^2 + q^{*2}). \quad (5.10)$$

Clearly, Eq. (5.9) describes an effective potential governing the radial motion, from which one can find r as a function of the proper time τ . One can then find $\hat{r}$ as a function of τ by integrating the equation

$$\frac{d\hat{r}}{d\tau} = \frac{\vec{\mathbb{J}}}{mr^2} \times \hat{r}, \quad (5.11)$$

which follows from Eqs. (5.7) and (5.8). This completely determines the position $\vec{x} = r\hat{r}$ of the particle as a function of time.

It is worth understanding the geometry behind Eq. (5.11). Crucially, the particle's trajectory lies on a cone defined by $\vec{\mathbb{J}} \cdot \hat{r} = -\tilde{\kappa}$, known as Poincaré cone [584]. The cone's half-angle from the $\hat{\mathbb{J}}$ -axis is given by $\pi/2 + \xi$, where

$$\xi := \sin^{-1} \left(\frac{\tilde{\kappa}}{|\vec{\mathbb{J}}|} \right). \quad (5.12)$$

Note that ξ approaches zero when the limit of vanishing interaction so that the cone degenerates to a flat plane. Now the interpretation of Eq. (5.11) is obvious: the particle revolves around the axis of the Poincaré cone with angular velocity $|\vec{\mathbb{J}}|/mr^2$. In consequence, determining the radial motion from the effective potential fixes the entire trajectory.

5.2 Self-Dual Dyon as the Hydrogen Atom

A SD dyon is a dyon whose electric and magnetic charges satisfy the relation $Q = iQ^*$. It sources a SD electromagnetic field: $\vec{E} = i\vec{B}$.

An ASD dyon is a dyon whose electric and magnetic charges satisfy the relation $Q = -iQ^*$. It sources an ASD electromagnetic field: $\vec{E} = -i\vec{B}$.

Together, they are referred to as chiral dyons. Clearly, these chiral limits are the extremal cases in which the parameter γ in Eq. (5.10) becomes zero.

From now on, we study the motion of the test particle in the background of a chiral dyon. As mentioned earlier, this may be seen as a rather bizarre pathway from a conservative point of view. Due to the imaginary magnetic charge, the particle experiences a complex-valued Lorentz force. Hence its velocity becomes complex, and even its position becomes complex; otherwise there is no solution to the classical EoM. However, this complexified limit turns out to be intriguing enough to pursue, at least as a formal curiosity.

First of all, upon imposing the chiral limit $\gamma \rightarrow 0$, Eq. (5.9) boils down to

$$\gamma = 0 \quad \implies \quad \frac{m}{2} \left(\frac{dr}{d\tau} \right)^2 + \frac{\vec{\mathbb{J}}^2}{2mr^2} + \frac{\mathbb{E}\kappa}{mr} = \frac{\mathbb{E}^2 - m^2}{2m}. \quad (5.13)$$

Remarkably, this is isomorphic to the effective potential equation for the *non-relativistic* hydrogen atom (i.e., Kepler problem).

The nonrelativistic hydrogen atom is famously known to exhibit the Laplace-Runge-Lenz (LRL) vector as a conserved quantity. Motivated by this fact, we perform brute-force computation to find that Eq. (5.3) implies

$$\frac{d}{d\tau} \left(\vec{p} \times \vec{\mathbb{J}} \right) = \frac{\gamma - \mathbb{E}\kappa r}{mr^3} \frac{d\hat{r}}{d\tau}. \quad (5.14)$$

As a result, it follows that

$$\vec{\tilde{A}} := \vec{p} \times \vec{\tilde{J}} + E\kappa \hat{r} \implies \frac{d\vec{\tilde{A}}}{d\tau} = \frac{\gamma}{mr^3} (\vec{p}r - m\dot{r}\vec{x}). \quad (5.15)$$

Therefore, a new set of conserved quantities emerges in the chiral limit $\gamma \rightarrow 0$:

$$\gamma = 0 \implies \frac{d\vec{\tilde{A}}}{d\tau} = 0. \quad (5.16)$$

It is amusing how the LRL vector admits a relativistic reincarnation via complexifications and the formal chiral limits.

In sum, we identify seven conserved quantities: E , $\vec{\tilde{J}}$, $\vec{\tilde{A}}$. There are two algebraic relations relating them. On $\gamma = 0$, we find

$$\vec{\tilde{J}} \cdot \vec{\tilde{A}} = -E\kappa\tilde{\kappa}, \quad \vec{\tilde{A}}^2 = (E^2 - m^2)(\vec{\tilde{J}}^2 + \kappa^2) + E^2\kappa^2. \quad (5.17)$$

Therefore the number of independent conserved quantities is five, which is $2 \times 3 - 1$ in a six-dimensional phase space. Hence the system is *maximally superintegrable*. A further computation shows that the dynamical symmetry algebra is nicely closed, describing $\mathfrak{so}(4)$ for $E^2 < m^2$, $\mathfrak{so}(1, 3)$ for $E^2 > m^2$ (or $\mathfrak{so}(4, \mathbb{C})$ in the complexified category), and $\mathfrak{iso}(3)$ for $E^2 = m^2$.

In these concrete senses, we conclude that the dynamics of a relativistic test particle in the background of a chiral dyon is isomorphic to that of the nonrelativistic Hydrogen atom or Kepler problem.

5.3 Maximal Superintegrability from Killing-Yano Tensors

In the above exposition, we have intentionally implemented the $1 + 3$ split to be pedagogical; after all, the background stipulates a preferred time direction. However, it should be pointed out that there is also a more elegant approach that retains the four-vector notation.

Without loss of generality, we now assume the background of a SD dyon with electric charge $Q/2$ and magnetic charge $-iQ/2$. Let u^μ be the timelike Killing vector encoding the stationary direction. In the relativistic notation, the electromagnetic field in Eq. (5.1) describes

$$F_{\mu\nu}(x) = \frac{Q}{8\pi|\vec{x}|^3} *Y_{\mu\nu}^+(x). \quad (5.18)$$

where we have defined SD and ASD two-forms $Y_{\mu\nu}^\pm$ as

$$*Y_{\mu\nu}^\pm(x) = (u \wedge x)_{\mu\nu}^\pm = \frac{1}{2} (u_\mu x_\nu - x_\mu u_\nu) \mp \frac{i}{2} \varepsilon_{\mu\nu\rho\sigma} u^\rho x^\sigma. \quad (5.19)$$

The $\pm$ superscript denotes the SD/ASD projection. These essentially encode the position three-vector $\vec{x}$ via each $\mathfrak{sl}(2, \mathbb{C})$ sector of $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$: $(\vec{x} \cdot \vec{\sigma})^{\dot{\alpha}}_{\dot{\beta}}$ and $(\vec{x} \cdot \vec{\sigma})^{\beta}_{\alpha}$. If $Y_{\mu\nu}(x) = Y_{\mu\nu}^+(x) + Y_{\mu\nu}^-(x)$, we find

$$\partial_{\rho} Y_{\mu\nu}(x) = \varepsilon_{\mu\nu\rho\sigma} u^{\sigma} = \partial_{[\rho} Y_{\mu\nu]}(x). \quad (5.20)$$

This is the defining equation for a Killing-Yano [585] tensor. Moreover, it is easy to see that it “commutes” with the field strength in Eq. (5.18) as

$$F^{\mu}_{\rho}(x) Y^{\rho}_{\nu}(x) = Y^{\mu}_{\rho}(x) F^{\rho}_{\nu}(x). \quad (5.21)$$

In generic electromagnetic fields, the dyonic test particle with charges $(q, q^{\star})$ exhibits the EoM

$$\begin{aligned} p^{\mu} &= m \frac{dx^{\mu}}{d\tau}, \quad m \frac{dp_{\mu}}{d\tau} = q F_{\mu\nu}(x) p^{\nu} - q^{\star} *F_{\mu\nu}(x) p^{\nu}, \\ &= q_+ F_{\mu\nu}^+(x) p^{\nu} + q_- F_{\mu\nu}^-(x) p^{\nu}, \end{aligned} \quad (5.22)$$

where $p^2 + m^2 = 0$ and $q_{\pm} = q \mp iq^{\star}$. In SD backgrounds, this reduces to

$$p^{\mu} = m \frac{dx^{\mu}}{d\tau}, \quad \frac{dp^{\mu}}{d\tau} = \frac{q_+}{m} F^{\mu}_{\nu}(x) p^{\nu}. \quad (5.23)$$

Provided the two conditions in Eqs. (5.20) and (5.21), Eq. (5.23) implies

$$\frac{d}{d\tau} \left(Y^{\mu}_{\rho}(x) p^{\rho} \right) = \frac{q_+}{m} F^{\mu}_{\nu}(x) \left(Y^{\nu}_{\rho}(x) p^{\rho} \right). \quad (5.24)$$

This means that p^{μ} and $Y^{\mu}_{\rho}(x) p^{\rho}$ exhibit the same cyclotron precession angular frequency, $q_+ F^{\mu}_{\nu}(x)/m$. See also Refs. [273, 585–595].

Meanwhile, consider the ASD two-forms of Plebański [315, 339]:

$$\begin{aligned} \Sigma_1 &= i(dx \wedge dt + i dy \wedge dz), \\ \Sigma_2 &= i(dy \wedge dt + i dz \wedge dx), \\ \Sigma_3 &= i(dz \wedge dt + i dx \wedge dy). \end{aligned} \quad (5.25)$$

These describe a trio of two-forms, $(\Sigma_a)_{\mu\nu}$ for $a = 1, 2, 3$, and correspond to $(\sigma_a)_{\alpha}^{\beta}$ in the spinor notation, so $*(\Sigma_a)_{\mu\nu} = -i(\Sigma_a)_{\mu\nu}$. Notably, these trivially serve as Killing-Yano tensors [272, 273] and define a quaternionic structure as [271–273]

$$(\Sigma_a)^{\mu}_{\rho} (\Sigma_b)^{\rho}_{\nu} = -\delta_{ab} \delta^{\mu}_{\nu} + \varepsilon^c_{ab} (\Sigma_c)^{\mu}_{\nu}. \quad (5.26)$$

Moreover, they also commute with the field strength in Eq. (5.18) as

$$(\Sigma_a)^{\mu}_{\rho} F^{\rho}_{\nu}(x) = F^{\mu}_{\rho}(x) (\Sigma_a)^{\rho}_{\nu}, \quad (5.27)$$

as any ASD element commutes with a SD element in the Lorentz algebra.

As a result, the classical EoM in Eq. (5.23) also imply

$$\frac{d}{d\tau} \left((\Sigma_a)^\mu{}_\rho p^\rho \right) = \frac{q_+}{m} F^\mu{}_\nu(x) \left((\Sigma_a)^\nu{}_\rho p^\rho \right), \quad (5.28)$$

so we now find that a triple of four-vectors, p^μ , $Y^\mu{}_\rho(x)p^\rho$, and $(\Sigma_a)^\mu{}_\rho p^\rho$ exhibit a synchronized precession behavior. Clearly they describe three axes of a common Lorentz frame attached to the particle.

From this geometric realization, it is clear that the following scalar quantities are dynamically conserved:

$$\mathbb{C} = -p_\mu Y^\mu{}_\rho(x) Y^\rho{}_\nu(x) p^\nu = (\vec{x} \times \vec{p})^2, \quad (5.29)$$

$$\mathbb{K}_a = -p_\mu (\Sigma_a)^\mu{}_\rho Y^\rho{}_\nu(x) p^\nu = \vec{p} \times (\vec{x} \times \vec{p}) + i p^0 \vec{x} \times \vec{p}. \quad (5.30)$$

Of course, there are also conserved quantities due to the Killing vectors. It is left as an exercise to check that they reproduce the energy $\mathbb{E}$ and the angular momentum $\vec{\mathbb{J}}$ in Eqs. (5.6) and (5.7), for which it will be helpful to adopt the approach in Ref. [588]. With this understanding, we plug in $p^0 = \mathbb{E} - \kappa/r$ and $\vec{x} \times \vec{p} = \vec{\mathbb{J}} + \tilde{\kappa} \hat{r}$ according to Eqs. (5.6) and (5.7). Then Eqs. (5.29) and (5.30) boil down to

$$\mathbb{C} = \vec{\mathbb{J}}^2 + \kappa^2, \quad \mathbb{K}_a = \mathbb{A}_a + i \mathbb{E} \mathbb{J}_a, \quad (5.31)$$

on the support of $\kappa = i \tilde{\kappa}$. While $\mathbb{C}$ —the Carter [586] constant—does not yield a new conserved quantity, $\mathbb{K}_a$ does: the LRL vector identified earlier.

In conclusion, we have reproduced the conserved charges found in Sec. 5.2 in the Lorentz-covariant notation. The underlying geometric structures are identified as the Killing-Yano tensor $Y_{\mu\nu}(x)$ and the ASD Plebański two-forms $(\Sigma_a)_{\mu\nu}$. It is left as an exercise to understand how the quaternionic structure in Eq. (5.26) boils down to the symmetry algebra.

Notably, the same geometric structures arise for the SD Taub-NUT black hole as well. This forms the basis of the statement [266] that the SD Taub-NUT black hole is secretly the hydrogen atom.

More details can be found in the next chapter, where it is shown that the SD Taub-NUT black hole is the double copy of the SD dyon. Specifically, we discover a new metric for the SD Taub-NUT solution by double copying the Kerr-Schild gauge potential of the SD dyon.

SELF-DUAL BLACK HOLE FROM DOUBLE COPY

We derive a previously unknown KS metric for the SD Taub-NUT solution by classical double copy. We show its equivalence with the well-known Gibbons-Hawking instanton metric by means of an explicit complex coordinate transformation. Our KS metric develops a semi-infinite string defect as the Misner string, much like the original metric. It also establishes two more classical double copy correspondences to the SD dyon in electromagnetism, one of which is a novel nonlocal kind based on propagators between dyonic matter.

Contents of Secs. 6.1 to 6.6 are adapted from Ref. [163], J.-H. Kim, “Single Kerr-Schild metric for Taub-Newman-Unti-Tamburino instanton,” *Phys. Rev. D* **111** no. 2, (2025) L021703, [arXiv:2405.09518 \[hep-th\]](#).

6.1 Introduction

Recently, an exciting program dubbed classical double copy [161–174, 181–185] has refined analogies between gravity and electromagnetism into concrete mappings between classical solutions while being rooted in the study of scattering amplitudes [86–89, 327]. The prime example is the KS double copy, established in the seminal work [160].¹ It is shown that a stationary KS solution $g = \eta + \Phi \ell \otimes \ell$ in GR defines a stationary solution in Maxwell theory by the gauge potential $A = \Phi \ell$, which takes a “single copy” of the null one-form ℓ . Conversely, one envisions the gravitational counterpart $g = \eta + \Phi \ell \otimes \ell$ of a stationary solution $A = \Phi \ell$ in electromagnetism, by taking a “double copy” of ℓ . Well-known instances of this correspondence are Schwarzschild to point charge and Kerr to a rotating solution dubbed “ $\sqrt{\text{Kerr}}$ ” [98, 160, 180, 202].

From this angle, it is natural to expect that the Taub-NUT solution [596, 597] could correspond to a dyon, given the interpretation of the NUT charge as the gravitomagnetic monopole [172, 294, 474–476, 486, 489, 491]. Unfortunately, only a *double* KS metric $g = \eta + \Phi_1 \ell_1 \otimes \ell_1 + \Phi_2 \ell_2 \otimes \ell_2$ has been known to exist for the Taub-NUT solution: a classic result due to Plebański and

¹KS metrics give an abelian sector of GR in the sense of the effective linearization property [177, 178]. In this light, the KS double copy may be viewed as probing the double copy correspondence between the abelian sectors of YM theory and GR.

Demiański [331, 332, 598]. Consequently, the anticipated correspondence between the Taub-NUT solution and the electromagnetic dyon had to be achieved only by an alternative construction using the double KS ansatz [181].

Remarkably, the present letter points out that a direct application of the original KS double copy is viable for the *SD* Taub-NUT solution. This is the extremal case of the Taub-NUT solution that has recently gained a revived interest [260, 261, 263, 265, 266].

Crucially, we discover that the SD Taub-NUT solution admits a *single* KS metric: $g = \eta + \Phi \ell \otimes \ell$. First, a KS metric is obtained by KS double copying the gauge potential of the SD dyon in electromagnetism. We then show that it describes the SD Taub-NUT solution by explicitly constructing a coordinate transformation to the well-known Gibbons-Hawking [579, 580] instanton metric. To our understanding, this gives the first instance of the classical double copy program yielding a new result in GR (a previously unknown metric).

Next, the physical interpretation of the metric as a gravitational dyon is established within the KS description. Most importantly, we confirm the characteristic *Misner string* geometry: the gravitational analog [474, 475] of the Dirac string [454, 455]. Notably, finding its distributional source [475, 476] becomes a significantly simpler problem in KS coordinates, thanks to the effective linearization property [176–178].

Lastly, the classical double copy is established in two more ways. Our KS description concretely validates the Weyl double copy proposed by Refs. [182–184], which has been stated without referencing an explicit metric. We then propose a new exotic form of classical double copy employing a *nonlocal* operator, which arises from propagators between dyonic matter [456, 599–602].

In this chapter, we use the field theorists’ natural unit in which $8\pi G = 1$.

6.2 Derivation of Self-Dual Taub-NUT Solution from Double Copy

In electromagnetism, a dyon is a point particle carrying both electric and magnetic charges (recall Eq. (5.1)). The SD dyon is a dyon with electric and magnetic charges, say, Q and $-iQ$. Its gauge potential can be given as

$$A = \frac{Q}{4\pi R} \left(-dt - iR(1 - \cos\theta) d\phi \right), \quad (6.1)$$

where we have set up spherical coordinates (R, θ, ϕ) and oriented the Dirac string [454, 455] along the negative z -axis (south pole). This describes a com-

plexified solution where the field strength $F = dA$ is SD: $*F = +iF$.

Interestingly, adding a total derivative $(Q/4\pi)d\log(R+z)$ gauge transforms Eq. (6.1) to a KS form:

$$A = \frac{Q}{4\pi R} \left(-dt + \frac{x+iy}{R+z} dx + \frac{y-ix}{R+z} dy + dz \right). \quad (6.2)$$

Here, $(x, y, z) = (R \sin \theta \cos \phi, R \sin \theta \sin \phi, R \cos \theta)$ are the Cartesian coordinates. Importantly, the terms inside the bracket comprise a null one-form.

With this understanding, the KS double copy [160] associates Eq. (6.2) to the following line element, by literally taking “two copies” of the null one-form:

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 + \frac{M}{4\pi R} \left(-dt + \frac{x+iy}{R+z} dx + \frac{y-ix}{R+z} dy + dz \right)^2. \quad (6.3)$$

Here, the charge Q is mapped to a mass parameter M .

Remarkably, Eq. (6.3) derives a new metric for the SD Taub-NUT solution, whose mass and NUT charge are M and $-iM$. To prove this, we simply construct an explicit coordinate transformation to a previously known metric.

6.3 The Diffeomorphism

6.3.1 Direct Construction

It is helpful to employ complex coordinates $\zeta = (x+iy)/(R+z)$ and $\tilde{\zeta} = (x-iy)/(R+z)$ that arise from stereographically projecting the two-sphere with respect to the south pole, so Eq. (6.3) translates to

$$ds^2 = -dt^2 + dR^2 + R^2 \frac{4d\tilde{\zeta}d\zeta}{(1+\tilde{\zeta}\zeta)^2} + \frac{M}{4\pi R} \left(-dt + dR - R \frac{2\tilde{\zeta}d\zeta}{1+\tilde{\zeta}\zeta} \right)^2. \quad (6.4)$$

By writing down an ansatz at each order in the expansion in $M/4\pi$, we could perturbatively construct a diffeomorphism whose resummation gives

$$\begin{aligned} t &\mapsto t + \frac{M}{4\pi}, & \zeta &\mapsto \zeta, \\ R &\mapsto R + \frac{M}{4\pi}, & \tilde{\zeta} &\mapsto \frac{1}{1+(M/4\pi R)(1+\tilde{\zeta}\zeta)} \tilde{\zeta}. \end{aligned} \quad (6.5)$$

Applying Eq. (6.5) to Eq. (6.4) gives

$$ds^2 = \begin{pmatrix} -\left(1 + \frac{M}{4\pi R}\right)^{-1} \left(dt + \frac{M}{4\pi R} \left(dR - R \frac{2\tilde{\zeta}d\zeta}{1+\tilde{\zeta}\zeta} \right) \right)^2 \\ + \left(1 + \frac{M}{4\pi R}\right) \left(dR^2 + R^2 \frac{4d\tilde{\zeta}d\zeta}{(1+\tilde{\zeta}\zeta)^2} \right) \end{pmatrix}, \quad (6.6)$$

which is a Gibbons-Hawking metric [579, 580] for the SD Taub-NUT solution with mass M . This diffeomorphism is surely invertible; it becomes self-evident soon that the inverse map is simply Eq. (6.5) with M replaced with $-M$.

As a reminder, the Gibbons-Hawking line element reads [579, 580]

$$ds^2 = -\frac{1}{V} \left(dt + a \right)^2 + V \left(dx^2 + dy^2 + dz^2 \right). \quad (6.7)$$

Eq. (6.7) defines a SD, Ricci-flat metric if V and a are scalar and one-form fields on the three-dimensional space of (x, y, z) such that

$$-dV = i *_3 da. \quad (6.8)$$

Here, $*_3$ denotes the Hodge dual of the three-dimensional flat metric $dx^2 + dy^2 + dz^2$. Eq. (6.8) means that V and a can be interpreted as describing a stationary SD electromagnetic field. Diffeomorphisms of the form $t \mapsto t + \chi(x, y, z)$ correspond to the gauge transformations $a \sim a + d\chi$.

The SD Taub-NUT solution is included in the Gibbons-Hawking class as a single-centered, asymptotically locally flat instance. When the spherical coordinates are used for the flat three-space, it arises by taking

$$V = 1 + \frac{M}{4\pi R} \implies -dV = \frac{M}{4\pi R^2} dR, \quad da = \frac{-iM}{4\pi} \sin\theta d\theta \wedge d\phi. \quad (6.9)$$

It is easy to see that Eq. (6.6) precisely encodes Eq. (6.9).

To sum up, we have provided a *constructive* proof that the KS and Gibbons-Hawking metrics in Eqs. (6.3) and (6.6) are diffeomorphic. Consequently, we establish that the SD Taub-NUT solution admits a construction from the KS double copy. Previously, these facts were only recently *conjectured* by Ref. [174] in split signature.

An important remark is in order. The astute reader may have noticed an unusual feature of the diffeomorphism in Eq. (6.5): it describes a *complexified diffeomorphism*,² since ζ is frozen while $\tilde{\zeta}$ gets scrambled: hence the notation $\tilde{\zeta}$ instead of $\bar{\zeta}$. This is justified as a legitimate operation in the current context because the Lorentzian-signature SD Taub-NUT solution is

²Such a complexification is exactly what is practiced in Plebański [331], Sec. 7, and Plebański and Demiański [332], Sec. 13. Mathematically, this means to work in the holomorphic category of manifolds, where the “barred” coordinates play no role. See the clarification which Penrose makes in Ref. [312]. See also Refs. [313–315, 328–330] for related discussions and reviews.

inherently a complex saddle as a SD solution [261]. Just as the SD dyon gauge potential described in Eq. (6.2) is a solution to complexified Maxwell's equations due to the imaginary magnetic charge $-iQ$, the SD Taub-NUT metric in Eq. (6.3) solves complexified Einstein's equations governing holomorphic metrics [258, 261, 314, 315, 328]. As a consequence of this interpretation, ζ and $\tilde{\zeta}$ can be transformed independently as *holomorphic* coordinates.

6.3.2 Kerr-Schild to Gibbons-Hawking via Null Geodesic Flow

In fact, the explicit diffeomorphism in Eq. (6.5) simply turns out to be a flow along the null congruence associated with the KS metric:

$$(t, x, y, z) \mapsto \left(t + \frac{M}{4\pi}, x + \frac{M}{4\pi} \frac{x + iy}{|\vec{x}| + z}, y + \frac{M}{4\pi} \frac{y - ix}{|\vec{x}| + z}, z + \frac{M}{4\pi} \right). \quad (6.10)$$

This fact can be explained as follows. Firstly, suppose a vector field ℓ^μ in complexified Minkowski space defines a geodesic congruence: $\ell^\mu{}_{,\nu} \ell^\nu = 0$. The time- ϵ flow generated by this vector field is simply $x^\mu \mapsto x^\mu + \epsilon \ell^\mu(x)$, as geodesics are straight lines in flat space. The pull-back of the flat metric $\eta = \eta_{\mu\nu} dx^\mu \otimes dx^\nu$ is $\eta + 2\epsilon \ell_{(\mu,\nu)} dx^\mu \otimes dx^\nu + \epsilon^2 \eta_{\rho\sigma} \ell^\rho{}_{,\mu} \ell^\sigma{}_{,\nu} dx^\mu \otimes dx^\nu$.

The null vector field (NVF) associated with the KS metric of the SD Taub-NUT solution, $\ell^\mu = (1, \zeta, -i\zeta, 1)$, is stationary, geodesic, and shear-free. With the self-duality condition, they together imply that $2\partial_\nu \ell^\mu = \rho \tilde{m}^\mu m_\nu$ when described with a null tetrad $(\ell^\mu, n^\mu = 2u^\mu - \ell^\mu, m^\mu, \tilde{m}^\mu)$ where $u^\mu := \delta^\mu_0$. The complex expansion computes to $\rho = 1/r$.

Thus, it follows that the flat metric transforms to $\eta + \epsilon \rho \tilde{m} \odot m$ if the geodesic flow $x^\mu \mapsto x^\mu + \epsilon \ell^\mu(x)$ is used as a diffeomorphism, where $\odot$ denotes the symmetrized tensor product. In turn, the KS metric $\eta + \phi \ell^2$ transforms to

$$\eta + \epsilon \rho \tilde{m} \odot m + \phi' \ell^2, \quad (6.11)$$

where $\phi'(x) := \phi(x + \epsilon \ell(x))$.

Meanwhile, using $\eta = -2u \odot \ell + \ell^2 + \tilde{m} \odot m$, the Gibbons-Hawking metric $-(1 + \phi)^{-1} (u - \phi(\ell - u))^2 + (1 + \phi)(u^2 + \eta)$ boils down to

$$\eta + \phi \tilde{m} \odot m + (\phi^{-1} + 1)^{-1} \ell^2. \quad (6.12)$$

Therefore, the KS metric is mapped to the Gibbons-Hawking metric if we can flow along the geodesic congruence such that $\phi^{-1} + 1 = \phi'^{-1}$ and $\epsilon \rho = \phi$. Clearly, $\epsilon = M/4\pi$ satisfies both of these conditions.

6.4 Physical Interpretation

We should now examine the physical validity of the KS metric in Eq. (6.3). To begin with, we point out that Eq. (6.3) *itself* is quite unusual as a KS metric. While the KS potential is simply $\Phi = M/4\pi r$, the null one-form ℓ involves a *semi-infinite string defect* along the negative z -axis, which is exactly how the gravitomagnetic charge is encoded in this metric:

$$\ell = -dt + \frac{x+iy}{r+z}dx + \frac{y-ix}{r+z}dy + dz. \quad (6.13)$$

This is intriguing, as the null one-form is usually regular wherever the KS potential is regular—which is especially true for the double KS construction of Taub-NUT [181].

We expect that the interpretation of this string defect should be the *Misner string* [474, 475], just as in the original metric in Eq. (6.6): the gravitational analog of the Dirac string, inputting a gravitomagnetic flux of $-im$ at the origin. To show this, we recall that there are at least three senses in which the Taub-NUT metric [474, 475] describes a gravitational analog of a dyon: (a) the Misner string is invisible as a coordinate artifact if the gravitational analog of the Dirac quantization condition holds [474], (b) the Misner string implements a flux tube of time monodromy [172, 294, 486], and (c) the distributional stress-energy of the string describes an energy-less thin solenoid of mass current [475, 476]. Our goal now is to show that all of these properties hold in the KS description as well.

Firstly, we derive the Dirac quantization condition. For the SD dyon gauge potential in Eq. (6.2), adding a term $(Q/2\pi)d\log\zeta$ repositions its Dirac string along the positive z -axis (north pole). This encodes the complexified group-valued transition function $\zeta^{ieQ/2\pi}$, where e is the electromagnetic coupling. Demanding its single-valuedness on the “equator” [25] implies $ieQ \in 2\pi\mathbb{Z}$.

The gravitational Dirac quantization can be derived in a similar fashion, i.e., by hiding the strings with overlapping coordinate patches [172, 294, 474, 486]. In the KS description, we find that the diffeomorphism

$$t \mapsto t + \frac{M}{2\pi} \log \zeta, \quad \tilde{\zeta} \mapsto \frac{1 - (M/4\pi r)(1 + \tilde{\zeta}\zeta)/\tilde{\zeta}\zeta}{1 + (M/4\pi r)(1 + \tilde{\zeta}\zeta)} \tilde{\zeta} \quad (6.14)$$

with r and ζ fixed transforms the KS metric in Eq. (6.4) to another KS metric:

$$ds^2 = -dt^2 + dr^2 + r^2 \frac{4d\tilde{\zeta}d\zeta}{(1 + \tilde{\zeta}\zeta)^2} + \frac{M}{4\pi r} \left(-dt + dr + r \frac{2d\zeta}{\zeta(1 + \tilde{\zeta}\zeta)} \right)^2, \quad (6.15)$$

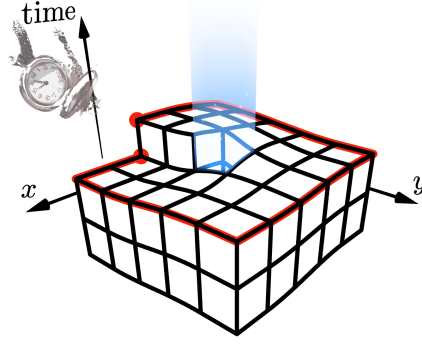


Figure 6.1: The Misner string is a “dislocation” in spacetime, inducing time monodromy as an AB phase.

which develops a string defect along the positive z -axis (north pole). For this diffeomorphism to be single-valued within the overlapping region, the time coordinate should be periodic as $t \sim t + iM$, given the $2\pi i$ periodicity of $\log \zeta$. Hence the energy e of a test particle is quantized as $ieM \in 2\pi\mathbb{Z}$. This is precisely the gravitational Dirac quantization condition due to the NUT charge $-iM$.

Secondly, we study the near-string geometries in an explicit manner. Let $\xi := x + iy$, $\tilde{\xi} := x - iy$ be complex coordinates for the (complexified) x - y plane, and suppose a positive infinitesimal ϵ . Applying the replacement $z \mapsto -1/\epsilon + z$ to the gauge potential in Eq. (6.2) and then taking the limit $\epsilon \rightarrow 0$, we obtain $A = (Q/2\pi)d\log \tilde{\xi}$. This precisely describes an infinite Dirac string generated by a large gauge transformation [31].

For the line element in Eq. (6.3), the same limiting procedure leaves out a divergent part that can be compensated by a shift $\xi \mapsto \xi + ((M/\pi)(1/\epsilon - z) - 3M^2/4\pi^2)/\tilde{\xi}$. Amusingly, the resulting line element is simply the Minkowski line element, but with a multivalued time:

$$t' = t + \frac{M}{2\pi} \log \tilde{\xi}. \quad (6.16)$$

Namely, in the vicinity of the string, the geometry approaches the image of empty spacetime under a large diffeomorphism. Here, a Sagnac interferometer measures a monodromy in time as the gravitational AB phase [483, 486, 489, 490], localized along the z -axis (as $\tilde{\xi} = 0$). This precisely describes an infinite Misner string as a thin tube of gravitomagnetic flux $-iM$.³

³One might make a parallel between the field strength $F = dA$ and the torsion two-form ddt' (cf. Refs. [460, 519, 520]) associated with these two strings, which are distributional.

Thirdly, we derive the distributional stress-energy tensor of this string. Among various approaches [603–605], we particularly consider the “harmonic-gauge linearization” of Ref. [176] which stationary KS metrics enjoy as a luxury. Provided the congruence of $\ell^\mu \partial_\mu$ is geodesic and shear-free (GSF), this approach identifies $T^{\mu\nu} = u^{(\mu} J^{\nu)}$ as the source for the double copy solution [176]. Here J^μ is the single copy source, while $u^\mu = \delta^\mu_0$ is the unit timelike Killing vector. For our case, the GSF property of the congruence associated with the null one-form in Eq. (6.13) can be established from an identity

$$\partial_\mu \ell_\nu = \frac{1}{2R} \left(\eta_{\mu\nu} + \ell_{(\mu} (2u - \ell)_{\nu)} - i \varepsilon_{\mu\nu\rho\sigma} u^\rho \ell^\sigma \right), \quad (6.17)$$

while the single copy source J^μ easily follows by acting on the Laplacian $\square := -\eta^{\mu\nu} \partial_\mu \partial_\nu$ to Eq. (6.1). Applying the formula $T^{\mu\nu} = u^{(\mu} J^{\nu)}$ then immediately reproduces Bonnor’s stress-energy tensor [475, 476], describing a point mass m attached to a massless semi-infinite rotating rod.

To sum up, we have convinced ourselves from multiple angles that the KS metric in Eq. (6.3) indeed exhibits the geometrical and physical characteristics of a Taub-NUT spacetime arising from the Misner string. This analysis demonstrates that the KS description of the solution is not merely a formal construct but physical. Moreover, the quick derivation of the distributional source demonstrates its usefulness as well.

6.5 More Double Copy Correspondences

Finally, for interested readers, we discuss alternative classical double copy relations that can be drawn between SD Taub-NUT and SD dyon. The possible options are Weyl double copy [182–184], operator KS double copy [160, 161, 169], and SD double copy [160, 327].

To this end, we first note that our KS metric for the SD Taub-NUT solution describes an elegant formula in the spinor notation: Eq. (6.3) translates to

$$ds^2 = ds_{\text{flat}}^2 + \frac{M}{4\pi R} \left(\eta_\alpha \tilde{o}_{\dot{\alpha}} dx^{\dot{\alpha}\alpha} \right)^2, \quad (6.18)$$

where the null one-form splits into two spinors as

$$\ell^\mu = (\sigma^\mu)_{\alpha\dot{\alpha}} \tilde{o}^{\dot{\alpha}} \eta^\alpha, \quad \tilde{o}^{\dot{\alpha}} = \begin{pmatrix} 1 \\ \zeta \end{pmatrix}, \quad \eta^\alpha = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (6.19)$$

Here, the SD spinor $\tilde{o}^{\dot{\alpha}}$ describes a “square root” of the position three-vector $\vec{x}$ in essence, while the ASD spinor η^α is a constant reference controlling the

direction of the Misner string. It can be seen that the former encodes physical information, whereas the latter encodes pure gauge information.

To elaborate, computation shows that the principal spinors for the KS metric in Eq. (6.3) are $\tilde{o}^{\dot{\alpha}} = (1, \zeta)$ and $\tilde{\iota}^{\dot{\alpha}} = (-\tilde{\zeta}, 1)/(1 + \tilde{\zeta}\zeta)$, the former of which is precisely the SD spinor in Eq. (6.19). It is interesting to note that the *real* combinations $[\tilde{o}_{\dot{\alpha}}]^* \tilde{o}_{\dot{\beta}}$, $[\tilde{\iota}_{\dot{\alpha}}]^* \tilde{\iota}_{\dot{\beta}}$ of these principal spinors exactly reproduce the null one-forms $-dt \pm dr$ associated with the KS metrics of the Schwarzschild solution as envisioned in Refs. [174, 183, 184].⁴

6.5.1 Weyl Double Copy

The *Weyl double copy* [182–184] seeks to directly construct the Weyl curvature of four-dimensional algebraically special spacetimes by “squaring” electromagnetic field strengths via a choice of a scalar field (“zeroth copy field”) while not referencing a metric nor a gauge potential. It has been formulated in both position [182] and dual twistor [183, 184] spaces.

Below, we establish that the SD Taub-NUT solution is the Weyl double copy of the SD dyon in both position space and dual twistor space senses.

First of all, direct computation based on Eqs. (6.2) and (6.3) shows that the SD curvature tensors (field strength and Weyl) of SD dyon and SD Taub-NUT solutions are given by

$$F_{\dot{\alpha}\dot{\beta}} \propto \frac{Q}{4\pi R^2} \tilde{o}_{(\dot{\alpha}} \tilde{\iota}_{\dot{\beta})}, \quad C_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} \propto \frac{M}{4\pi R^3} \tilde{o}_{(\dot{\alpha}} \tilde{o}_{\dot{\beta}} \tilde{\iota}_{\dot{\gamma}} \tilde{\iota}_{\dot{\delta})}, \quad (6.20)$$

respectively. Of course, the ASD curvature tensors vanish. The Weyl double copy in position space refers to the squaring relation [182]

$$C_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}} = F_{(\dot{\alpha}\dot{\beta}} F_{\dot{\gamma}\dot{\delta})} / S. \quad (6.21)$$

We see that Eq. (6.21) holds for Eq. (6.20) with the choice $S \propto 1/R$.

Second of all, we see that the above S , $F_{\dot{\alpha}\dot{\beta}}$, and $C_{\dot{\alpha}\dot{\beta}\dot{\gamma}\dot{\delta}}$ satisfy the spin- h zero-rest-mass equations in flat background for $h = 0, 1, 2$, respectively. As such, they arise from the Penrose transform [607] as

$$\phi_{\dot{\alpha}_1 \dots \dot{\alpha}_{2h}}(x) = \oint \frac{\pi_{\dot{\gamma}} d\pi^{\dot{\gamma}}}{2\pi i} \frac{\pi_{\dot{\alpha}_1} \dots \pi_{\dot{\alpha}_{2h}}}{Q(\pi, \pi x)^{h+1}}. \quad (6.22)$$

⁴Cf. the GSF spinor equation [341, 606] is chiral.

Here, $Q(\pi, \pi x)$ evaluates the following dual twistor quadric [184, 608–610] on the dual twistor line:

$$Q(W) = \omega^\alpha \delta_{\alpha\dot{\alpha}} \pi^{\dot{\alpha}}, \quad \delta_{\alpha\dot{\alpha}} \propto u_\mu (\sigma^\mu)_{\alpha\dot{\alpha}}, \quad (6.23)$$

where $W^A = (\pi_{\dot{\alpha}}, \omega^\alpha)$. Note that the zeros of $Q(\pi, \pi x)$ encode the principal spinors $\tilde{o}_{\dot{\alpha}}, \tilde{l}_{\dot{\alpha}}$ as per the Kerr theorem [255, 606, 611]. From this analysis, we identify a relation between dual twistor space data,⁵

$$\mathcal{M}_h(W) = Q(W)^{-h-1} \implies \mathcal{M}_2(W) = (\mathcal{M}_1(W))^2 / \mathcal{M}_0(W). \quad (6.24)$$

Eq. (6.24) is the squaring relation for the Weyl double copy in dual twistor space [183, 184].

Previously, the above squaring relations for the SD Taub-NUT solution had been speculated on in Refs. [183, 184]. The argument was that the SD Taub-NUT solution would describe the “SD part” of Schwarzschild, so its SD Weyl curvature and principal spinors might be exactly equated to those of the Schwarzschild solution. Although physically plausible, it should be pointed out that this claim is strictly speaking a statement whose validity is restricted within the linearized level. Otherwise, it is a highly nontrivial assertion in light of the nonlinearity of GR demanding an extra effort for justification.

The clarification which our explicit KS metric in Eq. (6.3) brings is that Eq. (6.20) has in fact described the *exact* Weyl tensor of the SD Taub-NUT solution at the fully nonlinear level. The Cartesian coordinates $x^{\dot{\alpha}\alpha}$ of the flat background employed in the Penrose transform in Eq. (6.22) can be identified with the KS coordinates for the exact curved SD Taub-NUT geometry.⁶ Consequently, the Weyl double copy construction of the SD Taub-NUT solution is validated at the fully nonlinear orders.

Generally speaking, it seems that the validity of the Weyl double copy could only be strictly ensured at the nonlinear orders when it can be assisted by an underlying KS structure at the metric level, in which case one can appeal to the effective linearization property [177, 178].

⁵See Ref. [612] for a scattering amplitudes interpretation, where the dual twistor data via the Penrose transform is identified as the three-point scattering amplitude for absorbing one negative-helicity massless quantum, half-Fourier transformed to the dual twistor space.

⁶Note that the “position vector” $\vec{x}$ which one identifies from the Killing spinor as $\tilde{\chi}^{\dot{\alpha}}_{\dot{\beta}} = \frac{1}{2} (\vec{x} \cdot \vec{\sigma})^{\dot{\alpha}}_{\dot{\beta}}$ [261] also describes the (spatial part of) our KS coordinates.

6.5.2 Nonlocal Operator Kerr-Schild Double Copy

Another known instance of a classical double copy construction is the operator KS double copy [160, 161, 169]. The operator KS double copy promotes the null one-form ℓ_μ in the KS ansatz to a differential operator $\hat{k}_\mu$, which may factorize as $-\frac{1}{2}(\sigma^\mu)_{\alpha\dot{\alpha}}\hat{k}_\mu = o_\alpha\hat{k}_{\dot{\alpha}}$ so that the fields are manifestly put in the lightcone gauge [160, 161, 169].

Intriguingly, we find that an operator KS double copy for the SD Taub-NUT solution is viable only in an exotic *nonlocal* form. Eqs. (6.2) and (6.3) arise precisely if the operator $\hat{k}_{\dot{\alpha}}$ is given by

$$\hat{k}_{\dot{\alpha}} = \left(-\frac{\partial_x + i\partial_y}{\partial_z}, 1 \right), \quad (6.25)$$

which is nonlocal. From $\hat{k}_{\dot{\alpha}}(1/r) = (1/r)\tilde{o}_{\dot{\alpha}}$ and $\hat{k}_{\dot{\alpha}}\hat{k}_{\dot{\beta}}(1/r) = (1/r)\tilde{o}_{\dot{\alpha}}\tilde{o}_{\dot{\beta}}$, it follows that a double copy correspondence between SD dyon gauge potential and the KS graviton field $h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu}$ of SD Taub-NUT is realized as

$$A_{\alpha\dot{\alpha}} = o_\alpha\hat{k}_{\dot{\alpha}}\frac{q}{4\pi r}, \quad h_{\alpha\dot{\alpha}\beta\dot{\beta}} = o_\alpha o_\beta \hat{k}_{\dot{\alpha}}\hat{k}_{\dot{\beta}}\frac{m}{4\pi r}. \quad (6.26)$$

Notably, this new form of classical double copy admits a nice physical explanation from an electric-magnetic dual perturbation theory. The key fact is that the SD dyon can be viewed as an object in the “extended” Maxwell theory where the violation of Bianchi identity by *magnetic matter* $J^\star$ is allowed by the introduction of the dual gauge potential $A^\star$ [455]: $F = dA + *dA^\star$. In this picture, the distributional source for the SD dyon is

$$\begin{aligned} J^\mu &= \partial_\nu F^{\mu\nu} = qu^\mu \delta^{(3)}(\vec{x}), \\ J^{\star\mu} &= -\partial_\nu *F^{\mu\nu} = -iqu^\mu \delta^{(3)}(\vec{x}). \end{aligned} \quad (6.27)$$

We may redescribe these in the “helicity basis” as $J_+^\mu = J^\mu - iJ^{\star\mu} = 0$ and $J_-^\mu = J^\mu + iJ^{\star\mu} = 2qu^\mu \delta^{(3)}(\vec{x})$, while having $A_\mu^\pm := \frac{1}{2}(A_\mu \pm iA_\mu^\star)$ for the potentials.

As a dynamical field theory, a formulation of this extended Maxwell theory has been given by Zwanziger [456, 599–602]. We note that the propagators of this theory can be summarized into a single formula in the spinor notation:

$$\frac{\hat{\Delta}_{\alpha\dot{\alpha}\beta\dot{\beta}}^{+-}}{\square} = \frac{n_{\alpha\dot{\beta}}\partial_{\beta\dot{\alpha}}}{-n\cdot\partial} \frac{1}{\square}, \quad (6.28)$$

where n is an auxiliary four-vector that traces back to the Dirac string. Eq. (6.28) describes the position-space propagator from $A_{\beta\dot{\beta}}^-$ to $A_{\alpha\dot{\alpha}}^+$, while the $\langle A^- A^- \rangle$

and $\langle A^+ A^+ \rangle$ propagators vanish. A simple calculation then shows that the SD dyon gauge potential in Eq. (6.2) is precisely reproduced by this propagator, if one takes $n_{\alpha\dot{\alpha}} = -o_\alpha \bar{o}_{\dot{\alpha}}$ for the auxiliary vector so that $\hat{k}_{\dot{\alpha}} = -\partial_{1\dot{\alpha}}/\partial_{0\dot{0}}$.

Amusingly, the graviton field arises in the same fashion by using the linearized gravity equivalent of Zwanziger propagator, which is given by doubling Eq. (6.28) as is advocated in Ref. [602]. In this case, the sources in the two-potential theory are $T_+^{\mu\nu} = 0$ and $T_-^{\mu\nu} = 2mu^\mu u^\nu \delta^{(3)}(\vec{x})$. The resulting linearized metric perturbation happens to be also the full metric perturbation by taking the KS form [176–178].

In summary, the double copy structure proposed in Eq. (6.26) can be traced back to propagators between dyonic matter in electrodynamics and linearized gravity:

$$A_{\alpha\dot{\alpha}}^+ = \frac{\hat{\Delta}_{\alpha\dot{\alpha}\beta\dot{\beta}}^{+-}}{\square} J_-^{\beta\dot{\beta}}, \quad h_{\alpha\dot{\alpha}\beta\dot{\beta}}^+ = \frac{\hat{\Delta}_{\alpha\dot{\alpha}\gamma\dot{\gamma}}^{+-} \hat{\Delta}_{\beta\dot{\beta}\delta\dot{\delta}}^{+-}}{\square} T_-^{\gamma\dot{\gamma}\delta\dot{\delta}}. \quad (6.29)$$

A few comments are in order. Firstly, it is instructive to translate the propagator in Eq. (6.28) into the vector notation, where one can identify a combination of operators $(\vec{n} \times \vec{\partial} / \vec{n} \cdot \vec{\partial})$ connecting between the Gilbertian and Ampèrian implementations [613] of multipoles. This implies that the nonlocality of the operator in Eq. (6.25) has encoded the Dirac string structures. Secondly, our construction could be justified from the on-shell perspective. The “numerator” of the off-shell propagator in Eq. (6.28) correctly factorizes into the null polarization vectors so that the double copy of sources from $J_-^\mu = 2qu^\mu \delta^{(3)}(\vec{x})$ to $T_-^{\mu\nu} = 2mu^\mu u^\nu \delta^{(3)}(\vec{x})$ encodes the squaring relation of the massive on-shell amplitudes [206, 602, 614, 615]. In this perspective, the nonlocality traces back to that of the x -factor. Lastly, it is interesting to apply this approach to construct generic Taub-NUT spacetimes, in which case one naturally obtains a double KS metric perturbation.

6.5.3 Self-Dual Double Copy

Computation shows that our KS coordinates are also the Plebański coordinates [314, 320] for the second heavenly equation, on which the SD double copy [160, 161, 327] is based. The SD double copy could draw fascinating conclusions about the SD dyon and Taub-NUT solutions in the context of Moyal deformation of the heavenly equation [158, 159, 323–326]. Strangely, however, the SD Taub-NUT metric is not straightforwardly the SD double copy of SD dyon in terms of the Plebański scalar field: $A_{\alpha\dot{\alpha}} = o_\alpha \partial_{0\dot{\alpha}}(q\zeta/4\pi)$,

$h_{\alpha\dot{\alpha}\beta\dot{\beta}} = o_{\alpha}o_{\beta}\partial_{0\dot{\alpha}}\partial_{0\dot{\beta}}(m(2r+z)\zeta^2/24\pi)$. This puzzle deserves further investigation, in comparison with the Eguchi-Hanson case [161].

6.6 Conclusions

In this work, we derived a KS metric for the SD Taub-NUT solution by classical double copy and validated it by an explicit complexified coordinate transformation. The KS metric exhibits the Misner string defect, supporting its physical interpretation. Overall, we established three concrete senses in which the SD Taub-NUT solution is the gravitational counterpart of the SD dyon in electromagnetism: KS, Weyl, and nonlocal operator KS classical double copies. The last one adds to the inventory of classical double copies as a new entry, whose physical origin is propagators between dyonic matter.

The distinctive charm of this work lies in the exciting development along the very first three equations. The usual punchline of classical double copy has been starting with a gravitational solution and then deducing the electromagnetic counterpart. This work, in contrast, takes a reverse path: we constructed an electromagnetic solution and then applied the double copy to yield a new statement in GR.

It is also insightful to frame this work in the angle of “classical double copy of gravitational instantons.” The only known instance has been Ref. [161]’s work on the Eguchi-Hanson instanton solution [616–618], which traces back to Refs. [365, 619]. This work adds another instance to the discourse, namely the Taub-NUT instanton. It is amusing that the correspondence between SD objects in electromagnetism and gravity which the Gibbons-Hawking ansatz [579, 580] had suggested a long time ago becomes strengthened from new perspectives by the advent of the classical double copy program.

The KS metric of the SD Taub-NUT solution conveys an aesthetic elegance in terms of its minimalist and spinorial nature. We anticipate that this new metric will contribute to future studies on the SD Taub-NUT solution, both conceptually and practically. For further consequences or applications, we suggest the following directions: asymptotic symmetries [620, 621], integrability of test-particle motion [272, 582], black hole perturbation theory [240, 261, 266, 622], exact graviton propagators [623–625], or self-force calculations [626–629] in an idealized setup. Especially, explorations in the scattering context are likely to shed insights on the Kerr black hole scattering for an in-

tegrable subsector, on account of the Newman-Janis shift [202]: the “SD part” of Kerr is a SD Taub-NUT solution [174, 256, 257, 260, 261, 630].

6.A Double Kerr-Schild Metric

Consider the following double KS metric:

$$ds^2 = \left(\begin{array}{l} -dt^2 + dx^2 + dy^2 + dz^2 \\ + \frac{M_1}{4\pi|\vec{x}|} \left(-dt + \frac{x+iy}{|\vec{x}|+z} dx + \frac{y-ix}{|\vec{x}|+z} dy + dz \right)^2 \\ + \frac{M_2}{4\pi|\vec{x}|} \left(-dt + \frac{x-iy}{|\vec{x}|-z} dx + \frac{y+ix}{|\vec{x}|-z} dy - dz \right)^2 \end{array} \right). \quad (6.30)$$

Eq. (6.30) is a double KS metric for the SD Taub-NUT solution with mass $M_1 + M_2$ according to Plebański and Demiański [332] (which arises by a zooming-in procedure taking $a \rightarrow \infty$).

It can be explicitly shown that Eq. (6.30) is diffeomorphic to a Gibbons-Hawking metric with mass $M_1 + M_2$. Explicitly, apply the diffeomorphism

$$\begin{aligned} t &\mapsto t + \frac{M_1}{4\pi} + \frac{M_2}{4\pi}, \\ x &\mapsto x + \frac{M_1}{4\pi} \frac{x+iy}{|\vec{x}|+z} + \frac{M_2}{4\pi} \frac{x-iy}{|\vec{x}|-z}, \\ y &\mapsto y + \frac{M_1}{4\pi} \frac{y-ix}{|\vec{x}|+z} + \frac{M_2}{4\pi} \frac{y+ix}{|\vec{x}|-z}, \\ z &\mapsto z + \frac{M_1}{4\pi} - \frac{M_2}{4\pi}, \end{aligned} \quad (6.31)$$

which induces $|\vec{x}| \mapsto |\vec{x}| + (M_1 + M_2)/4\pi$. Evidently, Eq. (6.31) arises by taking the sum of the two NVFs involved in Eq. (6.30). Computation shows that Eq. (6.31) maps Eq. (6.30) to the Gibbons-Hawking metric for

$$V = 1 + \frac{(M_1 + M_2)}{4\pi|\vec{x}|}, \quad a = a_1 + a_2, \quad (6.32)$$

where

$$a_1 = \frac{M_1}{4\pi|\vec{x}|} \left(\frac{x+iy}{|\vec{x}|+z} dx + \frac{y-ix}{|\vec{x}|+z} dy + dz \right), \quad (6.33a)$$

$$a_2 = \frac{M_2}{4\pi|\vec{x}|} \left(\frac{x-iy}{|\vec{x}|-z} dx + \frac{y+ix}{|\vec{x}|-z} dy - dz \right). \quad (6.33b)$$

In spherical coordinates, Eq. (6.33) boils down to

$$a_1 = \frac{M_1}{4\pi} \left(\frac{1}{R} dR - \tan \frac{\theta}{2} d\theta - i(1 - \cos \theta) d\phi \right), \quad (6.34a)$$

$$a_2 = \frac{M_2}{4\pi} \left(\frac{1}{R} dR + \cot \frac{\theta}{2} d\theta + i(1 + \cos \theta) d\phi \right), \quad (6.34b)$$

from which it is clear that $da = d(a_1 + a_2) = -i((M_1 + M_2)/4\pi) \sin \theta d\theta \wedge d\phi$.

A corollary is that Eq. (6.30) for $M_2 = -M_1$ provides a double KS representation of empty spacetime.

6.B A Puzzle Regarding Two Classical Double Copies

The amplitudes-level double copy of BCJ [87] implies that the perturbative (Duff [631]) constructions of classical solutions in YM theory and gravity can exhibit a correspondence through replacing the color Lie algebra with a kinematic Lie algebra, whose identity is known in the SD sector [327] but not beyond. In the meantime, the classical double copy originally proposed by Monteiro and O’Connell [160] has resorted to the effective linearization property [177, 178] of KS metrics.

To our best understanding, an explicit demonstration of the former implying the latter has been not given so far. First, the KS double copy program has not discussed a faithful diagram-by-diagram mapping at the nonlinear orders. Second, such a mapping is unlikely to be identified easily after all, due to the mysterious status of the kinematic algebra beyond the SD sector. Hence to address the question of whether the amplitudes double copy implies the classical double copy, it could be more promising to first focus on the SD sector (cf. Ref. [632]).

In this context, Ref. [163] identified a puzzle that there seems to be a disagreement between the amplitudes double copy and the classical double copy. On the one hand, Ref. [163]’s KS metric shows that the SD Taub-NUT solution is the classical double copy [160] of the SD dyon, which describes a correspondence between type [1,1] and type D solutions. On the other hand, the amplitudes double copy via the second heavenly equation [160] works by taking the same Plebański scalar for both gauge theory and gravity up to attaching a constant color factor. Ref. [163] explicitly constructs the Plebański scalars for the SD dyon and the SD Taub-NUT solutions and finds mismatch.

Taking lessons from a recent work [175], we realize that the mapping stip-

ulated by the amplitudes double copy generically describes a correspondence between type [2] (null Maxwell field) and type D solutions. This is because Tod [365]’s construction essentially concerns the linearized case of the second heavenly equation. This validates the mismatch pointed out in this chapter [163]. Consequently, we learn that all SD black holes seem to have “two square roots.”

NEWMAN-JANIS ALGORITHM FROM TAUB-NUT INSTANTONS

It is shown that the Kerr metric represents the nonlinear superposition of self-dual and anti-self-dual Taub-NUT instantons. This promotes the Newman-Janis algorithm to a rigorous derivation of the Kerr metric with a definite physical origin. In the same way, the Kerr-Newman and charged Kerr-Taub-NUT solutions are systems of Taub-NUT instantons and chiral dyons.

This chapter reproduces the contents of Ref. [164], J.-H. Kim, “Newman-Janis algorithm from Taub-Newman-Unti-Tamburino instantons,” *Phys. Rev. Lett.* **136** no. 23, (2026) 231401, [arXiv:2412.19611 \[gr-qc\]](#).

7.1 Introduction

The NJA [202] is a mystery in the history of relativity. Discovered about sixty years ago, the NJA aims to derive the rotating black hole solution of Kerr [633] by applying a “complex coordinate transformation” to the static black hole solution of Schwarzschild. Apparently, this procedure is a sequence of ad hoc manipulations. Each element in the Schwarzschild metric must be promoted to specific combinations of complex variables, which are then transformed by a rule of unknown physical origin. Hence the NJA has stood merely as a formal mathematical trick.

Such a lack of clear understanding is unsettling because the NJA holds the historical significance as the very method that led to the discovery of the Kerr-Newman metric [179]: the charged rotating black hole solution.

Despite early skepticism—epitomized by characterizations such as a “fluke” [243] or “methods which transcend logic” [634]—research efforts have persisted over decades toward a satisfactory justification [98, 163, 174, 207–209, 226, 242–260, 329, 330, 479, 634–673]. These works may be classified into three categories, as outlined below.

The first category could be named *classical explanations*. Here, the goal is to justify the NJA from geometrical perspectives. This means to resolve inherent ambiguities, identify hidden assumptions, and explore simplifications

or generalizations. Notable contributions in this direction include works by Talbot [242], Drake and Szekeres [243], Gürses and Gürsey [246], Flaherty [247, 330], Rajan and Visser [248], and Giampieri [249]. The original article [202] also provides a brief justification along this direction: an allowed possibility in Kerr theorem [611]. These approaches could be agnostic about physical origins, however.

The second category describes *the works of Newman himself* [252–259, 329, 330], which portray a more serious stance. Emphasis is put on the fact that the NJA realizes the spin of Kerr black hole via an imaginary displacement into complexified spacetime. The NJA is treated not as a mere mathematical curiosity but a signal of a deeper physical structure, such as a Hodge duality on angular momenta as mass dipole moments [252]. A fundamental unification of spin and spacetime is envisioned from the construction of “complex center of mass” [252–259, 330, 672].

Lastly, the third category describes the *modern explanation* emerged through works [98, 206–209]. Here, one adopts a particle physicist’s perspective and imagines observing the Kerr black hole by throwing gravitons (i.e., through gravitational waves) from far away, as if one detects the internal structure of a proton by firing electrons in a particle collider. In terms of scattering cross-sections, it is revealed that the Kerr black hole behaves like an ideally pointlike object when the incident gravitons are all circularly polarized in either right or left handedness, i.e., self-dual (SD) or anti-self-dual (ASD) [98, 207–212, 221, 222]. Thus, despite the black hole’s extended geometrical features such as the ring singularity or the horizon, its “graviton X-ray image” displays just two objects: a point that absorbs SD gravitons only and another point that absorbs ASD gravitons only. This stunning simplicity is identified as the basis of NJA [98], as the coordinates of such points precisely predict Newman’s complex centers of mass by taking complex values.

The modern explanation shows that the NJA is not a mere mathematical construct but rather a representation of some physical structure: the NJA is recast as a statement about scattering amplitudes, a gauge-invariant physical observable. That said, it did not decipher every aspect of the NJA in its original, relativist form as a statement about bulk metrics. The conversion of the scattering amplitudes to bulk metrics is based on perturbative methods [142, 176–178, 631], so a resummed picture is lacking.

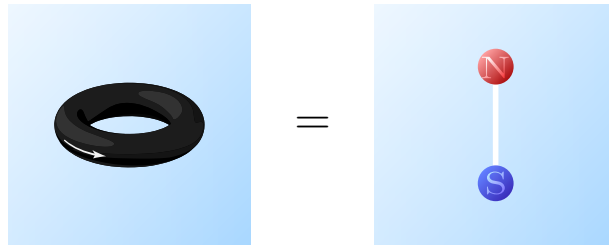


Figure 7.1: In electromagnetism, a current loop behaves like a pair of opposite monopoles: Ampèrian and Gilbertian dipoles [613]. This letter reveals a gravitational version of this duality.

A similar caveat applies to the fact attributed to Crawley, Guevara, Miller, and Strominger [260] or Ghezelbash, Mann, and Sorkin [630], which is limited to the SD sector and thus fall short of fully explaining the nonlinear aspect of NJA with both SD and ASD components.

In this letter, we provide a principled derivation of the Kerr metric that faithfully reproduces the NJA in its original form. This not only identifies a definite physical origin for structures observed in the classical explanations and Newman’s works but also reveals a fully nonlinear, nonperturbative picture completing the modern explanation.

The key new finding is that the Kerr metric describes a pair of SD and ASD Taub-Newman-Unti-Tamburino (NUT) instantons, which are known to be gravitational analogs of chiral dyons. These instantons are exactly the ideally pointlike objects in the field theorists’ graviton X-ray image. The Kerr metric is derived by establishing a mathematical theorem about nonlinear superposition of Kerr-Schild (KS) spacetimes. This builds upon a crucial recent finding [163] and a modern framework known as KS double copy [160]. Our construction generalizes to charged Kerr-Taub-NUT solutions, thus validating the historical derivation [179] of the Kerr-Newman metric as well.

7.2 Intuition and Overview

The prototype of this chapter grew out of a moment of pure imagination that struck the present author some years ago. In electromagnetism, a magnetic dipole arises from not only an Ampèrian current loop but also a static monopole pair, like the N and S poles of a bar magnet (Fig. 7.1). By analogy, the question was whether the ring singularity of Kerr black hole can be dualized into a pair of gravitomagnetic monopoles.

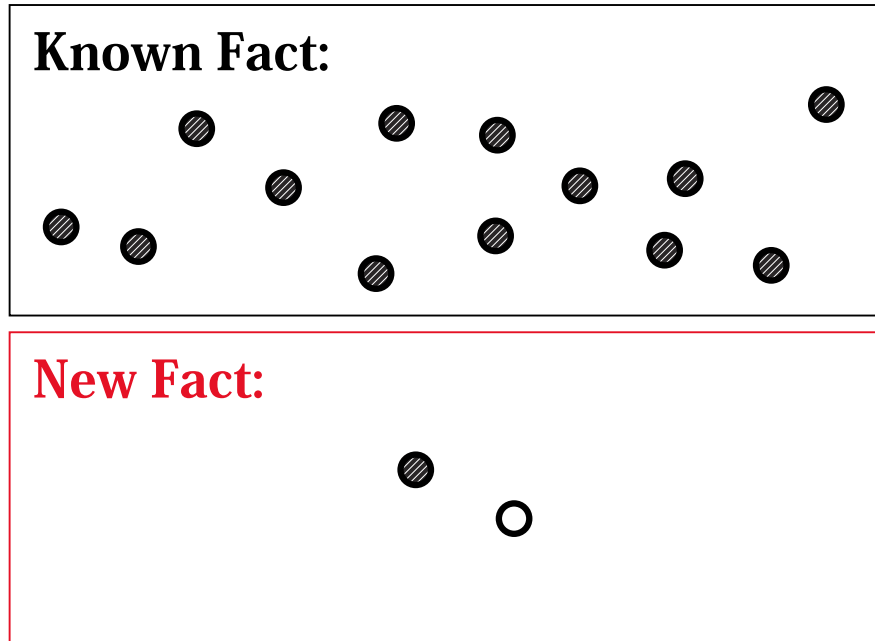


Figure 7.2: The established fact due to Gibbons and Hawking [579, 580] concerns the nonlinear superposition of Taub-NUT instantons of the same chirality. The new fact proved in this work concerns the nonlinear superposition of two Taub-NUT instantons of the opposite chirality.

It is known [331, 474–476, 479, 485, 674] that the gravitational analog of the monopole is the Taub-NUT [596, 597] solution. This is a stationary vacuum solution characterized by two parameters: mass M and magnetic mass N , which are gravitoelectric and gravitomagnetic charges, respectively. The metric of the Taub-NUT solution develops a semi-infinite line defect as the gravitational Dirac string, dubbed Misner string [474, 475].

A modern framework known as KS double copy [160] has elevated such analogies to exact mathematical correspondences between stationary vacuum solutions in Maxwell theory and GR. For example, the Coulomb solution is the counterpart of the Schwarzschild solution. It has been recently shown [163] that this correspondence applies to the Taub-NUT solution as well, yet only in chiral (SD or ASD) limits in which $M = \pm iN$.

The chiral limits of the Taub-NUT solution are referred to as Taub-NUT instantons. Historically, they were introduced by the seminal works of Gibbons and Hawking [579, 580] as paradigmatic instances of gravitational instantons; hence the name.

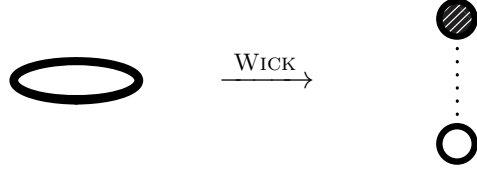


Figure 7.3: The ring singularity of Kerr black hole factorizes into a pair of SD and ASD Taub-NUT instantons.

A previously *known fact* is that any number of Taub-NUT instantons of the same chirality can be put together to yield a multi-centered exact solution [579, 580]. The *new fact* we prove is that SD and ASD Taub-NUT instantons can form a two-centered exact solution: opposite chiralities. Remarkably, this two-centered solution is nothing other than the Kerr solution.

7.3 Factorization of Kerr

As an introduction, we begin with an innocuous mathematical fact about a quartic polynomial. Consider the zero locus

$$(x^2 + y^2 + z^2 - a^2)^2 + (2az)^2 = 0, \quad (7.1)$$

where a is a constant. Eq. (7.1) describes a *ring* of radius a in the three-dimensional space $(x, y, z) \in \mathbb{R}^3$. Intriguingly, this ring can also appear as a set of *two disjoint points*. Suppose we allow x, y, z in Eq. (7.1) to take complex values so that we obtain a complex algebraic variety in the affine space $\mathbb{C}^3$. Then the ring is the slice of this variety by the real section $x, y, z \in \mathbb{R}$. However, consider the slice $x, y, z \in i\mathbb{R}$. Wick-rotating x, y, z in Eq. (7.1) gives

$$(x^2 + y^2 + z^2 + a^2)^2 - (2az)^2 = 0, \quad (7.2)$$

which factorizes to $(x^2 + y^2 + (z + a)^2)(x^2 + y^2 + (z - a)^2) = 0$. This admits just two points as solutions: $(0, 0, \pm a)$.

The astute reader will point out that Eq. (7.1) is the very equation of the ring singularity of the Kerr black hole. The Kerr metric is characterized by mass M and ring radius a . In KS coordinates, for instance, it reads

$$\begin{aligned} & -dt^2 + dx^2 + dy^2 + dz^2 \\ & + \frac{2Mr^3}{r^4 + a^2z^2} \left(-dt + \frac{rx - ay}{r^2 + a^2} dx + \frac{ry + ax}{r^2 + a^2} dy + \frac{z}{r} dz \right)^2. \end{aligned} \quad (7.3)$$

Here, r is a function of x, y, z implicitly defined by

$$\frac{x^2 + y^2}{r^2 + a^2} + \frac{z^2}{r^2} = 1. \quad (7.4)$$

The contour surfaces of r are oblate spheroids, which degenerate to a disc of radius a at $r = 0$. The boundary of this disc is exactly the ring singularity shown in Eq. (7.1).

By analytic continuation, we regard Eq. (7.3) as a holomorphic metric on a complex four-manifold solving complexified vacuum Einstein's equations (cf. Refs. [314, 315, 331, 332]). Then the total Wick rotation, $x^\mu \mapsto ix^\mu$, is an allowed coordinate change. The resulting line element is

$$dt^2 - dx^2 - dy^2 - dz^2 + \frac{2iM\tilde{r}^3}{\tilde{r}^4 - a^2z^2} \left(dt - \frac{\tilde{r}x + iay}{\tilde{r}^2 - a^2} dx - \frac{\tilde{r}y - iax}{\tilde{r}^2 - a^2} dy - \frac{z}{\tilde{r}} dz \right)^2, \quad (7.5)$$

where $\tilde{r}$ is a function of the transformed x, y, z :

$$\frac{x^2 + y^2}{\tilde{r}^2 - a^2} + \frac{z^2}{\tilde{r}^2} = 1. \quad (7.6)$$

The level sets of $\tilde{r}$ are now prolate spheroids. The line element in Eq. (7.5) is ill-defined over a “needle” along $x, y = 0$ and $z \in [-a, a]$, which is the degenerate prolate spheroid at $\tilde{r} = a$. Curvature invariants imply that the boundary of this needle is only truly singular, which exactly describes the set of two points specified by Eq. (7.2).

Direct computation shows that these point singularities are Taub-NUT instantons of opposite chiralities. By taking $z \mapsto z \pm a$ and sending $a \rightarrow \infty$, one finds that the metric in the vicinity of the upper/lower tip of the needle is exactly the SD/ASD Taub-NUT metric of Ref. [163].

Direct computation also shows that the needle is a Misner string that transports the magnetic mass flux from one instanton to another. The metric near any point on the needle can be represented as $(dt - 2iM d\log(x - iy))^2 - dx^2 - dy^2 - dz^2$ ¹, which describes the image of flat spacetime under a large diffeomorphism. Such a line defect of time monodromy is the very notion of Misner string as a gravitomagnetic flux tube [163, 172, 486, 521, 566, 674].

¹The line element diverges when zooming to an arbitrary point $z = \eta a \in (-a, a)$. However, it can be compensated by a diffeomorphism $(x + iy) \mapsto (x + iy) - 2iM((1 - \eta^2)a - 2iM - 2\eta z)/(x - iy)$, $(x - iy) \mapsto (x - iy)$.

In sum, we have explicitly shown that the Kerr metric, as a *whole*, represents a pair of SD and ASD Taub-NUT instantons. In Fig. 7.3, the instantons are visualized as blobs while the Misner string is drawn as a dotted line:

$$\text{Kerr} = \bullet + \circ. \quad (7.7)$$

It is easily verified that their masses, magnetic masses, and center positions are given as below (if measured with respect to the slice of mostly-positive metric signature):

$$\begin{cases} \text{SD instanton, } \bullet : (M/2, -iM/2) \text{ at } \vec{x} = +i\vec{a}, \\ \text{ASD instanton, } \circ : (M/2, +iM/2) \text{ at } \vec{x} = -i\vec{a}. \end{cases} \quad (7.8)$$

Note that the magnetic mass dipole moment computes to $\vec{J} = (-iM/2)(2i\vec{a}) = M\vec{a}$, which is precisely the angular momentum of the Kerr black hole. In fact, the static configuration in Eq. (7.8) reproduces all the black hole's spin-induced mass multipole moments for finite $\vec{a}$, which are known to take the same magnitude with alternating signs [200–204, 256]².

The separation $2i\vec{a}$ has to be pure-imaginary for $\vec{J}$ to be real, since Taub-NUT instantons carry imaginary magnetic mass in Lorentzian signature. In fact, we will learn that the imaginary coordinates $\pm i\vec{a}$ are exactly the imaginary displacements of the NJA. Therefore, we realize that the NJA is a statement about the curvature singularity of Kerr solution when taken as a complex saddle.

It should be stressed that our analysis above does not count on any methods of linearization or perturbative expansions. It is readily performed in other coordinate systems as well, such as the Boyer-Lindquist system.

To our knowledge, the above explicit proof of the exact equivalence between Kerr and a dipolar configuration of Taub-NUT instantons is new, although an argument appears in Ref. [675]. Our result is also disparate from the “black dihole” of Refs. [676–688].

7.4 Nonlinear Superposition Theorem

Next, we regard the Taub-NUT instantons as *parts* and show that they can be assembled together to form the Kerr black hole:

$$\bullet + \circ = \text{Kerr}. \quad (7.9)$$

² $(M/2)(ia)^{2k} + (M/2)(ia)^{2k} = (-1)^k Ma^{2k}$ for electric 2^{2k} -pole moments and $(-iM/2)(ia)^{2k+1} + (iM/2)(-ia)^{2k+1} = (-1)^k Ma^{2k+1}$ for magnetic 2^{2k+1} -pole moments.

A crucial feature of Einstein's gravity theory, however, is nonlinearity of field equations. Hence generally speaking, it is virtually impossible to assemble two vacuum solutions together to obtain another by a simple process. The $+$ symbol in Eq. (7.9) signifies a *nonlinear* sum.

Yet remarkably, the Taub-NUT instanton solution admits a KS metric [163] as mentioned earlier. For instance, the KS metric of a SD Taub-NUT instanton is given in an elegant formula in the spinor notation:

$$ds^2 = ds_{\text{flat}}^2 + \frac{M}{|\vec{x}|} \left(\eta_\alpha \tilde{o}_{\dot{\alpha}} dx^{\dot{\alpha}\alpha} \right)^2. \quad (7.10)$$

Here, the ASD spinor η_α is an arbitrary constant reference that controls the direction of the Misner string (as an ASD null plane), whereas the SD spinor $\tilde{o}_{\dot{\alpha}}$ describes a “square root” of the position three-vector $\vec{x}$ ³. That is, the gauge and physical data are respectively stored in the ASD and SD spinors. The Weyl tensor is purely SD. See Ref. [163] for the derivation of Eq. (7.10) from the well-known Gibbons-Hawking metric [579, 580].

Notably, KS metrics are known to enjoy an *effective linearization* property: the exact vacuum Einstein's equations are solved if the linearized equations are solved [176–178]. Via this effective linearization, two vacuum KS metrics can be assembled into another as shown below.

First of all, a NVF ℓ in complexified Minkowski space is *canonical* if it is stationary, GSF, and unit-normalized as $\ell^0 = 1$ with respect to the time direction. One can treat canonical NVFs as projective objects, due to the fixed normalization $\ell^0 = 1$.

Each canonical NVF is associated with two solutions. The *single copy solution* is the electromagnetic gauge potential $A_\mu = \theta \ell_\mu$. The *double copy solution* is the metric $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon \theta \ell_\mu \ell_\nu$, where ϵ is an auxiliary parameter. Provided that $\theta = \partial_\mu \ell^\mu$ is the expansion of the congruence of ℓ , A_μ solves vacuum Maxwell's equations in flat spacetime while $g_{\mu\nu}$ solves vacuum Einstein's equations.

The above set of definitions is our formulation of the framework known as KS double copy [160].

Next, we present what we call the *superposition lemma*. Suppose two canon-

³Concretely, $\tilde{o}^{\dot{\alpha}}$ is the eigenspinor of the Killing spinor $\tilde{\chi}^{\dot{\alpha}}{}_{\dot{\beta}} = \frac{1}{2}(\vec{x} \cdot \vec{\sigma})^{\dot{\alpha}}{}_{\dot{\beta}}$ whose eigenvalue is $+|\vec{x}|/2$.

ical NVFs ℓ_1 and ℓ_2 with expansions θ_1 and θ_2 . Then

$$\ell^{\dot{\alpha}\alpha} \propto \ell_1^{\dot{\alpha}\beta} \delta_{\beta\dot{\beta}} \ell_2^{\dot{\beta}\alpha}, \quad \tilde{\ell}^{\dot{\alpha}\alpha} \propto \ell_2^{\dot{\alpha}\beta} \delta_{\beta\dot{\beta}} \ell_1^{\dot{\beta}\alpha} \quad (7.11)$$

are canonical NVFs with expansions θ and $\tilde{\theta}$ such that

$$\theta_1 \ell_{1\mu} + \theta_2 \ell_{2\mu} = \theta \ell_\mu + \tilde{\theta} \tilde{\ell}_\mu + \partial_\mu \chi, \quad (7.12)$$

$$\theta_1 \ell_{1\mu} \ell_{1\nu} + \theta_2 \ell_{2\mu} \ell_{2\nu} = \theta \ell_\mu \ell_\nu + \tilde{\theta} \tilde{\ell}_\mu \tilde{\ell}_\nu + 2 \partial_{(\mu} \xi_{\nu)}, \quad (7.13)$$

and $\theta_1 + \theta_2 = \theta + \tilde{\theta}$. Here, $\delta_{\beta\dot{\beta}} = (\sigma^0)_{\beta\dot{\beta}}$ is the stationary direction boiled down into the spinor notation. χ and ξ_ν are some scalar and one-form fields.

Suppose $\tilde{\theta}$ happens to vanish. Then Eq. (7.12) implies that the single copy solution of ℓ describes the superposition of the single copy solutions of ℓ_1 and ℓ_2 . Similarly, Eq. (7.13) implies that the double copy solution of ℓ describes the nonlinear superposition of the double copy solutions of ℓ_1 and ℓ_2 . First, the linear sum $\theta_1 \ell_{1\mu} \ell_{1\nu} + \theta_2 \ell_{2\mu} \ell_{2\nu}$ is gauge-equivalent to $\theta \ell_\mu \ell_\nu$ in linearized gravity. Second, $\theta \ell_\mu \ell_\nu$ uplifts to an exact solution by the effective linearization property [176–178] of KS metrics.

The *nonlinear superposition theorem* is the following: if two canonical NVFs ℓ_1 and ℓ_2 are respectively associated with SD and ASD single copy solutions, then the double copy solution of ℓ in Eq. (7.11) describes the nonlinear superposition of the double copy solutions of ℓ_1 and ℓ_2 . This is because $\tilde{\theta}$ vanishes.

The proofs are detailed in the supplemental material⁴, but we shall remark here that the chiral nature of Eq. (7.11) plays an important role: the SD (ASD) spinor index $\dot{\alpha}$ (α) is controlled by the SD (ASD) solution's NVF ℓ_1 (ℓ_2). Also, Eq. (7.11) translates into the vector notation as

$$\ell^\mu = \frac{\ell_1^\mu + \ell_2^\mu + \ell_1 \cdot \ell_2 u^\mu + i \varepsilon^\mu{}_{\nu\rho\sigma} u^\nu \ell_1^\rho \ell_2^\sigma}{2 + \ell_1 \cdot \ell_2}, \quad (7.14)$$

which describes a highly nonlinear formula.

7.5 Derivation of Kerr Metric

The nonlinear superposition theorem provides a *derivation* of Kerr metric (without quotation marks).

First, note that the SD Taub-NUT instanton metric in Eq. (7.10) is the double copy solution of the canonical NVF $\ell^{\dot{\alpha}\alpha} \propto \tilde{\partial}^{\dot{\alpha}} \eta^\alpha$, whose single copy solution is SD [163].

⁴The supplemental material includes Refs. [341, 344–346, 606].

Second, prepare a SD Taub-NUT instanton centered at $\vec{x}_1$ and an ASD Taub-NUT instanton centered at $\vec{x}_2$. Their canonical NVFs are $\ell_1^{\dot{\alpha}\alpha} \propto \tilde{o}_1^{\dot{\alpha}} \eta^\alpha$ and $\ell_2^{\dot{\alpha}\alpha} \propto \tilde{\eta}^{\dot{\alpha}} o_2^\alpha$, where $\tilde{o}_1^{\dot{\alpha}}(\vec{x}) = \tilde{o}^{\dot{\alpha}}(\vec{x} - \vec{x}_1)$ and $o_2^\alpha(\vec{x}) = o^\alpha(\vec{x} - \vec{x}_2)$ are “square roots” of $\vec{x} - \vec{x}_{1,2}$. The expansions are $\theta_{1,2} = |\vec{x} - \vec{x}_{1,2}|^{-1}$, as should be clear from Eq. (7.10).

Third, construct their nonlinear superposition via the nonlinear superposition theorem. Eq. (7.11) gives

$$\ell^{\dot{\alpha}\alpha} \propto \tilde{o}_1^{\dot{\alpha}} \eta^\beta \delta_{\beta\dot{\beta}} \tilde{\eta}^{\dot{\beta}} o_2^\alpha \propto \tilde{o}_1^{\dot{\alpha}} o_2^\alpha, \quad (7.15)$$

which simply transvects the physical spinors $\tilde{o}_1^{\dot{\alpha}}$ and o_2^α . The expansion is $\theta = \theta_1 + \theta_2 = |\vec{x} - \vec{x}_1|^{-1} + |\vec{x} - \vec{x}_2|^{-1}$.

Finally, the new vacuum spacetime is obtained as

$$\partial_{\text{flat}}^2 - M \left(\frac{1}{|\vec{x} - \vec{x}_1|} + \frac{1}{|\vec{x} - \vec{x}_2|} \right) \left(\frac{\tilde{o}_1^{\dot{\alpha}} o_2^\alpha}{o_2^\gamma \delta_{\gamma\dot{\gamma}} \tilde{o}_1^{\dot{\gamma}}} \partial_{\alpha\dot{\alpha}} \right)^2, \quad (7.16)$$

where we have written down the inverse metric for a direct comparison with Ref. [202] while setting $\epsilon = M$.

Remarkably, the vacuum solution in Eq. (7.16) is exactly the Kerr solution with mass M and angular momentum $\vec{J} = -iM(\vec{x}_1 - \vec{x}_2)/2$, centered around $(\vec{x}_1 + \vec{x}_2)/2$.

In particular, take $\vec{x}_1 = +i\vec{a}$ and $\vec{x}_2 = -i\vec{a}$ for $\vec{a} = (0, 0, a)$. Then Eq. (7.16) exactly coincides with the Kerr inverse metric obtained in the very article [202].

This establishes a mathematically rigorous derivation of the Kerr solution as the nonlinear superposition of SD and ASD Taub-NUT instantons.

7.6 Newman-Janis Algorithm Deciphered

It remains to explicate that the above derivation faithfully reproduces the NJA. Consider the inverse metric of the Schwarzschild solution, identified as the nonlinear superposition of SD and ASD Taub-NUT instantons overlapped at the same point:

$$\partial_{\text{flat}}^2 - M \left(\frac{1}{|\vec{x}|} + \frac{1}{|\vec{x}|} \right) \left(\frac{\tilde{o}^{\dot{\alpha}} o^\alpha}{o^\gamma \delta_{\gamma\dot{\gamma}} \tilde{o}^{\dot{\gamma}}} \partial_{\alpha\dot{\alpha}} \right)^2. \quad (7.17)$$

Each term here acquires a unique meaning as per the nonlinear superposition theorem, so there exists a unique way of deforming Eq. (7.17) such that the

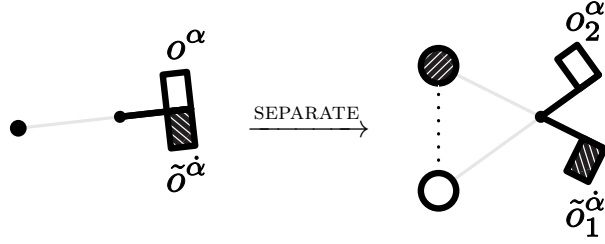


Figure 7.4: In an accurate sense, the NJA first decomposed a Schwarzschild black hole into SD (●) and ASD (○) Taub-NUT instantons and then moved them independently.

instantons get displaced to $\vec{x}_1 = +i\vec{a}$ and $\vec{x}_2 = -i\vec{a}$. Eq. (7.15) dictates the transformation of the spinors: the SD (ASD) spinor shifts by $+i\vec{a}$ ($-i\vec{a}$). The KS potential is mandated to be the total expansion: $\theta_1 + \theta_2 = |\vec{x} - i\vec{a}|^{-1} + |\vec{x} + i\vec{a}|^{-1}$. Thus, one unambiguously generates the Kerr inverse metric in Eq. (7.16) from the Schwarzschild inverse metric in Eq. (7.17). See Fig. 7.4 for a visualization.

This operation precisely reproduces the NJA in its very original formulation [202], as elaborated in the supplemental material. In particular, Eqs. (4) and (7) of Ref. [202] are *literally* reproduced upon using spheroidal coordinates.

In Ref. [202], the $\pm i\vec{a}$ imaginary shifts of the spinors (null tetrad) and the “complexification” of the KS potential $2/|\vec{x}| \rightarrow 2\text{Re}|\vec{x} - i\vec{a}|^{-1}$ are posited as ad hoc rules. The nonlinear superposition theorem shows that this set of replacements, i.e., the NJA, is a strict implication of the fact that the Schwarzschild-Kerr family of solutions are systems of SD and ASD Taub-NUT instantons.

This refined understanding implies the following clarifications on the NJA.

First, the NJA is *not* a coordinate transformation as originally conceived [202]. The instanton pairs with zero and nonzero separations, i.e., Schwarzschild and Kerr, are never diffeomorphic.

Second, the NJA is not about one object as originally conceived [256, 257], but *two*. It does not merely shifts a Schwarzschild black hole but instead *splits* it into SD and ASD parts and then *separates* them, as is illustrated in Fig. 7.4. The displacements $\vec{x}_1, \vec{x}_2$ are also *independent*, although a reality condition can be imposed a posteriori.

Third, it is important to explicitly address the nonlinear gravitational interaction between the SD and ASD instantons. Otherwise the NJA is not fully explained nor justified as a method that generates an exact solution in Ein-

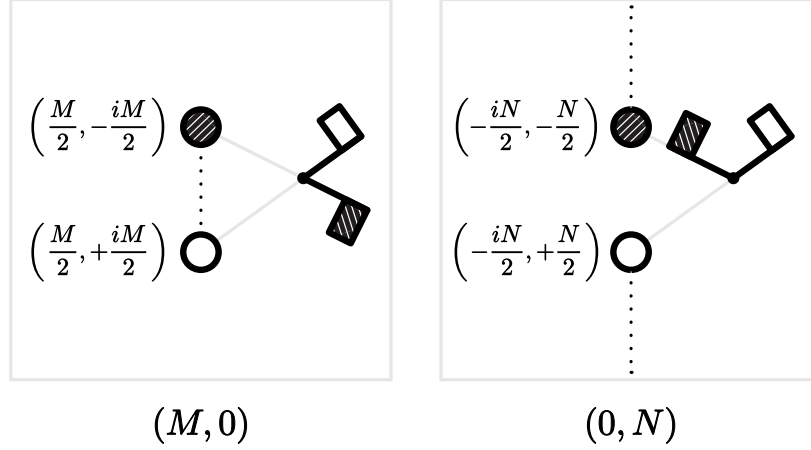


Figure 7.5: Implementing pure mass (*left*) and magnetic mass (*right*) with Taub-NUT instantons.

stein gravity. Newman’s complex center of mass analysis [252, 256, 257], for instance, identifies one worldline from the SD sector (in linearized gravity) and does not elaborate much on the validity of “plus complex conjugate” in the nonlinear theory. Works [260, 630] have shown that the SD Kerr-Taub-NUT solution describes a SD Taub-NUT instanton, which is not a statement about the Kerr solution which the NJA concerns. To reiterate, the NJA is about *two* Taub-NUT instantons, not just one SD instanton.

The technical structure of the formula in Eq. (7.14) may have been effectively observed in works [244, 246, 248], but the instanton pair interpretation is not realized there.

7.7 Newman-Janis Algorithm for Charged Kerr-Taub-NUT Solution

Finally, we construct more general black holes. First of all, the electric-magnetic dual version of Kerr solution arises as

$$- \bullet + \circ = \text{Kerr}^*. \quad (7.18)$$

The nonlinear superposition theorem admits no freedom for relative coefficients, but there is a binary choice for the NVF per each instanton. In particular, the KS metric of SD instanton can be realized in either outgoing ($\ell^{\dot{\alpha}\alpha} \propto \bar{\partial}^{\dot{\alpha}}\eta^\alpha$) or ingoing ($\ell^{\dot{\alpha}\alpha} \propto \tilde{\iota}^{\dot{\alpha}}\eta^\alpha$) conventions. Switching to ingoing flips the sign of the expansion, and in this sense the instantons can be subtracted as in Eq. (7.18).

Explicitly, the metric for $\ell^{\dot{\alpha}\alpha} \propto \tilde{l}_1^{\dot{\alpha}} o_2^\alpha$ is

$$dt^2 - dx^2 - dy^2 - dz^2 - \frac{2Na\tilde{r}z}{\tilde{r}^4 - a^2z^2} \left(dt + \frac{\tilde{r}}{a} \left(\frac{zx + i\tilde{r}y}{\tilde{r}^2 - z^2} dx + \frac{zy - i\tilde{r}x}{\tilde{r}^2 - z^2} dy - dz \right) \right)^2, \quad (7.19)$$

which describes a massless black hole with magnetic mass N and gravitoelectric dipole moment $-Na$. The exercise of zooming to each part of this metric can be repeated to explicitly verify the nonlinear superposition; see Fig. 7.5.

By adding up the two KS perturbations in Eqs. (7.5) and (7.19), one obtains the generic three-parameter Kerr-Taub-NUT solution in the double KS form [181, 331, 598, 689]⁵. The two canonical NVFs $\tilde{o}_1^{\dot{\alpha}} o_2^\alpha$ and $\tilde{l}_1^{\dot{\alpha}} o_2^\alpha$ share a common spinor o_2^α , making them mutually orthogonal.

Meanwhile, the Maxwell stress-energy tensor $T^{\dot{\alpha}\alpha\dot{\beta}\beta} = \tilde{F}^{\dot{\alpha}\dot{\beta}} F^{\alpha\beta}$ [690] vanishes for SD field strength ($F^{\alpha\beta} = 0$). Hence the single copy and double copy solutions of $\ell^{\dot{\alpha}\alpha} \propto \tilde{o}_1^{\dot{\alpha}} \eta^\alpha$ together define a solution in Einstein-Maxwell theory: a SD Taub-NUT instanton endowed with SD electromagnetic charge. By using such Einstein-Maxwell configurations as building blocks, a generalization of the nonlinear superposition theorem given in the supplemental material derives the Kerr-Newman solution, stipulating that the KS potential exactly needs to be modified as

$$\frac{M}{|\vec{x} - \vec{x}_1|} + \frac{M}{|\vec{x} - \vec{x}_2|} - \frac{Q^2}{|\vec{x} - \vec{x}_1||\vec{x} - \vec{x}_2|}. \quad (7.20)$$

This reproduces the complexification prescription posited in Ref. [179]. The Q^2 term in Eq. (7.20) reflects static electromagnetic interaction between SD and ASD dyons. Working in the similar fashion, one derives the five-parameter subclass of Plebański-Demiański [331, 332, 691] solutions with mass, magnetic mass, ring radius, electric charge, and magnetic charge, in the double KS form.

7.8 Summary

This chapter elevates the NJA to a rigorous derivation of rotating black hole solutions with a definite origin: factorizations into SD and ASD Taub-NUT instantons and chiral dyons.

This resolution of an old mystery will spark new perspectives on modern theoretical studies of rotating black holes: hidden symmetries [233, 266, 586,

⁵In the SD limit, $N \rightarrow -iM$, this double KS metric describes a SD Taub-NUT instanton, consistently with Refs. [260, 630].

593] ⁶, perturbation theory [240, 261], scattering amplitudes [98, 206–212, 221, 222], and effective dynamics [92, 93, 95, 176, 207–209, 692–694].

7.A Nonlinear Superposition Theorem

The idea of patching SD and ASD spacetimes together has been a long-standing aspiration [344–346].⁷ Here, we tackle this problem in a narrow setup: *stationary KS spacetimes*.

The complexified Minkowski space $\mathbb{M}$ is the linear space $\mathbb{C}^4$ equipped with a holomorphic flat metric η with a signature- $(-, +, +, +)$ real section. Let x^μ be Cartesian coordinates. Let u^μ be a unit vector such that $u^2 = \eta_{\mu\nu} u^\mu u^\nu = -1$; this is the characterization of the stationary direction $u^\mu = \delta^\mu_0$ in the main article. All indices are raised and lowered with the flat metric η .

Definition 1.1—A null vector field (NVF) ℓ on $\mathbb{M}$ is a vector field $\ell \in \Gamma(T\mathbb{M})$ such that $\ell^2 = \eta_{\mu\nu} \ell^\mu \ell^\nu = 0$.

Definition 1.2—A NVF ℓ on $\mathbb{M}$ is **canonical** iff it is stationary, geodesic, shear-free, and unit-normalized as $-u \cdot \ell = 1$.

Proposition 1.3—The first-order derivative $\partial_\mu \ell_\nu$ of a canonical NVF ℓ takes the following general form:

$$\partial_\mu \ell_\nu = \frac{\theta}{2} \left(\eta_{\mu\nu} + \ell_{(\mu} (2u - \ell)_{\nu)} \right) + \frac{\psi}{2} \varepsilon_{\mu\nu\rho\sigma} u^\rho \ell^\sigma. \quad (7.21)$$

Here, the scalar fields θ and ψ are referred to as the **expansion** and **twist** of ℓ , defined as $\theta = \partial_\mu \ell^\mu$ and $\psi = u_\mu \varepsilon^{\mu\nu\rho\sigma} \ell_\nu \partial_\rho \ell_\sigma$. The **complex expansions** of ℓ refer to $\rho^\pm = (\theta \pm i\psi)/2$.

(Proposition 1.3) is easily verified. For instance, the shear-free property arises from the fact that $\eta_{\mu\nu} + \ell_\mu u_\nu + u_\mu \ell_\nu - \ell_\mu \ell_\nu$ and $\varepsilon_{\mu\nu\rho\sigma} u^\rho \ell^\sigma$ in Eq. (7.21) are tensors lying within the two-dimensional plane transverse to both u and ℓ , spanning the trace-only and rotation modes. Ref. [176] makes a fruitful use of the formula in Eq. (7.21).

Proposition 1.4—From Eq. (7.21) and its integrability $\partial_\mu(\partial_\nu \ell_\rho) = \partial_\nu(\partial_\mu \ell_\rho)$, it follows that $\ell^\mu \partial_\mu \rho^\pm = -(\rho^\pm)^2$, $\ell^\mu \ell^\nu \partial_\mu \partial_\nu \rho^\pm = 2(\rho^\pm)^3$, and $\partial^2 \rho^\pm = 0$. The first equation is the Sachs equation [341, 606] in flat background.

⁶The Killing-Yano tensor of Kerr decomposes into the conformal Killing-Yano tensors of the instantons.

⁷The author thanks Maciej Dunajski for this anecdote.

Definition 2.1—For each canonical NVF ℓ on $\mathbb{M}$, the **single copy solution** is the abelian gauge connection $A_\mu = \theta \ell_\mu$ while the **double copy solution** is the metric $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon \theta \ell_\mu \ell_\nu$, where ϵ is a constant parameter.

Proposition 2.2—The single copy and double copy solutions of a canonical NVF solve vacuum flat-background Maxwell's equations and vacuum Einstein's equations in the holomorphic category, respectively.

Definition 2.3—A canonical NVF ℓ is **self-dual** (SD) iff its expansion is nonvanishing and the field strength $F_{\mu\nu} = \partial_\mu(\theta \ell_\nu) - \partial_\nu(\theta \ell_\mu)$ of its single copy solution is SD. A canonical NVF ℓ is **anti-self-dual** (ASD) iff its expansion is nonvanishing and its field strength is ASD.

Proposition 2.4—A canonical NVF is SD iff $\rho^+ \neq 0$ and $\rho^- = 0$. A canonical NVF is ASD iff $\rho^+ = 0$ and $\rho^- \neq 0$.

Proposition 2.5—The double copy solution of a SD (ASD) canonical NVF exhibits SD (ASD) curvature: $*R_{\mu\nu\rho\sigma} = \pm i R_{\mu\nu\rho\sigma}$.

(Proposition 2.5) can be shown by employing the orthonormal coframe $e^A_\mu = \delta^A_\mu (\delta^\mu_\nu + \epsilon \theta \ell^\mu \ell_\nu / 2)$, in which case the spin connection $\gamma_{AB\rho}$ is

$$-\frac{\epsilon}{2} \left(F_{\mu\nu} \delta^\mu_A \delta^\nu_B \ell_\rho + \left(\rho^+ \Pi^+_{ABCD} + \rho^- \Pi^-_{ABCD} \right) \left(\delta^C_\rho + \delta^C_\kappa u^\kappa \ell_\rho \right) \left(\delta^D_\sigma A^\sigma \right) \right), \quad (7.22)$$

where the SD and ASD projectors are $\Pi^\pm_{ABCD} = \frac{1}{2}(\eta_{AC}\eta_{BD} - \eta_{AD}\eta_{BC} \mp i\varepsilon_{ABCD})$.

Lemma 3.1 (*Little Superposition Lemma*)—Suppose a collection of canonical NVFs ℓ_i ($i = 1, 2, \dots, n$) such that $\sum_i c_i \theta_i = 0$ for a set of constants c_i . If there exists a scalar function χ such that $\sum_i c_i \theta_i \ell_{i\mu} = \partial_\mu \chi$, then there exists a one-form field ξ_μ such that $\sum_i c_i \theta_i \ell_{i\mu} \ell_{i\nu} = 2\partial_{(\mu} \xi_{\nu)}$. Explicitly, $\xi_\mu = \chi u_\mu - \sum_i c_i \ell_{i\mu}$.

Lemma 3.2 (*Superposition Lemma*)—If $\ell_{1,2}$ are canonical NVFs with complex expansions $\rho_{1,2}^\pm$, then

$$\ell^{\dot{\alpha}\alpha} = \frac{\ell_1^{\dot{\alpha}\beta} \delta_{\beta\dot{\beta}} \ell_2^{\dot{\beta}\alpha}}{1 + \ell_1 \cdot \ell_2 / 2}, \quad \tilde{\ell}^{\dot{\alpha}\alpha} = \frac{\ell_2^{\dot{\alpha}\beta} \delta_{\beta\dot{\beta}} \ell_1^{\dot{\beta}\alpha}}{1 + \ell_1 \cdot \ell_2 / 2} \quad (7.23)$$

are canonical NVFs with respective complex expansions $(\rho^+, \rho^-) = (\rho_1^+, \rho_2^-)$ and $(\tilde{\rho}^+, \tilde{\rho}^-) = (\tilde{\rho}_2^+, \tilde{\rho}_1^-)$, satisfying

$$\begin{aligned} \theta \ell_\mu + \tilde{\theta} \tilde{\ell}_\mu &= \theta_1 \ell_{1\mu} + \theta_2 \ell_{2\mu} + \partial_\mu \chi \\ \implies \theta \ell_\mu \ell_\nu + \tilde{\theta} \tilde{\ell}_\mu \tilde{\ell}_\nu &= \theta_1 \ell_{1\mu} \ell_{1\nu} + \theta_2 \ell_{2\mu} \ell_{2\nu} + 2\partial_{(\mu} \xi_{\nu)} \end{aligned} \quad (7.24)$$

with $\chi = \log((1 + \ell_1 \cdot \ell_2 / 2)^2)$ and $\xi_\mu = \chi u_\mu - \ell_\mu - \tilde{\ell}_\mu + \ell_{1\mu} + \ell_{2\mu}$. Here, $\delta_{\alpha\dot{\alpha}} = -u_\mu(\sigma^\mu)_{\alpha\dot{\alpha}}$.

The GSF condition plays an important role in (Lemma 3.1). (Lemma 3.2) follows by direct computation and (Lemma 3.1), where the chiral nature of Eq. (7.23) is crucial.

Theorem 3.3 (*Nonlinear Superposition Theorem*)—A SD canonical NVF ℓ_1 and an ASD canonical NVF ℓ_2 together define an exact solution $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon(\theta_1 + \theta_2)\ell_\mu\ell_\nu$ to Einstein's equations via $\ell^{\dot{\alpha}\alpha} = \ell_1^{\dot{\alpha}\beta}\delta_{\beta\dot{\beta}}\ell_2^{\dot{\beta}\alpha}/(1 + \ell_1 \cdot \ell_2 / 2)$.

Proof is straightforward from (Lemma 3.2) and (Definition 2.3): $(\tilde{\rho}^+, \tilde{\rho}^-) = (\tilde{\rho}_2^+, \tilde{\rho}_1^-)$ vanishes. This describes the nonlinear superposition of SD and ASD vacuum metrics in the sense that the Riemann tensor of $\eta_{\mu\nu} + \epsilon\theta\ell_\mu\ell_\nu$ equals the sum of the Riemann tensors of $\eta_{\mu\nu} + \epsilon\theta_1\ell_{1\mu}\ell_{1\nu}$ and $\eta_{\mu\nu} + \epsilon\theta_2\ell_{2\mu}\ell_{2\nu}$ at leading order in the parameter ϵ .

Lemma 4.1 (*Maxwell Stress-Energy via Modifying the KS Potential*)—If ℓ is a canonical NVF on $\mathbb{M}$ with complex expansions $\rho^\pm$, then the gauge potential $A_\mu = \theta\ell_\mu$ and the metric $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon(\theta - 2\alpha\rho^+\rho^-)\ell_\mu\ell_\nu$ together define a solution to $G^\mu{}_\nu[g] = \epsilon\alpha T^\mu{}_\nu[A, g]$ and $\nabla^\nu F_{\mu\nu} = 0$: Einstein-Maxwell theory in the holomorphic category. Here, $G^\mu{}_\nu[g]$ is the Einstein tensor associated with the metric g , $T^\mu{}_\nu[A, g]$ is the Maxwell stress-energy tensor, ∇ is the Levi-Civita connection of the metric g , $F_{\mu\nu} = 2\nabla_{[\mu}A_{\nu]} = 2\partial_{[\mu}A_{\nu]}$ is the field strength, and α is a constant parameter.

Theorem 4.2 (*Nonlinear Superposition for Einstein-Maxwell Solutions*)—A SD canonical NVF ℓ_1 and an ASD canonical NVF ℓ_2 together define an exact solution $A_\mu = (\theta_1 + \theta_2)\ell_\mu$, $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon(\theta_1 + \theta_2 - 2\alpha\theta_1\theta_2)\ell_\mu\ell_\nu$ to Einstein-Maxwell theory via $\ell^{\dot{\alpha}\alpha} = \ell_1^{\dot{\alpha}\beta}\delta_{\beta\dot{\beta}}\ell_2^{\dot{\beta}\alpha}/(1 + \ell_1 \cdot \ell_2 / 2)$. Here, Q is a constant parameter such that $\epsilon\alpha = Q^2/2$.

For deriving the Kerr-Newman solution, one rescales the gauge potential by $Q/8\pi\epsilon_0$ so that the electromagnetic charges of the SD and ASD solutions become $(Q/2, \mp iQ/2)$. Consequently, the constant parameters are taken as $\epsilon = GM/c^2$ and $\epsilon\alpha = (8\pi G/c^4) \cdot (1/\mu_0) \cdot (Q/8\pi\epsilon_0)^2/c^2 = Gk_e Q^2/2c^4$. Hence, in the relativists' natural unit ($c=1$, $G=1$, $k_e=1$), one finds $\epsilon = M$ and $\epsilon\alpha = Q^2/2$ as stated in the main article.

Theorem 5.1 (*Effective Linearization*)—A metric of the form $g_{\mu\nu} = \eta_{\mu\nu} + \phi\ell_\mu\ell_\nu$ with $\ell^2 = 0$ solves the exact vacuum Einstein's equations if it solves the

linearized vacuum Einstein's equations [176–178].

7.B Spheroidal Coordinates

The main article describes the inverse metrics in Eqs. (7.16) and (7.17) in the spinor language. The original paper [202], on the other hand, uses the null tetrad language in spheroidal coordinates. We show that they are equivalent.

Null Tetrad.—Given two spinors $\tilde{o}_1^{\dot{\alpha}}$ and o_2^{α} and $\delta_{\alpha\dot{\alpha}} = -u_{\mu}(\sigma^{\mu})_{\alpha\dot{\alpha}}$, a null tetrad in flat spacetime is formed by

$$\begin{aligned} \ell^{\dot{\alpha}\alpha} &= \frac{\tilde{o}_1^{\dot{\alpha}} o_2^{\alpha}}{\langle o_2 | \delta | \tilde{o}_1 \rangle}, \quad n^{\circ\dot{\alpha}\alpha} = \frac{\tilde{l}_2^{\dot{\alpha}} \iota_1^{\alpha}}{\langle o_2 | \delta | \tilde{o}_1 \rangle}, \quad m^{\dot{\alpha}\alpha} = \frac{\tilde{o}_1^{\dot{\alpha}} \iota_1^{\alpha}}{\langle o_2 | \delta | \tilde{o}_1 \rangle}, \quad \tilde{m}^{\dot{\alpha}\alpha} = \frac{\tilde{l}_2^{\dot{\alpha}} o_2^{\alpha}}{\langle o_2 | \delta | \tilde{o}_1 \rangle} \\ \implies \eta^{\mu\nu} &= -\ell^{(\mu} n^{\circ\nu)} + m^{(\mu} \tilde{m}^{\nu)}, \end{aligned} \quad (7.25)$$

where $\tilde{l}_{2\dot{\alpha}} = -o_2^{\alpha} \delta_{\alpha\dot{\alpha}}$ and $\iota_{1\alpha} = -\delta_{\alpha\dot{\alpha}} \tilde{o}_1^{\dot{\alpha}}$ so that $[\tilde{o}_1 \tilde{l}_2] = \langle o_2 | \delta | \tilde{o}_1 \rangle$ and $\langle \iota_1 o_2 \rangle = \langle o_2 | \delta | \tilde{o}_1 \rangle$.

For example, the principal spinors of the Schwarzschild solution are given by $\tilde{o}^{\dot{\alpha}} = (1, (x+iy)/(|\vec{x}|+z))$ and $o^{\alpha} = (1, (x-iy)/(|\vec{x}|+z))$. Taking $\tilde{o}_1^{\dot{\alpha}} = \tilde{o}^{\dot{\alpha}}$ and $o_2^{\alpha} = o^{\alpha}$ in Eq. (7.25) gives $\ell^{\mu} = (1, x/|\vec{x}|, y/|\vec{x}|, z/|\vec{x}|)$, which is the outgoing canonical NVF for the KS metric of the Schwarzschild solution.

The physical spinor of SD/ASD Taub-NUT instanton equals the SD/ASD principal spinor of Schwarzschild [163].

The main article sets $\tilde{o}_1^{\dot{\alpha}}$ and o_2^{α} as the physical spinors of the SD and ASD Taub-NUT instantons at $\vec{x}_1 = +i\vec{a}$ and $\vec{x}_2 = -i\vec{a}$ with $\vec{a} = (0, 0, a)$. Explicitly, this means to take $\tilde{o}_1^{\dot{\alpha}} = (1, \tilde{\zeta})$, $o_2^{\alpha} = (1, \zeta)$, $\tilde{l}_2^{\dot{\alpha}} = (-\tilde{\zeta}, 1)$, $\iota_1^{\alpha} = (-\zeta, 1)$, and $1/\langle o_2 | \delta | \tilde{o}_1 \rangle = (r+z)/2r$, where

$$\begin{aligned} \tilde{\zeta} &= \frac{x+iy}{|\vec{x}-i\vec{a}|+(z-ia)} = \frac{x+iy}{r+z} \frac{r}{r-ia}, \\ \zeta &= \frac{x-iy}{|\vec{x}+i\vec{a}|+(z+ia)} = \frac{x-iy}{r+z} \frac{r}{r+ia}. \end{aligned} \quad (7.26)$$

Here, r denotes the function of x, y, z defined in Eq. (7.4). A branch cut prescription [98, 257] sets $|\vec{x} \mp i\vec{a}| = \sqrt{(\vec{x} \mp i\vec{a})^2} = r \mp iaz/r$. The computation of the resulting flat null tetrad due to Eq. (7.25) is straightforward.

Spheroidal Coordinates.—A nice choice of angle variables (θ, φ) simplifies the result as $\ell^{\mu} = (1, \sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$: $x \pm iy = (r \mp ia) \sin \theta e^{\pm i\varphi}$, $z = r \cos \theta$. In fact, (r, θ, φ) is known as the twisted oblate spheroidal coordinate system. As a result, the components of ℓ are simply $(1, 1, 0, 0)$ in the

(t, r, θ, φ) coordinate system. Thus, by introducing the retarded time $u = t - r$, the null tetrad is

$$\begin{aligned} \ell &= \partial_r, & m &= \frac{e^{i\varphi}}{r + ia \cos \theta} \left(ia \sin \theta (\partial_u - \partial_r) + \partial_\theta + \frac{i}{\sin \theta} \partial_\varphi \right), \\ n^\circ &= 2\partial_u - \partial_r, & \tilde{m} &= \frac{e^{-i\varphi}}{r - ia \cos \theta} \left(-ia \sin \theta (\partial_u - \partial_r) + \partial_\theta - \frac{i}{\sin \theta} \partial_\varphi \right). \end{aligned} \quad (7.27)$$

The null tetrad $(\ell, n, m, \tilde{m})$ for the curved inverse metric in Eq. (7.16) follows by perturbing $n^\circ \rightarrow n = n^\circ + M\theta\ell$:

$$n = 2\partial_u - \partial_r + M \left(\frac{1}{r - ia \cos \theta} + \frac{1}{r + ia \cos \theta} \right) \partial_r. \quad (7.28)$$

Eqs. (7.27) and (7.28) are literally Eq. (7) of Ref. [202]. Taking the limit $a \rightarrow 0$ exactly reproduces Eq. (4) of Ref. [202].

The NJA inverts this limit. Ref. [202] views it as a “complex coordinate transformation” due to the suggestiveness of the spheroidal coordinates. However, the main article clarifies that the NJA is not a coordinate transformation.

By repeating the above exercise, one systematically derives the NJA for Kerr-Newman, Kerr*, Kerr-Taub-NUT, and charged Kerr-Taub-NUT (five-parameter Plebański-Demiański) solutions. We emphasize that what facilitates such a systematic and universal derivation of NJAs is the very instanton pair structure elucidated in the main article. In other words, the origin of the NJA is the factorization of black holes into Taub-NUT instantons and chiral dyons.

Note that this derivation of the NJA works in any signature: Euclidean, Lorentzian, Kleinian, mostly-plus, mostly-minus, etc. For instance, one can employ twisted prolate hyperboloidal coordinate system $x \pm iy = (\tilde{r} \pm ja) \sin \theta e^{\pm j\varphi}$, $z = \tilde{r} \cos \theta$, where j denotes the unit split complex number ($j^2 = 1$).

GEODESIC DEVIATION TO ALL ORDERS

We establish an in-in formalism for geodesic deviation as an alternative to Synge calculus, based on a covariant calculus of differential forms in tangent bundle. This derives the exact Lagrangian and equations governing the finite geodesic deviation between a free-falling test particle and an arbitrary observer, in terms of infinite sums whose coefficients are products of binomial coefficients. Explicit expressions are provided up to tenth order, finding agreements with the previous fourth-order result.

This section reproduces the contents of Ref. [695], J.-H. Kim, “Geodesic deviation to all orders via a tangent bundle formalism,” *General Relativity and Gravitation* **58** no. 7, (2026) 75, [arXiv:2509.23600 \[gr-qc\]](#).

8.1 Introduction

The GDE is a foundational topic in GR, commonly covered in textbooks [20, 291, 696–700]. First derived in 1927 by Levi-Civita and Synge [701–705], this equation describes the evolution of the *infinitesimal* separation between two nearby free-falling test particles. Explicitly, it reads

$$\frac{D^2 y^\mu}{d\tau^2} = -R^\mu{}_{\nu\rho\sigma} u^\nu y^\rho u^\sigma, \quad (8.1)$$

where y^μ describes the infinitesimal separation as a vector and u^μ describes the unit-normalized four-velocity.

The generalization of the GDE to the case of *finite* separations has been a topic of research in the literature [706–722]. Here, the objective is to find corrections to the GDE perturbatively in the orders of the separation vector y , covariantly defined as a tangent to the geodesic segment joining the two particles. Namely, the right-hand side of Eq. (8.1) is augmented with terms involving nonlinear powers of y , coupled to the Riemann tensor $R^\mu{}_{\nu\rho\sigma}$ and its derivatives. Physically, this captures the higher-order tidal effects due to the non-infinitesimal size of y .

The higher-order extensions of the GDE have found fruitful physical applications [718, 722]: gravitational wave detectors, astrophysical jets, measurement

of spacetime curvature, and an analytical treatment of eccentric relativistic orbits [710–717, 723–731].

The explicit GDEs through $\mathcal{O}(y^3)$ and $\mathcal{O}(y^4)$ were obtained by works [706–709] in the 1970s and Vines [718] in 2014, respectively. Moreover, Vines [718] also provided formulae for the all-orders extension of the GDE and its Lagrangian formulation, though in terms of tensor expressions that are not fully expanded out in terms of Riemann curvature and its derivatives. To find the explicit equation or Lagrangian at an order, one has to solve a group of interrelated recursion relations in Appendix 3 of Ref. [718], which arise in the context of various and intricate identities of the Synge bitensor formalism [626, 732–734].

In this work, we intend to revisit this problem from a different framework. The achievements are the following. Firstly, we develop an alternative to the Synge formalism in which the relevant tensor expressions are computed from a covariant calculus of differential forms. Secondly, we provide the exact all-orders formula for the so-called Jacobi propagators [229, 735–737], fully expanded out in terms of Riemann tensor and its derivatives. Our exact expressions are infinite sums whose coefficients are products of binomial coefficients, showing agreements with Vines [718]. Consequently, we provide the explicit GDE and its Lagrangian up to $\mathcal{O}(y^{10})$.

The key idea of our approach is to formulate geodesic deviation as an initial-value problem like in works [738–743]. Namely, we specify an *in-in* boundary condition for the geodesic: point x and a tangent vector y at x . This is to be contrasted with the Synge bitensor formalism [626, 732–734] where a geodesic segment is characterized by its two endpoints x and z as an *in-out* boundary condition.

8.2 Geodesic Deviation in Tangent Bundle

Let $(\mathcal{M}, g)$ be a d -dimensional real-analytic manifold with local coordinates x^μ and metric $g_{\mu\nu}$, which we call spacetime. Its tangent bundle, $T\mathcal{M}$, can be viewed as a $2d$ -dimensional manifold with local coordinates (x^μ, y^μ) , based on its local trivialization by the coordinate vector fields.

Coordinate transformations of $T\mathcal{M}$ are restricted in the form $(x^\mu, y^\mu) \mapsto (f^\mu(x), f^\mu{}_{,\nu}(x)y^\nu)$, where $f^\mu(x)$ describes a set of real-analytic functions in a local patch of $\mathcal{M}$. Importantly, the notion of covariance in $T\mathcal{M}$ is defined with

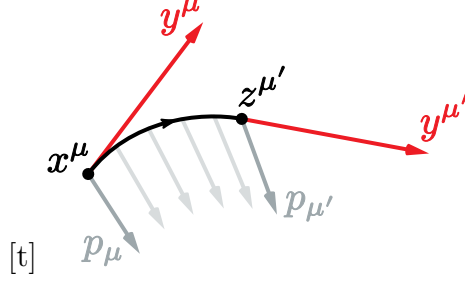


Figure 8.1: A geodesic segment joins two spacetime points x (observer) and z (test particle). The tangent vectors are respectively denoted as y^μ and $y^{\mu'}$, which are normalized such that x and z are related by unit-time geodesic flows.

respect to these coordinate transformations.

In particular, the following vector field in $T\mathcal{M}$ is *invariant* under such coordinate transformations:

$$N = y^\mu \frac{\partial}{\partial x^\mu} - \Gamma^\mu_{\rho\sigma}(x) y^\rho y^\sigma \frac{\partial}{\partial y^\mu}, \quad (8.2)$$

where $\Gamma^\mu_{\rho\sigma}(x)$ denote the Christoffel symbols. This can be easily seen by considering its interior products with a one-form basis in $T\mathcal{M}$ that transform covariantly:

$$\iota_N dx^\mu = y^\mu, \quad \iota_N Dy^\mu = 0. \quad (8.3)$$

Here, D denotes the covariant exterior derivative: $Dy^\mu = dy^\mu + \Gamma^\mu_{\nu\rho}(x) y^\nu dx^\rho$. In light of its invariance, N defines a structure characteristic of the tangent bundle. In the mathematical jargon, it describes a horizontal vector field due to the Ehresmann notion of a connection [381, 744].

Crucially, N holds the very significance as the *generator of geodesic deviation*. To see this, consider the first-order formulation of the geodesic equation:

$$\begin{aligned} \frac{d}{d\eta} X^\mu(\eta) &= Y^\mu(\eta), \\ \frac{d}{d\eta} Y^\mu(\eta) &= -\Gamma^\mu_{\rho\sigma}(X(\eta)) Y^\rho(\eta) Y^\sigma(\eta). \end{aligned} \quad (8.4)$$

Evidently, the explicit components of N in Eq. (8.2) encode Eq. (8.4). This implies that the power series solution of Eq. (8.4) is given by $X^\mu(\eta) = e^{\eta N} x^\mu$ and $Y^\mu(\eta) = e^{\eta N} y^\mu$ for initial conditions $X^\mu(0) = x^\mu$ and $Y^\mu(0) = y^\mu$, where N is understood as a differential operator. Namely, the time- η flow of N solves the geodesic equation.

In particular, consider the unit-time flow:

$$X^\mu(1) = e^N x^\mu, \quad Y^\mu(1) = e^N y^\mu. \quad (8.5)$$

Borrowing the notation in the Synge bitensor formalism [626, 732–734], we rephrase Eq. (8.5) as the following:

$$z^{\mu'} = \delta^{\mu'}_\mu (e^N x^\mu), \quad y^{\mu'} = \delta^{\mu'}_\mu (e^N y^\mu), \quad (8.6)$$

where $\delta^{\mu'}_\mu$ is Kronecker delta. Namely, we assign unprimed and primed indices to objects at the original (x) and deviated (z) points, respectively. The rationale might be that indices represent the behavior under coordinate transformations, and tensors at x and z transform with different Jacobian factors. Fig. 8.1 provides a spacetime picture that visualizes Eq. (8.6).

The unit-time flow of N can be regarded as a diffeomorphism in $T\mathcal{M}$. By the very definition of the Lie derivative, the pullback of a tensor α in $T\mathcal{M}$ is given by $e^{\mathcal{L}_N} \alpha$. Practically, this pullback replaces every x with $e^N x$ and y with $e^N y$. For instance, the following identity holds:

$$dz^{\mu'} = \delta^{\mu'}_\mu (e^{\mathcal{L}_N} dx^\mu). \quad (8.7)$$

Notably, for differential forms, Lie derivatives are neatly computed by using the Cartan magic formula:

$$\mathcal{L}_N = d\iota_N + \iota_N d. \quad (8.8)$$

We leave it as an exercise to check Eq. (8.7) at each order in y by using Eq. (8.8).

8.3 Geodesic Deviation in A Direct Sum Bundle

For pedagogical reasons, let us also consider a slight generalization of the above construction. Suppose the direct sum of the tangent and cotangent bundles: $\mathcal{P} = T\mathcal{M} \oplus T^*\mathcal{M}$, where local coordinates are (x^μ, y^μ, p_μ) . Eventually, this $3d$ -dimensional space will serve as our geometrical arena for a first-order formulation of GDE. In this larger bundle, the vector field N is uniquely defined by $\iota_N dx^\mu = y^\mu$, $\iota_N Dy^\mu = 0$, and $\iota_N Dp_\mu = 0$.

It is not difficult to see that this N is the generator of geodesic deviation and parallel transport. In particular, consider the analog of the first-order differential equations in Eq. (8.4) to find that $p_{\mu'} = (e^N p_\mu) \delta^\mu_{\mu'}$ is the covector

p_μ parallel-transported to the point z along the geodesic; see Fig. 8.1. This finding is succinctly stated as

$$p_{\mu'} = p_\mu W^\mu_{\mu'}, \quad (8.9)$$

where $W^\mu_{\mu'}$ denotes the *parallel propagator*: the Wilson line of the Levi-Civita connection computed about the geodesic between x and z . Namely, $p_\mu \mapsto p_{\mu'} = p_\mu W^\mu_{\mu'}$ describes the isomorphism between the cotangent spaces at x and z facilitated by the geodesic parallel transport.

Again, $(x, y, p) \mapsto e^{\eta N}(x, y, p)$ describes the time- η flow in $\mathcal{P}$ generated by the vector field N . Due to this geometrical interpretation, the unit-time geodesic deviation and parallel transport is given by the exponentiated Lie derivative $e^{\mathcal{L}_N}$.

For a concrete example, consider a one-form $p_\mu dx^\mu$, i.e., the trivial extension of the cotangent bundle's tautological one-form to $\mathcal{P}$. Then we have

$$p_{\mu'} dz^{\mu'} = e^{\mathcal{L}_N}(p_\mu dx^\mu), \quad (8.10)$$

the right-hand side of which can be computed order-by-order by series-expanding $e^{\mathcal{L}_N}$. We leave it as an exercise to carry out this computation and check consistency between the left-hand and right-hand sides.

8.4 Covariant Lie Derivative and Dressing Identity

The computations utilizing $\mathcal{L}_N$, however, are not very efficient. One encounters unoccupied connection coefficients arising from the components of N , so covariance at the original point x is not manifested in intermediate steps.

In this light, we introduce a shorthand notation,

$$\mathcal{L}_N^D = D\iota_N + \iota_N D, \quad (8.11)$$

defined on any tensor-valued differential form. This describes a covariantized analog of the Lie derivative; more information might be found in Refs. [745–747].

Crucially, since $p_\mu dx^\mu$ is a scalar-valued one-form carrying no free indices, using D instead of d makes no difference. Hence we can equivalently use $\exp(\mathcal{L}_N^D)$ in Eq. (8.10). It is a privilege of exponentiated operators that they are distributable as

$$e^{\mathcal{L}_N^D}(p_\mu dx^\mu) = (e^{\mathcal{L}_N^D} p_\mu)(e^{\mathcal{L}_N^D} dx^\mu) = p_\mu(e^{\mathcal{L}_N^D} dx^\mu), \quad (8.12)$$

where the last equality follows from $\mathcal{L}_N^D p_\mu = \iota_N D p_\mu = 0$. Therefore, from Eqs. (8.9), (8.10), and (8.12), it follows that

$$W^\mu_{\mu'} dz^{\mu'} = e^{\mathcal{L}_N^D} dx^\mu. \quad (8.13)$$

In the same way, it can be shown that the $\exp(\mathcal{L}_N^D)$ of a tensor-valued differential form computes its value at the deviated point, followed by the parallel-transportation back to the original point via dressing by the Wilson lines. This fact will be referred to as the *dressing identity*.

8.5 Recursion for Jacobi Propagators

Having set up the foundations of our formalism, we now concern explicit evaluations. In particular, it follows from Eq. (8.11) that the right-hand side of Eq. (8.13) evaluates as

$$dx^\mu + Dy^\mu + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} ((\iota_N D)^{\ell-2} \iota_N R^\mu{}_\nu) y^\nu, \quad (8.14)$$

where $R^\mu{}_\nu$ denotes the Riemann curvature two-form such that $D^2 y^\mu = R^\mu{}_\nu y^\nu$. This computation is illustrated in Fig. 8.2 as a curved deformation of the sequence of differential forms due to the Cartan magic formula. As a tensor at the point x , Eq. (8.14) eventually boils down to the following form:

$$W^\mu_{\mu'} dz^{\mu'} = e^{\mathcal{L}_N^D} dx^\mu = X^\mu{}_\sigma dx^\sigma + Y^\mu{}_\sigma Dy^\sigma. \quad (8.15)$$

The objective now is to find the tensors $X^\mu{}_\sigma$ and $Y^\mu{}_\sigma$ explicitly. Note that $W^{\mu'}{}_\mu X^\mu{}_\sigma$ and $W^{\mu'}{}_\mu Y^\mu{}_\sigma$ are exactly what are known as Jacobi propagators (denoted as $K^{\mu'}{}_\sigma$ and $H^{\mu'}{}_\sigma$ in Refs. [718, 748]), where $W^{\mu'}{}_\mu$ is the inverse of $W^\mu{}_{\mu'}$. We find it preferable to peel off the Wilson line, for which the path-ordered exponential formula is well-known and amenable.

To this end, one needs to compute $(\iota_N D)^{\ell-2} \iota_N R^\mu{}_\nu$ for $\ell \geq 2$. For $\ell = 3$, one finds $\iota_N D \iota_N R^\mu{}_\nu = y^\kappa R^\mu{}_{\nu\rho\sigma;\kappa}(x) y^\rho dx^\sigma + R^\mu{}_{\nu\rho\sigma}(x) y^\rho Dy^\sigma$. When one hits this with yet another $\iota_N D$, the covariant exterior derivative D can act on Dy^σ to generate another Riemann tensor. As a result, one finds two single-Riemann terms and one double-Riemann term at $\ell = 4$ (see Sec. 8.A). In the same fashion, higher concatenations of Riemann tensors arise at higher orders.

A recursive structure can be identified in these calculations. First of all, it

$$\begin{array}{ccccc}
dx^\mu & \xrightarrow{D\iota_N} & Dy^\mu & \longrightarrow & 0 \\
\iota_N D \downarrow & & \downarrow & & \\
0 & & (\iota_N R^\mu{}_\nu) y^\nu & \longrightarrow & 0 \\
& & \downarrow & & \\
& & (\iota_N D \iota_N R^\mu{}_\nu) y^\nu & \longrightarrow & 0 \\
& & \downarrow & & \\
& & \vdots & &
\end{array}$$

Figure 8.2: A sequence of one-forms originating from dx^μ via the covariant Lie derivative $\mathcal{L}_N^D = D\iota_N + \iota_N D$. Note that $(D\iota_N)(\iota_N D) = 0$, while Ddx^μ vanishes for zero torsion.

follows that

$$\begin{aligned}
(\iota_N D)^{\ell-2} \iota_N R^\mu{}_\nu y^\nu &= \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} c^{\alpha_1, \dots, \alpha_p} (Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu{}_\sigma dx^\sigma \\
&+ \sum_{p=1}^{\lfloor \frac{\ell-1}{2} \rfloor} \sum_{\alpha \in \Omega_p(\ell-1)} c'^{\alpha_1, \dots, \alpha_p} (Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu{}_\sigma Dy^\sigma,
\end{aligned} \tag{8.16}$$

where α runs over ordered partitions such that

$$\begin{aligned}
\alpha &= (\alpha_1, \alpha_2, \dots, \alpha_p) \in \Omega_p(\ell) \\
\iff \alpha_1 + \alpha_2 + \dots + \alpha_p &= \ell, \quad \alpha_i \geq 2.
\end{aligned} \tag{8.17}$$

Note that the total number of such partitions at each ℓ , i.e., $\sum_{p=1}^{\lfloor \ell/2 \rfloor} |\Omega_p(\ell)|$, is the $(\ell-1)^{\text{th}}$ Fibonacci number. In Eq. (8.16), the “ Q -tensors” are defined as

$$\begin{aligned}
(Q_\ell)^\mu{}_\sigma &:= y^{\kappa_1} \cdots y^{\kappa_\ell} R^\mu{}_{\kappa_1 \kappa_2 \sigma; \kappa_3; \dots; \kappa_\ell}(x) \\
\implies (Q_\ell)_{\mu\sigma} &= (Q_\ell)_{\sigma\mu}, \quad (Q_\ell)^\mu{}_\sigma y^\sigma = 0.
\end{aligned} \tag{8.18}$$

The recursion relations for the coefficients in Eq. (8.16) are easily found from identifying the following action of $\iota_N D$:

$$\begin{aligned}
(Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu{}_\sigma dx^\sigma &\mapsto \left[\sum_{i=1}^p (Q_{\alpha_1} \cdots Q_{\alpha_{i-1}} Q_{\alpha_i+1} Q_{\alpha_{i+1}} \cdots Q_{\alpha_p})^\mu{}_\sigma dx^\sigma \right] \\
&+ (Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu{}_\sigma Dy^\sigma,
\end{aligned} \tag{8.19a}$$

$$\begin{aligned}
(Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu{}_\sigma Dy^\sigma &\mapsto \left[\sum_{i=1}^p (Q_{\alpha_1} \cdots Q_{\alpha_{i-1}} Q_{\alpha_i+1} Q_{\alpha_{i+1}} \cdots Q_{\alpha_p})^\mu{}_\sigma Dy^\sigma \right] \\
&+ (Q_{\alpha_1} \cdots Q_{\alpha_p} Q_2)^\mu{}_\sigma dx^\sigma.
\end{aligned} \tag{8.19b}$$

With the proper identification of the boundary conditions ¹, the solution is determined as ²

$$c^{\alpha_1, \dots, \alpha_p} = \prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2}, \quad c'^{\alpha_1, \dots, \alpha_p} = \prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 1}{\alpha_i - 2}, \quad (8.20)$$

which describes products of binomial coefficients. Finally, plugging in Eq. (8.20) to Eq. (8.16), we arrive at the following formula for $e^{\mathcal{L}_N^D} dx^\mu$ from which the tensors X and Y in Eq. (8.15) are readily read off:

$$\begin{aligned} e^{\mathcal{L}_N^D} dx^\mu &= dx^\mu + Dy^\mu \\ &+ \sum_{\ell=2}^{\infty} \frac{1}{\ell!} \sum_{p=1}^{[\ell/2]} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right] \\ &\quad \left((Q_{\alpha_1} \cdots Q_{\alpha_p})^\mu dx^\sigma + (\alpha_p - 2) (Q_{\alpha_1} \cdots Q_{\alpha_{p-1}} Q_{\alpha_p - 1})^\mu_\sigma Dy^\sigma \right). \end{aligned} \quad (8.21)$$

See Sec. 8.B.1 for explicit enumerations.

The explicit solution for the Jacobi propagators in Eq. (8.21) has not been spelled out in the literature to our best knowledge, though Appendix 3 of Vines [718] has identified a set of relevant recursion relations by building upon Ottewill and Wardell [749] and Dixon [748].

8.6 The Lagrangians

We are now ready to derive the all-orders GDE and its Lagrangian. When described with the primed variables, the first-order action of a free-falling test particle is

$$\int d\sigma \left[p_{\mu'} \frac{dz^{\mu'}}{d\sigma} - \frac{e}{2} \left(g^{\mu'\nu'}(z) p_{\mu'} p_{\nu'} + m^2 \right) \right], \quad (8.22)$$

where e is the einbein (a Lagrange multiplier), and m is the rest mass. To describe this particle from an observer's worldline, $\sigma \mapsto x^\mu(\sigma)$, we identify that the first term in Eq. (8.22) originates from the one-form $p_{\mu'} dz^{\mu'} = p_\mu (\exp(\mathcal{L}_N^D) dx^\mu)$. Consequently, Eq. (8.22) boils down to

$$\int d\sigma \left[p_\mu \left(X^\mu{}_\nu \frac{dx^\nu}{d\sigma} + Y^\mu{}_\nu \frac{Dy^\nu}{d\sigma} \right) - \frac{e}{2} (p^2 + m^2) \right], \quad (8.23)$$

¹(a) $c^{\alpha_1, \alpha_2, \dots, \alpha_p} = 0$, $c'^{\alpha_1, \alpha_2, \dots, \alpha_p} = 0$ if any of $\alpha_1, \alpha_2, \dots, \alpha_p$ equals one. (b) $c^2 = 1$, $c'^2 = 0$.

²The generating function $G(\zeta_1, \dots, \zeta_p) = \sum_{\alpha} c^{\alpha_1, \dots, \alpha_p}(\zeta_1)^{\alpha_1} \cdots (\zeta_p)^{\alpha_p}$ is determined as $G(\zeta_1, \dots, \zeta_p) = ((\zeta_p)^2 / (\zeta_1 + \cdots + \zeta_p - 1)) \cdot (\zeta_1 / (\zeta_1 - 1))^2 \cdot (\zeta_2 / (\zeta_1 + \zeta_2 - 1))^2 \cdots (\zeta_{p-1} / (\zeta_1 + \cdots + \zeta_{p-1} - 1))^2$.

where $p^2 = g^{\mu\nu}(x)p_\mu p_\nu$ since Wilson lines due to the Levi-Civita connection respect the metric. By integrating out p_μ , we also find a second-order Lagrangian:

$$\int d\sigma \left[\frac{1}{2e} \left(X \frac{dx}{d\sigma} + Y \frac{Dy}{d\sigma} \right)^2 - \frac{m^2 e}{2} \right]. \quad (8.24)$$

Adopting an invariant measure of time $d\tau = med\sigma$ for $m \neq 0$ reproduces Vines [718]’s action for affinely parameterized worldlines (isochronous correspondence [709, 718]):

$$m \int d\tau \frac{1}{2} \left((X u + Y v)^2 - 1 \right). \quad (8.25)$$

Here, we have denoted $u^\mu := dx^\mu/d\tau$ and $v^\mu := Dy^\mu/d\tau$.

Eqs. (8.23)-(8.25) provide exact first-order and second-order Lagrangian formulations of the all-orders geodesic deviation, where the deviation y^μ is defined as a vector attached to the observer’s worldline. We clarify that the observer’s worldline is introduced as a nondynamical reference [718], while y^μ and p_μ are dynamical variables ³.

The Lagrangian in Eq. (8.25) is explicitly given up to $\mathcal{O}(y^{10})$ in Sec. 8.B.2, showing perfect agreement with the previous $\mathcal{O}(y^5)$ result due to Vines [718].

Note that the second-order actions in Eqs. (8.24) and (8.25) could have been directly obtained by implementing our formalism simply in the tangent bundle $T\mathcal{M}$, instead of employing the extended bundle $\mathcal{P}$.

8.7 The All-Orders Geodesic Deviation Equation

The all-orders GDE follows by varying the above Lagrangians. Otherwise, it can also be derived from our formalism in the following way.

Consider the first-order formulation of the free-falling EoM, associated with Eq. (8.22):

$$\frac{dz^{\mu'}}{d\sigma} = e p^{\mu'}, \quad \frac{Dp^{\mu'}}{d\sigma} = 0. \quad (8.26)$$

The idea is to re-covariantize Eq. (8.26) at the observer’s position, x^μ , via the geodesic Wilson line dressing. Earlier, we have obtained $W^\mu{}_{\mu'} dz^{\mu'} = X^\mu{}_\nu dx^\nu + Y^\mu{}_\nu Dy^\nu$ in Eq. (8.15). This applies to the first equation in Eq. (8.26). Taking

³It might be interesting to understand the transformations to other correspondences as gauge transformations in the first-order setup.

a similar approach for the second equation as well, we obtain a first-order formulation of the all-orders GDE:

$$ep^\mu = X^\mu{}_\nu \frac{dx^\nu}{d\sigma} + Y^\mu{}_\nu \frac{Dy^\nu}{d\sigma}, \quad (8.27a)$$

$$\frac{Dp^\mu}{d\sigma} = - \left(\dot{X}^\mu{}_{\nu\sigma} \frac{dx^\sigma}{d\sigma} + \dot{Y}^\mu{}_{\nu\sigma} \frac{Dy^\sigma}{d\sigma} \right) p^\nu. \quad (8.27b)$$

For obtaining Eq. (8.27b), we have applied the dressing identity to the one-form Dp^μ :

$$W^\mu{}_{\mu'} Dp^{\mu'} = e^{\mathcal{L}_N^D} Dp^\mu = Dp^\mu + \dot{X}^\mu{}_{\nu\sigma} p^\nu dx^\sigma + \dot{Y}^\mu{}_{\nu\sigma} p^\nu Dy^\sigma. \quad (8.28)$$

The tensors $\dot{X}^\mu{}_{\nu\sigma}$ and $\dot{Y}^\mu{}_{\nu\sigma}$ are given as

$$\dot{X}^\mu{}_{\nu\sigma} = \sum_{\ell=2}^{\infty} \frac{1}{(\ell-1)!} \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right] (\dot{Q}_{\alpha_1})^\mu{}_{\nu\kappa} (Q_{\alpha_2} \cdots Q_{\alpha_p})^\kappa{}_\sigma, \quad (8.29a)$$

$$\dot{Y}^\mu{}_{\nu\sigma} = \sum_{\ell=2}^{\infty} \frac{1}{(\ell-1)!} \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right] (\alpha_p - 2) (\dot{Q}_{\alpha_1})^\mu{}_{\nu\kappa} (Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_p - 1})^\kappa{}_\sigma, \quad (8.29b)$$

which simply replaces $\ell!$ in Eq. (8.21) to $(\ell-1)!$ regarding the combinatorial factors. Here, we have defined, for $\ell \geq 2$,

$$\begin{aligned} (\dot{Q}_\ell)^\mu{}_{\nu\sigma} &:= y^{\kappa_1} \cdots y^{\kappa_{\ell-1}} R^\mu{}_{\nu\kappa_1\sigma; \kappa_2 \cdots \kappa_{\ell-1}}(x) \\ \implies \quad \begin{cases} (\dot{Q}_\ell)_{\mu\nu\sigma} = -(\dot{Q}_\ell)_{\nu\mu\sigma}, & (\dot{Q}_\ell)_{\mu\nu\sigma} y^\sigma = 0, \\ (\dot{Q}_\ell)_{[\mu\nu\sigma]} = 0, & (\dot{Q}_\ell)^\mu{}_{\nu\sigma} y^\nu = (Q_\ell)^\mu{}_\sigma. \end{cases} \end{aligned} \quad (8.30)$$

Combining Eqs. (8.27a) and (8.27b), the second-order formulation of the all-orders GDE in the isochronous correspondence is found as

$$\begin{aligned} -Y^\mu{}_\nu \frac{Dv^\nu}{d\tau} &= \left(\nabla_\rho X^\mu{}_\sigma + \dot{X}^\mu{}_{\nu\rho} X^\nu{}_\sigma \right) u^\rho u^\sigma \\ &\quad + 2 \left(\nabla_\rho Y^\mu{}_\sigma + \dot{X}^\mu{}_{\nu\rho} Y^\nu{}_\sigma \right) u^\rho v^\sigma \\ &\quad + \left(\frac{\partial}{\partial y^\rho} Y^\mu{}_\sigma + \dot{Y}^\mu{}_{\nu\rho} Y^\nu{}_\sigma \right) v^\rho v^\sigma, \end{aligned} \quad (8.31)$$

where it should be understood that ∇_ρ will be acted only on the Riemann tensors inside the Q -tensors.

By inverting the matrix $Y^\mu{}_\nu$ on the left-hand side and using Eq. (8.31), we obtained the explicit GDE in the form $-Dv^\mu/d\tau = \dots$ up to $\mathcal{O}(y^{10})$. Moreover, we also verified that the GDE obtained in this way is exactly reproduced from varying the second-order Lagrangian in Eq. (8.25), up to $\mathcal{O}(y^5)$. Here, we spell out the explicit GDE up to $\mathcal{O}(y^5)$ due to limited space:⁴

$$\begin{aligned}
-\frac{Dv}{d\tau} = & \dot{Q}_2(u, u) + \frac{1}{2!} \left[\left(\dot{Q}_3(u, u) + \nabla_u(Q_2u) \right) + 4\dot{Q}_2(v, u) \right] \\
& + \frac{1}{3!} \left[\left(\dot{Q}_4(u, u) + \nabla_u(Q_3u) + \dot{Q}_2(u, Q_2u) + 3\dot{Q}_2(Q_2u, u) - 2Q_2\dot{Q}_2(u, u) \right) \right. \\
& \quad \left. + \left(2\nabla_v(Q_2u) + 6\dot{Q}_3(v, u) \right) + 4\dot{Q}_2(v, v) \right] \\
& + \frac{1}{4!} \left[\left(\dot{Q}_5(u, u) + \nabla_u(Q_4u) + (\nabla_u Q_2)(Q_2u) + Q_2\nabla_u(Q_2u) - 2Q_2\nabla_u(Q_2u) \right. \right. \\
& \quad \left. \left. + 6\dot{Q}_3(Q_2u, u) + 3\dot{Q}_3(u, Q_2u) + \dot{Q}_2(u, Q_3u) + 4\dot{Q}_2(Q_3u, u) \right. \right. \\
& \quad \left. \left. - 2Q_2\dot{Q}_3(u, u) - 2Q_3\dot{Q}_2(u, u) \right) \right. \\
& \quad \left. + \left(4\nabla_u(Q_3v) + 8\dot{Q}_4(v, u) + 8\dot{Q}_2(v, Q_2u) + 8\dot{Q}_2(Q_2v, u) - 8Q_2\dot{Q}_2(v, u) \right) \right. \\
& \quad \left. + \left(2\nabla_v(Q_2v) + 10\dot{Q}_3(v, v) \right) \right] \\
& + \frac{1}{5!} \left[\left(\dot{Q}_6(u, u) + \nabla_u(Q_5u) + 10\dot{Q}_4(Q_2u, u) + 10\dot{Q}_3(Q_3u, u) + 5\dot{Q}_2(Q_4u, u) \right. \right. \\
& \quad \left. \left. + 5\dot{Q}_2(Q_2Q_2u, u) + Q_2\nabla_u(Q_3u) + 3Q_3\nabla_u(Q_2u) \right. \right. \\
& \quad \left. \left. + (\nabla_u Q_2)Q_3 + 3(\nabla_u Q_3)Q_2 - \frac{10}{3}Q_2\nabla_u(Q_3u) - 5Q_3\nabla_u(Q_2u) \right. \right. \\
& \quad \left. \left. + \dot{Q}_2(u, Q_4u) + 4\dot{Q}_3(u, Q_3u) + 6\dot{Q}_4(u, Q_2u) + 10\dot{Q}_2(Q_2u, Q_2u) \right. \right. \\
& \quad \left. \left. + \dot{Q}_2(u, Q_2Q_2u) - \frac{10}{3}Q_2\dot{Q}_4(u, u) - 10Q_2\dot{Q}_2(Q_2u, u) - 5Q_3\dot{Q}_3(u, u) \right. \right. \\
& \quad \left. \left. - 3Q_4\dot{Q}_2(u, u) + \frac{7}{3}Q_2Q_2\dot{Q}_2(u, u) - \frac{10}{3}Q_2\dot{Q}_2(u, Q_2u) \right) \right. \\
& \quad \left. + \left(10\dot{Q}_5(v, u) + 6\nabla_u(Q_4v) + 2Q_2\nabla_u(Q_2v) + 2(\nabla_u Q_2)(Q_2v) \right. \right. \\
& \quad \left. \left. - \frac{20}{3}Q_2\nabla_u(Q_2v) + 20\dot{Q}_3(Q_2v, u) + 20\dot{Q}_2(Q_3v, u) + 10\dot{Q}_2(v, Q_3u) \right. \right. \\
& \quad \left. \left. + 30\dot{Q}_3(v, Q_2u) - 20Q_2\dot{Q}_3(v, u) - 20Q_3\dot{Q}_2(v, u) \right) \right. \\
& \quad \left. + \left(6\nabla_v(Q_3v) + 18\dot{Q}_4(v, v) + 6\dot{Q}_2(Q_2v, v) - 2Q_2\dot{Q}_2(v, v) \right) \right. \\
& \quad \left. + \left(10\dot{Q}_2(v, Q_2v) - \frac{40}{3}Q_2\dot{Q}_2(v, v) \right) \right] \\
& + \mathcal{O}(y^6).
\end{aligned}
\tag{8.32}$$

We have color-coded terms due to Y^{-1} and adopted a condensed notation:

⁴The $\mathcal{O}(y^4)$ GDE provided by Vines [718] describes -1 (instead of 4) for the coefficient of the term $Q_2(Q_3u, u)$ and -8 (instead of 8) for the coefficient of the terms $\dot{Q}_2(v, Q_2u)$ and $\dot{Q}_2(Q_2v, u)$. As our $\mathcal{O}(y^5)$ Lagrangian in Eq. (8.40d) agrees flawlessly with Eq. (5) in Vines, and since it is explicitly verified in the ancillary file `Low.nb` provided with Ref. [695] that the variation of the $\mathcal{O}(y^5)$ Lagrangian implies our $\mathcal{O}(y^4)$ GDE, we suppose the above mismatched coefficients are typos.

$Dv/d\tau \rightarrow Dv^\mu/d\tau$, $\dot{Q}_3(v, u) \rightarrow (\dot{Q}_3)^\mu{}_{\rho\sigma} v^\rho u^\sigma$, $Q_2 \dot{Q}_2(u, u) \rightarrow (Q_2)^\mu{}_\nu (\dot{Q}_2)^\nu{}_{\rho\sigma} u^\rho u^\sigma$, $\nabla_v(Q_2 u) \rightarrow v^\rho (\nabla_\rho(Q_2)^\mu{}_\sigma) u^\sigma$, $(\nabla_u Q_2)(Q_2 u) \rightarrow u^\rho (\nabla_\rho(Q_2)^\mu{}_\sigma) (Q_2)^\sigma{}_\kappa u^\kappa$, etc. The sources of the minus signs on the right-hand side are either Y^{-1} or the Riemann tensor identities used for simplifying the quadratic-in- v part.

8.8 Zero-Torsion Identities

One may notice that the behavior of $\dot{X}^\mu{}_{\nu\rho}$ and $\dot{Y}^\mu{}_{\nu\rho}$ in Eq. (8.31) is suggestive of connection coefficients. In fact, they arise from the conjugation $e^{\mathcal{L}_N^D} D e^{-\mathcal{L}_N^D} = e^{[\mathcal{L}_N^D, \cdot]} D$ of the covariant exterior derivative. In turn, it can be seen that they encode the covariant derivative at the deviated point: see Eq. (8.53).

On a related note, the torsion-free condition for the Levi-Civita connection, as $Ddx^\mu = \Gamma^\mu{}_{\rho\sigma} dx^\sigma \wedge dx^\rho = 0$, implies $0 = (e^{\mathcal{L}_N^D} D e^{-\mathcal{L}_N^D})(e^{\mathcal{L}_N^D} dx^\mu) = D(X^\mu{}_\sigma dx^\sigma + Y^\mu{}_\sigma Dy^\sigma) + (\dot{X}^\mu{}_{\nu\rho} dx^\rho + \dot{Y}^\mu{}_{\nu\rho} Dy^\rho) \wedge (X^\nu{}_\sigma dx^\sigma + Y^\nu{}_\sigma Dy^\sigma)$, which unpacks into three identities:

$$\nabla_{[\rho} X^\mu{}_{\sigma]} + \dot{X}^\mu{}_{\nu[\rho} X^\nu{}_{\sigma]} = -\frac{1}{2} Y^\mu{}_\lambda R^\lambda{}_{\nu\rho\sigma}, \quad (8.33a)$$

$$\nabla_\rho Y^\mu{}_\sigma + \dot{X}^\mu{}_{\nu\rho} Y^\nu{}_\sigma = \frac{\partial}{\partial y^\rho} X^\mu{}_\sigma + \dot{Y}^\mu{}_{\nu\rho} X^\nu{}_\sigma, \quad (8.33b)$$

$$\frac{\partial}{\partial y^{[\rho}} Y^\mu{}_{\sigma]} + \dot{Y}^\mu{}_{\nu[\rho} Y^\nu{}_{\sigma]} = 0. \quad (8.33c)$$

Especially, we have made use of Eq. (8.33b) in Eq. (8.31) to simplify the computation of the term linear in both u and v . The ancillary file `Low.nb` provided with Ref. [695] provides an explicit check of Eqs. (8.33a) and (8.33c) up to $\mathcal{O}(y^5)$ or $\mathcal{O}(y^4)$, which exploits various identities about the Riemann tensor. Also, note that more identities follow in a similar fashion by conjugating $D^2 = R$, $[D, R] = 0$, etc.

8.9 Summary and Outlook

In this work, we revisited the problem of finding the all-orders-exact GDE for finite separations by formulating geodesic deviation and transport as a flow along a vector field in tangent bundle. The technique of covariant Lie derivative then systematically defines and generates various bitensors with the parallel propagators peeled off, directly producing manifestly covariantized expressions at the original point. This achieves an in-in formalism for geodesic deviation that serves as an alternative to the Synge calculus.

The explicit outcomes are the all-orders formula for the Jacobi propagators in Eq. (8.21) as well as the second-order GDE and its Lagrangian obtained up to $\mathcal{O}(y^{10})$.

Our tangent-bundle formalism for geodesic deviation is a versatile tool and can find applications in various areas of physics, besides classical gravitation: covariant worldline perturbation theory [750–764], quantum field theory [282], and sigma models [391, 764–767], for instance. Sec. 8.C implements our formalism for gauge-covariant translations in nonabelian gauge theories.

Furthermore, we realize that our innocuous attempt to remaster the fundamental subject of all-orders-in-deviation GDE surprisingly connects to a persistent and seemingly disparate problem in modern gravitational physics: finding the all-orders-in-spin EoM for the Kerr black hole in its effective point-particle description [93, 94, 97, 98, 207–209, 226]. By implementing a probe counterpart of the Newman-Janis algorithm [202], we have found that the all-orders GDE for an imaginary deviation can deduce a part of the black hole’s all-orders-in-spin EoM, yielding a consistent nonlinear completion of the Mathisson-Papapetrou-Dixon [227–229] equations. The details have been given in Refs. [768–770].

8.A The “Organic Chemistry” of Covariant Lie Derivative Calculus

In this appendix, we devise a variant of the Penrose graphical notation [771–773] with the following rules.

$$\begin{aligned}
 \mu - \circ &= dx^\mu \\
 \mu - \bullet &= y^\mu \\
 \mu - \odot &= Dy^\mu
 \end{aligned}
 \quad
 \begin{aligned}
 \begin{array}{c} \mu \\ \diagup \\ \bullet \\ \diagdown \\ \nu \end{array}
 &=
 \begin{array}{c} \nu \\ \diagup \\ \bullet \\ \diagdown \\ \mu \end{array}
 = R^\mu{}_{\nu\rho\sigma}(x)
 \end{aligned}
 \quad (8.34)$$

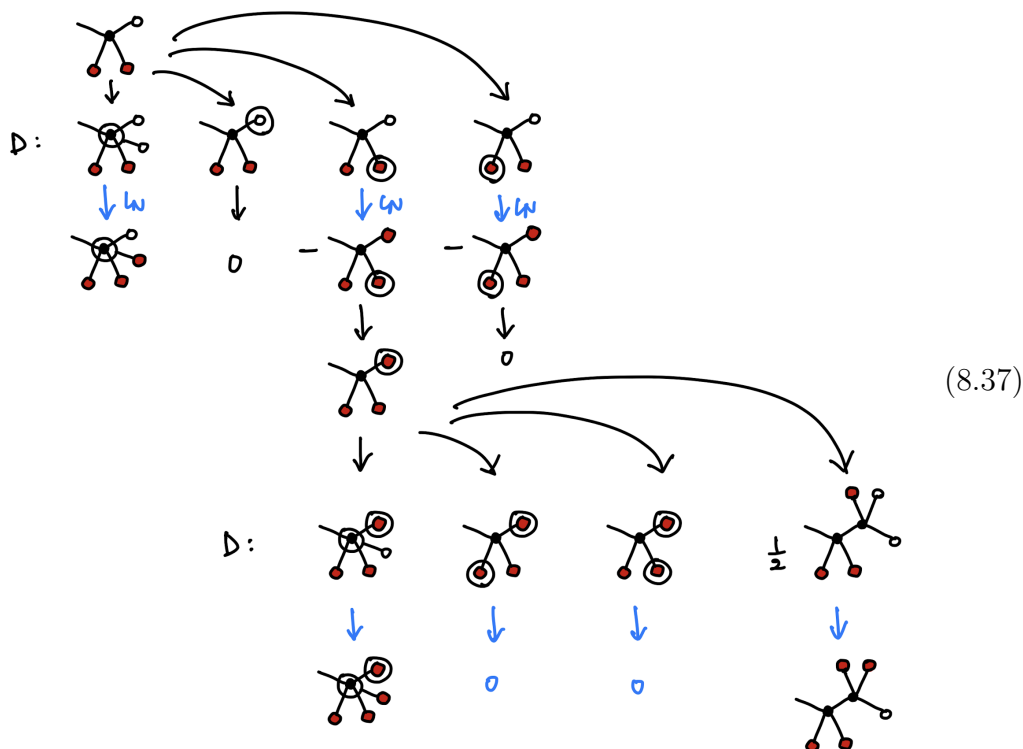
The higher covariant derivatives of the Riemann tensor will be denoted as the following.

$$\begin{aligned}
 \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \mu \quad \sigma \\ \alpha_k \end{array}
 &=
 \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \mu \quad \sigma \\ \bullet \quad \dots \quad \bullet \\ \alpha_k \end{array}
 = R^\mu{}_{\nu\rho\sigma;\kappa_1\dots\kappa_k}(x) y^\nu y^\rho y^{\kappa_1} \dots y^{\kappa_k}
 \end{aligned}
 \quad (8.35)$$

The relevant “molecular backbones” are the “carbon” (as Riemann tensor) chains of the following form.



The computation of $\iota_N D(\iota_N R^\mu{}_\nu)$ and $(\iota_N D)^2(\iota_N R^\mu{}_\nu)$ proceeds as the following.



The ordering for the differential forms should be clear from the orientations designated in Eq. (8.34). The interior product ι_N is a “reduction reaction” that substitutes a “hydroxyl group” (as dx) with a “hydrogen” (as y). Observe the mechanism for the “carbon polymerization” (as Q -tensor concatenation).

8.B Explicit Results

8.B.1 The Jacobi Propagators

The Jacobi propagators with the parallel propagator peeled off are given in Eq. (8.21). It is then trivial to enumerate them explicitly up to any desired

order, say $\mathcal{O}(y^9)$. With the definition of the Q -tensors in Eq. (8.18), they are

$$Dy^\mu = Dy^\mu, \quad (8.38a)$$

$$(\iota_N R^\mu{}_\nu) y^\nu = (Q_2)^\mu{}_\sigma dx^\sigma, \quad (8.38b)$$

$$(\iota_N D \iota_N R^\mu{}_\nu) y^\nu = (Q_3)^\mu{}_\sigma dx^\sigma + (Q_2)^\mu{}_\sigma Dy^\sigma, \quad (8.38c)$$

$$((\iota_N D)^2 \iota_N R^\mu{}_\nu) y^\nu = (Q_4 + Q_2 Q_2)^\mu{}_\sigma dx^\sigma + (2Q_3)^\mu{}_\sigma Dy^\sigma, \quad (8.38d)$$

$$\begin{aligned} ((\iota_N D)^3 \iota_N R^\mu{}_\nu) y^\nu &= (Q_5 + 3Q_3 Q_2 + Q_2 Q_3)^\mu{}_\sigma dx^\sigma \\ &\quad + (3Q_4 + Q_2 Q_2)^\mu{}_\sigma Dy^\sigma, \end{aligned} \quad (8.38e)$$

$$\begin{aligned} ((\iota_N D)^4 \iota_N R^\mu{}_\nu) y^\nu &= (Q_6 + 6Q_4 Q_2 + 4Q_3 Q_3 + Q_2 Q_4)^\mu{}_\sigma dx^\sigma \\ &\quad + (4Q_5 + 4Q_3 Q_2 + 2Q_2 Q_3)^\mu{}_\sigma Dy^\sigma \\ &\quad + (Q_2 Q_2 Q_2)^\mu{}_\sigma dx^\sigma, \end{aligned} \quad (8.38f)$$

$$\begin{aligned} ((\iota_N D)^5 \iota_N R^\mu{}_\nu) y^\nu &= (Q_7 + 10Q_5 Q_2 + 10Q_4 Q_3 + 5Q_3 Q_4 + Q_2 Q_5)^\mu{}_\sigma dx^\sigma \\ &\quad + (5Q_6 + 10Q_4 Q_2 + 10Q_3 Q_3 + 3Q_2 Q_4)^\mu{}_\sigma Dy^\sigma \\ &\quad + (5Q_3 Q_2 Q_2 + 3Q_2 Q_3 Q_2 + Q_2 Q_2 Q_3)^\mu{}_\sigma dx^\sigma \\ &\quad + (Q_2 Q_2 Q_2)^\mu{}_\sigma Dy^\sigma, \end{aligned} \quad (8.38g)$$

$$\begin{aligned} ((\iota_N D)^6 \iota_N R^\mu{}_\nu) y^\nu &= (Q_8 + 15Q_6 Q_2 + 20Q_5 Q_3 + 15Q_4 Q_4 + 6Q_3 Q_5 + Q_2 Q_6)^\mu{}_\sigma dx^\sigma \\ &\quad + (6Q_7 + 20Q_5 Q_2 + 30Q_4 Q_3 + 18Q_3 Q_4 + 4Q_2 Q_5)^\mu{}_\sigma Dy^\sigma \\ &\quad + \left(\begin{array}{c} 15Q_4 Q_2 Q_2 + 18Q_3 Q_3 Q_2 + 6Q_2 Q_4 Q_2 \\ + 6Q_3 Q_2 Q_3 + 4Q_2 Q_3 Q_3 + Q_2 Q_2 Q_4 \end{array} \right)^\mu{}_\sigma dx^\sigma \\ &\quad + (6Q_3 Q_2 Q_2 + 4Q_2 Q_3 Q_2 + 2Q_2 Q_2 Q_3)^\mu{}_\sigma Dy^\sigma \\ &\quad + (Q_2 Q_2 Q_2 Q_2)^\mu{}_\sigma dx^\sigma, \end{aligned} \quad (8.38h)$$

$$\begin{aligned} ((\iota_N D)^7 \iota_N R^\mu{}_\nu) y^\nu &= (Q_9 + 21Q_7 Q_2 + 35Q_6 Q_3 + 35Q_5 Q_4 + 21Q_4 Q_5 + 7Q_3 Q_6 + Q_2 Q_7)^\mu{}_\sigma dx^\sigma \\ &\quad + (7Q_8 + 35Q_6 Q_2 + 70Q_5 Q_3 + 63Q_4 Q_4 + 28Q_3 Q_5 + 5Q_2 Q_6)^\mu{}_\sigma Dy^\sigma \\ &\quad + \left(\begin{array}{c} 35Q_5 Q_2 Q_2 + 63Q_4 Q_3 Q_2 + 42Q_3 Q_4 Q_2 + 10Q_2 Q_5 Q_2 \\ + 21Q_4 Q_2 Q_3 + 28Q_3 Q_3 Q_3 + 10Q_2 Q_4 Q_3 \\ + 7Q_3 Q_2 Q_4 + 5Q_2 Q_3 Q_4 + Q_2 Q_2 Q_5 \end{array} \right)^\mu{}_\sigma dx^\sigma \\ &\quad + \left(\begin{array}{c} 21Q_4 Q_2 Q_2 + 28Q_3 Q_3 Q_2 + 10Q_2 Q_4 Q_2 \\ + 14Q_3 Q_2 Q_3 + 10Q_2 Q_3 Q_3 + 3Q_2 Q_2 Q_4 \end{array} \right)^\mu{}_\sigma Dy^\sigma \\ &\quad + (7Q_3 Q_2 Q_2 Q_2 + 5Q_2 Q_3 Q_2 Q_2 + 3Q_2 Q_2 Q_3 Q_2 + Q_2 Q_2 Q_2 Q_3)^\mu{}_\sigma dx^\sigma \\ &\quad + (Q_2 Q_2 Q_2 Q_2)^\mu{}_\sigma Dy^\sigma. \end{aligned} \quad (8.38i)$$

Eqs. (8.38a)-(8.38e) agree exactly with Eq. (83) of Ref. [718].

8.B.2 The Lagrangian

The Lagrangian for the all-orders GDE in isochronous correspondence is given in Eq. (8.25):

$$\begin{aligned} L &= \frac{m}{2}(Xu + Yv)^2 - \frac{m}{2}, \\ &= m \left(\frac{u^2 - 1}{2} + u \cdot v \right) + \frac{1}{2}mv^2 + m \sum_{\ell=2}^{\infty} \frac{1}{\ell!} \mathcal{L}_{\ell}. \end{aligned} \quad (8.39)$$

In the last expression, the bracketed terms can be discarded as a constant plus a total derivative. The term $\frac{1}{2}mv^2$, on the other hand, is the standard kinetic energy. Hence it remains to spell out the “interaction Lagrangian” $\mathcal{L}_{\ell}$ at each order ℓ , which follows from the explicit Jacobi propagators by straightforward algebra:

$$\mathcal{L}_2 = u Q_2 u, \quad (8.40a)$$

$$\mathcal{L}_3 = u Q_3 u + v \left(4Q_2 \right) u, \quad (8.40b)$$

$$\mathcal{L}_4 = u \left(Q_4 + 4Q_2 Q_2 \right) u + v \left(6Q_3 \right) u + v \left(4Q_2 \right) v, \quad (8.40c)$$

$$\mathcal{L}_5 = u \left(Q_5 + 14Q_3 Q_2 \right) u + v \left(8Q_4 + 16Q_2 Q_2 \right) u + v \left(10Q_3 \right) v, \quad (8.40d)$$

$$\begin{aligned} \mathcal{L}_6 &= u \left(Q_6 + 22Q_4 Q_2 + 14Q_3 Q_3 + 16Q_2 Q_2 Q_2 \right) u \\ &\quad + v \left(10Q_5 + 50Q_3 Q_2 + 30Q_2 Q_3 \right) u + v \left(18Q_4 + 16Q_2 Q_2 \right) v, \end{aligned} \quad (8.40e)$$

$$\begin{aligned} \mathcal{L}_7 &= u \left(Q_7 + 32Q_5 Q_2 + 50Q_4 Q_3 + 62Q_3 Q_2 Q_2 + 66Q_2 Q_3 Q_2 \right) u \\ &\quad + v \left(12Q_6 + 108Q_4 Q_2 + 108Q_3 Q_3 + 52Q_2 Q_4 + 64Q_2 Q_2 Q_2 \right) u \\ &\quad + v \left(28Q_5 + 112Q_3 Q_2 \right) v, \end{aligned} \quad (8.40f)$$

$$\begin{aligned} \mathcal{L}_8 &= u \left(\begin{aligned} &Q_8 + 44Q_6 Q_2 + 82Q_5 Q_3 + 50Q_4 Q_4 \\ &+ 114Q_4 Q_2 Q_2 + 302Q_3 Q_3 Q_2 + 62Q_3 Q_2 Q_3 \\ &+ 174Q_2 Q_4 Q_2 + 64Q_2 Q_2 Q_2 Q_2 \end{aligned} \right) u \\ &\quad + v \left(\begin{aligned} &14Q_7 + 196Q_5 Q_2 + 266Q_4 Q_3 + 210Q_3 Q_4 + 84Q_2 Q_5 \\ &+ 238Q_3 Q_2 Q_2 + 308Q_2 Q_3 Q_2 + 126Q_2 Q_2 Q_3 \end{aligned} \right) u \\ &\quad + v \left(40Q_6 + 220Q_3 Q_3 + 272Q_4 Q_2 + 64Q_2 Q_2 Q_2 \right) v. \end{aligned} \quad (8.40g)$$

Eqs. (8.40a)-(8.40d) agree flawlessly with Eq. (5) in Vines [718].

8.B.3 The Geodesic Deviation Equation

The GDE, in the second-order formulation, is given in Eq. (8.31). The definition of $X^\mu{}_\sigma$ and $Y^\mu{}_\sigma$ is given in Eq. (8.21). The definition of $\dot{X}^\mu{}_{\nu\rho}$ and $\dot{Y}^\mu{}_{\nu\rho}$ is given in Eq. (8.29). For the reader's sake, we explicitly spell them out at low orders:

$$X^\mu{}_\sigma = \delta^\mu{}_\sigma + \frac{1}{2!}(Q_2)^\mu{}_\sigma + \frac{1}{3!}(Q_3)^\mu{}_\sigma + \frac{1}{4!}(Q_4 + Q_2Q_2)^\mu{}_\sigma \quad (8.41a)$$

$$+ \frac{1}{5!}(Q_5 + 3Q_3Q_2 + Q_2Q_3)^\mu{}_\sigma + \mathcal{O}(y^6),$$

$$Y^\mu{}_\sigma = \delta^\mu{}_\sigma + \frac{1}{3!}(Q_2)^\mu{}_\sigma + \frac{1}{4!}(2Q_3)^\mu{}_\sigma + \frac{1}{5!}(3Q_4 + Q_2Q_2)^\mu{}_\sigma, \quad (8.41b)$$

$$+ \frac{1}{6!}(4Q_5 + 4Q_3Q_2 + 2Q_2Q_3)^\mu{}_\sigma + \mathcal{O}(y^6)$$

$$\dot{X}^\mu{}_{\nu\sigma} = \frac{1}{1!}(\dot{Q}_2)^\mu{}_{\nu\sigma} + \frac{1}{2!}(\dot{Q}_3)^\mu{}_{\nu\sigma} + \frac{1}{3!}\left((\dot{Q}_4)^\mu{}_{\nu\sigma} + (\dot{Q}_2)^\mu{}_{\nu\lambda}(Q_2)^\lambda{}_\sigma\right) \quad (8.41c)$$

$$+ \frac{1}{4!}\left((\dot{Q}_5)^\mu{}_{\nu\sigma} + 3(\dot{Q}_3)^\mu{}_{\nu\lambda}(Q_2)^\lambda{}_\sigma + (\dot{Q}_2)^\mu{}_{\nu\lambda}(Q_3)^\lambda{}_\sigma\right) + \mathcal{O}(y^5),$$

$$\dot{Y}^\mu{}_{\nu\sigma} = \frac{1}{2!}(\dot{Q}_2)^\mu{}_{\nu\sigma} + \frac{1}{3!}(2\dot{Q}_3)^\mu{}_{\nu\sigma} + \frac{1}{4!}\left(3(\dot{Q}_4)^\mu{}_{\nu\sigma} + (\dot{Q}_2)^\mu{}_{\nu\lambda}(Q_2)^\lambda{}_\sigma\right) \quad (8.41d)$$

$$+ \frac{1}{5!}\left(4(\dot{Q}_5)^\mu{}_{\nu\sigma} + 4(\dot{Q}_3)^\mu{}_{\nu\lambda}(Q_2)^\lambda{}_\sigma + 2(\dot{Q}_2)^\mu{}_{\nu\lambda}(Q_3)^\lambda{}_\sigma\right) + \mathcal{O}(y^5).$$

Next, we need to compute the inverse $(Y^{-1})^\mu{}_\nu$ of the Jacobi propagator $Y^\mu{}_\nu$ in Eq. (8.41b), which is viable by geometric series expansion around $\delta^\mu{}_\nu$:

$$Y^{-1} = \mathbb{1} - \frac{1}{3!}\left(Q_2\right) - \frac{1}{4!}\left(2Q_3\right) - \frac{1}{5!}\left(3Q_4 - \frac{7}{3}Q_2Q_2\right) \quad (8.42)$$

$$- \frac{1}{6!}\left(4Q_5 - 8Q_2Q_3 - 6Q_3Q_2\right)$$

$$- \frac{1}{7!}\left(5Q_6 - 11Q_4Q_2 - 25Q_3Q_3 - 18Q_2Q_4 + \frac{31}{3}Q_2Q_2Q_2\right)$$

$$- \frac{1}{8!}\left(6Q_7 - \frac{52}{3}Q_5Q_2 - 54Q_4Q_3 - 66Q_3Q_4 - \frac{100}{3}Q_2Q_5\right.$$

$$\left. + 34Q_3Q_2Q_2 + \frac{124}{3}Q_2Q_3Q_2 + \frac{146}{3}Q_2Q_2Q_3\right).$$

Finally, the right-hand side of Eq. (8.31) should be computed. The algebraic terms, $\dot{X}^\mu{}_{\nu\rho}X^\nu{}_\sigma$, $\dot{X}^\mu{}_{\nu\rho}Y^\nu{}_\sigma$, $\dot{Y}^\mu{}_{\nu\rho}X^\nu{}_\sigma$, and $\dot{Y}^\mu{}_{\nu\rho}Y^\nu{}_\sigma$, are straightforward to evaluate from Eq. (8.41). The evaluation of the differential terms, however,

is relatively less trivial. The issue in particular is the evaluation of the y -derivative. When a y -derivative hits a Q -tensor, it inserts v in all possible positions in the string of y -vectors. Such terms can be gathered and simplified by commuting covariant derivatives and using Bianchi identities of the Riemann tensor in various ways.

For the part linear in both u and v , this issue is simply avoided by virtue of the zero-torsion identity in Eq. (8.33b). Hence it suffices to massage the part quadratic in v , which is the third term in the right-hand side of Eq. (8.31).

As the result of this simplification, we obtain the final expression as a sum of terms that strictly conform to the following particular tensor structures:

$$\text{Derivative : } (Q_{\gamma_1} \cdots Q_{\gamma_r})^\mu{}_\nu (Q_\ell)^\nu{}_{\kappa|\rho} (Q_{\alpha_1} \cdots Q_{\alpha_p})^\kappa{}_\sigma, \quad (8.43a)$$

$$\text{Branching : } (Q_{\gamma_1} \cdots Q_{\gamma_r})^\mu{}_\nu (\dot{Q}_\ell)^\nu{}_{\lambda\kappa} (Q_{\alpha_1} \cdots Q_{\alpha_p})^\kappa{}_\sigma (Q_{\beta_1} \cdots Q_{\beta_q})^\lambda{}_\rho. \quad (8.43b)$$

Here, we have denoted

$$(Q_\ell)^\mu{}_{\sigma|\rho} := y^{\kappa_1} \cdots y^{\kappa_{\ell-1}} R^\mu{}_{\kappa_1 \kappa_2 \sigma; \kappa_3; \dots; \kappa_{\ell-1}; \rho}(x) \quad \text{for } \ell \geq 3. \quad (8.44)$$

Crucially, the free index ρ in Eq. (8.44), which can be contracted with u or v , for instance, is made to describe the last (outmost) covariant derivative acting on the Riemann tensor. This is always possible by commuting covariant derivatives, while employing the Bianchi identities facilitates rewriting the remaining terms strictly in terms of the Q - and $\dot{Q}$ -tensors.

The final results are contained in the ancillary Mathematica notebook `A11.nb` provided with Ref. [695].

8.C Gauge-Covariant Translations in Nonabelian Gauge Theory

It should be remarked that our framework applies to not only gravity but also nonabelian gauge theories. Suppose a nonabelian gauge theory in a d -dimensional flat spacetime $\mathbb{M}$ with gauge group G . Let $A^i_j = A^i_{j\mu}(x) dx^\mu$ be the gauge connection and let $F^i_j = dA^i_j + A^i_k \wedge A^k_j$ be its curvature, where $i, j, \dots = 1, 2, \dots, N$ are the fundamental indices. For example, suppose a vector bundle E over $\mathbb{M}$ whose typical fiber is $\mathbb{C}^N$. This bundle is coordinatized by x^μ and ψ^i , associated respectively with the base and fiber.

Our tangent bundle formalism then considers the larger bundle $\mathcal{P} = T\mathbb{M} \oplus E$, which is locally isomorphic to $\mathbb{R}^{2d} \times \mathbb{C}^N$. Local trivialization equips $\mathcal{P}$ with coordinates x^μ , y^μ , and ψ^i . In $\mathcal{P}$, the generator of gauge-covariant translations

is given by the vector field

$$N = y^\rho \left(\frac{\partial}{\partial x^\rho} - A^i_{j\rho}(x) \psi^j \frac{\partial}{\partial \psi^i} \right). \quad (8.45)$$

Again, this describes a horizontal vector field due to the Ehresmann [744] notion of the connection A . Recalling Eq. (8.4), one derives the set of first-order differential equations it encodes, which leads to the conclusion that it generates gauge-covariant translations: $x^\mu \mapsto x^\mu + y^\mu$, $y^\mu \mapsto y^\mu$, and $\psi^i \mapsto \psi^{i'} = W^{i'}_i \psi^i$. Here, the Wilson line $W^{i'}_i$ is given by the path-ordered exponential,

$$\text{P exp} \left(- \int_0^1 ds A_\rho(x + ys) y^\rho \right), \quad (8.46)$$

which describes the parallel transport along the straight path from x to $x + y$. Under gauge transformations, Eq. (8.46) transforms bilocally.

Now consider the “ $\mathcal{L}_N^D$ sequence” of the one-form $D\psi^i = d\psi^i + A^i_{j\rho}(x) \psi^j dx^\rho$.

$$\begin{array}{ccc} D\psi^i & \xrightarrow{D\iota_N} & 0 \\ \iota_N D \downarrow & & \\ \iota_N F^i_j \psi^j & \longrightarrow & 0 \\ \downarrow & & \\ \iota_N D \iota_N F^i_j \psi^j & \longrightarrow & 0 \\ \downarrow & & \\ \vdots & & \end{array} \quad (8.47)$$

This derives

$$e^{\mathcal{L}_N^D} D\psi^i = D\psi^i + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} (\iota_N D)^{\ell-1} \iota_N F^i_j \psi^j, \quad (8.48a)$$

$$= D\psi^i + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} \left((P_\ell)^i_{j\sigma} dx^\sigma + (\ell-1) (P_{\ell-1})^i_{j\sigma} dy^\sigma \right) \psi^j, \quad (8.48b)$$

where the “ P -tensors” are defined for $\ell \geq 1$ as

$$(P_\ell)^i_{j\sigma} := y^{\kappa_1} \cdots y^{\kappa_\ell} F^i_{j\kappa_1\sigma; \kappa_2; \dots; \kappa_\ell}(x) \implies (P_\ell)^i_{j\sigma} y^\sigma = 0. \quad (8.49)$$

The ancillary file `P.nb` provided with Ref. [695] verifies Eq. (8.48b) in a direct fashion up to $\mathcal{O}(y^{10})$. For a physical application, Eq. (8.48) can be used for deriving the Wong equations [774] of a color-charged particle as seen by an arbitrary observer.

To be further concrete, the dressing identity $\delta^{i'}_i e^{\mathcal{L}_N}(D\psi^i) = W^{i'}_i(e^{\mathcal{L}_N^D} D\psi^i)$ unpacks the content of Eq. (8.48b) as

$$\begin{aligned} d\psi^{i'} + A^{i'}_{j\rho}(z) \psi^{j'} dz^\rho \\ = W^{i'}_i \left(d\psi^i + A^i_{j\rho}(x) \psi^j dx^\rho \right) \\ + W^{i'}_i \sum_{\ell=1}^{\infty} \frac{1}{\ell!} \left((P_\ell)^i_{j\sigma} dx^\sigma + (\ell-1) (P_{\ell-1})^i_{j\sigma} dy^\sigma \right) \psi^j, \end{aligned} \quad (8.50)$$

where the deviated coordinates z^μ are understood as $x^\mu + y^\mu$. This reveals that Eq. (8.48b) encodes

$$W^{i'}_{i'} A^{i'}_{j\rho}(z) W^{j'}_j + W^{i'}_{i'} \frac{\partial}{\partial x^\rho} W^{i'}_j = A^i_{j\rho}(x) + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} (P_\ell)^i_{j\rho}, \quad (8.51a)$$

$$W^{i'}_{i'} A^{i'}_{j\rho}(z) W^{j'}_j + W^{i'}_{i'} \frac{\partial}{\partial y^\rho} W^{i'}_j = \sum_{\ell=2}^{\infty} \frac{1}{\ell!} (\ell-1) (P_{\ell-1})^i_{j\rho}. \quad (8.51b)$$

Especially, Eq. (8.51a) describes a “gauge transformation” of the connection $A^{i'}_{j\rho}(z)$ at the deviated point z , via the Wilson line to the original point x . In this sense, its right-hand side describes an avatar of the gauge connection. In fact, the sum of P -tensors in Eqs. (8.51a) and (8.51b), which computes the difference between the connection at z dragged back to x and the connection at x , can be used for studying the Fock-Schwinger gauge [775].

Note that all variables in Eqs. (8.51a) and (8.51b) are understood as functions of x and y , by the very construction of our formalism. Also, Eqs. (8.51a) and (8.51b) can be reproduced from the following formula that describes generic variations of the Wilson line $W(s_2, s_1) = \text{P exp}(-\int_{s_1}^{s_2} ds A_\rho(\gamma(s)) \dot{\gamma}^\rho(s))$ about an arbitrary contour $s \mapsto \gamma^\mu(s)$:

$$\begin{aligned} W(0, 1) \delta W(1, 0) &= A_\sigma(\gamma(0)) \delta \gamma^\sigma(0) - W(0, 1) A_\sigma(\gamma(1)) \delta \gamma^\sigma(1) W(1, 0) \\ &+ \int_0^1 ds \dot{\gamma}^\rho(s) W(0, s) F_{\rho\sigma}(\gamma(s)) W(s, 0) \delta \gamma^\sigma(s). \end{aligned} \quad (8.52)$$

Note that the identities in Eq. (8.51) are the consequences of the conjugation $e^{\mathcal{L}_N^D} D e^{-\mathcal{L}_N^D}$; recall the discussion around Eq. (8.33). In the same fashion, more identities follow by conjugating $D^2 = F$, $[D, F] = 0$, etc. by $e^{\mathcal{L}_N^D}$.

Lastly, the identities analogous to Eq. (8.51) in Riemannian geometry are

$$W^\mu_{\mu'} \Gamma^{\mu'}_{\nu\rho'}(z) W^{\nu'}_{\nu} W^{\rho'}_{\kappa} X^\kappa_{\rho} + W^\mu_{\mu'} \left(\frac{\partial}{\partial x^\rho} - \Gamma^\kappa_{\lambda\rho}(x) y^\lambda \frac{\partial}{\partial y^\kappa} \right) W^{\mu'}_{\nu} \quad (8.53a)$$

$$= \Gamma^\mu_{\nu\rho}(x) + \dot{X}^\mu_{\nu\rho},$$

$$W^\mu_{\mu'} \Gamma^{\mu'}_{\nu\rho'}(z) W^{\nu'}_{\nu} W^{\rho'}_{\kappa} Y^\kappa_{\rho} + W^\mu_{\mu'} \frac{\partial}{\partial y^\rho} W^{\mu'}_{\nu} = \dot{Y}^\mu_{\nu\rho}, \quad (8.53b)$$

which can be deduced from Eq. (8.28). Notice the presence of the Jacobi propagators X^κ_{ρ} and Y^κ_{ρ} on the left-hand sides, which reflects the fact that the curve for the Wilson lines depends on the gravitational fields unlike in gauge theory.

PROBE NEWMAN-JANIS ALGORITHM

We implement a probe counterpart of NJA, which Wick rotates the all-orders geodesic deviation equation into a part of exact spinning-particle EoM. Consequently, the gravitational dynamics of the Kerr black hole in its point-particle effective theory is completely constrained in the SD sector for a hidden symmetry, implying the spin exponentiation of same-helicity gravitational Compton amplitudes to all multiplicities.

This section reproduces the contents of Ref. [768], J.-H. Kim, “Note on the Kerr spinning-particle equations of motion,” [arXiv:2512.23697 \[gr-qc\]](#).

9.1 Introduction

Suppose a free-falling probe particle in space, followed by an observer. Their separation is famously described by the geodesic deviation equation. Yet, typical textbooks describe this equation only in the limit of infinitesimal separation. The exact equation valid to all orders in *separation* and *curvature* was investigated by Vines [718]. Recently, we have revisited this topic from an alternative formalism [695].

It seems that this foundational subject of geodesic deviation is intimately tied to a pressing problem in contemporary gravitational physics: the exact dynamics of the Kerr black hole as a probe object.

Suppose a Kerr black hole in the presence of mild gravitational perturbations; it may travel through macroscopic gravitational waves or constitute an inspiral-phase binary system. It is well-established [92–97] that the black hole can be treated as a relativistic spinning point particle in the *effective theory* sense, provided separation of scales. In this effective description, the persistent problem has been determining the exact EoM for the Kerr black hole in the probe limit, to all orders in the so-called *spin length* (spin per mass, encoding the ring radius) and *curvature*. At the linear order in curvature, the EoM have been completely fixed to all orders in spin length [226, 673]. Frustratingly, ambiguities arise from the very quadratic order in curvature: in layman’s terms, physicists in this 21st century do not know the “ $\vec{F} = m\vec{a}$ ” of spinning black

holes. This agony has gained a considerable attention lately and has been scrutinized in terms of scattering amplitudes [98, 206–225].

The widespread expectation is that the Kerr black hole will exhibit the *simplest* EoM among all massive spinning objects in four dimensions, in light of its simplicity uncovered in a modern analysis [98, 206–209] that traces back to the NJA [202]. The NJA, introduced by the very article “Note on the Kerr Spinning-Particle Metric” in 1965 [202], is a method that derives the Kerr solution by displacing the Schwarzschild solution along a complexified direction in a sense. Crucially, this reinterprets spin as an imaginary deviation [252, 253]. Due to Ref. [164], the NJA now holds the status of a rigorous derivation of the Kerr solution with a definite origin.

This article shows that a unique answer exists for the simplest spinning-particle EoM to all orders in spin and curvature, in SD spacetimes. This Wick rotates the *deviation vector* in the exact geodesic deviation equation to the *spin length pseudovector* in exact spinning-particle EoM, as an incarnation of NJA at the probe level.

Our proposal is distinguished in terms of a hidden symmetry in the presence of Killing-Yano tensors. In perturbation theory, it implies the property known as spin exponentiation [98, 206–212] at all graviton multiplicities.

9.2 Schwarzschild Equations of Motion

The free-fall EoM of a scalar probe is given in the first-order formulation as

$$p^{\mu'} = \frac{dz^{\mu'}}{d\tau}, \quad \frac{Dp^{\mu'}}{d\tau} = 0, \quad (9.1)$$

where D denotes the covariant differential. $z^{\mu'}$ describes the position, while $p^{\mu'}$ describes the momentum. We have put primes on the indices with hindsight.

Now introduce an arbitrary observer at position x^μ . Ref. [695] shows that Eq. (9.1) is equivalent to

$$p^\mu = \alpha^\mu{}_\nu \frac{dx^\nu}{d\tau} + \beta^\mu{}_\nu \frac{Dy^\nu}{d\tau}, \quad \frac{Dp^\mu}{d\tau} = \Omega^\mu{}_\nu p^\nu, \quad (9.2)$$

where $\alpha^\mu{}_\nu$, $\beta^\mu{}_\nu$, and $\Omega^\mu{}_\nu$ are tensors composed of the Riemann tensor $R^\mu{}_{\nu\rho\sigma}(x)$, its covariant derivatives, and the vector y^μ , etc., all transforming tensorially at x . Also, $\Omega_{\mu\nu} = -\Omega_{\nu\mu}$ so that p^2 is conserved.

The idea is to construct a geodesic joining between x and z , whose tangent at x is y^μ :

$$z^{\mu'} = \delta^{\mu'}_{\mu} \left(x^\mu + y^\mu - \frac{1}{2} \Gamma^\mu_{\rho\sigma}(x) y^\rho y^\sigma + \mathcal{O}(y^3) \right). \quad (9.3a)$$

In the well-known Synge formalism [626, 732–734], the parallel propagator about the geodesic path is given by

$$W^{\mu'}_{\nu} = \delta^{\mu'}_{\nu} - \delta^{\mu'}_{\mu} \Gamma^\mu_{\nu\rho}(x) y^\rho + \mathcal{O}(y^2), \quad (9.3b)$$

so a vector p^μ based at x is parallel-transported to z as $p^{\mu'} = W^{\mu'}_{\nu} p^\nu$. Especially, the geodesic's tangent vector at z is $y^{\mu'} = W^{\mu'}_{\nu} y^\nu$. Multiplying Eq. (9.1) by the inverse $W^{\mu}_{\mu'}$ of $W^{\mu'}_{\mu}$ re-covariantizes tensors at z with respect to the point x , hence deriving Eq. (9.2).

Ref. [695] provides all-orders formulae for α^μ_{ν} , β^μ_{ν} , and Ω^μ_{ν} explicitly as power series in y :

$$\begin{aligned} \alpha^\mu_{\nu} = & \delta^\mu_{\nu} + \frac{1}{2} y^\rho y^\sigma R^\mu_{\rho\sigma\nu}(x) + \frac{1}{6} y^\rho y^\sigma y^\kappa R^\mu_{\rho\sigma\nu;\kappa}(x) \\ & + \frac{1}{24} y^\rho y^\sigma y^\kappa y^\lambda R^\mu_{\rho\sigma\nu;\kappa\lambda}(x) \\ & + \frac{1}{24} y^\rho y^\sigma y^\kappa y^\lambda R^\mu_{\rho\sigma\xi}(x) R^\xi_{\kappa\lambda\nu}(x) + \mathcal{O}(y^5), \end{aligned} \quad (9.4)$$

for instance. Note the nonlinearity in curvature.

Eq. (9.2) is the first-order EoM of a free-falling probe as seen by an arbitrary observer. The dynamical variables are y^μ and p^μ , while x^μ is a nondynamical reference. The second-order version is worked out in Ref. [695].

Eq. (9.2) characterizes the exact dynamics of a minimally coupled scalar probe, which can model a Schwarzschild black hole in its point-particle effective theory.

9.3 “Derivation” of Kerr Equations of Motion

Now we implement a three-step procedure that derives a distinct class of exact spinning-particle EoM in four dimensions.

Firstly, take the Schwarzschild EoM in Eq. (9.2) for an imaginary (Wick-rotated) deviation:

$$y^\mu \mapsto i y^\mu. \quad (9.5)$$

The result reads

$$p^\mu = \alpha^\mu{}_\nu \frac{dx^\nu}{d\tau} + i\beta^\mu{}_\nu \frac{Dy^\nu}{d\tau}, \quad (9.6a)$$

$$\frac{Dp^\mu}{d\tau} = \Omega^\mu{}_\nu p^\nu, \quad (9.6b)$$

with the understanding that the expressions for $\alpha^\mu{}_\nu$, $\beta^\mu{}_\nu$, and $\Omega^\mu{}_\nu$ should change accordingly by analytic continuation of power series: for instance, $\alpha^\mu{}_\nu$ in Eq. (9.6a) equals $\delta^\mu{}_\nu - \frac{1}{2}y^\rho y^\sigma R^\mu{}_{\rho\sigma\nu}(x) - \frac{i}{6}y^\rho y^\sigma y^\kappa R^\mu{}_{\rho\sigma\nu;\kappa}(x) + \mathcal{O}(y^4)$.

Secondly, stipulate an additional pair of equations,

$$\frac{Dy^\mu}{d\tau} = \Omega^\mu{}_\nu y^\nu, \quad (9.6c)$$

which demands that the precession of y^μ is synchronized with that of the momentum p^μ in Eq. (9.6b).

Let us pause for a second to clarify the interpretations. The Wick rotation in Eq. (9.5) implements the idea of NJA at the level of probe EoM, so y^μ is now interpreted as the spin length pseudovector. Its definition reads

$$y^\mu = -*S^{\mu\nu}p_\nu/p^2, \quad (9.7)$$

where $S^{\mu\nu}$ is the spin angular momentum tensor. Eq. (9.7) is equivalent to imposing $S^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma}y_\rho p_\sigma$ and $p \cdot y = 0$.

Consequently, all of x^μ , p^μ , and y^μ had to be interpreted as dynamical variables of a spinning particle, requiring a total of 12 equations. The additional stipulation in Eq. (9.6c) was exactly for this purpose. Physically, such a synchronization of orbital (p^μ) and spin (y^μ) precessions will encode a spinning/complexified analog of equivalence principle proposed in Ref. [672].

Finally, we enter the last step of our procedure. The 12 equations in Eqs. (9.6a), (9.6b), and (9.6c) together define well-posed spinning-particle EoM, only if the variables x^μ, p^μ, y^μ are complexified. This is because the Wick rotation in Eq. (9.5) induces imaginary units inside the tensors $\alpha^\mu{}_\nu$, $\beta^\mu{}_\nu$, $\Omega^\mu{}_\nu$. Therefore, Eq. (9.6) describes the EoM of a spinning particle in *complexified spacetime*.

Thus, we propose that each imaginary unit in Eq. (9.6) is eliminated by trading off with a Hodge star $*$ on a Riemann tensor, which is justified if the complexified spacetime is taken as *SD*: $i \mapsto *$ via $iR_{\mu\nu\rho\sigma} = *R_{\mu\nu\rho\sigma}$. In this

manner, Eq. (9.6) is uplifted to a set of completely real equations that dictates the time evolution of x^μ, p^μ, y^μ as real variables. Details will be elaborated later.

Parity property of y^μ implies that the linear-in-curvature part of Eq. (9.6) reproduces the unity [202, 203, 256] of spin-induced multipole moments of the Kerr black hole, if uplifted in this precise fashion. This provides a justification. Intuitively, we have turned the localized orbital momentum in geodesic deviation (yielding unity of mass multipoles) into a spin angular momentum.

In sum, we have proposed a probe counterpart of NJA that uniquely derives a well-posed instance of spinning-particle EoM in *heaven*, which will then be readily uplifted to *earth*. Here, we have used technical terminologies coined by Newman [258, 328] and Plebański [314, 315]:

$$\begin{aligned}\text{HEAVEN} &= \text{SD SPACETIME}, \\ \text{EARTH} &= \text{REAL SPACETIME}.\end{aligned}\tag{9.8}$$

To clarify, heaven is a complex-analytic four-manifold with a holomorphic metric, while earth is a real-analytic four-manifold with a pseudo-Riemannian metric.

Remarkably, the following hold for Eq. (9.6) in heaven:

- (A) The linear-in-curvature part of Eq. (9.6) describes the established linear coupling for Kerr.
- (B) The dynamics due to Eq. (9.6) features exact (all-orders) conserved quantities in the presence of Killing vectors and Killing-Yano tensors.
- (C) Eq. (9.6) admits a Lagrangian formulation.

In these senses, Eq. (9.6) is an exact and consistent nonlinear completion of the linearized Kerr spinning-particle EoM, characterized by hidden symmetry. Hence we propose that Eq. (9.6) is *the* exact Kerr spinning-particle EoM in SD spacetimes. Below, we refer to Eq. (9.6) as the *heavenly Kerr spinning-particle EoM*.

9.4 Hidden Symmetry

Let us begin with the proof of proposition (B). First of all, recall the equivalence between Eqs. (9.1) and (9.2) via Eq. (9.3). Due to the Wick rotation in

Eq. (9.5), Eq. (9.3) is now redefined as

$$z^{\mu'} = \delta^{\mu'}_{\mu} \left(x^{\mu} + iy^{\mu} + \frac{1}{2} \Gamma^{\mu}_{\rho\sigma}(x) y^{\rho} y^{\sigma} + \mathcal{O}(y^3) \right), \quad (9.9a)$$

$$W^{\mu'}_{\nu} = \delta^{\mu'}_{\nu} - i \delta^{\mu'}_{\mu} \Gamma^{\mu}_{\nu\rho}(x) y^{\rho} + \mathcal{O}(y^2). \quad (9.9b)$$

Eq. (9.9a) describes a complexified geodesic for which the complexified parallel propagator is Eq. (9.9b). These are well-established ideas in differential geometry: adapted complex structure [743, 776–782]. See Ref. [226] for a reinvention.

Via Eq. (9.9), Eq. (9.6) is equivalent to

$$p^{\mu'} = \frac{dz^{\mu'}}{d\tau}, \quad \frac{Dp^{\mu'}}{d\tau} = 0, \quad \frac{Dy^{\mu'}}{d\tau} = 0, \quad (9.10)$$

where $p^{\mu'} = W^{\mu'}_{\mu} p^{\mu}$ and $y^{\mu'} = W^{\mu'}_{\mu} y^{\mu}$ via Eq. (9.9b). Again, multiplying by $W^{\mu'}_{\mu}$ yields Eq. (9.10) from Eq. (9.6).

By the very construction of the probe NJA, the first two equations in Eq. (9.10) are simply the free-fall equations in Eq. (9.1) with the revised definitions of $z^{\mu'}$ and $p^{\mu'}$ due to Eq. (9.9). Consequently, a part of the heavenly Kerr spinning-particle EoM is literally the Schwarzschild scalar-particle EoM, albeit imaginary-shifted.

The last equation in Eq. (9.10), on the other hand, is a remarkable restatement of the orbit-spin synchronization. It is amusing to see how the black hole's spin precession parameters such as the gravimagnetic ratio $\kappa = 1$ [235–237] are precisely reformulated as parallel transport (covariant constancy) of $p^{\mu'}, y^{\mu'}$ along the complex worldline $z^{\mu'}$.

Next, recall the following well-known facts [273, 585–595] about free-fall motion with position x^{μ} and momentum p^{μ} . First, a vector field $X^{\mu}(x)$ is Killing if $X^{\mu;\rho}(x) = X^{[\mu;\rho]}(x)$, which implies constancy of $p_{\mu} X^{\mu}(x)$. Second, a two-form $Y_{\mu\nu}(x)$ is Killing-Yano if $Y_{\mu\nu;\rho}(x) = Y_{[\mu\nu;\rho]}(x)$, which implies covariant constancy of $Y^{\mu}_{\nu}(x) p^{\nu}$. In turn, the scalar combination $(Y(x)p)^2 = -p_{\mu} Y^{\mu}_{\rho}(x) Y^{\rho}_{\nu}(x) p^{\nu}$ is strictly constant along the worldline.

Via complexification, these facts are immediately inherited to the Kerr spinning particle:

$$\frac{d}{d\tau} \left(p_{\mu'} X^{\mu'}(z) \right) = p_{\mu'} X^{\mu'}_{;\rho'}(z) p^{\rho'} = 0, \quad (9.11)$$

$$\frac{D}{d\tau} \left(Y^{\mu'}_{\nu'}(z) p^{\nu'} \right) = Y^{\mu'}_{\nu';\rho'}(z) p^{\nu'} p^{\rho'} = 0. \quad (9.12)$$

Meanwhile, the spinning particle features yet another variable that is covariantly constant: $y^{\mu'}$ in Eq. (9.10). Consequently, $p^{\mu'}$, $Y^{\mu'}_{\nu'}(z)p^{\nu'}$, and $y^{\mu'}$ are parallel-transported altogether along the complex worldline $z^{\mu'}$, so any scalar product between them is strictly constant.

In this way, it holds true that the following are exact, meaning all-orders in spin and curvature, conserved quantities of the Kerr spinning particle:

$$\mathbb{Q} = p_{\mu'} X^{\mu'}(z), \quad (9.13)$$

$$\mathbb{R} = p_{\mu'} Y^{\mu'}_{\nu'}(z) y^{\nu'}, \quad (9.14)$$

$$\mathbb{C} = -p_{\mu'} Y^{\mu'}_{\rho'}(z) Y^{\rho'}_{\nu'}(z) p^{\nu'}. \quad (9.15)$$

Eqs. (9.14) and (9.15) are all-orders completions of the Rüdiger [783, 784] and Carter [586] constants, respectively.

Explicitly, Eq. (9.13) can be re-covariantized at x as

$$p_{\mu} W^{\mu}_{\mu'} X^{\mu'}(z) = \sum_{\ell=0}^{\infty} \frac{i^{\ell}}{\ell!} p_{\mu} X^{\mu}_{;\rho_1 \dots \rho_{\ell}}(x) y^{\rho_1} \dots y^{\rho_{\ell}}, \quad (9.16)$$

yielding an infinite expansion in the spin length. When written in this form, the conservation becomes quite nontrivial to check: the use of identities regarding α^{μ}_{ν} , β^{μ}_{ν} , Ω^{μ}_{ν} is mandated at each order. This highlights the power of the spin-resummed formulation facilitated by Eq. (9.10). The same analysis applies to Eqs. (9.14) and (9.15) as well.

Furthermore, heaven is stipulated to be SD, so its ASD spinor bundle is flat. As a result, the ASD Plebański two-forms [315, 339] are covariantly constant: $(\Sigma^a)_{\mu\nu}$ for $a = 1, 2, 3$. These serve as Killing-Yano tensors [272, 273] and define a quaternionic structure as $(\Sigma_a)^{\mu}_{\rho} (\Sigma_b)^{\rho}_{\nu} = -\delta_{ab} \delta^{\mu}_{\nu} + \varepsilon^c_{ab} (\Sigma_c)^{\mu}_{\nu}$ [271–273]. Consequently, we now identify $p^{\mu'}$, $Y^{\mu'}_{\nu'}(z)p^{\nu'}$, $(\Sigma_a)^{\mu'}_{\nu'}(z)p^{\nu'}$, $y^{\mu'}$ being covariantly constant altogether, so any scalar product between them is a constant of motion, such as

$$\mathbb{K}_a = p_{\mu'} (\Sigma_a)^{\mu'}_{\rho'}(z) Y^{\rho'}_{\nu'}(z) p^{\nu'}. \quad (9.17)$$

A physically relevant example is the SD Taub-Newman-Unti-Tamburino (NUT) background, which describes a gravitational instanton [579, 580]. It is now well-understood [163, 164, 174, 256, 257, 260, 261, 630, 675, 785] to an explicit extent [163, 164] that this background is the SD sector of the Kerr background, such that the same Killing-Yano tensor is shared [163, 164]. By

using the Kerr-Schild metric provided by Ref. [163], Eq. (9.17) evaluates to $\vec{\mathbb{K}} = \vec{p} \times (\vec{x} \times \vec{p}) - ip_0(\vec{x} \times \vec{p})$,¹ which amusingly reincarnates the LRL vector (cf. Refs. [266, 272, 582, 692, 786–789]).

By incorporating time translation and rotation invariances as well, we establish that the motion of the Kerr spinning-particle in the SD sector of the Kerr background is *superintegrable*, with hidden symmetry $\mathfrak{so}(4)$:² that of Kepler problem. This is a new result on the all-orders dynamics of the Kerr-Kerr system. Previous works have truncated the probe’s spin order [233, 790–792] or NJ (Newman-Janis) shifted only the background [266, 267].

To summarize, we have shown that exact symmetries arise for our proposed Kerr spinning particle by the very structure of the probe NJA. The conserved quantities of the spinning probe can arise as direct NJ shifts of the conserved quantities of the scalar probe. For the explicit example of physical relevance, we have found a sufficient number of functionally independent conserved quantities, establishing superintegrability.

9.5 Lagrangian Formulation

Next, we prove proposition (C). We find an explicit first-order worldline action,

$$\int p_{\mu'} \left(dz^{\mu'} - i Dy^{\mu'} + * \Theta^{\mu'}{}_{\nu'} y^{\nu'} \right) - \mathcal{C}, \quad (9.18)$$

where we have omitted constraint terms ($\mathcal{C}$) suitably encoding the conditions such as $p^2 = -m^2$ and $p \cdot y = 0$. Computation shows that the saddle of Eq. (9.18) is precisely Eq. (9.10); see Chap. X for details.

In Eq. (9.18), $\Theta^{\mu'}{}_{\nu'}$ describes the “infinitesimal angle” as a one-form, coupled to $S^{\nu'}{}_{\mu'} = *(y \wedge p)^{\nu'}{}_{\mu'}$ to realize the kinetic term for spin. The covariant exterior derivative D acts as $D\Theta^{\mu'}{}_{\nu'} = \frac{1}{2} R^{\mu'}{}_{\nu'\rho'\sigma'}(z) dz^{\rho'} \wedge dz^{\sigma'}$, implying

$$D \left(-i Dy^{\mu'} + * \Theta^{\mu'}{}_{\nu'} y^{\nu'} \right) = 0 \quad (9.19)$$

for the SDity. Via Eq. (9.19), Eq. (9.18) is equivalent to

$$\int p_{\mu} \left(\alpha^{\mu}{}_{\nu} dx^{\nu} + i \beta^{\mu}{}_{\nu} Dy^{\nu} - i Dy^{\mu} + * \Theta^{\mu}{}_{\nu} y^{\nu} \right) - \mathcal{C}, \quad (9.20)$$

which re-covariantizes a one-form at the point x [695]. Both Eqs. (9.18) and (9.20) retrieve the action of a free massive spinning particle [793, 794] in the flat limit.

¹Also, the three-vector index of $\mathbb{K}_a$ is global since we gauge-fixed the ASD spinor bundle by zeroing the spin connection coefficients.

²with suitable complexification

Eq. (9.18) is a complexified action dictating the evolution of complexified variables $z^{\mu'}$, $p^{\mu'}$, $y^{\mu'}$. When converted to Eq. (9.20), it can be uplifted to a purely real action dictating the evolution of real variables x^{μ} , p^{μ} , y^{μ} .

9.6 Linearized Coupling

Finally, we prove proposition (A) by showing that dropping all terms nonlinear in the curvature in Eq. (9.20) reproduces the well-known Levi-Steinhoff action [97] up to iterating EoM. The only important term is $p_{\mu} \alpha^{\mu}_{\nu} dx^{\nu}$, as $\beta^{\mu}_{\nu} - \delta^{\mu}_{\nu} = \mathcal{O}(R^1 y^2)$ [695] and $Dy^{\mu}/d\tau = \Omega^{\mu}_{\nu} y^{\nu} = \mathcal{O}(R^1 y^2)$ on EoM. As is glimpsed in Eq. (9.4), the $\mathcal{O}(R^1)$ part of α^{μ}_{ν} in heaven is given by [695]

$$\delta^{\mu}_{\nu} + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} {}^{\ell}R^{\mu}_{\kappa_1 \kappa_2 \nu; \kappa_3; \dots; \kappa_{\ell}}(x) y^{\kappa_1} \dots y^{\kappa_{\ell}}, \quad (9.21)$$

after the Wick rotation in Eq. (9.5). Via $dx^{\mu}/d\tau = p^{\mu} + \mathcal{O}(R^1 y^2)$ and $S^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} y_{\rho} p_{\sigma}$, the relevant parts of the action boil down to $\int p_{\mu} \alpha^{\mu}_{\nu} p^{\nu} - \frac{1}{2} S^{\mu\nu} {}^* \Theta_{\mu\nu} - \mathcal{C}$, reproducing the Levi-Steinhoff action [97] for the Kerr multipole coefficients $C_{\ell} = 1$ [98, 206–209] upon plugging in Eq. (9.21).

In fact, the unity $C_{\ell} = 1$ of multipole coefficients can be shown directly from the EoM by matching Ω^{μ}_{ν} in Eq. (9.6) with the universal template for spin precession EoM [673] extending the Mathisson-Papapetrou-Dixon [227–229] equations. Invariantly, it is verified by the spin exponentiation for one graviton (see Eq. (9.22)).

9.7 Spin Exponentiation at All Multiplicities

Finally, let us explore all orders in perturbation theory:

- (D) Positive-helicity graviton Compton amplitudes due to the action in Eq. (9.18) exhibit spin exponentiation at all multiplicities.

No physical explanation has existed for this property to the best of our knowledge, although its consistency at the amplitudes level has been known [210–212]. In our case, the structure of the probe NJA is highly constrained for the hidden symmetries in the SD sector, whose implication is spin exponentiation at all multiplicities.

To prove proposition (D), we expand the gravitational fields around Minkowski background, yet after switching to the tetrad formalism. In tetrad formalism, the constraints and Hamiltonian (put in $\mathcal{C}$ in Eq. (9.18)) take the same form as

in the free theory. Also, the spin connection describes antisymmetric indices as $\gamma^{AB}{}_{\rho} = -\gamma^{BA}{}_{\rho}$, on which the Hodge star acts to yield $+i$ in heaven.

Crucially, this shows that the coupling to the spin connection identically vanish in Eq. (9.18)—consistently with Eq. (9.19)—so that the only graviton coupling is via the coframe perturbation, $h^A{}_{\mu} = e^A{}_{\mu} - \delta^A{}_{\mu}$: the interaction Lagrangian describes a single term, $p_{A'} h^A{}_{\mu'}(z) \dot{z}^{\mu'}$. This is identical to the interaction Lagrangian $p_A h^A{}_{\mu}(x) \dot{x}^{\mu}$ of the minimally coupled scalar particle in tetrad formalism, via complexifications $p_A \mapsto p_{A'}$, $x^{\mu} \mapsto z^{\mu'}$. Consequently, it can be diagrammatically proven that all Feynman diagrams for the Kerr spinning-particle are mere complexifications of those of the Schwarzschild scalar particle. This implies that the amplitudes for $(n-2)$ gravitons satisfy

$$\mathcal{M}_{\text{Kerr}}(3^+ \cdots n^+) = \mathcal{M}_{\text{Sch}}(3^+ \cdots n^+) e^{(\sum k) \cdot a}, \quad (9.22)$$

where $(\sum k) = k_3 + \cdots + k_n$ is the total momentum. Here, the asymptotic free trajectory is $z^{\mu'} = -ia^{\mu'} + mu^{\mu'}\tau$ for constant spin $a^{\mu'}$ and velocity $u^{\mu'}$ parameters³.

It suffices to work in heaven for showing Eq. (9.22) because heaven consists of positive-helicity gravitons in the incoming convention [342, 343]. Also, it should be clear that any real uplift of the action in Eq. (9.20) implies the all-multiplicity spin exponentiation for both positive- and negative-helicity gravitons. One should be able to reproduce the all-multiplicity spin exponentiation with any such real uplift, as scattering amplitudes are (worldline) field redefinition invariant.

9.8 Uplift to Earth

Eventually, we elaborate on the uplifting procedure, which provides the extension to generic backgrounds with *both SD and ASD modes*. For instance, recall $\alpha^{\mu}{}_{\nu}$ in Eq. (9.4) and suppose the Wick rotation in Eq. (9.5) is applied. On the support of SDity, each imaginary unit i is absorbed to a Riemann tensor. Clearly, ambiguities arise from $\mathcal{O}(y^4)$ due to the nonlinear-in-Riemann term: which Riemann tensor are we absorbing i into, $R^{\mu}{}_{\rho\sigma\xi}(x)$ or $R^{\xi}{}_{\kappa\lambda\nu}(x)$?

We dub this ambiguity, in distributing the Hodge stars over nonlinear strings of Riemann tensors, the *star-shifting ambiguity*. Its invariant content at $\mathcal{O}(R^{n-2})$

³We are sorry that our convention for the spin length pseudovector flips sign from the standard convention: $y^{\mu} = -a^{\mu}$, so $S_{12} = \varepsilon_{1203} p^0 a^3 = (+1)(+m)a^3$ in the rest frame.

in the Lagrangian can be substantiated as contact deformations of the n -graviton Compton amplitude.

Take $n = 4$, for instance. By plugging in the straight-line trajectory, we learn that it suffices to compare between the following two terms in the plane-wave background $R_{\mu\nu\rho\sigma}(x) = \epsilon_3 \varphi_{3\mu\nu}^+ \varphi_{3\rho\sigma}^+ e^{ik_3 x} + \epsilon_4 \varphi_{4\mu\nu}^- \varphi_{4\rho\sigma}^- e^{ik_4 x}$, where φ_3^+ and φ_4^- are gauge-invariant polarizations for the positive- and negative-helicity gravitons:

$$S^{\mu\nu}((a \cdot \partial)^{j_1} * R_{\mu\nu a \kappa})((a \cdot \partial)^{j_2} R^\kappa_{a\rho\sigma}) S^{\rho\sigma}, \quad (9.23a)$$

$$S^{\mu\nu}((a \cdot \partial)^{j_1} R_{\mu\nu a \kappa})((a \cdot \partial)^{j_2} * R^\kappa_{a\rho\sigma}) S^{\rho\sigma}. \quad (9.23b)$$

Upon extracting the part multilinear in ϵ_3 and ϵ_4 , Eqs. (9.23a) and (9.23b) are equal in magnitude but *opposite in sign*: $\pm(\varphi_3^+ S)(\varphi_4^- S)((k_3 a)^{j_1} (k_4 a)^{j_2} - (k_4 a)^{j_1} (k_3 a)^{j_2}) (a \varphi_3^+ \varphi_4^- a)$. Thus, each time one shifts the Hodge star from one Riemann tensor to another, one deforms the Compton amplitude by a contact term if $j_1 \neq j_2$.

In fact, one can simply add an explicit worldline operator in the Lagrangian that involves at least one SD and one ASD curvatures. Yet, such an explicit deformation may seem more artificial than star-shifting.

Most generally, we conclude that the uplift of the heavenly Kerr EoM and its Lagrangian to earth is ambiguous due to the possibility of adding such mixed-chirality terms: *earthly deformations*, so to speak.⁴ Star-shifting is a natural instance of earthly deformations.

9.9 Compton Amplitude

Note that star-shifting ambiguities begin at $\mathcal{O}(y^4)$. For instance, we had obtained the spurious-pole-free mixed-helicity gravitational Compton amplitude from an earthly Kerr Lagrangian:

$$\frac{\mathcal{M}_{\text{Kerr}}(3^+ 4^-)}{\mathcal{M}_{\text{Sch}}(3^+ 4^-) e^{-\alpha/2} e^{\beta/2}} = 1 + \gamma + \gamma^2 \left(\frac{\gamma + \beta}{\alpha + \beta} E_2(\alpha) - \frac{\gamma - \alpha}{\alpha + \beta} E_2(-\beta) \right), \quad (9.24)$$

where α, β, γ are linear-in-spin parameters such that the trivial spin exponentiation is $\mathcal{M}_{\text{Sch}}(3^+ 4^-) e^{-\alpha/2} e^{\beta/2}$ and the AHH amplitude with the spurious

⁴Despite this ambiguity and despite the possible mismatches with black hole perturbation theory computations, our explicit result on an earthly uplift is at least practically useful: it provides the ansatz that automatically guarantees the same-helicity spin exponentiation to all multiplicities, on top of which one can add mixed-chirality operators systematically.

pole [206] is $\mathcal{M}_{\text{Sch}}(3^+4^-)e^{-\alpha/2}e^{\beta/2}e^\gamma$, and $E_2(\xi) := (e^\xi - \xi - 1)/\xi^2$. Notably, Eq. (9.24) deviates from AHH already from $\mathcal{O}(\gamma^4)$, although the strict necessity of contact deformations arises from $\mathcal{O}(\gamma^5)$ [206].⁵

9.10 Interpretation

The NJA had been in a mysterious status for decades. Despite initial criticisms [243, 634], constant attempt was made for explanations [202, 242, 243, 246–249, 330, 611]. Especially, Newman himself envisioned a fundamental unification of spin and spacetime into a complex geometry, in which the spinning particle truly becomes a scalar particle [252–254, 254–259, 329, 330]. This idea was not only pursued at the level of sourced fields but also at the level of probe particles: “complex center of mass.” An attempt for EoM appears in Ref. [259].

This chapter vividly realizes Newman’s dream in terms of explicit and rigorous frameworks of geodesic deviation [695] and adapted complex structure [743, 776–782]. The holomorphic worldline $z^{\mu'}$ is Newman’s complex center of mass, realized in fully nonlinear gravity. However, our analysis on earthly deformations clarifies that Newman’s proposal of the Kerr spinning particle being equivalent to a scalar particle is valid only in the SD sector. To reiterate, the probe NJA pinpoints a unique dynamics *within* heaven. This point is made further evident in Chap. X, where an explicit ASD group of terms ruins the equivalence between Eqs. (9.18) and (9.20) in earth.

A similar realization has been critical in grasping the full content of the NJA [164]. The SD limit of the Kerr-Taub-NUT solution is diffeomorphic to the SD Taub-NUT solution [260]. In Kerr-Schild coordinates [163, 164], this diffeomorphism is just a translation by $+i\vec{a}$. Crucially, however, the non-SD Kerr solution describes *two* centers at $\pm i\vec{a}$ in the complexified description: a pair of SD *and* ASD Taub-NUT solutions [164]. Thus, the equivalence between Kerr and a pointlike object is lost beyond the SD sector.

Based on this repertoire that applies for both our probe NJA and the NJA, it seems that a natural interpretation for the holomorphic worldline $z^{\mu'}$ in Eq. (9.3a) is the worldline of the ASD Taub-NUT instanton constituting the Kerr black hole. Surely, the anti-holomorphic worldline could have also been obtained by geodesically deviating in the direction of $-iy^\mu$, in which case the SD Taub-NUT instanton is reached.

⁵A similar phenomenon had occurred in the electromagnetic analog [795].

This interpretation is nicely consistent with spin exponentiation, disclosing an underlying physical picture: spin exponentiation is the helicity selection rule [261, 263, 264] of Taub-NUT instantons. It also provides a physical interpretation for the worldsheet structure speculated in Ref. [226]: the Misner string (gravitational Dirac string) as a flux tube of NUT charge joining the SD and ASD instantons.

Due to the nonlinearity of gravity inherent in Eq. (9.3a), there is a difficulty in simultaneously manifesting the nonlinear NJ shifts (all-multiplicity spin exponentiations, spin resummations) for both SD (positive-helicity) and ASD (negative-helicity) sectors. Still, we shall make maximum use of the superintegrability of the SD sector shown in Eq. (9.17) by perturbing around the holomorphic variable z^μ : a *googly* [347–350] approach to Kerr spinning-particle effective theory.

9.11 Conclusion

We proposed the probe counterpart of NJA to uniquely pinpoint a remarkable instance of a massive spinning particle in SD spacetimes, which exhibits exact symmetries in the presence of Killing vectors and Killing-Yano tensors. It is shown that this distinguished point in the effective theory landscape of all massive spinning particles enjoys the privileges of superintegrability in SD Taub-NUT background and spin exponentiation at every graviton multiplicity. It is also determined that the uplift to real spacetimes is ambiguous precisely by mixed-chirality deformations, corresponding to contact deformations for Compton amplitudes.

This work provides a partial answer to the persistent question, “What principle in the effective theory characterizes Kerr among all massive spinning objects?” Previously, the speculations had described a *simplicity* of linearized interactions [98, 206–209] or its continuation [210–212], a *pattern* suggested at low orders [213, 214], or the *gauge redundancy* in the higher-spin formalism [217, 218]. However, we identified an exact hidden *symmetry* embedded in nonlinear interactions, thus demonstrating that a physical principle can directly address the question.

THE KERR EFFECTIVE ACTION

Contents of this chapter are adapted from Ref. [769], J.-H. Kim, “The Kerr two-twistor particle,” [arXiv:2602.19495 \[gr-qc\]](#).

10.1 Introduction

The twistor particle program [377, 796–808] was a movement in the ’70s to ’80s that applied twistor theory to particle physics. In particular, massive particles were implemented as systems of two or more twistors, whose internal symmetries were associated with color or flavor.

In a modern reboot of this program [673, 794], the two-twistor implementation is adopted such that the $SU(2)$ internal symmetry describes the massive little group instead, in accordance with Ref. [206]. This revival has also expanded the scope from particle physics to astrophysics, based on the point-particle effective theory of macroscopic bodies [92–97] such as neutron stars or black holes.

The two-twistor system should be able to realize interacting particles; see Refs. [673, 802, 809–812] for previous attempts. The problem of direct relevance to the current PM gravity community is to implement the Kerr black hole coupled to curved spacetime, capturing all its 2^ℓ -pole moments for $\ell = 0, 1, 2, \dots$ [97, 176, 202, 203, 207–209, 256] as well as further nonlinear-in-curvature couplings.

In this chapter, we describe how an attempt toward curved massive twistor theory had resulted in a candidate effective point-particle Lagrangian for the Kerr black hole to all orders in spin and curvature. This construction was originally obtained in 2022 by the present author.

This derivation implements an intrinsic feature of twistor particle theory that spin is literally an imaginary deviation in terms of complexified incidence relation [252, 813, 814]. As a result, the NJA [202] is realized for dynamical particles. A differential-geometric formulation arises in terms of the generator of geodesic deviation, N , and an almost-complex structure, J .

10.2 Flat

10.2.1 Free Particle

The twistor space $\mathbb{T}$ is the Kähler vector space $\mathbb{C}^4$ with $(2, 2)$ -signature metric; we consider the product $\mathbb{T} \times \mathbb{T}$ [802, 814]. Linear coordinates on $\mathbb{T} \times \mathbb{T}$ are Z_A^I . The index $A = 0, 1, 2, 3$ is a Dirac spinor while $I = 0, 1$ is an $\mathrm{SU}(2)$ spinor. The Weyl blocks are

$$Z_A^I = \begin{pmatrix} \lambda_\alpha^I \\ i\mu^{\dot{\alpha}I} \end{pmatrix}, \quad \bar{Z}_I^A = \begin{pmatrix} -i\bar{\mu}_I^\alpha & \bar{\lambda}_{I\dot{\alpha}} \end{pmatrix}, \quad (10.1)$$

where $\bar{Z}_I^A = [Z_A^I]^*$. The symplectic form is

$$\omega^\circ = i d\bar{Z}_I^A \wedge dZ_A^I = d\mu^{\dot{\alpha}I} \wedge d\bar{\lambda}_{I\dot{\alpha}} + d\bar{\mu}_I^\alpha \wedge d\lambda_\alpha^I. \quad (10.2)$$

The conformal group $\mathrm{SU}(2, 2)$ acts from the left while the massive little group $\mathrm{SU}(2)$ acts from the right. The $\mathrm{U}(1)$ which $\bar{Z}_I^A Z_A^I$ generates is merely gauge [673, 794].

Consider the quadric $-\frac{1}{2} I^{AB} Z_A^I Z_B^J \epsilon_{IJ} = \det(\lambda) = c$ in $\mathbb{T} \times \mathbb{T}$, which we denote as Δ_c . Here, I^{AB} is the infinity twistor [814]. For mathematical precision, one may define the massive twistor space as $\mathbb{MT} = (\mathbb{T} \times \mathbb{T}) \setminus \Delta_0$. Eq. (10.2) shows that $\mathbb{MT}$ is the cotangent bundle of the space $\lambda_\alpha^I \in \mathrm{GL}(2, \mathbb{C})$ of Ref. [206]’s massive spinor-helicity variables.

Symplectic reduction derives a 12-dimensional constrained phase space $\times_* = \Delta_m/\mathbb{C}$, realizing the 3 translational and 3 rotational degrees of freedom of a massive spinning body with rest mass $m > 0$ [673, 794]¹.

The Poincaré group is included as a subgroup of the $\mathrm{SU}(2, 2)$. The Poincaré charges are found as

$$\begin{aligned} p_{\alpha\dot{\alpha}} &= -\lambda_\alpha^I \bar{\lambda}_{I\dot{\alpha}}, \\ j^{\dot{\alpha}}_{\dot{\beta}} &= \mu^{\dot{\alpha}I} \bar{\lambda}_{I\dot{\beta}} - \frac{1}{2} \delta^{\dot{\alpha}}_{\dot{\beta}} \mu^{\dot{\gamma}I} \bar{\lambda}_{I\dot{\gamma}}. \end{aligned} \quad (10.3)$$

The Poincaré-invariant symplectic potential is

$$\theta^\circ = \frac{i}{2} \left(\bar{Z}_I^A dZ_A^I - d\bar{Z}_I^A Z_A^I \right), \quad (10.4)$$

such that $\omega^\circ = d\theta^\circ$.

¹It is easy to incorporate the Regge trajectory [793]; see Refs. [673, 794].

10.2.2 Twistor Magic

A remarkable feature of the twistor particle framework is that an imaginary shift of the two-twistor's co-incident position derives a spinning particle from a scalar particle.

As is well-known, the relation between twistor space and spacetime $\mathbb{M} = (\mathbb{R}^4, \eta)$ is established by the incidence relation [611, 814], the massive adaptation of which reads

$$\mu^{\dot{\alpha}I} = x^{\dot{\alpha}\alpha} \lambda_{\alpha}^I. \quad (10.5)$$

Geometrically, Eq. (10.5) means that the two twistors $Z_A^{I=0,1}$ are co-incident at the same spacetime point $x \in \mathbb{M}$. By plugging in Eq. (10.5) to Eq. (10.4), one obtains

$$\theta^{\circ} = p_{\alpha\dot{\alpha}} dx^{\dot{\alpha}\alpha}, \quad (10.6)$$

which is the symplectic potential of a scalar particle.

Now consider the complexification of Eq. (10.5):

$$\mu^{\dot{\alpha}I} = \left(x^{\dot{\alpha}\alpha} + iy^{\dot{\alpha}\alpha} \right) \lambda_{\alpha}^I. \quad (10.7)$$

This shifts the co-incident point of the twistors $Z_A^{I=0,1}$ to a point in complexified Minkowski space $\widehat{\mathbb{M}} = (\mathbb{C}^4, \widehat{\eta})$, where $\widehat{\eta}$ is the holomorphic extension of the Lorentzian metric η . By plugging in Eq. (10.7) to Eq. (10.4), one obtains

$$\theta^{\circ} = p_{\alpha\dot{\alpha}} dx^{\dot{\alpha}\alpha} + iy^{\dot{\alpha}\alpha} \left(\lambda_{\alpha}^I d\bar{\lambda}_{I\dot{\alpha}} - d\lambda_{\alpha}^I \bar{\lambda}_{I\dot{\alpha}} \right), \quad (10.8)$$

which is precisely the symplectic potential of a massive spinning particle [673, 794, 802]. Inserting Eq. (10.7) in Eq. (10.3) shows that $y^{\dot{\alpha}\alpha}$ describes the spin (Pauli-Lubański) pseudovector normalized in the unit of length. Hence, to put in a well-recognized notation [97, 793] for the reader's sake, the transition from Eq. (10.6) to Eq. (10.8) describes ²

$$pdx \mapsto pdx + \frac{1}{2} S \Lambda d\Lambda. \quad (10.9)$$

The insight that the complexified Minkowski space in twistor theory unifies spacetime x^{μ} and spin y^{μ} as real and imaginary parts traces back to Refs. [252, 814], whose explicit demonstration for dynamical massless particles is given in Ref. [813]. Here, we have provided the explicit demonstration for dynamical massive particles.

² Note that the U(1) gauge generator evaluates to $\bar{Z}_I^A Z_A^I = -p_{\alpha\dot{\alpha}} y^{\dot{\alpha}\alpha}$ on Eq. (10.7). In Eq. (10.9), we have imposed $-p_{\alpha\dot{\alpha}} y^{\dot{\alpha}\alpha} = 0$.

10.2.3 Spinspace-time

The complexified Minkowski space in this context is dubbed spinspace-time [672]:

$$\widehat{\mathbb{M}} \cong T\mathbb{M}. \quad (10.10)$$

As a real manifold, spinspace-time is the tangent bundle $T\mathbb{M}$ of the space-time $\mathbb{M}$, with base coordinates $x^{\dot{\alpha}\alpha}$ and fiber coordinates $y^{\dot{\alpha}\alpha}$. As a complex manifold, spinspace-time is equipped with complex coordinates

$$z^{\dot{\alpha}\alpha} = x^{\dot{\alpha}\alpha} + iy^{\dot{\alpha}\alpha}, \quad (10.11)$$

which are holomorphic due to the almost-complex structure $dx^{\dot{\alpha}\alpha} \mapsto dy^{\dot{\alpha}\alpha}$, $dy^{\dot{\alpha}\alpha} \mapsto -dx^{\dot{\alpha}\alpha}$. (In this chapter, we wish to formulate complex structures as isomorphisms of the cotangent bundle rather than of the tangent bundle.)

Smooth tensor fields on spacetime are analytically continued to smooth tensor fields on spinspace-time. Their Cauchy-Riemann flow arises by the exponentiated Lie derivative $\exp(i\mathcal{L}_N)$, where $N \in \Gamma(T\mathbb{M})$:

$$N = y^{\dot{\alpha}\alpha} \frac{\partial}{\partial x^{\dot{\alpha}\alpha}}. \quad (10.12)$$

In particular, the complexified metric is $\widehat{\eta} = \exp(i\mathcal{L}_N)\eta$.

As explicated in Refs. [252, 672], the physical origin of Eq. (10.11) is a Hodge duality in the decomposition of angular momentum for massive particles,

$$j = (x \wedge p) + *(y \wedge p). \quad (10.13)$$

The SD part of Eq. (10.13) is the SD part of $(z \wedge p)$, as $*$ is sent to $+i$. The complexification in Eq. (10.7) precisely arises in this way via $j^{\dot{\alpha}}_{\dot{\beta}}$ in Eq. (10.3).

10.2.4 Correspondence Space

In massless twistor theory, the incidence relation is mathematically formalized as the double fibration. Let us achieve its massive equivalent.

Let $S_2\widehat{\mathbb{M}}$ be a trivial $\mathrm{GL}(2, \mathbb{C})$ bundle over $\widehat{\mathbb{M}}$. Fig. 10.1 concerns two maps $\varphi : (z^{\dot{\alpha}\alpha}; \lambda_{\alpha}^I) \mapsto (\lambda_{\alpha}^I, iz^{\dot{\alpha}\alpha}\lambda_{\alpha}^I)$ and $\pi_2 : (z^{\dot{\alpha}\alpha}; \lambda_{\alpha}^I) \mapsto z^{\dot{\alpha}\alpha}$ from it. The first is an invertible diffeomorphism. The second is the bundle projection. This provides a mathematical formalization of the massive incidence relation in Eq. (10.7), establishing the relation between massive twistor space $\widehat{\mathbb{MT}}$ and spinspace-time $\widehat{\mathbb{M}}$.

$$\begin{array}{ccccc}
& S_2\widehat{\mathbb{M}} & \xleftarrow{\gamma} & (T \oplus S_2)\mathbb{M} & \\
\varphi \swarrow & & \searrow \pi_2 & & \searrow \pi_1 \\
\times & & \widehat{\mathbb{M}} & & \mathbb{M}
\end{array}$$

Figure 10.1: Flat massive twistor theory.

When $\widehat{\mathbb{M}}$ is regarded as a real manifold as per Eq. (10.10), $S_2\widehat{\mathbb{M}}$ is viewed as the direct sum bundle $(T \oplus S_2)\mathbb{M}$ over spacetime $\mathbb{M}$. This is represented as a map $\gamma : (x^{\dot{\alpha}\alpha}; y^{\dot{\alpha}\alpha}, \lambda_\alpha^I) \mapsto (x^{\dot{\alpha}\alpha} + iy^{\dot{\alpha}\alpha}; \lambda_\alpha^I)$ in Fig. 10.1.

The space

$$\mathbb{K} = (T \oplus S_2)\mathbb{M} \cong S_2\widehat{\mathbb{M}} \quad (10.14)$$

serves as the massive analog of the correspondence space, which has the same dimension as $\mathbb{MT}$. Crucially, $\mathbb{K}$ inherits the Kähler geometry of $\mathbb{MT}$ through φ and γ .

First, $\mathbb{K}$ is equipped with an almost-complex structure

$$J : (dx^{\dot{\alpha}\alpha}, dy^{\dot{\alpha}\alpha}, d\lambda_\alpha^I) \mapsto (dy^{\dot{\alpha}\alpha}, -dx^{\dot{\alpha}\alpha}, -i d\lambda_\alpha^I). \quad (10.15)$$

This corresponds to the complex structure $dZ_A^I \mapsto -i dZ_A^I$ of $\mathbb{MT}$.

Second, $\mathbb{K}$ is equipped with the symplectic potential in Eq. (10.8). This is the pullback of the symplectic potential of $\mathbb{MT}$ given in Eq. (10.4) by $\varphi \circ \gamma$. (By abuse of notation, we have denoted both symplectic potentials as θ° .)

As a result, $\mathbb{K}$ is a Kähler manifold. It is left as an exercise to derive the pullback of the metric by $\varphi \circ \gamma$ and check the Kähler triple relation within $\mathbb{K}$.

We also equip $\mathbb{K}$ with a vector field $y^{\dot{\alpha}\alpha} \partial / \partial x^{\dot{\alpha}\alpha} \in \Gamma(T\mathbb{K})$, which is again denoted as N by abuse of notation.

Note that working in the massive correspondence space $\mathbb{K}$ is what Refs. [811, 813] refer to the so-called “hybrid” description. Namely, $\mathbb{K}$ is the phase space of a free massive spinning body described in terms of spacetime position $x^{\dot{\alpha}\alpha}$, spin length pseudovector $y^{\dot{\alpha}\alpha}$, and the spin frame λ_α^I encoding momentum and Eulerian angles.

10.2.5 N and J for Newman-Janis

The twistor magic can be implemented in the massive correspondence space $\mathbb{K}$. Namely, the spinning-particle symplectic potential arises from the scalar-

particle symplectic potential as

$$\theta^\circ = \left(1 + J \iota_N d\right) (p_{\alpha\dot{\alpha}} dx^{\dot{\alpha}\alpha}). \quad (10.16)$$

Here, ι_N denotes the interior product with respect to N while J is the almost-complex structure in Eq. (10.15).

Eq. (10.16) precisely transcribes the complexification of the co-incidence relation in Eq. (10.7). Firstly, $\iota_N d$ implements a deviation $x \mapsto x + y$ in the manner that conforms to the polarization choice. Then J implements $\pm i$ factors such that the twistors Z_A^I are co-incident at $x + iy$ and the dual twistors $\bar{Z}_I^A$ are co-incident at $x - iy$.

Note that J implements Hodge duality on the massive spinor-helicity fibers: $+i$ for the right-handed $\bar{\lambda}_{I\dot{\alpha}}$ and $-i$ for the left-handed λ_α^I . Physically, this converts the electric mass dipole of a deviated scalar particle to the magnetic mass dipole, i.e., spin, of the spinning particle:

$$J(-dp_{\alpha\dot{\alpha}} y^{\dot{\alpha}\alpha}) = i y^{\dot{\alpha}\alpha} \left(\lambda_\alpha^I d\bar{\lambda}_{I\dot{\alpha}} - d\lambda_\alpha^I \bar{\lambda}_{I\dot{\alpha}} \right). \quad (10.17)$$

Eq. (10.17) is an incarnation of the Hodge duality between orbital and spin angular momenta in Eq. (10.13).

10.3 Earth

The desire of this chapter is to implement the twistor magic in curved backgrounds. However, we do not have a definition of “curved massive twistor space” yet.

Recall the scaffolding of curved twistor theory [309, 310, 313] (cf. App. A.3). One may start by discussing the geometry of the correspondence space within the conventional mathematical languages of general relativity. Then the “radical” twistor-space picture will arise afterwards via the self-duality condition as integrability along α -surfaces.

We shall follow the same approach. First of all, we have to realize the curved massive correspondence space $\mathcal{K}$ as the typical phase space of a massive spinning particle in textbook general relativity. After establishing this “conservative” description, we may then envision how a notion of curved massive twistor space $\mathbb{MT}$ could emerge.

10.3.1 Curved Correspondence Space

In textbook general relativity, spacetime $(\mathcal{M}, g)$ is a real-analytic pseudo-Riemannian four-manifold. Local trivialization of $T\mathcal{M}$ describes the frame $E_m = E^\mu_m(x) \partial_\mu$ and coframe $e^m = e^m_\mu(x) dx^\mu$ such that $\langle e^m, E_n \rangle = \delta^m_n$. Here, $m, n, \dots = 0, 1, 2, 3$ are local Lorentz indices while $\mu, \nu, \dots = 0, 1, 2, 3$ are spacetime indices.

The decomposition $T\mathcal{M} \cong (S^+ \otimes S^-)\mathcal{M}$ into SD and ASD spinor bundles describes that a local Lorentz index is a pair of spinor indices as $e^m \leftrightarrow e^{\dot{\alpha}\alpha}$. The Levi-Civita connection ∇ of g splits to $\mathbf{SL}(2, \mathbb{C})$ -valued connections $\bar{\gamma}$ and γ on $S^\pm\mathcal{M}$. The structure equations are $0 = De^{\dot{\alpha}\alpha} = de^{\dot{\alpha}\alpha} - \bar{\gamma}^{\dot{\alpha}}_{\dot{\beta}} \wedge e^{\dot{\beta}\alpha} + e^{\dot{\alpha}\beta} \wedge \gamma_\beta^\alpha$ and $R_\alpha^\beta = d\gamma_\alpha^\beta + \gamma_\alpha^\gamma \wedge \gamma_\gamma^\beta$, where D denotes the covariant exterior derivative with respect to ∇ .

Physically, the degrees of freedom of a massive spinning particle in $(\mathcal{M}, g)$ are described by x^μ , y^m , and λ_α^I . Here, the index form declares the behaviors under coordinate and local Lorentz transformations. The position variable x^μ is supposed to describe coordinates on $\mathcal{M}$. The spin length pseudovector y^m and the spin frame λ_α^I are supposed to be local Lorentz degrees of freedom.

Therefore, the curved massive correspondence space should be defined as the direct sum bundle

$$\mathcal{K} = (T \oplus S_2)\mathcal{M}, \quad (10.18)$$

equipped with base coordinates $x^\mu \in \mathbb{R}^4$ and fiber coordinates $y^m \in \mathbb{R}^4$, $\lambda_\alpha^I \in \mathbf{GL}(2, \mathbb{C})$. Here, $S_2\mathcal{M}$ is a principal $\mathbf{GL}(2, \mathbb{C})$ -bundle over $\mathcal{M}$ which is isomorphic to an open subbundle of $(S^- \oplus S^-)\mathcal{M}$.

The transition functions for $\mathcal{K}$ as a manifold are restricted to the following specific forms:

$$\begin{aligned} (x^\mu, y^m, \lambda_\alpha^I) &\mapsto (f^\mu(x), y^m, \lambda_\alpha^I), \\ (x^\mu, y^m, \lambda_\alpha^I) &\mapsto (x^\mu, \Omega^m_n(x) y^n, \Omega_\alpha^\beta(x) \lambda_\beta^I). \end{aligned} \quad (10.19)$$

These encode spacetime diffeomorphisms and local Lorentz transformations, respectively. Here, $\Omega^m_n(x) \in \mathbf{SO}(1, 3)$ arises from $\Omega_\alpha^\beta(x) \in \mathbf{SL}(2, \mathbb{C})$.

The curved massive correspondence space is equipped with a triple of geometrical structures:

$$(\mathcal{K}, N, J, \omega). \quad (10.20)$$

$N \in \Gamma(T\mathcal{K})$ is a vector field, $J : T^*\mathcal{K} \rightarrow T^*\mathcal{K}$ is an almost-complex structure, and $\omega \in Z^2(\mathcal{K}) \subset \Omega^2(\mathcal{K})$ is a symplectic structure. N and J are invariant under the restricted coordinate transformations of $\mathcal{K}$ stipulated in Eq. (10.19).

First, N is uniquely defined by the interior products

$$\iota_N : (e^m, Dy^m, D\lambda_\alpha^I) \mapsto (y^m, 0, 0). \quad (10.21)$$

It is easy that $N = y^m \tilde{E}_m$, where $\tilde{E}_m \in \Gamma(T\mathcal{K})$ is the horizontal lift [381, 744] of the frame vector field $E_m \in \Gamma(T\mathcal{M})$ with respect to ∇ . It follows that N is the generator of geodesic deviation and transport [695].

Second, J is uniquely defined as the map such that

$$J : (e^m, Dy^m, D\lambda_\alpha^I) \mapsto (Dy^m, -e^m, -i D\lambda_\alpha^I). \quad (10.22)$$

Third, $\omega \in Z^2(\mathcal{K})$ is defined as a symplectic form that approaches the following two-form $\omega^\bullet \in \Omega^2(\mathcal{K})$ in the limit of vanishing curvature of ∇ , namely $\lim_{R \rightarrow 0} \omega = \omega^\bullet$:

$$\begin{aligned} \omega^\bullet = & \bar{\lambda}_{I\dot{\alpha}} (e^{\dot{\alpha}\alpha} - i Dy^{\dot{\alpha}\alpha}) \wedge D\lambda_\alpha^I - D\bar{\lambda}_{I\dot{\alpha}} \wedge (e^{\dot{\alpha}\alpha} + i Dy^{\dot{\alpha}\alpha}) \lambda_\alpha^I \\ & + 2i y^{\dot{\alpha}\alpha} D\lambda_\alpha^I \wedge D\bar{\lambda}_{I\dot{\alpha}}. \end{aligned} \quad (10.23)$$

Here, we define $p_{\alpha\dot{\alpha}} = -\lambda_\alpha^I \bar{\lambda}_{I\dot{\alpha}}$.

It should be clear that N in Eq. (10.21), J in Eq. (10.22), and $\omega^\bullet$ in Eq. (10.23) respectively reduce to N in Eq. (10.12), J in Eq. (10.15), and the exterior derivative of Eq. (10.8) in the limit of vanishing curvature of ∇ , in which case $\mathcal{K} \cong \mathbb{K}$. They are the very covariantizations of the free theory's N , J , and $\omega^\circ = d\theta^\circ$ due to ∇ .

When ∇ is curved, ω must differ from $\omega^\bullet$ since $d\omega^\bullet \neq 0$. In this sense the particle's symplectic structure necessarily develops a curvature correction.

The structures N and J could also develop curvature corrections in principle, but it will suffice to deform just the symplectic structure for our physical purposes.

The particle's 12-dimensional physical phase space arises by symplectic reduction of $\mathcal{K}$ by gauge generators $-p_m y^m \approx 0$ and $p^2 + m^2 \approx 0$.

10.3.2 Traditional Minimal Coupling

By choosing the symplectic structure ω differently, one realizes various gravitational couplings of the massive spinning particle in the phase space $\mathcal{K}$. In particular, the choice of ω amounts to specifying the particle's action.

$$\begin{array}{ccccc}
p_m e^m & \xrightarrow{d\iota_N} & d(p_m y^m) & \longrightarrow & 0 \\
\downarrow \iota_N d & & \downarrow & & \\
J \swarrow & -Dp_m y^m & \longrightarrow & 0 & \\
& \downarrow & & & \\
p_m \star \Theta^m_n y^n + W_0 d\psi \rightarrow 0 & p_m (\iota_N R^m_n) y^n & \longrightarrow & 0 & \\
\downarrow & \downarrow & & & \\
p_m (\iota_N \star R^m_n) y^n \longrightarrow 0 & p_m (\iota_N D \iota_N R^m_n) y^n & \longrightarrow & 0 & \\
\downarrow & \downarrow & & & \\
p_m (\iota_N D \iota_N \star R^m_n) y^n \rightarrow 0 & \vdots & & & \\
\downarrow & & & & \\
\vdots & & & &
\end{array}$$

Figure 10.2: The “ $\iota_N d/J$ sequence” for Kerr. A tree of one-forms emanates from the scalar-particle symplectic potential, $p_m e^m$.

For example, the choice dubbed “minimal coupling” in the traditional sense defines the symplectic potential on $\mathcal{K}$ by directly covariantizing Eq. (10.8):

$$\theta_{(0)} = p_m e^m + i y^{\dot{\alpha}\alpha} \left(\lambda_\alpha^I D \bar{\lambda}_{I\dot{\alpha}} - D \lambda_\alpha^I \bar{\lambda}_{I\dot{\alpha}} \right). \quad (10.24)$$

In this case, the symplectic form is $\omega_{(0)} = d\theta_{(0)} = \omega^\bullet + \omega'_{(0)}$, where $\omega'_{(0)} = p_m \star R^m_n y^n$. $\star$ denotes the Hodge star acting on the internal (local Lorentz) indices. Note that it could be instructive to rewrite Eq. (10.24) as

$$\theta_{(0)} = p_m e^m + p_m \star \Theta^m_n y^n + W_0 d\psi, \quad (10.25)$$

where Θ^m_n is a Lorentz-valued Maurer-Cartan form such that $D\Theta^m_n = R^m_n$, ψ is a $U(1)$ angle, and $W_0 = -p_m y^m$.

The resulting Hamiltonian EoM are immediate by the technique of covariant symplectic perturbations. They reproduce the Mathisson-Papapetrou-Dixon [227–229] equations, which describe the quadrupolar coupling [230–238] $C_2 = 0$. However, it is well-known that the Kerr black hole carries $C_2 = 1$ [176, 202, 203, 207–209, 256]. Therefore, the symplectic potential in Eq. (10.24) fails to describe the Kerr black hole.

10.3.3 N and J for Newman-Janis, with Curvature

In spirit of the NJA, we ask whether the Kerr symplectic potential can be constructed in $\mathcal{K}$ solely from the Schwarzschild symplectic potential $p_m e^m$ and the invariant structures N and J .

Eq. (10.16) has implemented the twistor magic in the flat correspondence

space $\mathbb{K}$. Its covariantization is

$$\theta_{(0)} = \left(1 + J\iota_N d\right)(p_m e^m), \quad (10.26)$$

which unfortunately evaluates to Eq. (10.24). A natural modification is

$$\theta_{(1)} = \left(\cos(\mathcal{L}_N) + \text{sinc}(\mathcal{L}_N) J\iota_N d\right)(p_m e^m). \quad (10.27)$$

Eq. (10.27) faithfully implements the concept of NJA: imaginary shifts along spin direction. Since N is the generator of geodesic deviation, the imaginary deviations arise by $\exp(\pm i\mathcal{L}_N) = \cos(\mathcal{L}_N) \pm i \text{sinc}(\mathcal{L}_N) \mathcal{L}_N$. Eq. (10.27) implements the $\pm i$ factors with the almost-complex structure J while replacing a $\mathcal{L}_N$ with $\iota_N d$ as a direct generalization of the flat construction in Eq. (10.16).

Crucially, the repeated action of $\mathcal{L}_N$ does not truncate at a finite order in the presence of curvature. As per the Cartan magic formula, the repeated action of $\mathcal{L}_N = \iota_N d + d\iota_N$ is diagrammatically represented as a planar grid where each node is a differential form and each edge is an arrow representing the action of either $\iota_N d$ or $d\iota_N$, such that $(\iota_N d)(d\iota_N) = 0$ and $(d\iota_N)(\iota_N d) = 0$. As shown in Fig. 10.2, this grid does not truncate if $R^m_n \neq 0$.

Hence we shall regard that $\cos(\mathcal{L}_N)$ and $\text{sinc}(\mathcal{L}_N)$ were secretly present in Eq. (10.16) as well, albeit trivialized. In curved backgrounds, they ensure the unity $C_\ell = 1$ of even and odd multipoles coefficients. That is, Eq. (10.27) correctly captures the linear gravitational coupling of Kerr established through Refs. [176, 202, 203, 207–209, 256].

In fact, Eq. (10.27) pinpoints a unique gravitational coupling to all orders in spin and curvature. From Fig. 10.2, it can be seen that Eq. (10.27) can be represented as

$$\theta_{(1)} = \theta_{(0)} + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} p_m ((\iota_N D)^{\ell-2} \iota_N \star^\ell R^m_n) y^n, \quad (10.28)$$

where $\star^\ell$ means to act on the internal Hodge star ℓ times. By methods in differential geometry [695], Eq. (10.28) is explicitly computed as

$$\begin{aligned} \theta_{(1)} = \theta_{(0)} + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right] \\ \left(\begin{aligned} &((\star^\ell Q_{\alpha_1}) Q_{\alpha_2} \cdots Q_{\alpha_p})^m_s e^s \\ &+ (\alpha_p - 2) ((\star^\ell Q_{\alpha_1}) Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_p})^m_s D y^s \end{aligned} \right). \end{aligned} \quad (10.29)$$

where $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_p) \in \Omega_p(\ell)$ runs over ordered partitions such that $\alpha_1 + \alpha_2 + \dots + \alpha_p = \ell$ and $\alpha_i \geq 2$. The so-called Q -tensors [695] are defined as

$$(\star^\ell Q_j)^m_s = \star^\ell R^m_{r_1 r_2 s; r_3; \dots; r_j}(x) y^{r_1} \dots y^{r_j}. \quad (10.30)$$

The nonlinear-in-Riemann terms in Eq. (10.29) are fine-tuned to yield remarkable physical consequences in SD backgrounds, as we have explained in Ref. [768].

10.4 Heaven

10.4.1 Heavenly Portal to Curved Spacetime

Eq. (10.27) and its explicit evaluation in Eq. (10.29) provide a manifestly covariant, gauge-invariant, and reparametrization-invariant construction of the Kerr action in generic, real spacetimes. It is based on two invariant geometric structures in the spinning-particle phase space: N and J .

However, it is based on the typical languages of general relativity. Apparently, the spinspace and massive twistor descriptions are lost. This means that the curved analogs of the left part of the diagram in Fig. 10.1 are yet to be constructed.

The first step toward the “radical” reformulations is to identify the map γ . It can be discovered by traveling to SD spacetimes, i.e., *heaven* in the terminologies of Newman [258, 328] and Plebański [314, 315].

The spacetime curvature R^m_n splits to SD and ASD parts as $\bar{R}^{\dot{\alpha}}_{\dot{\beta}}$ and R_{α}^{β} . Consider the (formal) complexified limit in which the latter vanishes, so $\star R^m_n = +i R^m_n$. Then Eq. (10.28) equals

$$\theta_{(1)} = \theta_{(0)} + \sum_{\ell=2}^{\infty} \frac{i^\ell}{\ell!} p_m ((\iota_N D)^{\ell-2} \iota_N R^m_n) y^n. \quad (10.31)$$

Notably, Eq. (10.31) is nearly equal to the symplectic potential of a scalar particle, geodesically deviated in a pure-imaginary direction iy (cf. Ref. [695]):

$$e^{i\mathcal{L}_N}(p_m e^m) = p_m e^m + i p_m D y^m + \sum_{\ell=2}^{\infty} \frac{i^\ell}{\ell!} p_m ((\iota_N D)^{\ell-2} \iota_N R^m_n) y^n. \quad (10.32)$$

That is, Eqs. (10.31) and (10.32) differ only by the combination

$$\theta_{G2A} = p_m \star \Theta^m_n y^n - i p_m D y^m + W_0 d\psi, \quad (10.33)$$

precisely since the SD limit facilitates replacing J in Eq. (10.27) with $+i$ except at the dipolar spin order $\ell = 1$.

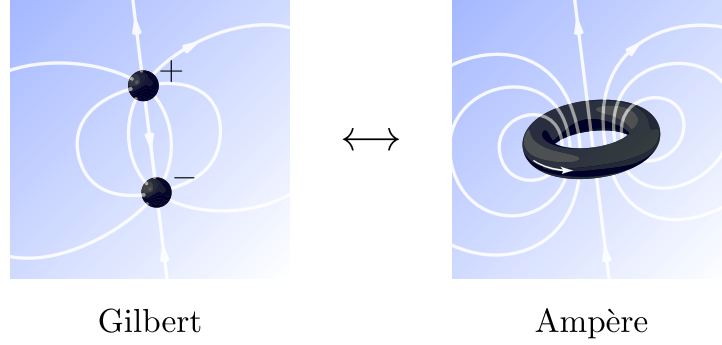


Figure 10.3: Gilbert-Ampère duality.

Therefore, our Kerr particle is indistinguishable from an imaginary-deviated Schwarzschild particle in SD backgrounds, up to a slight subtlety at the dipolar order.

To understand the origin of this subtlety from a physicist’s intuition, note that Eq. (10.33) replaces an imaginary *electric mass dipole* term $ip_m Dy^m$ with the *magnetic mass dipole* (i.e., spin) term $p_m \star \Theta^m_n y^n$. This reflects a kinematics-level difference between spinning and non-spinning particles. Specifically, there is a physical sense in which Eq. (10.33) replaces a Gilbertian dipole (displaced charges) with an Ampèrian dipole (current loop), as illustrated in Fig. 10.3. In SD backgrounds, magnetic charge is indistinguishable from i times electric charge. Thus the Ampèrian dipole $p_m \star \Theta^m_n y^n$ gets replaced with i times the Gilbertian dipole $p_m Dy^m$, and upon this very mechanism spin becomes i times deviation—which is the gist of the NJA, in fact [164].

Hence, physically speaking, the almost-complex structure J has implemented a duality between Gilbertian and Ampèrian multipoles. This observation can be rephrased as electric-magnetic or Hodge duality; recall the discussions around Eqs. (10.16) and (10.17).

With this insight, we rewrite the Kerr symplectic potential in the SD limit as

$$\theta_{(1)} = e^{i\mathcal{L}_N}(p_m e^m) + \theta_{\text{G2A}} = e^{i\mathcal{L}_N}(p_m e^m + \theta_{\text{G2A}}), \quad (10.34)$$

where the second line follows from the fact that θ_{G2A} in Eq. (10.33) carries no gravitational charge in the SD limit such that $d\theta_{\text{G2A}} = 0$; it describes a sort of a topological defect.

Remarkably, Eq. (10.34) shows that the Kerr symplectic potential in SD spacetime is a one-form localized at a “complex spacetime point” whose coor-

$$\begin{array}{ccccc}
& & \mathcal{K} & \xleftarrow{\gamma} & (T^{1,0} \oplus S_2^{1,0} \oplus S_2^{1,0})\mathcal{M} \\
& \swarrow \varphi & & \searrow \pi_2 & \searrow \pi_1 \\
\mathcal{MT} & & & \mathcal{M}_z & \mathcal{M}_x
\end{array}$$

Figure 10.4: Curved massive twistor theory.

dinates are

$$e^{iN}x^\mu = x^\mu + iy^\mu + \frac{1}{2}\Gamma^\mu_{\rho\sigma}(x)y^\rho y^\sigma + \mathcal{O}(y^3). \quad (10.35)$$

Eq. (10.35) generalizes Eq. (10.11) in a nonlinear and generally covariant fashion: *curved spinspace*.

10.4.2 Curved Spinspace

As shown in Sec. 10.A, spinspace exists for any real-analytic spacetime. To expedite our discussion, however, we may want to directly jump to a complexified setup regarding its relevance to the SD limit and twistorial ideas.

From now on, spacetime is a complex-analytic four-manifold $(\mathcal{M}, g)$ equipped with a holomorphic metric g .

Spinspace is the holomorphic tangent bundle $\widehat{\mathcal{M}} = T^{1,0}\mathcal{M}$, which has base coordinates x^μ and fiber coordinates y^μ . This means that coordinate transformations on $\mathcal{M}$ induce $(x^\mu, y^\mu) \mapsto (f^\mu(x), f^\mu_{,\nu}(x)y^\nu)$, where $f^\mu(x)$ are holomorphic functions. Spinspace is equipped with the holomorphic vector field $N \in T^{1,0}\widehat{\mathcal{M}}$ that is horizontal with respect to the Levi-Civita connection ∇ of g and satisfies $\iota_N dx^\mu = y^\mu$.

The time- i flow of N is a map $\gamma : \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$:

$$z^{\mu'} = \delta^{\mu'}_{\mu}(e^{iN}x^\mu), \quad y^{\mu'} = \delta^{\mu'}_{\mu}(e^{iN}y^\mu). \quad (10.36)$$

Here, we are adopting the primed index notation of Synge calculus [626, 695, 732–734]. $z^{\mu'}$ are the coordinates of the point in $\mathcal{M}$ that is reached by geodesically deviating from x along the direction of y , so $\text{Exp} : T^{1,0}\mathcal{M} \rightarrow \mathcal{M} : (x, y) \mapsto z$ is the holomorphic exponential map; $y^{\mu'} = W^{\mu'}_{\mu} y^\mu$ describe the tangent to the geodesic segment at the endpoint z ; $W^{\mu'}_{\mu}$ is the parallel propagator from x to z .

In sum, we identify two coordinate charts on $\widehat{\mathcal{M}} = T^{1,0}\mathcal{M}$. The “defining” chart is $(x^\mu; y^\mu)$; the other chart $(z^{\mu'}; y^{\mu'})$ will be referred to as a “googly” chart.

10.4.3 Curved Correspondence Space

For the complex spacetime $(\mathcal{M}, g)$, the curved massive correspondence space is the holomorphic direct sum bundle

$$\mathcal{K} = (T^{1,0} \oplus S_2^{1,0} \oplus S_2^{1,0})\mathcal{M}, \quad (10.37)$$

equipped with base coordinates $x^\mu \in \mathbb{C}^4$ and fiber coordinates $y^m \in \mathbb{C}^4$, $\lambda_\alpha^I, \tilde{\lambda}_{I\dot{\alpha}} \in \mathrm{GL}(2, \mathbb{C})$. Its transition functions are restricted in terms of holomorphic functions $f^\mu(x) \in \mathbb{C}^4$ and $\Omega_\alpha^\beta(x), \tilde{\Omega}^{\dot{\alpha}}_{\dot{\beta}}(x) \in \mathrm{SL}(2, \mathbb{C})$ (cf. Eq. (10.19)).

The curved correspondence space is equipped with the holomorphic vector field $N \in \Gamma(T^{1,0}\mathcal{K})$ that is horizontal with respect to ∇ and satisfies $\iota_N dx^\mu = y^\mu$, which we again denote as N by abuse of notation.

Similarly, the time- i flow of N is denoted as γ . This is the map $\gamma : \mathcal{K} \rightarrow \mathcal{K}$ such that $x^\mu \mapsto z^{\mu'} = (\exp(x, y))^{\mu'}$, $y^\mu \mapsto y^{\mu'} = W^{\mu'}_\mu y^\mu$, and $\lambda_\alpha^I \mapsto \lambda_{\alpha'}^I = W_{\alpha'}^\alpha \lambda_\alpha^I$, $\tilde{\lambda}_{I\dot{\alpha}} \mapsto \tilde{\lambda}_{I\dot{\alpha}'} = \tilde{\lambda}_{I\dot{\alpha}} W^{\dot{\alpha}}_{\dot{\alpha}'}$. Here, the parallel propagator $W^{\mu'}_\mu$ decomposes into $\mathrm{SL}(2, \mathbb{C})$ blocks. Note that it is metric-preserving as $g_{\mu\nu}(z) W^{\mu'}_\mu W^{\nu'}_\nu = g_{\mu\nu}(x)$.

The defining chart on $\mathcal{K}$ is $(x^\mu; y^\mu, \lambda_\alpha^I, \tilde{\lambda}_{I\dot{\alpha}})$. The googly chart on $\mathcal{K}$ is $(z^{\mu'}; y^{\mu'}, \lambda_{\alpha'}^I, \tilde{\lambda}_{I\dot{\alpha}'})$. The former chart establishes the projection $\pi_1 : \mathcal{K} \rightarrow \mathcal{M} : (x^\mu; y^\mu, \lambda_\alpha^I, \tilde{\lambda}_{I\dot{\alpha}}) \mapsto x^\mu$. The latter chart establishes the projection $\pi_2 : \mathcal{K} \rightarrow \mathcal{M} : (z^{\mu'}; y^{\mu'}, \lambda_{\alpha'}^I, \tilde{\lambda}_{I\dot{\alpha}'}) \mapsto z^{\mu'}$. As a result, $\mathcal{K}$ fibers on $\mathcal{M}$ in two different ways. If a disambiguation is needed, we denote the former and latter versions of $\mathcal{M}$ as $\mathcal{M}_x$ and $\mathcal{M}_z$, respectively.

Note that $\widehat{\mathcal{M}} = T^{1,0}\mathcal{M}_x \cong \mathcal{M}_z \times \mathcal{M}_{\tilde{z}}$, where z and $\tilde{z}$ arise by $e^{+iN}x$ and $e^{-iN}x$. In the flat limit $\mathcal{M}_x \rightarrow \mathbb{M}$, $\mathcal{M}_z$ and $\mathcal{M}_{\tilde{z}}$ are precisely the holomorphic and anti-holomorphic subspaces of $\widehat{\mathbb{M}} = \mathbb{M}^\mathbb{C}$, $\widehat{\mathbb{M}}^{1,0}$ and $\widehat{\mathbb{M}}^{0,1}$.

Lastly, $\mathcal{K}$ is not only equipped with the horizontal vector field $N \in \Gamma(T^{1,0}\mathcal{K})$ but also $J : T_{1,0}^*\mathcal{K} \rightarrow T_{1,0}^*\mathcal{K}$ such that $J^2 = -\mathrm{id}$ and $\omega \in \Omega^{2,0}(\mathcal{K})$ such that $d\omega = 0$: $(\mathcal{K}, N, J, \omega)$. N and J are invariant under the restricted coordinate transformations of $\mathcal{K}$ stipulated above.

The Kerr symplectic potential arises by the same formula in Eq. (10.27), yet now $\theta^{(1)} \in \Omega^{1,0}(\mathcal{K})$.

10.4.4 Chiral Action and Motion in Heaven

If the holomorphic metric g is SD, then Eq. (10.34) implies that the Kerr symplectic potential is $\theta_{(1)} = \gamma^* \theta_{(1)}^+ + d(iW_0)$, where

$$\theta_{(1)}^+ = -\tilde{\lambda}_{I\dot{\alpha}'} e^{\dot{\alpha}\alpha} \lambda_\alpha^I - 2i \tilde{\lambda}_{I\dot{\alpha}'} y^{\dot{\alpha}\alpha} d\lambda_\alpha^I. \quad (10.38)$$

For simplicity, we have set the ASD spin connection coefficients to zero in a local patch so that the ASD Wilson line $W_{\alpha'}^\beta$ is trivial; hence the primes on left-handed spinor indices have been dropped.

Eq. (10.38) provides a chiral formulation of the Kerr particle, treating λ_α^I and $\tilde{\lambda}_{I\dot{\alpha}'}$ asymmetrically. This formulation makes it evident that the symplectic form is isomorphic to the free theory's symplectic form ω° via the map $\varrho : (dz^{\dot{\alpha}\alpha}, d\bar{z}^{\dot{\alpha}\alpha}, d\lambda_\alpha^I, d\tilde{\lambda}_{I\dot{\alpha}}) \mapsto (e^{\dot{\alpha}\alpha}, e^{\dot{\alpha}\alpha} - 2iDy^{\dot{\alpha}\alpha}, d\lambda_\alpha^I, D\tilde{\lambda}_{I\dot{\alpha}})$. Hence there is no covariant symplectic perturbation, and the resulting Hamiltonian EoM are just isomorphic to free theory:

$$\begin{aligned} e^{m'}_{\mu'}(z) \dot{z}^{\mu'} &= p^{m'}, \quad \dot{\lambda}_\alpha^I = 0, \\ Dy^{m'}/d\tau &= 0, \quad D\tilde{\lambda}_{I\dot{\alpha}'}/d\tau = 0. \end{aligned} \quad (10.39)$$

Remarkably, the dynamics of the Kerr black hole in heaven is thus simply geodesic motion with parallel transportation of the local Lorentz degrees of freedom $y^{m'}, \tilde{\lambda}_{I\dot{\alpha}'}, \lambda_\alpha^I$ on the complex worldline $z^{\mu'}$ ³.

Eq. (10.39) concretely verifies the spinning, complexified analog of equivalence principle that uniquely characterizes spinning black holes in SD external fields, which the current author proposed through works [672, 673]: the left-handed spin frame λ_α^I is frozen, i.e., does not precess at all, in SD backgrounds.

When transcribed to the defining chart (covariantized at x^μ in spacetime), Eq. (10.39) yields an all-orders-in-spin extension of Mathisson-Papapetrou-Dixon equations with $C_\ell = 1$ for all ℓ . Moreover, Eq. (10.39) implies exact hidden symmetries in the presence of Killing and Killing-Yano tensors by dynamically NJ shifting the conserved quantities of the Schwarzschild probe [768].

It is also easily seen that the holomorphic coordinates $z^{\mu'}$ of the Kerr black hole's curved spinspace $\widehat{\mathcal{M}}$ remain Poisson-commutative in SD gravity.

³A similar idea seems to have been conceived in Ref. [259] in the context of $\mathcal{H}$ -space.

10.4.5 Curved Massive Twistor Space

Lastly, we expand the holomorphic coframe around a flat background as $e^{\dot{\alpha}\alpha} = dz^{\dot{\alpha}\alpha} + h^{\dot{\alpha}\alpha}$. The physical relevance of this setup is perturbation theory, for instance. Concretely, employ the second Plebański coordinates [314, 320] to let α remain as the “rigid” $\mathrm{SL}(2, \mathbb{C})$ index. Then Eq. (10.38) becomes

$$-\tilde{\lambda}_{I\dot{\alpha}'} d(z^{\dot{\alpha}\alpha} \lambda_{\alpha}^I) + \tilde{\lambda}_{I\dot{\alpha}'} (z^{\dot{\alpha}\alpha} - 2iy^{\dot{\alpha}\alpha}) d\lambda_{\alpha}^I + \theta'_{(1)}, \quad (10.40)$$

where $\theta' = -\tilde{\lambda}_{I\dot{\alpha}'} h^{\dot{\alpha}\alpha} \lambda_{\alpha}^I = p_{m'} h^{m'} = p_{m'} h^{m'}_{\mu'}(z) dz^{\mu'}$. This suggests to identify the “mu variables” as

$$\mu^{\dot{\alpha}I} = z^{\dot{\alpha}\alpha} \lambda_{\alpha}^I, \quad \tilde{\mu}_I^{\alpha} = \tilde{\lambda}_{I\dot{\alpha}'} (z^{\dot{\alpha}\alpha} - 2iy^{\dot{\alpha}\alpha}), \quad (10.41)$$

in which case Eq. (10.40) and its exterior derivative become

$$\begin{aligned} \theta_{(1)}^+ &= -\tilde{\lambda}_{I\dot{\alpha}'} d\mu^{\dot{\alpha}I} + \tilde{\mu}_I^{\alpha} d\lambda_{\alpha}^I + \theta'_{(1)}, \\ \omega_{(1)} &= d\mu^{\dot{\alpha}I} \wedge d\tilde{\lambda}_{I\dot{\alpha}'} + d\tilde{\mu}_I^{\alpha} \wedge d\lambda_{\alpha}^I + d\theta'_{(1)}. \end{aligned} \quad (10.42)$$

The free-theory part of Eq. (10.42) is identical to ω° in Eq. (10.2) upon unpriming the right-handed spinor indices. Eq. (10.42)—eventually—verifies the Kerr symplectic structure in SD spacetime we derived in Ref. [673] from a bootstrap approach based on the Kähler geometry (zig-zag structure) of massive twistor space and the Souriau insight [815, 816] that deformations of particle symplectic structure implement couplings to external fields.

Eq. (10.42) establishes that the coupling of the Kerr particle to SD gravity can be formulated as a deformation of the symplectic structure on massive twistor space:

$$(\mathrm{MT}, \omega^{\circ}) \rightarrow \mathcal{MT} \cong (\mathrm{MT}, \omega^{\circ} + d\theta'_{(1)}). \quad (10.43)$$

Crucially, this deformation is via a $(1, 0)$ -perturbation $\theta'_{(1)}$ at the level of symplectic potential such that the Poisson commutativity of holomorphic spin-spacetime coordinates [672] is preserved: $\{z^{\mu'}, z^{\nu'}\} = 0$. Physically, this encodes that the gravitoelectromagnetic interaction action is supported on the NJ shifted worldline $z^{\mu'}$ for the SD (incoming positive-helicity [342, 343]) sector.

In this context, Eq. (10.41) is identified as the deformed massive incidence relation, from which the scaffolding of curved massive twistor theory arises.

Namely, Eq. (10.41) defines a diffeomorphism $\varphi : \mathcal{K} \rightarrow \mathcal{MT}$ and completes the map of our journey shown in Fig. 10.4.

We may note that φ seems to encode a deformation of complex structure as well when approached from the holomorphic complex structure of curved spinspace-time $\widehat{\mathcal{M}}$. The original complex coordinates of $\widehat{\mathcal{M}} \cong \mathcal{M}_z \times \mathcal{M}_{\bar{z}}$ are $z^{\mu'}$ and $\bar{z}^{\mu''}$ that describe $e^{\pm iN}x$. However, the deformed massive incidence relation in Eq. (10.41) identifies $z^{\mu'} - 2iy^{\mu'}$ as the “ z -bar” variable:

$$w^{\mu'} = z^{\mu'} - 2iy^{\mu'} = z^{\mu'} - 2i\sigma^{\mu'}(z, \bar{z}). \quad (10.44)$$

Here, $\sigma(z, \bar{z})$ describes the complexified Synge world function [626, 732–734], $\sigma : \mathcal{M}_z \times \mathcal{M}_{\bar{z}} \rightarrow \mathbb{C}$. Crucially, $w^{\mu'}$ differs from $\bar{z}^{\mu''}$ for nonzero curvature.

In massless curved twistor theory, the celebrated nonlinear graviton construction of Penrose [313] formulates SD gravity as deformations of the complex structure in twistor space. This describes deformations of the twistor lines in terms of the deformed incidence relation.

Our proposal above envisions curved massive twistor theory from a deformed massive incidence relation relevant to perturbation theory, which is reminiscent yet not fully equivalent to massless curved twistor theory. It may be also possible to consider a formulation that follows the ideas of massless curved twistor theory more closely, in which case the symplectic structure might be fixed.

10.4.6 Googly Kerr Action

In Eq. (10.29), we have already obtained the Kerr action in generic spacetimes with both SD and ASD modes, i.e., *earth* in the terminologies of Newman [258, 328] and Plebański [314, 315].

We then took an excursion to SD backgrounds, i.e., *heaven*, to discover the nonlinear NJ shift in Eq. (10.35). The key punchline is that the *NJ shift manifests in heaven* as the localization of the Kerr action on the holomorphic worldline $z^{\mu'}$.

It should be stressed that there is an obstruction in simultaneously manifesting the NJ shifts in earth for both SD and ASD sectors. This obstruction is due to the very nonlinearity of gravity and poses a massive analog of the googly [347–350] problem. Concretely, it is impossible to write the earthly Kerr action as a sum of two separate worldline actions localized respectively at $z^{\mu'}$ and $\bar{z}^{\mu''}$, since there exist nonlinear mixing between SD and ASD curvatures.

See also Sec. 10.B, where the same problem is found for the case of non-abelian gauge theory.

The best we can do is to perturb away from the SD sector, which is a typical twistor-theorist move. Assuming the complexified setup, the Kerr symplectic potential in Eq. (10.27) can be written as

$$\theta_{(1)} = p_m e^m + \frac{e^{i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_{\text{G2A}}^+ + \frac{e^{-i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_{\text{G2A}}^-, \quad (10.45)$$

where we have defined

$$\theta_{\text{G2A}}^\pm := \frac{1}{2} \left(p_m D y^m \mp i (p_m \star \Theta^m{}_n y^n + W_0 d\psi) \right). \quad (10.46)$$

Meanwhile, the symplectic potential $e^{i\mathcal{L}_N}(p_m e^m)$ of the imaginary-deviated Schwarzschild particle is

$$p_m e^m + \frac{e^{i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_{\text{G2A}}^+ + \frac{e^{-i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_{\text{G2A}}^-. \quad (10.47)$$

The difference between Eqs. (10.46) and (10.47) derives

$$\begin{aligned} \theta_{(1)} &= e^{i\mathcal{L}_N} \left(p_m e^m + \frac{e^{-2i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_{\text{G2A}}^- \right), \quad (10.48) \\ &= e^{i\mathcal{L}_N} \left(\theta_{(0)} - i p_m D y^m + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} p_m ((\iota_N D)^{\ell-2} \iota_N R^{-m}{}_n) y^n \right), \\ &= p_{m'} e^{m'} + p_{m'} \star \Theta^{m'}{}_{n'} y^{n'} - i p_{m'} D y^{m'} + W_0 d\psi \\ &\quad + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{\sum_{j=i}^p \alpha_j - 2}{\alpha_i - 2} \right] \\ &\quad \left(((Q_{\alpha_1}^-) Q_{\alpha_2} \cdots Q_{\alpha_p})^{m'}{}_{s'} e^{s'} \right. \\ &\quad \left. + (\alpha_p - 2) ((Q_{\alpha_1}^-) Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_{p-1}})^{m'}{}_{s'} D y^{s'} \right), \end{aligned}$$

where $R^\pm := \frac{1}{2}(R \mp i \star R)$ and $(Q_j^\pm) := \frac{1}{2}(Q_j \mp i \star Q_j)$. To be precise, the first line is the pullback of the second line by the diffeomorphism $\gamma : \mathcal{K} \rightarrow \mathcal{K}$.

Eq. (10.48) provides a googly formulation of the Kerr action. It describes *both* SD and ASD couplings as an action localized on the *holomorphic* worldline. It reduces to Eq. (10.34) in the SD limit. It treats SD and ASD modes, and the $z^{\mu'}$ and $\bar{z}^{\mu''}$ worldlines, asymmetrically.

In actual applications, Eq. (10.48) facilitates a perturbation theory in which the NJ shift (as spin exponentiation [206–209]) of scattering amplitudes is manifestly guaranteed in the SD sector to all graviton multiplicities [768], on top of which the ASD perturbations are systematically added on.

10.5 Interpretation

In Eq. (10.48), $(e^{-2i\mathcal{L}_N} - 1)/(-2i\mathcal{L}_N) = \int_0^1 d\eta e^{-2i\eta\mathcal{L}_N}$ inserts the NJ shifts of θ_{G2A}^- along the geodesic segment joining the two points z and $\tilde{z}$. Hence Eq. (10.48) states that the Kerr black hole describes a dangling open string attached to a point mass at z . This string is charged under ASD gravitons ($d\theta_{G2A}^- = p_m R^{-m}{}_n y^n$) but is invisible to SD fields as remarked earlier. In fact, recalling our earlier discussions on Gilbertian and Ampèrian dipoles, it precisely describes the ASD combination of electric and magnetic mass dipole moments. Thus, we identify its interpretation as an ASD Misner string.

Similarly, it can be seen that Eq. (10.27) describes two point masses joined by a thin gravitomagnetic flux tube: an ordinary Misner string⁴. The two pictures, googly and real, are equivalent via identifying a pair of opposite-sign masses as a line defect of electric mass dipole.

The above analysis elicits a provocative idea that the Kerr black hole describes a system of SD and ASD dyons. At least in the effective theory of Kerr black holes, Eqs. (10.34) and (10.48) show that this statement holds true as the physical origin of the NJ shift. The question is whether it holds in the ultraviolet theory, i.e., general relativity, as a nonperturbative fact.

The persistent endeavor of the current author around this question had resulted in works [163, 164], which established that the Kerr metric represents the exact nonlinear superposition of SD and ASD Taub-NUT solutions as a holomorphic saddle, from which the NJA [202] is faithfully reproduced.

The worldlines $z^{\mu'}$ and $\tilde{z}^{\mu''}$ are the infrared avatars of the Taub-NUT instantons. The worldline actions due to Eqs. (10.27) and (10.48) insert the Misner string along the geodesic between the two instantons. As long as the geodesicity of the Misner string (or locality in worldline proper time) can be assumed, Eq. (10.27), equivalently Eq. (10.48), defines the unique answer for the effective action that describes a dynamical pair of SD and ASD Taub-NUT instantons. Regarding this point, it will be helpful to investigate if the Misner string serves as a genuinely topological surface defect in dynamical, full (non-SD) gravity (cf. Sec. 3.5), in which case the Misner worldsheet can be realized in any shape to predict the same physical observables.

⁴This string structure was noticed in Ref. [226], but the physical interpretations and googly formulation are not provided there. Ref. [226] also does not provide explicit expressions such as Eqs. (10.29) and (10.48) that expand out the action fully in terms of covariant worldline operators.

10.6 Conclusions

In this chapter, we derived an explicit all-orders worldline effective action for the Kerr black hole by generalizing an intrinsic feature of twistor particle theory in curved spacetime. This implements the NJA at the level of test-particle actions, in terms of differential-geometric elements N and J . Nonlinear, dynamical NJ shift manifests in SD backgrounds as localizations on the holomorphic worldline. The full gravitational interactions are systematically understood by perturbing around the SD sector. Clear physical interpretations are given and are substantiated by the ultraviolet description of the Kerr black hole.

This chapter is written in the geometrical language and wishes to deliver itself to the twistor theory community as well. This chapter attempts to make a small yet ambitious step towards “curved massive twistor theory,” which shall exist in light of the histories of massless twistor theory and twistor particle program. Some tentative identifications and definitions are given for the deformed massive incidence relation and curved massive twistor space, based on the particular case of the Kerr black hole as a massive twistor.

10.A Spinspace from Adapted Complex Structure

This appendix elaborates on the definition of spinspace for real spacetime.

In Eq. (10.10), we discussed two perspectives toward spinspace: real and complex manifolds. In the case of curved real-analytic spacetime $(\mathcal{M}, g)$, however, the latter picture is seemingly lost. The formula in Eq. (10.11) becomes utterly nonsensical, since x^μ are *coordinates* while y^μ are components of a *vector*. This represents a clash between two necessary features of “curved spinspace”: general covariance and holomorphy.

Luckily, the mathematical framework known as adapted complex structure [743, 776–782] provides a nice resolution, showing that a sufficiently small neighborhood of the zero section of the tangent bundle $T\mathcal{M}$ of a real-analytic $(\mathcal{M}, g)$ can be embedded into a complex manifold $\widehat{\mathcal{M}}$ via complexifying the exponential map (geodesic flow):

$$\Pi_i : T'\mathcal{M} \rightarrow \widehat{\mathcal{M}}. \quad (10.49)$$

Here, we have supposed a sufficiently narrow neighborhood $T'\mathcal{M}$ of the zero section in $T\mathcal{M}$.

$$\begin{array}{ccc}
& \alpha^{\mathbb{C}} \in \Omega^p(\widehat{\mathcal{M}}) & \\
\Pi_i^* \swarrow & & \nwarrow \text{a.c.} \\
\widehat{\alpha} \in \Omega^p(T'\mathcal{M}) & & \alpha \in \Omega^p(\mathcal{M})
\end{array}$$

Figure 10.5: The “central dogma” of adapted complex structure.

The idea is simple. The time- η exponential map sends the pair $(x, y) \in T'\mathcal{M}$ to the point in $\mathcal{M}$ whose coordinates are $w^\mu(x, y; \eta) = x^\mu + \eta y^\mu - \frac{\eta^2}{2} \Gamma^\mu_{\rho\sigma}(x) y^\rho y^\sigma + \mathcal{O}(\eta^3)$. This is a power series solution which converges in $T'\mathcal{M}$.² Upon a real-analytic coordinate transformation $x^\mu \mapsto f^\mu(x)$ on $\mathcal{M}$, the coordinates of the geodesically deviated point transform as $w^\mu(x, y; \eta) \mapsto f^\mu(w(x, y; \eta))$ for any η , by construction.

By analytic continuation, $z^\mu(x, y) := w^\mu(x, y; i)$ is well-defined within $T'\mathcal{M}$, whose explicit formula is exactly Eq. (10.35). By analyticity, the coordinate transformation on $\mathcal{M}$ induces $z^\mu(x, y) \mapsto f^{\mathbb{C}\mu}(z(x, y))$, where $f^{\mathbb{C}}$ is the analytic continuation of f . This shows that $z^\mu(x, y)$ defines a holomorphic coordinate chart on $T'\mathcal{M}$. Therefore, $T'\mathcal{M}$ becomes a complex manifold.

Physically speaking, this construction ensures that holomorphy is an observer-independent notion. This is important since holomorphy will be linked with self-duality by the NJ shift.

Now let us unravel the precise mathematical details. The time- η exponential map can be formalized as

$$\Pi_\eta = \pi \circ \Phi_\eta : T'\mathcal{M} \rightarrow \mathcal{M}, \quad (10.50)$$

where $\Phi_\eta : T'\mathcal{M} \rightarrow T\mathcal{M}$ denotes the time- η flow by the horizontal vector field $N \in \Gamma(T\mathcal{M})$ such that $\iota_N dx^\mu = y^\mu$, and $\pi : T\mathcal{M} \rightarrow \mathcal{M}$ is the bundle projection. The pullback Π_η^* maps analytic functions on $\mathcal{M}$ to analytic functions on $T'\mathcal{M}$, the explicit power series solutions for which arise by the exponentiated Lie derivative $e^{\eta \mathcal{L}_N}$.

By analytic continuation to $\eta \in \mathbb{C}$, one defines Π_i^* as a map from real-analytic functions on $\mathcal{M}$ to complex-analytic functions on $T'\mathcal{M}$, the application of which on coordinate functions establishes the map Π_i in Eq. (10.49) such that $\mathcal{M}$, as the zero section of $T\mathcal{M}$, is embedded in the complex-analytic manifold $\widehat{\mathcal{M}} = \mathcal{M}^{\mathbb{C}}$ as a totally real submanifold of maximal dimension [743, 817, 818].

The practical use of this formalism is facilitated by the fact that the ex-

ponentiated Lie derivative $e^{i\mathcal{L}_N}$ computes the pullback Π_i^* . Any differential p -form $\alpha \in \Omega^p(\mathcal{M})$ is transcribed to $\alpha^{\mathbb{C}} \in \Omega^p(\widehat{\mathcal{M}})$ by analytic continuation, which translates to $\widehat{\alpha} = \Pi_i^*[\alpha^{\mathbb{C}}] = e^{i\mathcal{L}_N}\alpha \in \Omega^p(T'\mathcal{M})$. This is the “central dogma” of adapted complex structure, depicted in Fig. 10.5. For instance, a zero-form $\phi \in C^\infty(\mathcal{M})$ exhibits the covariant Taylor expansion

$$\begin{aligned}\widehat{\phi}(x, y) &= \phi^{\mathbb{C}}(z(x, y)) = e^{i\mathcal{L}_N}\phi(x), \\ &= \sum_{\ell=0}^{\infty} \frac{i^\ell}{\ell!} \phi_{;\rho_1; \dots; \rho_\ell}(x) y^{\rho_1} \dots y^{\rho_\ell}.\end{aligned}\tag{10.51}$$

Note how general covariances coexist at both points x and z Eq. (10.51).

Curved spinspace-time is the complex-analytic four-manifold $\widehat{\mathcal{M}}$ that recasts the tangent bundle of spacetime via adapted complex structure, in the sense of Eq. (10.49).

Spacetime fields, such as $\phi(x)$ in Eq. (10.51), permeate into spinspace-time via analytic continuation: $\phi^{\mathbb{C}}(z)$. The spinspace-time field $\phi^{\mathbb{C}}(z)$ is represented in the “spacetime + spin” form, $\widehat{\phi}(x, y)$, by means of the operator $e^{i\mathcal{L}_N}$ ⁵.

Historically, it was Newman who first envisioned the concept of spinspace-time and its physical applications [252–259, 329, 330]. The curved spinspace-time constructions of this chapter transparently realizes Newman’s provisional insights in terms of concrete differential-geometric constructions.

10.B Nonabelian Root-Kerr Action

Here, we construct the $\sqrt{\text{Kerr}}$ symplectic structure coupled to nonabelian gauge theory. Suppose gauge group G and gauge algebra $\mathfrak{g}$. Let $a, b, \dots$ be the adjoint indices. Let $\mathcal{V}$ be a representation space assigned with indices $i, j, \dots$, on which the generators are $(t_a)^i_j$.

The color degrees of freedom can be implemented by phase spaces such as T^*G , $T^*\mathcal{V}$, etc. Take $T^*\mathcal{V}$ with complex coordinates ψ^i , for instance. The correspondence space is enlarged as $\mathbb{K}_{\text{col}} = (T \oplus S_2)\mathbb{M} \oplus E$, where E is a vector bundle over $\mathbb{M}$ whose typical fiber is $T^*\mathcal{V}$.

Let $A \in \Omega^1(\mathbb{M}; \mathfrak{g})$ be the nonabelian gauge connection. A complete, gauge-covariant basis of one-forms on $\mathbb{K}_{\text{col}}$ is $(dx^\mu, dy^\mu, d\lambda_\alpha^I, D\psi^i)$, where D is the gauge-covariant exterior derivative. The horizontal vector field $N \in \Gamma(\mathbb{K}_{\text{col}})$

⁵The latter description is perturbative in spin and may break down beyond the region $T'\mathcal{M}$ in $T\mathcal{M}$. However, it suffices to consider small enough spin for the physical purposes of this chapter.

$$\begin{array}{c}
i\bar{\psi}_i D\psi^i \xrightarrow{d\iota_N} 0 \\
\downarrow \iota_N d \\
\text{dual} \nearrow q_a(\iota_N F^a) \longrightarrow 0 \\
\downarrow \\
q_a(\iota_N * F^a) \longrightarrow 0 \quad q_a(\iota_N D \iota_N F^a) \rightarrow 0 \\
\downarrow \quad \downarrow \\
q_a(\iota_N D \iota_N * F^a) \rightarrow 0 \quad \vdots \\
\downarrow \quad \downarrow \\
\vdots \quad \vdots
\end{array}$$

Figure 10.6: The “ $\iota_N d/*$ sequence” for $\sqrt{\text{Kerr}}$. A tree of one-forms emanates from the Coulomb particle’s color-sector symplectic potential, $i\bar{\psi}_i D\psi^i$.

is uniquely defined by the interior products $\iota_N : (dx^\mu, dy^\mu, d\lambda_\alpha^I, D\psi^i) \mapsto (y^\mu, 0, 0, 0)$. It follows that N is the generator of gauge-covariant translations [695].

The Coulomb particle is described by the color-sector symplectic potential

$$\check{\theta}_{(0)} = i\bar{\psi}_i D\psi^i = i\bar{\psi}_i d\psi^i + q_a A^a, \quad (10.52)$$

where $q_a = i\bar{\psi}_i (t_a)^i_j \psi^j$ is the adjoint color charge. A deviated Coulomb particle is described by $e^{\mathcal{L}_N} \check{\theta}_{(0)}$:

$$\cos(\mathcal{L}_N)(i\bar{\psi}_i D\psi^i) + \text{sinc}(\mathcal{L}_N)(q_a \iota_N F^a). \quad (10.53)$$

In Eq. (10.53), the $\cos(\mathcal{L}_N)$ term implements two electric charges separated apart at $x \pm iy$. The $\text{sinc}(\mathcal{L}_N)$, on the other hand, inserts a thin electric flux tube between these electric charges. We apply electric-magnetic duality on the $\text{sinc}(\mathcal{L}_N)$ term via Hodge dual on the field strength:

$$\check{\theta}_{(1)} = \cos(\mathcal{L}_N)(i\bar{\psi}_i D\psi^i) + \text{sinc}(\mathcal{L}_N)(q_a \iota_N * F^a). \quad (10.54)$$

As a result, we obtain the symplectic potential describing two electric charges at $x \pm iy$ joined by a magnetic flux tube, i.e., a Dirac string (cf. Ref. [186]). Upon Gilbert-Ampère duality, this describes a pair of SD and ASD dyons. This pair is nothing other than the (dynamical version of) $\sqrt{\text{Kerr}}$ solution: repeat the exercise in Ref. [164].

Eq. (10.54) pinpoints a unique gauge theory interaction to all orders in spin and gauge coupling. From Fig. 10.6, it can be seen that Eq. (10.54) can be represented as

$$\check{\theta}_{(1)} = \check{\theta}_{(0)} + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} q_a ((\iota_N D)^{\ell-1} \iota_N * F^a), \quad (10.55)$$

where $*^\ell$ means to act on the internal Hodge star ℓ times. Eq. (10.55) parallels Eq. (10.28) by identifying spin angular momentum as “the gravitational charge” and Riemann curvature two-form as the “gravitational field strength”: gravity as a gauge theory of Lorentz group.

By methods in differential geometry [695], Eq. (10.55) evaluates to

$$\check{\theta}_{(1)} = i\bar{\psi}_i D\psi^i + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} q_a \left((*^\ell P_\ell)^a{}_\sigma dx^\sigma + (\ell-1) (*^\ell P_{\ell-1})^a{}_\sigma dy^\sigma \right), \quad (10.56)$$

where the so-called P -tensors [695] are defined as

$$(*^\ell P_j)^a{}_\sigma = *^\ell F^a{}_{\rho_1\sigma;\rho_2;\dots;\rho_j}(x) y^{\rho_1} \dots y^{\rho_j}. \quad (10.57)$$

Eq. (10.56) provides the action of the $\sqrt{\text{Kerr}}$ particle coupled to generic nonabelian gauge field configurations (earth). It is manifestly real and parity-symmetric and is based on the conventional “spacetime + spin” picture.

NJ shift manifests in heaven. On the support of self-duality $*F^a = +iF^a$, Eq. (10.55) becomes

$$\check{\theta}_{(1)} = e^{i\mathcal{L}_N} \check{\theta}_{(0)} = i\bar{\psi}_{i'} D\psi^{i'}, \quad (10.58)$$

which localizes on $z^\mu = x^\mu + iy^\mu$. Here, we have complexified $\mathcal{V}$ (as a real vector space) to $\mathcal{V}^\mathbb{C}$.

There is an obstruction in simultaneously manifesting the NJ shifts in earth for both SD and ASD sectors. This obstruction is due to the very nonlinearity of nonabelian gauge theory and poses a massive analog of the googly problem. Concretely, it is impossible to write the earthly $\sqrt{\text{Kerr}}$ action as a sum of two separate worldline actions localized respectively at z^μ and $\bar{z}^\mu$, since P_ℓ inevitably exhibits nonlinear mixing between SD and ASD modes from $\ell \geq 2$.

Note that there is no such googly problem in abelian gauge theory, as the gauge potential cleanly splits into SD and ASD parts and the $\sqrt{\text{Kerr}}$ symplectic potential is linear in fields.

Still, we can manifest the spinspace localization on either z^μ or $\bar{z}^\mu$. By mimicking Eq. (10.45) in terms of $((e^{i\mathcal{L}_N} - 1)/\mathcal{L}_N) q_a(\iota_N F^{\pm a})$, it follows that

$$\begin{aligned} \check{\theta}_{(1)} &= e^{i\mathcal{L}_N} \left(i\bar{\psi}_i D\psi^i + \frac{e^{-2i\mathcal{L}_N} - 1}{\mathcal{L}_N} q_a(\iota_N F^{-a}) \right), \\ &= i\bar{\psi}_{i'} D\psi^{i'} + \sum_{\ell=1}^{\infty} \frac{(-2i)^\ell}{\ell!} q_{a'} \left((P_\ell^-)^{a'}{}_\sigma dz^\sigma + (\ell-1) (P_{\ell-1}^-)^{a'}{}_\sigma dy^\sigma \right), \end{aligned} \quad (10.59)$$

where $F^{\pm a} := \frac{1}{2}(F^a \mp i*F^a)$ and $(P_\ell^\pm) := \frac{1}{2}(P_\ell \mp i*P_\ell)$.

Eq. (10.59) provides the “radical” spinspace-time formulation of the $\sqrt{\text{Kerr}}$ action, i.e., a googly formulation. It describes *both* SD and ASD couplings as an action localized on the *holomorphic* worldline z^μ . It reduces to Eq. (10.58) in the SD limit. It treats SD and ASD modes, and the z^μ and $\bar{z}^\mu$ worldlines, asymmetrically. In perturbation theory, it manifests spin exponentiation of the positive-helicity (but not negative-helicity) Compton amplitudes.

THE KERR-NEWMAN EFFECTIVE ACTION

An all-orders worldline effective action for Kerr-Newman black hole is achieved in twistor particle theory. Exact hidden symmetries are identified in SD backgrounds.

This chapter reproduces the contents of Ref. [770], J.-H. Kim, “The Kerr-Newman two-twistor particle,” [arXiv:2603.07537 \[gr-qc\]](#).

11.1 Introduction

An important topic in the current gravitational physics literature has been the construction of worldline effective actions [92–97] of spinning black holes. Recently, Ref. [769] has derived an exact first-order action of the Kerr black hole and provided its explicit all-orders formula, by using massive twistor theory [377, 796, 801] and the tangent bundle formalism for geodesic deviation [695].

Ref. [769]’s derivation of the Kerr action provides the test-particle counterpart of the NJA [768, 769]. The NJA is a solution-generating technique that rederived the Kerr metric by performing a complex-geometrical transformation on the Schwarzschild metric [202]. This trick then facilitated the very historical derivation of the Kerr-Newman solution [179]. According to a modern view [164], the NJA encodes that the Kerr solution is diffeomorphic to a pair of SD and ASD Taub-NUT instantons when taken as a holomorphic saddle.

The present paper generalizes Ref. [769]’s construction to derive the first-order action of the Kerr-Newman black hole coupled to background gravitational and electromagnetic fields. Again, the systematic formalism of Ref. [695] proves instrumental in obtaining explicit formulae for the action. We identify a Hamiltonian system that can represent the exact, all-orders dynamics of the Kerr-Newman black hole in its point-particle effective theory.

It is found that the conclusions of Ref. [769] are preserved. Nonlinear NJ shift manifests in SD backgrounds. A chiral (“googly”) formulation of the Kerr-Newman action is viable for generic, non-SD external fields. In perturbation

theory, the spin exponentiation of same-helicity Compton amplitudes [98, 207, 208, 210–212] is manifested to all graviphoton multiplicities.

We also derive the classical EoM in SD backgrounds. We confirm the complexified, spinning analog of equivalence principle proposed by Ref. [672]. This implies dynamical hidden symmetries in the presence of Killing-Yano tensors, generalizing Ref. [768]’s previous result about the Kerr particle.

11.2 Geodesic Deviation in Einstein-Maxwell Geometry

First, we present the new application of Ref. [695]’s tangent bundle formalism for geodesic deviation.

Let $(\mathcal{M}, g)$ be a real-analytic pseudo-Riemannian four-manifold, and let $F \in \Omega^2(\mathcal{M})$ be a two-form. Let ∇ be the Levi-Civita connection. Let D be the covariant exterior derivative due to ∇ .

Let $N \in T(T\mathcal{M})$ be the horizontal lift of the tautological vector field of the tangent bundle $T\mathcal{M}$ with respect to ∇ . This is the geodesic spray, i.e., the generator of geodesic deviation in $T\mathcal{M}$.

We explicitly evaluate $(\iota_N d)^{\ell-1} \iota_N F = (\iota_N D)^{\ell-1} \iota_N F$ for all integers $\ell \geq 1$. The result is

$$\begin{aligned} (\iota_N D)^{\ell-1} \iota_N F &= (P_\ell)_\mu dx^\mu + (\ell-1)(P_{\ell-1})_\mu Dy^\mu \\ &+ \sum_{p=2}^{[(\ell+1)/2]} \sum_{\alpha \in \Omega_p(\ell+1)} \mathcal{K}_\alpha \left(\begin{aligned} &(P_{\alpha_1-1} Q_{\alpha_2} \cdots Q_{\alpha_p})_\mu dx^\mu \\ &+ (\alpha_p-2)(P_{\alpha_1-1} Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_p-1})_\mu Dy^\mu \end{aligned} \right). \end{aligned} \quad (11.1)$$

This is a new calculation, not presented in our previous works [695, 769]. Yet, our notations are the same. Each $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_p) \in \Omega_p(\ell)$ is an ordered partition such that $\alpha_1 + \alpha_2 + \dots + \alpha_p = \ell$ and $\alpha_i \geq 2$. The coefficient $\mathcal{K}_\alpha$ for $\alpha \in \Omega_p(\ell)$ is a product of binomial coefficients:

$$\mathcal{K}_\alpha = \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right]. \quad (11.2)$$

For any $\alpha = (\alpha_1, \dots, \alpha_p)$ such that $\min\{\alpha_1, \dots, \alpha_p\} < 2$, we set $\mathcal{K}_\alpha = 0$. The P - and Q -tensors [695] are

$$(P_\ell)_\mu = F_{\rho_1 \mu; \rho_2; \dots; \rho_\ell}(x) y^{\rho_1} y^{\rho_2} \cdots y^{\rho_\ell}, \quad (11.3a)$$

$$(Q_\ell)^\mu{}_\sigma = R^\mu{}_{\rho_1 \rho_2 \sigma; \rho_3; \dots; \rho_\ell}(x) y^{\rho_1} y^{\rho_2} y^{\rho_3} \cdots y^{\rho_\ell}, \quad (11.3b)$$

while P_ℓ and Q_ℓ are set to zero for $\ell = 0$ and $\ell = 0, 1$, respectively.

$$\begin{array}{ccccc}
& & A & \xrightarrow{d\iota_N} & 0 \\
& & \downarrow \iota_N d & & \\
& & \iota_N F & \longrightarrow & 0 \\
& \swarrow \text{dual} & \downarrow & & \\
\iota_N * F & \longrightarrow & 0 & \quad \quad & \iota_N D \iota_N F \longrightarrow 0 \\
\downarrow & & \downarrow & & \downarrow \\
\iota_N D \iota_N * F & \longrightarrow & 0 & \quad \quad & \vdots \\
\downarrow & & \downarrow & & \\
\vdots & & \vdots & &
\end{array}$$

Figure 11.1: The “ $\mathcal{L}_N$ sequence” for Einstein-Maxwell geometry.

Eq. (11.1) is readily proven by mathematical induction, based on the recursive structure illustrated in Fig. 11.2. This applies the “organic chemistry” intuition of Ref. [695].

Amusingly, Eq. (11.1) can be quickly seen by a “covariant color-kinematics” [90] mapping, which views gravity as a gauge theory of the Lorentz group. Namely, Eq. (11.1) can be deduced from the formula for $(\iota_N D)^{\ell-1} \iota_N R^m_n$ given in Ref. [695, 769]. By viewing the antisymmetric pair of local Lorentz indices as the adjoint index of the Lorentz group as a gauge group, $(\iota_N D)^{\ell-1} \iota_N R^m_n$ corresponds to $(\iota_N D)^{\ell-1} \iota_N F^a$ where F^a is a nonabelian field strength. Taking the abelian limit then reproduces Eq. (11.1).

11.3 The Kerr-Newman Action

Second, we implement the probe counterpart of NJA, in the version of either Ref. [769] (twistorial, top-down) or Ref. [768] (relatively bottom-up), to derive the exact effective action of the Kerr-Newman black hole in the presence of external gravitational and electromagnetic fields.

The gravitational part has been already provided by Ref. [769]: the Kerr two-twistor particle. In this chapter, we simply need to add the electromagnetic interaction action. Let $A \in \Omega^1(\mathcal{M})$ be the electromagnetic gauge potential, and let $F = dA \in Z^2(\mathcal{M}) \subset \Omega^2(\mathcal{M})$ now denote the electromagnetic field strength.

The “ $\mathcal{L}_N$ sequence” for A is shown in Fig. 11.1. The recipe of Refs. [768, 769] applies Hodge duality ℓ times on the 2^ℓ -pole moment term in this sequence. This determines the electromagnetic interaction symplectic potential. Conse-

quently, the full symplectic potential of the Kerr-Newman particle is

$$\begin{aligned} \theta_{\text{KN}} = & \left(\cos(\mathcal{L}_N) + \text{sinc}(\mathcal{L}_N) J \iota_N d \right) (p_\mu dx^\mu) \\ & + \left(\cos(\mathcal{L}_N) + \text{sinc}(\mathcal{L}_N) \iota_N * d \right) (A_\mu(x) dx^\mu). \end{aligned} \quad (11.4)$$

For simplicity, we have set the electric charge q to a unit value; to reinstate it, rescale $A \mapsto qA$. We are now working in the massive correspondence space $\mathcal{K}$ defined in Ref. [769]. Due to limited space, we leave the details to Ref. [769]. Recall that $\mathcal{K}$ is equipped with the geometric data $N \in \Gamma(T\mathcal{K})$ and $J : T^*\mathcal{K} \rightarrow T^*\mathcal{K}$, where $J^2 = -\text{id}$.

Eq. (11.1) facilitates the explicit evaluation of Eq. (11.4):

$$\begin{aligned} \theta_{\text{KN}} = & \theta_{(0)} + A_\mu(x) dx^\mu \\ & + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} \sum_{p=1}^{[\ell/2]} \sum_{\alpha \in \Omega_p(\ell)} \mathcal{K}_\alpha p_\mu \left(\begin{aligned} & ((*^\ell Q_{\alpha_1}) Q_{\alpha_2} \cdots Q_{\alpha_p})^\mu{}_\sigma dx^\sigma \\ & + (\alpha_p - 2) ((*^\ell Q_{\alpha_1}) Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_{p-1}})^\mu{}_\sigma Dy^\sigma \end{aligned} \right) \\ & + \sum_{\ell=1}^{\infty} \frac{1}{\ell!} \left((*^\ell P_\ell)_\mu dx^\mu + (\ell - 1) (*^\ell P_{\ell-1})_\mu Dy^\mu \right) \\ & + \sum_{\ell=2}^{\infty} \frac{1}{\ell!} \sum_{p=2}^{[(\ell+1)/2]} \sum_{\alpha \in \Omega_p(\ell+1)} \mathcal{K}_\alpha \left(\begin{aligned} & ((*^\ell P_{\alpha_1-1}) Q_{\alpha_2} \cdots Q_{\alpha_p})_\mu dx^\mu \\ & + (\alpha_p - 2) ((*^\ell P_{\alpha_1-1}) Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_{p-1}})_\mu Dy^\mu \end{aligned} \right), \end{aligned} \quad (11.5)$$

Here, $*^\ell P_j$ and $*^\ell Q_j$ means to act on the Hodge star ℓ times on the electromagnetic field strength F and the Riemann tensor R inside P_j and Q_j .

Again, $\theta_{(0)}$ is defined in Ref. [769]: the covariantization of the free-theory symplectic potential of the two-twistor particle. For the reader's sake, we reproduce it below:

$$\theta_{(0)} = p_\mu dx^\mu + i y^{\dot{\alpha}\alpha} \left(\lambda_\alpha^I D \tilde{\lambda}_{I\dot{\alpha}} - D \lambda_\alpha^I \tilde{\lambda}_{I\dot{\alpha}} \right). \quad (11.6)$$

By incorporating the first-class constraints $\phi_0 = \frac{1}{2}(p^2 + m^2)$ and $\phi_1 = -p \cdot y$ via Lagrange multipliers, Eq. (11.5) concretely defines an all-orders first-order worldline action in which each term is fully and explicitly evaluated as a manifestly covariant worldline operator.

It should be clear that Eq. (11.5) nicely reduces to the Kerr and $\sqrt{\text{Kerr}}$ symplectic potentials in Ref. [769]. Also, it is straightforward that the nonabelian Kerr-Newman particle is obtained from Eq. (11.5) by simple replacements $A_\mu(x) dx^\mu \mapsto i \bar{\psi}_i D \psi^i$ and $(P_\ell)_\sigma \mapsto q_a (P_\ell)^a{}_\sigma$ (see the appendix of Ref. [769] for the relevant setup).

11.4 Nonlinear Newman-Janis Shift

Third, we impose SDity condition on both electromagnetic and Riemannian curvatures to reveal the dynamical, nonlinear NJ shift: $*F_{\mu\nu} = iF_{\mu\nu}$, $*R^\mu{}_{\nu\rho\sigma} = iR^\mu{}_{\nu\rho\sigma}$.

To this end, we simply replace $*^\ell$ with i^ℓ in Eq. (11.5). Recalling the explanations given in Ref. [769], it is seen that this replacement results in

$$\theta_{\text{KN}} \sim e^{i\mathcal{L}_N} \left(p_\mu dx^\mu + A_\mu(x) dx^\mu + \theta_{\text{G2A}} \right), \quad (11.7)$$

where $\sim$ signifies equivalence modulo total derivative. Specifically, we have discarded $i \text{sinc}(\mathcal{L}_N) d(A_\mu(x) y^\mu)$.

Eq. (11.7) is nothing but a minimally coupled charged scalar particle—i.e., the Reissner-Nordström particle—geodesically deviated into complex space-time, up to the “Gilbert-to-Ampère” replacement [769] implemented by θ_{G2A} which reflects a kinematical detail:

$$\theta_{\text{G2A}} = i y^{\dot{\alpha}\alpha} \left(\lambda_\alpha^I D \tilde{\lambda}_{I\dot{\alpha}} - D \lambda_\alpha^I \tilde{\lambda}_{I\dot{\alpha}} \right) - i p_\mu D y^\mu. \quad (11.8)$$

Clearly, this incarnates the spirit of the NJA [202] and its very historical application [179] to the Reissner-Nordström solution for the derivation of the Kerr-Newman solution.

The exponentiated Lie derivative $e^{i\mathcal{L}_N}$ in Eq. (11.7) describes the map γ defined in Ref. [769] that brings us to the “primed chart” $(z^{\mu'}, y^{\mu'}, \lambda_{\alpha'}^I)$ on the (complexified) massive correspondence space $\mathcal{K}$. With this understanding, Eq. (11.7) boils down to

$$\theta_{\text{KN}}^+ = p_{\mu'} dz^{\mu'} - 2i \tilde{\lambda}_{I\dot{\alpha}'} y^{\dot{\alpha}\alpha'} D \lambda_{\alpha'}^I + A_{\mu'}(z) dz^{\mu'}, \quad (11.9)$$

where we have discarded a total derivative just as in Ref. [769]. Here, $z^{\mu'} = \delta^{\mu'}_\mu (x^\mu + i y^\mu + \frac{1}{2} \Gamma^\mu{}_{\rho\sigma}(x) y^\rho y^\sigma + \mathcal{O}(y^3))$ are the holomorphic coordinates of curved spinspace-time.

Reviving the analysis in Ref. [768], one sees that the worldline perturbation theory due to Eq. (11.9) manifests the same-helicity (positive-helicity, in particular) spin exponentiation of the graviphotonic Compton amplitudes at all multiplicities and any massless species.

11.5 Heavenly Equations of Motion

The classical EoM due to the symplectic potential in Eq. (11.9) and the first-class constraints ϕ_0 and ϕ_1 are easy to derive by the technique of covariant

symplectic perturbations [764]. By fixing their Lagrange multipliers respectively as $\kappa^0(\tau) = 1$ and $\kappa^1(\tau) = 0$, we find

$$\begin{aligned} \dot{z}^{\mu'} &= p^{\mu'}, \quad \frac{Dy^{\mu'}}{d\tau} = F^{\mu'}_{\nu'}(z) y^{\nu'}, \\ \frac{D\lambda_{\alpha'}^I}{d\tau} &= 0, \quad \frac{D\tilde{\lambda}_{I\alpha'}}{d\tau} = \tilde{\lambda}_{I\beta'} \tilde{F}^{\dot{\beta}'}_{\dot{\alpha}'}(z). \end{aligned} \quad (11.10)$$

This shows that the motion of our Kerr-Newman particle describes a complex geodesic worldline $z^{\mu'}$ in spinspace-time on which the left-handed spin frame $\lambda_{\alpha'}^I$ is covariantly constant (parallel-transported), while the right-handed spin frame $\tilde{\lambda}_{I\alpha'}$ and the spin length pseudovector $y^{\mu'}$ are transported according to the Lorentz force law due to the dynamically NJ shifted field strength.

Based on the fact that the left-handed spinor bundle is flat for the moment, $D\lambda_{\alpha'}^I/d\tau = 0$ explicitly verifies the spinning, complexified analog of equivalence principle, which was proposed by Ref. [672] as the property characterizing spinning black holes. Namely, there is no ASD component in the spin precession, in SD backgrounds.

When re-covariantized at x^μ in spacetime (i.e., transcribed to the defining chart of $\mathcal{K}$), Eq. (11.10) yields an all-orders extension of the Bargmann-Michel-Telegdi (BMT) [819] and MPD [227–229] equations with all the multipole coefficients, both electromagnetic and gravitational, set to unity. This is indeed the desired behavior for the Kerr-Newman particle. Crucial point here is that the unity of the gravitational multipole coefficients is achieved because we have utilized geodesic deviation in generating the spin-curvature couplings in the action.

Similarly, it is also easily seen that the holomorphic coordinates $z^{\mu'}$ of the Kerr black hole's curved spinspace-time are Poisson-commutative in this SD limit:

$$\{z^{\mu'}, z^{\nu'}\} = 0. \quad (11.11)$$

11.6 Hidden Symmetry

Fourth, let us suppose the presence of a Killing vector $X \in \Gamma(T\mathcal{M})$ as well as a Killing-Yano tensor $Y \in \Omega^2(\mathcal{M})$. These shall satisfy

$$\mathcal{L}_X g = 0, \quad \mathcal{L}_X F = 0, \quad Y^\mu{}_\rho F^\rho{}_\nu = F^\mu{}_\rho Y^\rho{}_\nu, \quad (11.12)$$

and $Y_{\mu\nu;\rho} = Y_{[\mu\nu;\rho]}$. The last condition describes a commuting property, which holds for the Kerr-Newman background by the type-D property [588] and

also for its SD sector: the SD charged Taub-Newman-Unti-Tamburino (NUT) solution in which $F^\mu{}_\nu \propto Y^{+\mu}{}_\nu/|\vec{x}|^3$. See Ref. [164] and also Ref. [261]; recall Chap. V.

As is well-known [588], the Reissner-Nordström particle, governed by the EoM $\dot{x}^\mu = p^\mu$, $Dp^\mu/d\tau = F^\mu{}_\nu(x)p^\nu$, enjoys the conserved quantities

$$\mathbb{Q} = p_\mu X^\mu(x) + \alpha(x), \quad (11.13a)$$

$$\mathbb{C} = -p_\mu Y^\mu{}_\rho(x) Y^\rho{}_\nu(x) p^\nu. \quad (11.13b)$$

In Eq. (11.13a), $\alpha \in C^\infty(\mathcal{M})$ is a scalar field such that $d\alpha = -\iota_X F$ [588], whose local existence is guaranteed because $-d\iota_X F = \iota_X dF = 0$. Concretely, one can take $\alpha = \iota_X A$ if one is willing to make an explicit gauge choice such that $\mathcal{L}_X A = 0$. To derive Eq. (11.13b), one notes that the covariant precession of the vector $Y^\mu{}_\nu(x)p^\nu$ is the same as p^μ by virtue of the commuting property in Eq. (11.12).

Ref. [768] has established the bold proposal that exact (all-orders in spin and coupling) conserved charges of the spinning black hole probe directly arise by dynamically NJ shifting the conserved quantities of the non-spinning black hole probe, in the SD sector. By appreciating this idea, we identify the following exact conserved charges for our Kerr-Newman probe:

$$\mathbb{Q} = p_{\mu'} X^{\mu'}(z) + \alpha(z), \quad (11.14a)$$

$$\mathbb{R} = p_{\mu'} Y^{\mu'}{}_{\nu'}(z) y^{\nu'}, \quad (11.14b)$$

$$\mathbb{C} = -p_{\mu'} Y^{\mu'}{}_{\rho'}(z) Y^{\rho'}{}_{\nu'}(z) p^{\nu'}. \quad (11.14c)$$

In the explicit gauge choice for the electromagnetic gauge potential, one finds $\alpha(z) = A_{\mu'}(z) X^{\mu'}(z)$. It is easy to verify that $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ in Eq. (11.14) are all conserved by the EoM in Eq. (11.10). Note that $Dp^{\mu'}/d\tau = F^{\mu'}{}_{\nu'}(z)p^{\nu'}$. The covariant precession behaviors exhibited by $p^{\mu'}$, $y^{\mu'}$, and $Y^{\mu'}{}_{\nu'}(z)p^{\nu'}$ are all identical.

The conserved charges in Eq. (11.14) are sufficient to establish *integrability* of the Kerr-Newman probe dynamics in SD Einstein-Maxwell backgrounds admitting at least two Killing vectors and a Killing-Yano tensor. In Ref. [768], such integrability has been identified as the property that characterizes black holes among all possible massive spinning objects. While Ref. [768] has substantiated this proposal by the Kerr particle, this work reports the same conclusion for the Kerr-Newman particle.

In fact, one can establish *superintegrability* by incorporating the ASD Plebański two-forms [315], just as in Ref. [768]:

$$\mathbb{K}_a = p_{\mu'}(\Sigma_a)^{\mu'}_{\rho'}(z) Y^{\rho'}_{\nu'}(z) p^{\nu'}. \quad (11.15)$$

It should also be highlighted that the symmetry algebra formed by $\mathbb{Q}$, $\mathbb{C}$, and $\mathbb{K}_a$ is *identical* to the non-spinning case up to complexification, by virtue of the very important property that $\{z^{\mu'}, z^{\nu'}\} = 0$ (Eq. (11.11)) which is characteristic of black hole spacetime.

In short, we identify superintegrability in the dynamics of the Kerr-Newman black hole binary system at the zeroth self-force order, yet in the SD sector.

11.7 Curved Massive Twistor Space

Fifth, we identify the “mu variables” to return to the massive twistor space.

To this end, we trivialize the ASD spin connection coefficients, employ tetrad formalism, and expand the holomorphic coframe around a flat background as $e^{\dot{\alpha}\alpha} = dz^{\dot{\alpha}\alpha} + h^{\dot{\alpha}\alpha}$ which may utilize the second Plebański coordinates [314]. Then Eq. (11.9) becomes

$$\theta_{\text{KN}}^+ = -\tilde{\lambda}_{I\dot{\alpha}'} dz^{\dot{\alpha}\alpha} \lambda_{\alpha}^I - 2i \tilde{\lambda}_{I\dot{\alpha}'} y^{\dot{\alpha}\alpha} d\lambda_{\alpha}^I + \theta'_{\text{KN}}. \quad (11.16)$$

We have discarded a total derivative, while

$$\theta'_{\text{KN}} = \left(-\tilde{\lambda}_{I\dot{\alpha}'} h^{\dot{\alpha}\alpha}_{\mu'}(z) \lambda_{\alpha}^I + A_{\mu'}(z) \right) dz^{\mu'}. \quad (11.17)$$

As a result, the same identification for the mu variables as in Ref. [769] are made:

$$\mu^{\dot{\alpha}I} = z^{\dot{\alpha}\alpha} \lambda_{\alpha}^I, \quad \tilde{\mu}_I^{\alpha} = \tilde{\lambda}_{I\dot{\alpha}'} (z^{\dot{\alpha}\alpha} - 2iy^{\dot{\alpha}\alpha}). \quad (11.18)$$

Eventually, we find

$$\begin{aligned} \theta_{\text{KN}}^+ &= -\tilde{\lambda}_{I\dot{\alpha}'} d\mu^{\dot{\alpha}I} + \tilde{\mu}_I^{\alpha} d\lambda_{\alpha}^I + \theta'_{\text{KN}}, \\ \omega_{\text{KN}} &= d\mu^{\dot{\alpha}I} \wedge d\tilde{\lambda}_{I\dot{\alpha}'} + d\tilde{\mu}_I^{\alpha} \wedge d\lambda_{\alpha}^I + d\theta'_{\text{KN}}. \end{aligned} \quad (11.19)$$

Eq. (11.19) identifies the notion of curved massive twistor space in terms of a deformation of the symplectic structure on massive twistor space:

$$(\text{MT}, \omega^{\circ}) \rightarrow (\mathcal{MT}, \omega^{\circ} + d\theta'_{\text{KN}}). \quad (11.20)$$

This deformation is via a $(1,0)$ -perturbation θ'_{KN} at the level of symplectic potential such that the Poisson commutativity of holomorphic spinspace-time coordinates [672] is preserved as Eq.(11.11). Physically, this encodes that the gravitoelectromagnetic interaction action is supported on the NJ shifted worldline $z^{\mu'}$ for the SD (incoming positive-helicity [342, 343]) sector.

As remarked in Ref. [769], the above recipes are motivated by the practical use of our framework in perturbation theory, so other pathways toward the definition of curved massive twistor space might exist.

11.8 Googly Formulation

Lastly, our final task is to derive the chiral (“googly”) formulation of the Kerr-Newman action. This computes the generic, non-SD action described in Eq.(11.5) in terms of the complex, primed variables $(z^{\mu'}, y^{\mu'}, \lambda_{\alpha'}^I)$ by considering the map γ of Ref. [769].

To this end, we revive the gymnastics in Ref. [769]. Define

$$\theta_1^\pm = \frac{1}{2} \left(p_\mu D y^\mu \pm i y^{\dot{\alpha}\alpha} \left(\lambda_\alpha^I D \tilde{\lambda}_{I\dot{\alpha}} - D \lambda_\alpha^I \tilde{\lambda}_{I\dot{\alpha}} \right) \right) + \iota_N F^\pm, \quad (11.21)$$

where the $\pm$ superscripts describe the projections by $\frac{1}{2}(1 \mp i*)$. Then it follows that

$$\theta_{\text{KN}} = p_\mu dx^\mu + A_\mu(x) dx^\mu + \frac{e^{i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_1^+ + \frac{e^{-i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_1^-, \quad (11.22a)$$

$$e^{i\mathcal{L}_N} \left(p_\mu dx^\mu + A_\mu(x) dx^\mu \right) \sim p_\mu dx^\mu + A_\mu(x) dx^\mu + \frac{e^{i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_1^+ + \frac{e^{i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_1^-. \quad (11.22b)$$

The difference between Eqs. (11.22a) and (11.22b)—Kerr-Newman and “Reissner-Nordström + iy ”—derives

$$\begin{aligned} \theta_{\text{KN}} &\sim e^{i\mathcal{L}_N} \left(p_\mu dx^\mu + \frac{e^{-2i\mathcal{L}_N} - 1}{\mathcal{L}_N} \theta_1^- \right), \\ &= e^{i\mathcal{L}_N} \left(\begin{aligned} &\theta_{(0)} - i p_\mu D y^\mu \\ &+ \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} p_\mu \left((\iota_N D)^{\ell-2} \iota_N R^{-\mu}{}_\nu \right) y^\nu \\ &+ \sum_{\ell=1}^{\infty} \frac{(-2i)^\ell}{\ell!} (\iota_N d)^{\ell-1} \iota_N F^- \end{aligned} \right), \end{aligned} \quad (11.23)$$

which leads to

$$\begin{aligned}
\theta_{\text{KN}} \sim & p_{\mu'} dz^{\mu'} - 2i \tilde{\lambda}_{I\dot{\alpha}'} y^{\dot{\alpha}'\alpha'} D\lambda_{\alpha'}^I + A_{\mu'}(z) dz^{\mu'} \\
& + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} \sum_{p=1}^{\lfloor \ell/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell)} \mathcal{K}_\alpha p_{\mu'} \left((Q_{\alpha_1}^- Q_{\alpha_2} \cdots Q_{\alpha_p})^{\mu'}_{\sigma'} dz^{\sigma'} \right. \\
& \left. + (\alpha_p - 2) (Q_{\alpha_1}^- Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_{p-1}})^{\mu'}_{\sigma'} Dy^{\sigma'} \right) \\
& + \sum_{\ell=1}^{\infty} \frac{(-2i)^\ell}{\ell!} \left((P_\ell^-)_{\sigma'} dz^{\sigma'} + (\ell-1) (P_{\ell-1}^-)_{\sigma'} Dy^{\sigma'} \right) \\
& + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} \sum_{p=2}^{\lfloor (\ell+1)/2 \rfloor} \sum_{\alpha \in \Omega_p(\ell+1)} \mathcal{K}_\alpha \left((P_{\alpha_1-1}^- Q_{\alpha_2} \cdots Q_{\alpha_p})_{\mu'} dz^{\mu'} \right. \\
& \left. + (\alpha_p - 2) (P_{\alpha_1-1}^- Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_{p-1}})_{\mu'} Dy^{\mu'} \right).
\end{aligned} \tag{11.24}$$

Again, we find the structure that the first entry in each string of P or Q tensors develops the ASD projection. Eq. (11.24) reproduces Eq. (11.9) in the SD limit. As a friendly reminder, our index notation [695] implies

$$(P_\ell)_{\mu'} = F_{\rho'_1 \mu'; \rho'_2; \dots; \rho'_\ell}(z) y^{\rho'_1} y^{\rho'_2} \cdots y^{\rho'_\ell}, \tag{11.25a}$$

$$(Q_\ell)^{\mu'}_{\sigma'} = R^{\mu}_{\rho'_1 \rho'_2 \sigma'; \rho'_3; \dots; \rho'_\ell}(z) y^{\rho'_1} y^{\rho'_2} y^{\rho'_3} \cdots y^{\rho'_\ell}. \tag{11.25b}$$

Eq. (11.24) provides the googly formulation of the Kerr-Newman action. To reiterate, it describes the completely general case of non-SD gravitoelectromagnetic field configurations but formulates the SD and ASD modes on an unequal footing. It provides a worldline action localized on the holomorphic worldline $z^{\mu'}$, the perturbation theory of which manifests the same-helicity spin exponentiation for incoming positive-helicity photons and gravitons. The photon coupling is always linear. Graviphoton contact vertices arise iff the incoming photon carries negative helicity. When examined in the spin length expansion, the first graviphoton contact vertex appears at $\mathcal{O}(y^3)$.

11.9 Interpretation

We end with quick remarks on the physical interpretation by recalling the comments given in Ref. [769]. Eq. (11.5) represents two charged masses joined by a Dirac-Misner string (line defect of magnetic and NUT fluxes). Eq. (11.24) represents a dangling ASD Dirac-Misner string attached to a charged mass. The shape of the Dirac-Misner string happens to be geodesic for each slice of simultaneity defined by the worldline proper time. For the Dirac string part, this is completely justified based on its well-established status as a topological surface operator.

By virtue of the transparent decoding of the NJA revealed by Ref. [164], we readily realize that this string structure is the precise IR counterpart of the actual Dirac-Misner string that exists in the UV description of the Kerr-Newman solution.¹ Namely, it is shown in Ref. [164] that the Kerr-Newman solution as a holomorphic saddle in Einstein-Maxwell theory represents a pair of SD and ASD charged Taub-NUT instantons. In conclusion, the holomorphic worldline $z^{\mu'}$ is the IR avatar of the worldline of the ASD charged Taub-NUT instanton that constitutes the Kerr-Newman black hole.

The googly action in Eq. (11.24) perturbs around the SD sector—the sector in which we, at least, have a full understanding on the dynamics in terms of remarkable properties such as the superintegrability due to Eqs. (11.14) and (11.15) or the same-helicity spin exponentiation. The googly agenda means to approach the complicated dynamics of the Kerr-Newman probe from its exactly solvable part, which should sound reasonable.

Note that Ref. [226] had speculated on a worldsheet structure, although no clear physical interpretation nor explicit formulae such as Eqs. (11.5) and (11.24) were provided.

¹Here, IR refers to the point-particle effective theory, while UV refers to GR.

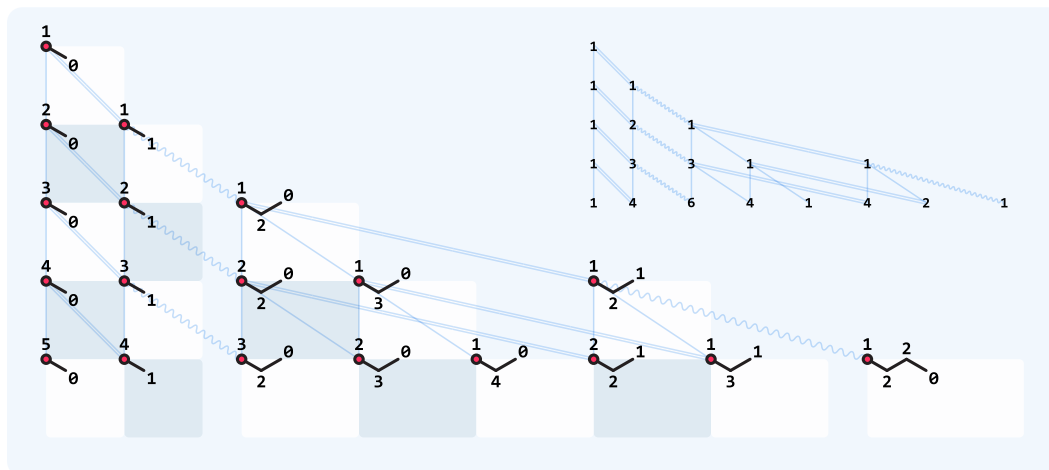


Figure 11.2: The “organic chemistry” of Einstein-Maxwell geometry. Molecules at each row follow from those on the former row by the action of $\iota_N D$. Single lines represent the reaction $(P_{\alpha_1} Q_{\alpha_2} \cdots Q_{\alpha_p}) dx \mapsto \sum_{\beta_1 + \cdots + \beta_p = 1} (P_{\alpha_1 + \beta_1} Q_{\alpha_2 + \beta_2} \cdots Q_{\alpha_p + \beta_p}) dx$ with $\beta_1, \dots, \beta_p \geq 0$. Double lines represent the reaction $(PQ \cdots Q) dx \mapsto (PQ \cdots Q) Dy$. Squiggly lines represent the “polymerization” reaction $(PQ \cdots Q) Dy \mapsto (PQ \cdots Q)(Q_2 dx)$. See Fig. 11.3 and also Chap. VIII. The smaller diagram in the upper right corner displays the coefficient of each molecule.

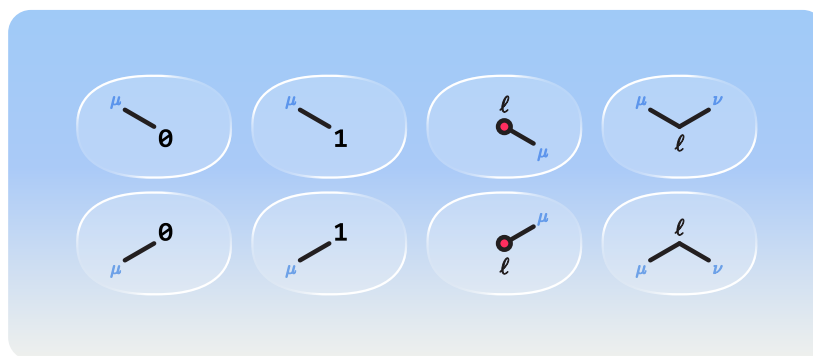


Figure 11.3: The graphical notation used in Fig. 11.2. Elements in the first row denote dx^μ , Dy^μ , $(P_\ell)_\mu$, and $(Q_\ell)^{\mu\nu}$, respectively. Elements in the second row denote dx_μ , Dy_μ , $(P_\ell)^\mu$, and $(Q_\ell)_{\mu\nu}$, respectively.

SPINSPACETIME

We show that Poincaré invariance directly implies the existence of a complexified Minkowski space whose real and imaginary directions unify spacetime and spin, which we dub spinspace. Despite the intrinsic noncommutativity of spin, spinspace exhibits mutually commuting holomorphic coordinates.

Its twistorial construction can derive the NJ shift property of spinning black holes by massive half-Fourier transforming complexified on-shell kinematics, which encode a spinning analog of equivalence principle.

This section reproduces the contents of Ref. [672], J.-H. Kim, “Asymptotic spinspace,” *Phys. Rev. D* **111** no. 10, (2025) 105011, [arXiv:2309.11886 \[hep-th\]](#).

12.1 Introduction

In the 70s, Ezra Ted Newman investigated the curious idea of unifying spacetime and spin into a complex geometry [252–259]. While offering a unique perspective on relativistic angular momenta, it provided a suggestive argument for the NJ [179, 202] derivation of spinning black hole solutions where spin turns into an imaginary deviation. However, it remains uncertain if the NJ shift really follows solely from this complex geometry, since Newman’s argument implicitly counts on an assumption that the black hole would be localized in the complex for an unknown reason. On a related note, it has also remained unclear whether this concept is necessarily limited to black holes or not. Unfortunately, this line of ideas appears to be no longer actively revived in the current literature.

In this chapter, we aim to give a dedicated spotlight on this “forgotten lore” and unveil its relevance to the modern program of describing spinning black holes in terms of scattering amplitudes [98, 206–212, 214, 218, 219, 226, 673, 795]. As the very first step towards this modern resurrection, we endow it with a name: “spinspace.” We provide a complete definition of spinspace that unifies various perspectives [252–259, 330, 807, 810, 820–830]. Especially, we highlight its characteristic commutator structure, which, to our knowledge, was not investigated in Newman’s works [252–259]. Consequently,

we refine Newman’s ideas to precise statements. In particular, we envision a new derivation of the NJ shift from scattering amplitudes, which may trigger an extension of the on-shell/twistor diagrams [831–837] to massive states.

Our central findings are twofold. Firstly, we show that spinspace *universally* arises in any massive system exhibiting global Poincaré symmetry: particles and fields in asymptotically flat spacetimes, or even systems that may lack a spacetime formulation. We construct real variables x^μ, y^μ directly from the Poincaré charges. Then Poincaré symmetry dictates their commutators to be

$$[z, z] = 0, \quad [z, \bar{z}] \neq 0, \quad [\bar{z}, \bar{z}] = 0, \quad (12.1)$$

if $z^\mu = x^\mu + iy^\mu$ and $\bar{z}^\mu = x^\mu - iy^\mu$. These complex coordinates unify spacetime and spin as real and imaginary parts, which can be seen from the following split of the total angular momentum into orbital and spin parts:

$$J = (x \wedge p) + *(y \wedge p) \implies J^+ = (z \wedge p)^+. \quad (12.2)$$

Here, an intrinsic association between chirality and holomorphy is implied, as the SD part of the total angular momentum, J^+ , arises solely from the holomorphic coordinates z as $* \rightarrow +i$. Therefore, a complexified Minkowski space exhibiting a fascinating geometry arises, whose complex coordinates $z^\mu, \bar{z}^\mu$ (1) realize a particular “off-diagonal” form of commutators as in Eq. (12.1) and (2) are inherently linked with chirality as in Eq. (12.2). This provides the complete characterization of spinspace.

Secondly, we specialize this framework in the scattering problem of massive spinning states from asymptotically free regions. To this end, we adopt a twistorial implementation of spinspace, in which case the off-diagonal form of Eq. (12.1) originates from the oscillator commutator algebra (Kähler geometry) [378, 814, 838] of twistor space:

$$[Z, Z] = 0, \quad [Z, \bar{Z}] \neq 0, \quad [\bar{Z}, \bar{Z}] = 0. \quad (12.3)$$

In this description, the NJ shift (as localizations of scattering amplitudes at complex positions [98, 206–212, 214]) is directly derived from complexified on-shell kinematics in Fig. 12.2 via *massive* half-Fourier transforms. As they provide a massive counterpart of the maximally helicity violating (MHV) kinematics [836] and describe a spinning analog of equivalence principle, a novel perspective is provided for the minimality [98, 206–209] of spinning black holes.

In conclusion, spinspace-time is a universal framework for any massive Poincaré-invariant system, in which the minimality of spinning black holes is strikingly manifest through its distinctive complex-geometrical structures. This refined understanding fully resolves the questions left by Newman's works and allows us to extend and generalize its curious story with exciting future applications.

12.2 Reconstructing Spacetime from Poincaré Symmetry

In any system enjoying global Poincaré symmetry, there shall exist Poincaré charges:

$$\begin{aligned} [J^{\mu\nu}, J^{\rho\sigma}] &= i(-4\delta^{[\mu}_{[\kappa}\eta^{\nu]\rho}\delta^{\sigma]}_{\lambda]})J^{\kappa\lambda}, \\ [J^{\mu\nu}, p_\rho] &= ip_\sigma(-2\eta^{\sigma[\mu}\delta^{\nu]}_{\rho}), \quad [p_\mu, p_\nu] = 0. \end{aligned} \quad (12.4)$$

Additionally, the notion of mass dimension is accompanied in any physical situation. This amounts to having an operator D such that

$$[J^{\mu\nu}, D] = 0, \quad [p_\mu, D] = +ip_\mu, \quad (12.5)$$

assigning dimensions 0 and +1 to J and p , respectively.

Suppose the system is massive so that its configurations do not realize $p^2 = 0$. Then, define

$$x^\mu = J^{\mu\nu}p_\nu/p^2 - Dp^\mu/p^2, \quad (12.6)$$

as a function of J , p , and D . By straightforward calculations, it follows that x^μ exactly transforms like a position vector in Minkowski space under the Poincaré group action: $[x^\mu, J^{\rho\sigma}] = i(-2\eta^{\mu[\rho}\delta^{\sigma]}_{\nu})x^\nu$, $[x^\mu, p_\rho] = i\delta^\mu_{\rho}$. In this manner, a Minkowski space emerges from Poincaré symmetry.

It should be clear that this construction applies to *any* massive system with global Poincaré symmetry: a particle, a group of particles, a field, a group of fields, or even a system not prescribed in a spacetime formulation. The prime example is a massive system in an asymptotically flat spacetime, in which case our argument asserts that a *flat bulk* emerges from the Poincaré charges at asymptotic infinity.

A few remarks are in order. First, suppose we have only assumed the strict Poincaré algebra in Eq. (12.4). Then $J^{\mu\nu}p_\nu/p^2$ behaves almost, but not perfectly, like a position vector. Amusingly, taking Eq. (12.6) then as an ansatz, imposing the desired transformation behavior of x^μ mandates the commutation relations in Eq. (12.5), so the same construction is reproduced. Next, at

the technical level, the above calculations will closely parallel those carried out in Refs. [839, 840]. However, Refs. [839, 840] specifically restricted their scope to a free particle, whereas our very point here is the universality of this construction.

12.3 Spin-induced Spacetime Fuzziness

Next, we observe a puzzling feature in this notion of an emergent flat spacetime: it is noncommutative as

$$[x^\mu, x^\nu] = -\frac{i}{p^2} S^{\mu\nu}, \quad (12.7)$$

which directly follows from Eqs. (12.4) and (12.6). Here $S^{\mu\nu}$ denotes the transverse projection of $J^{\mu\nu}$ such that $p_\mu S^{\mu\nu} = 0$. Namely, it is the spin angular momentum which by definition is the part of the Lorentz charge $J^{\mu\nu}$ unexplainable by the orbital angular momentum computed with spacetime coordinates: $J^{\mu\nu} = 2x^{[\mu}p^{\nu]} + S^{\mu\nu}$.

This peculiarity is inevitable in the sense that Eq. (12.6) is the only Poincaré-covariant and real spacetime coordinates that can be directly constructed from $J^{\mu\nu}$, p_μ , and D , as implied by the analysis provided in the supplemental material. Indeed, the noncommutativity in Eq. (12.7) has been observed in a considerable amount of work [793, 807, 810, 821–830, 839, 841–850] from various angles and from numerous spinning particle models.

12.4 Unification of Spacetime and Spin

In the co-moving frame of the momentum, Eq. (12.7) boils down to $[x^i, x^j] = (i\hbar/m^2c^2)\varepsilon^{ij}_k S^k$, which implies a noncommutativity scale of $\Delta x \sim s^{1/2}(\hbar/mc)$ for spin s . Here, we have temporarily restored the fundamental constants. Meanwhile, recall the famous commutator, $[S^i, S^j] = i\hbar\varepsilon^{ij}_k S^k$. Amusingly, these right-hand sides can be canceled by composing complex combinations $x^i \mp iS^i/mc$, precisely because the imaginary unit squares to -1 .

Let us implement this solution to the noncommutativity puzzle in a fully Poincaré-covariant way. Define the spin length pseudovector, which describes the Pauli-Lubanski pseudovector normalized in the units of length:

$$y^\mu = -*J^{\mu\nu}p_\nu/p^2 \implies S^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma}y_\rho p_\sigma. \quad (12.8)$$

Then Eq. (12.4) implies that y^μ transforms like a tangent vector: $[y^\mu, J^{\rho\sigma}] = i(-2\eta^{\mu[\rho}\delta^{\sigma]}_\nu)y^\nu$, $[y^\mu, p_\rho] = 0$. In turn, for $z^\mu = x^\mu + iy^\mu$, Eqs. (12.4) and (12.5)

imply

$$\begin{aligned} [z^\mu, z^\nu] &= [\bar{z}^\mu, \bar{z}^\nu] = 0, \\ [z^\mu, \bar{z}^\nu] &= -\frac{2}{p^2} \left(2y^{(\mu} p^{\nu)} + i \varepsilon^{\mu\nu\rho\sigma} y_\rho p_\sigma \right). \end{aligned} \quad (12.9)$$

Remarkably, z^μ themselves are *commutative*.

In conclusion, a complexified Minkowski space [611],¹ equipped with the commutators in Eq. (12.9) featuring commutative holomorphic coordinates, can be constructed in any massive system with global Poincaré symmetry. This complexified Minkowski space is the *spinspace*.

One might wonder if the use of the imaginary unit, seemingly a bit radical, is really necessary. However, as we have shown in a bootstrap analysis in the supplemental material, this is *the only Poincaré-covariant solution* to the noncommutativity puzzle. This uniqueness seems to have not been realized explicitly, although proposals for the complex-valued coordinates have existed [820, 821].

Thus, it appears that Poincaré invariance intrinsically entails a curious complex geometry, which we then must pay attention to. Especially, it is easily checked that the Poincaré group action preserves the complex structure: it does not mix up holomorphic and anti-holomorphic variables. Hence there is a possibility for this complex structure, or holomorphy, of spinspace acquiring a physical significance as a frame-independent notion.

12.5 Newman's Derivation of Spinspace

Indeed, holomorphy in spinspace is intrinsically associated with *chirality*.² This point is emphasized in Newman's original derivation of spinspace [252], sketched in Eq. (12.2). To recapitulate, Newman derives the complex coordinates by observing that the SD and ASD parts of the angular momentum are $J^\pm = ((x \pm iy) \wedge p)^\pm$, where $(\alpha \wedge \beta)^\pm$ denotes the SD/ASD projection of the bivector $(\alpha \wedge \beta)^{\mu\nu} = \alpha^\mu \beta^\nu - \alpha^\nu \beta^\mu$. Crucially, the SD part J^+ depends solely on the holomorphic position, while the ASD part J^- depends solely on the *anti*-holomorphic position. In this light, the imaginary unit in $x \pm iy$ has

¹Mathematically, this is the space $\mathbb{C}^4$ equipped with the flat metric $\eta^{\mathbb{C}}$ whose signature reduces to Lorentzian on the real section $z^\mu = \bar{z}^\mu$ [350]. One may regard it as the tangent bundle of Minkowski space equipped with the adapted complex structure [743, 776–779].

²The easiest way of seeing this is to realize that y^μ is a pseudovector: parity flip induces complex conjugation, $z^\mu \leftrightarrow \bar{z}^\mu$.

originated from a Hodge (electric-magnetic [830]) duality between orbital and spin angular momenta, $(x \wedge p)$ and $*(y \wedge p)$.

Newman [252]’s approach to spinspace-time is rather independent from the approach in the previous sections where the focus is put on the commutator structure [820, 821]. However, we emphasize that both perspectives capture essential defining features of spinspace-time: the geometry of spinspace-time is completely characterized only if *both* (1) the commutator structure in Eq. (12.1) (more precisely Eq. (12.9)) and (2) the association between holomorphy and chirality tracing back to Eq. (12.2) are understood.

These two unique geometrical features lead to remarkable physical implications. In fact, they can be demonstrated nicely in a twistorial implementation of spinspace-time, which we now elaborate on.

12.6 Spinspace-time from Massive Twistor Space

The astute reader will realize that the development so far strongly resonates with the very spirit of twistor theory. First of all, spacetime is regarded as a secondary construct [378, 380, 814]. And not to mention, a complex-geometrical structure is disclosed. Further, the $[z^\mu, \bar{z}^\nu]$ bracket takes a simple form in the spinor notation, as will be revealed shortly in Eq. (12.12). In addition, $D = -p_\mu x^\mu$ in Eq. (12.5) and (12.6) is the dilatation charge. These all point to an intimate connection to twistor theory where complex geometry, spinors, and conformal algebra play central roles.

A definition of twistor space is the vector space $\mathbb{C}^4$ equipped with a $(2, 2)$ -signature Hermitian form, serving as the linear representation space of $U(2, 2)$ [378, 814]. Geometrically, a twistor represents a null ray in complexified Minkowski space [378, 611, 814]. To implement the mass, one considers two copies of twistor space, $\mathbb{C}^8 \cong \mathbb{C}^4 \times \mathbb{C}^4$ [377, 796, 802], where $U(2, 2)$ acts from the left and $U(2)$ acts from the right with a shared $U(1)$. In essence, this realizes a massive momentum as a sum of two null momenta. This space $\mathbb{C}^8$ is the *massive twistor space*. According to a modern view developed in Refs. [673, 794], which differs a bit from the original interpretation taken in the 70’s twistor particle program [377, 796], the right $SU(2)$ is the massive little group while the shared $U(1)$ is gauge.

Having defined the massive twistor space, the next step is to construct its spinspace-time coordinates explicitly. Let $Z_A^I = (\lambda_\alpha^I, i\mu^{\dot{\alpha}I})$ and $\bar{Z}_I^A =$

$(-i\bar{\mu}_I{}^\alpha, \bar{\lambda}_{I\dot{\alpha}})$ be holomorphic and anti-holomorphic coordinates of the massive twistor space. Here $A, B, \dots$ are $SU(2, 2)$ (Dirac spinor) indices, while $I, J, \dots$ are $SU(2)$ indices. This space is a Kähler vector space [378, 814, 838], so the commutation relations are given by the oscillator algebra as sketched in Eq. (12.3): $[Z_A{}^I, \bar{Z}_J{}^B] = \delta_A{}^B \delta_J{}^I$, to be explicit.

The $U(2, 2)$ generators are given by $G_A{}^B = Z_A{}^I \bar{Z}_I{}^A$, the Weyl block decomposition of which gives

$$p_{\alpha\dot{\alpha}} = -\lambda_\alpha{}^I \bar{\lambda}_{I\dot{\alpha}}, \quad G^{\dot{\alpha}}{}_{\dot{\beta}} = \mu^{\dot{\alpha}I} \bar{\lambda}_{I\dot{\beta}}. \quad (12.10)$$

The first equation is the famous “massive spinor-helicity” decomposition of the massive momentum [206, 377, 796, 851, 852]. The second equation unifies dilatation and SD angular momentum: $G^{\dot{\alpha}}{}_{\dot{\beta}} = J^{\dot{\alpha}}{}_{\dot{\beta}} + \frac{1}{2} \delta^{\dot{\alpha}}{}_{\dot{\beta}} (D + i\tilde{D})$. A new ingredient is $\tilde{D}$, which generates the $U(1)$ gauge group. It has zero commutator with any other generators and induces a central extension from $SU(2, 2)$ to $U(2, 2)$.

With this understanding, we translate our earlier formula for spinspace coordinates into the spinor notation. Notably, it takes a strikingly simple form:

$$z^{\dot{\alpha}\alpha} = -G^{\dot{\alpha}}{}_{\dot{\beta}} (p^{-1})^{\dot{\beta}\alpha}. \quad (12.11)$$

Here, $(p^{-1})^{\dot{\alpha}\alpha} = \bar{\epsilon}^{\dot{\alpha}\dot{\beta}} \epsilon^{\alpha\beta} p_{\beta\dot{\beta}} / \det(p)$ denotes the inverse of $p_{\alpha\dot{\alpha}}$ as a square matrix, defined for $p^2 \neq 0$. In turn, the conformal algebra directly implies that

$$[z^{\dot{\alpha}\alpha}, z^{\dot{\beta}\beta}] = 0, \quad [z^{\dot{\alpha}\alpha}, \bar{z}^{\dot{\beta}\beta}] = i(z - \bar{z})^{\dot{\alpha}\beta} (p^{-1})^{\dot{\beta}\alpha}, \quad (12.12)$$

reproducing Eq. (12.9) up to the extension $y^\mu \mapsto y^\mu - \tilde{D} p^\mu / p^2$.³ Finally, applying Eq. (12.11) to the charges in Eq. (12.10), the spinspace coordinates for the massive twistor space are found as

$$z^{\dot{\alpha}\alpha} = \mu^{\dot{\alpha}I} (\lambda^{-1})_I{}^\alpha \implies \mu^{\dot{\alpha}I} = z^{\dot{\alpha}\alpha} \lambda_\alpha{}^I. \quad (12.13)$$

Interestingly, Eq. (12.13) reinvents the incidence relation of twistor theory [252, 611, 814] for the massive case. Its interpretation is that the two complexified null rays represented by the twistors $Z_A{}^{I=0,1}$ are “co-incident” at a point $z^{\dot{\alpha}\alpha}$ in complexified Minkowski space.

³Eqs. (12.11) and (12.12) apply generally to any Hamiltonian system in asymptotically flat spacetime, where one can find conformal generators from the conformal killing vectors (even though the Hamiltonian may not enjoy the full conformal symmetry).

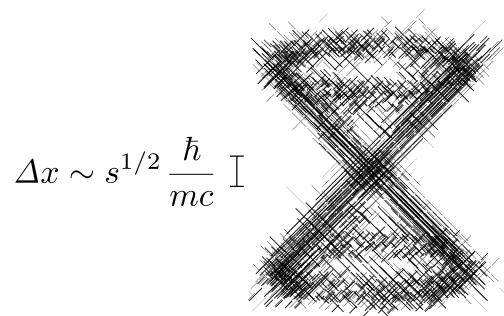


Figure 12.1: Lightcone in twistor theory according to Penrose [378, 462]. Penrose advocates a point of view that there is an absurdity inherent in the notion of spacetime already at the Compton scale of elementary particles [377, 379, 380]. Accordingly, the very philosophy of twistor theory is to take spacetime points as secondary constructs emerging from intersections between null rays.

In conclusion, the spinspace of massive twistor space is simply the complexified Minkowski space which its incidence relation in Eq. (12.13) defines.⁴ Intriguingly, this implies that the noncommutativity in Eq. (12.7) could be reinterpreted as the fuzziness which spacetime points acquires in twistor theory as light rays become fundamental objects [378, 380, 807, 810, 814, 822–829]: see Fig. 12.1.

The twistorial picture associates fascinating geometric origins to the defining features of spinspace. First, the off-diagonal commutators (Eq. (12.1)) have signaled the Kähler geometry of twistor space (Eq. (12.3)). Second, holomorphy in spinspace corresponds to the holomorphy of twistors. Third, the link between holomorphy and chirality in spinspace (Eq. (12.2)) traces back to that of twistor space. Note that this link is deeply endemic in twistor space: the *holomorphic* Z_A^I contains the *left-handed* spinor-helicity variable λ_α^I .

12.7 S-Matrix in Massive Twistor Space

Eventually, we study scattering in asymptotically flat spacetimes. This setup concerns the spinspace which Poincaré charges in asymptotically free regions reconstruct, together with an asymptotic massive twistor space as a *phase space*.

As well-known, a massive on-shell scattering state is represented by the massive spinor-helicity variables $\lambda_\alpha^I, \tilde{\lambda}_{I\dot{\alpha}}$ [206]. The massive twistor space can be

⁴Note that a double fibration picture could be pursued to formulate this relationship in a more mathematically precise sense, analogously to the massless case.

viewed as the cotangent bundle of the space $\mathbb{C}^4$ of massive spinor-helicity variables, in light of the commutators $[\lambda_\alpha^I, \bar{\mu}_J^\beta] = i\delta_\alpha^\beta \delta_J^I$, $[\tilde{\lambda}_{I\dot{\alpha}}, \mu^{\dot{\beta}J}] = i\delta_I^J \delta_{\dot{\alpha}}^{\dot{\beta}}$.

The massive twistor space admits various polarizations, offering different bases for the scattering states. For instance, the scattering states can be realized in the coherent state basis (Kähler polarization),

$$Z|Z_1\rangle = |Z_1\rangle Z_1, \quad \langle \bar{Z}_2|\bar{Z} = \bar{Z}_2\langle \bar{Z}_2|, \quad (12.14)$$

or in the spinor-helicity basis,

$$\lambda|\lambda_1\tilde{\lambda}_1\rangle = |\lambda_1\tilde{\lambda}_1\rangle\lambda_1, \quad \tilde{\lambda}|\lambda_1\tilde{\lambda}_1\rangle = |\lambda_1\tilde{\lambda}_1\rangle\tilde{\lambda}_1, \quad (12.15)$$

where we have started to omit indices to avoid clutter. The overlaps are given by $\langle \bar{Z}_2|Z_1\rangle = e^{\bar{Z}_2 Z_1}$ and $\langle \lambda_2\tilde{\lambda}_2|\lambda_1\tilde{\lambda}_1\rangle = \delta^{(4)}(\lambda_1 - \lambda_2)\delta^{(4)}(\tilde{\lambda}_1 - \tilde{\lambda}_2)$, where $\bar{Z}_2 Z_1$ abbreviates $(\bar{Z}_2)_I{}^A (Z_1)_A{}^I$. The transformations between these two bases are given by the half-Fourier transforms, which one may define concretely by analytically continuing to the $(2, 2)$ -signature [349, 835]:

$$\begin{aligned} |Z_1\rangle &= |\lambda_1\mu_1\rangle = \int [d^4\tilde{\lambda}] e^{i\tilde{\lambda}\mu_1} |\lambda_1\tilde{\lambda}\rangle, \\ \langle \bar{Z}_2| &= \langle \tilde{\lambda}_2\bar{\mu}_2| = \int [d^4\lambda] e^{-i\bar{\mu}_2\lambda} \langle \lambda\tilde{\lambda}_2|. \end{aligned} \quad (12.16)$$

In turn, S-matrix elements in the spinor-helicity basis can be converted to the massive twistor basis: $\langle \lambda_2\tilde{\lambda}_2|S|\lambda_1\tilde{\lambda}_1\rangle \rightarrow \langle \bar{Z}_2|S|Z_1\rangle$. To our best knowledge, such half-Fourier transforms have been actively pursued only for massless legs [349, 612, 835].

12.8 What Defines Spinning Black Holes

Now we ask the following question: What is the simplest S-matrix in massive twistor space? For the case of massive-massive-massless three-particle scattering with equal masses, the on-shell kinematics exhibit two *extremal* cases that freeze either of the left-handed or right-handed massive spinor-helicity variables. Let us call these *zig* and *zag*⁵ kinematics and denote them respectively as white and black blobs as shown in Fig. 12.2, analogously to the massless on-shell diagram notation in Refs. [831–837]. Then momentum conservation fully refines their delta function supports as

$$\begin{aligned} \text{zig} : \quad & \delta^{(4)}(\lambda_1 - \lambda_2) \delta^{(4)}(\tilde{\lambda}_1 - \lambda_1^{-1}k_3 - \tilde{\lambda}_2), \\ \text{zag} : \quad & \delta^{(4)}(\lambda_1 - k_3\tilde{\lambda}_2^{-1} - \lambda_2) \delta^{(4)}(\tilde{\lambda}_1 - \tilde{\lambda}_2), \end{aligned} \quad (12.17)$$

⁵This terminology is inspired by Penrose [462], Chapter 25.2: The zigzag picture of the electron, pp. 628–633.

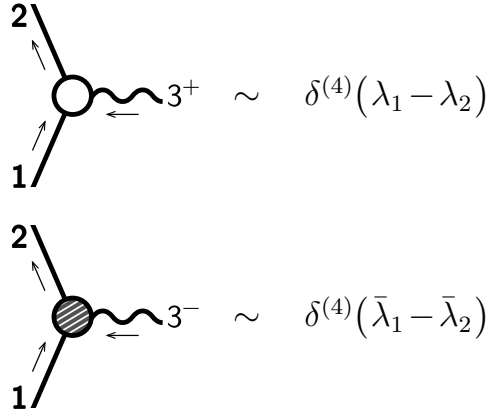


Figure 12.2: The building blocks of “massive on-shell diagrams,” describing extremal cases of complexified on-shell kinematics.

where $k_{3\alpha\dot{\alpha}}$ is the massless momentum of the third leg. We have abbreviated contracted indices as $(\lambda_1^{-1}k_3)_{I\dot{\alpha}} = (\lambda_1^{-1})_I{}^\alpha k_{3\alpha\dot{\alpha}}$. These are *complexified* kinematics ⁶, so $\tilde{\lambda}_{1,2}$ here would be more appropriately denoted as $\tilde{\lambda}_{1,2}$.

Furthermore, we stipulate a link between chirality and holomorphy: the *zig* and *zag* kinematics are realized for *positive* and *negative* helicities of the massless leg, respectively. With this postulate, the zig and zag kinematics embody the scattering amplitudes statement of an analog of equivalence principle in half-flat backgrounds, which we dub *spinning equivalence principle*.

Suppose a spinning object traveling in a SD background spacetime. What would be its simplest motion? By definition, the left-handed spinor bundle is flat, which one can locally trivialize. If the gravitational forces are to arise uniquely from this geometry, the object’s left-handed Lorentz frame would remain constant in local laboratories. This deduces the spinning equivalence principle: the left-handed (right-handed) frame of a spinning object is covariantly constant under dynamical evolution in SD (ASD) backgrounds.

Of course, in a modern understanding, the equivalence principle is rather a statement about the minimality of a test object, which can be broken by non-minimal couplings. For example, a scalar particle endowed with nonzero non-minimal Wilson coefficients will not follow the geodesic trajectory as desired by the equivalence principle. Similarly, the spinning equivalence principle rather

⁶Cf. Penrose’s argument in Refs. [312, 313] that the notion of a “single-graviton spacetime” is inherently complex.

proposes the most ideal behavior which a spinning object may exhibit.

Remarkably, we find that such an ideal behavior is exhibited by *black holes*. To see this, consider the massive half-Fourier transform of the zig kinematics to the twistor *coherent state* basis ⁷:

$$\begin{aligned} & \exp\left(-i\bar{\mu}_2\lambda_1\right) \exp\left(i(\lambda_1^{-1}k_3 + \tilde{\lambda}_2)\mu_1\right) \\ &= \langle \bar{Z}_2 | Z_1 \rangle \exp\left(i k_3 \mu_1 \lambda_1^{-1}\right). \end{aligned} \quad (12.18)$$

By the massive incidence relation in Eq. (12.13), the exponential factor computes to $e^{ik_3 z_1} = e^{ik_3 x_1} e^{-k_3 y_1}$, where the factor $e^{-k_3 y_1}$ ⁸ for receiving positive-helicity force carriers is known to implement the NJ shift property at the amplitudes level [98, 207–209]. Here, it is crucial that the “imaginary impact parameter” y_1^μ describes spin due to the very (re)interpretation of complex coordinates in twistor theory as spinspace-time coordinates, as is deduced in Eq. (12.13). Similarly, half-Fourier transforming the zag kinematics yields the factor $e^{ik_3 \bar{z}_1} = e^{ik_3 x_1} e^{k_3 y_1}$, encoding the NJ shift for receiving negative-helicity force carriers.

In sum, spinning black holes are characterized by the extremal complexified on-shell kinematics in Fig. 12.2, which encode the spinning equivalence principle. Notably, we could have also assumed any multiplicity of massless quanta, building up arbitrary half-flat backgrounds. This derives the same-helicity spin exponentiation [210–212] at all multiplicities via the half-Fourier transform. Eventually, this case would arise from a gluing of the same-helicity on-shell diagrams. It will be interesting if a smart gluing is viable for mixed helicities as well, given the spurious pole problem [206, 209, 214].

A few remarks are in order. Firstly, in the above derivation of NJ shift, the off-diagonal commutator structure of the massive twistor space (as the structure of half-Fourier transforms), namely the Kähler geometry, has played an important role. To make this point more evident, consider the following identity:

$$\langle \lambda_2 \tilde{\lambda}_2 | \exp\left(i k_3 \mu \lambda^{-1}\right) = \langle \lambda_2, \tilde{\lambda}_2 + \lambda_1^{-1} k_3 |. \quad (12.19)$$

⁷The punchline here parallels that of Ref. [835]. Also, a hidden assumption will be that the in and out states for the black hole are given in the twistor coherent states.

⁸Here, the minus sign is due to our conventions $\varepsilon_{0123} = +1$ and $S^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} y_\rho p_\sigma$: $y^\mu = -a^\mu$.

Overlapping with $|\lambda_1 \tilde{\lambda}_1\rangle$ reproduces Eq. (12.18) as the matrix element of $e^{ik_3 z}$ in the twistor basis. Crucially, the *holomorphic* operator $e^{ik_3 z}$ shifts the *anti-holomorphic* variable $\tilde{\lambda}$, precisely due to the off-diagonal structure.

Another essential role is played by the intrinsic association between chirality and holomorphy. To see this, suppose we knew that the object is a spinning black hole from the beginning. While sitting at the asymptotic infinity with an experimentalist mindset, we “measure” the S-matrix to deduce what background this black hole had interacted with. If the S-matrix is *holomorphic*, then, by the off-diagonal structure, the *anti-holomorphic* variable $\tilde{\lambda}$ solely exhibits precession. But crucially, the *anti-holomorphic* spinor-helicity variable is *right-handed*. Due to this very endemic link between holomorphy and handedness, the background must have been filled with *right-handed*, i.e., SD fields.

On a related note, the exponent in Eq. (12.19) describes the SD angular momentum $J^+ = (z \wedge p)^+$ times a factor, providing a twistorial realization of the equations employed in Ref. [207].

For generic massive spinning objects, the kinematics is not extremal. For example, λ and $\tilde{\lambda}$ are shifted symmetrically (cf. Ref. [98]) when the object’s spin-induced multipole moments [97] vanish from the quadrupole order [673]. The complexified kinematics we have studied is sensitive to all multipole moments, while Newman’s explorations [252–259] focused only on the dipole coupling (gyromagnetic ratio).

Lastly, we were able to deduce the essential physics just by focusing on the delta function support, similarly as in Ref. [836]. A full-blown theory should not only incorporate the mx^h prefactors [206] but also properly formulate the independent massive little group orbits. The first-quantized massive twistor particle [673, 794] could provide a reference, which concretely reproduces the delta function supports considered above [673].

12.9 Conclusions

In this work, we established the notion of spinspace-time as a universal consequence of Poincaré symmetry. Its characteristic geometric structures, which trace back to those of twistor space, revealed that spinning black holes obey a spinning analog of equivalence principle. The Newman-Janis shift factors are effortlessly obtained by massive half-Fourier transforming the extremal

on-shell kinematics in Fig. 12.2.

This revival of a curious concept should trigger various future directions. Firstly, the theory of bulk spinspace-time will be developed, paralleling the historical development of curved twistor theory [853]. To this end, one expects a curved-spacetime generalization or reincarnation of Newman and Winicour [252]’s Hodge duality between orbital and spin angular momenta in Eq. (12.2).⁹ Secondly, applications to PM gravity will be pursued. Lastly, the massive on-shell diagrams and their gluing should be investigated, reminiscently of the massless story [349, 831–837]. It will be exciting if a complete on-shell framework, incorporating both massless and massive legs, can serve as a powerful tool for both theoretical and practical directions.

⁹Note that we have realized this general-relativistic “orbit-spin duality” in Chaps. IX and X: the local orbital angular momentum due to geodesic deviation turns into a spin angular momentum, giving rise to the general-relativistic holomorphic spinspace-time coordinates in the SD sector. In fact, we had developed and established the complete mathematical and physical theory of curved spinspace-time from this perspective of orbit-spin duality, which, unfortunately, had to be omitted from this thesis due to lack of space.

EXACT BLACK HOLE EQUATIONS OF MOTION

We demonstrate how the framework of spinspace-time facilitates a systematic derivation of all-orders-in-spin EoM of spinning black holes in their point-particle effective theory. A model-independent derivation of BMT equations and their all-orders-in-spin extensions is demonstrated. A model-independent derivation of $\sqrt{\text{Kerr}}$ EoM is provided in general, non-SD external electromagnetic fields. Applications of the spinspace-time framework to dynamical symmetries are also discussed. A dynamical NJ shift derives exact, all-orders conserved charges of the spinning black hole probe from the conserved charges of the non-spinning probe. A maximally superintegrable subsector is identified for the $\sqrt{\text{Kerr}}\text{-}\sqrt{\text{Kerr}}$ binary problem.

Contents of this section are adapted from Ref. [854], J.-H. Kim and S. Lee, “Universality in relativistic spinning particle models,” [arXiv:2603.27353 \[hep-th\]](#).

13.1 The Universal Poisson Bracket Relation of Spinning Particles

This section reviews a set of crucial facts shown by Ref. [672], which enables direct identifications of the universal geometrical structures of massive spinning-particle phase spaces. An additional analysis on symplectic leaves is provided.

13.1.1 Derivation from Poincaré Symmetry

Suppose any Hamiltonian formulation of a massive physical system that enjoys global Poincaré symmetry. Ref. [672] shows that one can always construct functions $\hat{x}^\mu$, $\hat{y}^\mu$, p_μ on its phase space $\mathbb{P}$ such that $p \cdot \hat{x} = 0 = p \cdot \hat{y}$ and

$$\begin{aligned} \{\hat{x}^\mu, \hat{x}^\nu\} &= \frac{1}{-p^2} \left(\hat{x}^\mu p^\nu - p^\mu \hat{x}^\nu + \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma \right), \\ \{\hat{x}^\mu, \hat{y}^\nu\} &= \frac{1}{-p^2} \hat{y}^\mu p^\nu, \quad \{\hat{y}^\mu, \hat{y}^\nu\} = \frac{1}{-p^2} \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma, \\ \{\hat{x}^\mu, p_\nu\} &= \delta^\mu_\nu - \frac{p^\mu p_\nu}{p^2}, \quad \{\hat{y}^\mu, p_\nu\} = 0, \quad \{p_\mu, p_\nu\} = 0. \end{aligned} \tag{13.1}$$

Here, $\{f, g\}$ denotes the Poisson bracket between functions f and g on the phase space $\mathbb{P}$. This result applies to not only discrete systems but also continuous systems: a compact mass distribution on flat spacetime with a conserved

stress-energy $T^{\mu\nu}$, for instance. Yet, the interest of this work will be restricted specifically to one-particle systems.

Firstly, Poincaré symmetry implies that one can construct the Poincaré charges as functions on the phase space $\mathbb{P}$: momentum p_μ and total angular momentum $J^{\mu\nu}$. They must satisfy the following Poisson brackets as per the definition of the Poincaré algebra:

$$\begin{aligned}\{J^{\mu\nu}, J^{\rho\sigma}\} &= J^{\mu\rho}\eta^{\nu\sigma} - J^{\mu\sigma}\eta^{\nu\rho} + J^{\nu\sigma}\eta^{\mu\rho} - J^{\nu\rho}\eta^{\mu\sigma}, \\ \{J^{\mu\nu}, p_\rho\} &= \delta^\mu_\rho p^\nu - \delta^\nu_\rho p^\mu, \\ \{p_\mu, p_\nu\} &= 0.\end{aligned}\tag{13.2}$$

Here, $\eta^{\mu\nu}$ is the flat inverse metric of signature $(-, +, +, +)$. The system being massive means that its classical states are not allowed to approach the zero locus of p^2 in $\mathbb{P}$.

Secondly, *define* $\hat{x}^\mu$ and $\hat{y}^\mu$ as

$$\hat{x}^\mu := \frac{1}{p^2} J^{\mu\nu} p_\nu, \quad \hat{y}^\mu := \frac{1}{-p^2} *J^{\mu\nu} p_\nu, \tag{13.3}$$

where $*J^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}J_{\rho\sigma}$ describes the Hodge star in the convention $\varepsilon_{0123} = +1$. By direct computation, one shows that the Poincaré algebra in Eq. (13.2) implies the Poisson brackets in Eq. (13.1) between $\hat{x}^\mu$, $\hat{y}^\mu$, and p_μ . Also, $p \cdot \hat{x} = 0 = p \cdot \hat{y}$ by the antisymmetry of $J^{\mu\nu}$ and $*J^{\mu\nu}$. This concludes the proof.

On account of the well-known definition of the Pauli-Lubanski pseudovector, the variable $\hat{y}^\mu$ in Eq. (13.3) describes the spin pseudovector normalized in units of length. In this work, we refer to this $\hat{y}^\mu$ as *spin length pseudovector*.

In fact, $\hat{y}^\mu$ can evaluate to zero in some systems; take a free relativistic scalar particle formulated in the cotangent bundle of Minkowski space, for instance. In this work, we concern systems with nontrivial $\hat{y}^\mu$ only.

13.1.2 Partly Reduced Formulation

Physically, the notion of mass dimension must exist in the phase space $\mathbb{P}$. Especially, the momentum p_μ carries mass dimension $+1$ while the angular momentum $J^{\mu\nu}$ carries mass dimension 0 . This implies that the dilatation charge D would also exist as a function on the phase space $\mathbb{P}$, such that the Poisson bracket with D measures the mass dimension:

$$\{p_\mu, D\} = +p_\mu, \quad \{J^{\mu\nu}, D\} = 0. \tag{13.4}$$

Ref. [672] shows that Eqs. (13.2) and (13.4) together imply the existence of functions x^μ , $\hat{y}^\mu$, and p_μ on $\mathbb{P}$ such that $p \cdot \hat{y} = 0$ and

$$\begin{aligned} \{x^\mu, x^\nu\} &= \frac{1}{-p^2} \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma = \{\hat{y}^\mu, \hat{y}^\nu\}, \\ \{x^\mu, \hat{y}^\nu\} &= \frac{1}{-p^2} \left(\hat{y}^\mu p^\nu + p^\mu \hat{y}^\nu \right), \\ \{x^\mu, p_\nu\} &= \delta^\mu{}_\nu, \quad \{\hat{y}^\mu, p_\nu\} = 0, \quad \{p_\mu, p_\nu\} = 0. \end{aligned} \quad (13.5)$$

The proof is straightforward. Define

$$x^\mu := \frac{1}{p^2} \left(J^{\mu\nu} p_\nu - D p^\mu \right), \quad (13.6)$$

which adds an additional term to $\hat{x}^\mu$. Then direct computation verifies Eq. (13.5).

In terms of x^μ , $\hat{y}^\mu$, and p_μ , the total angular momentum $J^{\mu\nu}$ and the dilation charge D are expressed as

$$J^{\mu\nu} = x^\mu p^\nu - p^\mu x^\nu + \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma, \quad D = -p \cdot x. \quad (13.7)$$

From Eq. (13.7) and the Poisson bracket $\{x^\mu, p_\nu\} = \delta^\mu{}_\nu$ in Eq. (13.5), it is evident that the interpretation of x^μ is a reconstructed notion of flat spacetime coordinates.

Again, the above result applies to any massive system. In the context of effective point-particle description of compact bodies, x^μ in Eq. (13.6) describes the Poincaré-covariant center coordinates. Accordingly, Eq. (13.7) describes the covariant split of the total angular momentum $J^{\mu\nu}$ to orbital ($x^\mu p^\nu - p^\mu x^\nu$) and spin ($\varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma$) parts.

13.1.3 Symplectic Leaves

Lastly, let us comprehend the above results from a mathematical perspective.

A Poisson manifold is a manifold equipped with a Poisson bracket $\{f, g\} = \Pi(df, dg)$ between functions f and g , which arises from a bivector Π . A symplectic manifold is a manifold equipped with a symplectic form ω , which is a closed nondegenerate two-form. Symplectic manifolds are Poisson manifolds, but the converse is not true. This is because the Poisson bivector Π can be degenerate, unlike the symplectic form ω .

Yet, any Poisson manifold is foliated by *symplectic leaves*, which are (connected) symplectic submanifolds of maximal dimension. The dimension of symplectic leaves is determined by the rank of the Poisson bivector Π .

Provided that $\hat{y}^\mu$ is nontrivial, $\hat{x}^\mu$, $\hat{y}^\mu$, and p_μ in Eq. (13.1) together describe a 10-dimensional Poisson submanifold embedded in the phase space $\mathbb{P}$. Its symplectic leaves are 8-dimensional. To see this, regard the Poisson brackets in Eq. (13.1) as components of an antisymmetric matrix and compute its rank: 8. Furthermore, examining its kernel shows that each of these symplectic leaves is the locus of

$$-p^2 = m^2, \quad -p^2 \hat{y}^2 = w^2, \quad (13.8)$$

where m and w are constants. Physically, the first equation computes the *rest mass* m whereas the second equation computes the *spin magnitude* w .

In this way, it follows that one can always identify an 8-dimensional symplectic submanifold $\mathbf{P}_8(m, w)$ inside any phase space $\mathbb{P}$ of a massive system with nontrivial $\hat{y}^\mu$. This symplectic submanifold will be referred to as the *8-dimensional universal phase space* of the massive spinning system, on which the rest mass and the spin magnitude are fixed to constant values m and w :

$$\mathbb{P} \supset \mathbf{P}_8(m, w). \quad (13.9)$$

In turn, 8 sets the minimal bound for the dimension of a massive spinning phase space. It achieves three translational degrees of freedom and one spinning degree of freedom. It can be seen that $\mathbf{P}_8(m, w)$ is homeomorphic to $\mathbb{R}^6 \times S^2$, which describes the coadjoint orbit [815, 855, 856] of the Poincaré group [815, 840, 857, 858]. It follows that $\mathbf{P}_8(m, w)$ is equivalent to Souriau's "8-dimensional space of motion" reviewed in Ref. [859].

Similarly, x^μ , $\hat{y}^\mu$, and p_μ in Eq. (13.5) together describe an 11-dimensional Poisson submanifold in $\mathbb{P}$ whose symplectic leaves are 10-dimensional. Computation shows that each symplectic leaf is the locus of $-p^2 \hat{y}^2 = w^2$. This identifies the *10-dimensional universal phase space* $\mathbf{P}_{10}(w)$, on which the spin magnitude is fixed to a constant value w :

$$\mathbb{P} \supset \mathbf{P}_{10}(w). \quad (13.10)$$

It can be seen that $\mathbf{P}_{10}(w)$ is homeomorphic to $\mathbb{R}^8 \times S^2$.

It is possible to reproduce the 8-dimensional universal phase space in Eq. (13.9) from the 10-dimensional universal phase space in Eq. (13.10):

$$\mathbf{P}_8(m, w) = \left\{ (x, \hat{y}, p) \in \mathbf{P}_{10}(w) \mid p^2 + m^2 = 0 \right\} / \sim. \quad (13.11)$$

This describes a quotient along one-parameter orbits in $\mathbf{P}_{10}(w)$,

$$(x^\mu, \hat{y}^\mu, p_\mu) \sim (x^\mu + k p^\mu, \hat{y}^\mu, p_\mu), \quad (13.12)$$

which is the Hamiltonian flow of $p^2 + m^2$. Physically, Eq. (13.12) implements a translation of the reconstructed spacetime coordinates in Eq. (13.6) along the momentum direction.

Mathematically, quotients of the form Eq. (13.11) could be called *symplectic quotients*: impose a constraint *and* quotient by its Hamiltonian flow [860, 861]. The process of reducing a symplectic manifold to a smaller one via symplectic quotient will be referred to as *symplectic reduction*. Clearly, symplectic reduction always reduces the dimension by an even number.

The physicists' toolkit for symplectic reduction is known as *Dirac bracket* [862–864]. In this work, we use Dirac brackets to explicitly realize symplectic quotients as symplectic submanifolds.

It is a simple exercise to show that the universal phase space $\mathbf{P}_8(m, w)$ in Eq. (13.9) is a symplectic submanifold of $\mathbf{P}_{10}(w)$ in Eq. (13.10), by constructing the Dirac bracket that fixes $p^2 + m^2$ and $p \cdot x$ to zero within $\mathbf{P}_{10}(w)$:

$$\mathbb{P} \supset \mathbf{P}_{10}(w) \supset \mathbf{P}_8(m, w). \quad (13.13)$$

Surely, this reduction recipe applies universally in every massive spinning system.

13.2 Universality in Interacting Theory: Electromagnetism

Sec. 13.1 has explicitly reviewed that any model of a massive spinning particle describes physical variables $(x^\mu, \hat{y}^\mu, p_\mu)$ that exhibit the following universal Poisson bracket relation in the *free theory*:

$$\begin{aligned} \{x^\mu, x^\nu\}^\circ &= \frac{1}{-p^2} \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma = \{\hat{y}^\mu, \hat{y}^\nu\}^\circ, \\ \{x^\mu, \hat{y}^\nu\}^\circ &= \frac{1}{-p^2} \left(\hat{y}^\mu p^\nu + p^\mu \hat{y}^\nu \right), \\ \{x^\mu, p_\nu\}^\circ &= \delta^\mu{}_\nu. \end{aligned} \quad (13.14)$$

Our objective now is to generalize such universal structures in the *interacting theory*, e.g., with electromagnetic interactions. The output is model-agnostic derivations of the physical EoM of massive spinning particles coupled to electromagnetism.

13.2.1 Couplings as Symplectic Perturbations

Our point of departure is the celebrated fact in symplectic geometry that a manifestly gauge-invariant coupling of particles to electromagnetic fields is viable by modifying the symplectic structure in terms of the field strength. This construction is due to Souriau [815] and also traces back to Feynman [816]. Historically, this manifestly gauge-invariant method has served as an important milestone in the study of particle mechanics in the symplectic framework [865–869]. Unfortunately, it appears to be less recognized in recent literature, so we shall provide a brief review (see also Ref. [764]).

For the sake of concreteness, let us suppose a simple example: a massive scalar particle in Minkowski space. This is a Hamiltonian system defined on the cotangent bundle $T^*\mathbb{M}$. In the free theory, this phase space is equipped with the symplectic form and Hamiltonian

$$\omega^\circ = dp_\mu \wedge dx^\mu, \quad \phi_0 = \frac{1}{2m} (p^2 + m^2). \quad (13.15)$$

We have started to put the accent $^\circ$ to explicitly indicate free-theory objects. The canonical Poisson bracket reads

$$\{x^\mu, x^\nu\}^\circ = 0, \quad \{x^\mu, p_\nu\}^\circ = \delta^\mu{}_\nu, \quad \{p_\mu, p_\nu\}^\circ = 0. \quad (13.16)$$

The Souriau-Feynman method of electromagnetic coupling modifies the geometric data in Eq. (13.15) as

$$\omega = dp_\mu \wedge dx^\mu + qF, \quad \phi_0 = \frac{1}{2m} (p^2 + m^2), \quad (13.17)$$

where $F = \frac{1}{2} F_{\mu\nu}(x) dx^\mu \wedge dx^\nu$ is the field strength two-form, and q is the electric charge. Crucially, the Hamiltonian is left unchanged while the symplectic form is perturbed as

$$\omega = \omega^\circ + \omega', \quad (13.18)$$

with $\omega' = qF$. As a result, the Poisson bracket in Eq. (13.16) is altered as

$$\{x^\mu, x^\nu\} = 0, \quad \{x^\mu, p_\nu\} = \delta^\mu{}_\nu, \quad \{p_\mu, p_\nu\} = qF_{\mu\nu}(x). \quad (13.19)$$

To see why Eq. (13.17) achieves the electromagnetic coupling, one can derive the EoM:

$$\dot{x}^\mu = \{x^\mu, \phi_0\} = \frac{p^\mu}{m}, \quad \dot{p}_\mu = \{p_\mu, \phi_0\} = \frac{q}{m} F_{\mu\nu}(x) p^\nu. \quad (13.20)$$

Eq. (13.20) exactly describes the EoM of a charged scalar particle in electromagnetic background. The modified $\{p_\mu, p_\nu\}$ bracket in Eq. (13.19) implements the Lorentz force law in Eq. (13.20). As a result, the intuition for noncanonical bracket $\{p_\mu, p_\nu\} \neq 0$ is the cyclotron precession exhibited by the momentum p_μ in the external electromagnetic field.

Crucially, this derivation of the Lorentz force law does not invoke the electromagnetic gauge potential at all, as is emphasized by both Souriau [815] and Feynman [816]. As a result, p_μ is the gauge-invariant, physical momentum (also known as kinetic momentum).

The only consistency requirement in this framework is the closure of the modified symplectic form in Eq. (13.17) [815], which holds via the Bianchi-type field equations $dF = 0$ on the background [816]. Physically, this closure amounts to the *Liouville property* of classical time evolution: conservation of classical probability (as phase space measure) [764, 870].

If one insists on keeping the canonical Poisson brackets, one can perform a gauge-dependent coordinate transformation in the phase space (a change of worldline field basis) to the canonical momenta. In this case the Hamiltonian is perturbed and solely encodes the electromagnetic interactions, while the symplectic structure is brought back to the free-theory form.

Albeit canonical, this formulation exhibits evident disadvantages in both conceptual and practical aspects. First, gauge invariance is grossly obscured. The canonical momentum is not a gauge-invariant observable. Second, the coupling is nonlinear. In perturbation theory, there arises a spurious quartic vertex whose sole role is to fix gauge invariance. Third, the derivation of the Lorentz force law from the Hamiltonian EoM becomes not so simple.

In this work, we always fix the Hamiltonian in the free-theory form such that the external field couplings are implemented solely by modifications of the symplectic structure (declaration of worldline field basis). When the symplectic form is modified in the form of Eq. (13.18), we say that the two-form ω' is a *symplectic perturbation*.

We end with a useful formula. The pointwise inverses of ω° and ω define the Poisson bivectors Π° and Π of the free and interacting theories, respectively. Their relation is

$$\omega = \omega^\circ + \omega' \implies \Pi = \Pi^\circ - \Pi^\circ \omega' \Pi. \quad (13.21)$$

In writing down Eq. (13.21), we have treated ω , ω° , ω' , Π , Π° like antisym-

metric matrices.

13.2.2 Model-Agnostic Derivation of BMT Equations

By combining the ideas in Secs. 13.1 and 13.2.1, we establish universal recipes of coupling massive spinning particles to electromagnetism.

Universal Recipe for Minimal Coupling Suppose a free massive spinning particle formulated in a symplectic manifold $\mathbb{P}$. Let ω° be the symplectic form given to $\mathbb{P}$. Time evolution describes a Hamiltonian vector field on $\mathbb{P}$, which we denote as T° .

The result in Sec. 13.1.2 implies that there exist functions $(x^\mu, \hat{y}^\mu, p_\mu)$ on $\mathbb{P}$ that satisfy the universal Poisson brackets in Eq. (13.14). Without loss of generality, we assume that these functions evolve under the free-theory time evolution as

$$T^\circ[x^\mu] = \frac{p^\mu}{m}, \quad T^\circ[\hat{y}^\mu] = 0, \quad T^\circ[p_\mu] = 0. \quad (13.22)$$

Physically, this means that the free particle follows a straight-line trajectory while carrying a constant spin pseudovector.

As an application of the Souriau-Feynman framework in Sec. 13.2.1, we suppose the following symplectic perturbation on $\mathbb{P}$:

$$\omega' = qF = \frac{1}{2} qF_{\rho\sigma}(x) dx^\rho \wedge dx^\sigma. \quad (13.23)$$

The formula in Eq. (13.21) then implies that the Poisson bivector is modified as

$$\Pi(df, dg) = \Pi^\circ(df, dg) - \Pi^\circ(df, dx^\rho) qF_{\rho\sigma}(x) \Pi(dx^\sigma, dg). \quad (13.24)$$

Iterating Eq. (13.24) will provide an infinite series expansion for $\{f, g\} = \Pi(df, dg)$, which governs how the Poisson bracket is modified under the symplectic perturbation: $\{f, g\} = \{f, g\}^\circ - \{f, x^\rho\}^\circ qF_{\rho\sigma}(x) \{x^\sigma, g\}^\circ + \{f, x^\rho\}^\circ qF_{\rho\sigma}(x) \{x^\sigma, x^\kappa\}^\circ qF_{\kappa\lambda}(x) \{x^\lambda, g\}^\circ - \dots$.

Let T be the time-evolution Hamiltonian vector field in the interacting theory such that the Hamiltonian EoM reads $\dot{f} = T[f]$. Eqs. (13.22) and (13.24) together imply that

$$\dot{f} = T^\circ[f] - \{f, x^\rho\}^\circ qF_{\rho\sigma}(x) \dot{x}^\sigma. \quad (13.25)$$

Plugging in the physical variables $(x^\mu, \hat{y}^\mu, p_\mu)$ to f in Eq. (13.25), we find

$$\begin{aligned}\dot{x}^\mu &= \frac{p^\mu}{m} - \frac{1}{m^2} \left(\varepsilon^{\mu\rho\kappa\lambda} \hat{y}_\kappa p_\lambda \right) \left(q F_{\rho\sigma}(x) \dot{x}^\sigma \right), \\ \dot{\hat{y}}^\mu &= \frac{1}{m^2} \left(\hat{y}^\mu p^\rho + p^\mu \hat{y}^\rho \right) \left(q F_{\rho\sigma}(x) \dot{x}^\sigma \right), \\ \dot{p}^\mu &= q F^\mu{}_\nu(x) \dot{x}^\nu,\end{aligned}\tag{13.26}$$

where we have used Eq. (13.14) to evaluate $\{x^\mu, x^\rho\}^\circ$, $\{\hat{y}^\mu, x^\rho\}^\circ$, and $\{p_\mu, x^\rho\}^\circ$.

Eq. (13.26) defines the classical EoM of a consistent Hamiltonian system, as the modified symplectic form due to Eq. (13.23) is closed. What is its physical interpretation, then?

BMT Equations The BMT [819] equations dictate the dynamics of charged massive spinning particles in external electromagnetic fields at the linear order in electric charge q , in the regime where derivative effects can be completely ignored: $F(x) \gg \hat{y} \partial F(x), \hat{y}^2 \partial^2 F(x), \dots$. In our conventions, they are

$$\dot{x}^\mu = \frac{p^\mu}{m} - \frac{q}{m} \left[\frac{g}{2} *F^\mu{}_\nu(x) + \left(1 - \frac{g}{2} \right) \hat{\delta}^\mu{}_\kappa *F^\kappa{}_\nu(x) \right] \hat{y}^\nu + \dots, \tag{13.27a}$$

$$\dot{\hat{y}}^\mu = \frac{q}{m} \left[F^\mu{}_\nu(x) - \left(1 - \frac{g}{2} \right) \hat{\delta}^\mu{}_\kappa F^\kappa{}_\nu(x) \right] \hat{y}^\nu + \dots, \tag{13.27b}$$

$$\dot{p}^\mu = \frac{q}{m} F^\mu{}_\nu(x) p^\nu + \dots, \tag{13.27c}$$

where the ellipses signify that terms of $\mathcal{O}(q^{1+n})$ or $\mathcal{O}(\partial^n F)$ are discarded from the right-hand sides for $n \geq 1$.

Many references present just Eqs. (13.27b) and (13.27c) without explicitly stating the relation between p^μ and $\dot{x}^\mu$. However, it should be emphasized that Eq. (13.27a) must be specified together with Eqs. (13.27b) and (13.27c) to fully define the time evolution and ensure its consistency, especially when working in the Hamiltonian formulation.

In Eq. (13.27), the constant parameter g is the gyromagnetic ratio, also known as the g -factor. The gyromagnetic ratio controls the spin-induced magnetic dipole moment. The specific value $g = 0$ has been identified as the minimal coupling in the traditional sense, in which case the BMT equations in

Eq. (13.27) read

$$\dot{x}^\mu = \frac{p^\mu}{m} - \frac{q}{m} \hat{\delta}^\mu{}_\kappa * F^\kappa{}_\nu(x) \hat{y}^\nu + \cdots, \quad (13.28a)$$

$$\dot{\hat{y}}^\mu = -\frac{q}{m} \hat{p}^\mu \hat{p}_\kappa F^\kappa{}_\nu(x) \hat{y}^\nu + \cdots, \quad (13.28b)$$

$$\dot{p}^\mu = \frac{q}{m} F^\mu{}_\nu(x) p^\nu + \cdots. \quad (13.28c)$$

Notably, the EoM derived in Eq. (13.26) yield the $g=0$ BMT equations in Eq. (13.28) through plugging in $\dot{x}^\mu = p^\mu/m + \mathcal{O}(q^1)$ and using an identity involving two epsilon tensors. Therefore, the $g=0$ BMT equations arise in any massive spinning particle model by prescribing the symplectic perturbation in Eq. (13.23) realized at the physical center x^μ .

Universal Recipe for Dipolar Coupling To implement the dipolar coupling in our universal framework, consider the following symplectic perturbation:

$$\omega' = \frac{1}{2} q F_{\mu\nu}(x) dx^\mu \wedge dx^\nu + q d \left(c_1 \hat{y}^\rho * F_{\rho\sigma}(x) dx^\sigma \right). \quad (13.29)$$

Here, c_1 is a constant. Eq. (13.29) is the most general symplectic perturbation that can contribute to the $\mathcal{O}(q^1 \hat{y}^1)$ part of the EoM. It is *bootstrapped* by the physical principles of

- (a) Liouville property,
 - (b) Poincaré invariance,
 - (c) Gauge invariance,
 - (d) Parity invariance.
- (13.30)

By applying the method demonstrated in Eq. (13.25), one finds that the $\mathcal{O}(q^1 y^1)$ contribution to the resulting EoM arises from the combination

$$-\frac{q}{m} \left[\{f, x^\rho\}^\circ F_{\rho\sigma}(x) p^\sigma + c_1 \{f, \hat{y}^\rho\}^\circ * F_{\rho\sigma}(x) p^\sigma \right]. \quad (13.31)$$

By examining Eq. (13.31) for $f = (x^\mu, \hat{y}^\mu, p_\mu)$, one exactly reproduces the BMT equations in Eq. (13.27) provided the identification

$$c_1 = g/2. \quad (13.32)$$

Therefore, the BMT equations with the generic gyromagnetic ratio g arise in any massive spinning particle model by prescribing the symplectic perturbation in Eq.(13.29) in terms of the physical center x^μ and spin length pseudovector $\hat{y}^\mu$.

13.2.3 Higher Multipoles and Beyond

Linear Couplings As is glimpsed in Eq. (13.29), our universal framework in Sec. 13.2 readily generalizes to all multipole orders. For electromagnetism, an ansatz that conforms to the physical principles in Eq. (13.30) is

$$\begin{aligned} \omega = \omega^\circ + \frac{1}{2} q F_{\mu\nu}(x) dx^\mu \wedge dx^\nu \\ + q d \left[\sum_{\ell=1}^{\infty} \frac{c_\ell}{\ell!} *^\ell F_{\rho_1\sigma, \rho_2\cdots\rho_\ell}(x) \hat{y}^{\rho_1} \hat{y}^{\rho_2} \cdots \hat{y}^{\rho_\ell} dx^\sigma \right], \end{aligned} \quad (13.33)$$

where $*^\ell$ means to apply the Hodge star ℓ times. Eq.(13.33) achieves the most general linear electromagnetic coupling, where the Wilson coefficient c_ℓ parameterizes the spin-induced 2^ℓ -pole moment. Note that the Hodge star implements parity invariance. Thus c_{2k} are electric moments, while c_{2k+1} are magnetic moments.

Nonlinear Couplings The above multipole coefficients c_ℓ dictate only the linear couplings to external fields. At the nonlinear level, there are broad possibilities for generalization. We may want to illuminate two categories of terms, in particular:

1. Ordinary terms (dx), such as $(q^2/m^2)(\hat{y}^\mu F_{\mu\rho}(x) F^\rho{}_\nu(x) \hat{y}^\nu)(p_\kappa dx^\kappa)$.
2. Spin-derivative terms ($d\hat{y}$), such as $q F_{\mu\nu}(x) \hat{y}^\mu d\hat{y}^\nu$ or $\hat{y}^\rho q F_{\mu\nu,\rho}(x) \hat{y}^\mu d\hat{y}^\nu$.

Here, we have enumerated perturbations on the symplectic potential.

The systematic enumeration of all possible terms at a given order is an important problem, although it goes beyond the present work's scope. Generally speaking, this counting will depend on the field basis choice (symplectic versus Hamiltonian perturbations), while computing scattering amplitudes can provide some guidance. As a reminder, our symplectic perturbation approach specifies a particular worldline field basis by fixing the mass-shell constraint in the free-theory form.

The above two classes of terms play different roles in worldline perturbation theory. Consider the diagrammatic computation of two-quanta Compton am-

plitudes, for instance. The example term $(q^2/m^2)(\hat{y}^\mu F_{\mu\rho}(x)F^\rho{}_\nu(x)\hat{y}^\nu)(p_\kappa dx^\kappa)$ contributes to the amplitude as a *contact vertex*. In contrast, the example terms in the spin-derivative category contribute essentially via *exchange channels* due to propagation of worldline fluctuations, as the background worldline will exhibit $\dot{y} = 0$.

This point should also justify the identification of the above spin-derivative terms as “nonlinear couplings.” Although they can be nominally linear in the curvature tensors at face value, they are still nonlinear in the sense that they do not contribute to the three-point amplitudes with one massless quantum. Their effects on the classical EoM, however, could be just as significant as the ordinary terms at the same curvature order.

As shown by Refs. [768–770], spin-derivative terms are *necessary* for achieving black hole couplings (see Eq. (13.46)). Also, they may not be so fundamentally different than the ordinary terms in the spinspace-time perspective, as spin is (the imaginary part of) spacetime.

Definition of Black Hole Coupling Physically, the nonlinear couplings will encode various tidal deformability effects, taking different values for each astrophysical object.

A particularly intriguing case is black holes, which have attracted considerable attention in the current literature. The determination of the exact black hole coupling to all orders has been a long-standing problem. The central question reads

“ *What defines black holes among all massive spinning objects in the point-particle effective theory?* ”

The problem stands unambiguously solved at least in the following subsectors.

1. In **LINEARIZED GRAVITY**: The Kerr black hole is defined by the *unity of multipole coefficients*, $C_\ell = 1$ for all $\ell = 2, 3, 4, \dots$ [200, 202, 204, 205] (cf. Refs. [98, 206–209]).
2. In **SELF-DUAL GRAVITY**: The Kerr black hole is defined by a *superintegrability* in type-D backgrounds with Killing-Yano tensors [768] (cf. Refs. [210–212]).

We have reviewed this development in Sec. 13.A. If a worldline model passes the above tests (a) and (b), then it qualifies as an effective description of the

Kerr black hole according to our best theoretical understanding at the current moment.

Criterion (a) is necessary for reproducing the NJ [202] property in linearized gravity. The NJ shift describes that the Kerr black hole is in some sense a Schwarzschild black hole displaced into “complex spacetime” along an imaginary direction set by its spin length pseudovector. Criterion (b) arises via a nonlinear generalization of this property in curved backgrounds [768].

The problem of constructing worldline models for black holes may have been approached in model-specific manners so far: spherical top model [226], massive twistor model [769, 770], fermionic model [871], etc. Meanwhile, Ref. [768] has explicated the model-independent nature of the problem by introducing a probe counterpart of the NJA [202] in terms of the universal variables $(x, \hat{y}, p)$ and providing the definition of black holes in terms of global symmetries (criterion (b) above).

In this work, we hope to illuminate this model-independent aspect again. For a quick demonstration, we will concern the electromagnetic analog of the Kerr black hole: The zero-gravitation limit $G_N \rightarrow 0$ of the Kerr-Newman solution, dubbed the $\sqrt{\text{Kerr}}$ solution [98, 180]. The $\sqrt{\text{Kerr}}$ solution exhibits the unity of multipole coefficients [180, 202, 256]: $c_\ell = 1$ for all $\ell = 1, 2, 3, \dots$. Its dynamics as a probe enjoys an analogous superintegrability.

13.3 Simplicity of Black Hole Dynamics

13.3.1 Hidden Complex Geometry of the Universal Phase Space

Recall Sec. 13.1.2, where we reviewed the fact [672] that the universal Poisson bracket relation in Eq. (13.5) holds as a direct implication of Poincaré symmetry.

In fact, we can now disclose to the reader that the universal Poisson bracket relation admits a remarkably simpler presentation [672]. Consider the complex combination,

$$z^\mu = x^\mu + i\hat{y}^\mu. \quad (13.34)$$

Straightforward algebra shows that Eq. (13.5) is equivalent to

$$\{z^\mu, z^\nu\}^\circ = 0, \quad \{z^\mu, p_\nu\}^\circ = \delta^\mu_\nu, \quad \{p_\mu, p_\nu\}^\circ = 0, \quad (13.35a)$$

$$\{z^\mu, \bar{z}^\nu\}^\circ = \frac{1}{p^2} \left(\delta^\mu_\rho \delta^\nu_\sigma + \delta^\mu_\sigma \delta^\nu_\rho + i\varepsilon^{\mu\nu}{}_{\rho\sigma} \right) (z^\rho - \bar{z}^\rho) p^\sigma, \quad (13.35b)$$

which fits on two lines. Many components are set to zero. This rewriting not only provides a succinct summary of Eq. (13.5) but also reveals that the components of z^μ in Eq. (13.34) are Poisson-commutative, despite the peculiar noncommutativity of the physical center coordinates x^μ in Eq. (13.5).

Following [672], we refer to z^μ as *spinspace-time coordinates*, as they unify spacetime x^μ and spin $\hat{y}^\mu$ as real and imaginary parts. Mathematically, they are the holomorphic coordinates for the complexified Minkowski space $\mathbb{M}^\mathbb{C} = (\mathbb{C}^4, \eta^\mathbb{C})$ whose real section is the flat spacetime $\mathbb{M} = (\mathbb{R}^4, \eta)$. The property that

$$\{z^\mu, z^\nu\}^\circ = 0, \quad \{z^\mu, \bar{z}^\nu\}^\circ \neq 0, \quad \{\bar{z}^\mu, \bar{z}^\nu\}^\circ = 0 \quad (13.36)$$

is referred to as the zig-zag structure, as the nicknames “zig” and “zag” are used for referring to “holomorphic” (z) and “anti-holomorphic” ($\bar{z}$), respectively [672].

It should be evident that spinspace-time and its zig-zag structure exist *universally* in every special-relativistic massive spinning system.

13.3.2 Inherent Link Between Holomorphy and Self-Duality

The idea of spinspace-time traces back to Newman and Winicour [252].¹ The observation was that the self-dual part of the total angular momentum in Eq. (13.7) is

$$J = (x \wedge p) + *(\hat{y} \wedge p) \implies J^+ = (z \wedge p)^+. \quad (13.37)$$

Here, the $+$ superscript signifies self-dual projection by $\frac{1}{2}(1 - i*)$. The Hodge duality $*$ in Eq. (13.37) turns into the imaginary unit $+i$ in the self-dual sector, so holomorphic coordinates $z^\mu = x^\mu + i\hat{y}^\mu$ arise from the self-dual angular momentum J^+ . In the same way, anti-holomorphic coordinates $\bar{z}^\mu = x^\mu - i\hat{y}^\mu$ arise from the anti-self-dual angular momentum J^- . Importantly, an inherent association arises between *holomorphy and self-duality* in spinspace-time, consistently with the fact that $\hat{y}^\mu$ is a *pseudovector*.

Newman and Winicour [252]’s discussion on spinspace-time, however, does not examine its Poisson structure in Eq. (13.35). The modern reboot of the spinspace-time program [672] proposes that one should pay attention to the

¹See also Refs. [253–257, 259, 329, 330], which portray Newman’s serious take on spinspace-time.

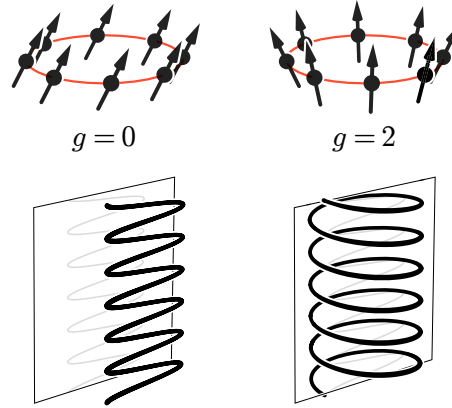


Figure 13.1: Spinspace-time offers a new angle on spin precession. The Dirac gyro-magnetic ratio $g = 2$ can be geometrically characterized as the synchronization of orbital and spin precessions in external fields. When seen from spinspace-time, such a motion appears as an isotropic spiral trajectory, pointing to the equal blending of all $dx, d\hat{y}$ components in $dz \wedge dz$. In contrast, the traditional minimal coupling $g = 0$ translates to a flattened sinusoidal curve, which points to the reality of $dx \wedge dx$. Spinspace-time identifies $g = 2$ as the simplest coupling.

Poisson structure of spinspace-time and its zig-zag property, as they are physical, universal structures.

For instance, it is instructive to employ the spinor notation, in which case Eq. (13.35) is boiled down to a further succinct form:

$$\{z^{\dot{\alpha}\alpha}, p_{\beta\dot{\beta}}\}^{\circ} = \delta^{\dot{\alpha}}_{\dot{\beta}} \delta_{\beta}^{\alpha}, \quad \{z^{\dot{\alpha}\alpha}, \bar{z}^{\dot{\beta}\beta}\}^{\circ} = \frac{1}{-p^2} (z - \bar{z})^{\dot{\alpha}\beta} p^{\dot{\beta}\alpha}. \quad (13.38)$$

Eq. (13.38) is a one-line summary of the entirety of universal Poisson brackets in Eq. (13.5).

The chiral nature of the zig-zag bracket $\{z^{\dot{\alpha}\alpha}, \bar{z}^{\dot{\beta}\beta}\}^{\circ}$ in Eq. (13.38) reflects the aforementioned association between holomorphy and self-duality, which dictates the precession behavior of spinning particles under influence of external fields and derives the probe-level NJ shift from an ideal spin precession behavior [672, 673].

13.3.3 The Simplest Symplectic Perturbation

Spinspace-time opens a new chapter in relativity, offering a unique perspective on spinning-particle mechanics in four dimensions. It is now not only *space* and *time* but also *spin* that are unified. This invites a different way of thinking: spin is the imaginary part of spacetime. Spacetime and spin may not be fundamentally distinct. Fig. 13.1 invites this shift.

An electromagnetic field on spacetime takes the form $\frac{1}{2} F_{\mu\nu}(x) dx^\mu \wedge dx^\nu$. Spacetime $\mathbb{M}$ is embedded in spinspace $\mathbb{M}^\mathbb{C}$ as the real section. Spacetime fields would somehow permeate into spinspace. This will describe $dz \wedge dz$, $dz \wedge d\bar{z}$, or $d\bar{z} \wedge d\bar{z}$ components.

For charged scalar particles, electromagnetic fields are symplectic perturbations (Sec. 13.2.1). For charged spinning particles, *spinspace fields* will be symplectic perturbations.

For scalar particles, $\omega' = \frac{1}{2} q F_{\mu\nu}(x) dx^\mu \wedge dx^\nu$ is said to be the minimal coupling because it is the simplest symplectic perturbation that one can possibly write down. For spinning particles, the simplest symplectic perturbation on *spinspace* will declare the true minimal coupling.

The spacetime field splits into self-dual and anti-self-dual parts as

$$\frac{1}{2} F_{\mu\nu}(x) dx^\mu \wedge dx^\nu = \frac{1}{2} F_{\mu\nu}^+(x) dx^\mu \wedge dx^\nu + \frac{1}{2} F_{\mu\nu}^-(x) dx^\mu \wedge dx^\nu. \quad (13.39)$$

When studying how spacetime fields permeate into spinspace, it can be helpful to examine their self-dual and anti-self-dual parts separately.

Recalling the earlier association, suppose the self-dual field is *holomorphic* on spinspace:²

$$\omega' = \frac{1}{2} q F_{\mu\nu}^+(z) dz^\mu \wedge dz^\nu. \quad (13.40)$$

Holomorphic symplectic perturbations are the simplest symplectic perturbations on spinspace, in the sense that it minimally deforms the universal Poisson brackets in Eq.(13.36). Holomorphic symplectic perturbations preserve the Poisson-commutativity of holomorphic coordinates, due to the very zig-zag structure in Eq. (13.36) (recall Eq. (13.21)):

$$\{z^\mu, z^\nu\} = \{z^\mu, z^\nu\}^\circ - \{z^\mu, z^\rho\}^\circ q F_{\rho\sigma}^+(z) \{z^\sigma, z^\nu\} = 0. \quad (13.41)$$

In this manner, only the zag-zag bracket is deformed by the zig symplectic perturbation:

$$\{z^\mu, z^\nu\} = 0, \quad \{z^\mu, \bar{z}^\nu\} \neq 0, \quad \{\bar{z}^\mu, \bar{z}^\nu\} \neq 0. \quad (13.42)$$

We should understand the physics due to the simplest symplectic perturbation in Eq.(13.40). By straightforward application of the methodology

²Note that the argument of $F_{\mu\nu}^+(z)$ here had to be also holomorphic, due to the principle of symplecticity (closure $d\omega' = 0$).

in Sec. 13.2.2, the EoM are derived from the universal Poisson brackets in Eq. (13.35)—equivalently Eq. (13.38), if one likes to work with spinors—that

$$\dot{z}^\mu = \hat{p}^\mu, \quad \dot{\bar{z}}^\mu = \hat{p}^\mu - \frac{2iq}{m} F^{+\mu}{}_\nu(z) \hat{y}^\nu, \quad \dot{p}^\mu = \frac{q}{m} F^{+\mu}{}_\nu(z) p^\nu. \quad (13.43)$$

We see that the EoM for the zig coordinates are completely ignorant of the external field, due to the very Poisson-commutativity shown in Eq. (13.41): $\dot{z}^\mu = \hat{p}^\mu$.

Eq. (13.43) translates to

$$\dot{z}^\mu = \hat{p}^\mu, \quad \dot{\hat{y}}^\mu = \frac{q}{m} F^{+\mu}{}_\nu(z) \hat{y}^\nu, \quad \dot{p}^\mu = \frac{q}{m} F^{+\mu}{}_\nu(z) p^\nu, \quad (13.44)$$

which describes that the orbital ($\hat{p}^\mu$) and spin ($\hat{y}^\mu$) precessions are perfectly synchronized by the cyclotron angular frequency $(q/m) F^{+\mu}{}_\nu(z)$. Recalling the preliminary exploration in Fig. 13.1, this synchronization implies the gyromagnetic ratio $g = 2$, which is $c_1 = 1$.³ In fact, we will see shortly that all multipole coefficients are unity: $c_\ell = 1$ for all $\ell = 1, 2, 3, \dots$.

13.3.4 The Simplest Real Symplectic Perturbation

The symplectic perturbation in Eq. (13.40) is formal, giving rise to a complexified Poisson structure as in Eq. (13.42). For a real dynamics, we add the missing complex conjugate so that the anti-self-dual field is restored:

$$\omega' = \frac{1}{2} q F_{\mu\nu}^+(z) dz^\mu \wedge dz^\nu + \frac{1}{2} q F_{\mu\nu}^-(\bar{z}) d\bar{z}^\mu \wedge d\bar{z}^\nu. \quad (13.45)$$

By methods in differential geometry [695], it follows that Eq. (13.45) describes [769, 770]⁴

$$\begin{aligned} \omega' = & \frac{1}{2} q F_{\mu\nu}(x) dx^\mu \wedge dx^\nu + q d \left[\sum_{\ell=1}^{\infty} \frac{1}{\ell!} *^\ell F_{\rho_1\sigma, \rho_2\cdots\rho_\ell}(x) \hat{y}^{\rho_1} \hat{y}^{\rho_2} \cdots \hat{y}^{\rho_\ell} dx^\sigma \right], \\ & + q d \left[\sum_{\ell=1}^{\infty} \frac{(\ell-1)}{\ell!} *^\ell F_{\rho_1\sigma, \rho_2\cdots\rho_{\ell-1}}(x) \hat{y}^{\rho_1} \hat{y}^{\rho_2} \cdots \hat{y}^{\rho_{\ell-1}} d\hat{y}^\sigma \right]. \end{aligned} \quad (13.46)$$

³If self-dual fields were anti-holomorphic in spinspace-time, then the particle exhibits $g = -2$, which deviates from the black hole behavior. See also Ref. [672] for a physical interpretation of the synchronization of orbital and spin precessions for gravitational interactions (“spinning equivalence principle”).

⁴To elaborate, the potential one-form for Eq. (13.45) can be represented as $\theta' = \cos(\mathcal{L}_N)(A_\mu(x) dx^\mu) + \text{sinc}(\mathcal{L}_N)(*F_{\mu\nu}(x) \hat{y}^\mu dx^\nu)$ via $F^\pm = (1 \mp i*)F/2$, where N is a vector field such that $\iota_N dx^\mu = \hat{y}^\mu$, $\iota_N d\hat{y}^\mu = 0$, and $\iota_N dp_\mu = 0$.

Compare Eq. (13.46) with Eq. (13.33). For the linear couplings, we identify the unity $c_\ell = 1$ of multipole moments. Thus Eq. (13.46) passes the first test of spinning black holes, criterion (a). Furthermore, Eq. (13.46) also describes nonlinear couplings via *spin-derivative terms*, which are necessary for passing the second test of spinning black holes.⁵

The message here is that the simplest symplectic perturbation from spin-spacetime directly yields the simplest (black hole) coupling. This fact is obscured in typical spacetime-based approaches; it is a unique insight that spin-spacetime can provide. Also, it should be clear that this derivation of a black hole coupling is model independent.

The traditional minimal coupling, $c_\ell = 0$, arises by the simplest symplectic perturbation from the standards of spacetime: $\frac{1}{2} q F_{\mu\nu}(x) dx^\mu \wedge dx^\nu$. The spin-spacetime Poisson brackets in Eq. (13.35) shows that this grossly complicates the dynamics of the particle, as $dx \wedge dx$ contains all the $dz \wedge dz$, $dz \wedge d\bar{z}$, and $d\bar{z} \wedge d\bar{z}$ components.

13.3.5 Universal Derivation of Root-Kerr Equations of Motion

Via Eqs. (13.21) and (13.35), it is easy to derive the complete $\sqrt{\text{Kerr}}$ EoM due to the real symplectic perturbation in Eq. (13.45):

$$\begin{aligned} \dot{p}_\mu &= qF_{\mu\nu}^+(z) \dot{z}^\nu + qF_{\mu\nu}^-(\bar{z}) \dot{\bar{z}}^\nu, \\ \dot{z}^\mu &= \hat{p}^\mu - \frac{2iq}{m^2} \left(\hat{y}^\mu p^\nu + p^\mu \hat{y}^\nu - i\varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma \right) F_{\nu\kappa}^-(\bar{z}) \dot{\bar{z}}^\kappa. \end{aligned} \quad (13.47)$$

The EoM for $\bar{z}^\mu$ follow by complex conjugation since the dynamics is real. The above equations have been known from the massive twistor model [795]. Here, we have reproduced them in the model-agnostic fashion. Note that Eq. (13.47) describes *all-orders-in-spin* EoM.

It is left as an exercise to deduce the EoM for x^μ and y^μ by taking real and imaginary parts. One may use an identity $\varepsilon^{\mu\nu\rho\sigma} \phi_{\nu\kappa}^\pm u^\kappa = \mp 3i \phi^{\pm[\rho\sigma} u^{\mu]}$ to remove the epsilon tensors if wanted, where $\phi_{\mu\nu}^+$ ($\phi_{\mu\nu}^-$) is self-dual (anti-self-dual).

Overall, spinspace is a powerful tool for making predictions on the all-orders-in-spin dynamics of spinning objects.

⁵For instance, it can be diagrammatically shown [770] that the resulting same-helicity Compton amplitudes exhibit spin exponentiation for all multiplicities. Thus Eq. (13.45) is the symplectic perturbation of the $\sqrt{\text{Kerr}}$ spinning particle up to mixed-chirality terms (earthly deformations [768]) that vanish in the self-dual limit, such as $(q^2/m^2)(\hat{y}F^+(x)F^-(x)\hat{y})(pdx)$.

As demonstrated in Ref. [673], spinspace is also useful for deriving the all-orders-in-spin EoM of particles with generic multipole moments. In this case, one perturbs away from the black hole coupling by redefining the Wilson coefficients on the complex worldline.

13.4 Dynamical Symmetries and Conserved Charges

This section demonstrates how model-ignorant conserved charges are directly uncovered from the universal Poisson and covariant Poisson bracket relations. For simplicity, we limit our attention to spacetime isometries.

13.4.1 Symmetry in Hamiltonian Mechanics

Let us first review the definition of symmetry in Hamiltonian mechanics. Suppose a Hamiltonian system defined by a symplectic form ω and a Hamiltonian H on a phase space. A vector field V implements an infinitesimal symmetry transformation on this system iff both of the following conditions are met:

$$\text{Symplectic Condition : } \mathcal{L}_V \omega = 0, \quad (13.48a)$$

$$\text{Hamiltonian Condition : } \mathcal{L}_V H = V[H] = 0. \quad (13.48b)$$

Due to the Cartan formula and the closure of ω , the symplectic condition translates to $d(\iota_V \omega) = 0$, which implies local existence of a function Q such that

$$\iota_V \omega = -dQ \iff V = \{ \cdot, Q \}. \quad (13.49)$$

Here, $\{ \cdot, f \} = \omega^{-1}(\cdot, df)$ denotes the Hamiltonian vector field of f . This shows that symmetries in Hamiltonian mechanics are realized as Hamiltonian actions on the phase space. In other words, we have approached the Noether theorem in the symplectic geometry language: the function Q is the Noether charge.

On the other hand, the Hamiltonian condition encodes compatibility with the system's time-evolution generator (or constraints if any). Plugging in Eq. (13.49), one finds that it encodes Poisson commutativity of Q and H :

$$\{H, Q\} = V[H] = 0. \quad (13.50)$$

We shall also note that the above definition does not invoke the symplectic potential θ . Famously, the symplectic potential involves redundancies $\theta \sim$

$\theta + \Delta\theta$ for any closed one-form $\Delta\theta$ on the phase space. Thus it cannot provide an invariant definition of symmetry.

Instead, the role of the symplectic potential in this discussion would be a customary gadget that provides a convenient formula for the Noether charge. By using the freedom $\theta \sim \theta + \Delta\theta$, one may find a particular θ such that $\mathcal{L}_V\theta = 0$. In this case, the Cartan formula asserts $\iota_V\omega = -d\iota_V\theta$, leading to a concrete identification of the Noether charge as

$$\mathcal{L}_V\theta = 0 \implies Q = \iota_V\theta. \quad (13.51)$$

In fact, such a symmetry-preserving symplectic potential θ will give rise to a natural, symmetry-adapted phase space action.

13.4.2 Symmetry-Preserving Symplectic Perturbations

Next, we develop a general theory of symmetries in the framework of symplectic perturbations. Suppose $(\mathbb{P}, \omega^\circ, H)$ defines the free theory of a Hamiltonian system. Suppose a vector field V on $\mathbb{P}$ implements an infinitesimal symmetry transformation on $(\mathbb{P}, \omega^\circ, H)$, whose Noether charge is Q° :

$$\iota_V\omega^\circ = -dQ^\circ, \quad \{H, Q^\circ\}^\circ = V[H] = 0. \quad (13.52)$$

A closed two-form ω' on $\mathbb{P}$ is a *symmetry-preserving symplectic perturbation* iff

$$\mathcal{L}_V\omega' = 0. \quad (13.53)$$

Again, through the Cartan formula and $d\omega' = 0$, Eq. (13.53) translates to $d(\iota_V\omega') = 0$, which implies local existence of a function Q' such that

$$\iota_V\omega' = -dQ'. \quad (13.54)$$

Via Eq. (13.52), Eq. (13.54) implies that

$$\iota_V\omega = -dQ, \quad \{H, Q\} = V[H] = 0 \quad (13.55)$$

where $Q = Q^\circ + Q'$ and $\omega = \omega^\circ + \omega'$.

The implication of Eqs. (13.52) and (13.55) is that V , the *same* vector field from the free theory (ω°, H) , implements an infinitesimal symmetry transformation on the interacting theory (ω, H) as well, yet via the modified Noether charge $Q = Q^\circ + Q'$ in Eq. (13.55):

$$V = \{ \quad, Q^\circ \}^\circ = \{ \quad, Q \}. \quad (13.56)$$

Further, suppose the free theory admits a number of symmetry charges forming an algebra. The above analysis shows that the symmetry actions on the phase space (as the vector fields) remain the same under symmetry-preserving symplectic perturbations. As a result, the symmetry algebra remains the same in the interacting theory.

13.4.3 Scalar Particle in Electromagnetism

For a concrete demonstration of the above abstract formalization, we could revisit the familiar example of the charged scalar particle described in Sec. 13.2.1.

The free theory is defined in Eq. (13.15), which supposes the phase space $T^*\mathbb{M}$ with coordinates x^μ and p_μ . The symplectic form is $\omega^\circ = dp_\mu \wedge dx^\mu$, while the mass-shell constraint $\phi_0 = \frac{1}{2}(\eta^{\mu\nu}p_\mu p_\nu + m^2)$ serves as the Hamiltonian. A well-known fact is that

$$V = K^\mu(x) \frac{\partial}{\partial x^\mu} - p_\nu K^\nu{}_{,\mu}(x) \frac{\partial}{\partial p_\mu} \quad (13.57)$$

generates a symmetry transformation on $(T^*\mathbb{M}, \omega^\circ, \phi_0)$ iff $K^\mu(x)\partial/\partial x^\mu$ is a Killing vector of flat spacetime $\mathbb{M} = (\mathbb{R}^4, \eta)$. To elaborate, the symplectic condition is trivially satisfied by the ansatz in Eq. (13.57), identifying the Noether charge as

$$Q^\circ = p_\mu K^\mu(x). \quad (13.58)$$

The Hamiltonian condition then yields the Killing equation in flat spacetime,

$$\mathcal{L}_K \eta = 0 \iff K_{\mu,\nu}(x) = -K_{\nu,\mu}(x). \quad (13.59)$$

Now consider an electromagnetic field strength F on flat spacetime such that

$$\mathcal{L}_K F = 0. \quad (13.60)$$

Via the Cartan formula and the closure $dF = 0$, Eq. (13.60) implies the local existence of a scalar field α such that [588]

$$\iota_K F = -d\alpha \iff K^\mu(x) F_{\mu\nu}(x) = -\alpha_{,\nu}(x). \quad (13.61)$$

Crucially, the condition in Eq. (13.60) implies that $\omega' = qF$ (now viewed as a two-form on $\mathbb{P}$ via the pullback of the bundle projection) is a symmetry-preserving symplectic perturbation. Namely, it is easy to show that $\mathcal{L}_V \omega' = 0$

by $\mathcal{L}_K F = 0$. According to Eq. (13.55), the correction to the Noether charge can be taken as $Q' = q\alpha$, as $-dQ' = \iota_V \omega' = -q d\alpha$. Therefore, the exact Noether charge in the interacting theory is

$$Q = p_\mu K^\mu(x) + \alpha(x). \quad (13.62)$$

Eq. (13.62) is precisely the conserved charge of the charged scalar probe [588].

To derive the perhaps more widely-recognized formula $Q = (p_\mu + qA_\mu(x))K^\mu(x)$, note that the scalar field α in Eq. (13.61) can be explicitly given as follows if one is willing to make an explicit gauge choice:

$$\mathcal{L}_K A = 0 \implies \alpha(x) = qA_\mu(x)K^\mu(x). \quad (13.63)$$

13.4.4 Spinning Particles in Electromagnetism

Now we are ready to derive the *universal* (model-ignorant) conserved charges of massive spinning particles in electromagnetism.

Suppose any Hamiltonian formulation of a free massive spinning particle based on a phase space $(\mathbb{P}, \omega^\circ)$. Suppose infinitesimal symmetry V exists for this particle. Our ansatz for the symmetry action on the physical variables is

$$\begin{aligned} V[x^\mu] &= \{x^\mu, Q^\circ\}^\circ = K^\mu(x), \\ V[\hat{y}^\mu] &= \{\hat{y}^\mu, Q^\circ\}^\circ = K^\mu{}_{,\nu}(x)\hat{y}^\nu, \\ V[p_\mu] &= \{p_\mu, Q^\circ\}^\circ = -p_\nu K^\nu{}_{,\mu}(x). \end{aligned} \quad (13.64)$$

To validate this ansatz, we apply the two conditions in Eq. (13.48). First, the symplectic condition is solved by taking the Noether charge as

$$Q^\circ = p_\mu K^\mu(x) - \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} \hat{y}_\rho p_\sigma K_{\mu,\nu}(x), \quad (13.65)$$

which reproduces Eq. (13.64) via the universal Poisson bracket relation in Eq. (13.14) provided that $K^\mu(x)$ exhibits linearity: $K^\mu{}_{,\rho\sigma}(x) = 0$. Second, the Hamiltonian condition is satisfied if $K_{\mu,\nu}(x) = -K_{\nu,\mu}(x)$, as Eq. (13.64) implies that the symmetry action on the mass-shell constraint $\phi_0 = \frac{1}{2}(p^2 + m^2)$ is $V[\phi_0] = -p_\nu K^\nu{}_{,\mu}(x)p^\mu$. Crucially, since Q° in Eq. (13.65) is composed solely of the physical variables $(x^\mu, \hat{y}^\mu, p_\mu)$, it must Poisson-commute with any gauge constraints of the model. Consequently, the Hamiltonian action of Eq. (13.65) passes both the symplectic and Hamiltonian tests of symmetry in Eq. (13.48).

Note that this fact could be shown without writing down an explicit formula for V (which will depend on the microscopic implementations, i.e., the details of the model).

It follows that the symplectic perturbation $\omega' = qF$ due to the universal minimal coupling recipe in Eq. (13.23) is symmetry-preserving as long as the electromagnetic background F exhibits symmetry à la Eq. (13.60). Since Eq. (13.64) implies $\iota_V dx^\mu = V[x^\mu] = K^\mu(x)$, we have (with abuse of notation regarding the pullbacks due to bundle projection)

$$\begin{aligned}\mathcal{L}_V \omega' &= d\iota_V \omega' = qd\left(K^\mu(x)F_{\mu\nu}(x)dx^\nu\right), \\ &= qd\iota_K F = q\mathcal{L}_K F = 0.\end{aligned}\tag{13.66}$$

Therefore, the same vector field V still defines a symmetry in the interacting theory.

It remains to find the Noether charge. Since $-dQ' = \iota_V \omega' = q\iota_K F = -qd\alpha$, so we can take $Q' = q\alpha(x)$. Thus, we conclude that minimally coupled massive spinning particles in electromagnetism universally exhibit the conserved charge

$$Q = p_\mu K^\mu(x) - \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\hat{y}_\rho p_\sigma K_{\mu,\nu}(x) + q\alpha(x),\tag{13.67}$$

where the gauge choice $\mathcal{L}_K A = 0$ will take $\alpha(x) = A_\mu(x)K^\mu(x)$.

To reiterate, the Poisson commutativity of Eq. (13.67) with the various constraints of the particle is automatic since the infinitesimal symmetry transformation V is left unchanged. Still, it is a nice and interesting exercise to reconfirm $\{H, Q\} = 0$ by direct computation, by using the exact universal EoM derived in Eq. (13.26), for instance.

Incorporating the multipolar couplings is not difficult. For instance, the non-minimal coupling in Eq. (13.29) adds $\theta' = (g/2)q *F_{\mu\nu}(x)\hat{y}^\mu dx^\nu$ on the symplectic potential. Computation using Eq. (13.64) shows that $\mathcal{L}_V \theta' = 0$ by $\mathcal{L}_K *F = 0$. Here, $\mathcal{L}_K$ preserves the Hodge star because it preserves the flat metric. This shows that θ' is a symmetry-preserving perturbation on the symplectic potential. Based on our earlier discussion around Eq. (13.51), this implies that the universal conserved charge with the generic gyromagnetic ratio g is

$$\begin{aligned}Q &= p_\mu K^\mu(x) - \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\hat{y}_\rho p_\sigma K_{\mu,\nu}(x) \\ &\quad + q\alpha(x) + \frac{g}{2}q *F_{\mu\nu}(x)\hat{y}^\mu K^\nu(x).\end{aligned}\tag{13.68}$$

13.5 Dynamical Newman-Janis Shift of Conserved Charges

As a glimpse toward the applications of the above EoM, we provide a reproduction of the fact [770] that an idealized version of the spinning black hole binary problem exhibits superintegrability via exact hidden symmetries. We follow the model-independent presentation due to Ref. [768]. This discussion confirms the criterion (b), confirming that Eq. (13.46) qualifies as an effective point-particle description of $\sqrt{\text{Kerr}}$.

The dream of a theoretical relativist is to understand the exact dynamics of the Kerr-Kerr black hole binary system. However, the problem is too hard. As a first simplification, we suppose that one of the black holes is significantly heavier than the other: the zeroth self-force limit. As a second simplification, we suppose the electromagnetic analog: the $\sqrt{\text{Kerr}}$ - $\sqrt{\text{Kerr}}$ binary problem.

As a result, we study the motion of a $\sqrt{\text{Kerr}}$ probe in the background of the $\sqrt{\text{Kerr}}$ solution. Since Newman and Janis [202], it has been available that the field strength of the $\sqrt{\text{Kerr}}$ solution takes the form

$$F_{\mu\nu}(x) = \frac{Q}{8\pi|\vec{x}-i\vec{a}|^3} *Y_{\mu\nu}^+(x) + \frac{Q}{8\pi|\vec{x}+i\vec{a}|^3} *Y_{\mu\nu}^-(x), \quad (13.69)$$

where Q is the charge parameter, a^μ is a spacelike pseudovector encoding the ring radius, and u^μ is a timelike unit vector defining the stationary direction. We have denoted

$$*Y_{\mu\nu}^\pm(x) = (u \wedge (x \mp ia))_{\mu\nu}^\pm, \quad (13.70)$$

which encodes the position three-vector from imaginary centers $\pm ia$. As disclosed by Ref. [164], these centers are precisely self-dual and anti-self-dual dyons.

The problem is still too hard. With both self-dual and anti-self-dual fields, the geometric series expansion of the Poisson bivector does not truncate at a finite order, so Eq. (13.47) describes time derivatives on its right-hand sides in contrast to Eq. (13.43).

Hence we simplify the problem again by dropping the anti-self-dual part of the field configuration in Eq. (13.69):

$$F_{\mu\nu}(x) = \frac{Q}{8\pi|\vec{x}-i\vec{a}|^3} *Y_{\mu\nu}^+(x) \implies *F_{\mu\nu}(x) = +iF_{\mu\nu}(x). \quad (13.71)$$

This extracts the self-dual part of the $\sqrt{\text{Kerr}}$ background in the precise sense; see Ref. [163] for the gauge potential. It is not difficult to see that Eq. (13.71)

describes

$$\vec{E}(\vec{x}) = i\vec{B}(\vec{x}) = \frac{Q}{8\pi} \frac{\vec{x} - i\vec{a}}{|\vec{x} - i\vec{a}|^3}, \quad (13.72)$$

which is the electromagnetic field of a dyon with electric charge $Q/2$ and magnetic charge $-iQ/2$: a self-dual dyon. The magnetic dipole moment of $\sqrt{\text{Kerr}}$ arises precisely like a *static* magnet made of monopoles: $Q\vec{a} = (-iQ/2)(2i\vec{a})$. See Ref. [164] for more details.

In short, we study the motion of a $\sqrt{\text{Kerr}}$ probe in the background of a self-dual dyon. The relevance to the original problem is that the self-dual dyon is a part of $\sqrt{\text{Kerr}}$.

In the self-dual dyon background, we concern the combination

$$Y_{\mu\nu}(x) = Y_{\mu\nu}^+(x) + Y_{\mu\nu}^-(x) = \varepsilon_{\mu\nu\rho\sigma} x^\rho u^\sigma - (u \wedge a)_{\mu\nu}. \quad (13.73)$$

It turns out that this defines a Killing-Yano tensor [585] such that

$$\begin{aligned} Y_{\mu\nu,\rho}(x) &= \varepsilon_{\mu\nu\rho\sigma} u^\sigma = Y_{[\mu\nu,\rho]}(x), \\ F^\mu{}_\rho(x) Y^\rho{}_\nu(x) &= Y^\mu{}_\rho(x) F^\rho{}_\nu(x). \end{aligned} \quad (13.74)$$

The first equation is the definition statement, while the second equation describes a commuting property with the field strength.

In this background, the $\sqrt{\text{Kerr}}$ probe obeys the complexified EoM in Eq. (13.44):

$$\dot{z}^\mu = \hat{p}^\mu, \quad \dot{p}^\mu = \frac{q}{m} F^\mu{}_\nu(z) p^\nu, \quad \dot{\hat{y}}^\mu = \frac{q}{m} F^\mu{}_\nu(z) \hat{y}^\nu, \quad (13.75)$$

Notably, the first two equations here are identical to those of the charged scalar particle up to the complexification $x \rightarrow z$, which originated from the Poisson commutativity of holomorphic spinspace-time in Eq. (13.41). Thus the well-known discourse on the scalar particle [588] is straightforwardly recycled as

$$\begin{aligned} \frac{d}{d\tau} \left(Y^\mu{}_\nu(z) p^\nu \right) &= Y^\mu{}_\nu(z) F^\nu{}_\rho(z) p^\rho + Y^\mu{}_{\nu,\rho}(z) p^\nu \hat{p}^\rho, \\ &= F^\mu{}_\nu(z) Y^\nu{}_\rho(z) p^\rho, \end{aligned} \quad (13.76)$$

showing that the precessions of p and $Y(z)p$ are synchronized.

Crucially, the precession of the spin length pseudovector $\hat{y}$ is also synchronized with p . Now we find three vectors singing in unison, p , $Y(z)p$, and $\hat{y}$. It is then immediate that

$$R = p_\mu Y^\mu{}_\nu(z) \hat{y}^\nu, \quad C = -p_\mu Y^\mu{}_\rho(z) Y^\rho{}_\nu(z) p^\nu \quad (13.77)$$

are conserved quantities: the all-orders-in-spin completions of the Rüdiger [872, 873] and Carter [586] constants.

Moreover, the self-dual dyon background exhibits isometries $\mathbb{R} \times \text{SO}(3)$. Again, we immediately deduce the $\sqrt{\text{Kerr}}$ probe's conserved charge by direct dynamical NJ shift of the Coulomb probe's conserved charge in Eq. (13.62):

$$Q = p_\mu K^\mu(z) + \alpha(z). \quad (13.78)$$

The symmetry algebra between these NJ shifted Killing charges are the same as in the non-spinning case, crucially because of the Poisson commutativity $\{z^\mu, z^\nu\} = 0$ that we have had to emphasize.

Using Eqs. (13.77) and (13.78), one can show that the dynamics is exactly solvable (integrable). See Refs. [768, 770] to grasp more details for the super-integrability.

In summary, we have learned that the self-dual sector of the $\sqrt{\text{Kerr}}\text{-}\sqrt{\text{Kerr}}$ binary problem is exactly solvable, based on direct dynamical NJ shifts of the non-spinning probe's conserved charges as well as the synchronization of orbital and spin precessions.

This is yet an idealized discussion. How should one approach the original problem of the $\sqrt{\text{Kerr}}\text{-}\sqrt{\text{Kerr}}$ binary system, then?

A reasonable pathway may be perturbing away from the self-dual sector where exact all-orders expressions for the conserved charges are known: the googly agenda [768].

13.6 Conclusion and Outlook

In this chapter, we have demonstrated the utility of spinspace in deriving exact spinning-particle EoM. Spinspace facilitates systematic all-orders extensions of the BMT equations. Moreover, spinspace proves to be a powerful tool for deriving and studying black hole EoM. The inherent association between chirality and holomorphy that it entails discovers hidden simplicities of spinning black hole mechanics. The spinspace framework also provides a model-independent approach to conserved charges. We established dynamical NJ shift of conserved charges for spinning black hole probes and concretely (re)demonstrated the maximal superintegrability in a SD subsector of the black hole binary problem, as a response to Chap. V where the simplicity of the SD sector is framed with the keyword “hydrogen.”

Further promising developments have been underway. The theory of curved spinspace-time is viable. The derivation of spinning black hole EoM in YM and gravity backgrounds can utilize the “googly” idea, which is the chiral approach of understanding the full nonlinear dynamics demonstrated in Sec. 2.5 applied to spinning particles. In the spinspace-time framework, this means to perturb away from the holomorphic worldline where SD dynamics resums onto (“perturbatively add zag on zig”). Compton amplitudes are computed via worldline methods, again by implementing a chiral perturbation theory in the perturbative googly spirit; see Chap. XIV.

13.A Definition of Black Hole Coupling

Eventually, we are now in a position to review the developments surrounding the question,

“ *What defines black holes* among all massive
spinning objects in the point-particle effective theory? ”

We will only cover the IR approaches within the point-particle effective theory, although a large volume of the current literature pursues the determination of black hole coupling from the UV (i.e., matching with GR).

1. In *linearized gravity*, the Kerr black hole is defined by the *unity of multipole coefficients* (i.e., *three-point spin exponentiation*).

This result is due to the classic works [200–205] in the ’60s and ’70s: the Kerr solution is shown to exhibit $C_\ell = 1$ for all $\ell = 1, 2, 3, \dots$.

According to the traditional definition, this seems like a grossly non-minimal coupling. However, a modern analysis [98, 206–209] shows that it is the simplest coupling in terms of a large simplification of the three-point graviton scattering amplitudes (cf. Ref. [874]). This simplicity is represented as a complexification of the impact parameter that adds spin length as an imaginary component and is dubbed the *three-point spin exponentiation*.

To elaborate, these three-point amplitudes describe the process of the black hole (as a point-particle source) sourcing a single massless quantum. Thus, they are synonymous to the linearized part of the Kerr metric [142, 176, 631]. In this manner, it is shown that the three-point spin exponentiation is the NJ shift in linearized gravity [98].

The NJ shift refers to the property discovered by Newman and Janis [202] in 1965 that the Kerr black hole is secretly a Schwarzschild black hole displaced into “complex spacetime” in some rough sense, in which case the spin length (as ring radius) describes the imaginary direction. This trick was the very method that enabled the historical discovery of the Kerr-Newman solution [179]. The aspects of NJ shift in linearized gravity have been well-studied from the modern scattering amplitudes perspective by Refs. [98, 206–209], while its nonlinear aspect has been recently elucidated by Ref. [164].

2. In *SD gravity*, the Kerr black hole is defined by the *n-point spin exponentiation*.

At the nonlinear orders, it has been unclear what principle in the point-particle effective theory defines spinning black holes. Meanwhile, explorations via scattering amplitudes investigated the n -point Compton scattering process in which the Kerr black hole receives $(n-2)$ gravitons, where $n \geq 3$. This research showed that the n -point gravitational Compton amplitudes can exhibit the spin exponentiation property as a theoretically allowed possibility, iff all the helicities of the gravitons are the same [210–212].⁶

Therefore, by extending the definition of the simplicity of scattering amplitudes, it has been believed that this “ n -point spin exponentiation” for all $n = 3, 4, 5, 6, \dots$ may define the black hole coupling at all nonlinear orders within the *same-helicity* sector as a beautiful possibility, yet without a necessary physical reason.

This scattering amplitudes statement is equivalent to the proposition that the point-particle effective action of the Kerr black hole is unique (up to world-line field redefinitions) in background SD spacetimes. A SD spacetime describes a formal complexified limit of spacetime in which the Riemann tensor becomes SD: $*R_{\mu\nu\rho\sigma} = +iR_{\mu\nu\rho\sigma}$. A well-established fact is that the same-helicity gravitational Compton amplitudes due to incoming positive-helicity gravitons describe scattering processes in SD background spacetimes; see Figure 1 in Ref. [310] and also Refs. [342, 343].

This unique effective action has been rigorously established in Refs. [768, 769]. The lesson is that the n -point spin exponentiation for positive-helicity

⁶The mixed-helicity Compton amplitudes are plagued by a contact term issue, first noticed in Ref. [206].

gravitons amounts to a nonlinear generalization of NJ shift for black hole probes in SD spacetimes.

In sum, three-point spin exponentiation is NJ shift in linearized gravity, while n -point spin exponentiation is nonlinear NJ shift in SD gravity.

It has been recently shown [768] that hidden symmetries (superintegrability) in the SD sector implies the spin exponentiation property of same-helicity Compton amplitudes to all multiplicities [768].⁷ This result has promoted the n -point spin exponentiation from a mere beauty statement to a physical assertion based on global symmetries.

3. In *full gravity*, the Kerr black hole is defined by (?).

The unique characterization in the SD sector does not pinpoint the complete black hole couplings in full nonlinear gravity due to mixed-helicity interactions. The extent of this ambiguity is explicitly examined in the analysis of Ref. [768].

So far, the construction of the exact worldline effective theory of the Kerr black hole may have been based on model-specific perspectives. An early sketch was given in Ref. [226] by extending the Levi-Steinhoff action in the spherical top model, which identified the NJ shift as an important clue. This work provided valuable intuitions and insights, although the subtleties of spin gauge redundancy, geodesic deviation, and integrability of complex structures are not clearly addressed while explicit expressions are limited to the linear-in-Riemann order. In the meantime, Ref. [673] provided an approach based on the Kähler geometry of massive twistor space, unearthing a unique geometrical perspective on the NJ shift that was also demonstrated in Ref. [672].

Eventually, Refs. [769, 770] have established rigorous constructions of the exact worldline actions for Kerr and Kerr-Newman black holes in the massive twistor model, providing their explicit formulae to all orders via the systematic formalism of Ref. [695]. As clearly portrayed in Ref. [769], this construction envisions a curved generalization of the magical feature of twistor particle theory that spin is literally an imaginary deviation in terms of the complexified incidence relation [252, 813, 814]. The dynamical NJ shift is implemented by a solid mathematical framework known as adapted complex structure [743, 777, 780], refining Ref. [226]’s earlier sketch.

⁷Ref. [768] also provided a physical interpretation of the n -point spin exponentiation: the helicity selection rule [261] exhibited by the SD Taub-Newman-Unti-Tamburino solution, which constitutes part of the Kerr black hole [164].

While Refs. [769, 770] paved a top-down pathway to the problem from massive twistor theory, Ref. [768] provided a complementary, bottom-up view that emphasizes model independence. Ref. [768] defines an EoM-level implementation of the NJA in terms of the universal variables $(x, \hat{y}, p)$ of the spinning probes. A corollary is the aforementioned hidden symmetry in the SD sector. Notably, the conserved charges of the spinning probe are directly obtained by dynamically NJ shifting the conserved charges of the scalar probe.

In this universal manner, Ref. [768] also suggests a natural proposal for uplifting the SD sector couplings to the full nonlinear sector in a rather model-independent fashion. In this case, the principle that defines black holes is “orbit-spin duality,” a generalization of an old idea due to Newman and Winicour [252] in curved backgrounds [672, 768, 769]. A dedicated article will appear soon [875] for an elaboration on this point.

BLACK HOLE COMPTON AMPLITUDES

We compute the classical Compton scattering amplitudes of the nonabelian $\sqrt{\text{Kerr}}$ and Kerr solutions receiving two gluons or gravitons in YM theory or GR, to all orders in spin. Our amplitudes are completely free of any spurious pole and exhibit correct factorizations for all helicity configurations. They are systematically structured in terms of truncated exponential functions. In fact, our nonabelian $\sqrt{\text{Kerr}}$ amplitude seems to be unique when restricting to a certain class of ansatz due to Newton’s interpolation formula.

Crucially, we provide a Lagrangian derivation of these amplitudes based on explicit local-in-worldline-time point-particle effective actions that exactly resum onto holomorphic or anti-holomorphic worldlines in SD or ASD backgrounds, respectively. These resummations realize the probe NJ shift, boiling down to the same-helicity spin exponentiation to all orders in perturbation theory. At the technical level, we implement the worldline formalism in massive twistor space in the backgrounds of nonlinearly superposed plane waves in YM theory and GR while making a fruitful use of axial gauge and covariant CK duality.

Physically, the holomorphic worldline is naturally interpreted as the infrared counterpart of the ASD dyon or Taub-NUT instanton constituting $\sqrt{\text{Kerr}}$ and Kerr, based on a helicity selection rule. Accordingly, we implement a chiral perturbation theory in which the worldline actions are represented by perturbing away from the holomorphic worldline. In this formalism, spin exponentiation is manifestly guaranteed in the positive-helicity sector to all orders in perturbation theory. The negative-helicity modes are then perturbatively added in a systematic fashion. Notably, our formalism provides a direct realization of the twistor theorists’ aspiration of achieving $\sqrt{\text{Kerr}}$ or Kerr by perturbing away from ASD dyon or ASD Taub-NUT: “MHV-ing spinning black holes,” to say.

Our framework does not preclude adding mixed-chirality contact deformations for Kerr. In any case, our explicit worldline actions and their chiral presentations establish a unique and powerful formalism that facilitates a systematic addition of mixed-chirality contact deformations without spoiling the same-helicity spin exponentiation at any perturbative order.

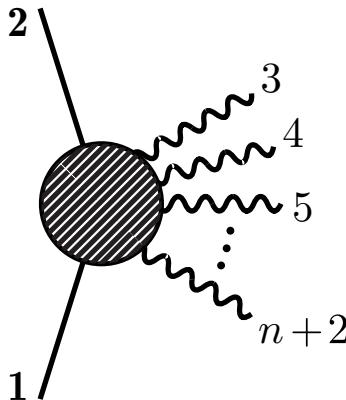


Figure 14.1: The Compton scattering amplitude for massless multiplicity n , which we refer to as “ n -Compton amplitude.” The wavy lines represent massless quanta; the straight lines are massive legs. Note that we always presume equal masses for the massive legs in this thesis, since our scope is strictly restricted to conservative physics. This chapter investigates the cases where the straight lines represent spinning black holes in their point-particle effective theory.

14.1 Introduction

In Sec. 1.2.3, we addressed the startling fact in contemporary relativity that no physicist is aware of the exact EoM of spinning black holes in their point-particle effective theory, even in the probe limit. This is a fairly simple question, at least conceptually, and describes a one-body problem in background, nondynamical geometry. In particular, any hope toward tackling the spinning black hole two-body problem is obstructed by this fundamental issue already.

As also mentioned in Sec. 1.2.3, this problem was first formulated in an invariant fashion by AHH in 2017 [206]. Presuming sufficiently weak external fields, the classical gravitational dynamics of an object as a test particle is completely characterized by specifying its gravitational n -Compton scattering amplitude for all graviton multiplicities $n = 1, 2, 3, \dots$ in the classical limit (see Fig. 14.1). Evidently, this characterization is not only gauge invariant but also field basis invariant. Said in another way, an object as a test particle is uniquely defined by a consistent collection of n -Compton amplitudes. This consistency must include conditions such as correct residues on physical poles and the absence of unphysical poles, in particular.

When one holds only a partial knowledge on the Compton data, the object will be only unambiguously identified up to a certain perturbative order. For example, it could suffice to restrict the multiplicity n to 1 and 2 if one’s goal is to study gravitational physics up to the 2PM order and not beyond.

AHH [206] provided a complete classification of 1-Compton amplitudes for equal masses. It was then established that the “simplest” gravitational 1-Compton amplitude defines the Kerr black hole in linearized gravity [98, 207–209]. In fact, this rephrases the unity $C_\ell = 1$ of Kerr multipole coefficients known from classic works [200–205]. After taking the classical limit for spin, this simplicity describes the celebrated “spin exponentiation” [207, 208] property. This means that the 1-Compton amplitude of the Kerr black hole, say $\mathcal{M}_{\text{spinning}}(3^\pm)$ receiving a graviton $3^\pm$, is related to the 1-Compton amplitude $\mathcal{M}_{\text{scalar}}(3^\pm)$ of the minimally coupled scalar particle (“Schwarzschild black hole”) as

$$\mathcal{M}_{\text{spinning}}(3^\pm) = \mathcal{M}_{\text{scalar}}(3^\pm) e^{\pm k_3 \cdot a}, \quad (14.1)$$

where $a^\mu = -y^\mu$ encodes the spin length of the Kerr black hole.

The natural aspiration of the amplitudes theorists was then to completely bootstrap the Compton data for the Kerr black hole, in which case the probe gravitational dynamics of the Kerr black hole (or more precisely, the point-particle representing it in the effective theory) is uniquely determined to all orders in perturbation theory. It was found that the n -Compton amplitude can be uniquely constructed by gluing the 1-Compton amplitude via recursion [210–212], iff all the incoming graviton helicities are the same.¹ Specifically, the spin exponentiation in Eq. (14.1) is generalized to

$$\begin{aligned} \mathcal{M}_{\text{spinning}}(3^+ \dots n^+) &= \mathcal{M}_{\text{scalar}}(3^+ \dots n^+) e^{+(k_3 + \dots + k_n) \cdot a}, \\ \mathcal{M}_{\text{spinning}}(3^- \dots n^-) &= \mathcal{M}_{\text{scalar}}(3^- \dots n^-) e^{-(k_3 + \dots + k_n) \cdot a}. \end{aligned} \quad (14.2)$$

Crucially, however, AHH [206] had shown that the gluing of the positive-helicity and negative-helicity 1-Compton amplitudes of Kerr by recursion gives a nonsensical 2-Compton amplitude, which is explicitly

$$\mathcal{M}_{\text{spinning}}^{(\text{AHH})}(3^+ 4^-) = \mathcal{M}_{\text{scalar}}(3^+ 4^-) e^{+k_3 \cdot a - k_4 \cdot a} e^\gamma. \quad (14.3)$$

The first exponential factor, $e^{+k_3 \cdot a - k_4 \cdot a}$, describes the expected spin exponentiations. On the contrary, the second exponential factor, e^γ , arises in terms of a

¹In the relativist terminology, this means that the probe gravitational dynamics of Kerr could possibly be completely fixed in purely SD backgrounds. In Chap. IX [768], we have added a relativist axis on this discourse and formulated the probe NJA that uniquely determines the dynamics of Kerr in SD backgrounds. This probe NJA was then shown to be equivalent to (i.e., is the relativist statement of) the same-helicity spin exponentiation.

linear-in-spin parameter γ (a^μ dotted with a certain null vector) that diverges in the backward limit $\theta \rightarrow \pi$ as $\tan \frac{\theta}{2}$; see, e.g., Ref. [221]. Hence the amplitude diverges without any exchange of a physical intermediate state. It follows that the n -Compton amplitude will suffer from the same sort of problem when any of the incoming helicity is different from another.

For general helicity h , the minimally coupled $\mathcal{M}_{\text{scalar}}(3^+4^-)$ contains the suppressing factor $\cos^{2h} \frac{\theta}{2}$. Hence, to become physical, the AHH 2-Compton amplitude in Eq. (14.3) must gain contact deformations such that the power of the problematic parameter γ gets topped off at $2h$ or becomes even smaller. Said in another way, the AHH result implies that the probe dynamics of spinning black holes could possibly be defined without ambiguity up to $\mathcal{O}(a^{2h})$ at most, which is $\mathcal{O}(a^4)$ in the case of gravity.²

Since 2017, a substantial body of work has been devoted to the problem of determining the 2-Compton amplitude for Kerr, demonstrating various perspectives such as a pattern suggested at low orders [213, 214], a gauge redundancy [217–219], black hole perturbation theory [220–223], or string amplitudes [224, 225].

In this chapter, we provide a Lagrangian derivation of the 2-Compton amplitudes for helicities $h = 1$ and 2 , based on the explicit constructions of $\sqrt{\text{Kerr}}$ and Kerr worldline effective actions in Chaps. IX and X [768–770]. Since our actions are local in worldline time and exhibit correct linearized coupling, we expect that the results will show correct factorization without spurious poles by construction. Our actions are also constructed for manifest resummations onto a holomorphic (anti-holomorphic) worldline in SD (ASD) backgrounds, so the same-helicity spin exponentiation is guaranteed to all orders. Our mixed-helicity 2-Compton amplitudes show a unified structure in terms of Hermite interpolation of the problematic factor e^γ of AHH.

In principle, the nonlinearity of YM theory and GR hinders simultaneously manifesting the spin exponentiation in the SD and ASD sectors. To this end, we implement a “googly” solution: perturbing away from the SD sector. We will see how the use of an axial gauge turns out to be fruitful regarding this

²Note that this does not preclude the possibility of contact deformations starting at an earlier order. The same assessment had applied to the same-helicity spin exponentiation in Eq. (14.2) as well, since, strictly speaking, the amplitudes-oriented derivation in Refs. [210–212] describes a proposal for a possible ideal behavior. The same-helicity spin exponentiation had previously been only a beauty argument before Ref. [768], which showed its necessity for a hidden dynamical symmetry in the SD sector: a physical principle, not mere aesthetics.

point.

Our derivations might also appear as top-down, based on the dynamical generalization of the factorization of $\sqrt{\text{Kerr}}$ or Kerr into a chiral dyon or Taub-NUT pair in Chap. VII [164]. The holomorphic worldline is the infrared avatar of the ASD dyon or Taub-NUT constituting $\sqrt{\text{Kerr}}$ or Kerr.

14.2 Summary of Results

This section documents our predictions on the all-orders-in-spin classical Compton scattering amplitudes of $\sqrt{\text{Kerr}}$ and Kerr receiving two photons, gluons, or gravitons in Maxwell theory, YM theory, or GR. Our amplitudes are free of any spurious pole and exhibit correct factorizations.

For same-helicity configurations, we find guaranteed spin exponentiation. For the opposite-helicity configuration, we record the “relative amplitude,”

$$\frac{\mathcal{M}_{\text{spinning}}(3^+4^-)}{\mathcal{M}_{\text{scalar}}(3^+4^-)e^{-\alpha/2}e^{\beta/2}}. \quad (14.4)$$

The relevant kinematic invariants are α, β, γ , which are dimensionless linear-in-spin parameters such that the trivial spin exponentiation is $\mathcal{M}_{\text{spinning}}(3^+4^-) = \mathcal{M}_{\text{scalar}}(3^+4^-)e^{-\alpha/2}e^{\beta/2}$ while the AHH [206] ansatz with the spurious pole is $\mathcal{M}_{\text{spinning}}(3^+4^-) = \mathcal{M}_{\text{scalar}}(3^+4^-)e^{-\alpha/2}e^{\beta/2}e^\gamma$. Explicitly,

$$\alpha := -2k_3 \cdot a, \quad \beta := -2k_4 \cdot a, \quad \gamma := (-2k_3 \cdot p) \frac{w \cdot a}{w \cdot p}, \quad (14.5)$$

where k_3, k_4 are the massless wavenumbers, p is the massive momentum, and $w_{\alpha\dot{\alpha}} \propto 4_\alpha \bar{3}_{\dot{\alpha}}$. We also define the “truncated exponential” functions as

$$E_h(\xi) := \sum_{\ell=h}^{\infty} \frac{\xi^{\ell-h}}{\ell!} = \frac{1}{\xi^h} \left(e^\xi - \sum_{\ell=0}^{h-1} \frac{\xi^\ell}{\ell!} \right). \quad (14.6)$$

Maxwell Theory

$$1 + \gamma \quad (14.7a)$$

YM Theory (nonabelian contribution, color-stripped)

$$1 + \gamma \left(\frac{\gamma + \beta}{\alpha + \beta} E_1(\alpha) - \frac{\gamma - \alpha}{\alpha + \beta} E_1(-\beta) \right) \quad (14.7b)$$

GR

$$1 + \gamma + \gamma^2 \left(\frac{\gamma + \beta}{\alpha + \beta} E_2(\alpha) - \frac{\gamma - \alpha}{\alpha + \beta} E_2(-\beta) \right) \quad (14.7c)$$

The YM and GR results in Eqs. (14.7b) and (14.7c) were obtained by the present author in the year 2024. The YM result was computed during academic travels in Europe in August and September. The GR result was computed in November and finalized by early December. These results were then immediately shared with several colleagues in December, including Clifford Cheung, Jung-Wook Kim, Sangmin Lee, Donal O'Connell, Julio Parra-Martinez, Justin Vines, and Zihan Zhou. They were publicly presented in a talk [876] at Institut des Hautes Études Scientifiques on January 15, 2025.

14.3 Properties

14.3.1 Absence of Spurious Poles

The fact that our amplitudes are free of any spurious pole can be comprehended in a systematic fashion as follows. The point of departure is an intriguing identity that rewrites the AHH relative amplitude:³

$$\begin{aligned} e^\gamma &= \frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \\ &+ (\gamma - \alpha)(\gamma + \beta) \sum_{j=0}^{\infty} \gamma^j \frac{E_{j+1}(\alpha) - E_{j+1}(-\beta)}{\alpha + \beta}. \end{aligned} \quad (14.8)$$

Based on Eq. (14.8), we can consider a truncation of the AHH answer up to a finite power $(h+1) = 1, 2, 3, \dots$ in the parameter γ :

$$\begin{aligned} \text{AHH}[h] &:= \frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \\ &+ (\gamma - \alpha)(\gamma + \beta) \sum_{j=0}^{h-1} \gamma^j \frac{E_{j+1}(\alpha) - E_{j+1}(-\beta)}{\alpha + \beta}. \end{aligned} \quad (14.9)$$

Said in another way, $\text{AHH}[h]$ gives a contact deformation of the AHH Compton amplitude in terms of the difference $\text{AHH}[h] - \text{AHH}[\infty] = (\gamma - \alpha)(\gamma + \beta) \sum_{j=h}^{\infty} \gamma^j (E_{j+1}(\alpha) - E_{j+1}(-\beta)) / (\alpha + \beta)$, where $\text{AHH}[\infty] = e^\gamma$. Also, it should be clear that $\text{AHH}[h]$ is completely regular at $\alpha + \beta = 0$.

Remarkably, we find that our relative amplitudes in Eqs. (14.7b) and (14.7c) are $\text{AHH}[1]$ and $\text{AHH}[2]$, respectively. This should be clear from the fact that Eq. (14.9) admits an alternative representation,

$$\text{AHH}[h] = \left(\sum_{\ell=0}^{h-1} \frac{\gamma^\ell}{\ell!} \right) + \gamma^h \left(\frac{\gamma + \beta}{\alpha + \beta} E_h(\alpha) - \frac{\gamma - \alpha}{\alpha + \beta} E_h(-\beta) \right). \quad (14.10)$$

³The second line in Eq. (14.8) computes a second divided difference $\frac{1}{\alpha + \beta} \left(\frac{e^\alpha - e^\gamma}{\alpha - \gamma} - \frac{e^\gamma - e^{-\beta}}{\gamma + \beta} \right)$ for the exponential function.

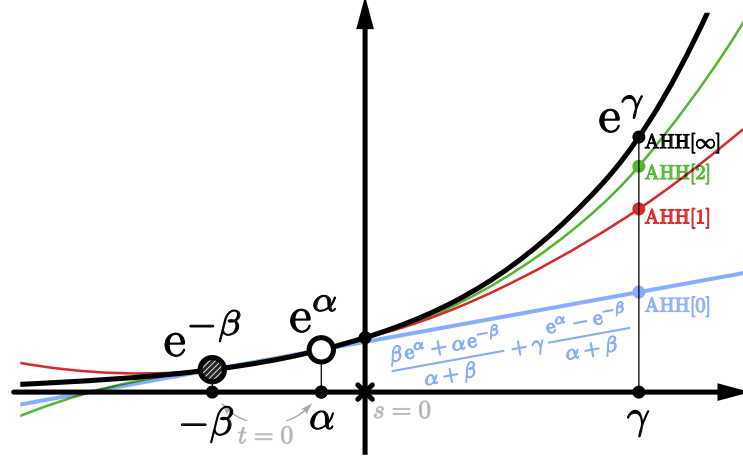


Figure 14.2: The truncation in Eq. (14.9) arises in the context of Hermite interpolation. AHH[0] represents the unique line through $(-\beta, e^{-\beta})$ and (α, e^{α}) . AHH[1] represents the unique parabola through $(-\beta, e^{-\beta})$, (α, e^{α}) , and $(0, 1)$. AHH[2] represents the unique cubic through $(-\beta, e^{-\beta})$, (α, e^{α}) , and $(0, 1)$ that is tangent to e^{γ} at $(0, 1)$.

Consequently, our amplitudes are free from the notorious spurious pole at $\gamma = \infty$. Let us elaborate on this point below.

For helicity h , the relative amplitude develops the notorious spurious pole if it involves a power of γ greater than $2h$, which is the content of the famous AHH result [206]. Within the above particular truncated class, the helicity-1 relative amplitude can in principle maximally extend to AHH[1], whereas the helicity-2 relative amplitude can in principle maximally extend to AHH[3]. In particular, the expansion in spin length gives

$$\text{AHH}[1] = 1 + \frac{\gamma^1}{1!} + \frac{\gamma^2}{2!} + \frac{1}{3!} \left(\gamma^2(\alpha - \beta) + \gamma\alpha\beta \right) + \mathcal{O}(a^4), \quad (14.11)$$

$$\text{AHH}[2] = 1 + \frac{\gamma^1}{1!} + \frac{\gamma^2}{2!} + \frac{\gamma^3}{3!} + \frac{1}{4!} \left(\gamma^3(\alpha - \beta) + \gamma^2\alpha\beta \right) + \mathcal{O}(a^5),$$

$$\text{AHH}[3] = 1 + \frac{\gamma^1}{1!} + \frac{\gamma^2}{2!} + \frac{\gamma^3}{3!} + \frac{\gamma^4}{4!} + \frac{1}{5!} \left(\gamma^4(\alpha - \beta) + \gamma^3\alpha\beta \right) + \mathcal{O}(a^6),$$

so the contact deformations are introduced from $\mathcal{O}(a^3)$, $\mathcal{O}(a^4)$, and $\mathcal{O}(a^5)$ for AHH[1], AHH[2], AHH[3], respectively.

Our helicity-2 relative amplitude in Eq. (14.7c) is AHH[2], which means that it introduces an additional contact deformation that is not strictly mandatory, at γ^4 . This could appear as a peculiarity, but we may note that the Maxwell answer in Eq. (14.7a) also has entailed such an extra deformation, as is first

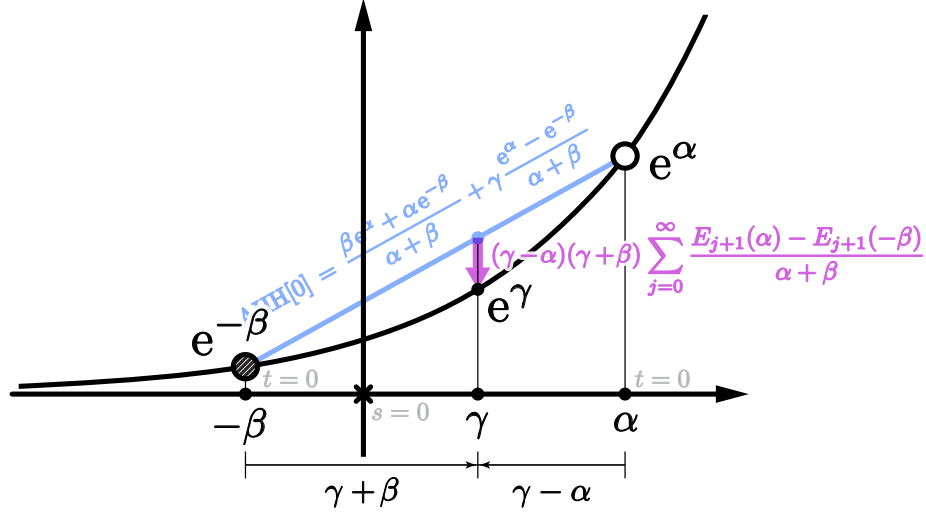


Figure 14.3: A visualization of the identity in Eq. (14.8). This uses the same geometric interpretation as Fig. 14.2.

noted in Ref. [795]. In fact, it might be possible to argue AHH[1] and AHH[2] from the order of nonlocality involved in the Dirac/Misner strings.⁴ On a related note, it will be interesting if a double copy relation could be established, or a useful integral representation of the relative amplitudes reflecting the open (Dirac/Misner, not fundamental) string structure is viable.

More generally, one can ask if the AHH[h] class of ansatz arises uniquely as the result of a bootstrap. Note that the identity in Eq. (14.8) admits a nice geometric interpretation: see Figs. 14.2 and 14.3.

14.3.2 Correct Residues on Physical Poles

The fact that our amplitudes exhibit correct residues on physical poles can be shown as follows (cf. Sec 3.1 of Ref. [221]). To clarify, we strictly work in the classical limit where the massless wavenumbers k_3, k_4 take macroscopic values. With the definitions

$$K_3 := 2k_3 \cdot p, \quad K_4 := 2k_4 \cdot p, \quad t := -2k_3 \cdot k_4, \quad (14.12)$$

the Mandelstam identity gives $K_3 + K_4 = \hbar t$ so that, in the classical limit,

$$K_3 = -K_4 \quad (14.13)$$

while t can take any value.

⁴Note that $E_h(\xi) = \int_0^1 d\eta_1 \int_0^{\eta_1} d\eta_2 \int_0^{\eta_2} d\eta_3 \cdots \int_0^{\eta_{h-1}} d\eta_h e^{\eta_h \xi}$ describes the exponential factor integrated h times. It will also be interesting to formulate orbit-spin duality at the amplitudes level.

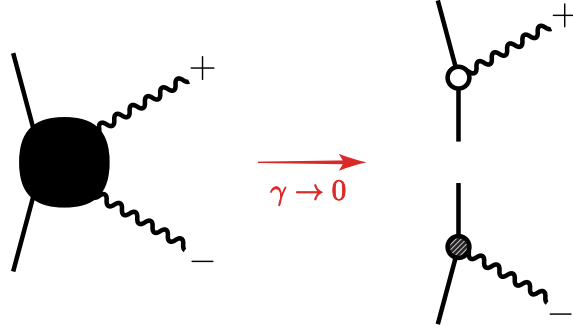


Figure 14.4: s -channel factorization of black hole Compton amplitudes.

To begin with, suppose the generic form of the spinning amplitude,

$$\mathcal{M}_{\text{spinning}}(3^+4^-) = \mathcal{M}_{\text{scalar}}(3^+4^-) e^{-\alpha/2} e^{\beta/2} \mathcal{R}(\alpha, \beta, \gamma), \quad (14.14)$$

where the relative amplitude $\mathcal{R}(\alpha, \beta, \gamma)$ is presumed to be regular at $\gamma = 0$.

Firstly, the s -channel factorization corresponds to the limit $K_3 \rightarrow 0$. We import the factorization of the scalar amplitude in the limit $K_3 \rightarrow 0$,

$$\mathcal{M}_{\text{scalar}}(3^+4^-) \sim \mathcal{M}_{\text{scalar}}(3^+) \frac{1}{K_3} \mathcal{M}_{\text{scalar}}(4^-). \quad (14.15)$$

Secondly, the t -channel factorization corresponds to the limit $t \rightarrow 0$, which will be only discussed for YM and GR amplitudes. By complexification, this boils down to either $\langle 34 \rangle = 0$ or $[34] = 0$. In these cases, the scalar amplitude factorizes as

$$\langle 34 \rangle = 0 : \quad \mathcal{M}_{\text{scalar}}(3^+4^-) \sim \mathcal{M}_{\text{scalar}}(5^-) \frac{1}{-t} \mathcal{M}_{\text{massless}}(3^+4^-5^+), \quad (14.16a)$$

$$[34] = 0 : \quad \mathcal{M}_{\text{scalar}}(3^+4^-) \sim \mathcal{M}_{\text{scalar}}(5^+) \frac{1}{-t} \mathcal{M}_{\text{massless}}(3^+4^-5^-), \quad (14.16b)$$

where $\mathcal{M}_{\text{massless}}(3^+4^-5^\pm)$ are the three-point gluon or graviton amplitudes (MHV and anti-MHV):

$$|3\rangle \propto |4\rangle \propto |5\rangle \quad \leftrightarrow \quad \mathcal{M}_{\text{massless}}(3^+4^-5^+) \propto \frac{[35]^4}{[34][45][35]}, \quad (14.17a)$$

$$[3] \propto [4] \propto [5] \quad \leftrightarrow \quad \mathcal{M}_{\text{massless}}(3^+4^-5^-) \propto \frac{\langle 45 \rangle^4}{\langle 34 \rangle \langle 45 \rangle \langle 35 \rangle}. \quad (14.17b)$$

We are now ready to discuss the factorizations of the generic spinning amplitude $\mathcal{M}_{\text{spinning}}(3^+4^-) = \mathcal{M}_{\text{scalar}}(3^+4^-) e^{-\alpha/2} e^{\beta/2} \mathcal{R}(\alpha, \beta, \gamma)$ in Eq. (14.14).

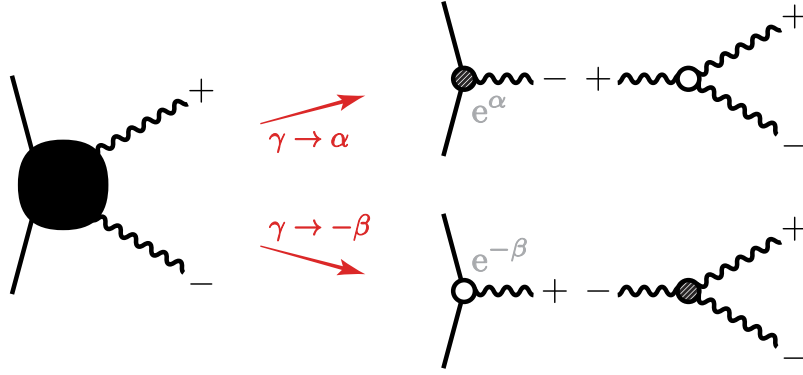


Figure 14.5: t -channel factorizations of black hole Compton amplitudes.

Firstly, the s -channel factorization entails

$$K_3 \rightarrow 0 \implies \gamma = -K_3 \frac{w \cdot a}{w \cdot p} \rightarrow 0. \quad (14.18)$$

With this understanding, Eq. (14.15) implies that the leading asymptotic of $\mathcal{M}_{\text{spinning}}(3^+4^-)$ in the limit $K_3 \rightarrow 0$ is given by

$$\left(\mathcal{M}_{\text{scalar}}(3^+) e^{-\alpha/2} \right) \frac{1}{K_3} \left(\mathcal{M}_{\text{scalar}}(4^-) e^{\beta/2} \right) \mathcal{R}(\alpha, \beta, 0). \quad (14.19)$$

Crucially, we see that all of the three relative amplitudes in Eq. (14.7) satisfy

$$\mathcal{R}(\alpha, \beta, 0) = 1, \quad (14.20)$$

in which case Eq. (14.19) precisely gives the desired residue: the product of the spin-exponentiated three-point spinning amplitudes.

Secondly, the t -channel factorization entails⁵

$$\langle 34 \rangle = 0 \implies w_{\alpha\dot{\alpha}} \propto 3_{\alpha} \bar{3}_{\dot{\alpha}} \implies \gamma = (-2k_3 \cdot p) \frac{k_3 \cdot a}{k_3 \cdot p} = \alpha, \quad (14.21a)$$

$$[34] = 0 \implies w_{\alpha\dot{\alpha}} \propto 4_{\alpha} \bar{4}_{\dot{\alpha}} \implies \gamma = (+2k_4 \cdot p) \frac{k_4 \cdot a}{k_4 \cdot p} = -\beta. \quad (14.21b)$$

With this understanding, Eq. (14.16) implies that the leading asymptotics of $\mathcal{M}_{\text{spinning}}(3^+4^-)$ in the limit $t \rightarrow 0$ are given by

$$\left(\mathcal{M}_{\text{scalar}}(5^-) e^{(\alpha+\beta)/2} \right) \left(e^{-\alpha} \mathcal{R}(\alpha, \beta, \alpha) \right) \frac{1}{-t} \mathcal{M}_{\text{massless}}(3^+4^-5^+), \quad (14.22a)$$

$$\left(\mathcal{M}_{\text{scalar}}(5^+) e^{-(\alpha+\beta)/2} \right) \left(e^{+\beta} \mathcal{R}(\alpha, \beta, -\beta) \right) \frac{1}{-t} \mathcal{M}_{\text{massless}}(3^+4^-5^-), \quad (14.22b)$$

⁵Note also that the Britto-Cachazo-Feng-Witten (BCFW) [877, 878] shift of the massless momenta for the s -channel factorization induces $\alpha \mapsto \alpha - \gamma$ and $\beta \mapsto \beta + \gamma$.

for $\langle 34 \rangle = 0$ and $[34] = 0$, respectively. Crucially, both of the relative amplitudes in Eqs. (14.7b) and (14.7c) satisfy the conditions

$$\mathcal{R}(\alpha, \beta, \alpha) = e^\alpha, \quad \mathcal{R}(\alpha, \beta, -\beta) = e^{-\beta}, \quad (14.23)$$

in which case Eq. (14.22) precisely establishes the desired factorization into the spin-exponentiated three-point spinning amplitude times the massless MHV or anti-MHV amplitude. To elaborate, it is easy to see that plugging in $\gamma = \alpha$ or $\gamma = -\beta$ to $\text{AHH}[h]$ in Eq. (14.9) respectively gives e^α and $e^{-\beta}$, which can be also directly seen from the geometric interpretation given in Fig. 14.3.

The above discussion endows us with a clear breakdown of the $\text{AHH}[h]$ ansatz in Eq. (14.9) and explicates how restrictive it is. In general, the ansatz for the relative amplitude can be

$$\begin{aligned} \mathcal{R}(\alpha, \beta, \gamma) = & \left(\frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \right) \\ & + (\gamma - \alpha)(\gamma + \beta) \mathcal{Q}(\alpha, \beta, \gamma), \end{aligned} \quad (14.24)$$

where $\mathcal{Q}(\alpha, \beta, \gamma)$ is polynomial in γ . The first group of terms on the right-hand side gives a “linear interpolation” between e^α and $e^{-\beta}$, while $(\gamma - \alpha)(\gamma + \beta) \rightarrow 0$ when $\gamma \rightarrow \alpha$ or $-\beta$ so that the t -channel factorization condition in Eq. (14.23) is guaranteed. Imposing the s -channel condition in Eq. (14.20) then yields

$$\mathcal{R}(\alpha, \beta, 0) = 1 \quad \implies \quad \mathcal{Q}(\alpha, \beta, 0) = \frac{E_1(\alpha) - E_1(-\beta)}{\alpha + \beta}. \quad (14.25)$$

Hence $\mathcal{Q}(\alpha, \beta, \gamma)$ in Eq. (14.24) can be taken as

$$\mathcal{Q}(\alpha, \beta, \gamma) = \frac{E_1(\alpha) - E_1(-\beta)}{\alpha + \beta} + \sum_{j=1}^{\infty} \gamma^j q_j(\alpha, \beta). \quad (14.26)$$

Consequently, the residual freedom for helicity h boils down to a set of functions $q_j(\alpha, \beta)$ for $j = 1, 2, \dots$. Eventually, by demanding the absence of spurious poles as well, we find that the case $h = 1$ allows for *no residual freedom* whereas the case $h = 2$ allows for choosing two functions, $q_1(\alpha, \beta)$ and $q_2(\alpha, \beta)$.

14.3.3 Comparison with Literature

Arguably, our amplitudes in Eqs. (14.7b) and (14.7c) exhibit a degree of elegance and simplicity while pointing to the suggestive truncation structure as $\text{AHH}[h]$.

Our amplitudes, however, do not agree with the amplitudes obtained or advocated in a group of works [217–223]. We can highlight three points.

- The amplitudes in Refs. [217–223] seem to appear more complicated than our prediction when brought to the same kinematic basis.
- The amplitudes in Refs. [217–223], to our understanding, seem to attach a special meaning to the center of the black hole on the real spacetime. In our opinion, this dependence seems rather spurious or even unphysical in view of the factorization of black holes [164] shown in Chap. VII: there is nothing on the midpoint of the Dirac/Misner strings, which are gauge artifacts after all. Recall that the same-helicity spin exponentiations are none other than localizations on the SD/ASD Taub-NUT worldlines.
- The amplitudes in Refs. [217–223] are dependent on yet another kinematic invariant: the optical parameter,

$$\xi^{-1} := \frac{m^2 t}{K_3^2}. \quad (14.27)$$

As a result, the absence of the spurious pole is achieved through a more involved mechanism, counting on a coordinated cancellation between the poles in γ and the optical parameter, according to Sec. 3.3 of Ref. [221].

14.4 Methodology

This section outlines our methodology for obtaining the classical Compton amplitudes in Eq. (14.7). Technical details are left to App. B.

In short, we carry out worldline perturbation theories of the explicit first-order $\sqrt{\text{Kerr}}$ and Kerr actions in Chap. X, implemented in the massive twistor model. The massive twistor space is essentially a phase space for a massive spinning particle; recall Chap. XII.

For Compton scattering with gluons or gravitons, the universal Feynman rules of symplectic sigma models [763] (reviewed in App. B.1.3) imply that it suffices to compute the interaction symplectic potential θ' in the background of nonlinearly superposed plane waves—those obtained in Sec. 2.5.6—and the symplectic perturbation ω' in the linearized background of a single plane wave. The former gives a vertex of fluctuation degree (meaning the valence of worldline fluctuations involved) zero and interaction degree (meaning the number of massless legs involved) two. The latter gives a vertex of fluctuation degree one and interaction degree one. Vertices of the latter kind describe a linear coupling, so they can be simply obtained by directly complexifying (NJ shifting, spin exponentiating) the corresponding vertices in the spinless case.

14.4.1 Vertices and Master Formula for Compton Amplitude

Explicitly, given a sigma model with a symplectic target equipped with coordinates ζ^i that are Darboux with respect to the free theory, the vertices of fluctuation degree zero and one are respectively given by

$$\mathcal{V}^{(0)} = \int d\sigma \theta'_i(\zeta(\sigma)) \dot{\zeta}^i, \quad \mathcal{V}^{(1)}(\sigma) = \delta\zeta^i(\sigma) \omega'_{ij}(\zeta(\sigma)) \dot{\zeta}^j, \quad (14.28)$$

assuming expansion around a free theory saddle $\sigma \mapsto \zeta^i(\sigma)$ whose time derivative is constant. Plugging in the nonlinearly superposed plane wave backgrounds generically yields the specific time dependence,

$$\begin{aligned} \theta'_i(\zeta(\sigma)) &= \epsilon_3 \theta'^3_i e^{iK_3\sigma} + \epsilon_4 \theta'^4_i e^{iK_4\sigma} + \epsilon_3 \epsilon_4 \theta'^{34}_i e^{i(K_3+K_4)\sigma}, \\ \omega'_{ij}(\zeta(\sigma)) &= \epsilon_3 \omega'^3_{ij} e^{iK_3\sigma} + \epsilon_4 \omega'^4_{ij} e^{iK_4\sigma} + \epsilon_3 \epsilon_4 \omega'^{34}_{ij} e^{i(K_3+K_4)\sigma}. \end{aligned} \quad (14.29)$$

As a result, a universal master formula for the Compton amplitude arises that

$$\mathcal{M} = \theta'^{34}_i \dot{\zeta}^i + \omega'^3_{ik} \dot{\zeta}^k \omega'^4_{jl} \dot{\zeta}^l \langle \delta\zeta^i(K_3) \delta\zeta^j(-K_3) \rangle. \quad (14.30)$$

Here, $\langle \delta\zeta^i(K_3) \delta\zeta^j(-K_3) \rangle$ describes the worldline propagator in the frequency domain, incorporating all quadratic vertices in the free theory. Its causality prescription does not matter much at this order.⁶

14.4.2 Propagators

The worldline propagators in the time domain can be derived from frameworks described in App. B (see also Ref. [795]), using a free-theory saddle such that

$$z^\mu(\sigma) = z^\mu + 2p^\mu \sigma, \quad \bar{z}^\mu(\sigma) = \bar{z}^\mu + 2p^\mu \sigma, \quad p_\mu(\sigma) = p_\mu. \quad (14.31)$$

Here, z^μ , $\bar{z}^\mu$, and p_μ are *constants* specifying the background worldline; $x^\mu = (z^\mu + \bar{z}^\mu)/2$ is the impact parameter (b^μ in usual notation); $y^\mu = (z^\mu - \bar{z}^\mu)/2i$ is the spin length pseudovector ($-a^\mu$ in usual convention). With the definition

$$\langle \delta\zeta^i(K_3) \delta\zeta^j(K_4) \rangle = \int d\sigma_3 d\sigma_4 e^{iK_3\sigma_3} e^{iK_4\sigma_4} \langle \delta\zeta^i(\sigma_3) \delta\zeta^j(\sigma_4) \rangle, \quad (14.32)$$

it follows that

$$\langle \delta z^\mu(K_3) \delta p_\nu(K_4) \rangle = \frac{-i\delta^\mu_\nu}{K_3}, \quad (14.33a)$$

$$\langle \delta z^\mu(K_3) \delta z^\nu(K_4) \rangle = \langle \delta \bar{z}^\mu(K_3) \delta \bar{z}^\nu(K_4) \rangle = -\frac{2\eta^{\mu\nu}}{K_3^2}, \quad (14.33b)$$

$$\langle \delta z^\mu(K_3) \delta \bar{z}^\nu(K_4) \rangle = \frac{2}{m^2} \left(-iS^{\mu\nu} - 2y^{(\mu} p^{\nu)} \right) \frac{1}{K_3} - \frac{2\eta^{\mu\nu} + 4\hat{p}^\mu \hat{p}^\nu}{K_3^2}, \quad (14.33c)$$

⁶There exist concrete prescriptions in which Eq. (14.30) can be legitimately called a scattering amplitude (in connection with Chap. XII [672]) or holds a direct classical significance.

on the support of $K_3 + K_4 = 0$. Here, $S^{\mu\nu} = \varepsilon^{\mu\nu\rho\sigma} y_\rho p_\sigma$ and $\hat{p}^\mu = p^\mu/m$.⁷ When working with the color extension, the adjoint color variable describes

$$\langle \delta q_a(K_3) \delta q_b(K_4) \rangle = \frac{-i q_c f_{ab}^c}{K_3}, \quad (14.33d)$$

where q_c refers to the background value: $\mathbf{q}_a(\sigma) = q_a$. To clarify, all of δz^μ , $\delta \bar{z}^\mu$, δp_μ , and δq_a in Eq. (14.33) are fluctuations of composite variables.

14.5 Interpretation

The propagators in Eqs. (14.33b) and (14.33c) admit a physical interpretation in the spinspace-time framework. The *second-order* poles represent *inertia* as the proportionality between force and acceleration, as is supported by its diagrammatic origin as well. The *first-order* pole in Eq. (14.33c) is evidently an incarnation of the “zig-zag” spinspace-time bracket in Eqs. (12.9) or (13.35b), which will represent an *angular momentum* effect.

In sum, intuitively speaking, the above propagators characterize the precise behavior in which the zig and zag points “wobble” when each of them is “hit by a hammer” by external perturbations. Note that the zig and zag points are where the SD and ASD couplings resum onto. For instance, recall that the abelian $\sqrt{\text{Kerr}}$ action cleanly splits into the interaction actions coupling SD gauge potential to the zig worldline and ASD gauge potential to the zag worldline, the observation of which traces back to Ref. [226]. Overall, the perturbation theory encourages and prompts us to imagine the zig and zag points as actual objects.

Remarkably, a vivid picture emerges by the revelation in Chap. VII [164]: the factorization of Kerr into Taub-NUT pairs. The zig point z is precisely where the ASD Taub-NUT center is located at in the complexified spacetime, while the zag point $\bar{z}$ is precisely where the SD Taub-NUT center is located at. In fact, we call ASD Taub-NUT a “zig” and SD Taub-NUT a “zag.” In *zig-zag theory*, zigs and zags were particles that exhibit a helicity selection rule, whose existence was first speculated out of the zig-zag twistor particle program [673] on account of the holomorphy of SD symplectic perturbation. Zigs can only absorb positive-helicity massless quanta and thus can only source negative-helicity massless quanta. It was represented as a white blob, on account of

⁷In fact, it turns out that the “longitudinal” components of Eq. (14.33c) with respect to p^μ could be freely discarded, reminiscently of a Ward identity.

the standard on-shell diagrams notation: recall Fig. 12.2. Amusingly, Fig. 7.3 discloses that this white blob is *real* and actually resides inside the Kerr black hole. It turns out that the zig is none other than the ASD Taub-NUT instanton. Indeed, the ASD Taub-NUT instanton enjoys the proposed helicity selection rule for zig [261, 263, 264].

Eq. (14.33b) states the zig and zag have (their own) inertia. Eq. (14.33c) tells us that, when one hits zag by inputting an ASD wave, the zig first moves in a “*skew*” direction just as in the case of gyroscope motion—that of the spinning *bicycle wheel*—as the first-order effect (which is precisely the content of the zig-zag spinspace bracket), then starts to drift/accelerate gradually by the second-order effect. Again, this description is all about how the white and black blobs in Fig. 7.3 move and react to external forces as actual physical entities, or, more precisely, their avatars in the infrared theory.

Note that the zig and zag locations are also Newman’s “complex centers of mass” [252–259, 329, 330], in which case we learn that Newman’s provisional ideas have been completed into a concrete picture: a fully dynamical theory of the zig-zag system, defined beyond SD backgrounds.

It is instructive to consider the case of Maxwell theory. The $\sqrt{\text{Kerr}}$ solution is secretly a pair of a zig (ASD dyon) and a zag (SD dyon) [164, 202]. No complicated calculation is needed for showing that the ASD dyon exhibits the precise helicity selection rule of a zig. As a test particle, a zig sees only the SD part of external fields via its Lorentz force law $d\vec{p}/dt = q(1 - i\vec{v} \times)(\vec{E} + i\vec{B})$. As a background, a zig always sources an ASD field regardless of motion. For a static zig, for instance, the fields are $\vec{E} = q\vec{x}/4\pi|\vec{x}|^3$, $\vec{B} = iq\vec{x}/4\pi|\vec{x}|^3$.

14.6 Kinematic Bases for Relative Amplitudes

As another preliminary, let us also discuss the kinematic invariants and their relations for each helicity configurations. To begin with, suppose two massless momenta k_3, k_4 and their associated bivector polarizations φ_3, φ_4 , which can carry any helicity at this moment. Suppose a massive momentum $p^2 = -m^2$ and a spin length pseudovector y subject to the condition $p \cdot y = 0$. The goal is to construct invariants out of these tensors and identify a minimal basis. In particular, let us seek the minimal basis for dimensionless and little group neutral variables, on which the relative amplitude can depend.⁸

⁸Regarding such invariants at higher multiplicities, we had explored multi-photon scattering off abelian $\sqrt{\text{Kerr}}$ in Maxwell theory by taking twistorial variables as fundamental. It

First of all, the invariants can be organized in terms of the number of polarizations involved and the mass dimension, for instance. Schematically, they are kk , kp , ky , $k\varphi p$, $y\varphi k$, $y\varphi p$, $\varphi\varphi$, $p\varphi\varphi p$, $y\varphi\varphi p$, $y\varphi\varphi y$, etc., together with the “spin magnitude” y^2 . Of course, not all of these are independent. By utilizing the on-shell conditions on the bivector polarizations, it can be shown that

$$k_4\varphi_3p = -\frac{p\varphi_3\varphi_4p}{2A_4}(\gamma_{43}+\beta), \quad (14.34a)$$

$$k_3\varphi_4p = +\frac{p\varphi_3\varphi_4p}{2A_3}(\gamma_{34}-\alpha), \quad (14.34b)$$

$$+\frac{y\varphi_3k_4}{k_3y} = \frac{p\varphi_3\varphi_4p}{K_3A_3\beta}(\delta - (\gamma_{34}-\alpha)\beta), \quad (14.34c)$$

$$-\frac{y\varphi_4k_3}{k_4y} = \frac{p\varphi_3\varphi_4p}{K_3A_4\alpha}(\delta + (\gamma_{43}+\beta)\alpha), \quad (14.34d)$$

$$\varphi_3 \cdot \varphi_4 = \frac{p\varphi_3\varphi_4p}{m^2\xi} \left(\frac{(\gamma_{34}-\alpha)(\gamma_{43}+\beta)}{\delta} - 1 \right), \quad (14.34e)$$

$$y\varphi_3\varphi_4y = \frac{p\varphi_3\varphi_4p}{K_3^2}(\delta + \gamma_{43}\alpha - \gamma_{34}\beta + \alpha\beta), \quad (14.34f)$$

$$ty^2 = \frac{\gamma^2}{\xi} + \delta. \quad (14.34g)$$

Here, we have denoted

$$A_3 := y\varphi_3p, \quad A_4 := y\varphi_4p, \quad (14.35)$$

while $p\varphi_3\varphi_4p$ abbreviates $p^\rho\varphi_{3\rho}{}^\mu\varphi_{4\mu\sigma}p^\sigma$ and so on. Also, $\varphi_3 \cdot \varphi_4 := \frac{1}{2}\varphi_{3\mu\nu}\varphi_4^{\mu\nu}$. We have identified dimensionless and little group neutral variables,

$$\begin{aligned} \alpha &:= 2k_3 \cdot y, \quad \beta := 2k_4 \cdot y, \quad \gamma_{34} := K_3 \frac{y\varphi_3\varphi_4p}{p\varphi_3\varphi_4p}, \\ \delta &:= -2 \frac{A_3A_4t}{p\varphi_3\varphi_4p}, \quad \gamma_{43} := K_3 \frac{y\varphi_4\varphi_3p}{p\varphi_3\varphi_4p}, \end{aligned} \quad (14.36)$$

while using

$$K_3 := 2k_3 \cdot p, \quad \xi := \frac{K_3^2}{m^2t} \quad (14.37)$$

as a substitute for the Mandelstams in the classical limit as introduced earlier in Eqs. (14.12), (14.13), and (14.27). The relative amplitude will be a function of the dimensionless little group neutrals: $\alpha, \beta, \gamma_{34}, \gamma_{43}, \delta, \xi$ and ty^2 .

seems that the integrands can be represented in the Hodges diagram notation with infinity twistors and a slight massive extension, for which the massless dual twistors and twistors are respectively incident on the holomorphic or anti-holomorphic spacetime coordinates to systematically construct kinematic invariants. Amusingly, the γ factor arises via the contraction $(\bar{W}_3)^A(W_4)_A \propto [\bar{3}|z(\sigma_3)-\bar{z}(\sigma_4)|4\rangle$ between massless twistor and dual twistor.

14.6.1 Same-Helicity Configurations

The same-helicity configurations are characterized by the identity

$$\frac{1}{2}(\varphi_3\varphi_4 + \varphi_4\varphi_3)_{\mu\nu} = -\eta_{\mu\nu}(\varphi_3 \cdot \varphi_4), \quad (14.38)$$

which is easy to see via the spinor-helicity formalism.

The implications of Eq. (14.38) are the following. First, $\gamma_{34} + \gamma_{43} = 0$ since $p \cdot y = 0$. Hence we can set

$$\gamma_{34} = -\gamma_{43} =: \gamma. \quad (14.39)$$

Second, $p\varphi_3\varphi_4p = m^2(\varphi_3 \cdot \varphi_4)$, so Eq. (14.34e) gives

$$\delta = -\frac{(\gamma - \alpha)(\gamma - \beta)}{1 + \xi}. \quad (14.40)$$

Third, $y^2 = -(y\varphi_3\varphi_4y)/(\varphi_3 \cdot \varphi_4)$, which reproduces Eq. (14.34g). To conclude, the minimal basis for the relative same-helicity Compton amplitude can be given as $(\alpha, \beta, \gamma, \xi)$ with $\gamma_{34} = -\gamma_{43} = \gamma$, or equivalently $(\alpha, \beta, \gamma, \delta)$.

14.6.2 Mixed-Helicity Configuration

The mixed-helicity configuration, $(+, -)$, is characterized by

$$[\varphi_3, \varphi_4] = 0, \quad \varphi_3 \cdot \varphi_4 = 0, \quad (14.41)$$

which are again trivial in the spinor-helicity formalism.

First of all, $[\varphi_3, \varphi_4] = 0$ implies that

$$\gamma_{34} = \gamma_{43} =: \gamma. \quad (14.42)$$

Second of all, $\varphi_3 \cdot \varphi_4 = 0$ implies, via Eq. (14.34e), that

$$\delta = (\gamma - \alpha)(\gamma + \beta). \quad (14.43)$$

Thirdly, Eq. (14.34f) boils down to

$$y\varphi_3\varphi_4y = \frac{p\varphi_3\varphi_4p}{K_3^2} \gamma^2. \quad (14.44)$$

To conclude, the minimal basis for the relative mixed-helicity Compton amplitude can be given as $(\alpha, \beta, \gamma, \xi)$, with $\gamma_{34} = \gamma_{43} = \gamma$. It is easily seen that this γ reproduces the earlier definition in Eq. (14.5), via $(\varphi_3\varphi_4)_{\mu\nu} \propto w_\mu w_\nu$:

$$w_{\alpha\dot{\alpha}} \propto 4_\alpha \bar{3}_{\dot{\alpha}} \implies \gamma = K_3 \frac{w \cdot y}{w \cdot p}. \quad (14.45)$$

Note that the spin-magnitude y^2 is not an independent invariant in any helicity configurations, which should be clear from Eq. (14.34g) (cf. Ref. [221]). On a related note, our scope here is strictly restricted to conservative dynamics.

14.7 Nonabelian Root-Kerr Compton Amplitudes

We scatter the nonabelian $\sqrt{\text{Kerr}}$ particle, implemented in the phase space $\text{MT} \times \mathbb{C}^N$ where MT is the massive twistor space and $\mathbb{C}^N$ is the fundamental representation space of $\text{SU}(N)$, off the nonlinearly superposed background of two plane waves in $\text{G} = \text{SU}(N)$ YM theory.

14.7.1 The Two-Gluon Background

Earlier in Eq. (2.177), we had constructed the on-shell two-gluon geometry in YM theory while using an axial gauge condition. We adopted covariant CK duality [90], which crucially allows for a structured/principled organization of perturbation theory in terms of SD and ASD sectors. The field strength was given by (evaluating at the holomorphic position with hindsight)

$$F^{a'} = \frac{ig\epsilon_3}{\sqrt{2}} c_3^{a'} \varphi_3 e^{ik_3 z} + \frac{ig\epsilon_4}{\sqrt{2}} c_4^{a'} \varphi_4 e^{ik_4 z} + g^2 \epsilon_3 \epsilon_4 c_{34}^{a'} \left(\varphi_{34} - \varphi_3 \frac{\eta \varphi_4 k_3}{k_4 \eta} + \varphi_4 \frac{\eta \varphi_3 k_4}{k_3 \eta} \right) \frac{1}{t} e^{ik_{34} z}, \quad (14.46)$$

where $c_{34} := [c_3, c_4]$ by the $\text{SU}(N)$ Lie bracket while $\varphi_{34} := [\varphi_3, \varphi_4]$ by the $\text{so}(1, 3)$ Lie bracket.

In general, one could orient the reference vector η^μ along a generic direction and see if its dependence drops out from the final results. This will provide an explicit check of the gauge invariance of the Compton amplitudes. To simplify calculations, however, one can also consider making a clever choice of η^μ .

Let us postpone the verification of gauge invariance to Sec. 14.7.7. With hindsight, we find that the spin length vector y^μ of the *background* worldline qualifies as a perfect candidate for the reference vector, as it is nondynamical. In fact, the nonlocal operator $1/\eta \cdot \partial$ in axial gauge is naturally associated with the Dirac string, while we recall from Chap. VII [164] that the $\sqrt{\text{Kerr}}$ solution represents a pair of chiral dyons joined by a Dirac string oriented in the direction of iy^μ .

We refer to this particular gauge, $\eta^\mu \propto y^\mu$, as the “aligned axial gauge.” With this choice made, Eq. (14.46) boils down to

$$F^{a'} = \frac{ig\epsilon_3}{\sqrt{2}} c_3^{a'} \varphi_3 e^{ik_3 z} + \frac{ig\epsilon_4}{\sqrt{2}} c_4^{a'} \varphi_4 e^{ik_4 z} + g^2 \epsilon_3 \epsilon_4 c_{34}^{a'} \left(\varphi_{34} - \varphi_3 \frac{y \varphi_4 k_3}{k_4 y} + \varphi_4 \frac{y \varphi_3 k_4}{k_3 y} \right) \frac{1}{t} e^{ik_{34} z}, \quad (14.47)$$

where Eqs. (14.34c) and (14.34d) can be applied.

14.7.2 The Two-Gluon Symplectic Structure

The $\sqrt{\text{Kerr}}$ symplectic potential is given in Eq. (10.59), in the googly form:

$$\theta_{\sqrt{\text{Kerr}}} = i \bar{Z}_I^A dZ_A^I + i \bar{\psi}_{i'} D\psi^{i'} - 2i E_1(-2i \mathcal{L}_N)[q_{a'} \iota_N F^{-a'}], \quad (14.48)$$

where we see that the truncated exponential function E_1 is in play, in connection with the Dirac string. With the identification of the free theory as

$$\theta^\circ = i \bar{Z}_I^A dZ_A^I + i \bar{\psi}_{i'} d\psi^{i'}, \quad (14.49)$$

the interaction symplectic potential for $\sqrt{\text{Kerr}}$ is

$$\theta'_{\sqrt{\text{Kerr}}} = q_{a'} A^{a'} + \sum_{\ell=1}^{\infty} \frac{(-2i)^\ell}{\ell!} q_{a'} \left((P_\ell^-)^{a'}{}_\sigma dz^\sigma + (\ell-1) (P_{\ell-1}^-)^{a'}{}_\sigma dy^\sigma \right), \quad (14.50)$$

so $\theta_{\sqrt{\text{Kerr}}} = \theta^\circ + \theta'_{\sqrt{\text{Kerr}}}$. Crucially, Eq. (14.49) employs the primed color variables. In this perturbation scheme, the color phase space $\mathbb{C}^N$ takes $\psi^{i'}$ as coordinates.

To study the physics of the $\sqrt{\text{Kerr}}$ particle in the two-gluon background, one plugs in Eq. (14.46) to Eq. (14.50). To this end, let $\mathbf{F} = \delta\zeta^i \partial/\partial\zeta^i$ be a section of $T(\text{MT} \times \mathbb{C}^N)$. Let $\mathbf{T}$ be a Hamiltonian vector field on $\text{MT} \times \mathbb{C}^N$ such that

$$\iota_{\mathbf{T}} dz^\mu = 2p^\mu, \quad \iota_{\mathbf{T}} dy^\mu = 0, \quad \iota_{\mathbf{T}} dp_\mu = 0, \quad \iota_{\mathbf{T}} dq_{a'} = 0. \quad (14.51)$$

The astute reader will recognize that $\mathbf{F}$ gives the notion of fluctuation, whereas $\mathbf{T}$ refers to the time-evolution vector field of the free theory described in Eq. (14.31). With this formulation, the universal formulae in Eq. (14.28) amount to computing the following geometric expressions on the saddle trajectory:

$$\mathcal{V}^{(0)} \rightsquigarrow \iota_{\mathbf{T}} \theta' = \theta'(\mathbf{T}), \quad (14.52a)$$

$$\mathcal{V}^{(1)} \rightsquigarrow \iota_{\mathbf{T}} \iota_{\mathbf{F}} \omega' = \omega'(\mathbf{F}, \mathbf{T}). \quad (14.52b)$$

14.7.3 Vertices of Fluctuation Valence One

We begin with Eq. (14.52b). In this case, we simply need to work in the linearized theory. Consequently, we can simply focus on deriving the positive-helicity vertices. This is because the negative-helicity vertices are trivially related to the positive-helicity ones at the linearized order.

In any SD background, the infinite sum in Eq. (14.50) drops so that the interaction symplectic form is simply

$$\omega'_{\sqrt{\text{Kerr}}} = dq_{a'} \wedge A^{a'} + q_{a'} \wedge dA^{a'}. \quad (14.53)$$

For instance, plugging in the background of a single SD plane wave 3^+ gives

$$\omega'_{\sqrt{\text{Kerr}}}(\mathbf{F}, \mathbf{T}) \Big|_{\epsilon_3} = \sqrt{2}g \left((\delta q_{a'} c^{a'}) \frac{2A_3}{\alpha} + i(q \cdot c) \delta z^\mu (\varphi_3 p)_\mu \right) e^{ik_3 z} e^{iK_3 \sigma}. \quad (14.54)$$

From this calculation, we deduce the following vertices:

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta z^\mu \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta z^\mu = \sqrt{2}i(q \cdot c) (\varphi_3^+ p)_\mu e^{ik_3 z}, \quad (14.55a)$$

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta \bar{z}^\mu \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta \bar{z}^\mu = \sqrt{2}i(q \cdot c) (\varphi_4^- p)_\mu e^{ik_4 \bar{z}}, \quad (14.55b)$$

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta q_{a'} \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta q_{a'} = 2\sqrt{2}c^{a'} \frac{A_3^+}{\alpha} e^{ik_3 z}, \quad (14.55c)$$

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta q_{a'} \quad \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \delta q_{a'} = 2\sqrt{2}c^{a'} \frac{A_4^-}{\beta} e^{ik_4 \bar{z}}. \quad (14.55d)$$

Here, we have peeled off the YM coupling g and explicitly written down the fluctuation species, while omitting the factors $e^{iK_3 \sigma}$, $e^{iK_4 \sigma}$. It is left as a straightforward exercise to derive the negative-helicity vertices in Eqs. (14.55b) and (14.55d) directly from the exterior derivative of Eq. (14.50), in which case incorporating the dy terms is crucial. By construction, the vertices in Eq. (14.55) are identical to the scalar vertices obtained in Ref. [763] up to attaching the e^{ikz} , $e^{ik\bar{z}}$ factors.

14.7.4 Vertices of Fluctuation Valence Zero

The trickier part is Eq. (14.52a), which contracts Eq. (14.51) with Eq. (14.50). In this case, we need to extract the part of $\theta'(\mathbf{T})$ multilinear in both ϵ_3 and ϵ_4 . Luckily, the axial gauge with $\eta^\mu \propto y^\mu$ radically simplifies the extraction of this multilinear part of the P -tensors: $y \cdot D = y \cdot \partial$. As a result, we find

$$\theta'_{\sqrt{\text{Kerr}}}(\mathbf{T}) \Big|_{\epsilon_3 \epsilon_4} = q_{a'} \left(A^{a'}_{\mu}(z) - 2iE_1(\alpha + \beta) y^\rho F^{-a'}_{\rho\mu}(z) \right) \Big|_{\epsilon_3 \epsilon_4} 2p^\mu. \quad (14.56)$$

In fact, this can be directly deduced from Eq. (14.48): note how the E_1 structure is directly inherited down. By axial gauge, Eq. (14.56) boils down to

$$\begin{aligned} & -4iq_{a'} \left(\frac{1}{\alpha + \beta} F^{a'}_{\rho\sigma}(z) + E_1(\alpha + \beta) F^{-a'}_{\rho\sigma}(z) \right) \Big|_{\epsilon_3 \epsilon_4} y^\rho p^\sigma, \\ & = \frac{-4iq_{a'}}{\alpha + \beta} \left(F^{+a'}_{\rho\sigma}(z) + F^{-a'}_{\rho\sigma}(\bar{z}) \right) \Big|_{\epsilon_3 \epsilon_4} y^\rho p^\sigma, \end{aligned} \quad (14.57)$$

where the replacement of $e^{\alpha+\beta} F^{-a'}_{\rho\sigma}(z)$ with $F^{-a'}_{\rho\sigma}(\bar{z})$ is valid by virtue of the dependence $F^{-a'}_{\rho\sigma}(z)|_{\epsilon_3\epsilon_4} \sim e^{i(k_3+k_4)z}$.

Same-Helicity Configurations We had discussed in Eq. (2.180) that the collision of two SD plane waves in YM theory produces a purely SD mode. In the same way, the collision of two ASD plane waves in YM theory produces a purely ASD mode. Provided this fact, the particular simplified form of Eq. (14.57) in the aligned axial gauge implies that the $(+, +)$ and $(-, -)$ vertices will be identical up to swapping $z \leftrightarrow \bar{z}$. In this sense it happens to be that our perturbation theory is treating the plus and minus helicities symmetrically. To emphasize, however, this accidental simplification is crucially by virtue of our clever choice $\eta^\mu \propto y^\mu$ of the gauge condition: see Sec. 14.7.7.

For the $(+, +)$ case, plugging in Eq. (14.47) to Eq. (14.57) gives

$$\frac{-4ig^2(q \cdot c_{34})}{(\alpha + \beta)t} \left(p\varphi_3^+\varphi_4^+p - p\varphi_4^+\varphi_3^+p - A_3^+ \frac{y\varphi_4^+k_3}{k_4y} + A_4^+ \frac{y\varphi_3^+k_4}{k_3y} \right) e^{ik_{34}z}. \quad (14.58)$$

Using the kinematic identities in Eqs. (14.34a) and (14.34b) with the findings in Sec. 14.6.1, this boils down to

$$(q \cdot c_{34}) \left[-\frac{4ig^2(p\varphi_3^+\varphi_4^+p)}{K_3t} \right] \left(1 + \frac{\delta}{\alpha\beta} \right) e^{ik_{34}z}. \quad (14.59)$$

As a result, the same-helicity contact vertices are found as

$$\textcircled{3^+4^+} = (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^+\varphi_4^+p)}{K_3t} \right] \left(1 + \frac{\delta}{\alpha\beta} \right) e^{ik_{34}z}, \quad (14.60a)$$

$$\textcircled{3^-4^-} = (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^-\varphi_4^-p)}{K_3t} \right] \left(1 + \frac{\delta}{\alpha\beta} \right) e^{ik_{34}\bar{z}}. \quad (14.60b)$$

Mixed-Helicity Configuration We had discussed in Eq. (2.181) that the collision of SD and ASD plane waves in YM theory produces both SD and ASD modes. Hence in the mixed-helicity configuration, both terms in Eq. (14.57) are activated.

Noting that $[\varphi_3^+, \varphi_4^-] = 0$, plugging in Eq. (14.47) to Eq. (14.57) gives

$$\begin{aligned} & \frac{-4ig^2(q \cdot c_{34})}{(\alpha + \beta)t} \left(-A_3^+ \frac{y\varphi_4^-k_3}{k_4y} e^{ik_{34}z} + A_4^- \frac{y\varphi_3^+k_4}{k_3y} e^{ik_{34}\bar{z}} \right) \\ &= \frac{-4ig^2(q \cdot c_{34})}{(\alpha + \beta)t} \left(-A_3^+ \frac{y\varphi_4^-k_3}{k_4y} e^{-\beta} + A_4^- \frac{y\varphi_3^+k_4}{k_3y} e^{\alpha} \right) e^{ik_3z} e^{ik_4\bar{z}}. \end{aligned} \quad (14.61)$$

Again, we use the kinematic identities in Eqs. (14.34a) and (14.34b) with the findings in Sec. 14.6.2. The resulting mixed-helicity contact vertex is

$$\begin{array}{c} \blacksquare \\ 3^+ 4^- \end{array} = (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^+ \varphi_4^- p)}{K_3 t} \right] \left(\left(1 + \frac{\delta}{\alpha\beta} \right) \frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \right) e^{ik_3 z} e^{ik_4 \bar{z}}. \quad (14.62)$$

14.7.5 Derivation of Same-Helicity Compton Amplitudes

Provided the Feynman rules in Secs. 14.4, 14.7.4, and 14.7.3, we are finally ready to compute the $\sqrt{\text{Kerr}}$ Compton amplitudes. We begin with the same-helicity cases, in which case we may simply work out the $(+, +)$ configuration since the Feynman rules are symmetric in the aligned axial gauge.

Abelian Contribution By using the vertex in Eq. (14.55a) and the propagator in Eq. (14.33b), we immediately find

$$\begin{array}{c} \text{---} \text{---} \\ 3^+ \quad 4^+ \end{array} = (q \cdot c_3)(q \cdot c_4) \left[-\frac{4(p\varphi_3^+ \varphi_4^+ p)}{K_3^2} \right] e^{ik_{34} z}. \quad (14.63)$$

Upon stripping off the trivial product $(q \cdot c_3)(q \cdot c_4)$ of color factors, Eq. (14.63) is precisely the scalar Compton amplitude of abelian gauge theory shown in Eq. (14.109), spin-exponentiated to the zig position z .

Nonabelian Contribution By using the vertex in Eq. (14.55c) and the propagator in Eq. (14.33d), we find

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ 3^+ \quad 4^+ \end{array} = (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^+ \varphi_4^+ p)}{K_3 t} \right] \left(-\frac{\delta}{\alpha\beta} \right) e^{ik_{34} z}, \quad (14.64)$$

where the definition of δ in Eq. (14.36) is used. Via the core calculation

$$\left(1 + \frac{\delta}{\alpha\beta} \right) + \left(-\frac{\delta}{\alpha\beta} \right) = 1, \quad (14.65)$$

the sum of Eqs. (14.64) and (14.60a) is found as

$$\begin{array}{c} \text{---} \text{---} \text{---} \\ 3^+ \quad 4^+ \end{array} + \begin{array}{c} \text{---} \\ 3^+ 4^+ \end{array} = (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^+ \varphi_4^+ p)}{K_3 t} \right] e^{ik_{34} z}. \quad (14.66)$$

Upon dropping the color factor $(q \cdot c_{34})$, Eq. (14.66) is precisely the color-stripped scalar Compton amplitude of YM theory shown in Eq. (14.109), spin-exponentiated to the zig position z . Note that the calculations here are isomorphic to the calculations for the minimally coupled scalar particle in Ref. [763].

Conclusion To conclude, we have successfully derived the mixed-helicity $\sqrt{\text{Kerr}}$ Compton amplitude as

$$\begin{aligned} \mathcal{M}_{\sqrt{\text{Kerr}}}(3^+4^-) &= \left((q \cdot c_3)(q \cdot c_4) \left[-\frac{4(p\varphi_3^+\varphi_4^-p)}{K_3^2} \right] (1 + \gamma) \right. \\ &\quad \left. + (q \cdot c_{34}) \left[-\frac{4i(p\varphi_3^+\varphi_4^-p)}{K_3 t} \right] \left(\frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \right. \right. \\ &\quad \left. \left. + (\gamma - \alpha)(\gamma + \beta) \frac{E_1(\alpha) - E_1(-\beta)}{\alpha + \beta} \right) \right) e^{ik_3 z} e^{ik_4 \bar{z}}. \end{aligned} \quad (14.73)$$

14.7.7 Gauge Invariance

In Secs. 14.7.5 and 14.7.6, we have shown the derivation of the $\sqrt{\text{Kerr}}$ Compton amplitudes from the googly $\sqrt{\text{Kerr}}$ action while employing the aligned axial gauge: $\eta^\mu \propto y^\mu$. Remarkably, this particular choice of the reference vector rendered the perturbation theory symmetric on the helicities.

The role of this subsection is to describe the case when the reference vector η^μ is oriented along a generic direction. The objective is twofold. First, we establish the gauge invariance and overall consistency of our formalism by explicitly showing that the reference-dependent terms in the Feynman rules cancel each other in the amplitudes. Second, we explicitly clarify that the perturbation theories due to googly actions are supposed to treat the helicities asymmetrically, i.e., are inherently chiral in general. When the googly action is realized around the zig worldline, the minus-helicity spin exponentiation onto zag is nontrivial and rather provides a nice consistency check.

To this end, one should find the general formula for the P -tensors evaluated on the background in Eq. (14.46) with the generic reference η^μ . This is viable by carrying out brute-force calculations at a few low orders and then observing a pattern, for instance. By resummation, the vertices are found as follows. The linear vertices are

$$\begin{aligned} \textcircled{\text{O}} \text{---} \delta z^\mu &= \sqrt{2}i(q \cdot c)(\varphi_3^+ p)_\mu e^{ik_3 z}, \\ \textcircled{\text{O}} \text{---} \delta \bar{z}^\mu &= \sqrt{2}i(q \cdot c)(\varphi_4^- p)_\mu e^{ik_4 \bar{z}}, \\ \textcircled{\text{O}} \cdots \delta q_{a'} &= 2\sqrt{2}c^{a'} \frac{A_3^+}{\alpha} (1 + X_3) e^{ik_3 z}, \\ \textcircled{\text{O}} \cdots \delta q_{a'} &= 2\sqrt{2}c^{a'} \frac{A_4^-}{\beta} (1 + X_4 e^{-\beta}) e^{ik_4 \bar{z}}, \end{aligned} \quad (14.74)$$

The contact vertices, after stripping off $(q \cdot c_{34})$ and normalizing by the scalar Compton amplitude, give

$$\begin{aligned}
\bigcirc_{3^+4^+} &\rightsquigarrow \left(1 + \frac{\delta}{\alpha\beta} (1 + \textcolor{red}{X}_3)(1 + \textcolor{red}{X}_4)\right) e^{ik_3 z} e^{ik_4 z}, \\
\boxminus_{3^-4^-} &\rightsquigarrow \left(1 + \frac{\delta}{\alpha\beta} (1 + \textcolor{red}{X}_3 e^{-\alpha})(1 + \textcolor{red}{X}_4 e^{-\beta})\right) e^{ik_3 \bar{z}} e^{ik_4 \bar{z}}, \\
\blacksquare_{3^+4^-} &\rightsquigarrow \left(\text{AHH}[1] + \frac{\delta}{\alpha\beta} (1 + \textcolor{red}{X}_3)(1 + \textcolor{red}{X}_4 e^{-\beta})\right) e^{ik_3 z} e^{ik_4 \bar{z}}.
\end{aligned} \tag{14.75}$$

We color-code reference-dependent terms in red. It is easily seen that Eqs. (14.74) and (14.75) predict the same amplitudes as in Secs. 14.7.5 and 14.7.6. This establishes the gauge invariance of our framework. We have defined

$$\alpha' := 2k_3\eta, \quad \beta' := 2k_4\eta, \quad \textcolor{red}{X}_3 := \frac{\eta\varphi_3 y}{\alpha'} \frac{K_3}{A_3}, \quad \textcolor{red}{X}_4 := \frac{\eta\varphi_4 y}{\beta'} \frac{K_4}{A_4}, \tag{14.76}$$

so $X_3, X_4 \rightarrow 0$ as η^μ gets aligned with y^μ .

Clearly, the vertices in Eqs. (14.74) and (14.75) treat the plus and minus helicities on an unequal footing. In particular, the square-shaped vertices in Eqs. (14.74) and (14.75) fail to be localized at a single point, unlike the round-shaped ones. For instance, $(1 + \textcolor{red}{X}_4 e^{-\beta}) e^{ik_4 \bar{z}} = e^{ik_4 \bar{z}} + \textcolor{red}{X}_4 e^{ik_4 z}$ could be seen as describing a mixture of a vertex localizing at the zag position $\bar{z}$ and a vertex localizing at the zig position z . Similarly, the $(-, -)$ vertex in Eq. (14.75) could be seen as describing a mixture of interactions happening at (z, z) , $(z, \bar{z})$, $(\bar{z}, z)$, and $(\bar{z}, \bar{z})$ for the third and fourth waves, i.e., at all possible combination of locations. Overall, we see that the nonabelian $\sqrt{\text{Kerr}}$ particle always interacts with incoming SD waves via the zig position z but receives incoming ASD waves either at the zig position z or the zag position $\bar{z}$. This describes a mild form of nonlocality, over a discrete set of points.⁹

For the square vertex in Eq. (14.74) and the $(-, -)$ vertex in Eq. (14.75), this nonlocality will be considered a *gauge artifact*, since it completely disappears in the aligned axial gauge. For the $(-, +)$ contact vertex in Eq. (14.75), however, even the aligned axial gauge yields a superposition of multipoint localizations due to the nature of AHH[1]. It seems unclear whether there can exist a perturbation theory that represents the $(+, -)$ amplitude as a sum in which each term is local in spacetime.

⁹In any case, our $\sqrt{\text{Kerr}}$ particle seems to interact with the external environment always through either z or $\bar{z}$, and never x . The amplitudes advocated in Refs. [217–223], in contrast, seem to have a tendency to add $x = (z + \bar{z})/2$ as an additional interaction point.

Physically, the $\sqrt{\text{Kerr}}$ interaction action is rather a faithful implementation of the chiral dyon pair model. Amusingly, the nonabelian Dirac string term in Eq. (10.54),

$$\begin{aligned} \int \text{sinc}(\mathcal{L}_N)(q_a \iota_N * F^a) &= \int_{-1}^{+1} \frac{d\eta}{2} \int e^{i\eta \mathcal{L}_N}(q_a \iota_N * F^a), \\ &= \frac{1}{2i} \int q_b(\tau) W^b_a(X(\tau, 0), X(\tau, \eta)) * F^a_{\mu\nu}(X(\tau, \eta)) \frac{\partial X^\mu(\tau, \eta)}{\partial \eta} \frac{\partial X^\nu(\tau, \eta)}{\partial \tau} d\eta d\tau, \end{aligned} \quad (14.77)$$

could be compared to the topological surface operator in Chap. III, Eq. (3.31),

$$\int_S *F^a \lambda_a, \quad (14.78)$$

with λ_a made dynamical by parallel transportation of a dynamical worldline profile $q_a(\tau)$ across the worldsheet:

$$\lambda_a(\tau, \eta) = \frac{1}{2i} q_b(\tau) W^b_a(X(\tau, 0), X(\tau, \eta)). \quad (14.79)$$

Recalling the topological nature of the surface operator in Eq. (14.78), the same-helicity amplitudes would localize on a single worldline (see Ref. [226] for a prototype discussion in Maxwell theory).

The discrete nonlocality of the $(+, -)$ amplitude and its physical content stand out as interesting in this respect. Rather than attributing it directly to the Dirac worldsheet, we might wish to understand this nonlocality in complexified spacetime as a consequence of a dynamical, physical interaction between the nonabelian SD and ASD dyons, via inertia and the angular momentum effect.

14.8 Kerr Compton Amplitudes

We scatter the Kerr particle, in its massive twistor implementation, off the nonlinearly superposed background of two gravitational waves in GR.

We follow the same methodology as in Sec. 14.7. For reasons of space, we will simply perform the calculation in the aligned axial gauge, despite the fact that the gauge invariance of our formalism can be verified in generic axial gauges.

14.8.1 The Two-Graviton Background

Earlier in Eq. (2.186), we had constructed the on-shell two-graviton geometry in generic axial gauge. Here, we implement the aligned axial gauge. The spin

length vector y^μ at x and its parallel propagation to the zig point z , $y^{\mu'}$, must agree at the level of their background value by the very scattering setup. Our aligned axial gauge will be $\eta^{\mu'} \propto y^{\mu'}$ via *spacetime* (not tetrad) components.

For instance, let us reproduce Eq. (2.187) in the aligned axial gauge for the mixed-helicity Compton scattering:

$$\varphi_{34}^{\tilde{a}\tilde{b}'} = \varphi_3^{+\tilde{a}'}\varphi_3^{+\tilde{b}'}\left(\frac{y\varphi_4k_3}{k_4y}\right)^2 + \varphi_4^{-\tilde{a}'}\varphi_4^{-\tilde{b}'}\left(\frac{y\varphi_3k_4}{k_3y}\right)^2. \quad (14.80)$$

Again, the kinematic identities in Eqs. (14.34c) and (14.34d) will be applied.

14.8.2 The Two-Graviton Symplectic Structure

The Kerr symplectic potential coupled to full GR is shown in Eq. (10.29). By a worldline field redefinition, it can be equivalently presented as Eq. (10.48), which manifests the resummation of SD graviton couplings onto the holomorphic (zig) worldline in spinspace. This worldline field basis will be referred to as the googly basis.

Via discarding a total derivative (cf. the gymnastics in Sec. 10.4.4), Eq. (10.48) can be brought to

$$\begin{aligned} \theta_{\text{Kerr}} = & p_{m'}e^{m'} - 2i\bar{\lambda}_{I\dot{\alpha}'}y^{\dot{\alpha}\alpha'}D\lambda_{\alpha'}^I \\ & + (-2i)^2E_2(-2i\mathcal{L}_N)[p_{m'}(\iota_N R^{-m'}_{n'})y^{n'}], \end{aligned} \quad (14.81)$$

where $p_{\alpha\dot{\alpha}'} = -\lambda_{\alpha'}^I\tilde{\lambda}_{I\dot{\alpha}'}$ is a composite. To reiterate, Eq. (14.81) represents the exact Kerr symplectic potential in generic, non-SD backgrounds. The first term $p_{m'}e^{m'}$ in the right-hand side implements the mass monopole while the second term $-2i\bar{\lambda}_{I\dot{\alpha}'}y^{\dot{\alpha}\alpha'}D\lambda_{\alpha'}^I$ is associated with dipoles as discussed below Eq. (10.33). The last term, crucially involving the E_2 function, implements higher multipoles in terms of the ASD Q -tensors.

Notably, in this googly presentation, we see that SD gravitational modes couple to the Kerr particle only through the coframe $e^{m'}$: the SD spin connection completely drops out from the dipole part, while the higher multipole terms insert an explicit ASD projector on the curvature two-form $R^{m'}_{n'}$. Eventually, it follows that the perturbation theory in the SD sector is isomorphic to that of the minimally coupled scalar particle (symplectic potential p_me^m) via the zig complexification.

Concretely, to *define* perturbation theory, let $h^{m'} = e^{m'} - \delta^{m'}_{\mu'}dz^{\mu'}$ be the tetrad perturbation around the Kronecker vacuum. With this split, we identify

the free theory part of Eq. (14.81) as

$$\theta^\circ = -\tilde{\lambda}_{I\dot{\alpha}'} dz^{\dot{\alpha}'\alpha'} \lambda_{\alpha'}^I - 2i \tilde{\lambda}_{I\dot{\alpha}'} y^{\dot{\alpha}'\alpha'} d\lambda_{\alpha'}^I. \quad (14.82)$$

Based on this particular physical motivation, we now make the crucial identification of the “nominal” zag coordinates as

$$w^{\dot{\alpha}'\alpha'} := z^{\dot{\alpha}'\alpha'} - 2iy^{\dot{\alpha}'\alpha'}. \quad (14.83)$$

In fact, we had already encountered this idea in Eq. (10.41). Via the algebraic worldline field redefinition (construction of the mu variables)

$$\mu^{\dot{\alpha}'I} = z^{\dot{\alpha}'\alpha'} \lambda_{\alpha'}^I, \quad \tilde{\mu}_I^{\alpha'} = \tilde{\lambda}_{I\dot{\alpha}'} w^{\dot{\alpha}'\alpha'} = \tilde{\lambda}_{I\dot{\alpha}'} (z^{\dot{\alpha}'\alpha'} - 2iy^{\dot{\alpha}'\alpha'}), \quad (14.84)$$

Eq. (14.82) is brought to

$$\theta^\circ = -\tilde{\lambda}_{I\dot{\alpha}'} d\mu^{\dot{\alpha}'I} + \tilde{\mu}_I^{\alpha'} d\lambda_{\alpha'}^I, \quad (14.85)$$

which exactly describes the complexified free massive twistor model via the unpriming worldline field redefinition $\tilde{\lambda}_{I\dot{\alpha}'} \mapsto \tilde{\lambda}_{I\dot{\alpha}}$, $\mu^{\dot{\alpha}'I} \mapsto \mu^{\dot{\alpha}I}$, etc. Crucially, Eq. (14.85) identifies Darboux coordinates so that the free theory is quadratic.

Finally, the interaction symplectic potential arises by the difference between Eqs. (14.81) and (14.82), which computes to

$$\begin{aligned} \theta'_{\text{Kerr}} = & p_{m'} h^{m'} - 2i p_{m'} \gamma^{-m'}_{n'} y^{n'} \\ & + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} \sum_{p=1}^{\ell/2} \sum_{\alpha \in \Omega_p(\ell)} \left[\prod_{i=1}^p \binom{(\sum_{j=i}^p \alpha_j) - 2}{\alpha_i - 2} \right] \\ & \left(((Q_{\alpha_1}^-) Q_{\alpha_2} \cdots Q_{\alpha_p})^{m'}_{s'} e^{s'} \right. \\ & \left. + (\alpha_p - 2) ((Q_{\alpha_1}^-) Q_{\alpha_2} \cdots Q_{\alpha_{p-1}} Q_{\alpha_p - 1})^{m'}_{s'} D y^{s'} \right). \end{aligned} \quad (14.86)$$

For the purposes of this chapter, it suffices to truncate the Q -tensor tower up to the second order. As a result, the relevant terms inside Eq. (14.86) fall into two groups. The first group originates from a covariant tower of multipoles,

$$\begin{aligned} \theta'^{(Q)}_{\text{Kerr}} = & p_{m'} h^{m'} - 2i p_{m'} \gamma^{-m'}_{n'} y^{n'} \\ & + \sum_{\ell=2}^{\infty} \frac{(-2i)^\ell}{\ell!} p_{m'} \left((Q_{\ell}^-)^{m'}_{s'} e^{s'} + (\ell-2) (Q_{\ell-1}^-)^{m'}_{s'} D y^{s'} \right), \end{aligned} \quad (14.87)$$

whereas the second group inserts curvature-squared operators:

$$\theta'^{(QQ)}_{\text{Kerr}} = \sum_{\ell=4}^{\infty} \frac{(-2i)^\ell}{\ell!} \sum_{j=2}^{\ell-2} \binom{\ell-2}{j} p_{m'} \left((Q_{\ell-j}^- Q_j)^{m'}_{s'} e^{s'} + (j-2) (Q_{\ell-j}^- Q_{j-1})^{m'}_{s'} D y^{s'} \right). \quad (14.88)$$

Let us refer to the terms in Eqs. (14.87) and (14.88) as “level-one” and “level-two” terms, respectively. While Eq. (14.87) could be compared to Eq. (14.50), higher-level terms such as those in Eq. (14.88) seem characteristic to gravity.

Lastly, we identify the fluctuation and time evolution vector fields, $\mathbf{F}$ and $\mathbf{T}$. We define perturbation theory in the field basis in Eq. (14.85), so

$$\mathbf{F} = \delta\lambda_{\alpha'}^I \frac{\partial}{\partial\lambda_{\alpha'}^I} + \delta\mu^{\dot{\alpha}'I} \frac{\partial}{\partial\mu^{\dot{\alpha}'I}} + \delta\tilde{\lambda}_{I\dot{\alpha}'} \frac{\partial}{\partial\tilde{\lambda}_{I\dot{\alpha}'}} + \delta\tilde{\mu}_I^{\alpha'} \frac{\partial}{\partial\tilde{\mu}_I^{\alpha'}} \quad (14.89)$$

with $\delta\lambda_{\alpha'}^I, \delta\mu^{\dot{\alpha}'I}, \delta\tilde{\lambda}_{I\dot{\alpha}'}^{}, \delta\tilde{\mu}_I^{\alpha'}$ declared as fundamental. The time-evolution vector field is, in the conventions of this thesis,

$$\mathbf{T} = \det(\lambda)\det(\tilde{\lambda}) \left((\tilde{\lambda}^{-1})^{\dot{\alpha}'I} \frac{\partial}{\partial\mu^{\dot{\alpha}'I}} + (\lambda^{-1})_I^{\alpha'} \frac{\partial}{\partial\tilde{\mu}_I^{\alpha'}} \right), \quad (14.90)$$

which implies

$$\iota_{\mathbf{T}} dz^{m'} = 2p^{m'}, \quad \iota_{\mathbf{T}} dy^{m'} = 0, \quad \iota_{\mathbf{T}} dp_{m'} = 0. \quad (14.91)$$

Note that $z^{m'} := \delta^{m'}_{\mu'} z^{\mu'}$ by the flat background tetrad.

14.8.3 Vertices of Fluctuation Valence One

In any SD background, the Kerr interaction symplectic potential in Eq. (14.86) collapses to $p_{m'} h^{m'}$, which is isomorphic to the minimally coupled scalar particle's $p_m h^m$ via the zig complexification. In this case, the interaction symplectic form is simply

$$\omega'_{\text{Kerr}} = dp_{m'} \wedge h^{m'} + p_{m'} dh^{m'}. \quad (14.92)$$

Evaluating Eq. (14.92) in the background of a single SD graviton readily derives the positive-helicity linearized vertices as simple NJ shifts of the scalar vertices produced in Ref. [763]. By linearity, the negative-helicity linearized vertices are then also found to be isomorphic.

In this way, one deduces all linearized vertices of fluctuation degree one for the Kerr particle as

$$\begin{array}{c} \text{○} \\ 3^+ \end{array} \text{---} \delta z^{m'} = 2i \frac{A_3^+}{\alpha} (\varphi_3^+ p)_{m'} e^{ik_3 z}, \quad (14.93a)$$

$$\begin{array}{c} \text{⊗} \\ 4^- \end{array} \text{---} \delta w^{m'} = 2i \frac{A_4^-}{\beta} (\varphi_4^- p)_{m'} e^{ik_4 \bar{z}}, \quad (14.93b)$$

$$\begin{array}{c} \text{○} \\ 3^+ \end{array} \text{---} \delta p_{m'} = 4 \frac{A_3^+}{\alpha^2} (y\varphi_3^+)^{m'} e^{ik_3 z}, \quad (14.93c)$$

$$\begin{array}{c} \text{⊗} \\ 4^- \end{array} \text{---} \delta p_{m'} = 4 \frac{A_4^-}{\beta^2} (y\varphi_4^-)^{m'} e^{ik_4 \bar{z}}. \quad (14.93d)$$

Again, here we have peeled off the gravitational coupling κ and explicitly wrote down the fluctuation species, while omitting the factors $e^{iK_3\sigma}$, $e^{iK_4\sigma}$. To clarify, $\bar{z}^{m'}$ in $e^{ik\bar{z}}$ is defined as $z^{m'} - 2iy^{m'}$ with the background worldline's holomorphic impact parameter $z^{m'}$ and spin length vector $y^{m'}$.

14.8.4 Vertices of Fluctuation Valence Zero

To derive vertices of fluctuation valence zero, one contracts the interaction symplectic potential in Eq. (14.86) with the free-theory time evolution vector field in Eq. (14.90).

Level Two For Compton scattering, the curvature-squared operators in Eq. (14.88) directly boil down to *gauge-invariant* contact terms at each spin order. Hence they can be computed and treated separately. Simple calculation shows that Eq. (14.88) gives the vertices

$$\begin{array}{c} (2) \\ \bullet \\ 3^+4^- \end{array} \rightsquigarrow \left(-\delta\gamma^2 \mathcal{S}(\beta, \alpha) \right) e^{ik_{34}z}, \quad (14.94a)$$

$$\begin{array}{c} (2) \\ \bullet \\ 3^-4^- \end{array} \rightsquigarrow \left(-\delta \left(\delta + \alpha\beta - \gamma(\alpha + \beta) \right) \left(\mathcal{S}(\beta, \alpha) + \mathcal{S}(\alpha, \beta) \right) \right) e^{ik_{34}z}, \quad (14.94b)$$

after normalizing by the scalar Compton amplitude. Here, we have defined

$$\begin{aligned} \mathcal{S}(\beta, \alpha) &:= \sum_{\ell=4}^{\infty} \frac{1}{\ell!} \sum_{j=2}^{\ell-2} \binom{\ell-2}{j} \beta^{\ell-j-2} \alpha^{j-2}, \\ &= \frac{E_2(\alpha + \beta) - E_2(\beta) - \alpha E_2'(\beta)}{\alpha^2} = \int_0^1 d\eta \, \eta^2 (1 - \eta) e^{\eta\beta} E_2(\eta\alpha). \end{aligned} \quad (14.95)$$

Clearly, Eq. (14.95) directly transcribes the structure of the level-two Q -tensor tower in Eq. (14.88): the combinatorics of Jacobi propagators in Chap. VIII [695]. Note also the identity

$$E_h'(\xi) = \frac{1}{\xi} \left(E_{h-1}(\xi) - h E_h(\xi) \right), \quad (14.96)$$

which holds for $h = 1, 2, 3, \dots$ with $E_0(\xi) := e^\xi$.

Level One The level-one terms in Eq. (14.87) require relatively more complicated manipulations, but brute-force calculation shows that they give¹⁰

¹⁰In the first line of Eq. (14.97b), $(1 + \delta/\alpha\beta - \delta\gamma/\alpha\beta^2 + \delta\gamma/\alpha^2\beta)$ arises from the monopole part. The contribution from the dipole part is $\gamma^2(\gamma + \beta)/\alpha^2$, which happens to not involve a $1/(\alpha + \beta)$ factor. Importantly, the formula in Eq. (2.184a) involves a coframe $e^B{}_\nu$ that does not disappear by the axial gauge. If one mistakenly ignores this contribution, one obtains a wrong result $\gamma^2(\gamma + \beta)^2/\alpha^2(\alpha + \beta)$.

$$\textcircled{\text{O}}_{3^+4^+} \rightsquigarrow \left(1 + \frac{\delta}{\alpha\beta} + \frac{\delta\gamma}{\alpha\beta^2} + \frac{\delta\gamma}{\alpha^2\beta} \right) e^{ik_{34}z}, \quad (14.97a)$$

$$\textcircled{\text{O}}_{3^+4^-}^{(1)} \rightsquigarrow \left(1 + \frac{\delta}{\alpha\beta} - \frac{\delta\gamma}{\alpha\beta^2} + \frac{\delta\gamma}{\alpha^2\beta} + \frac{\gamma^2(\gamma+\beta)}{\alpha^2} E_1(\alpha+\beta) \right) e^{ik_{34}z}, \quad (14.97b)$$

$$\textcircled{\text{O}}_{3^-4^-}^{(1)} \rightsquigarrow \left(1 + \frac{\delta}{\alpha\beta} + \frac{\delta\gamma}{\alpha\beta^2} + \frac{\delta\gamma}{\alpha^2\beta} \right) e^{ik_{34}\bar{z}} + \left(\delta \left(\delta + \alpha\beta - \gamma(\alpha+\beta) \right) (\mathcal{S}(\beta, \alpha) + \mathcal{S}(\alpha, \beta)) \right) e^{ik_{34}z}. \quad (14.97c)$$

Again, we have normalized by the scalar Compton amplitude. We see that, even with the aligned axial gauge, the plus and minus helicities are treated asymmetrically in this perturbation theory, unlike in the earlier YM case. While the $(+, +)$ vertex in Eq. (14.97a) is local in spinspace-time, the $(-, -)$ vertex in Eq. (14.97c) is not localized at a single complexified point.

14.8.5 Derivation of Compton Amplitudes for All Helicities

Given the Feynman rules in Secs. 14.4, 14.8.4, and 14.8.3, we are finally ready to compute the Kerr Compton amplitudes for any helicity configuration.

Helicities $(+, +)$ From the vertices in Eqs. (14.93a), (14.93c), and (14.97a), the $(+, +)$ amplitude is computed as

$$\textcircled{\text{O}}_{3^+4^+} + \textcircled{\text{O}}_{3^+4^+} + \textcircled{\text{O}}_{3^+4^+} + \textcircled{\text{O}}_{3^+4^+} \rightsquigarrow e^{ik_{34}z}. \quad (14.98)$$

The core calculation here is

$$\left(-\frac{\delta}{\alpha\beta} \right) + \left(-\frac{\delta\gamma}{\alpha\beta^2} \right) + \left(-\frac{\delta\gamma}{\alpha^2\beta} \right) + \left(1 + \frac{\delta}{\alpha\beta} + \frac{\delta\gamma}{\alpha\beta^2} + \frac{\delta\gamma}{\alpha^2\beta} \right) = 1. \quad (14.99)$$

Note how the second-order and first-order propagators contribute in this calculation. We conclude that

$$\mathcal{M}_{\text{Kerr}}(3^+4^+) = \left[-\frac{4(p\varphi_3^+\varphi_4^+p)^2}{K_3^2 t} \right] e^{ik_{34}z}. \quad (14.100)$$

Helicities $(+, -)$ Next, we consider the case of prime interest: mixed helicities. From the vertices in Eqs. (14.93), (14.94a), and (14.97b), the $(+, -)$ amplitude is computed as the following.

$$\textcircled{\text{O}}_{3^+4^-} + \textcircled{\text{O}}_{3^+4^-} + \textcircled{\text{O}}_{3^+4^-} + \textcircled{\text{O}}_{3^+4^-}^{(1)} + \textcircled{\text{O}}_{3^+4^-}^{(2)} \quad (14.101)$$

The first three diagrams give

$$\left(-\frac{\delta}{\alpha\beta}(1+\gamma) + \frac{\delta\gamma}{\alpha\beta^2} - \frac{\delta\gamma}{\alpha^2\beta} \right) e^{ik_3z} e^{ik_4\bar{z}}, \quad (14.102a)$$

while the last two diagrams, via a nice cancellation of $\mathcal{S}(\alpha, \beta)$, give

$$\left(1 + \frac{\delta}{\alpha\beta} - \frac{\delta\gamma}{\alpha\beta^2} + \frac{\delta\gamma}{\alpha^2\beta} + \frac{\gamma^2(\gamma+\beta)}{\alpha^2} E_1(\alpha+\beta) \right) e^{-\beta} e^{ik_3z} e^{ik_4\bar{z}}. \quad (14.102b)$$

The sum of Eqs. (14.102a) and (14.102b) readily computes to $\text{AHH}[2] e^{ik_3z} e^{ik_4\bar{z}}$. Hence we conclude that

$$\begin{aligned} & \mathcal{M}_{\text{Kerr}}(3^+4^-) \\ &= \left[-\frac{4(p\varphi_3^+\varphi_4^-p)^2}{K_3^2 t} \right] \left(\begin{aligned} & \frac{\beta e^\alpha + \alpha e^{-\beta}}{\alpha + \beta} + \gamma \frac{e^\alpha - e^{-\beta}}{\alpha + \beta} \\ & + (\gamma - \alpha)(\gamma + \beta) \frac{E_1(\alpha) - E_1(-\beta)}{\alpha + \beta} \\ & + (\gamma - \alpha)(\gamma + \beta) \gamma \frac{E_2(\alpha) - E_2(-\beta)}{\alpha + \beta} \end{aligned} \right) e^{ik_3z} e^{ik_4\bar{z}}. \end{aligned} \quad (14.103)$$

Helicities $(-, -)$ Finally, we compute the $(-, -)$ amplitude as a rather nontrivial consistency check of our chiral perturbation theory. From the vertices in Eqs. (14.93b), (14.93d), (14.94b), and (14.97c), the $(-, -)$ amplitude is computed as the following.

$$\begin{array}{c} \text{---} \otimes \text{---} \otimes \text{---} \otimes \text{---} \\ 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \end{array} + \begin{array}{c} \text{---} \otimes \text{---} \otimes \text{---} \otimes \text{---} \\ 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \end{array} + \begin{array}{c} \text{---} \otimes \text{---} \otimes \text{---} \otimes \text{---} \\ 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \end{array} + \begin{array}{c} \text{---} \otimes \text{---} \otimes \text{---} \otimes \text{---} \\ 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \end{array} + \begin{array}{c} \text{---} \otimes \text{---} \otimes \text{---} \otimes \text{---} \\ 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \quad 3^- \quad 4^- \end{array} \quad (14.104)$$

While the first four diagrams achieve the same cancellation mechanism as in Eq. (14.99), it is clear that the $\mathcal{S}$ -function terms cancel nicely. Thus, we find

$$\mathcal{M}_{\text{Kerr}}(3^-4^-) = \left[-\frac{4(p\varphi_3^-\varphi_4^-p)^2}{K_3^2 t} \right] e^{ik_3z} e^{ik_4\bar{z}}. \quad (14.105)$$

Despite the fact that there is no symmetry between the $(+, +)$ and $(-, -)$ configurations at the level of individual Feynman rules, the total amplitude in Eq. (14.105) is exactly conjugate to the $(+, +)$ amplitude in Eq. (14.100).

Note also how the gauge-invariant contact vertex from level two, Eq. (14.94b), has canceled a part of the level-one vertex in Eq. (14.97c) that will be generically gauge dependent. In any case, gauge dependencies must cancel without invoking the level-two vertices at this order.

14.8.6 Manifest Reality; Earthly Deformations

Finally, suppose we carried out the perturbation theory with the earthly Kerr action in Eq. (10.29). This perturbation scheme treats the SD and ASD modes on an equal footing and reproduces the same amplitudes. Unfortunately, the same-helicity spin exponentiation is never manifest at the Lagrangian level and is portrayed as a nontrivial property relying on mystical cancellations.

Another evident point regarding this approach is that it utilizes the real section of complexified spacetime as a spurious “reference” structure. It declares the (geodetic) midpoint between the chiral dyons or Taub-NUT centers as a particularly distinguished point of a certain importance, the physical rationale of which may appear thin given the topological nature of the Dirac/Misner string. It should be understood that there is nothing in between the SD and ASD Taub-NUT instantons, so any wave factor in the final answer shall presumably describe either e^{ikz} or $e^{ik\bar{z}}$ (and not e^{ikx}). Of course, a balanced view will arise because one could also argue that the Kerr solution is supposed to represent a real spacetime (although by considering, e.g., all-plus Compton amplitudes one inevitably is embracing complexification).

In sum, there lies a tension between manifest spin exponentiation and manifest reality. The manifestly real Kerr Lagrangian is given in Eq. (10.29). The manifestly spin-exponentiating Kerr Lagrangian is given in Eq. (10.48).

In connection with this point, let us summarize and compare how the level-two terms contribute to the relative amplitude in each perturbation scheme. The contact vertex $\theta_{\text{Kerr}}^{(QQ)}(\mathbf{T})$ evaluates to

$$\text{Googly : } 2 \sum_{\ell=4}^{\infty} \frac{(-2i)^{\ell}}{\ell!} \sum_{j=2}^{\ell-2} \binom{\ell-2}{j} p_{m'} (Q_{\ell-j}^- Q_j)^{m' s'} p^{s'}, \quad (14.106a)$$

$$\text{Symmetric : } 2 \sum_{\ell=4}^{\infty} \frac{1}{\ell!} \sum_{j=2}^{\ell-2} \binom{\ell-2}{j} p_m ((\star^{\ell} Q_{\ell-j}) Q_j)^m{}_s p^s. \quad (14.106b)$$

Consequently, with

$$\begin{aligned} C_0 &:= 8A_3A_4(y\varphi_3\varphi_4y), \\ &= \begin{cases} \delta\gamma^2 & (\text{mixed}), \\ \delta(\delta + \alpha\beta - \gamma(\alpha + \beta)) & (\text{same}), \end{cases} \end{aligned} \quad (14.107)$$

the contributions to the relative amplitude are given by

$$\begin{aligned} \text{Googly, } 3^+4^+ &: 0, \\ \text{Googly, } 3^+4^- &: -C_0 \mathcal{S}(\beta, \alpha) e^{-\beta}, \end{aligned} \quad (14.108a)$$

$$\begin{aligned} \text{Googly, } 3^-4^- &: -C_0 (\mathcal{S}(\beta, \alpha) + \mathcal{S}(\alpha, \beta)) e^{-\alpha-\beta}, \\ \text{Symmetric, } 3^+4^+ &: -\frac{1}{16} C_0 (\mathcal{S}(-\frac{\beta}{2}, -\frac{\alpha}{2}) + \mathcal{S}(-\frac{\alpha}{2}, -\frac{\beta}{2})) e^{\alpha/2+\beta/2}, \\ \text{Symmetric, } 3^+4^- &: -\frac{1}{16} C_0 (\mathcal{S}(\frac{\beta}{2}, \frac{\alpha}{2}) + \mathcal{S}(-\frac{\alpha}{2}, -\frac{\beta}{2})) e^{\alpha/2-\beta/2}, \\ \text{Symmetric, } 3^-4^- &: -\frac{1}{16} C_0 (\mathcal{S}(\frac{\beta}{2}, \frac{\alpha}{2}) + \mathcal{S}(\frac{\alpha}{2}, \frac{\beta}{2})) e^{-\alpha/2-\beta/2}. \end{aligned} \quad (14.108b)$$

Note how the symmetric scheme induces the half-scaled factors $\alpha/2$, $\beta/2$ in both signs, instating a spurious expansion point in the middle of z and $\bar{z}$.

Said in another way, if the level-two operators are dropped at the Lagrangian level, the Compton amplitude gains a contact deformation by 0 , $+C_0 \mathcal{S}(\beta, \alpha)$, $+C_0 (\mathcal{S}(\beta, \alpha) + \mathcal{S}(\alpha, \beta))$, etc. Obviously, this must not happen in the googly scheme, as it violates reality. In the symmetric scheme, on the other hand, one might be allured to drop the QQ tower of operators by associating them with tidal deformability, although it is a grossly worldline field basis dependent operation. Crucially, this is also forbidden if the same-helicity spin exponentiation is a nonnegotiable feature of spinning black hole amplitudes. As discussed in Chap. IX [768], the best possibility is to add curvature-squared operators on the worldline that explicitly implements a mixed-chirality projection, such as $p_m (Q_{\ell-j}^+ Q_j^-)^m s p^s$. Since this deformation starts at $\mathcal{O}(a^4)$, it could be asked if there exists a natural deformation prescription that achieves AHH[3], or the amplitudes advocated in Refs. [217–223].

14.9 Conclusions

In this chapter, we provided a Lagrangian derivation of the nonabelian $\sqrt{\text{Kerr}}$ and Kerr Compton amplitudes in YM theory and GR, based on explicit effective worldline actions that guarantee the same-helicity spin exponentiation. Our results are all orders in spin, exhibit correct residues on physical factorization channels, and are completely free of unphysical poles, for all helicity configurations. They also obey a systematic organization in terms of truncated exponential functions and Hermite interpolation. In other words, we obtained a new concrete proposal regarding the longstanding puzzle of AHH [206].

The physical significance of this calculation is that it gives the gauge invariant and field basis invariant characterization of spinning black hole dynamics

up to the second order in coupling, in terms of scattering amplitudes. This finally addresses the persistent curiosity toward the exact spinning-black-hole EoM in a fully invariant fashion.

Conceptually, our construction implements a dynamical Taub-NUT or chiral dyon pair. At the technical level, this physical interpretation stipulates particular towers of curvature operators with certain fixed coefficients that have origins in the combinatorics behind geodesic or gauge-covariant translations. Arguably, our Lagrangians carry out a unique minimal off-shell extrapolation from the three-point/same-helicity couplings, in the sense that Dirac/Misner strings carry no physical degrees of freedom.

In the most critical assessment, our construction is no more than an idealized model of spinning black holes,¹¹ which is yet plausible in the sense of guaranteed same-helicity spin exponentiation to all orders in perturbation theory. To our knowledge, the explicit all-orders construction of worldline effective actions for spinning black holes that exhibit this property is new, and our mixed-helicity Compton amplitudes have not been proposed in the previous literature.

Even with this critical evaluation, we emphasize that our effective actions will serve as a *useful* starting point of future (higher-order, higher-multiplicity, all-orders-in-spin) worldline effective theory calculations for spinning black holes, if the same-helicity spin exponentiation is a nonnegotiable feature. The addition of mixed-chirality deformations can be carried out systematically.

A further critical investigation, in connection with black hole perturbation theory results [220–223], will be expected. It is interesting to imagine carrying out Kerr perturbation theory with the factorized Taub-NUT picture such that the boundary conditions at the ASD and SD Taub-NUT centers correspond to the behavior of the zig and zag points in the effective theory, although the role of the horizon may become less emphasized. In any case, it will be fruitful to promote cross-pollination between the UV picture of Chap. VII [164] and the IR picture of Chaps. IX and X [768, 769], as has already happened a number of times in this thesis.

Formalism-wise, we demonstrated a chiral way of comprehending Kerr by expanding around a holomorphic worldline onto which the SDGR couplings become resummed, whose physical interpretation is the effective theory avatar

¹¹the “zig-zag model,” so to speak

of ASD Taub-NUT. As mentioned several times, the idea itself, of achieving Kerr by perturbing away from SD or ASD Taub-NUT, is an old one and has been advocated by twistor theorists (see, e.g., Refs. [261, 266]). To our knowledge, however, its full-blown implementation in the worldline perturbation theory is new. On a related note, it should be clear that our explorations draw conceptual precursors and insights from the discourse on MHV amplitudes [836], first-quantized massive twistor models [879, 880], twistor particle program [377, 796–808], and the various provisional ideas of Newman [252–259, 329, 330].

14.A Scalar Compton Amplitudes

For reference, we record the classical Compton amplitudes for the minimally coupled scalar particle below.

Maxwell Theory

$$\mathcal{M}_{\text{scalar}} = -\frac{4(p\varphi_3\varphi_4p)}{K_3^2} \quad (14.109a)$$

YM Theory (nonabelian contribution, color-stripped)

$$\mathcal{M}_{\text{scalar}} = -\frac{4i(p\varphi_3\varphi_4p)}{K_3 t} \quad (14.109b)$$

GR

$$\mathcal{M}_{\text{scalar}} = -\frac{4(p\varphi_3\varphi_4p)^2}{K_3^2 t} \quad (14.109c)$$

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MORE ON SELF-DUAL SPACETIME

For interested readers, we provide a brief review of Plebański's second heavenly formalism and the basic scaffolding of curved twistor theory, in connection with Sec. 2.5.

A.1 Plebański's Second Heavenly Equation

This appendix reviews the derivation of Plebański's second heavenly equation governing SD spacetimes [314, 319–322].

When working in the complexified, holomorphic setup, spacetime is an oriented complex four-manifold $\mathcal{M}$ endowed with a holomorphic metric g . Based on the decomposition of its tangent bundle into SD and ASD spinor bundles,

$$T\mathcal{M} \cong S^+\mathcal{M} \otimes S^-\mathcal{M}, \quad (\text{A.1})$$

the Levi-Civita connection ∇ of g splits into SD and ASD parts, defining connections on $S^\pm\mathcal{M}$.

In this thesis, we defined that a spacetime $\mathcal{M}$ is SD if $S^-\mathcal{M}$ is flat. This means $\mathcal{M}$ is hyperkähler (structure group $\text{SL}(2, \mathbb{C}) \cong \text{Sp}(2, \mathbb{C})$) and is automatically Ricci-flat.

Let $e^{\dot{\alpha}\alpha}$ be the coframe due to a local trivialization of $T\mathcal{M}$. In a SD spacetime $\mathcal{M}$, the ASD spin connection coefficients can be locally set to zero. This statement translates to the closure of the Plebański [315, 339] area two-forms, $\Sigma^{\alpha\beta} := \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} e^{\dot{\alpha}\alpha} \wedge e^{\dot{\beta}\beta}$:

$$d\Sigma^{\alpha\beta} = 0. \quad (\text{A.2})$$

Let $o^\alpha \doteq \delta_0^\alpha$ and $\iota^\alpha \doteq \delta_1^\alpha$ be constant spinors forming an ASD dyad. Since $\Sigma^{\alpha\beta}$ is closed, there locally exists functions $q^{\dot{\alpha}}$ and $p_{\dot{\alpha}}$ such that

$$\Sigma^{01} = dq^{\dot{\alpha}} \wedge dp_{\dot{\alpha}}, \quad \Sigma^{11} = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} dq^{\dot{\alpha}} \wedge dq^{\dot{\beta}}, \quad (\text{A.3})$$

on account of the Darboux theorem [881]. These functions are the very second heavenly coordinates of Plebański [314]:

$$x^{\dot{\alpha}0} := p^{\dot{\alpha}}, \quad x^{\dot{\alpha}1} := q^{\dot{\alpha}}. \quad (\text{A.4})$$

Certainly, this coordinate choice is a sort of a gauge-fixing that trivializes a part of the self-duality conditions in Eq. (A.2), so it remains to examine the condition $d\Sigma^{00} = 0$.

To this end, note that the coframe that yields Eq. (A.3) takes the form

$$e^{\dot{\alpha}0} = \Lambda^{\dot{\alpha}}_{\dot{\beta}}(dp^{\dot{\beta}} + \kappa \Phi^{\dot{\beta}}_{\dot{\gamma}} dq^{\dot{\gamma}}), \quad e^{\dot{\alpha}1} = \Lambda^{\dot{\alpha}}_{\dot{\beta}} dq^{\dot{\beta}}, \quad (\text{A.5})$$

where $\Lambda^{\dot{\alpha}}_{\dot{\beta}}$ is $\text{SL}(2, \mathbb{C})$ -valued, $\Phi^{\dot{\alpha}}_{\dot{\beta}}$ is traceless, and κ is a customary constant. Indices are raised and lowered as $q^{\dot{\alpha}} = \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} q_{\dot{\beta}}$, $q_{\dot{\alpha}} = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} q^{\dot{\beta}}$. With Eq. (A.5) and $d\Sigma^{00} = 0$, the following can be shown. First of all, $\partial_{0\dot{\alpha}} \Phi^{\dot{\alpha}}_{\dot{\beta}} = 0$, which guarantees the local existence of a scalar function Φ such that $\Phi_{\dot{\alpha}\dot{\beta}} = \partial_{0\dot{\alpha}} \partial_{0\dot{\beta}} \Phi$. Second of all, the very equation

$$\square \Phi = \kappa \Phi_{\dot{\alpha}\dot{\beta}} \Phi^{\dot{\alpha}\dot{\beta}}. \quad (\text{A.6})$$

This derives the result that the sole dynamical geometry of SD spacetime is governed by Plebański's second heavenly equation [314].

In Eq. (A.6), $\square$ is the Laplacian associated with a flat metric $\eta = -2dp_{\dot{\alpha}} \odot dq^{\dot{\alpha}}$ on $\mathcal{M}$. To elaborate, this flat limit arises as the $\kappa \rightarrow 0$ limit of the curved metric which the coframe in Eq. (A.5) defines:

$$g = -2\epsilon_{\alpha\beta} \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} e^{\dot{\alpha}\alpha} \odot e^{\dot{\beta}\beta} = -2dp_{\dot{\alpha}} \odot dq^{\dot{\alpha}} + 2\kappa \Phi_{\dot{\alpha}\dot{\beta}} dq^{\dot{\alpha}} \odot dq^{\dot{\beta}}. \quad (\text{A.7})$$

Explicitly, the Laplacian operators associated with the flat metric η and the curved metric g are

$$\square := -2 \frac{\partial^2}{\partial p_{\dot{\alpha}} \partial q^{\dot{\alpha}}}, \quad \square_{\Phi} := \square - 2\kappa \Phi_{\dot{\alpha}\dot{\beta}} \frac{\partial^2}{\partial p_{\dot{\alpha}} \partial p_{\dot{\beta}}}, \quad (\text{A.8})$$

respectively. It follows that linearized perturbations around a solution Φ of Eq. (A.6) are harmonic with respect to the curved Laplacian $\square_{\Phi}$ [319].

Note that the coframe in Eq. (A.5) is unimodular, and the frame vector fields dual to Eq. (A.5) are

$$E_{0\dot{\alpha}} = (\Lambda^{-1})^{\dot{\beta}}_{\dot{\alpha}} \frac{\partial}{\partial p^{\dot{\beta}}}, \quad E_{1\dot{\alpha}} = (\Lambda^{-1})^{\dot{\beta}}_{\dot{\alpha}} \left(\frac{\partial}{\partial q^{\dot{\beta}}} - \kappa \Phi^{\dot{\gamma}}_{\dot{\beta}} \frac{\partial}{\partial p^{\dot{\gamma}}} \right). \quad (\text{A.9})$$

Note also that the SD Weyl tensor components are given by the fourth-order derivatives of the Plebański scalar Φ with respect to the p -coordinates.

A.2 Self-Dual Spacetime as a Cotangent Bundle

The coordinate choice in Eq. (A.4) is characterized by the Darboux formatting of the ASD two-forms Σ^{01} and Σ^{11} shown in Eq. (A.3). Hence diffeomorphisms of $\mathcal{M}$ shall be restricted such that Σ^{01} and Σ^{11} are preserved.

Consequently, the SD spacetime $\mathcal{M}$ is understood as the cotangent bundle $T^*\mathcal{N}$ of a two-dimensional symplectic manifold $\mathcal{N}$ [319, 321, 322]:

$$\mathcal{M} \cong T^*\mathcal{N}. \quad (\text{A.10})$$

Here, the base $\mathcal{N}$ is coordinatized by $q^{\dot{\alpha}}$ and endowed with $\Sigma^{11} = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} dq^{\dot{\alpha}} \wedge dq^{\dot{\beta}}$ as the symplectic form. The cotangent fibers are coordinatized by $p_{\dot{\alpha}}$ and soldered by $\Sigma^{01} = dq^{\dot{\alpha}} \wedge dp_{\dot{\alpha}}$.

Specifically, it can be seen that any vector field in $\mathcal{M}$ that preserves Σ^{01} and Σ^{11} is given by a linear combination of

$$X_u = -u^{\dot{\alpha}}(q) \frac{\partial}{\partial q^{\dot{\alpha}}} + p_{\dot{\beta}} u^{\dot{\beta}}{}_{\dot{\alpha}}(q) \frac{\partial}{\partial p_{\dot{\alpha}}}, \quad Y_v = v_{\dot{\alpha}}(q) \frac{\partial}{\partial p_{\dot{\alpha}}}, \quad (\text{A.11})$$

for $u, v \in C^\infty(\mathcal{N})$. Here, we have denoted

$$f_{\dot{\alpha}}(q) := \frac{\partial}{\partial q^{\dot{\alpha}}} f(q), \quad f_{\dot{\alpha}\dot{\beta}\dots}(q) := \frac{\partial}{\partial q^{\dot{\alpha}}} \frac{\partial}{\partial q^{\dot{\beta}}} f(q), \quad \dots, \quad (\text{A.12})$$

if $f(q)$ is a sole function of $q^{\dot{\alpha}}$. Clearly, X_u describes the uplift of a symplectomorphism on $\mathcal{N}$ while Y_v describes a certain translation of the zero section of $T^*\mathcal{N}$.

As a result, it follows that the residual diffeomorphisms form a Lie group $\text{SDiff}(\mathcal{N}) \ltimes (\Gamma(T^*\mathcal{N}), +)$, which describes a subgroup of $\text{SDiff}(\mathcal{M})$:

$$[X_u, X_v] = X_{u_{\dot{\alpha}}v^{\dot{\alpha}}}, \quad [X_u, Y_v] = Y_{u_{\dot{\alpha}}v^{\dot{\alpha}}}, \quad [Y_u, Y_v] = 0. \quad (\text{A.13})$$

Here, $\text{SDiff}(\mathcal{M})$ denotes the volume-preserving diffeomorphism group of $\mathcal{M}$, while $\text{SDiff}(\mathcal{N})$ describes the symplectomorphism group of $\mathcal{N}$.

The $2 + 2$ splicing of spacetime in Eq. (A.10) underlies the connection between SD gravity and two-dimensional models based on area-preserving diffeomorphisms [322, 325, 553].

In this context, consider how the infinitesimal coordinate transformations in Eq. (A.11) affect the Plebański scalar Φ . As the heavenly equation in Eq. (A.6) should be preserved, the change in the Plebański scalar Φ should be harmonic with respect to the curved Laplacian $\square_\Phi$.

For instance, take $\Lambda^{\dot{\alpha}}_{\dot{\beta}} = \delta^{\dot{\alpha}}_{\dot{\beta}}$ and compute $\mathcal{L}_{X_{\kappa u}} e^{\dot{\beta}\alpha} + \kappa u^{\dot{\alpha}}_{\dot{\beta}} e^{\dot{\beta}\alpha}$ and $\mathcal{L}_{Y_{\kappa v}} e^{\dot{\alpha}\alpha}$. The former is reproduced by $\Phi \mapsto \Phi + \delta_u \Phi$ while the latter is reproduced by $\Phi \mapsto \Phi + \delta_v \Phi$, where

$$\delta_u \Phi = \frac{1}{6} p_{\dot{\alpha}} p_{\dot{\beta}} p_{\dot{\gamma}} u^{\dot{\alpha}\dot{\beta}\dot{\gamma}}(q) + \kappa X_u[\Phi], \quad (\text{A.14a})$$

$$\delta_v \Phi = \frac{1}{2} p_{\dot{\alpha}} p_{\dot{\beta}} v^{\dot{\alpha}\dot{\beta}}(q) + \kappa Y_v[\Phi]. \quad (\text{A.14b})$$

Note that $e^{\dot{\alpha}\alpha} \mapsto (\delta^{\dot{\alpha}}_{\dot{\beta}} + \kappa u^{\dot{\alpha}}_{\dot{\beta}}) e^{\dot{\beta}\alpha}$ is undone by adjusting the SD frame $\Lambda^{\dot{\alpha}}_{\dot{\beta}}$.

Straightforward computation verifies that the variations in Eq. (A.14) are elements of $\ker(\square_{\Phi})$ but not $\ker(\square)$. Nonetheless, they leave the heavenly equation in Eq. (A.6) invariant.

When deriving Eq. (A.14), one integrates a second derivative $\partial^2/\partial p_{\dot{\alpha}} \partial p_{\dot{\beta}}$. The kernel for this operation describes

$$\delta_a \Phi = a(q), \quad \delta_b \Phi = p_{\dot{\alpha}} b^{\dot{\alpha}}(q), \quad (\text{A.15})$$

where $a, b \in C^{\infty}(\mathcal{N})$. The variations in Eq. (A.15) are elements of not only $\ker(\square)$ but also $\ker(\square_{\Phi})$. Thus, they leave each side of the heavenly equation in Eq. (A.6) invariant. Parenthetically, we might remark that Eq. (A.15) is analogous to the symmetry of Galileons [529].

Two solutions to the heavenly equation in Eq. (A.6) that can be mapped to each other via Eqs. (A.14) and (A.15) should be identified as the same solution, as they will describe the same spacetime geometry. In other words, Eqs. (A.14) and (A.15) describe the gauge redundancies of the heavenly equation in Eq. (A.6). See Ref. [319] for a further discussion.

A.3 Curved Twistor Theory

Finally, let us sketch the basics of curved twistor theory [310, 313, 340, 341]. In particular, in Sec. 2.5 (around Eq. (2.114)), we mentioned the Lax pair formulation of SDGR and its connection to curved twistor space. We should explicate on this point.

Recall that the Plebański approach to SD spacetime in App. A.1 utilizes a fixed ASD direction o_{α} , which essentially is a gauge fixing for SDGR: null axial gauge, i.e., lightcone gauge. On the contrary, the idea here is to treat all ASD directions on an equal footing by means of an auxiliary ASD spinor variable $\lambda_{\alpha} \in \mathbb{C}^2$, which can be thought of as projectively coordinatizing a Riemann sphere $\mathbb{CP}^1$.

A.3.1 Lax Pair Formulation of Self-Dual Gravity

There are two versions of this construction. The version of Plebański and Gindikin [314, 318] works in terms of differential forms. The version due to Mason and Newman [309, 316, 317] works in terms of vector fields.

The first version starts with a four-manifold $\mathcal{M}$ equipped with a nondegenerate volume form ν . If $\mathcal{M}$ admits a set of two-forms $\Sigma^{\alpha\beta}$ such that

$$d\Sigma(\lambda) = 0 \quad \text{and} \quad \Sigma(\lambda) \wedge \Sigma(\lambda) = 0 \quad \forall \lambda_\alpha \in \mathbb{C}^2, \quad (\text{A.16})$$

where $\Sigma(\lambda) := \lambda_\alpha \lambda_\beta \Sigma^{\alpha\beta}$, it can be shown that a coframe $e^{\dot{\alpha}\alpha}$ exists such that $\Sigma^{\alpha\beta} = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} e^{\dot{\alpha}\alpha} \wedge e^{\dot{\beta}\beta}$, which defines a SD metric¹ $g = \epsilon_{\alpha\beta} \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} e^{\dot{\alpha}\alpha} \odot e^{\dot{\beta}\beta}$ on $\mathcal{M}$.

The second version also starts with a four-manifold $\mathcal{M}$ equipped with a nondegenerate four-form ν , but assumes the existence of four linearly independent vector fields $V_{\alpha\dot{\alpha}}$ on it. For $\lambda_\alpha \in \mathbb{C}^2$, one then constructs the vector fields $L_{\dot{\alpha}} := \lambda^\alpha V_{\alpha\dot{\alpha}}$. If $L_{\dot{\alpha}}$ form a Lax pair such that

$$[L_{\dot{\alpha}}, L_{\dot{\beta}}] = 0 \quad \text{and} \quad \mathcal{L}_{L_{\dot{\alpha}}} \nu = 0 \quad \forall \lambda_\alpha \in \mathbb{C}^2, \quad (\text{A.17})$$

then the vector fields $E_{\alpha\dot{\alpha}} = f^{-1} V_{\alpha\dot{\alpha}}$ give an orthonormal frame of a SD Riemannian structure on $\mathcal{M}$ with $f^2 = i\nu(V_{0\dot{0}}, V_{0\dot{1}}, V_{1\dot{0}}, V_{1\dot{1}})$. Conversely, an SD spacetime $(\mathcal{M}, g)$ locally admits an orthonormal frame $E_{\alpha\dot{\alpha}}$ and a nonvanishing function f such that $L_{\dot{\alpha}} = f\lambda^\alpha E_{\alpha\dot{\alpha}}$ form a volume-preserving Lax pair as in Eq. (A.17). for some volume form ν . This is the Mason-Newman result that we had reviewed in Sec. 2.4.3, in fact.

The above two approaches can be related as $\tilde{\epsilon}^{\dot{\alpha}\dot{\beta}}(\iota_{L_{\dot{\alpha}}} \iota_{L_{\dot{\beta}}} \nu) = i\Sigma(\lambda)$ and $\iota_{L_{\dot{\alpha}}} \Sigma(\lambda) = 0$. In Plebański's second heavenly coordinates, the scaling factor is trivial as $f = 1$, so $L_{\dot{\alpha}} = \lambda^\alpha E_{\alpha\dot{\alpha}}$.

Crucially, the vanishing Lie bracket $[L_{\dot{\alpha}}, L_{\dot{\beta}}] = 0$ in Eq. (A.17) implies that translations along SD null directions in $\mathcal{M}$ are integrable for each fixed λ_α , forming SD null surfaces known as α -surfaces. The space of α -surfaces is the *twistor space* $\mathbb{PT}$. As will become self-evident, the twistor space has complex dimension three.

A.3.2 Deformed Incidence Relation

To establish the precise relation between the twistor space $\mathbb{PT}$ and the space-time $\mathcal{M}$, it is helpful to employ what is known as the correspondence space,

¹Note that this metric can also be directly obtained from $\Sigma^{\alpha\beta}$ via the Urbantke formula [339, 360].

the concept of which arises by promoting the auxiliary spinor variable λ_α to coordinates of an extended space.

The correspondence space $\mathcal{F} = \mathbb{P}S^-\mathcal{M}$ is the projectivization of the ASD spinor bundle, which arises by removing the zero section and stipulating the identification $\lambda_\alpha \sim r\lambda_\alpha$ for $r \in \mathbb{C}^*$. A function on $\mathcal{F}$ means a function $f(x, \lambda)$ on $S^-\mathcal{M}$ homogeneous in λ_α . It descends to the twistor space $\mathbb{PT}$ iff it is constant along the α -surfaces, meaning $L_{\dot{\alpha}}[f(x, \lambda)] = 0$ with $L_{\dot{\alpha}} = \lambda^\alpha E_{\alpha\dot{\alpha}} \in \Gamma(T\mathcal{F})$.

Moreover, one sees that $\Sigma(\lambda)$ in Eq. (A.16) defines a two-form on $\mathcal{F}$ that descends to the twistor space. In particular, it can be brought to a Darboux form $\Sigma(\lambda) = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} d_x F^{\dot{\alpha}}(x, \lambda) \wedge d_x F^{\dot{\beta}}(x, \lambda)$, where d_x denotes the “partial” exterior derivative that discards $d\lambda_\alpha$. Then $\iota_{L_{\dot{\gamma}}} \Sigma(\lambda) = 0$ gives

$$L_{\dot{\gamma}}[F^{\dot{\alpha}}(x, \lambda)] = 0, \quad (\text{A.18})$$

so $F^{\dot{\alpha}}(x, \lambda)$ gives a function on $\mathcal{F}$, with degree one in λ_α , that descends to $\mathbb{PT}$.

The incidence relation refers to the equation

$$\mu^{\dot{\alpha}} = F^{\dot{\alpha}}(x, \lambda). \quad (\text{A.19})$$

For each fixed λ_α and $\mu^{\dot{\alpha}}$, Eq. (A.19) defines an α -surface in $\mathcal{M}$. It follows that the pair $Z_A = (\lambda_\alpha, \mu^{\dot{\alpha}})$ provides projective coordinates for the twistor space $\mathbb{PT}$. Each point of $\mathbb{PT}$ represents the equivalence class $[Z]$ of $Z_A = (\lambda_\alpha, \mu^{\dot{\alpha}})$ under identifications by rescalings.

It is instructive to consider the twistor space of flat spacetime $\mathbb{M}$. Let $x^{\dot{\alpha}\alpha}$ be lightcone coordinates of $\mathbb{M}$. α -surfaces in $\mathbb{M}$ describe $x^{\dot{\alpha}\alpha} \sim x^{\dot{\alpha}\alpha} + y^{\dot{\alpha}}\lambda^\alpha$ for fixed λ^α and variable $y^{\dot{\alpha}}$, along which

$$\mu^{\dot{\alpha}} = x^{\dot{\alpha}\alpha} \lambda_\alpha \quad (\text{A.20})$$

are constant. Eq. (A.20) is the incidence relation for flat spacetime. The flat twistor space is

$$\mathbb{PT} = \left\{ [(\lambda_\alpha, \mu^{\dot{\alpha}})] \in \mathbb{CP}^3 \mid \lambda_\alpha \neq 0 \right\}. \quad (\text{A.21})$$

Upon introduction of spacetime curvature, the incidence relation in Eq. (A.20) develops a “general-relativistic correction” and gets deformed into the generic form in Eq. (A.19). In this sense, Eq. (A.19) is sometimes referred to as “deformed incidence relation” for an explicit disambiguation [310].

The celebrated nonlinear graviton construction of Penrose [313] states that, the curved twistor space $\mathbb{PT}$ of a SD spacetime $\mathcal{M}$ locally arises by deforming the complex structure of flat twistor space $\mathbb{PT}$. This deformation is known to utilize the holomorphic Poisson structure,

$$I = \tilde{\epsilon}^{\dot{\alpha}\dot{\beta}} \frac{\partial}{\partial \mu^{\dot{\alpha}}} \wedge \frac{\partial}{\partial \mu^{\dot{\beta}}} . \quad (\text{A.22})$$

The origin of Eq.(A.22) can be traced back to the fact that the Gindikin 2-form $\Sigma(\lambda) = \tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} d_x F^{\dot{\alpha}}(x, \lambda) \wedge d_x F^{\dot{\beta}}(x, \lambda)$ corresponds to $\tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} d\mu^{\dot{\alpha}} \wedge d\mu^{\dot{\beta}}$.

A.3.3 Double Fibration

More precisely, the geometry of the incidence relation is discussed in the picture known as double fibration, shown below.

$$\begin{array}{ccc} & \mathcal{F} & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ \mathbb{PT} & & \mathcal{M} \end{array} \quad (\text{A.23})$$

The map $\pi_1 : \mathcal{F} \rightarrow \mathbb{PT} : (x^\mu, [\lambda_\alpha]) \mapsto [(\lambda_\alpha, F^{\dot{\alpha}}(x, \lambda))]$ formalizes the incidence relation in Eq.(A.19). The map $\pi_2 : \mathcal{F} \rightarrow \mathcal{M} : (x^\mu, [\lambda_\alpha]) \mapsto x^\mu$ is the bundle projection. Via this double fibration, geometric data in twistor space is transcribed to spacetime and vice versa.

A well-known duality between twistor space and spacetime is discussed as follows. Each twistor $[Z] \in \mathbb{PT}$ defines the α -surface in spacetime as

$$\mathcal{S}_{[Z]} = \pi_2(\pi_1^{-1}([Z])) \subset \mathcal{M} . \quad (\text{A.24})$$

Meanwhile, each spacetime point $x \in \mathcal{M}$ defines a holomorphically embedded $\mathbb{CP}^1$ in the twistor space $\mathbb{PT}$, known as the twistor line $\mathcal{L}_x$:

$$\mathcal{L}_x := \pi_1(\pi_2^{-1}(x)) \subset \mathbb{PT} . \quad (\text{A.25})$$

In this context, the incidence relation provides an explicit parameterization of the twistor line: $[Z_A] = [(\lambda_\alpha, F^{\dot{\alpha}}(x, \lambda))]$ for $\lambda_\alpha \in \mathbb{C}^2$. Finally, it follows that

$$[Z] \in \mathcal{L}_x \subset \mathbb{PT} \iff x \in \mathcal{S}_{[Z]} \subset \mathcal{M} , \quad (\text{A.26})$$

meaning that an α -surface $\mathcal{S}_{[Z]}$ is incident at the spacetime point x iff the twistor $[Z]$ lies on the twistor line $\mathcal{L}_x$. The vivid picture behind this statement

is that the twistor line $\mathcal{L}_x$ essentially represents the lightcone, i.e., *sky* at x (as the space $\mathbb{CP}^1 \cong S^2$ of null directions). Evidently, a twistor $[Z]$ appears on the sky $\mathcal{L}_x$ for an observer at x iff x lies on the pathway of the light ray $[Z]$.

To conclude, the twistor space $\mathbb{PT}$ is the space of SD null surfaces in a SD spacetime $\mathcal{M}$. The precise relationship between $\mathbb{PT}$ and $\mathcal{M}$ arises via the correspondence space and the double fibration. This framework essentially traces back to the Lax pair formulation of SDGR. The twistor-space counterpart of spacetime point $x \in \mathcal{M}$ is the twistor line $\mathcal{L}_x$, which gets deformed into “non-straight curves” upon introduction of gravity. Accordingly, in twistor space, SDGR is formulated as deformations of the complex structure.

Twistor space finds various applications in physics. For instance, the Penrose transform [607] famously constructs massless on-shell fields (zero-rest-mass fields) from meromorphic functions on twistor space (more precisely, representatives of twistor cohomology classes) via contour integral [606, 814, 837].

A.4 Canonical Transformations on Fibers

The redundancies of the heavenly framework, discussed in App. A.2, corresponds to Hamiltonian vector fields in twistor space with respect to the holomorphic Poisson structure in Eq. (A.22). This Poisson structure is equipped to each fiber of $\pi_0 : \mathbb{PT} \rightarrow \mathbb{CP}^1 : [(\lambda_\alpha, \mu^{\dot{\alpha}})] \mapsto [\lambda_\alpha]$.

To approach this fact concretely, one may use an explicit formula for the deformed incidence relation in Plebański’s second heavenly coordinates [319, 320]:

$$F^{\dot{\alpha}}(x, \lambda) = x^{\dot{\alpha}\alpha} \lambda_\alpha + \kappa \frac{\partial \Phi}{\partial p_{\dot{\alpha}}} \frac{\lambda_0^2}{\lambda_1} + \kappa \frac{\partial \Phi}{\partial q_{\dot{\alpha}}} \frac{\lambda_0^3}{\lambda_1^2} + \cdots. \quad (\text{A.27})$$

This explicit formula describes a series in λ_0/λ_1 and is valid in the affine patch $\lambda_1 \neq 0$ of $\mathbb{CP}^1$. It arises in a framework known as Dunajski-Mason recursion [319, 882, 883].

Based on Eq. (A.27), one can directly investigate the transformation behavior of $F^{\dot{\alpha}}(x, \lambda)$ under the redundancies identified in App. A.2. Computation verifies that it amounts to a Poisson diffeomorphism with respect to Eq. (A.22) that transforms the μ -fibers of $\pi_0 : \mathbb{PT} \rightarrow \mathbb{CP}^1$ as

$$\tilde{\epsilon}_{\dot{\alpha}\dot{\beta}} \delta \mu^{\dot{\alpha}} d\mu^{\dot{\beta}} = dh(\mu), \quad (\text{A.28})$$

where the Hamiltonian, after pullback by $\pi_1 : \mathcal{F} \rightarrow \mathbb{P}\mathcal{T}$, exhibits the expansion $h(F(x, \lambda)) = \kappa \lambda_1^2 (h^{(0)}(x) - h^{(1)}(x)\zeta + h^{(2)}(x)\zeta^2 - h^{(3)}(x)\zeta^3 + \dots)$:

$$\begin{aligned}
 h^{(0)} &= u(q), \quad h^{(1)} = p_{\dot{\alpha}} u^{\dot{\alpha}}(q) + v(q), \\
 h^{(2)} &= \frac{1}{2} p_{\dot{\alpha}} p_{\dot{\beta}} u^{\dot{\alpha}\dot{\beta}}(q) + \kappa u_{\dot{\alpha}}(q) \Phi^{\dot{\alpha}} + p_{\dot{\alpha}} v^{\dot{\alpha}}(q) - b(q), \\
 h^{(3)} &= \frac{1}{6} p_{\dot{\alpha}} p_{\dot{\beta}} p_{\dot{\gamma}} u^{\dot{\alpha}\dot{\beta}\dot{\gamma}}(q) + \kappa X_u[\Phi] + \frac{1}{2} p_{\dot{\alpha}} p_{\dot{\beta}} v^{\dot{\alpha}\dot{\beta}}(q) + \kappa X_v[\Phi] \\
 &\quad - a(q) - p_{\dot{\alpha}} b^{\dot{\alpha}}(q).
 \end{aligned} \tag{A.29}$$

In fact, a quick derivation of Eq. (A.29) is viable by series-expanding $u(F(x, \lambda))$, $v(F(x, \lambda))$, etc.

WORLDLINE FORMALISMS

This appendix reviews the computation or definition of fundamental objects in scattering theory in first-quantized frameworks.

B.1 S-Matrix Element; Scattering Amplitudes

B.1.1 The Classic Worldline Formalism

The celebrated worldline formalism [750–761] provides a direct realization of the idea that Feynman diagrams visualize spacetime trajectories of quantum particles. In particular, it establishes that the transition amplitude (as partition function) for a first-quantized relativistic particle, moving from one spacetime point to another, derives an off-shell propagator in field theory. The formalism is also known for providing powerful pathways toward loop amplitudes or effective actions and has also been identified as the infinite-tension counterpart of string-theoretic computations [755–758].

To elaborate, the paradigmatic calculation in this formalism gives rise to the Schwinger parameter representation of the free off-shell propagator (see, e.g., Ref. [752–754]):

$$\frac{-i}{p^2 + m^2 - i\epsilon} = \int_0^\infty dT \, e^{-iT(p^2 + m^2)}. \quad (\text{B.1})$$

Here, the right-hand side arises by a faithful sum over histories for a worldline traveling from one spacetime point to another, which involves a sum over the moduli space of einbeins e on a finite interval $[0, 1]$. This refers to the path integral,

$$\int \frac{\mathcal{D}e}{\text{vol}(\text{Gauge})} \longrightarrow \int_0^\infty dT \int \mathcal{D}e \, \delta[e - 2T], \quad (\text{B.2})$$

where the path integral measure is crucially quotiented on account of the one-dimensional diffeomorphism redundancy on the worldline.

To extract tree-level scattering amplitudes in the worldline formalism, the conventional practice simply imports the LSZ reduction formula as an *external input* from field theory. For instance, the multi-photon Compton amplitudes are obtained by first computing the off-shell propagators in the backgrounds

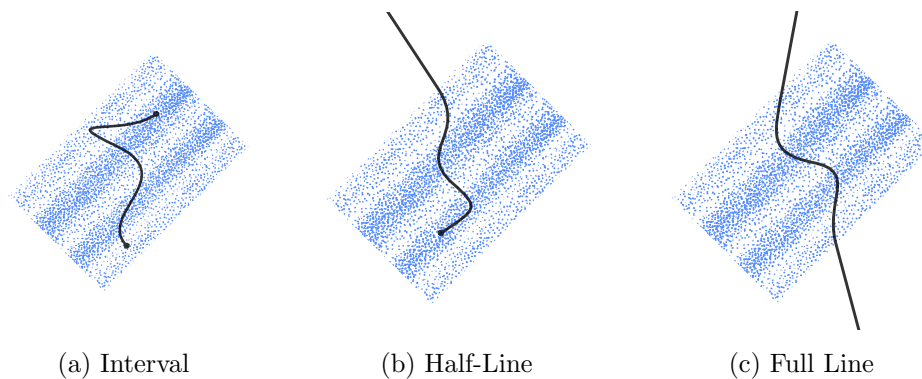


Figure B.1: Various worldline topologies for particle propagation in sandwich geometry. (Figure adapted from Ref. [763]).

of superposed electromagnetic plane waves and then amputating two external legs via LSZ reduction.

Computationally, however, this post-processing might not be entirely trivial in practice. Besides the technicalities of Gaussian and Fourier integrals, a central tension lies in the fact that the primary output of the worldline formalism is an off-shell propagator in position space, whereas scattering amplitudes are on-shell objects examined in momentum space. Thus, if one's goal is to obtain tree-level scattering amplitudes, the original workflow might have offered a slightly roundabout path, although conceptually faithful and direct.

B.1.2 Reduction by Topology

Regarding this point, it had been observed [884–889] that employing non-compact worldline topologies, such as half-line or full line, can automate the LSZ reductions. This observation traces back to Fradkin [885], according to Ref. [884], and has been revived in a modern literature as well [890, 891].

Notably, Ref. [894] provided a top-down derivation of this fact from the moduli space of einbein for each noncompact topology, while Ref. [763], released in the same year as Ref. [894], had also independently discovered the same conclusion through a physical motivation (an exploration of boundary conditions in the first-order formulation, in connection with manifest gauge invariance). Crucially, based on the geometric semantics of LSZ reduction as bulk-to-boundary propagation [289], it was realized that the first-quantized probability amplitude for the interval, half-line, and full line topologies can re-

spectively be interpreted as a bulk-to-bulk, bulk-to-boundary, and boundary-to-boundary propagator, the last of which is identified with scattering amplitude [763]. It follows that states escaping to the boundaries must be on-shell for consistency and finiteness of results.

The mantra goes as follows. What have escaped to the boundary are on-shell states. Bulk-to-bulk processes are virtual and are off-shell. LSZ reduction sends the bulk endpoints of an off-shell propagator to the boundaries, making the particle register the infinities at far past and future.

Fig.B.1 visualizes the three worldline topologies, interval $[0, 1]$, half-line $[0, \infty)$, and full line $(-\infty, +\infty)$. The recipe for the einbein path integral is given for each topology as follows.¹

For the interval topology, the recipe is well-known from classic literature (see, e.g., Ref. [754]) and is shown in Eq. (B.2). The moduli space of einbeins is one-dimensional and is coordinatized by the diffeomorphism-invariant total length $\int_0^1 e(\sigma) d\sigma = 2T$, and this T is dubbed the Schwinger proper time. In Eq. (B.2), $\delta[e - 2T]$ is the delta functional fixing the einbein to a constant profile, $e(\sigma) = 2T$. Consequently, one performs an ordinary integral over the Schwinger proper time T .

For the half-line topology, the recipe is [763, 894]

$$\int \frac{\mathcal{D}e}{\text{vol}(\text{Gauge})} \longrightarrow \int \mathcal{D}e \, \delta[e - e_\infty], \quad (\text{B.3})$$

which gauge-fixes the einbein to a nonzero constant e_∞ . This is because the moduli space collapses to a single point: all half-lines are diffeomorphic to each other.² In fact, by the very reparametrization invariance, the value of e_∞ is irrelevant: one can rescale it away to any other value, of which the partition function is completely ignorant. Consequently, one simply drops the einbein path integral entirely, in practice.

For the full line topology, the recipe reads [763, 894]

$$\int \frac{\mathcal{D}e}{\text{vol}(\text{Gauge})} \longrightarrow \frac{1}{\delta(0)} \int \mathcal{D}e \, \delta[e - e_\infty], \quad (\text{B.4})$$

where $\delta(0) = \int_{\mathbb{R}} d\sigma = \text{vol}(\mathbb{R})$ is the volume of the space of constant translations. In this case, the gauge quotient is bigger than the integral itself. While

¹Here, we ignore the Faddeev-Popov determinant for simplicity.

²This assumes sum over complete metrics such that the total length of the half-line is infinite, in which case the einbein has to asymptote to the nonzero constant e_∞ .

all full lines are diffeomorphic to each other, there is a residual gauge redundancy formed by diffeomorphisms that asymptote to a common constant value at both ends, the space of which is $\mathbb{R}$.³ Again, the value of the constant e_∞ in Eq. (B.4) can be arbitrary. Consequently, one drops the einbein path integral while also discarding an infinite volume factor $\delta(0)$.

Ref. [763] gives a friendly discussion on how the iconic calculation in Eq. (B.1) gets revisited with the noncompact worldlines. Concretely, suppose the computation of the massive off-shell propagator of a charged particle dressed with one photon leg. Let the momenta for the massive legs be p_1 and p_2 while k is the wavenumber for the massless leg.

In the classic version employing the interval topology, the core part of the computation describes

$$\int_0^\infty dT e^{-iTX_2} \int_0^1 d\sigma T e^{iTK\sigma} = \frac{-i}{X_2 - i\epsilon} \frac{-i}{X_2 - K - i\epsilon}, \quad (\text{B.5})$$

where we have denoted $X_2 = p_2^2 + m^2$ and $K = 2k \cdot p_2$. This produces two propagators for the two off-shell massive legs. Hence two LSZ reductions are performed by hand.

In comparison, the core part of the half-line approach describes

$$\int_0^\infty d\sigma e^{iK\sigma} = \frac{-i}{-K - i\epsilon}, \quad (\text{B.6})$$

provided that one imposes $X_2 = 0$ proactively. This produces one off-shell massive pole, namely $1/(p_1^2 + m^2)$. In this case, only one LSZ reduction needs to be performed manually.

Lastly, in the case of the full line topology, the core calculation describes

$$\frac{1}{\delta(0)} \int_{-\infty}^\infty d\sigma e^{iK\sigma} = \frac{\delta(K)}{\delta(0)} = 1, \quad (\text{B.7})$$

in which case one imposes both $p_1^2 + m^2 = 0$ and $p_2^2 + m^2 = 0$ from the beginning, so $K = 0$. Again, what have escaped to the boundary are on-shell states. Here, all the LSZ reductions are completely automated.⁴

³Here, we again presume sum over complete metrics and take the two ends of the full line as distinguishable.

⁴Yet, one comment is that, in the case of the conventional scalar particle, there is no free-theory saddle that changes the value of the momentum, so the calculation using full line topology could be a bit tricky or subtle to carry out. Notably, the massive twistor worldline avoids this issue in an amusing way.

Note that, when utilizing noncompact topologies, it is desirable to have definite-momentum states be realized at the asymptotic infinities. In fact, based on this connection between topologies and boundary conditions, the key motivation and idea of Ref. [763] was to utilize phase space actions to have momentum as a basic variable on the worldline, automating both LSZ reductions and Fourier transforms.

To sum up, the above geometrical discourse [763, 894, 895] achieves a direct formulation of LSZ reduction *within* first quantization.

When equipped with this automation of LSZ reduction, the worldline formalism becomes perfectly suited for computing Compton scattering amplitudes efficiently. In particular, Ref. [763] concretely showed that computing the boundary-to-boundary propagator in the backgrounds of nonlinearly superposed plane waves in YM theory and GR derives the gluon and graviton Compton scattering amplitudes of a massive probe, while also providing an explicit higher-order exploration in Maxwell theory up to six points.

B.1.3 Universal Feynman Rules for Phase Space Actions

The classic worldline formalism [750–762] had conventionally preferred working with configuration space (second-order) actions. On the other hand, Ref. [763] demonstrated the use of phase space (first-order) actions.

Specifically, Ref. [763] worked out the systematics of the perturbation theory of generic one-dimensional sigma models onto a symplectic target. The Feynman rules are organized in terms of elements of symplectic geometry. In particular, some key features that are helpful to remember are reported as follows.

1. The bare propagator encodes the free theory’s Poisson bracket.
2. Vertices arise from Hamiltonian and symplectic perturbations (dubbed Hamiltonian and symplectic vertices, respectively).
3. Vertices sourcing no worldline fluctuation compute the interaction action about the background trajectory.
4. Vertices sourcing one worldline fluctuation arise from corrections to the Hamiltonian EoM. Especially, the linearized symplectic vertex of valence one is quickly derived by a formula analogous to “Lorentz force.”
5. The symplectic vertices are classified into two types: regular and pinched.

To see these facts, suppose $(\theta^\circ, \omega^\circ = d\theta^\circ, H^\circ)$ and $(\theta, \omega = d\theta, H)$ define the free theory and the interacting theory on an even-dimensional phase space $\mathcal{P}$ in terms of first-order Lagrangians, equipped with coordinates ζ^i that are Darboux with respect to ω° :

$$L^\circ[\zeta] = \theta_i^\circ(\zeta) \dot{\zeta}^i - H^\circ(\zeta), \quad L[\zeta] = \theta_i(\zeta) \dot{\zeta}^i - H(\zeta). \quad (\text{B.8})$$

Ref. [763] finds that the all-order variation of $L[\zeta]$ can be given as

$$\begin{aligned} L[\zeta + \delta\zeta] = & L^\circ[\zeta] + L_{\text{bd}}[\zeta, \delta\zeta] + \delta\zeta^i \left(\omega_{ij}^\circ(\zeta) \dot{\zeta}^j - H_{,i}^\circ(\zeta) \right) \\ & + L_{\text{prop}}[\zeta, \delta\zeta] + L_{\text{sv}}[\zeta, \delta\zeta] + L_{\text{hv}}[\zeta, \delta\zeta], \end{aligned} \quad (\text{B.9})$$

where terms inside the bracket vanish upon plugging in a saddle of the free theory as a background, say $\sigma \mapsto \zeta^i(\sigma)$. The boundary term is chosen as

$$L_{\text{bd}}[\zeta, \delta\zeta] = \frac{d}{d\sigma} \left[\sum_{n=0}^{\infty} \frac{1}{(n+1)!} \theta_{j,i_1 \dots i_n}(\zeta) \delta\zeta^j \delta\zeta^{i_1} \dots \delta\zeta^{i_n} \right]. \quad (\text{B.10})$$

Propagator The worldline propagator is determined from the term

$$L_{\text{prop}}[\zeta, \delta\zeta] = \frac{1}{2} \omega_{ij}^\circ \delta\zeta^i \delta\zeta^j, \quad (\text{B.11})$$

which is quadratic in the worldline fluctuation $\delta\zeta^i$. Via Dirac's correspondence, it follows that the worldline propagator must satisfy

$$\lim_{\epsilon \rightarrow 0^+} \left(G^{ij}(\epsilon, 0) - G^{ji}(\epsilon, 0) \right) = -(\omega^{\circ-1})^{ij}, \quad (\text{B.12})$$

in any causality/boundary prescriptions. For example,

$$G^{ij}(\sigma_1, \sigma_2) = (\omega^{\circ-1})^{ij} \frac{1}{-\partial_{\sigma_1}} \delta(\sigma_1 - \sigma_2) \quad (\text{B.13})$$

qualifies as a Green's function, where the anti-derivative $(-\partial_{\sigma_1})^{-1} \delta(\sigma_1 - \sigma_2)$ is resolved into either of the following:

$$\Theta(\sigma_1, \sigma_2) = \begin{cases} -1/2 & \text{if } \sigma_1 > \sigma_2, \\ +1/2 & \text{if } \sigma_1 < \sigma_2, \end{cases} \quad (\text{B.14a})$$

$$\Theta_{>}(\sigma_1, \sigma_2) = \begin{cases} -1 & \text{if } \sigma_1 > \sigma_2, \\ 0 & \text{if } \sigma_1 < \sigma_2, \end{cases} \quad (\text{B.14b})$$

$$\Theta_{<}(\sigma_1, \sigma_2) = \begin{cases} 0 & \text{if } \sigma_1 > \sigma_2, \\ +1 & \text{if } \sigma_1 < \sigma_2. \end{cases} \quad (\text{B.14c})$$

In general, however, it is not always the case that $G^{ij}(\sigma_1, \sigma_2)$ can be written in the form in Eq. (B.13).

Vertices Suppose the free-interaction splits,

$$\theta = \theta^\circ + \theta', \quad H = H^\circ + H'. \quad (\text{B.15})$$

The Hamiltonian vertices arise from

$$\begin{aligned} L_{\text{hv}}[\zeta, \delta\zeta] \\ = -H'(\zeta) - H'_{,i}(\zeta)\delta\zeta^i - \sum_{n=2}^{\infty} \frac{1}{n!} H_{,i_1 i_2 \dots i_n}(\zeta) \delta\zeta^{i_1} \delta\zeta^{i_2} \dots \delta\zeta^{i_n}, \end{aligned} \quad (\text{B.16})$$

which should give familiar result. The symplectic vertices, on the other hand, arise from

$$L_{\text{sv}}[\zeta, \delta\zeta] = L_{\text{sv}}^{\text{regular}}[\zeta, \delta\zeta] + L_{\text{sv}}^{\text{pinched}}[\zeta, \delta\zeta], \quad (\text{B.17a})$$

$$L_{\text{sv}}^{\text{regular}}[\zeta, \delta\zeta] = \theta'_i(\zeta) \dot{\zeta}^i - \sum_{n=1}^{\infty} \frac{1}{n!} \dot{\zeta}^j \omega'_{ji_1, i_2 \dots i_{n-1}}(\zeta) \delta\zeta^{i_1} \dots \delta\zeta^{i_{n-1}}, \quad (\text{B.17b})$$

$$L_{\text{sv}}^{\text{pinched}}[\zeta, \delta\zeta] = - \sum_{n=2}^{\infty} \frac{n-1}{n!} \omega'_{ji_1, i_2 \dots i_{n-1}}(\zeta) \delta\zeta^j \delta\zeta^{i_1} \dots \delta\zeta^{i_{n-1}}. \quad (\text{B.17c})$$

In particular, the symplectic vertex with no worldline fluctuation is simply the interaction action due to the symplectic perturbation,

$$\bullet = \int d\sigma \theta'_i(\zeta(\sigma)) \dot{\zeta}^i(\sigma), \quad (\text{B.18})$$

whereas the symplectic vertex with $n = 1$ worldline fluctuation is the “Lorentz force due to symplectic perturbation”:

$$\delta\zeta^i(\sigma) \text{---}\bullet = \omega'_{ij}(\zeta(\sigma)) \dot{\zeta}^j(\sigma). \quad (\text{B.19})$$

Note that the derivation of Eqs. (B.18) and (B.19) itself does not require complicated calculation. For instance, Eq. (B.19) can be directly seen from the Hamiltonian EoM. Otherwise, consider the “**T**-and-**F** formalism” in Sec. 14.7.2: **T** gives the free theory’s time evolution vector field while **F** formalizes the fluctuation. The first-order variation of the interaction symplectic potential gives

$$\mathcal{L}_{\mathbf{F}}\theta' = \iota_{\mathbf{F}}\omega' + d(\iota_{\mathbf{F}}\theta') \implies \langle \mathcal{L}_{\mathbf{F}}\theta', \mathbf{T} \rangle = \iota_{\mathbf{T}}\iota_{\mathbf{F}}\omega' + \mathbf{T}[(\iota_{\mathbf{F}}\theta')]. \quad (\text{B.20})$$

Since $\mathbf{T}[(\iota_{\mathbf{F}}\theta')]$ gives a boundary term when integrated after pullback onto the background worldline, the vertex of fluctuation valence one is

$$\mathcal{V}^{(1)} = \iota_{\mathbf{T}}\iota_{\mathbf{F}}\omega' = \omega'(\mathbf{F}, \mathbf{T}), \quad (\text{B.21})$$

which precisely gives Eq. (B.19) after the pullback; recall Eq. (14.52b).

B.2 S-Symplectomorphism; Scattering Generators

The worldline formalism covered in App. B.1 computes scattering *amplitudes* as transition amplitudes of first-quantized particles. It implements a path integral calculation with definite initial and final conditions, and thus computes S-matrix elements with kets and bras:

$$\mathcal{M}(p_2, p_1) = \langle p_2 | \hat{T} | p_1 \rangle. \quad (\text{B.22})$$

By “classical amplitude,” it means the $\hbar \rightarrow 0$ limit of Eq. (B.22) if well-defined.

In this appendix, we review an alternative formalism that computes a different object:

$$\Omega = \lim_{\hbar \rightarrow 0} \mathcal{Q}^{-1}(i\hbar \log \hat{S}). \quad (\text{B.23})$$

This framework has been recently established through works [795, 870, 896–903], with an early identification in Ref. [673]⁵ and roots in the formalism of, e.g., Refs. [891–893]. In fact, it traces back to works [904–910] around the ’70–80s as well. The object Ω in Eq. (B.23) has been dubbed *classical scattering generator*. It provides a nice in-in formalism well-suited for classical problems.

Via Hamiltonian action, the classical scattering generator is shown to generate the map from the initial phase space to the final phase space in classical scattering, which we dub *S-symplectomorphism* [870, 901] or classical S-map. Said in another way, the scattering generator is an “effective Hamiltonian” that reproduces the entire interaction-picture time evolution from far past to far future within dimensionless unit time. The impulses of classical observables are computed via the pullback map of the S-symplectomorphism [901], which unfolds into the so-called nested bracket formula [795, 896, 897]:

$$\Delta f = \{f, \Omega\} + \frac{1}{2!} \{\{f, \Omega\}, \Omega\} + \frac{1}{3!} \{\{\{f, \Omega\}, \Omega\}, \Omega\} + \dots \quad (\text{B.24})$$

One of the landmark successes of the modern amplitudes program is indisputably its fruitful applications to classical physics. In particular, the celebrated formalism of Kosower, Maybee, and O’Connell [911] established the systematics of extracting classical observables from the S-matrix via the $\hbar \rightarrow 0$ limit. The S-symplectomorphism program seeks to provide a purely and manifestly classical framework, in which the scattering generator emerges as a new currency, while also greatly clarifying the relation with the quantum S-matrix.

⁵“a dequantization of the S-matrix will be a time-ordered action of the deformed Hamiltonian vector fields as infinitesimal diffeomorphisms on the physical phase space”

Historically, it seems that the idea of S-symplectomorphism⁶ had been proposed and investigated in works [904–910] over the '70s and '80s, under the name “(canonical) S-transformation.” Interesting applications can be found in, e.g., Refs. [908, 910].

To be precise, there are various physical situations in which the phase space is a degenerate Poisson manifold. In such cases, the terminologies “classical S-map” or “Poisson S-diffeomorphism” will be adequate, although one could always work in the symplectic setup in principle since every Poisson manifold admits a symplectic realization.

B.2.1 Purely Classical Definition

Concretely, suppose a classical system formulated on a phase space $\mathcal{P}$, endowed with a symplectic structure ω for simplicity. A principle of Hamiltonian mechanics reads that time evolution is a symplectomorphism. Suppose $U^\circ(t_1, t_2)$ and $U(t_1, t_2)$ are the time-evolution symplectomorphisms from the phase space at time t_2 to the phase space at time t_1 , due to the free and interacting theory of the system, respectively. We restrict our attention to perturbative interactions with a coupling parameter ϵ , so $\lim_{\epsilon \rightarrow 0} U(t_1, t_2) = U^\circ(t_1, t_2)$.

The S-symplectomorphism, realized at a fixed time t_0 of one's choice (usually $t_0 = 0$), is defined as

$$S = \lim_{t_\pm \rightarrow \pm\infty} \left(U^\circ(t_0, t_+) \circ U(t_+, t_-) \circ U^\circ(t_-, t_0) \right), \quad (\text{B.25})$$

provided a well-defined limit.

To see the physical significance of the formula in Eq. (B.25), we may borrow the framework of the Liouville equation in classical statistical mechanics, in which case the time-evolution symplectomorphisms act on the time-dependent probability distribution $\rho(t)$, representing the state of a classical ensemble, as

$$\rho(t_1) = \rho(t_2) \circ U(t_2, t_1). \quad (\text{B.26})$$

With Eq. (B.26), it can be seen that Eq. (B.25) implies

$$\tilde{\rho}(+\infty) = \tilde{\rho}(-\infty) \circ S^{-1} = S^*[\tilde{\rho}(-\infty)]. \quad (\text{B.27})$$

Firstly, we have denoted

$$\tilde{f}(t) := f(t) \circ U^\circ(t, t_0). \quad (\text{B.28})$$

⁶The terminology “symplectomorphism” is being preferred over “canonical transformation” here due to a standardized contextual meaning of “generator of canonical transformation” (as on-shell action, generating function).

This is the definition of the interaction picture in classical mechanics tracing back to Campbell [912]. Secondly, we have adopted the differential operator representation of diffeomorphisms [541] by the induced left action,⁷

$$S^*[f] = f \circ S^{-1}. \quad (\text{B.29})$$

With these understandings, Eq. (B.27) establishes that the induced left action S^* by the S-symplectomorphism S maps the initial classical state $\tilde{\rho}(-\infty)$ to the final state $\tilde{\rho}(+\infty)$ in the interaction picture. Recalling that the S-matrix $\hat{S}$ maps the initial quantum state $|\psi(-\infty)\rangle$ to the final quantum state $|\psi(+\infty)\rangle$, one sees that the S-symplectomorphism provides a classical counterpart of the S-matrix. As will be reviewed in App. B.2.3, the precise relation between the S-symplectomorphism and the S-matrix can be established [901] by means of the phase space formulation of quantum mechanics [913–919].

It is not difficult to see that the definition in Eq. (B.28) conforms to the expected concept of the interaction picture in classical mechanics. In fact, it simply means to plug in the free theory time evolution, such as the straight-line motion in nonrelativistic mechanics: $\tilde{f}(x, p; t) = f(x + pt/m, p; t)$.

In Eq. (B.27), S^* gives the differential operator representation of S [541], acting on $\tilde{\rho}(-\infty)$. The definition of the classical scattering generator Ω is

$$X_\Omega = \log S^* \implies S^* = \exp(X_\Omega), \quad (\text{B.30})$$

with the identification of vector fields with first-order differential operators. Namely, Ω is the generator of the S-symplectomorphism in the senses of Hamiltonian action and exponential map. The technical ambiguity in Ω due to addition of constant functions will be fixed shortly. Crucially, the exponential representation in Eq. (B.30) manifests symplecticity of classical scattering.

B.2.2 Concrete Definition and Computation by Magnus Series

To be more concrete, suppose the time-evolution symplectomorphisms $U^\circ(t_1, t_2)$ and $U(t_1, t_2)$ arise by Hamiltonian evolutions due to H° and $H(t)$, respectively. Suppose the free-interaction split

$$H(t) = H^\circ + \epsilon V(t). \quad (\text{B.31})$$

As is well-known, the Liouville equation is given by

$$\rho(t_1) = U(t_1, t_2)^*[\rho(t_2)] \iff \dot{\rho}(t) = X_{H(t)}[\rho(t)]. \quad (\text{B.32})$$

⁷We use this notation within this appendix; this is the pullback by S^{-1} .

Here, $X_f = \{f, \quad\}$ denotes the Hamiltonian vector field of f .

In Eq. (B.32), the time-evolution diffeomorphism $U(t_1, t_2)$ is concretely represented as a *differential operator* $U(t_1, t_2)^*$, which can be explicitly computed by Dyson series [920].

After implementing the interaction picture, it can be shown that

$$\tilde{\rho}(t) = \epsilon X_{\tilde{V}(t)}[\tilde{\rho}(t)], \quad (\text{B.33})$$

where $\tilde{V}(t) := V(t) \circ U^\circ(t, t_0)$. Eqs. (B.27) and (B.33) lead to the Dyson series representation of the S-symplectomorphism,

$$S^* = \text{T exp} \left(\epsilon \int dt X_{\tilde{V}(t)} \right). \quad (\text{B.34})$$

Computing Eq. (B.34) could yet be not practically ideal, since it generically produces a giant collection of differential operators of all degree and will violate symplecticity perturbatively upon truncation at each finite order.

Since Eq. (B.34) represents a symplectomorphism, its actual content simply boils down to a single Hamiltonian function, namely the classical scattering generator Ω in Eq. (B.30):

$$X_\Omega = \log \text{T exp} \left(\epsilon \int dt X_{\tilde{V}(t)} \right). \quad (\text{B.35})$$

Notably, the Magnus series [921] facilitates the explicit evaluation of Ω as

$$\begin{aligned} \Omega = & \epsilon \int dt_1 \tilde{V}(t_1) + \frac{\epsilon^2}{2} \int_{t_1 > t_2} d^2t \{ \tilde{V}(t_1), \tilde{V}(t_2) \} \\ & + \frac{\epsilon^3}{6} \int_{t_1 > t_2 > t_3} d^3t \left(\{ \tilde{V}(t_1), \{ \tilde{V}(t_2), \tilde{V}(t_3) \} \} \right. \\ & \left. + \{ \tilde{V}(t_3), \{ \tilde{V}(t_2), \tilde{V}(t_1) \} \} \right) + \dots, \end{aligned} \quad (\text{B.36})$$

where the default integral domains are $(-\infty, +\infty)$. To obtain Eq. (B.36), one uses the Lie algebra homomorphism $[X_f, X_g] = X_{\{f, g\}}$ and also selects the natural default value $\Omega \rightarrow 0$ in the $\epsilon \rightarrow 0$ limit given the perturbative context.

In sum, the above exposition establishes a purely classical definition of the S-symplectomorphism S and the classical scattering generator Ω , concretely and explicitly by means of Dyson and Magnus series. Notably, the classical scattering generator serves as an efficient data type that is easy to handle while encoding the full information of classical scattering.

Diagrammatically, Eq. (B.36) amounts to using the retarded Green's function arising from the Poisson bracket between free-theory trajectories [897].

Although we have assumed a Hamiltonian deformation scenario in Eq. (B.31) for concreteness, the definitions in Eqs. (B.25) and (B.30) are coordinate-free, implying worldline field basis invariance. The symplectic deformation scenario stands much subtler, but it is being observed by examples that the same diagrammatic rules can be used (see, e.g., Ref. [899]).

B.2.3 Via Dequantization

The phase space formulation of quantum mechanics [913–919] is based on the idea of “quantization map,” which associates each function on a phase space $\mathcal{P}$ with an operator acting on a Hilbert space $\mathcal{H}$:

$$\mathcal{Q} : C^\infty(\mathcal{P}) \rightarrow \text{Ops}(\mathcal{H}). \quad (\text{B.37})$$

For simplicity and concreteness, we also restrict our interest to the cases in which this quantization map is invertible:

$$\mathcal{Q}^{-1} : \text{Ops}(\mathcal{H}) \rightarrow C^\infty(\mathcal{P}). \quad (\text{B.38})$$

As a result, each operator $\hat{f}$ is associated with a unique phase space function $f = \mathcal{Q}^{-1}(\hat{f})$, known as the *symbol* of $\hat{f}$. The inverse map $\mathcal{Q}^{-1}$ is colloquially referred to as the dequantization map.

Well-known examples of the quantization/dequantization maps include the Weyl and Wigner transforms [915, 916] as well as Glauber-Sudarshan P -map [922, 923] and Husimi Q -representation [924]. For spinning systems, elegant formalisms exist via compact phase spaces (see, e.g., Ref. [925]), while we underscore the fact that the massive twistor formalism arguably provides the best framework for quantizing/dequantizing relativistic spin in four dimensions via its coherent states and Darboux nature.

The phase space formulation of quantum mechanics describes the category of noncommutative manifolds, for which the star product is given by

$$f = \mathcal{Q}^{-1}(\hat{f}), \quad g = \mathcal{Q}^{-1}(\hat{g}) \implies f \star g = \mathcal{Q}^{-1}(\hat{f}\hat{g}). \quad (\text{B.39})$$

Automorphisms of the noncommutative ring of smooth functions (i.e., maps preserving the pointwise addition and the star product of smooth functions) will be identified as fuzzy diffeomorphisms.

The fuzzy Poisson bracket is defined as

$$\{f, g\}^\star = \frac{1}{i\hbar} \mathcal{Q}^{-1}([\hat{f}, \hat{g}]) = \frac{1}{i\hbar} (f \star g - g \star f). \quad (\text{B.40})$$

The fuzzy Hamiltonian vector field is defined as

$$X_f^\star = \{f, \cdot\}^\star \implies [X_f^\star, X_g^\star] = X_{\{f,g\}^\star}^\star, \quad (\text{B.41})$$

which is a derivation on the noncommutative ring. In general, a map on $C^\infty(\mathcal{P})$ will be called a fuzzy vector field on $\mathcal{P}$ if it is a derivation on the noncommutative ring. Concretely, this could describe a giant sum of differential operators of all degree.

The quantum scattering generator refers to the logarithm of the S-matrix:

$$\hat{\Omega} = i\hbar \log \hat{S} \in \text{Ops}(\mathcal{H}) \implies \hat{S} = \exp\left(\frac{1}{i\hbar} \hat{\Omega}\right). \quad (\text{B.42})$$

In the phase space formulation, its symbol is given by

$$\Omega^\star := \mathcal{Q}^{-1}(i\hbar \log \hat{S}). \quad (\text{B.43})$$

Consider the adjoint action of the S-matrix on quantum observables:

$$\text{Ad}_{\hat{S}} \hat{\mathcal{O}} = \exp\left(\frac{1}{i\hbar} \text{ad}_{\hat{\Omega}}\right) \hat{\mathcal{O}}. \quad (\text{B.44})$$

Upon dequantization, it follows that

$$\mathcal{Q}^{-1}(\text{Ad}_{\hat{S}} \hat{\mathcal{O}}) = \exp(X_{\Omega^\star}^\star)[\mathcal{O}]. \quad (\text{B.45})$$

Here, $\mathcal{O} = \mathcal{Q}^{-1}(\hat{\mathcal{O}})$. Since $X_{\Omega^\star}^\star$ is a derivation on the noncommutative ring, i.e., is a fuzzy vector field, its exponentiation in Eq. (B.45) defines an automorphism, i.e., a fuzzy diffeomorphism:

$$S^\star := \exp(X_{\Omega^\star}^\star) \in \text{Diff}^\star(\mathcal{P}). \quad (\text{B.46})$$

This means that

$$S^\star[f + g] = S^\star[f] + S^\star[g], \quad S^\star[f \star g] = S^\star[f] \star S^\star[g]. \quad (\text{B.47})$$

We refer to $S^\star$ in Eq. (B.46) as the fuzzy S-diffeomorphism. This is essentially the phase space formulation of the (adjoint) S-matrix.

In typical situations, the symbol of the quantum scattering generator admits a well-defined classical limit, dubbed classical scattering generator (reproducing Eq. (B.23)):

$$\Omega := \lim_{\hbar \rightarrow 0} \Omega^\star = \lim_{\hbar \rightarrow 0} \mathcal{Q}^{-1}(i\hbar \log \hat{S}). \quad (\text{B.48})$$

Accordingly, the automorphism in Eq. (B.46) also admits a well-defined classical limit, say S^* :

$$S^* := \lim_{\hbar \rightarrow 0} S^\star \in \text{Diff}(\mathcal{P}, \{ \cdot, \cdot \}). \quad (\text{B.49})$$

Since the classical limit of the fuzzy Poisson bracket in Eq. (B.40) defines the Poisson bracket of the classical theory, S^* in Eq. (B.49) is an automorphism of the Poisson algebra, i.e., represents a Poisson diffeomorphism S . Explicitly, Eq. (B.46) shows that it is the exponentiation of the Hamiltonian vector field of the classical scattering generator:

$$S^* = \exp(X_\Omega) \in \text{Diff}(\mathcal{P}, \{ \cdot, \cdot \}). \quad (\text{B.50})$$

In sum, the S-symplectomorphism S is the dequantization of the adjoint action of the S-matrix, up to the differential operator representation.

Finally, the impulse of quantum observables in quantum scattering is defined from the adjoint action of the inverse S-matrix:⁸

$$\Delta \hat{\mathcal{O}} = \hat{S}^{-1} \hat{\mathcal{O}} \hat{S} - \hat{\mathcal{O}} = \text{Ad}_{\hat{S}^{-1}} \hat{\mathcal{O}} - \hat{\mathcal{O}}. \quad (\text{B.51})$$

It should be clear that the dequantization of Eq. (B.51) is

$$\Delta \mathcal{O} = (S^{-1})^*[\mathcal{O}] - \mathcal{O} = e^{-X_\Omega}[\mathcal{O}] - \mathcal{O} = e^{\{ \cdot, \Omega \}}[\mathcal{O}] - \mathcal{O}, \quad (\text{B.52})$$

which reproduces Eq. (B.24) when expanded. That is, the impulse of classical observables in a classical scattering process is computed via the pullback of the S-symplectomorphism. This fact should be clear by recalling Eq. (B.29), in which case Eq. (B.52) is brought to

$$\Delta \mathcal{O} = \mathcal{O} \circ S - \mathcal{O}. \quad (\text{B.53})$$

B.2.4 In Deformation Quantization

The classical scattering generator is well-defined on Poisson manifolds [870]. Flipping the perspective, one could consider starting from the classical scattering map S and $\hbar$ -deforming it into a quantum S-diffeomorphism $S^\star$, without presuming the existence of the operators $\hat{S}$ and $\hat{\Omega}$ on a concrete Hilbert space.

In formal (not necessarily strict) deformation quantization [336, 926–931], $S^\star$ and $\Omega^\star$ will be defined as formal power series in $\hbar$. This essentially means to ignore the convergence properties, and for physicists, work perturbatively in the loop expansion yielding asymptotic series.

⁸To be more precise, the quantum impulse should be defined as the expectation value of this $\Delta \hat{\mathcal{O}}$ via a density matrix. The symbol of the density matrix evolves by the dequantization of the adjoint S-matrix, i.e., the fuzzy S-diffeomorphism.